

Table 13.1 Some Dominant and Recessive Traits in Humans

Recessive Traits	Phenotypes	Dominant Traits	Phenotypes
Albinism	Lack of melanin pigmentation	Middigital hair	Presence of hair on middle segment of fingers
Alkaptonuria	Inability to metabolize homogenistic acid	Brachydactyly	Short fingers
Red-green color blindness	Inability to distinguish red or green wavelengths of light	Huntington's disease	Degeneration of nervous system, starting in middle age
Cystic fibrosis	Abnormal gland secretion, leading to liver degeneration and lung failure	Phenylthiocarbamide (PTC) sensitivity	Ability to taste PTC as bitter
Duchenne muscular dystrophy	Wasting away of muscles during childhood	Camptodactyly	Inability to straighten the little finger
Hemophilia	Inability to form blood clots	Hypercholesterolemia (the most common human Mendelian disorder—1 in 500)	Elevated levels of blood cholesterol and risk of heart attack
Sickle cell anemia	Defective hemoglobin that causes red blood cells to curve and stick together	Polydactyly	Extra fingers and toes

To explain these results, Mendel proposed a simple model. It has become one of the most famous models in the history of science, containing simple assumptions and making clear predictions. The model has five elements:

1. Parents do not transmit physiological traits directly to their offspring. Rather, they transmit discrete information about the traits, what Mendel called “factors.” These factors later act in the offspring to produce the trait. In modern terms, we would say that information about the alternative forms of characters that an individual expresses is *encoded* by the factors that it receives from its parents.
2. Each individual receives two factors that may code for the same trait or for two alternative traits for a character. We now know that there are two factors for each character present in each individual because these factors are carried on chromosomes, and each adult individual is *diploid*. When the individual forms gametes (eggs or sperm), they contain only one of each kind of chromosome (see chapter 12); the gametes are *haploid*. Therefore, only one factor for each character of the adult organism is contained in the gamete. Which of the two factors ends up in a particular gamete is randomly determined.
3. Not all copies of a factor are identical. In modern terms, the alternative forms of a factor, leading to alternative forms of a character, are called **alleles**. When two haploid gametes containing exactly the same allele of a factor fuse during fertilization to form a zygote, the offspring that develops from that zygote is said to be **homozygous**; when the two haploid gametes contain different alleles, the individual offspring is **heterozygous**.

In modern terminology, Mendel’s factors are called **genes**. We now know that each gene is composed of a particular DNA nucleotide sequence (see chapter 3). The particular location of a gene on a chromosome is referred to as the gene’s **locus** (plural, loci).

4. The two alleles, one contributed by the male gamete and one by the female, do not influence each other in any way. In the cells that develop within the new individual, these alleles remain discrete. They neither blend with nor alter each other. (Mendel referred to them as “uncontaminated.”) Thus, when the individual matures and produces its own gametes, the alleles for each gene segregate randomly into these gametes, as described in element 2.
5. The presence of a particular allele does not ensure that the trait encoded by it will be expressed in an individual carrying that allele. In heterozygous individuals, only one allele (the dominant one) is expressed, while the other (recessive) allele is present but unexpressed. To distinguish between the presence of an allele and its expression, modern geneticists refer to the totality of alleles that an individual contains as the individual’s **genotype** and to the physical appearance of that individual as its **phenotype**. The phenotype of an individual is the observable outward manifestation of its genotype, the result of the functioning of the enzymes and proteins encoded by the genes it carries. In other words, the genotype is the blueprint, and the phenotype is the visible outcome.

These five elements, taken together, constitute Mendel’s model of the hereditary process. Many traits in humans also exhibit dominant or recessive inheritance, similar to the traits Mendel studied in peas (table 13.1).

When Mendel crossed two contrasting varieties, he found all of the offspring in the first generation exhibited one (dominant) trait, and none exhibited the other (recessive) trait. In the following generation, 25% were pure-breeding for the dominant trait, 50% were hybrid for the two traits and exhibited the dominant trait, and 25% were pure-breeding for the recessive trait.

How Mendel Interpreted His Results

Does Mendel's model predict the results he actually obtained? To test his model, Mendel first expressed it in terms of a simple set of symbols, and then used the symbols to interpret his results. It is very instructive to do the same. Consider again Mendel's cross of purple-flowered with white-flowered plants. We will assign the symbol P to the dominant allele, associated with the production of purple flowers, and the symbol p to the recessive allele, associated with the production of white flowers. By convention, genetic traits are usually assigned a letter symbol referring to their more common forms, in this case "P" for purple flower color. The dominant allele is written in upper case, as P ; the recessive allele (white flower color) is assigned the same symbol in lower case, p .

In this system, the genotype of an individual that is true-breeding for the recessive white-flowered trait would be designated pp . In such an individual, both copies of the allele specify the white-flowered phenotype. Similarly, the genotype of a true-breeding purple-flowered individual would be designated PP , and a heterozygote would be designated Pp (dominant allele first). Using these conventions, and denoting a cross between two strains with $\times$, we can symbolize Mendel's original cross as $pp \times PP$.

The F₁ Generation

Using these simple symbols, we can now go back and re-examine the crosses Mendel carried out. Because a white-flowered parent (pp) can produce only p gametes, and a pure purple-flowered (homozygous dominant) parent (PP) can produce only P gametes, the union of an egg and a sperm from these parents can produce only heterozygous Pp offspring in the F₁ generation. Because the P allele is dominant, all of these F₁ individuals are expected to have purple flowers. The p allele is present in these heterozygous individuals, but it is not phenotypically expressed. This is the basis for the latency Mendel saw in recessive traits.

The F₂ Generation

When F₁ individuals are allowed to self-fertilize, the P and p alleles segregate randomly during gamete formation. Their subsequent union at fertilization to form F₂ individuals is also random, not being influenced by which alternative alleles the individual gametes carry. What will the F₂ individuals look like? The possibilities may be visualized in a simple diagram called a **Punnett square**, named after its originator, the English geneticist Reginald Crundall Punnett (figure 13.13). Mendel's model, ana-

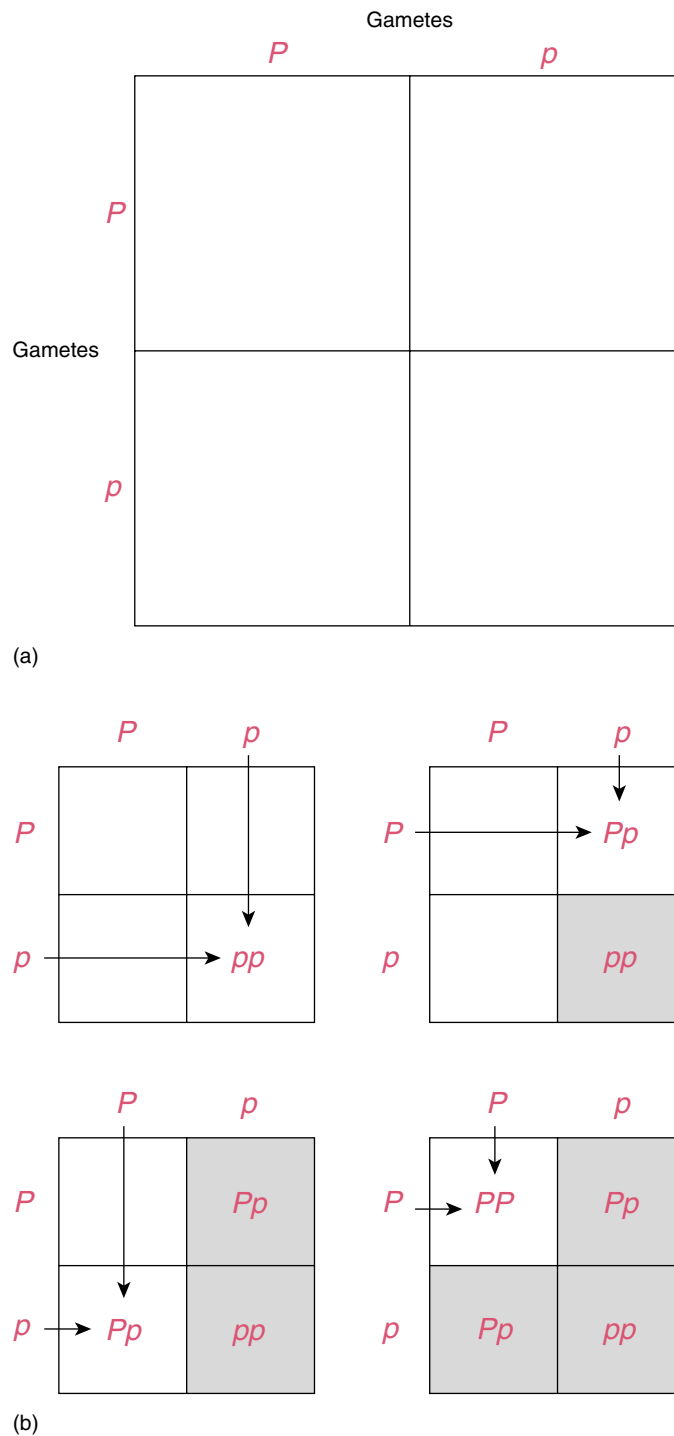


FIGURE 13.13

A Punnett square. (a) To make a Punnett square, place the different possible types of female gametes along one side of a square and the different possible types of male gametes along the other. (b) Each potential zygote can then be represented as the intersection of a vertical line and a horizontal line.

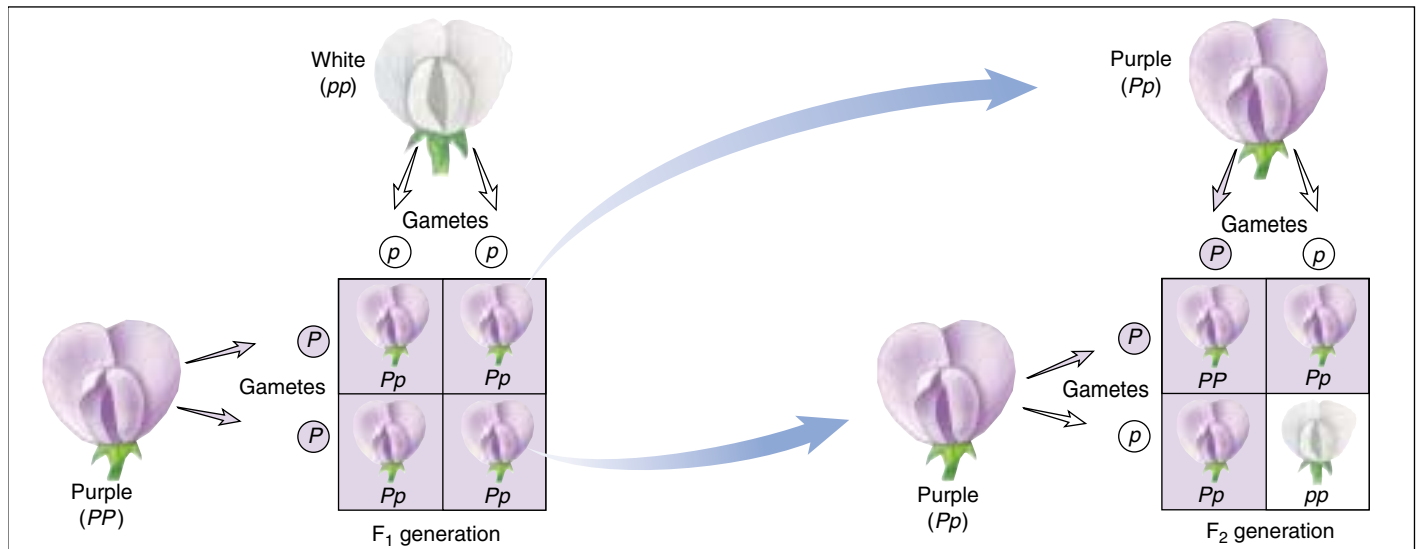


FIGURE 13.14

Mendel's cross of pea plants differing in flower color. All of the offspring of the first cross (the F_1 generation) are Pp heterozygotes with purple flowers. When two heterozygous F_1 individuals are crossed, three kinds of F_2 offspring are possible: PP homozygotes (purple flowers); Pp heterozygotes (also purple flowers); and pp homozygotes (white flowers). Therefore, in the F_2 generation, the ratio of dominant to recessive phenotypes is 3:1. However, the ratio of genotypes is 1:2:1 (1 PP : 2 Pp : 1 pp).

lyzed in terms of a Punnett square, clearly predicts that the F_2 generation should consist of $\frac{3}{4}$ purple-flowered plants and $\frac{1}{4}$ white-flowered plants, a phenotypic ratio of 3:1 (figure 13.14).

The Laws of Probability Can Predict Mendel's Results

A different way to express Mendel's result is to say that there are three chances in four ($\frac{3}{4}$) that any particular F_2 individual will exhibit the dominant trait, and one chance in four ($\frac{1}{4}$) that an F_2 individual will express the recessive trait. Stating the results in terms of probabilities allows simple predictions to be made about the outcomes of crosses. If both F_1 parents are Pp (heterozygotes), the probability that a particular F_2 individual will be pp (homozygous recessive) is the probability of receiving a p gamete from the male ($\frac{1}{2}$) times the probability of receiving a p gamete from the female ($\frac{1}{2}$), or $\frac{1}{4}$. This is the same operation we perform in the Punnett square illustrated in figure 13.13. The ways probability theory can be used to analyze Mendel's results is discussed in detail on page 251.

Further Generations

As you can see in figure 13.14, there are really three kinds of F_2 individuals: $\frac{1}{4}$ are pure-breeding, white-flowered individuals (pp); $\frac{1}{2}$ are heterozygous, purple-flowered individuals (Pp); and $\frac{1}{4}$ are pure-breeding, purple-flowered individuals (PP). The 3:1 phenotypic ratio is really a disguised 1:2:1 genotypic ratio.

Mendel's First Law of Heredity: Segregation

Mendel's model thus accounts in a neat and satisfying way for the segregation ratios he observed. Its central assumption—that alternative alleles of a character segregate from each other in heterozygous individuals and remain distinct—has since been verified in many other organisms. It is commonly referred to as **Mendel's First Law of Heredity**, or the **Law of Segregation**. As you saw in chapter 12, the segregational behavior of alternative alleles has a simple physical basis, the alignment of chromosomes at random on the metaphase plate during meiosis I. It is a tribute to the intellect of Mendel's analysis that he arrived at the correct scheme with no knowledge of the cellular mechanisms of inheritance; neither chromosomes nor meiosis had yet been described.

The Testcross

To test his model further, Mendel devised a simple and powerful procedure called the **testcross**. Consider a purple-flowered plant. It is impossible to tell whether such a plant is homozygous or heterozygous simply by looking at its phenotype. To learn its genotype, you must cross it with some other plant. What kind of cross would provide the answer? If you cross it with a homozygous dominant individual, all of the progeny will show the dominant phenotype whether the test plant is homozygous or heterozygous. It is also difficult (but not impossible) to distinguish between the two possible test plant genotypes by crossing with a heterozygous individual. However, if you cross the test plant with a homozygous recessive individual, the two possible test plant genotypes will give totally different results (figure 13.15):

- Alternative 1: unknown individual homozygous dominant (PP). $PP \times pp$: all offspring have purple flowers (Pp)
- Alternative 2: unknown individual heterozygous (Pp). $Pp \times pp$: $\frac{1}{2}$ of offspring have white flowers (pp) and $\frac{1}{2}$ have purple flowers (Pp)

To perform his testcross, Mendel crossed heterozygous F_1 individuals back to the parent homozygous for the recessive trait. He predicted that the dominant and recessive traits would appear in a 1:1 ratio, and that is what he observed. For each pair of alleles he investigated, Mendel observed phenotypic F_2 ratios of 3:1 (see figure 13.14) and testcross ratios very close to 1:1, just as his model predicted.

Testcrosses can also be used to determine the genotype of an individual when two genes are involved. Mendel carried out many two-gene crosses, some of which we will discuss. He often used testcrosses to verify the genotypes of particular dominant-appearing F_2 individuals. Thus, an F_2 individual showing both dominant traits ($A_ B_$) might have any of the following genotypes: $AABB$, $AaBB$, $AABb$, or $AaBb$. By crossing dominant-appearing F_2 individuals with homozygous recessive individuals (that is, $A_ B_ \times aabb$), Mendel was able to determine if either or both of the traits bred true among the progeny, and so to determine the genotype of the F_2 parent:

$AABB$	trait A breeds true	trait B breeds true
$AaBB$	_____	trait B breeds true
$AABb$	trait A breeds true	_____
$AaBb$	_____	_____

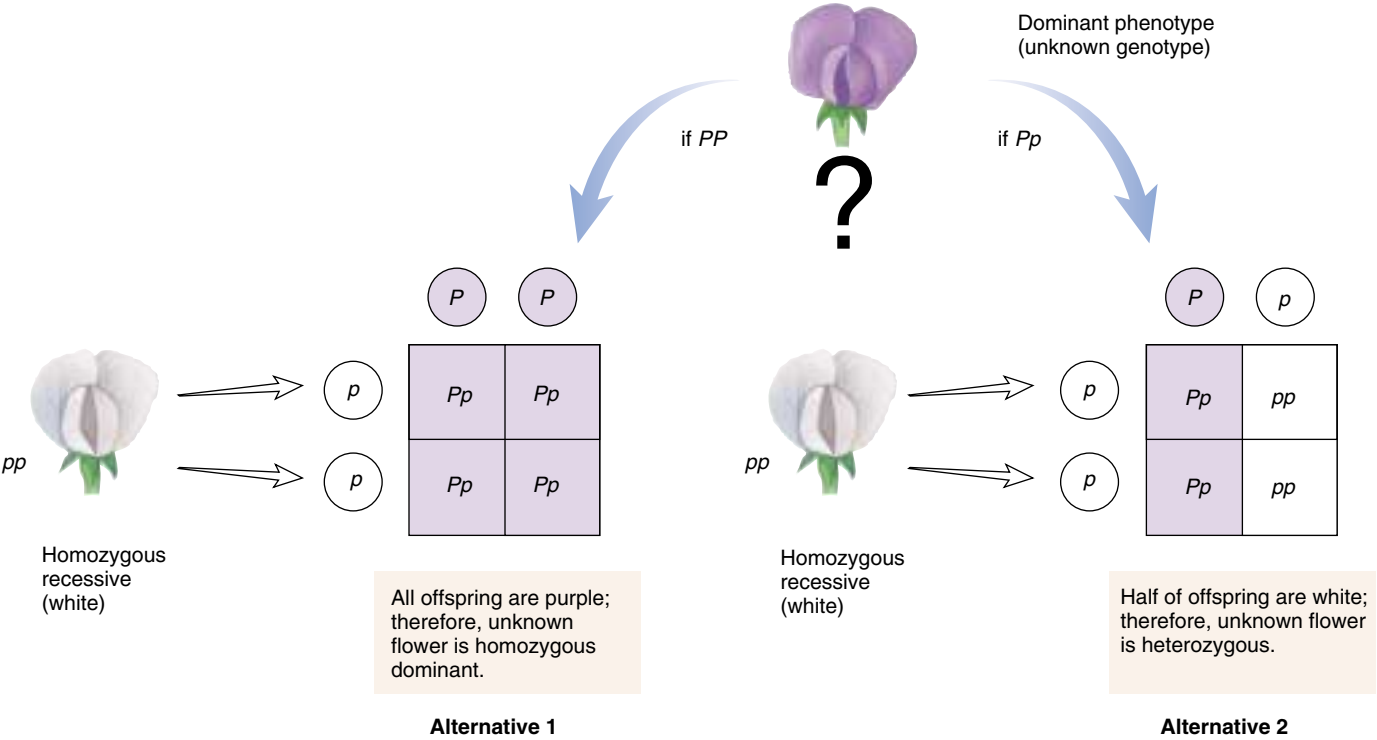


FIGURE 13.15
A testcross. To determine whether an individual exhibiting a dominant phenotype, such as purple flowers, is homozygous or heterozygous for the dominant allele, Mendel crossed the individual in question with a plant that he knew to be homozygous recessive, in this case a plant with white flowers.

Probability and Allele Distribution

Many, although not all, alternative alleles produce discretely different phenotypes. Mendel's pea plants were tall or dwarf, had purple or white flowers, and produced round or wrinkled seeds. The eye color of a fruit fly may be red or white, and the skin color of a human may be pigmented or albino. When only two alternative alleles exist for a given character, the distribution of phenotypes among the offspring of a cross is referred to as a **binomial distribution**.

As an example, consider the distribution of sexes in humans. Imagine that a couple has chosen to have three children. How likely is it that two of the children will be boys and one will be a girl? The frequency of any particular possibility is referred to as its **probability** of occurrence. Let p symbolize the probability of having a boy at any given birth and q symbolize the probability of having a girl. Since any birth is equally likely to produce a girl or boy:

$$p = q = \frac{1}{2}$$

Table 13.A shows eight possible gender combinations among the three children. The sum of the probabilities of the eight possible combinations must equal one. Thus:

$$p^3 + 3p^2q + 3pq^2 + q^3 = 1$$

The probability that the three children will be two boys and one girl is:

$$3p^2q = 3 \times (\frac{1}{2})^2 \times (\frac{1}{2}) = \frac{3}{8}$$

To test your understanding, try to estimate the probability that two parents heterozygous for the recessive allele producing albinism (a) will have one albino child in a family of three. First, set up a Punnett square:

		Father's Gametes	
		A	a
Mother's Gametes	A	AA	Aa
	a	Aa	aa

You can see that one-fourth of the children are expected to be albino (aa). Thus, for any given birth the probability of an albino child is $\frac{1}{4}$. This probability can be symbolized by q . The probability of a nonalbino child is $\frac{3}{4}$, symbolized by p . Therefore, the probability that there will be one albino child among the three children is:

$$3p^2q = 3 \times (\frac{3}{4})^2 \times (\frac{1}{4}) = \frac{27}{64}, \text{ or } 42\%$$

This means that the chance of having one albino child in the three is 42%.

Table 13.A Binomial Distribution of the Sexes of Children in Human Families

Composition of Family	Order of Birth	Calculation	Probability
3 boys	bbb	$p \times p \times p$	p^3
2 boys and 1 girl	bbg	$p \times p \times q$	p^2q
	bgb	$p \times q \times p$	p^2q
	gbb	$q \times p \times p$	p^2q
1 boy and 2 girls	ggb	$q \times q \times p$	pq^2
	gbg	$q \times p \times q$	pq^2
	bgg	$p \times q \times q$	pq^2
3 girls	ggg	$q \times q \times q$	q^3

Vocabulary of Genetics

allele One of two or more alternative forms of a gene.

diploid Having two sets of chromosomes, which are referred to as *homologues*. Animals and plants are diploid in the dominant phase of their life cycles as are some protists.

dominant allele An allele that dictates the appearance of heterozygotes. One allele is said to be dominant over another if a het-

erozygous individual with one copy of that allele has the same appearance as a homozygous individual with two copies of it.

gene The basic unit of heredity; a sequence of DNA nucleotides on a chromosome that encodes a polypeptide or RNA molecule and so determines the nature of an individual's inherited traits.

genotype The total set of genes present in the cells of an organism. This term is often also used to refer to the set of alleles at a single gene.

haploid Having only one set of chromosomes. Gametes, certain animals, protists and fungi, and certain stages in the life cycle of plants are haploid.

heterozygote A diploid individual carrying two different alleles of a gene on two homologous chromosomes. Most human beings are heterozygous for many genes.

homozygote A diploid individual carrying identical alleles of a gene on both homologous chromosomes.

locus The location of a gene on a chromosome.

phenotype The realized expression of the genotype; the observable manifestation of a trait (affecting an individual's structure, physiology, or behavior) that results from the biological activity of the DNA molecules.

recessive allele An allele whose phenotypic effect is masked in heterozygotes by the presence of a dominant allele.

Mendel's Second Law of Heredity: Independent Assortment

After Mendel had demonstrated that different traits of a given character (alleles of a given gene) segregate independently of each other in crosses, he asked whether different genes also segregate independently. Mendel set out to answer this question in a straightforward way. He first established a series of pure-breeding lines of peas that differed in just two of the seven characters he had studied. He then crossed contrasting pairs of the pure-breeding lines to create heterozygotes. In a cross involving different seed shape alleles (round, R , and wrinkled, r) and different seed color alleles (yellow, Y , and green, y), all the F_1 individuals were identical, each one heterozygous for both seed shape (Rr) and seed color (Yy). The F_1 individuals of such a cross are **dihybrids**, individuals heterozygous for both genes.

The third step in Mendel's analysis was to allow the dihybrids to self-fertilize. If the alleles affecting seed shape and seed color were segregating independently, then the probability that a particular pair of seed shape alleles would occur together with a particular pair of seed color alleles would be simply the product of the individual probabilities that each pair would occur separately. Thus, the probability that an individual with wrinkled green seeds ($rryy$) would appear in the F_2 generation would be equal to the probability of observing an individual with wrinkled seeds ($1/4$) times the probability of observing one with green seeds ($1/4$), or $1/16$.

Because the gene controlling seed shape and the gene controlling seed color are each represented by a pair of alternative alleles in the dihybrid individuals, four types of gametes are expected: RY , Ry , rY , and ry . Therefore, in the F_2 generation there are 16 possible combinations of alleles, each of them equally probable (figure 13.16). Of these, 9 possess at least one dominant allele for each gene (signified $R_Y_$, where the dash indicates the presence of either allele) and, thus, should have round, yellow seeds. Of the rest, 3 possess at least one dominant R allele but are homozygous recessive for color (R_yy); 3 others possess at least one dominant Y allele but are homozygous recessive for shape ($rrY_$); and 1 combination among the 16 is homozygous recessive for both genes ($rryy$). The hypothesis that color and shape genes assort independently thus predicts that the F_2 generation will display a 9:3:3:1 phenotypic ratio: nine individuals with round, yellow seeds, three with round, green seeds, three with wrinkled, yellow seeds, and one with wrinkled, green seeds (see figure 13.16).

What did Mendel actually observe? From a total of 556 seeds from dihybrid plants he had allowed to self-fertilize, he observed: 315 round yellow ($R_Y_$), 108 round green (R_yy), 101 wrinkled yellow ($rrY_$), and 32 wrinkled green ($rryy$). These results are very close to a 9:3:3:1 ratio (which would be 313:104:104:35). Consequently, the two genes appeared to assort completely independently of each other.

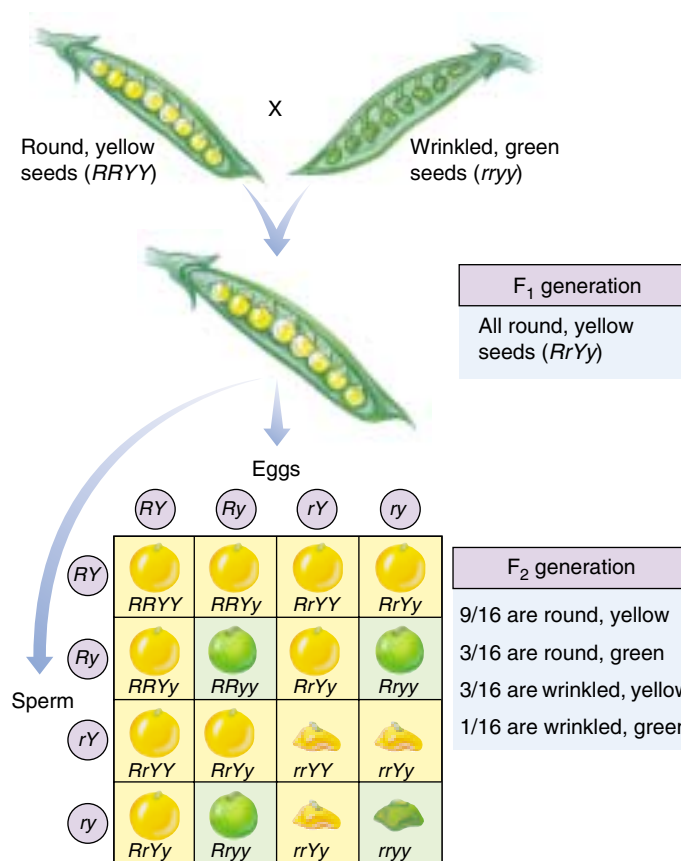


FIGURE 13.16
Analyzing a dihybrid cross. This Punnett square shows the results of Mendel's dihybrid cross between plants with round yellow seeds and plants with wrinkled green seeds. The ratio of the four possible combinations of phenotypes is predicted to be 9:3:3:1, the ratio that Mendel found.

Note that this independent assortment of different genes in no way alters the independent segregation of individual pairs of alleles. Round versus wrinkled seeds occur in a ratio of approximately 3:1 (423:133); so do yellow versus green seeds (416:140). Mendel obtained similar results for other pairs of traits.

Mendel's discovery is often referred to as **Mendel's Second Law of Heredity**, or the **Law of Independent Assortment**. Genes that assort independently of one another, like the seven genes Mendel studied, usually do so because they are located on different chromosomes, which segregate independently during the meiotic process of gamete formation. A modern restatement of Mendel's Second Law would be that *genes that are located on different chromosomes assort independently during meiosis*.

Mendel summed up his discoveries about heredity in two laws. Mendel's First Law of Heredity states that alternative alleles of a trait segregate independently; his Second Law of Heredity states that genes located on different chromosomes assort independently.

Mendelian Inheritance Is Not Always Easy to Analyze

Although Mendel's results did not receive much notice during his lifetime, three different investigators independently rediscovered his pioneering paper in 1900, 16 years after his death. They came across it while searching the literature in preparation for publishing their own findings, which closely resembled those Mendel had presented more than three decades earlier. In the decades following the rediscovery of Mendel, many investigators set out to test Mendel's ideas. However, scientists attempting to confirm Mendel's theory often had trouble obtaining the same simple ratios he had reported. Often, the expression of the genotype is not straightforward. Most phenotypes reflect the action of many genes that act sequentially or jointly, and the phenotype can be affected by alleles that lack complete dominance and the environment.

Continuous Variation

Few phenotypes are the result of the action of only one gene. Instead, most characters reflect the action of **polygenes**, many genes that act sequentially or jointly. When multiple genes act jointly to influence a character such as height or weight, the character often shows a range of small differences. Because all of the genes that play a role in determining phenotypes such as height or weight segregate independently of one another, one sees a gradation in the degree of difference when many individuals are examined (figure 13.17). We call this gradation **continuous variation**. The greater the number of genes that influence a character, the more continuous the expected distribution of the versions of that character.

How can one describe the variation in a character such as the height of the individuals in figure 13.17? Individuals range from quite short to very tall, with average heights more common than either extreme. What one often does is to group the variation into categories—in this case, by measuring the heights of the individuals in inches, rounding fractions of an inch to the nearest whole number. Each height, in inches, is a separate phenotypic category. Plotting the numbers in each height category produces a histogram, such as that in figure 13.17. The histogram approximates an idealized bell-shaped curve, and the variation can be characterized by the mean and spread of that curve.

Pleiotropic Effects

Often, an individual allele will have more than one effect on the phenotype. Such an allele is said to be **pleiotropic**. When the pioneering French geneticist Lucien Cuenot studied yellow fur in mice, a dominant trait, he was unable to obtain a true-breeding yellow strain by crossing individual yellow mice with each other. Individuals homozygous for the yellow allele died, because the yellow allele was

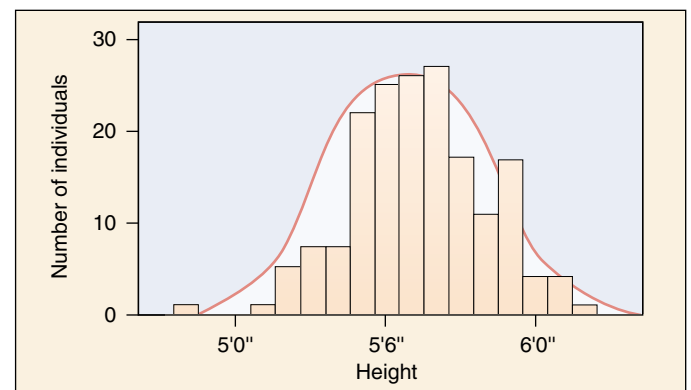
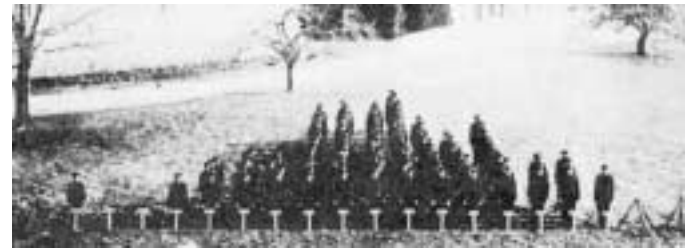


FIGURE 13.17

Height is a continuously varying trait. The photo shows variation in height among students of the 1914 class of the Connecticut Agricultural College. Because many genes contribute to height and tend to segregate independently of one another, the cumulative contribution of different combinations of alleles to height forms a *continuous* distribution of possible height, in which the extremes are much rarer than the intermediate values.

pleiotropic: one effect was yellow coat color, but another was a lethal developmental defect. A pleiotropic allele may be dominant with respect to one phenotypic consequence (yellow fur) and recessive with respect to another (lethal developmental defect). In pleiotropy, one gene affects many traits, in marked contrast to polygeny, where many genes affect one trait. Pleiotropic effects are difficult to predict, because the genes that affect a trait often perform other functions we may know nothing about.

Pleiotropic effects are characteristic of many inherited disorders, such as cystic fibrosis and sickle cell anemia, both discussed later in this chapter. In these disorders, multiple symptoms can be traced back to a single gene defect. In cystic fibrosis, patients exhibit clogged blood vessels, overly sticky mucus, salty sweat, liver and pancreas failure, and a battery of other symptoms. All are pleiotropic effects of a single defect, a mutation in a gene that encodes a chloride ion transmembrane channel. In sickle cell anemia, a defect in the oxygen-carrying hemoglobin molecule causes anemia, heart failure, increased susceptibility to pneumonia, kidney failure, enlargement of the spleen, and many other symptoms. It is usually difficult to deduce the nature of the primary defect from the range of a gene's pleiotropic effects.

Lack of Complete Dominance

Not all alternative alleles are fully dominant or fully recessive in heterozygotes. Some pairs of alleles instead produce a heterozygous phenotype that is either intermediate between those of the parents (incomplete dominance), or representative of both parental phenotypes (codominance). For example, in the cross of red and white flowering Japanese four o'clocks described in figure 13.18, all the F_1 offspring had pink flowers—indicating that neither red nor white flower color was dominant. Does this example of incomplete dominance argue that Mendel was wrong? Not at all. When two of the F_1 pink flowers were crossed, they produced red-, pink-, and white-flowered plants in a 1:2:1 ratio. Heterozygotes are simply intermediate in color.

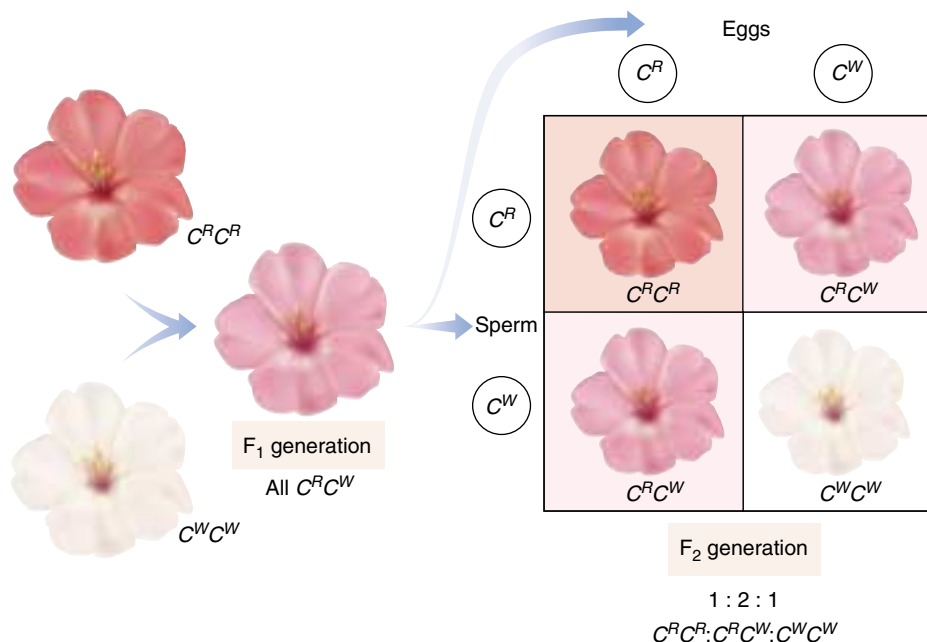


FIGURE 13.18

Incomplete dominance. In a cross between a red-flowered Japanese four o'clock, genotype $C^R C^R$, and a white-flowered one ($C^W C^W$), neither allele is dominant. The heterozygous progeny have pink flowers and the genotype $C^R C^W$. If two of these heterozygotes are crossed, the phenotypes of their progeny occur in a ratio of 1:2:1 (red:pink:white).

Environmental Effects

The degree to which an allele is expressed may depend on the environment. Some alleles are heat-sensitive, for example. Traits influenced by such alleles are more sensitive to temperature or light than are the products of other alleles. The arctic foxes in figure 13.19, for example, make fur pigment only when the weather is warm. Similarly, the *ch* allele in Himalayan rabbits and Siamese cats encodes a heat-sensitive version of tyrosinase, one of the enzymes mediating the production of melanin, a dark pigment. The *ch* version of the enzyme is inactivated at temperatures above about 33°C. At the surface of the body and head, the temperature is above 33°C and the tyrosinase enzyme is inactive, while it is more active at body extremities such as the tips of the ears and tail, where the temperature is below 33°C. The dark melanin pigment this enzyme produces causes the ears, snout, feet, and tail of Himalayan rabbits and Siamese cats to be black.



(a)



(b)

FIGURE 13.19

Environmental effects on an allele. (a) An arctic fox in winter has a coat that is almost white, so it is difficult to see the fox against a snowy background. (b) In summer, the same fox's fur darkens to a reddish brown, so that it resembles the color of the surrounding tundra. Heat-sensitive alleles control this color change.

Epistasis

In the tests of Mendel's ideas that followed the rediscovery of his work, scientists had trouble obtaining Mendel's simple ratios particularly with dihybrid crosses. Recall that when individuals heterozygous for two different genes mate (a dihybrid cross), four different phenotypes are possible among the progeny: offspring may display the dominant phenotype for both genes, either one of the genes, or for neither gene. Sometimes, however, it is not possible for an investigator to identify successfully each of the four phenotypic classes, because two or more of the classes look alike. Such situations proved confusing to investigators following Mendel.

One example of such difficulty in identification is seen in the analysis of particular varieties of corn, *Zea mays*. Some commercial varieties exhibit a purple pigment called anthocyanin in their seed coats, while others do not. In 1918, geneticist R. A. Emerson crossed two pure-breeding corn varieties, neither exhibiting anthocyanin pigment. Surprisingly, all of the F_1 plants produced purple seeds.

When two of these pigment-producing F_1 plants were crossed to produce an F_2 generation, 56% were pigment producers and 44% were not. What was happening? Emerson correctly deduced that two genes were involved in producing pigment, and that the second cross had thus been a dihybrid cross. Mendel had predicted 16 equally possible ways gametes could combine. How many of these were in each of the two types Emerson obtained? He multiplied the fraction that were pigment producers (0.56) by 16 to obtain 9, and multiplied the fraction that were not (0.44) by 16 to obtain 7. Thus, Emerson had a **modified ratio** of 9:7 instead of the usual 9:3:3:1 ratio.

Why Was Emerson's Ratio Modified? When genes act sequentially, as in a biochemical pathway, an allele expressed as a defective enzyme early in the pathway blocks the flow of material through the rest of the pathway. This makes it impossible to judge whether the later steps of the pathway are functioning properly. Such gene interaction, where one gene can interfere with the expression of another gene, is the basis of the phenomenon called **epistasis**.

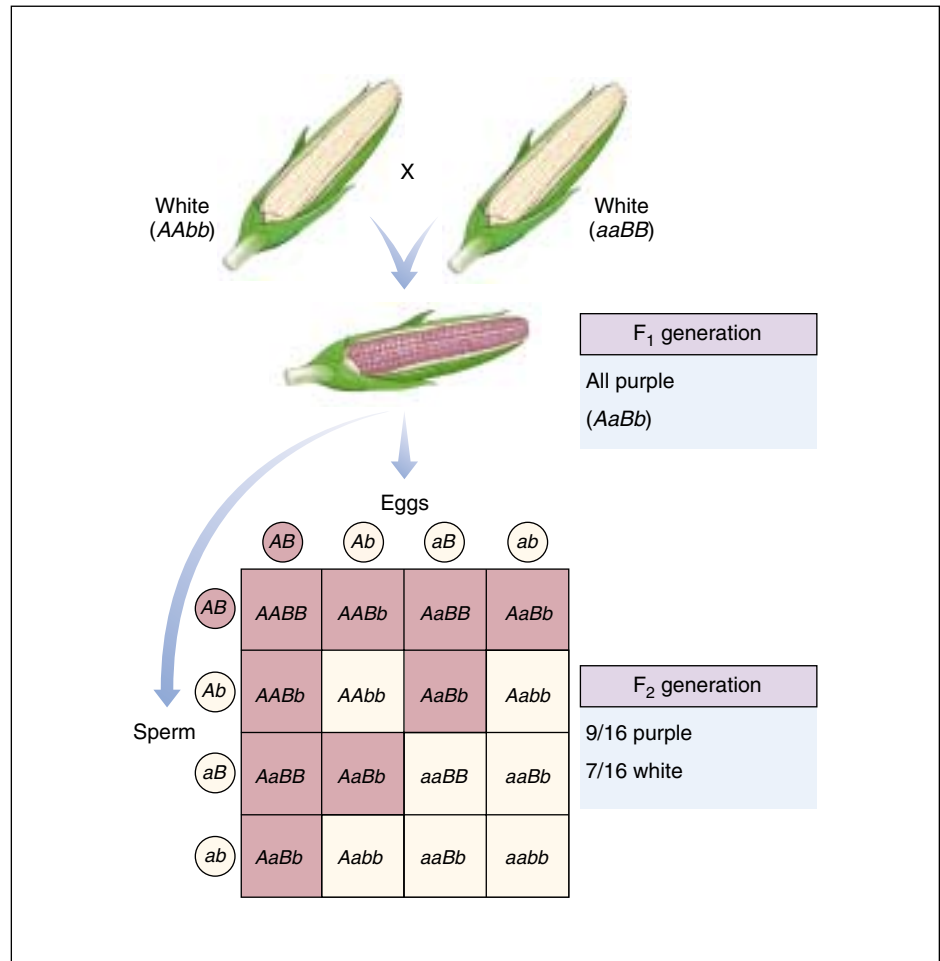
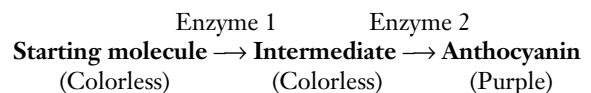


FIGURE 13.20

How epistasis affects grain color. The purple pigment found in some varieties of corn is the product of a two-step biochemical pathway. Unless both enzymes are active (the plant has a dominant allele for each of the two genes, A and B), no pigment is expressed.

The pigment anthocyanin is the product of a two-step biochemical pathway:



To produce pigment, a plant must possess at least one functional copy of each enzyme gene (figure 13.20). The dominant alleles encode functional enzymes, but the recessive alleles encode nonfunctional enzymes. Of the 16 genotypes predicted by random assortment, 9 contain at least one dominant allele of both genes; they produce purple progeny. The remaining 7 genotypes lack dominant alleles at either or both loci ($3 + 3 + 1 = 7$) and so are phenotypically the same (nonpigmented), giving the phenotypic ratio of 9:7 that Emerson observed. The inability to see the effect of enzyme 2 when enzyme 1 is nonfunctional is an example of epistasis.

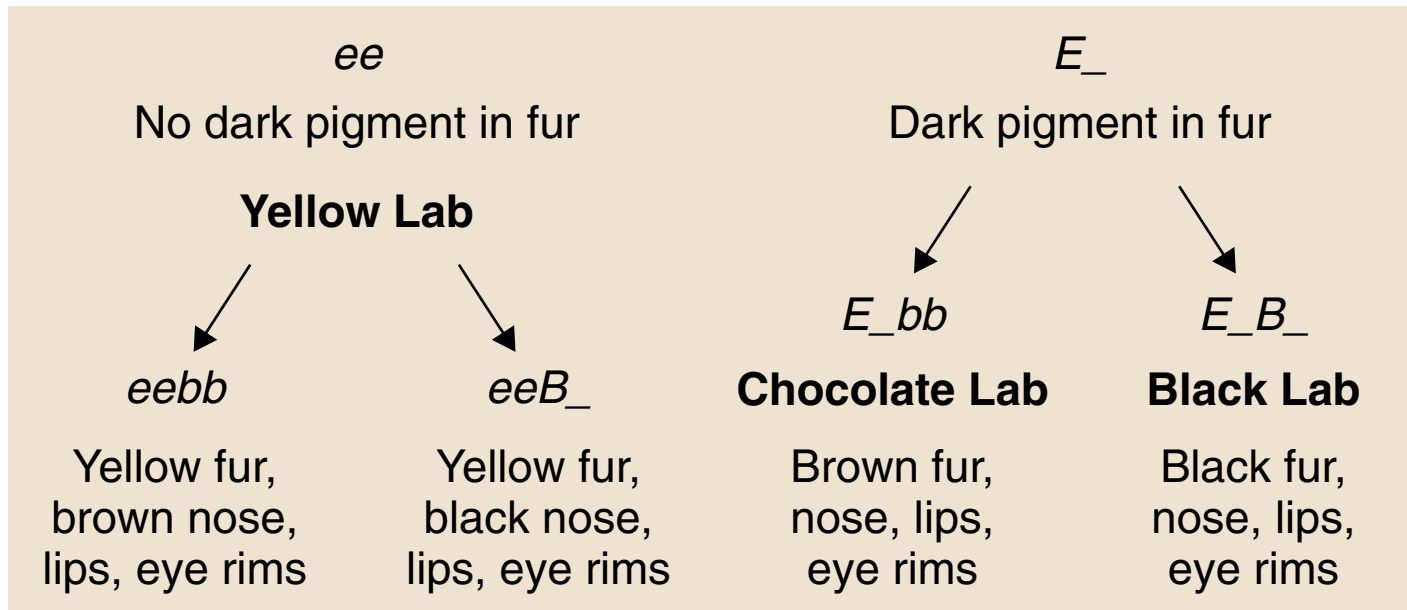


FIGURE 13.21

The effect of epistatic interactions on coat color in dogs. The coat color seen in Labrador retrievers is an example of the interaction of two genes, each with two alleles. The E gene determines if the pigment will be deposited in the fur, and the B gene determines how dark the pigment will be.

Other Examples of Epistasis

In many animals, coat color is the result of epistatic interactions among genes. Coat color in Labrador retrievers, a breed of dog, is due primarily to the interaction of two genes. The E gene determines if dark pigment (eumelanin) will be deposited in the fur or not. If a dog has the genotype ee , no pigment will be deposited in the fur, and it will be yellow. If a dog has the genotype EE or Ee ($E_$), pigment will be deposited in the fur.

A second gene, the B gene, determines how dark the pigment will be. This gene controls the distribution of melanosomes in a hair. Dogs with the genotype E_bb will have brown fur and are called chocolate labs. Dogs with the genotype $E_B_$ will have black fur. But, even in yellow dogs, the B gene does have some effect. Yellow dogs with

the genotype $eebb$ will have brown pigment on their nose, lips, and eye rims, while yellow dogs with the genotype $eeB_$ will have black pigment in these areas. The interaction among these alleles is illustrated in figure 13.21. The genes for coat color in this breed have been found, and a genetic test is available to determine the coat colors in a litter of puppies.

A variety of factors can disguise the Mendelian segregation of alleles. Among them are the continuous variation that results when many genes contribute to a trait, incomplete dominance and codominance that produce heterozygotes unlike either parent, environmental influences on the expression of phenotypes, and gene interactions that produce epistasis.

13.2 Human genetics follows Mendelian principles.

Random changes in genes, called mutations, occur in any population. These changes rarely improve the functioning of the proteins those genes encode, just as randomly changing a wire in a computer rarely improves the computer's functioning. Mutant alleles are usually recessive to other alleles. When two seemingly normal individuals who are heterozygous for such an allele produce offspring homozygous for the allele, the offspring suffer the detrimental effects of the mutant allele. When a detrimental allele occurs at a significant frequency in a population, the harmful effect it produces is called a **gene disorder**.

Most Gene Disorders Are Rare

Tay-Sachs disease is an incurable hereditary disorder in which the nervous system deteriorates. Affected children appear normal at birth and usually do not develop symptoms until about the eighth month, when signs of mental deterioration appear. The children are blind within a year after birth, and they rarely live past five years of age.

Tay-Sachs disease is rare in most human populations, occurring in only 1 of 300,000 births in the United States. However, the disease has a high incidence among Jews of Eastern and Central Europe (Ashkenazi), and among American Jews, 90% of whom trace their ancestry to Eastern and Central Europe. In these populations, it is estimated that 1 in 28 individuals is a heterozygous carrier of the disease, and approximately 1 in 3500 infants has the disease. Because the disease is caused by a recessive allele, most of the people who carry the defective allele do not themselves develop symptoms of the disease.

The Tay-Sachs allele produces the disease by encoding a nonfunctional form of the enzyme hexosaminidase A. This enzyme breaks down *gangliosides*, a class of lipids occurring within the lysosomes of brain cells (figure 13.22). As a result, the lysosomes fill with gangliosides, swell, and eventually burst, releasing oxidative enzymes that kill the cells. There is no known cure for this disorder.

Not All Gene Defects Are Recessive

Not all hereditary disorders are recessive. **Huntington's disease** is a hereditary condition caused by a dominant allele that leads to the progressive deterioration of brain cells (figure 13.23). Perhaps 1 in 24,000 individuals develops the disorder. Because the allele is dominant, every individual that carries the allele expresses the disorder. Nevertheless, the disorder persists in human populations because its symptoms usually do not develop until the affected individuals are more than 30 years old, and by that time most of those individuals have already had children. Consequently, the allele is often transmitted before the lethal condition develops. A person who is heterozygous for Huntington's

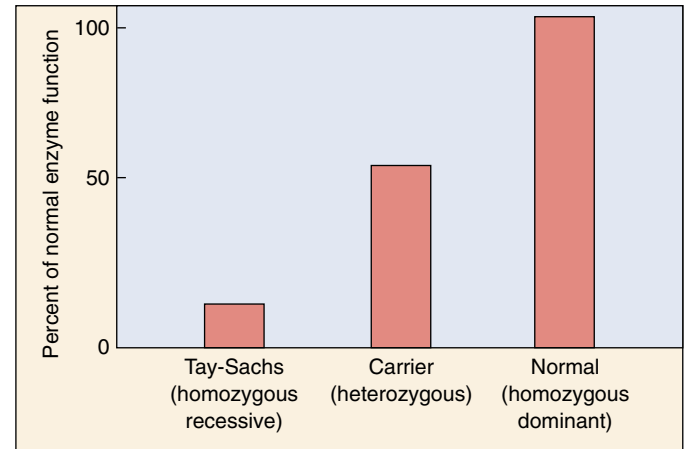


FIGURE 13.22

Tay-Sachs disease. Homozygous individuals (*left bar*) typically have less than 10% of the normal level of hexosaminidase A (*right bar*), while heterozygous individuals (*middle bar*) have about 50% of the normal level—enough to prevent deterioration of the central nervous system.

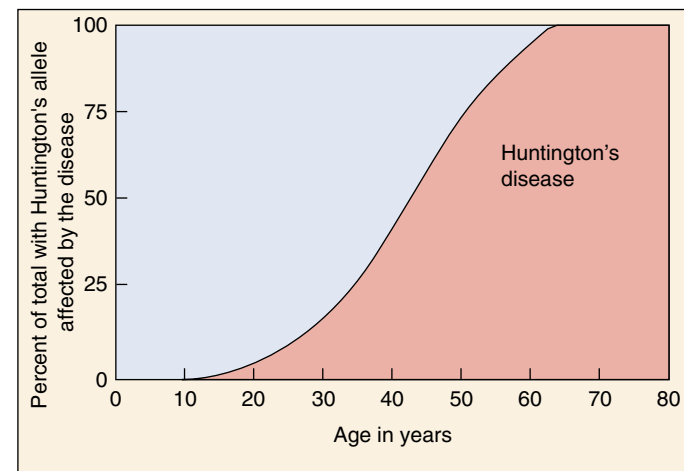


FIGURE 13.23

Huntington's disease is a dominant genetic disorder. It is because of the late age of onset of this disease that it persists despite the fact that it is dominant and fatal.

disease has a 50% chance of passing the *disease* to his or her children (even though the other parent does not have the disorder). In contrast, the carrier of a recessive disorder such as cystic fibrosis has a 50% chance of passing the *allele* to offspring and must mate with another carrier to risk bearing a child with the disease.

Most gene defects are rare recessives, although some are dominant.

Multiple Alleles: The ABO Blood Groups

A gene may have more than two alleles in a population, and most genes possess several different alleles. Often, no single allele is dominant; instead, each allele has its own effect, and the alleles are considered **codominant**.

A human gene with more than one codominant allele is the gene that determines ABO blood type. This gene encodes an enzyme that adds sugar molecules to lipids on the surface of red blood cells. These sugars act as recognition markers for the immune system. The gene that encodes the enzyme, designated *I*, has three common alleles: *I^A*, whose product adds galactose; *I^B*, whose product adds galactosamine; and *i*, which codes for a protein that does not add a sugar.

Different combinations of the three *I* gene alleles occur in different individuals because each person possesses two copies of the chromosome bearing the *I* gene and may be homozygous for any allele or heterozygous for any two. An individual heterozygous for the *I^A* and *I^B* alleles produces both forms of the enzyme and adds both galactose and galactosamine to the surfaces of red blood cells. Because both alleles are expressed simultaneously in heterozygotes, the *I^A* and *I^B* alleles are codominant. Both *I^A* and *I^B* are dominant over the *i* allele because both *I^A* or *I^B* alleles lead to sugar addition and the *i* allele does not. The different combinations of the three alleles produce four different phenotypes (figure 13.24):

1. Type A individuals add only galactosamine. They are either *I^AI^A* homozygotes or *I^Ai* heterozygotes.
2. Type B individuals add only galactose. They are either *I^BI^B* homozygotes or *I^Bi* heterozygotes.
3. Type AB individuals add both sugars and are *I^AI^B* heterozygotes.
4. Type O individuals add neither sugar and are *ii* homozygotes.

These four different cell surface phenotypes are called the **ABO blood groups** or, less commonly, the Landsteiner blood groups, after the man who first described them. As Landsteiner noted, a person's immune system can distinguish between these four phenotypes. If a type A individual receives a transfusion of type B blood, the recipient's immune system recognizes that the type B blood cells possess a "foreign" antigen (galactose) and attacks the donated blood cells, causing the cells to clump, or agglutinate. This also happens if the donated blood is type AB. However, if the donated blood is type O, no immune attack will occur, as there are no galactose antigens on the surfaces of blood cells produced by the type O donor. In general, any individual's immune system will tolerate a transfusion of type O blood. Because neither galactose nor galactosamine is foreign to type AB individuals (whose red blood cells have both sugars), those individuals may receive any type of blood.

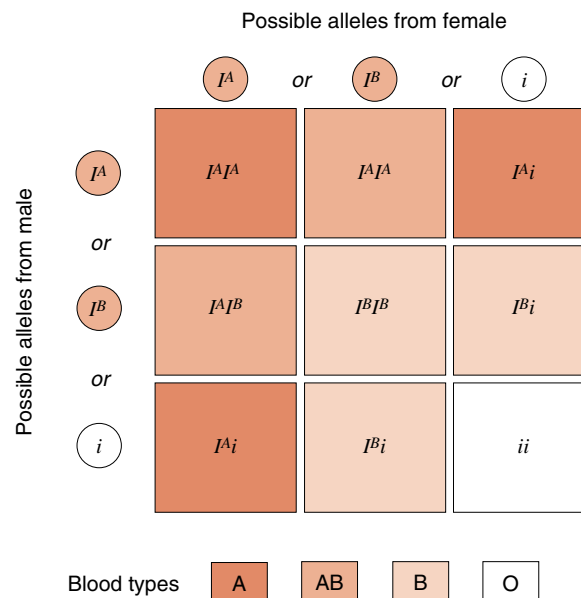


FIGURE 13.24
Multiple alleles control the ABO blood groups. Different combinations of the three *I* gene alleles result in four different blood type phenotypes: type A (either *I^AI^A* homozygotes or *I^Ai* heterozygotes), type B (either *I^BI^B* homozygotes or *I^Bi* heterozygotes), type AB (*I^AI^B* heterozygotes), and type O (*ii* homozygotes).

The Rh Blood Group

Another set of cell surface markers on human red blood cells are the **Rh blood group** antigens, named for the rhesus monkey in which they were first described. About 85% of adult humans have the Rh cell surface marker on their red blood cells, and are called Rh-positive. Rh-negative persons lack this cell surface marker because they are homozygous for the recessive gene encoding it.

If an Rh-negative person is exposed to Rh-positive blood, the Rh surface antigens of that blood are treated like foreign invaders by the Rh-negative person's immune system, which proceeds to make antibodies directed against the Rh antigens. This most commonly happens when an Rh-negative woman gives birth to an Rh-positive child (whose father is Rh-positive). At birth, some fetal red blood cells cross the placental barrier and enter the mother's bloodstream, where they induce the production of "anti-Rh" antibodies. In subsequent pregnancies, the mother's antibodies can cross back to the new fetus and cause its red blood cells to clump, leading to a potentially fatal condition called erythroblastosis fetalis.

Many blood group genes possess multiple alleles, several of which may be common.

Patterns of Inheritance Can Be Deduced from Pedigrees

When a blood vessel ruptures, the blood in the immediate area of the rupture forms a solid gel called a clot. The clot forms as a result of the polymerization of protein fibers circulating in the blood. A dozen proteins are involved in this process, and all must function properly for a blood clot to form. A mutation causing any of these proteins to lose their activity leads to a form of **hemophilia**, a hereditary condition in which the blood is slow to clot or does not clot at all.

Hemophilias are recessive disorders, expressed only when an individual does not possess any copy of the normal allele and so cannot produce one of the proteins necessary for clotting. Most of the genes that encode the blood-clotting proteins are on autosomes, but two (designated VIII and IX) are on the X chromosome. These two genes are sex-linked: any male who inherits a mutant allele of either of the two genes will develop hemophilia because his other sex chromosome is a Y chromosome that lacks any alleles of those genes.

The most famous instance of hemophilia, often called the Royal hemophilia, is a sex-linked form that arose in one of the parents of Queen Victoria of England (1819–1901; figure 13.25). In the five generations since Queen Victoria, 10 of her male descendants have had hemophilia. The present British royal family has escaped the disorder because Queen Victoria's son, King Edward VII, did not inherit the defective allele, and all the subsequent rulers of England are his descendants. Three of Victoria's nine children did receive the defective allele, however, and they carried it by marriage into many of the other royal families of Europe (figure 13.26), where it is still being passed to future generations—except in Russia, where all of the five children of Victoria's granddaughter Alexandra were killed soon after the Russian revolution in 1917. (Speculation that one daughter, Anastasia, survived ended in 1999 when DNA analysis confirmed the identity of her remains.)

Family pedigrees can reveal the mode of inheritance of a hereditary trait.



FIGURE 13.25

Queen Victoria of England, surrounded by some of her descendants in 1894. Of Victoria's four daughters who lived to bear children, two, Alice and Beatrice, were carriers of Royal hemophilia. Two of Alice's daughters are standing behind Victoria (wearing feathered boas): Princess Irene of Prussia (right), and Alexandra (left), who would soon become Czarina of Russia. Both Irene and Alexandra were also carriers of hemophilia.

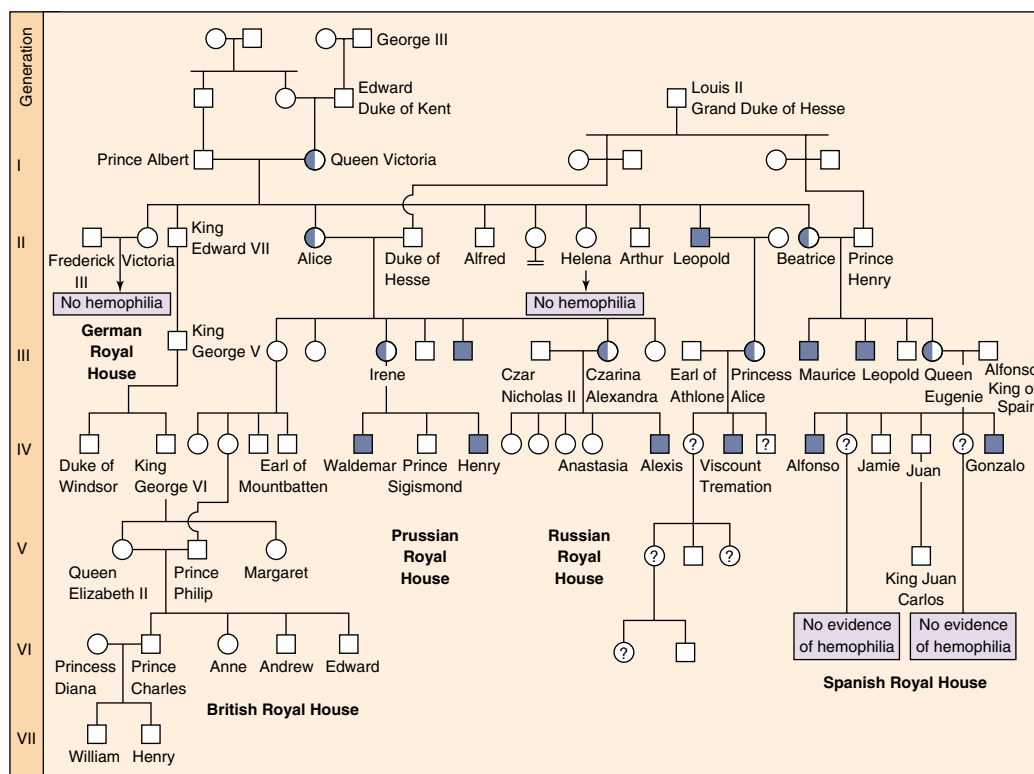


FIGURE 13.26

The Royal hemophilia pedigree. Queen Victoria's daughter Alice introduced hemophilia into the Russian and Austrian royal houses, and Victoria's daughter Beatrice introduced it into the Spanish royal house. Victoria's son Leopold, himself a victim, also transmitted the disorder in a third line of descent. Half-shaded symbols represent carriers with one normal allele and one defective allele; fully shaded symbols represent affected individuals.

Gene Disorders Can Be Due to Simple Alterations of Proteins

Sickle cell anemia is a heritable disorder first noted in Chicago in 1904. Afflicted individuals have defective molecules of hemoglobin, the protein within red blood cells that carries oxygen. Consequently, these individuals are unable to properly transport oxygen to their tissues. The defective hemoglobin molecules stick to one another, forming stiff, rod-like structures and resulting in the formation of sickle-shaped red blood cells (figure 13.27). As a result of their stiffness and irregular shape, these cells have difficulty moving through the smallest blood vessels; they tend to accumulate in those vessels and form clots. People who have large proportions of sickle-shaped red blood cells tend to have intermittent illness and a shortened life span.

The hemoglobin in the defective red blood cells differs from that in normal red blood cells in only one of hemoglobin's 574 amino acid subunits. In the defective hemoglobin, the amino acid valine replaces a glutamic acid at a single position in the protein. Interestingly, the position of the change is far from the active site of hemoglobin where the iron-bearing heme group binds oxygen. Instead, the change occurs on the outer edge of the protein. Why then is the result so catastrophic? The sickle cell mutation puts a very nonpolar amino acid on the surface of the hemoglobin protein, creating a "sticky patch" that sticks to other such patches—nonpolar amino acids tend to associate with one another in polar environments like water. As one hemoglobin adheres to another, ever-longer chains of hemoglobin molecules form.

Individuals heterozygous for the sickle cell allele are generally indistinguishable from normal persons. However, some of their red blood cells show the sickling characteristic when they are exposed to low levels of oxygen. The allele responsible for sickle cell anemia is particularly common among people of African descent; about 9% of African Americans are heterozygous for this allele, and about 0.2% are homozygous and therefore have the disorder. In some groups of people in Africa, up to 45% of all individuals are heterozygous for this allele, and 6% are homozygous. What factors determine the high frequency of sickle cell anemia in Africa? It turns out that heterozygosity for the sickle cell anemia allele increases



FIGURE 13.27
Sickle cell anemia. In individuals homozygous for the sickle cell trait, many of the red blood cells have sickle or irregular shapes, such as the cell on the far right.

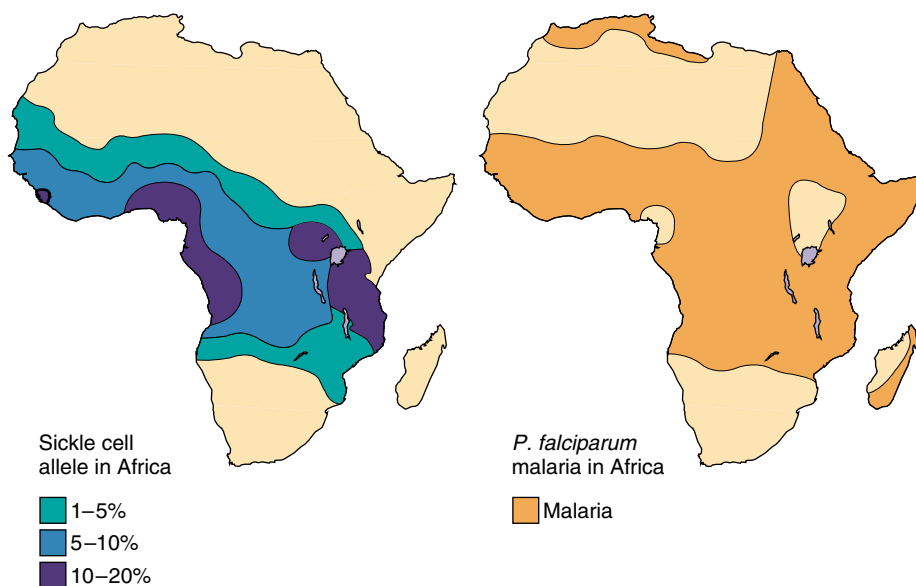


FIGURE 13.28
The sickle cell allele increases resistance to malaria. The distribution of sickle cell anemia closely matches the occurrence of malaria in central Africa. This is not a coincidence. The sickle cell allele, when heterozygous, increases resistance to malaria, a very serious disease.

resistance to malaria, a common and serious disease in central Africa (figure 13.28). We will discuss this situation in detail in chapter 21.

Sickle cell anemia is caused by a single-nucleotide change in the gene for hemoglobin, producing a protein with a nonpolar amino acid on its surface that tends to make the molecules clump together.

Table 13.2 Some Important Genetic Disorders

Disorder	Symptom	Defect	Dominant/ Recessive	Frequency among Human Births
Cystic fibrosis	Mucus clogs lungs, liver, and pancreas	Failure of chloride ion transport mechanism	Recessive	1/2500 (Caucasians)
Sickle cell anemia	Poor blood circulation	Abnormal hemoglobin molecules	Recessive	1/625 (African Americans)
Tay-Sachs disease	Deterioration of central nervous system in infancy	Defective enzyme (hexosaminidase A)	Recessive	1/3500 (Ashkenazi Jews)
Phenylketonuria	Brain fails to develop in infancy	Defective enzyme (phenylalanine hydroxylase)	Recessive	1/12,000
Hemophilia	Blood fails to clot	Defective blood clotting factor VIII	Sex-linked recessive	1/10,000 (Caucasian males)
Huntington's disease	Brain tissue gradually deteriorates in middle age	Production of an inhibitor of brain cell metabolism	Dominant	1/24,000
Muscular dystrophy (Duchenne)	Muscles waste away	Degradation of myelin coating of nerves stimulating muscles	Sex-linked recessive	1/3700 (males)
Hypercholesterolemia	Excessive cholesterol levels in blood, leading to heart disease	Abnormal form of cholesterol cell surface receptor	Dominant	1/500

Some Defects May Soon Be Curable

Some of the most common and serious gene defects result from single recessive mutations, including many of the defects listed in table 13.2. Recent developments in gene technology have raised the hope that this class of disorders may be curable. Perhaps the best example is **cystic fibrosis (CF)**, the most common fatal genetic disorder among Caucasians.

Cystic fibrosis is a fatal disease in which the body cells of affected individuals secrete a thick mucus that clogs the airways of the lungs. These same secretions block the ducts of the pancreas and liver so that the few patients who do not die of lung disease die of liver failure. There is no known cure.

Cystic fibrosis results from a defect in a single gene, called *cf*, that is passed down from parent to child. One in 20 individuals possesses at least one copy of the defective gene. Most carriers are not afflicted with the disease; only those children who inherit a copy of the defective gene from each parent succumb to cystic fibrosis—about 1 in 2500 infants.

The function of the *cf* gene has proven difficult to study. In 1985 the first clear clue was obtained. An investigator, Paul Quinton, seized on a commonly observed characteristic of cystic fibrosis patients, that their sweat is abnormally salty, and performed the following experiment. He isolated a sweat duct from a small piece of skin and placed it in a solution of salt (NaCl) that was three times as concentrated as the NaCl inside the duct. He then monitored the movement of ions. Diffusion tends to drive both the sodium (Na^+) and the chloride (Cl^-) ions into the duct because of the higher outer ion concentrations. In skin isolated from

normal individuals, Na^+ and Cl^- ions both entered the duct, as expected. In skin isolated from cystic fibrosis individuals, however, only Na^+ ions entered the duct—no Cl^- ions entered. For the first time, the molecular nature of cystic fibrosis became clear. Cystic fibrosis is a defect in a plasma membrane protein called CFTR (cystic fibrosis transmembrane conductance regulator) that normally regulates passage of Cl^- ions into and out of the body's cells. CFTR does not function properly in cystic fibrosis patients (see figure 4.8).

The defective *cf* gene was isolated in 1987, and its position on a particular human chromosome (chromosome 7) was pinpointed in 1989. In 1990 a working *cf* gene was successfully transferred via adenovirus into human lung cells growing in tissue culture. The defective cells were “cured,” becoming able to transport chloride ions across their plasma membranes. Then in 1991, a team of researchers successfully transferred a normal human *cf* gene into the lung cells of a living animal—a rat. The *cf* gene was first inserted into a cold virus that easily infects lung cells, and the virus was inhaled by the rat. Carried piggyback, the *cf* gene entered the rat lung cells and began producing the normal human CFTR protein within these cells! Tests of gene transfer into CF patients were begun in 1993, and while a great deal of work remains to be done (the initial experiments were not successful), the future for cystic fibrosis patients for the first time seems bright.

Cystic fibrosis, and other genetic disorders, are potentially curable if ways can be found to successfully introduce normal alleles of the genes into affected individuals.

13.3 Genes are on chromosomes.

Chromosomes: The Vehicles of Mendelian Inheritance

Chromosomes are not the only kinds of structures that segregate regularly when eukaryotic cells divide. Centrioles also divide and segregate in a regular fashion, as do the mitochondria and chloroplasts (when present) in the cytoplasm. Therefore, in the early twentieth century it was by no means obvious that chromosomes were the vehicles of hereditary information.

The Chromosomal Theory of Inheritance

A central role for chromosomes in heredity was first suggested in 1900 by the German geneticist Karl Correns, in one of the papers announcing the rediscovery of Mendel's work. Soon after, observations that similar chromosomes paired with one another during meiosis led directly to the **chromosomal theory of inheritance**, first formulated by the American Walter Sutton in 1902.

Several pieces of evidence supported Sutton's theory. One was that reproduction involves the initial union of only two cells, egg and sperm. If Mendel's model were correct, then these two gametes must make equal hereditary contributions. Sperm, however, contain little cytoplasm, suggesting that the hereditary material must reside within the nuclei of the gametes. Furthermore, while diploid individuals have two copies of each pair of homologous chromosomes, gametes have only one. This observation was consistent with Mendel's model, in which diploid individuals have two copies of each heritable gene and gametes have one. Finally, chromosomes segregate during meiosis, and each pair of homologues orients on the metaphase plate independently of every other pair. Segregation and independent assortment were two characteristics of the genes in Mendel's model.

A Problem with the Chromosomal Theory

However, investigators soon pointed out one problem with this theory. If Mendelian characters are determined by genes located on the chromosomes, and if the independent assortment of Mendelian traits reflects the independent assortment of chromosomes in meiosis, why does the number of characters that assort independently in a given kind of organism often greatly exceed the number of chromosome pairs the organism possesses? This seemed a fatal objection, and it led many early researchers to have serious reservations about Sutton's theory.

Morgan's White-Eyed Fly

The essential correctness of the chromosomal theory of heredity was demonstrated long before this paradox was re-

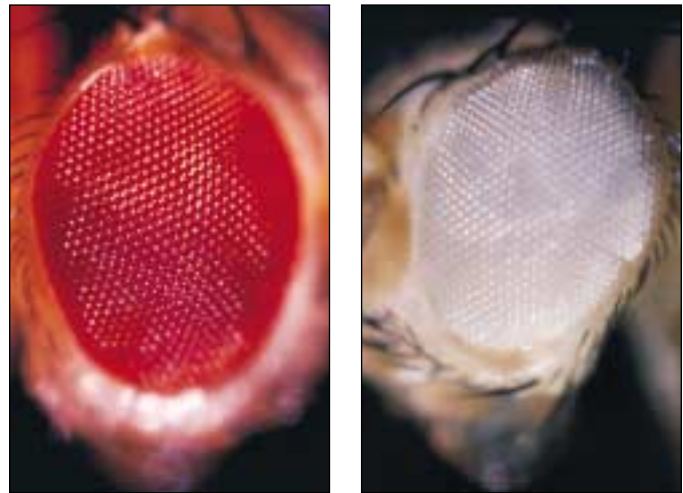


FIGURE 13.29

Red-eyed (normal) and white-eyed (mutant) *Drosophila*. The white-eyed defect is hereditary, the result of a mutation in a gene located on the X chromosome. By studying this mutation, Morgan first demonstrated that genes are on chromosomes.

solved. A single small fly provided the proof. In 1910 Thomas Hunt Morgan, studying the fruit fly *Drosophila melanogaster*, detected a **mutant** male fly, one that differed strikingly from normal flies of the same species: its eyes were white instead of red (figure 13.29).

Morgan immediately set out to determine if this new trait would be inherited in a Mendelian fashion. He first crossed the mutant male to a normal female to see if red or white eyes were dominant. All of the F_1 progeny had red eyes, so Morgan concluded that red eye color was dominant over white. Following the experimental procedure that Mendel had established long ago, Morgan then crossed the red-eyed flies from the F_1 generation with each other. Of the 4252 F_2 progeny Morgan examined, 782 (18%) had white eyes. Although the ratio of red eyes to white eyes in the F_2 progeny was greater than 3:1, the results of the cross nevertheless provided clear evidence that eye color segregates. However, there was something about the outcome that was strange and totally unpredicted by Mendel's theory—all of the white-eyed F_2 flies were males!

How could this result be explained? Perhaps it was impossible for a white-eyed female fly to exist; such individuals might not be viable for some unknown reason. To test this idea, Morgan testcrossed the female F_1 progeny with the original white-eyed male. He obtained both white-eyed and red-eyed males and females in a 1:1:1:1 ratio, just as Mendelian theory predicted. Hence, a female could have white eyes. Why, then, were there no white-eyed females among the progeny of the original cross?

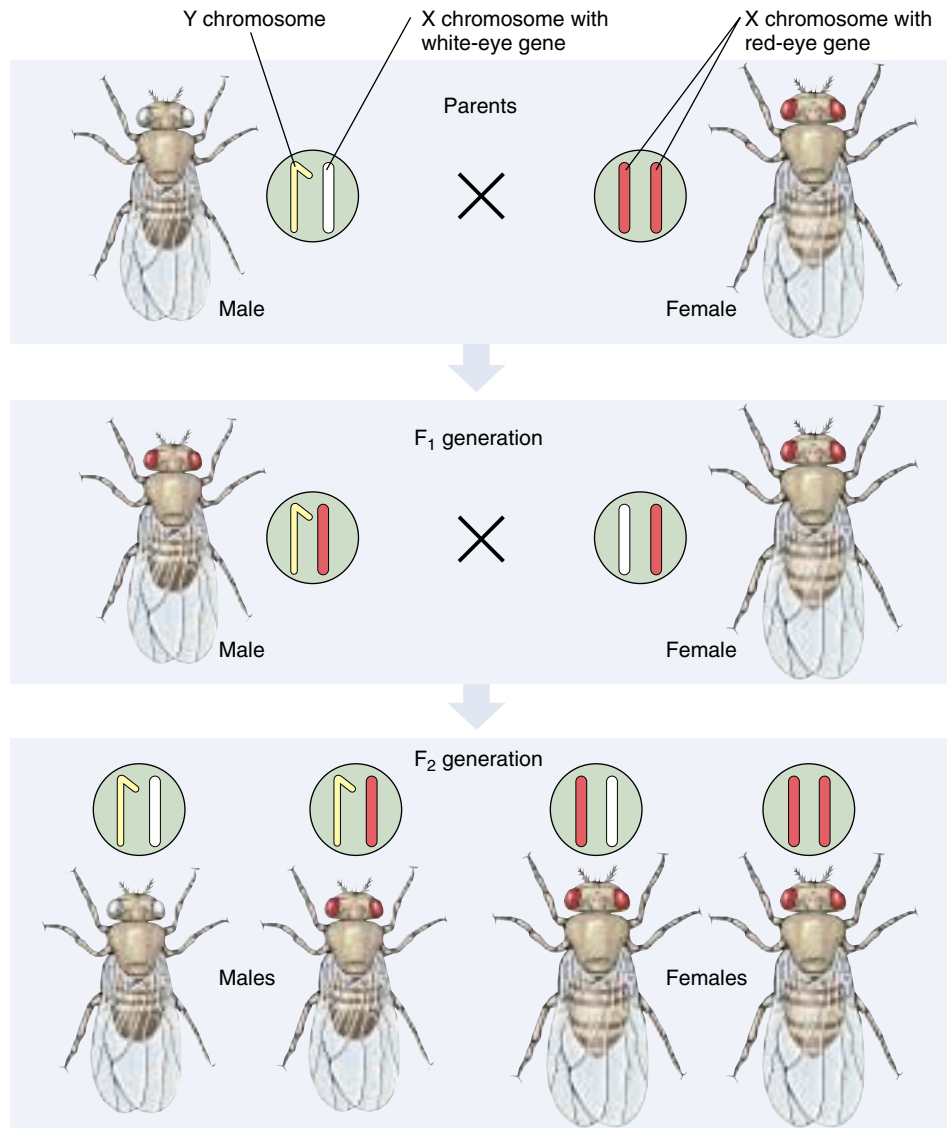


FIGURE 13.30
Morgan's experiment demonstrating the chromosomal basis of sex linkage in *Drosophila*. The white-eyed mutant male fly was crossed with a normal female. The F₁ generation flies all exhibited red eyes, as expected for flies heterozygous for a recessive white-eye allele. In the F₂ generation, all of the white-eyed flies were male.

Sex Linkage

The solution to this puzzle involved sex. In *Drosophila*, the sex of an individual is determined by the number of copies of a particular chromosome, the **X chromosome**, that an individual possesses. A fly with two X chromosomes is a female, and a fly with only one X chromosome is a male. In males, the single X chromosome pairs in meiosis with a dissimilar partner called the **Y chromosome**. The female thus produces only X gametes, while the male produces both X and Y gametes. When fertilization involves an X sperm, the result is an XX zygote, which develops into a female; when fertilization involves a Y sperm, the result is an XY zygote, which develops into a male.

The solution to Morgan's puzzle is that the gene causing the white-eye trait in *Drosophila* resides only on the X chromosome—it is absent from the Y chromosome. (We now know that the Y chromosome in flies carries almost no functional genes.) A trait determined by a gene on the X chromosome is said to be **sex-linked**. Knowing the

white-eye trait is recessive to the red-eye trait, we can now see that Morgan's result was a natural consequence of the Mendelian assortment of chromosomes (figure 13.30).

Morgan's experiment was one of the most important in the history of genetics because it presented the first clear evidence that the genes determining Mendelian traits do indeed reside on the chromosomes, as Sutton had proposed. The segregation of the white-eye trait has a one-to-one correspondence with the segregation of the X chromosome. In other words, Mendelian traits such as eye color in *Drosophila* assort independently because chromosomes do. When Mendel observed the segregation of alternative traits in pea plants, he was observing a reflection of the meiotic segregation of chromosomes.

Mendelian traits assort independently because they are determined by genes located on chromosomes that assort independently in meiosis.

Genetic Recombination

Morgan's experiments led to the general acceptance of Sutton's chromosomal theory of inheritance. Scientists then attempted to resolve the paradox that there are many more independently assorting Mendelian genes than chromosomes. In 1903 the Dutch geneticist Hugo de Vries suggested that this paradox could be resolved only by assuming that homologous chromosomes exchange elements during meiosis. In 1909, French cytologist F. A. Janssens provided evidence to support this suggestion. Investigating chiasmata produced during amphibian meiosis, Janssens noticed that of the four chromatids involved in each chiasma, two crossed each other and two did not. He suggested that this crossing of chromatids reflected a switch in chromosomal arms between the paternal and maternal homologues, involving one chromatid in each homologue. His suggestion was not accepted widely, primarily because it was difficult to see how two chromatids could break and rejoin at exactly the same position.

Crossing Over

Later experiments clearly established that Janssens was indeed correct. One of these experiments, performed in 1931 by American geneticist Curt Stern, is described in figure 13.31. Stern studied two sex-linked eye characters in *Drosophila* strains whose X chromosomes were visibly abnormal at both ends. He first examined many flies and identified those in which an exchange had occurred with respect to the two eye characters. He then studied the chromosomes of those flies to see if their X chromosomes had exchanged arms. Stern found that all of the individuals that had exchanged eye traits also possessed chromosomes that had exchanged abnormal ends. The conclusion was inescapable: genetic exchanges of characters such as eye color involve the physical exchange of chromosome arms, a phenomenon called **crossing over**. Crossing over creates new combinations of genes, and is thus a form of **genetic recombination**.

The chromosomal exchanges Stern demonstrated provide the solution to the paradox, because crossing over

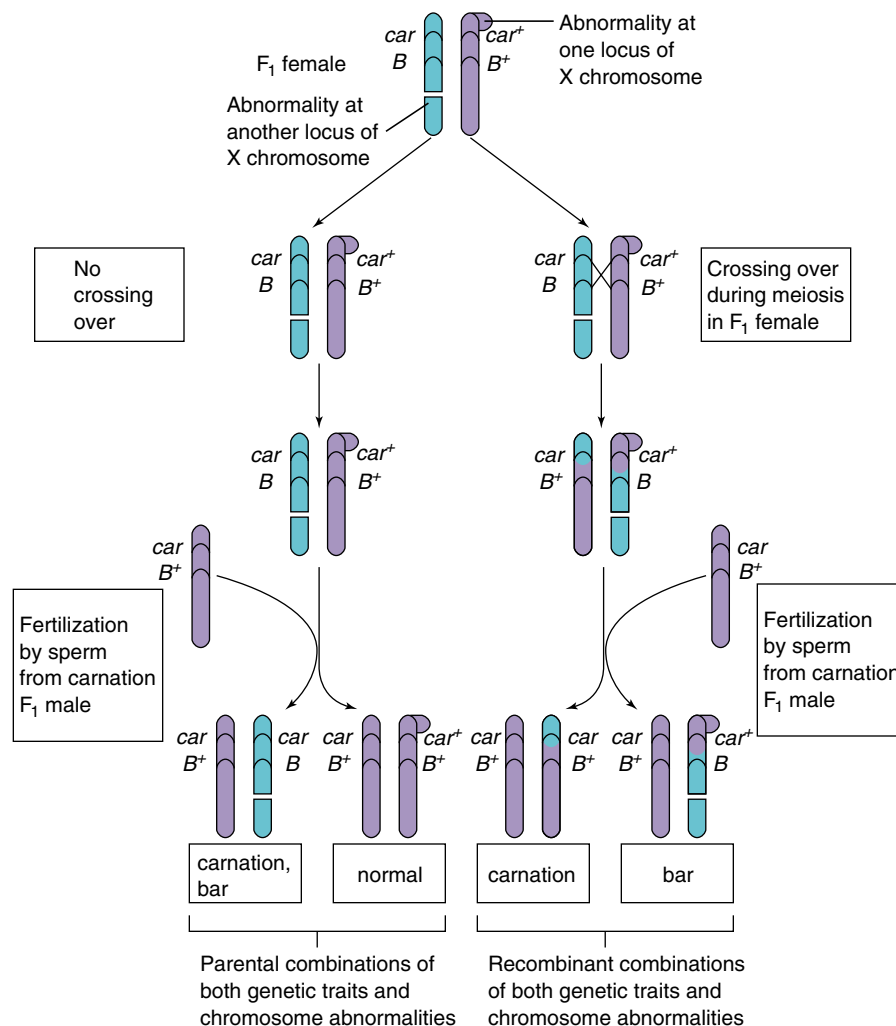


FIGURE 13.31

Stern's experiment demonstrating the physical exchange of chromosomal arms during crossing over. Stern monitored crossing over between two genes, the recessive carnation eye color (*car*) and the dominant bar-shaped eye (*B*), on chromosomes with physical peculiarities visible under a microscope. Whenever these genes recombined through crossing over, the chromosomes recombined as well. Therefore, the recombination of genes reflects a physical exchange of chromosome arms. The "+" notation on the alleles refers to the wild-type allele, the most common allele at a particular gene.

can occur between homologues anywhere along the length of the chromosome, in locations that seem to be randomly determined. Thus, if two different genes are located relatively far apart on a chromosome, crossing over is more likely to occur somewhere between them than if they are located close together. Two genes can be on the same chromosome and still show independent assortment if they are located so far apart on the chromosome that crossing over occurs regularly between them (figure 13.32).

Using Recombination to Make Genetic Maps

Because crossing over is more frequent between two genes that are relatively far apart than between two that are close together, the frequency of crossing over can be used to map the relative positions of genes on chromosomes. In a cross, the proportion of progeny exhibiting an exchange between two genes is a measure of the frequency of crossover events between them, and thus indicates the relative distance separating them. The results of such crosses can be used to construct a **genetic map** that measures distance between genes in terms of the frequency of recombination. One “map unit” is defined as the distance within which a crossover event is expected to occur in an average of 1% of gametes. A map unit is now called a **centimorgan**, after Thomas Hunt Morgan.

In recent times new technologies have allowed geneticists to create gene maps based on the relative positions of specific gene sequences called *restriction sites* because they are recognized by DNA-cleaving enzymes called restriction endonucleases. Restriction maps, discussed in chapter 19, have largely supplanted genetic recombination maps for detailed gene analysis because they are far easier to produce. Recombination maps remain the method of choice for genes widely separated on a chromosome.

The Three-Point Cross. In constructing a genetic map, one simultaneously monitors recombination among three or more genes located on the same chromosome, referred to as **syntenic** genes. When genes are close enough together on a chromosome that they do not assort independently, they are said to be **linked** to one another. A cross involving three linked genes is called a **three-point cross**. Data obtained by Morgan on traits encoded by genes on the X chromosome of *Drosophila* were used by his student A. H. Sturtevant, to draw the first genetic map (figure 13.33). By convention, the most common allele of a gene is often denoted with the symbol “+” and is designated as **wild type**. All other alleles are denoted with just the specific letters.

FIGURE 13.33

The first genetic map. This map of the X chromosome of *Drosophila* was prepared in 1913 by A. H. Sturtevant, a student of Morgan. On it he located the relative positions of five recessive traits that exhibited sex linkage by estimating their relative recombination frequencies in genetic crosses. Sturtevant arbitrarily chose the position of the *yellow* gene as zero on his map to provide a frame of reference. The higher the recombination frequency, the farther apart the two genes.

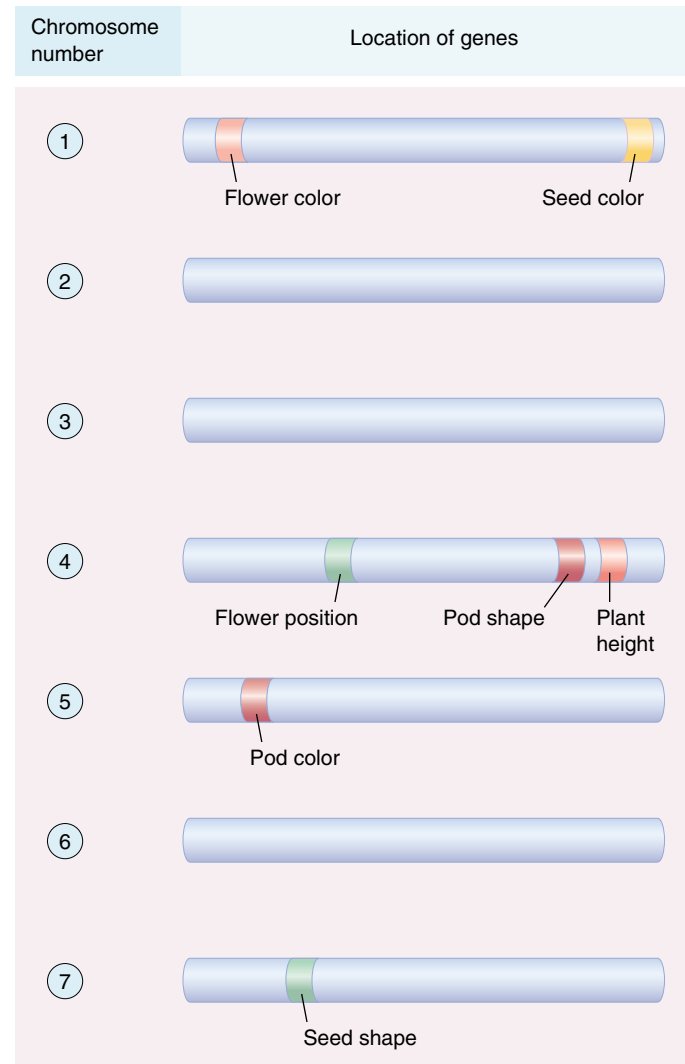
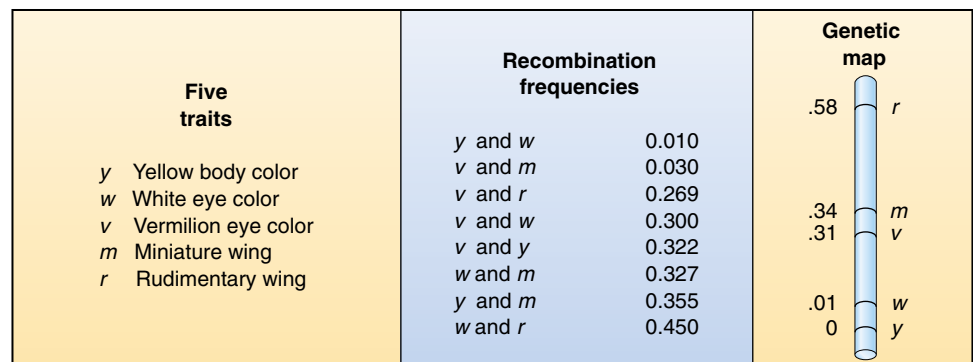
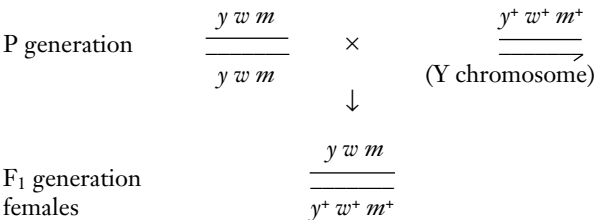


FIGURE 13.32

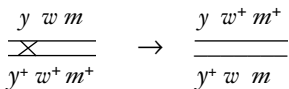
The chromosomal locations of the seven genes studied by Mendel in the garden pea. The genes for plant height and pod shape are very close to each other and rarely recombine. Plant height and pod shape were not among the characters Mendel examined in dihybrid crosses. One wonders what he would have made of the linkage he surely would have detected had he tested these characters.

Analyzing a Three-Point Cross. The first genetic map was constructed by A. H. Sturtevant, a student of Morgan's in 1913. He studied several traits of *Drosophila*, all of which exhibited sex linkage and thus were encoded by genes residing on the same chromosome (the X chromosome). Here we will describe his study of three traits: *y*, yellow body color (the normal body color is gray), *w*, white eye color (the normal eye color is red), and *m*, miniature wing (the normal wing is 50% longer).

Sturtevant carried out the mapping cross by crossing a female fly homozygous for the three recessive alleles with a normal male fly that carried none of them. All of the progeny were heterozygotes. Such a cross is conventionally represented by a diagram like the one that follows, in which the lines represent gene locations and + indicates the normal, or "wild-type" allele. Each female fly participating in a cross possesses two homologous copies of the chromosome being mapped, and both chromosomes are represented in the diagram. Crossing over occurs between these two copies in meiosis.



These heterozygous females, the F₁ generation, are the key to the mapping procedure. Because they are heterozygous, any crossing over that occurs during meiosis will, if it occurs between where these genes are located, produce gametes with different combinations of alleles for these genes—in other words, recombinant chromosomes. Thus, a crossover between the homologous X chromosomes of such a female in the interval between the *y* and *w* genes will yield recombinant [*y w*⁺] and [*y*⁺ *w*] chromosomes, which are different combinations than we started with. In the diagram below, the crossed lines between the chromosomes indicate where the crossover occurs. (In the parental chromosomes of this cross, *w* is always linked with *y* and *y*⁺ linked with *w*⁺.)



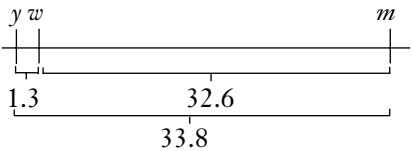
In order to see all the recombinant types that might be present among the gametes of these heterozygous flies, Sturtevant conducted a testcross. He crossed female heterozygous flies to males recessive for all three traits and examined the progeny. Because males contribute either a Y chromosome with no genes on it or an X chromosome with recessive alleles at all three loci, the male contribution does not disguise the potentially recombinant female chromosomes.

Table 13.3 summarizes the results Sturtevant obtained. The parents are represented by the highest number of progeny and the double crossovers (progeny in which two crossovers occurred) by the lowest number. To analyze his data, Sturtevant considered the traits in pairs and determined which involved a crossover event.

1. For the body trait (*y*) and the eye trait (*w*), the first two classes, [*y*⁺ *w*⁺] and [*y w*], involve no crossovers (they are parental combinations). In table 13.3, no progeny numbers are tabulated for these two classes on the "body-eye" column (a dash appears instead).
2. The next two classes have the same body-eye combination as the parents, [*y*⁺ *w*⁺] and [*y w*], so again no numbers are entered as recombinants under body-eye crossover type.
3. The next two classes, [*y*⁺ *w*] and [*y w*⁺], do *not* have the same body-eye combinations as the parent chromosomes, so the observed numbers of progeny are recorded, 16 and 12, respectively.
4. The last two classes also differ from parental chromosomes in body-eye combination, so again the observed numbers of each class are recorded, 1 and 0.
5. The sum of the numbers of observed progeny that are recombinant for body (*y*) and eye (*w*) is 16 + 12 + 1, or 29. Because the total number of progeny is 2205, this represents 29/2205, or 0.01315. The percentage of recombination between *y* and *w* is thus 1.315%, or 1.3 centimorgans.

To estimate the percentage of recombination between eye (*w*) and wing (*m*), one proceeds in the same manner, obtaining a value of 32.608%, or 32.6 centimorgans. Similarly, body (*y*) and wing (*m*) are separated by a recombination distance of 33.832%, or 33.8 centimorgans.

From this, then, we can construct our genetic map. The biggest distance, 33.8 centimorgans, separates the two outside genes, which are evidently *y* and *m*. The gene *w* is between them, near *y*.



The two distances 1.3 and 32.6 do not add up to 33.8 but rather to 33.9. The difference, 0.1, represents chromosomes in which two crossovers occurred, one between *y* and *w* and another between *w* and *m*. These chromosomes do not exhibit recombination between *y* and *m*.

Genetic maps such as this are key tools in genetic analysis, permitting an investigator reliably to predict how a newly discovered trait, once it has been located on the chromosome map, will recombine with many others.

Table 13.3 Sturtevant's Results							
	Phenotypes			Number of Progeny	Crossover Types		
	Body	Eye	Wing		Body-Eye	Eye-Wing	Body-Wing
Parental	y^+	w^+	m^+	758	—	—	—
	y	w	m	700	—	—	—
Single crossover	y^+	w^+	m	401	—	401	401
	y	w	m^+	317	—	317	317
	y^+	w	m	16	16	—	16
	y	w^+	m^+	12	12	—	12
Double crossover	y^+	w	m^+	1	1	1	—
	y	w^+	m	0	0	0	—
TOTAL				2205	29	719	746
Recombination frequency (%)					1.315	32.608	33.832

The Human Genetic Map

Genetic maps of human chromosomes (figure 13.34) are of great importance. Knowing where particular genes are located on human chromosomes can often be used to tell whether a fetus at risk of inheriting a genetic disorder actually has the disorder. The genetic-engineering techniques described in chapter 19 have begun to permit investigators to isolate specific genes and determine their nucleotide sequences. It is hoped that knowledge of differences at the gene level may suggest successful therapies for particular genetic disorders and that knowledge of a gene's location on a chromosome will soon permit the substitution of normal alleles for dysfunctional ones. Because of the great potential of this approach, investigators are working hard to assemble a detailed map of the entire human genome, the **Human Genome Project**, described in chapter 19. Initially, this map will consist of a "library" of thousands of small fragments of DNA whose relative positions are known. Investigators wishing to study a particular gene will first use techniques described in chapter 19 to screen this library and determine which fragment carries the gene of interest. They will then be able to analyze that fragment in detail. In parallel with this mammoth undertaking, the other, smaller genomes have already been sequenced, including those of yeasts and several bacteria. Progress on the human genome is rapid, and the full map is expected within the next 10 years.

Gene maps locate the relative positions of different genes on the chromosomes of an organism.

Traditionally produced by analyzing the relative amounts of recombination in genetic crosses, gene maps are increasingly being made by analyzing the sizes of fragments made by restriction enzymes.

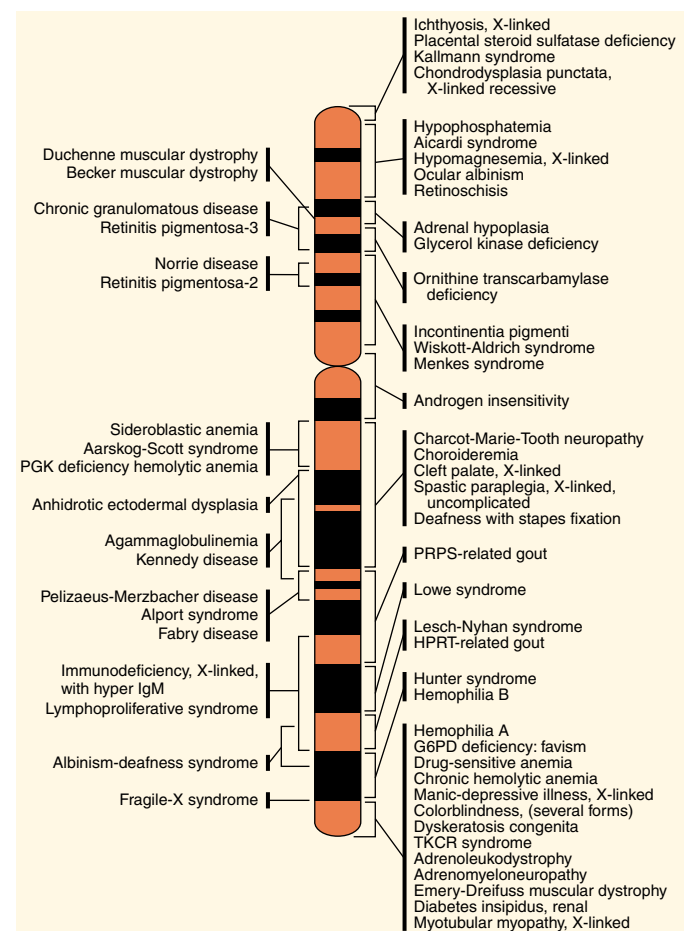


FIGURE 13.34

The human X chromosome gene map. Over 59 diseases have been traced to specific segments of the X chromosome. Many of these disorders are also influenced by genes on other chromosomes.

Human Chromosomes

Each human somatic cell normally has 46 chromosomes, which in meiosis form 23 pairs. By convention, the chromosomes are divided into seven groups (designated A through G), each characterized by a different size, shape, and appearance. The differences among the chromosomes are most clearly visible when the chromosomes are arranged in order in a karyotype (figure 13.35). Techniques that stain individual segments of chromosomes with different-colored dyes make the identification of chromosomes unambiguous. Like a fingerprint, each chromosome always exhibits the same pattern of colored bands.

Human Sex Chromosomes

Of the 23 pairs of human chromosomes, 22 are perfectly matched in both males and females and are called **autosomes**. The remaining pair, the **sex chromosomes**, consist of two similar chromosomes in females and two dissimilar chromosomes in males. In humans, females are designated XX and males XY. One of the sex chromosomes in the male (the Y chromosome) is highly condensed and bears few functional genes. Because few genes on the Y chromosome are expressed, recessive alleles on a male's single X chromosome have no *active* counterpart on the Y chromosome. Some of the active genes the Y chromosome does possess are responsible for the features associated with "maleness" in humans. Consequently, any individual with *at least* one Y chromosome is a male.

Sex Chromosomes in Other Organisms

The structure and number of sex chromosomes vary in different organisms (table 13.4). In the fruit fly *Drosophila*, females are XX and males XY, as in humans and most other vertebrates. However, in birds, the male has two Z chromosomes, and the female has a Z and a W chromosome. In some insects, such as grasshoppers, there is no Y chromosome—females are XX and males are characterized as XO (the O indicates the absence of a chromosome).

Sex Determination

In humans a specific gene located on the Y chromosome known as *SRY* plays a key role in development of male sexual characteristics. This gene is expressed early in development, and acts to masculinize genitalia and secondary sexual organs that would otherwise be female. Lacking a Y chromosome, females fail to undergo these changes.

Among fishes and in some species of reptiles, environmental changes can cause changes in the expression of this sex-determining gene, and thus of the sex of the adult individual.

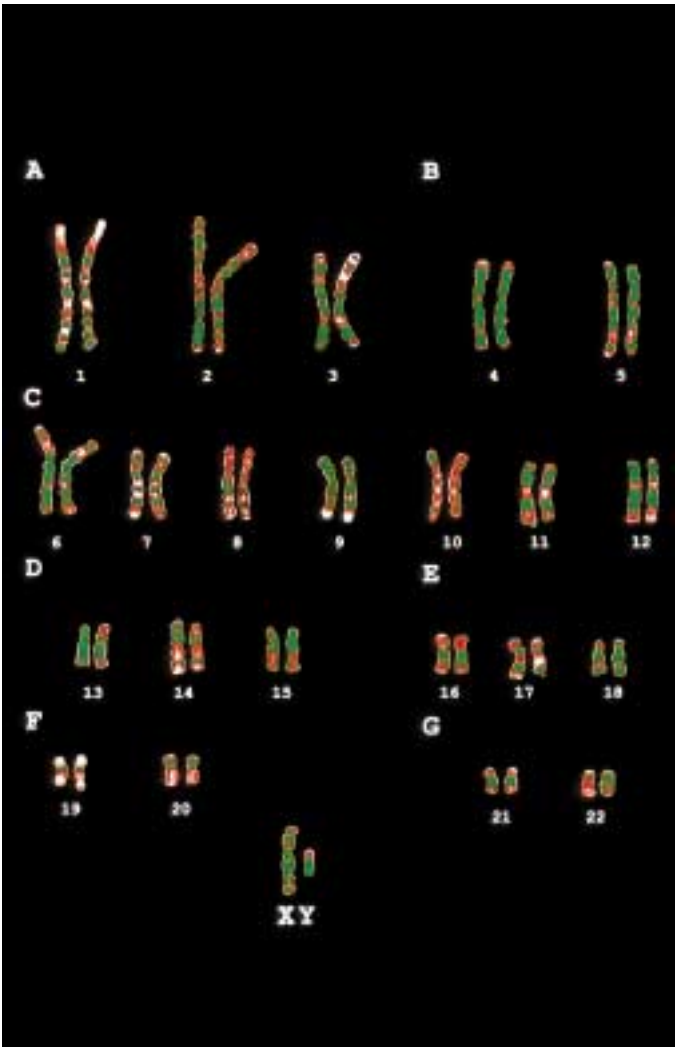


FIGURE 13.35
A human karyotype. This karyotype shows the colored banding patterns, arranged by class A–G.

Table 13.4 Sex Determination in Some Organisms		
	Female	Male
Humans, <i>Drosophila</i>	XX	XY
Birds	ZW	ZZ
Grasshoppers	XX	XO
Honeybees	Diploid	Haploid

Barr Bodies

Although males have only one copy of the X chromosome and females have two, female cells do not produce twice as much of the proteins encoded by genes on the X chromosome. Instead, one of the X chromosomes in females is inactivated early in embryonic development, shortly after the embryo's sex is determined. Which X chromosome is inactivated varies randomly from cell to cell. If a woman is heterozygous for a sex-linked trait, some of her cells will express one allele and some the other. The inactivated and highly condensed X chromosome is visible as a darkly staining **Barr body** attached to the nuclear membrane (figure 13.36).

X-inactivation is not restricted to humans. The patches of color on tortoiseshell and calico cats are a familiar result of this process. The gene for orange coat color is located on the X chromosome. The *O* allele specifies orange fur, and the *o* allele specifies black fur. Early in development, one X chromosome is inactivated in the cells that will become skin cells. If the remaining active X carries the *O* allele, then the patch of skin that results from that cell will have orange fur. If it carries the *o* allele, then the fur will be black. Because X-inactivation is a random process, the orange and black patches appear randomly in the cat's coat. Because only females have two copies of the X chromosome, only they can be heterozygous at the *O* gene, so almost all calico cats are females (figure 13.37). The exception is male cats that have the genotype XXY; the XXY genotype is discussed in the next section. The white on a calico cat is due to the action of an allele at another gene, the white spotting gene.

One of the 23 pairs of human chromosomes carries the genes that determine sex. The gene determining maleness is located on a version of the sex chromosome called Y, which has few other transcribed genes.

FIGURE 13.37

A calico cat. The coat coloration of this cat is due to the random inactivation of her X chromosome during early development. The female is heterozygous for orange coat color, but because only one coat color allele is expressed, she exhibits patches of orange and black fur.

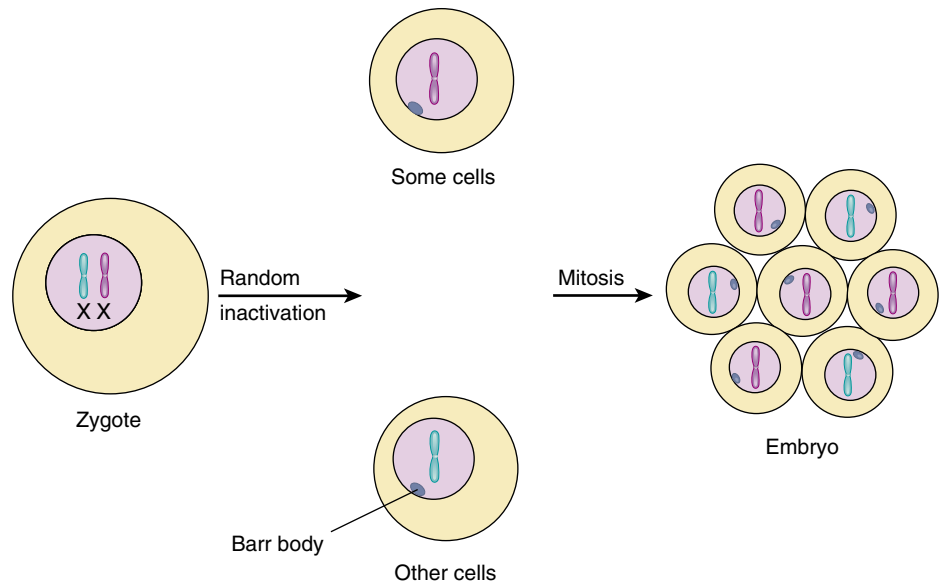


FIGURE 13.36

Barr bodies. In the developing female embryo, one of the X chromosomes (determined randomly) condenses and becomes inactivated. These condensed X chromosomes, called Barr bodies, then attach to the nuclear membrane.



Human Abnormalities Due to Alterations in Chromosome Number

Occasionally, homologues or sister chromatids fail to separate properly in meiosis, leading to the acquisition or loss of a chromosome in a gamete. This condition, called **primary nondisjunction**, can result in individuals with severe abnormalities if the affected gamete forms a zygote.

Nondisjunction Involving Autosomes

Almost all humans of the same sex have the same karyotype, for the same reason that all automobiles have engines, transmissions, and wheels: other arrangements don't work well. Humans who have lost even one copy of an autosome (called **monosomics**) do not survive development. In all but a few cases, humans who have gained an extra autosome (called **trisomics**) also do not survive. However, five of the smallest autosomes—those numbered 13, 15, 18, 21, and 22—can be present in humans as three copies and still allow the individual to survive for a time. The presence of an extra chromosome 13, 15, or 18 causes severe developmental defects, and infants with such a genetic makeup die within a few months. In contrast, individuals who have an extra copy of chromosome 21 or, more rarely, chromosome 22, usually survive to adulthood. In such individuals, the maturation of the skeletal system is delayed, so they generally are short and have poor muscle tone. Their mental development is also affected, and children with trisomy 21 or trisomy 22 are always mentally retarded.

Down Syndrome. The developmental defect produced by trisomy 21 (figure 13.38) was first described in 1866 by J. Langdon Down; for this reason, it is called **Down syndrome** (formerly “Down’s syndrome”). About 1 in every 750 children exhibits Down syndrome, and the frequency is similar in all racial groups. Similar conditions also occur in chimpanzees and other related primates. In humans, the defect is associated with a particular small portion of chromosome 21. When this chromosomal segment is present in three copies instead of two, Down syndrome results. In 97% of the human cases examined, all of chromosome 21 is present in three copies. In the other 3%, a small portion of chromosome 21 containing the critical segment has been added to another chromosome by a process called *translocation* (see chapter 18); it exists along with the normal two copies of chromosome 21. This condition is known as *translocation Down syndrome*.

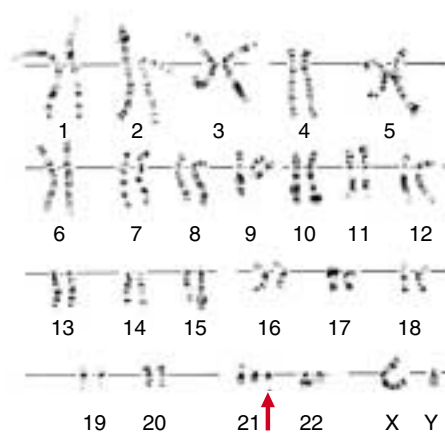


FIGURE 13.38

Down syndrome. As shown in this male karyotype, Down syndrome is associated with trisomy of chromosome 21. A child with Down syndrome sitting on his father’s knee.

Not much is known about the developmental role of the genes whose extra copies produces Down syndrome, although clues are beginning to emerge from current research. Some researchers suspect that the gene or genes that produce Down syndrome are similar or identical to some of the genes associated with cancer and with Alzheimer’s disease. The reason for this suspicion is that one of the human cancer-causing genes (to be described in chapter 18) and the gene causing Alzheimer’s disease are located on the segment of chromosome 21 associated with Down syndrome. Moreover, cancer is more common in children with Down syndrome. The incidence of leukemia, for example, is 11 times higher in children with Down syndrome than in unaffected children of the same age.

How does Down syndrome arise? In humans, it comes about almost exclusively as a result of primary nondisjunction of chromosome 21 during egg formation. The cause of these primary nondisjunctions is not known, but their incidence, like that of cancer, increases with age (figure 13.39). In mothers younger than 20 years of age, the risk of giving birth to a child with Down syndrome is about 1 in 1700; in mothers 20 to 30 years old, the risk is only about 1 in 1400. In mothers 30 to 35 years old, however, the risk rises to 1 in 750, and by age 45, the risk is as high as 1 in 16!

Primary nondisjunctions are far more common in women than in men because all of the eggs a woman will ever produce have developed to the point of prophase in meiosis I by the time she is born. By the time she has children, her eggs are as old as she is. In contrast, men produce new sperm daily. Therefore, there is a much greater chance for problems of various kinds, including those that cause primary nondisjunction, to accumulate over time in the gametes of women than in those of men. For this reason, the age of the mother is more critical than that of the father in couples contemplating childbearing.

Nondisjunction Involving the Sex Chromosomes

Individuals that gain or lose a sex chromosome do not generally experience the severe developmental abnormalities caused by similar changes in autosomes. Such individuals may reach maturity, but they have somewhat abnormal features.

The X Chromosome. When X chromosomes fail to separate during meiosis, some of the gametes that are produced possess both X chromosomes and so are XX gametes; the other gametes that result from such an event have no sex chromosome and are designated “O” (figure 13.40).

If an XX gamete combines with an X gamete, the resulting XXX zygote develops into a female with one functional X chromosome and two Barr bodies. She is sterile but usually normal in other respects. If an XX gamete instead combines with a Y gamete, the effects are more serious. The resulting XXY zygote develops into a sterile male who has many female body characteristics and, in some cases, diminished mental capacity. This condition, called *Klinefelter syndrome*, occurs in about 1 out of every 500 male births.

If an O gamete fuses with a Y gamete, the resulting OY zygote is nonviable and fails to develop further because humans cannot survive when they lack the genes on the X chromosome. If, on the other hand, an O gamete fuses with an X gamete, the XO zygote develops into a sterile female of short stature, with a webbed neck and immature sex organs that do not undergo changes during puberty. The mental abilities of an XO individual are in the low-normal range. This condition, called *Turner syndrome*, occurs roughly once in every 5000 female births.

The Y Chromosome. The Y chromosome can also fail to separate in meiosis, leading to the formation of YY gametes. When these gametes combine with X gametes, the XYY zygotes develop into fertile males of normal appearance. The frequency of the XYY genotype (Jacob’s syndrome) is about 1 per 1000 newborn males, but it is approximately 20 times higher among males in penal and mental institutions. This observation has led to the highly controversial suggestion that XYY males are inherently antisocial, a suggestion supported by some studies but not by others. In any case, most XYY males do not develop patterns of antisocial behavior.

Gene dosage plays a crucial role in development, so humans do not tolerate the loss or addition of chromosomes well. Autosome loss is always lethal, and an extra autosome is with few exceptions lethal too. Additional sex chromosomes have less serious consequences, although they can lead to sterility.

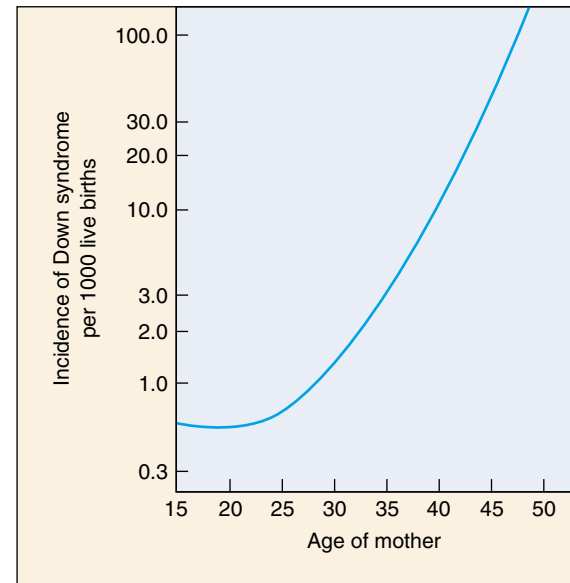


FIGURE 13.39
Correlation between maternal age and the incidence of Down syndrome. As women age, the chances they will bear a child with Down syndrome increase. After a woman reaches 35, the frequency of Down syndrome increases rapidly.

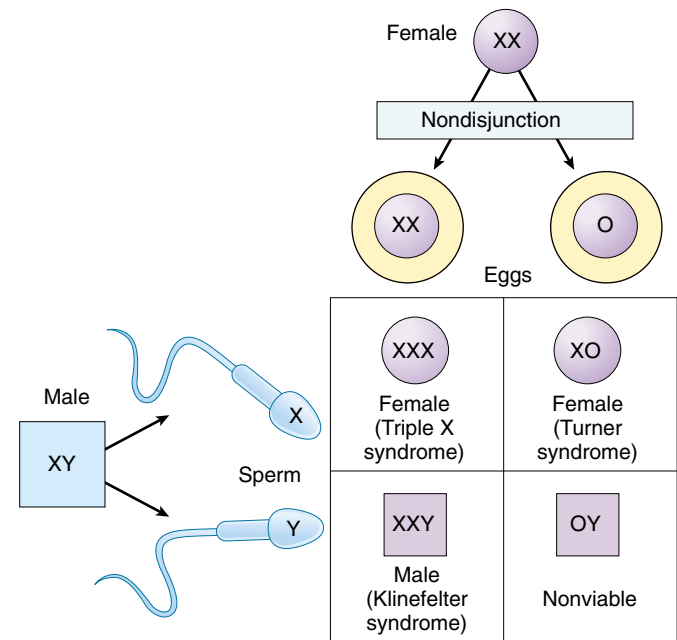


FIGURE 13.40
How nondisjunction can produce abnormalities in the number of sex chromosomes. When nondisjunction occurs in the production of female gametes, the gamete with two X chromosomes (XX) produces Klinefelter males (XXY) and XXX females. The gamete with no X chromosome (O) produces Turner females (XO) and nonviable OY males lacking any X chromosome.

Genetic Counseling

Although most genetic disorders cannot yet be cured, we are learning a great deal about them, and progress toward successful therapy is being made in many cases. In the absence of a cure, however, the only recourse is to try to avoid producing children with these conditions. The process of identifying parents at risk of producing children with genetic defects and of assessing the genetic state of early embryos is called *genetic counseling*.

If a genetic defect is caused by a recessive allele, how can potential parents determine the likelihood that they carry the allele? One way is through pedigree analysis, often employed as an aid in genetic counseling. By analyzing a person's pedigree, it is sometimes possible to estimate the likelihood that the person is a carrier for certain disorders. For example, if one of your relatives has been afflicted with a recessive genetic disorder such as cystic fibrosis, it is possible that you are a heterozygous carrier of the recessive allele for that disorder. When a couple is expecting a child, and pedigree analysis indicates that both of them have a significant probability of being heterozygous carriers of a recessive allele responsible for a serious genetic disorder, the pregnancy is said to be a **high-risk pregnancy**. In such cases, there is a significant probability that the child will exhibit the clinical disorder.

Another class of high-risk pregnancies is that in which the mothers are more than 35 years old. As we have seen, the frequency of birth of infants with Down syndrome increases dramatically in the pregnancies of older women (see figure 13.39).

When a pregnancy is diagnosed as being high-risk, many women elect to undergo *amniocentesis*, a procedure that permits the prenatal diagnosis of many genetic disorders. In the fourth month of pregnancy, a sterile hypodermic needle is inserted into the expanded uterus of the mother, removing a small sample of the amniotic fluid bathing the fetus (figure 13.41). Within the fluid are free-floating cells derived from the fetus; once removed, these cells can be grown in cultures in the laboratory. During amniocentesis, the position of the needle and that of the fetus are usually observed by means of *ultrasound*. The sound waves used in ultrasound are not harmful to mother or fetus, and they permit the person withdrawing the amniotic fluid to do so without damaging the fetus. In addition, ultrasound can be used to examine the fetus for signs of major abnormalities.

In recent years, physicians have increasingly turned to a new, less invasive procedure for genetic screening called **chorionic villi sampling**. In this procedure, the physician removes cells from the chorion, a membranous part of the placenta that nourishes the fetus. This procedure can be used earlier in pregnancy (by the eighth week) and yields results much more rapidly than does amniocentesis.

To test for certain genetic disorders, genetic counselors can look for three things in the cultures of cells obtained from amniocentesis or chorionic villi sampling. First, analysis of the karyotype can reveal aneuploidy (extra or missing chromosomes) and gross chromosomal alterations. Second, in many cases it is possible to test directly for the *proper functioning of enzymes* involved in genetic disorders. The lack of normal enzymatic activity signals the presence of the disorder. Thus, the lack of the enzyme responsible for breaking down phenylalanine signals PKU (phenylke-

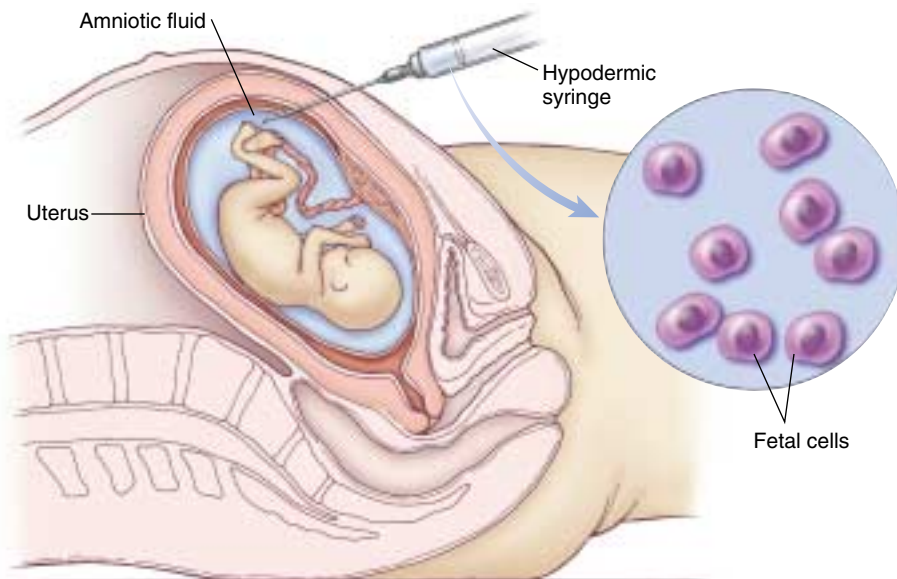


FIGURE 13.41

Amniocentesis. A needle is inserted into the amniotic cavity, and a sample of amniotic fluid, containing some free cells derived from the fetus, is withdrawn into a syringe. The fetal cells are then grown in culture and their karyotype and many of their metabolic functions are examined.

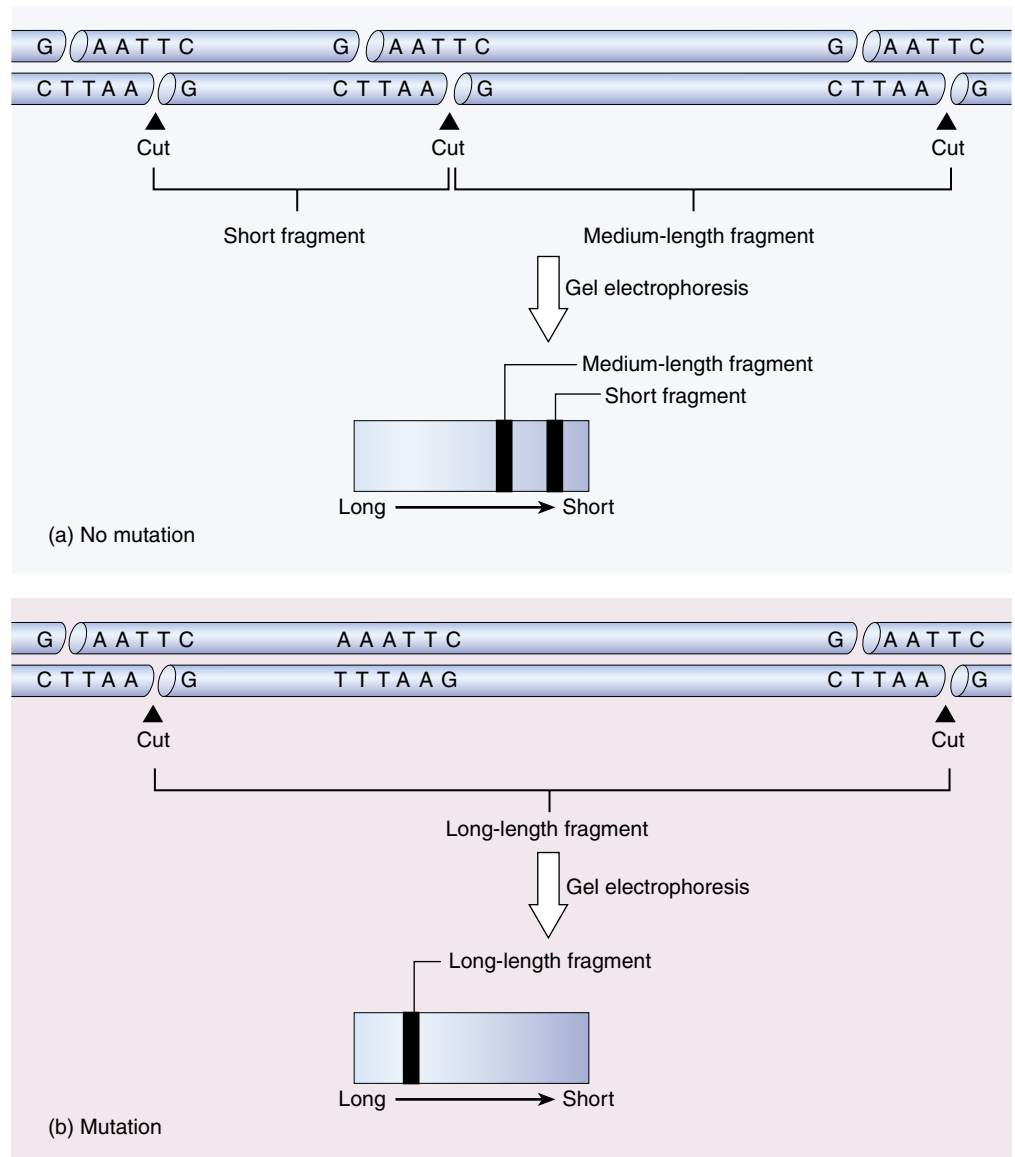


FIGURE 13.42

RFLPs. Restriction fragment length polymorphisms (RFLPs) are playing an increasingly important role in genetic identification. In (a), the restriction endonuclease cuts the DNA molecule in three places, producing two fragments. In (b), the mutation of a single nucleotide from G to A (see top fragment) alters a restriction endonuclease cutting site. Now the enzyme no longer cuts the DNA molecule at that site. As a result, a single long fragment is obtained, rather than two shorter ones. Such a change is easy to detect when the fragments are subjected to a technique called gel electrophoresis.

tonuria), the absence of the enzyme responsible for the breakdown of gangliosides indicates Tay-Sachs disease, and so forth.

Third, genetic counselors can look for an *association with known genetic markers*. For sickle cell anemia, Huntington's disease, and one form of muscular dystrophy (a genetic disorder characterized by weakened muscles), investigators have found other mutations on the same chromosomes that, by chance, occur at about the same place as the mutations that cause those disorders. By testing for the presence of these other mutations, a genetic counselor can identify individuals with a high probability of possessing the disorder-causing mutations. Finding such muta-

tions in the first place is a little like searching for a needle in a haystack, but persistent efforts have proved successful in these three disorders. The associated mutations are detectable because they alter the length of the DNA segments that restriction enzymes produce when they cut strands of DNA at particular places (see chapter 18). Therefore, these mutations produce what are called **restriction fragment length polymorphisms**, or **RFLPs** (figure 13.42).

Many gene defects can be detected early in pregnancy, allowing for appropriate planning by the prospective parents.



Summary

Questions

Media Resources

13.1 Mendel solved the mystery of heredity.

- Koelreuter noted the basic facts of heredity a century before Mendel. He found that alternative traits segregate in crosses and may mask each other's appearance. Mendel, however, was the first to quantify his data, counting the numbers of each alternative type among the progeny of crosses.
- By counting progeny types, Mendel learned that the alternatives that were masked in hybrids (the F_1 generation) appeared only 25% of the time in the F_2 generation. This finding, which led directly to Mendel's model of heredity, is usually referred to as the Mendelian ratio of 3:1 dominant-to-recessive traits.
- When two genes are located on different chromosomes, the alleles assort independently.
- Because phenotypes are often influenced by more than one gene, the ratios of alternative phenotypes observed in crosses sometimes deviate from the simple ratios predicted by Mendel.

1. Why weren't the implications of Koelreuter's results recognized for a century?
2. What characteristics of the garden pea made this organism a good choice for Mendel's experiments on heredity?
3. To determine whether a purple-flowered pea plant of unknown genotype is homozygous or heterozygous, what type of plant should it be crossed with?
4. In a dihybrid cross between two heterozygotes, what fraction of the offspring should be homozygous recessive for both traits?



- Exploration: Heredity in families



- Introduction to Classic Genetics
- Monohybrid Cross
- Dihybrid Cross



- Experiments: Probability and Hypothesis Testing in Biology

13.2 Human genetics follows Mendelian principles.

- Some genetic disorders are relatively common in human populations; others are rare. Many of the most important genetic disorders are associated with recessive alleles, which are not eliminated from the human population, even though their effects in homozygotes may be lethal.

5. Why is Huntington's disease maintained at its current frequency in human populations?



- Beyond Mendel



- On Science Article: Advances in Gene Therapy
- Experiment: Muller-Lethal Mutations in Populations

13.3 Genes are on chromosomes.

- The first clear evidence that genes reside on chromosomes was provided by Thomas Hunt Morgan, who demonstrated that the segregation of the white-eye trait in *Drosophila* is associated with the segregation of the X chromosome, which is involved in sex determination.
- The first genetic evidence that crossing over occurs between chromosomes was provided by Curt Stern, who showed that when two Mendelian traits exchange during a cross, so do visible abnormalities on the ends of the chromosomes bearing those traits.
- The frequency of crossing over between genes can be used to construct genetic maps.
- Primary nondisjunction results when chromosomes do not separate during meiosis, leading to gametes with missing or extra chromosomes. In humans, the loss of an autosome is invariably fatal.

6. When Morgan crossed a white-eyed male fly with a normal red-eyed female, and then crossed two of the red-eyed progeny, about $\frac{1}{4}$ of the offspring were white-eyed—but they were ALL male! Why?
7. What is primary nondisjunction? How is it related to Down syndrome?
8. Is an individual with Klinefelter syndrome genetically male or female? Why?

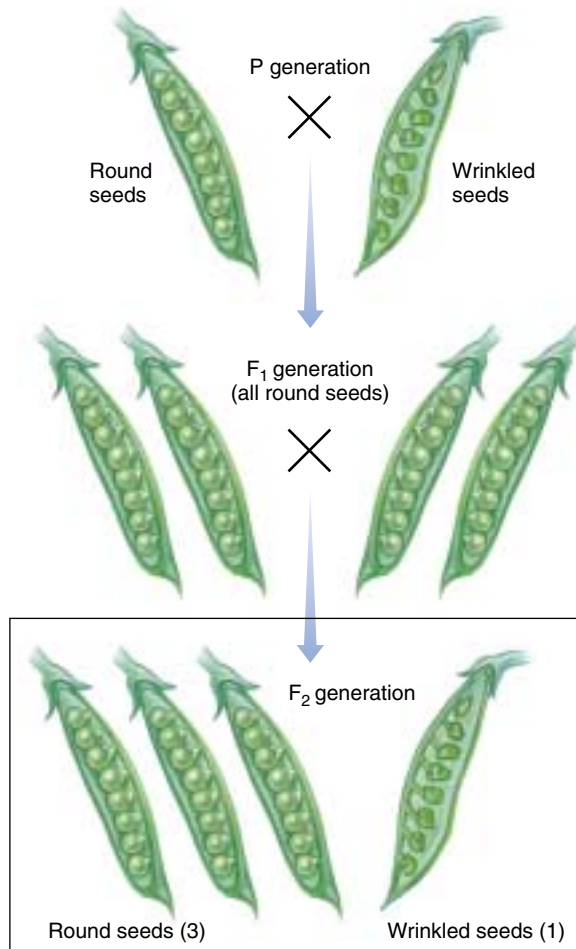


- Exploration: Down Syndrome
- Exploration: Constructing a Genetic Map
- Exploration: Gene Segregation within families
- Exploration: Making a Restriction Map
- Exploration: Cystic Fibrosis
- Recombination
- Introduction to Chromosomes Sex Chromosomes
- Abnormal Chromosomes



Mendelian Genetics Problems

1. The illustration below describes Mendel's cross of *wrinkled* and *round* seed characters. (Hint: Do you expect all the seeds in a pod to be the same?) What is wrong with this diagram?



2. The annual plant *Haplopappus gracilis* has two pairs of chromosomes (1 and 2). In this species, the probability that two characters *a* and *b* selected at random will be on the same chromosome is equal to the probability that they will both be on chromosome 1 ($\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$, or 0.25), plus the probability that they will both be on chromosome 2 (also $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$, or 0.25), for an overall probability of $\frac{1}{2}$, or 0.5. In general, the probability that two randomly selected characters will be on the same chromosome is equal to $\frac{1}{n}$ where *n* is the number of chromosome pairs. Humans have 23 pairs of chromosomes. What is the probability that any two human characters selected at random will be on the same chromosome?
3. Among Hereford cattle there is a dominant allele called *polled*; the individuals that have this allele lack horns. Suppose you acquire a herd consisting entirely of polled cattle, and you carefully determine that no

cow in the herd has horns. Some of the calves born that year, however, grow horns. You remove them from the herd and make certain that no horned adult has gotten into your pasture. Despite your efforts, more horned calves are born the next year. What is the reason for the appearance of the horned calves? If your goal is to maintain a herd consisting entirely of polled cattle, what should you do?

4. An inherited trait among humans in Norway causes affected individuals to have very wavy hair, not unlike that of a sheep. The trait, called *woolly*, is very evident when it occurs in families; no child possesses woolly hair unless at least one parent does. Imagine you are a Norwegian judge, and you have before you a woolly-haired man suing his normal-haired wife for divorce because their first child has woolly hair but their second child has normal hair. The husband claims this constitutes evidence of his wife's infidelity. Do you accept his claim? Justify your decision.
5. In human beings, Down syndrome, a serious developmental abnormality, results from the presence of three copies of chromosome 21 rather than the usual two copies. If a female exhibiting Down syndrome mates with a normal male, what proportion of her offspring would you expect to be affected?
6. Many animals and plants bear recessive alleles for *albinism*, a condition in which homozygous individuals lack certain pigments. An albino plant, for example, lacks chlorophyll and is white, and an albino human lacks melanin. If two normally pigmented persons heterozygous for the same albinism allele marry, what proportion of their children would you expect to be albino?
7. You inherit a racehorse and decide to put him out to stud. In looking over the stud book, however, you discover that the horse's grandfather exhibited a rare disorder that causes brittle bones. The disorder is hereditary and results from homozygosity for a recessive allele. If your horse is heterozygous for the allele, it will not be possible to use him for stud because the genetic defect may be passed on. How would you determine whether your horse carries this allele?
8. In the fly *Drosophila*, the allele for dumpy wings (*d*) is recessive to the normal long-wing allele (*d*⁺), and the allele for white eye (*w*) is recessive to the normal red-eye allele (*w*⁺). In a cross of *d*⁺*d*⁺*w*⁺*w* × *d*⁺*d**w**w*, what proportion of the offspring are expected to be "normal" (long wings, red eyes)? What proportion are expected to have dumpy wings and white eyes?
9. Your instructor presents you with a *Drosophila* with red eyes, as well as a stock of white-eyed flies and another stock of flies homozygous for the red-eye allele. You know that the presence of white eyes in *Drosophila* is caused by homozygosity for a recessive allele. How would you determine whether the single red-eyed fly was heterozygous for the white-eye allele?

10. Some children are born with recessive traits (and, therefore, must be homozygous for the recessive allele specifying the trait), even though neither of the parents exhibits the trait. What can account for this?
11. You collect two individuals of *Drosophila*, one a young male and the other a young, unmated female. Both are normal in appearance, with the red eyes typical of *Drosophila*. You keep the two flies in the same bottle, where they mate. Two weeks later, the offspring they have produced all have red eyes. From among the offspring, you select 100 individuals, some male and some female. You cross each individually with a fly you know to be homozygous for the recessive allele *sepia*, which produces black eyes when homozygous. Examining the results of your 100 crosses, you observe that in about half of the crosses, only red-eyed flies were produced. In the other half, however, the progeny of each cross consists of about 50% red-eyed flies and 50% black-eyed flies. What were the genotypes of your original two flies?
12. Hemophilia is a recessive sex-linked human blood disease that leads to failure of blood to clot normally. One form of hemophilia has been traced to the royal family of England, from which it spread throughout the royal families of Europe. For the purposes of this problem, assume that it originated as a mutation either in Prince Albert or in his wife, Queen Victoria.
 - a. Prince Albert did not have hemophilia. If the disease is a sex-linked recessive abnormality, how could it have originated in Prince Albert, a male, who would have been expected to exhibit sex-linked recessive traits?
 - b. Alexis, the son of Czar Nicholas II of Russia and Empress Alexandra (a granddaughter of Victoria), had hemophilia, but their daughter Anastasia did not. Anastasia died, a victim of the Russian revolution, before she had any children. Can we assume that Anastasia would have been a carrier of the disease? Would your answer be different if the disease had been present in Nicholas II or in Alexandra?
13. In 1986, *National Geographic* magazine conducted a survey of its readers' abilities to detect odors. About 7% of Caucasians in the United States could not smell the odor of musk. If neither parent could smell musk, none of their children were able to smell it. On the other hand, if the two parents could smell musk, their children generally could smell it, too, but a few of the children in those families were unable to smell it. Assuming that a single pair of alleles governs this trait, is the ability to smell musk best explained as an example of dominant or recessive inheritance?
14. A couple with a newborn baby is troubled that the child does not resemble either of them. Suspecting that a mix-up occurred at the hospital, they check the blood type of the infant. It is type O. As the father is type A and the mother type B, they conclude a mix-up must have occurred. Are they correct?
15. Mabel's sister died of cystic fibrosis as a child. Mabel does not have the disease, and neither do her parents. Mabel is pregnant with her first child. If you were a genetic counselor, what would you tell her about the probability that her child will have cystic fibrosis?
16. How many chromosomes would you expect to find in the karyotype of a person with Turner syndrome?
17. A woman is married for the second time. Her first husband has blood type A and her child by that marriage has type O. Her new husband has type B blood, and when they have a child its blood type is AB. What is the woman's blood genotype and blood type?
18. Two intensely freckled parents have five children. Three eventually become intensely freckled and two do not. Assuming this trait is governed by a single pair of alleles, is the expression of intense freckles best explained as an example of dominant or recessive inheritance?
19. Total color blindness is a rare hereditary disorder among humans. Affected individuals can see no colors, only shades of gray. It occurs in individuals homozygous for a recessive allele, and it is not sex-linked. A man whose father is totally color blind intends to marry a woman whose mother is totally color blind. What are the chances they will produce offspring who are totally color blind?
20. A normally pigmented man marries an albino woman. They have three children, one of whom is an albino. What is the genotype of the father?
21. Four babies are born in a hospital, and each has a different blood type: A, B, AB, and O. The parents of these babies have the following pairs of blood groups: A and B, O and O, AB and O, and B and B. Which baby belongs to which parents?
22. A couple both work in an atomic energy plant, and both are exposed daily to low-level background radiation. After several years, they have a child who has Duchenne muscular dystrophy, a recessive genetic defect caused by a mutation on the X chromosome. Neither the parents nor the grandparents have the disease. The couple sue the plant, claiming that the abnormality in their child is the direct result of radiation-induced mutation of their gametes, and that the company should have protected them from this radiation. Before reaching a decision, the judge hearing the case insists on knowing the sex of the child. Which sex would be more likely to result in an award of damages, and why?

Molecular Genetics

Can Cancer Tumors Be Starved to Death?

One of the most exciting recent developments in the war against cancer is the report that it might be possible to starve cancer tumors to death. Many laboratories have begun to look into this possibility, although it's not yet clear that the approach will actually work to cure cancer. One of the most exciting and frustrating things about watching a developing science story like this one is that you can't flip ahead and read the ending—in the real world of research, you never know how things are going to turn out.

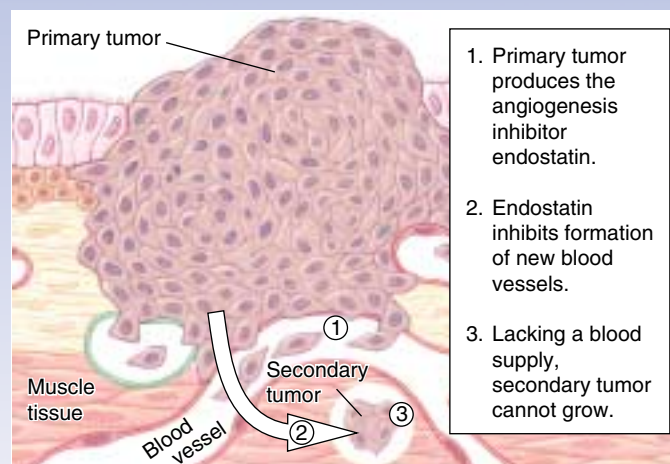
This story starts when a Harvard University researcher, Dr. Judah Folkman, followed up on a familiar observation made by many oncologists (cancer specialists), that removal of a primary tumor often leads to more rapid growth of secondary tumors. "Perhaps," Folkman reasoned, "the primary tumor is producing some substance that inhibits the growth of the other tumors." Such a substance could be a powerful weapon against cancer.

Folkman set out to see if he could isolate a chemical from primary tumors that inhibited the growth of secondary ones. Three years ago he announced he had found two. He called them angiostatin and endostatin.

To understand how these two proteins work, put yourself in the place of a tumor. To grow, a tumor must obtain from the body's blood supply all the food and nutrients it needs to make more cancer cells. To facilitate this necessary grocery shopping, tumors leak out substances into the surrounding tissues that encourage angiogenesis, the formation of small blood vessels. This call for more blood vessels insures an ever-greater flow of blood to the tumor as it grows larger.

When examined, Folkman's two cancer inhibitors turned out to be angiogenesis inhibitors. Angiostatin and endostatin kill a tumor by cutting off its blood supply. This may sound like an unlikely approach to curing cancer, but think about it—the cells of a growing tumor require a plentiful supply of food and nutrients to fuel their production of new cancer cells. Cut this off, and the tumor cells die, literally starving to death.

By producing factors like angiostatin and endostatin, the primary tumor holds back the growth of any competing



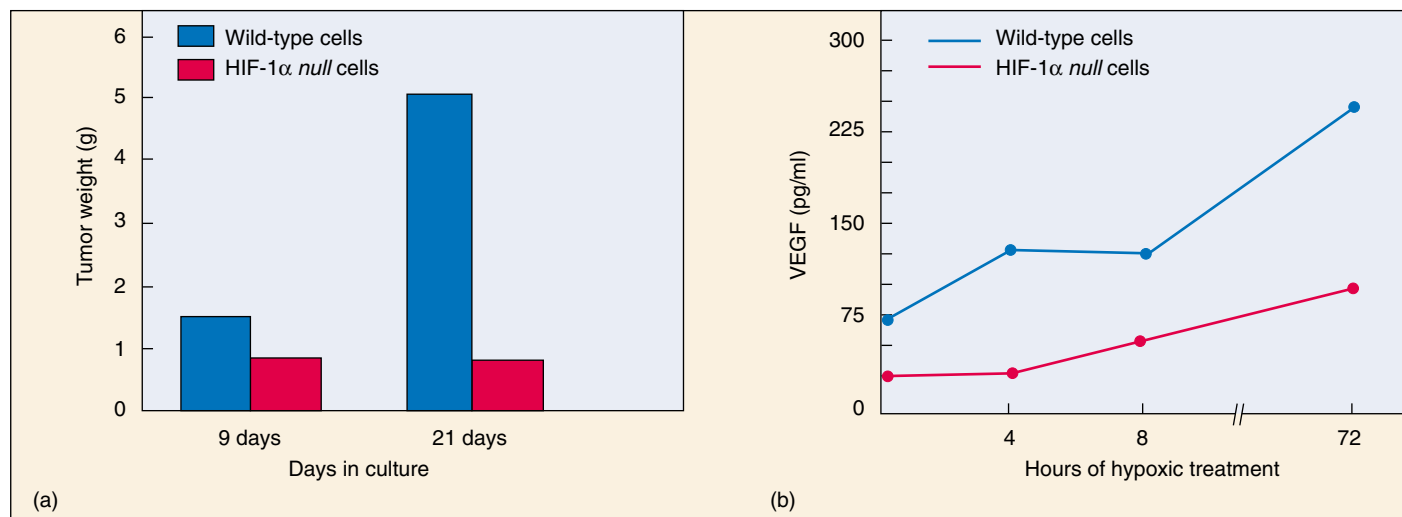
How primary tumors kill off the competition. Tumors require an ample blood supply to fuel their growth. The growth of new blood vessels is called angiogenesis. Inhibiting angiogenesis offers a possible way to block tumor growth.

tumors, allowing the primary tumor to hog the available resources for its own use (see above).

In laboratory tests the angiogenesis inhibitors caused tumors in mice to regress to microscopic size, a result that electrified researchers all over the world. Other scientists were soon trying to replicate this exciting result. Some have succeeded, others not. Five major laboratories have isolated their own angiogenesis inhibitors and published findings of antitumor activity. The National Cancer Institute is proceeding with tests of angiostatin and other angiogenesis inhibitors in humans. Preliminary results are encouraging. While not a cure-all for all cancers, angiogenesis inhibitors seem very effective against some, particularly solid-tumor cancers.

Gaining a better understanding of how tumors induce angiogenesis has become a high priority of cancer research. One promising line of research concerns hypoxia. As a solid tumor grows and outstrips its blood supply, its interior becomes hypoxic (oxygen depleted). In response to hypoxia, it appears that genes are turned on that promote survival under low oxygen pressure, including ones that increase blood flow to the tumor by promoting angiogenesis. Understanding this process may give important clues as to how angiogenesis inhibitors work to inhibit tumor growth.

So how does a lowering of oxygen pressure within a tumor promote blood vessel formation? Dr. Randall Johnson of the University of California, San Diego, is studying one important response by a tumor to hypoxia—the induction of a gene-specific transcription factor (that is, a protein that activates the transcription of a particular gene) that promotes angiogenesis. Called HIF-1, for *hypoxia inducible factor-1*, this transcription factor appears to induce the transcription of genes necessary for blood vessel formation.



Tumor growth in HIF-1α null cells and wild-type cells. (a) The size of tumors formed by the HIF-1α null cells were significantly smaller compared to those formed by wild-type cells. (b) HIF-1α null cells had significantly lower levels of VEGF protein production under hypoxic conditions compared to wild-type cells. VEGF promotes the formation of capillaries.

The Experiment

In order to examine the involvement of the hypoxia-inducible transcriptional factor (HIF-1) in angiogenesis, Johnson and his co-workers were faced with the problem that HIF-1 has many other effects on cell growth. To get a clear look at its role in angiogenesis, the researchers turned to embryonic stem cells. Embryonic stem cells are cells harvested from early embryos, before they have differentiated, while they are still capable of unlimited division. Because such stem cells have the capacity to form tumors (teratocarcinomas) when injected into certain kinds of mice, they offer a good natural laboratory in which to study how HIF-1 might influence cancer growth. The research team first prepared a mutant HIF-1 embryonic stem cell line in which the function of the transcription factor encoded by HIF-1 was completely destroyed or *null*.

The researchers then grew these HIF-1 *null* stem cells under hypoxic conditions. If HIF-1 genes indeed foster tumor growth in normal cells by promoting angiogenesis, then it would be expected that these *null* cells would be unable to promote tumor growth in this way.

The researchers tested the ability of *null* cells to promote tumor growth by injecting HIF-1α *null* cells into laboratory mice, and in control experiments injecting wild-type stem cells. The injected cells were allowed to grow and form tumors in both *null* and control host animals. The tumors that formed were then examined and measured for differences.

To get a closer look at what was really going on, the *null* and wild-type cells were compared in their ability to actually form new blood vessels. This was done by examining levels of mRNA of a growth factor that plays a key role in the formation and growth of blood vessels. This factor is a protein called vascular endothelial growth factor (VEGF). The levels of VEGF mRNA in the cells were determined by

hybridizing cDNA VEGF probes to mRNA isolated from tumors, and measuring in each instance how much tumor mRNA bound to the cDNA probe. In parallel studies, antibodies were used to determine levels of VEGF protein.

The Results

The researchers found that the *null* cells were greatly compromised in their ability to form tumors compared to the wild-type cells with the effects becoming more significant over time (see graph *a* above). Tumors were five times larger in wild-type cells than in the HIF-1 *null* cells after 21 days. Clearly knocking out HIF-1 retards tumor growth significantly.

This decrease in the size of tumors produced by *null* cells is further supported by the results of the VEGF protein analysis (see graph *b* above). Levels of the protein VEGF rise in wild-type cells under conditions of hypoxia, increasing the immediate availability of oxygen to the tumor by promoting capillary formation. The researchers found levels of VEGF protein were lower in *null* cell tumors, and responded to hypoxia at a lower rate.

Both the decrease in tumor size and the lower level of VEGF in the HIF-1 *null* cells supports the hypothesis that HIF-1 plays an essential role in promoting angiogenesis in a tumor, responding to a hypoxic condition by increasing the levels of VEGF.

Do the angiogenesis inhibitors like angiostatin, being tested as cancer cures, in fact act by inhibiting VEGF? The sorts of experiments being carried out in Johnson's laboratory, and in many other cancer centers, should soon cast light on this still-murky question.



To explore this experiment further, go to the Virtual Lab at www.mhhe.com/raven6/vlab5.mhtml

14

DNA: The Genetic Material

Concept Outline

14.1 What is the genetic material?

The Hammerling Experiment: Cells Store Hereditary Information in the Nucleus

Transplantation Experiments: Each Cell Contains a Full Set of Genetic Instructions

The Griffith Experiment: Hereditary Information Can Pass between Organisms

The Avery and Hershey-Chase Experiments: The Active Principle Is DNA

14.2 What is the structure of DNA?

The Chemical Nature of Nucleic Acids. Nucleic acids are polymers containing four nucleotides.

The Three-Dimensional Structure of DNA. The DNA molecule is a double helix, with two strands held together by base-pairing.

14.3 How does DNA replicate?

The Meselson-Stahl Experiment: DNA Replication Is Semiconservative

The Replication Process. DNA is replicated by the enzyme DNA polymerase III, working in concert with many other proteins. DNA replicates by assembling a complementary copy of each strand semidiscontinuously.

Eukaryotic DNA Replication. Eukaryotic chromosomes consist of many zones of replication.

14.4 What is a gene?

The One-Gene/One-Polypeptide Hypothesis. A gene encodes all the information needed to express a functional protein or RNA molecule.

How DNA Encodes Protein Structure. The nucleotide sequence of a gene dictates the amino acid sequence of the protein it encodes.

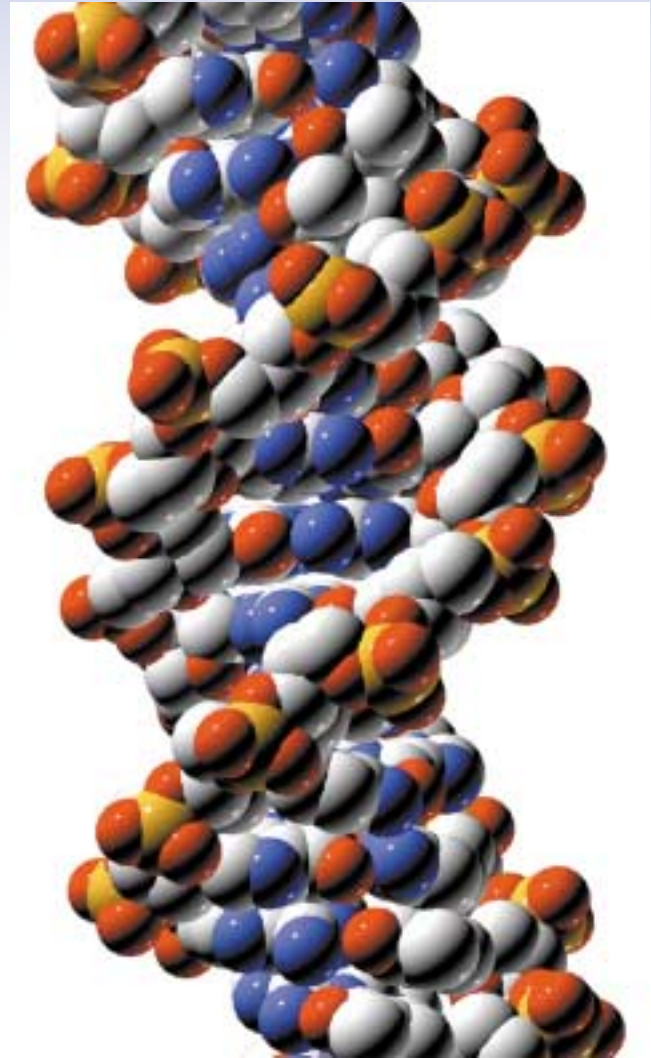


FIGURE 14.1

DNA. The hereditary blueprint in each cell of all living organisms is a very long, slender molecule called deoxyribonucleic acid (DNA).

The realization that patterns of heredity can be explained by the segregation of chromosomes in meiosis raised a question that occupied biologists for over 50 years: What is the exact nature of the connection between hereditary traits and chromosomes? This chapter describes the chain of experiments that have led to our current understanding of the molecular mechanisms of heredity (figure 14.1). The experiments are among the most elegant in science. Just as in a good detective story, each conclusion has led to new questions. The intellectual path taken has not always been a straight one, the best questions not always obvious. But however erratic and lurching the course of the experimental journey, our picture of heredity has become progressively clearer, the image more sharply defined.

14.1 What is the genetic material?

The Hammerling Experiment: Cells Store Hereditary Information in the Nucleus

Perhaps the most basic question one can ask about hereditary information is where it is stored in the cell. To answer this question, Danish biologist Joachim Hammerling, working at the Max Plank Institute for Marine Biology in Berlin in the 1930s, cut cells into pieces and observed the pieces to see which were able to express hereditary information. For this experiment, Hammerling needed cells large enough to operate on conveniently and differentiated enough to distinguish the pieces. He chose the unicellular green alga *Acetabularia*, which grows up to 5 cm, as a **model organism** for his investigations. Just as Mendel used pea plants and Sturtevant used fruit flies as model organisms, Hammerling picked an organism that was suited to the specific experimental question he wanted to answer, assuming that what he learned could then be applied to other organisms.

Individuals of the genus *Acetabularia* have distinct foot, stalk, and cap regions; all are differentiated parts of a single cell. The nucleus is located in the foot. As a preliminary experiment, Hammerling amputated the caps of some cells and the feet of others. He found that when he amputated the cap, a new cap regenerated from the remaining portions of the cell (foot and stalk). When he amputated the foot, however, no new foot regenerated from the cap and stalk. Hammerling, therefore, hypothesized that the hereditary information resided within the foot of *Acetabularia*.

Surgery on Single Cells

To test his hypothesis, Hammerling selected individuals from two species of the genus *Acetabularia* in which the caps look very different from one another: *A. mediterranea* has a disk-shaped cap, and *A. crenulata* has a branched, flower-like cap. Hammerling grafted a stalk from *A. crenulata* to a foot from *A. mediterranea* (figure 14.2). The cap that regenerated looked somewhat like the cap of *A. crenulata*, though not exactly the same.

Hammerling then cut off this regenerated cap and found that a disk-shaped cap exactly like that of *A. mediterranea* formed in the second regeneration and in every regeneration thereafter. This experiment supported Hammerling's hypothesis that the instructions specifying the kind of cap are stored in the foot of the cell, and that these instructions must pass from the foot through the stalk to the cap.

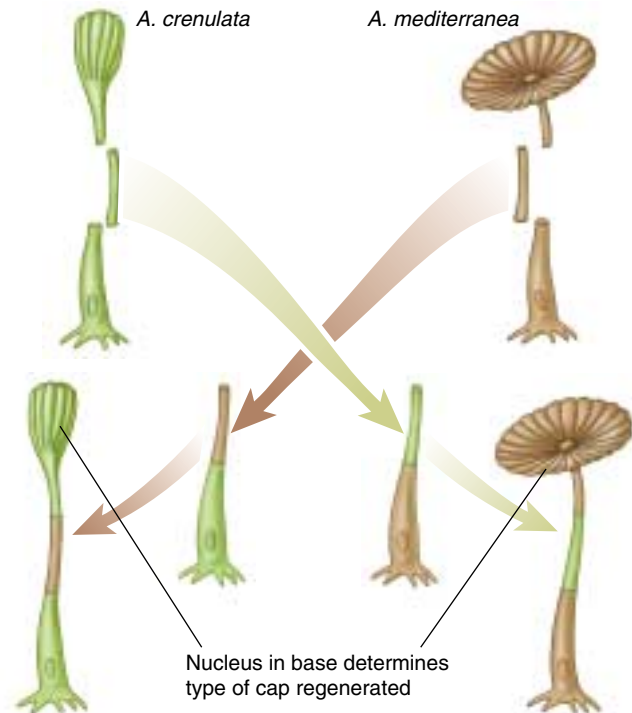


FIGURE 14.2

Hammerling's *Acetabularia* reciprocal graft experiment.

Hammerling grafted a stalk of each species of *Acetabularia* onto the foot of the other species. In each case, the cap that eventually developed was dictated by the nucleus-containing foot rather than by the stalk.

In this experiment, the initial flower-shaped cap was somewhat intermediate in shape, unlike the disk-shaped caps of subsequent generations. Hammerling speculated that this initial cap, which resembled that of *A. crenulata*, was formed from instructions already present in the transplanted stalk when it was excised from the original *A. crenulata* cell. In contrast, all of the caps that regenerated subsequently used new information derived from the foot of the *A. mediterranea* cell the stalk had been grafted onto. In some unknown way, the original instructions that had been present in the stalk were eventually “used up.” We now understand that genetic instructions (in the form of messenger RNA, discussed in chapter 15) pass from the nucleus in the foot upward *through the stalk* to the developing cap.

Hereditary information in *Acetabularia* is stored in the foot of the cell, where the nucleus resides.

Transplantation Experiments: Each Cell Contains a Full Set of Genetic Instructions

Because the nucleus is contained in the foot of *Acetabularia*, Hammerling's experiments suggested that the nucleus is the repository of hereditary information in a cell. A direct test of this hypothesis was carried out in 1952 by American embryologists Robert Briggs and Thomas King. Using a glass pipette drawn to a fine tip and working with a microscope, Briggs and King removed the nucleus from a frog egg. Without the nucleus, the egg did not develop. However, when they replaced the nucleus with one removed from a more advanced frog embryo cell, the egg developed into an adult frog. Clearly, the nucleus was directing the egg's development (figure 14.3).

Successfully Transplanting Nuclei

Can every nucleus in an organism direct the development of an entire adult individual? The experiment of Briggs and King did not answer this question definitively, because the nuclei they transplanted from frog embryos into eggs often caused the eggs to develop abnormally. Two experiments performed soon afterward gave a clearer answer to the question. In the first, John Gurdon, working with another species of frog at Oxford and Yale, transplanted nuclei from tadpole cells into eggs from which the nuclei had

been removed. The experiments were difficult—it was necessary to synchronize the division cycles of donor and host. However, in many experiments, the eggs went on to develop normally, indicating that the nuclei of cells in later stages of development retain the genetic information necessary to direct the development of all other cells in an individual.

Totipotency in Plants

In the second experiment, F. C. Steward at Cornell University in 1958 placed small fragments of fully developed carrot tissue (isolated from a part of the vascular system called the phloem) in a flask containing liquid growth medium. Steward observed that when individual cells broke away from the fragments, they often divided and developed into multicellular roots. When he immobilized the roots by placing them in a solid growth medium, they went on to develop normally into entire, mature plants. Steward's experiment makes it clear that, even in adult tissues, the nuclei of individual plant cells are "totipotent"—each contains a full set of hereditary instructions and can generate an entire adult individual. As you will learn in chapter 19, animal cells, like plant cells, can be totipotent, and a single adult animal cell can generate an entire adult animal.

Hereditary information is stored in the nucleus of eukaryotic cells. Each nucleus in any eukaryotic cell contains a full set of genetic instructions.

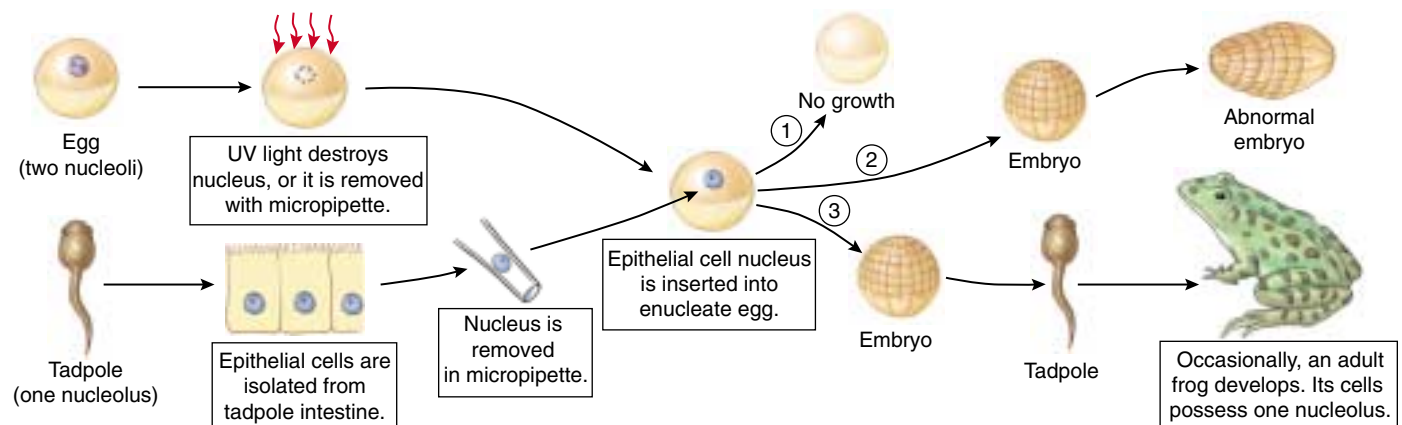


FIGURE 14.3

Briggs and King's nuclear transplant experiment. Two strains of frogs were used that differed from each other in the number of nucleoli their cells possessed. The nucleus was removed from an egg of one strain, either by sucking the egg nucleus into a micropipette or, more simply, by destroying it with ultraviolet light. A nucleus obtained from a differentiated cell of the other strain was then injected into this enucleate egg. The hybrid egg was allowed to develop. One of three results was obtained in individual experiments: (1) no growth occurred, perhaps reflecting damage to the egg cell during the nuclear transplant operation; (2) normal growth and development occurred up to an early embryo stage, but subsequent development was not normal and the embryo did not survive; and (3) normal growth and development occurred, eventually leading to the development of an adult frog. That frog was of the strain that contributed the nucleus and not of the strain that contributed the egg. Only a few experiments gave this third result, but they serve to clearly establish that the nucleus directs frog development.

The Griffith Experiment: Hereditary Information Can Pass between Organisms

The identification of the nucleus as the repository of hereditary information focused attention on the chromosomes, which were already suspected to be the vehicles of Mendelian inheritance. Specifically, biologists wondered how the **genes**, the units of hereditary information studied by Mendel, were actually arranged in the chromosomes. They knew that chromosomes contained both protein and deoxyribonucleic acid (DNA). Which of these held the genes? Starting in the late 1920s and continuing for about 30 years, a series of investigations addressed this question.

In 1928, British microbiologist Frederick Griffith made a series of unexpected observations while experimenting with pathogenic (disease-causing) bacteria. When he injected mice with a virulent strain of *Streptococcus pneumoniae* bacteria (then known as *Pneumococcus*), the mice died of blood poisoning. However, when he infected similar mice with a mutant strain of *S. pneumoniae* that lacked the virulent strain's polysaccharide coat, the mice showed no ill effects. The coat was apparently necessary for virulence. The normal pathogenic form of this bacterium is referred to as

the S form because it forms smooth colonies on a culture dish. The mutant form, which lacks an enzyme needed to manufacture the polysaccharide capsule, is called the R form because it forms rough colonies.

To determine whether the polysaccharide coat itself had a toxic effect, Griffith injected dead bacteria of the virulent S strain into mice; the mice remained perfectly healthy. As a control, he injected mice with a mixture containing dead S bacteria of the virulent strain and live coatless R bacteria, each of which by itself did not harm the mice (figure 14.4). Unexpectedly, the mice developed disease symptoms and many of them died. The blood of the dead mice was found to contain high levels of live, virulent *Streptococcus* type S bacteria, which had surface proteins characteristic of the live (previously R) strain. Somehow, the information specifying the polysaccharide coat had passed from the dead, virulent S bacteria to the live, coatless R bacteria in the mixture, permanently transforming the coatless R bacteria into the virulent S variety. **Transformation** is the transfer of genetic material from one cell to another and can alter the genetic makeup of the recipient cell.

Hereditary information can pass from dead cells to living ones, transforming them.

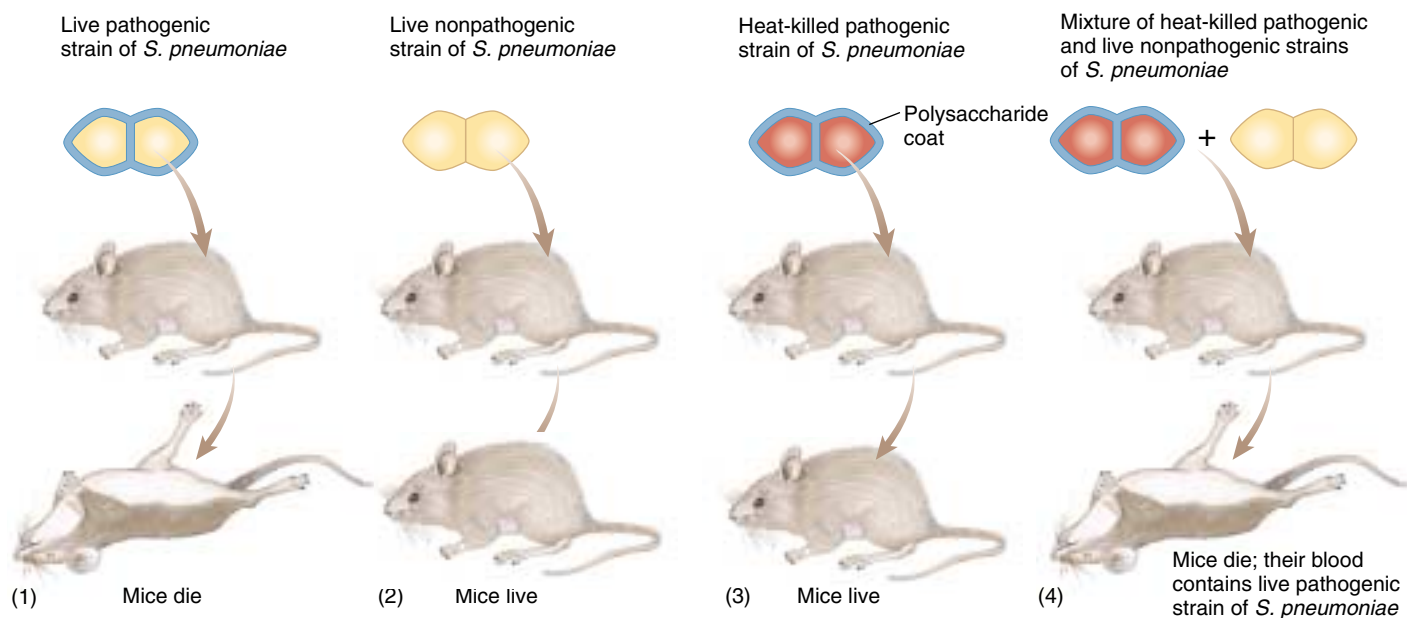


FIGURE 14.4

Griffith's discovery of transformation. (1) The pathogenicity of the bacterium *Streptococcus pneumoniae* kills many of the mice it is injected into. The bacterial cells are covered with a polysaccharide coat, which the bacteria themselves synthesize. (2) Interestingly, an injection of live, coatless bacteria produced no ill effects. However, the coat itself is not the agent of disease. (3) When Griffith injected mice with dead bacteria that possessed polysaccharide coats, the mice were unharmed. (4) But when Griffith injected a mixture of dead bacteria with polysaccharide coats and live bacteria without such coats, many of the mice died, and virulent bacteria with coats were recovered. Griffith concluded that the live cells had been "transformed" by the dead ones; that is, genetic information specifying the polysaccharide coat had passed from the dead cells to the living ones.

The Avery and Hershey-Chase Experiments: The Active Principle Is DNA

The Avery Experiments

The agent responsible for transforming *Streptococcus* went undiscovered until 1944. In a classic series of experiments, Oswald Avery and his coworkers Colin MacLeod and Maclyn McCarty characterized what they referred to as the “transforming principle.” They first prepared the mixture of dead S *Streptococcus* and live R *Streptococcus* that Griffith had used. Then Avery and his colleagues removed as much of the protein as they could from their preparation, eventually achieving 99.98% purity. Despite the removal of nearly all protein, the transforming activity was not reduced. Moreover, the properties of the transforming principle resembled those of DNA in several ways:

1. When the purified principle was analyzed chemically, the array of elements agreed closely with DNA.
2. When spun at high speeds in an ultracentrifuge, the transforming principle migrated to the same level (density) as DNA.
3. Extracting the lipid and protein from the purified transforming principle did not reduce its activity.
4. Protein-digesting enzymes did not affect the principle’s activity; nor did RNA-digesting enzymes.
5. The DNA-digesting enzyme DNase destroyed all transforming activity.

The evidence was overwhelming. They concluded that “a nucleic acid of the deoxyribose type is the fundamental unit of the transforming principle of *Pneumococcus* Type III”—in essence, that DNA is the hereditary material.

The Hershey–Chase Experiment

Avery’s results were not widely accepted at first, as many biologists preferred to believe that proteins were the repository of hereditary information. Additional evidence supporting Avery’s conclusion was provided in 1952 by Alfred Hershey and Martha Chase, who experimented with **bacteriophages**, viruses that attack bacteria. Viruses, described in more detail in chapter 33, consist of either DNA or RNA (ribonucleic acid) surrounded by a protein coat. When a *lytic* (potentially cell-rupturing) bacteriophage infects a bacterial cell, it first binds to the cell’s outer surface and then injects its hereditary information into the cell. There, the hereditary information directs the production of thousands of new viruses within the bacterium. The bacterial cell eventually ruptures, or lyses, releasing the newly made viruses.

To identify the hereditary material injected into bacterial cells at the start of an infection, Hershey and Chase used the bacteriophage T2, which contains DNA rather than RNA. They labeled the two parts of the viruses, the DNA and the protein coat, with different radioactive isotopes

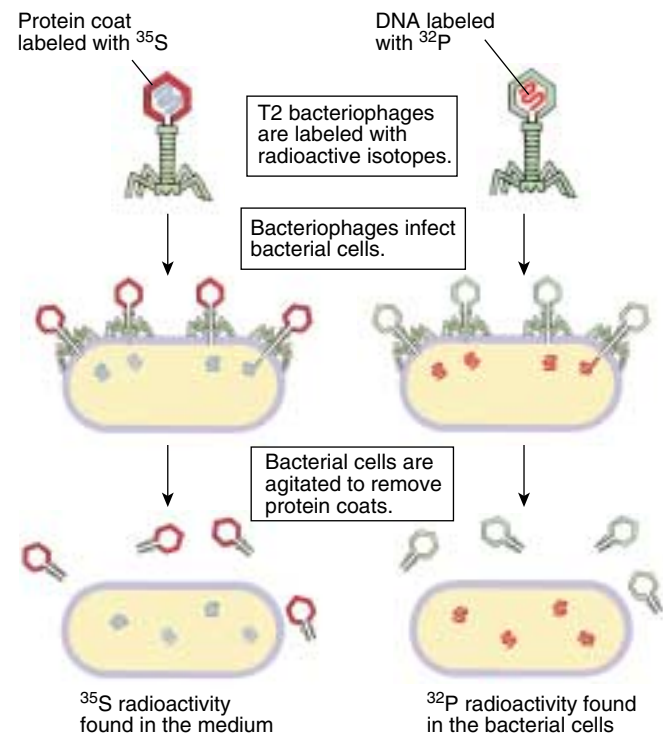


FIGURE 14.5

The Hershey and Chase experiment. Hershey and Chase found that ^{35}S radioactivity did not enter infected bacterial cells and ^{32}P radioactivity did. They concluded that viral DNA, not protein, was responsible for directing the production of new viruses.

that would serve as tracers. In some experiments, the viruses were grown on a medium containing an isotope of phosphorus, ^{32}P , and the isotope was incorporated into the phosphate groups of newly synthesized DNA molecules. In other experiments, the viruses were grown on a medium containing ^{35}S , an isotope of sulfur, which is incorporated into the amino acids of newly synthesized protein coats. The ^{32}P and ^{35}S isotopes are easily distinguished from each other because they emit particles with different energies when they decay.

After the labeled viruses were permitted to infect bacteria, the bacterial cells were agitated violently to remove the protein coats of the infecting viruses from the surfaces of the bacteria. This procedure removed nearly all of the ^{35}S label (and thus nearly all of the viral protein) from the bacteria. However, the ^{32}P label (and thus the viral DNA) had transferred to the interior of the bacteria (figure 14.5) and was found in viruses subsequently released from the infected bacteria. Hence, the hereditary information injected into the bacteria that specified the new generation of viruses was DNA and not protein.

Avery’s experiments demonstrate conclusively that DNA is Griffith’s transforming material. The hereditary material of bacteriophages is DNA and not protein.

14.2 What is the structure of DNA?

The Chemical Nature of Nucleic Acids

A German chemist, Friedrich Miescher, discovered DNA in 1869, only four years after Mendel's work was published. Miescher extracted a white substance from the nuclei of human cells and fish sperm. The proportion of nitrogen and phosphorus in the substance was different from that in any other known constituent of cells, which convinced Miescher that he had discovered a new biological substance. He called this substance "nuclein," because it seemed to be specifically associated with the nucleus.

Levene's Analysis: DNA Is a Polymer

Because Miescher's nuclein was slightly acidic, it came to be called **nucleic acid**. For 50 years biologists did little research on the substance, because nothing was known of its function in cells. In the 1920s, the basic structure of nucleic acids was determined by the biochemist P. A. Levene, who found that DNA contains three main components (figure 14.6): (1) phosphate (PO_4) groups; (2) five-carbon sugars; and (3) nitrogen-containing bases called **purines** (adenine, A, and guanine, G) and **pyrimidines** (thymine, T, and cytosine, C; RNA contains uracil, U, instead of T). From the roughly equal proportions of these components, Levene concluded correctly that DNA and RNA molecules are made of repeating units of the three components. Each unit, consisting of a sugar attached to a phosphate group and a base, is called a **nucleotide**. The identity of the base distinguishes one nucleotide from another.

To identify the various chemical groups in DNA and RNA, it is customary to number the carbon atoms of the base and the sugar and then refer to any chemical group attached to a carbon atom by that number. In the sugar, four of the carbon atoms together with an oxygen atom form a five-membered ring. As illustrated in figure 14.7, the carbon atoms are numbered 1' to 5', proceeding clockwise from the oxygen atom; the prime symbol (') indicates that the number refers to a carbon in a sugar rather than a base. Under this numbering scheme, the phosphate group is attached to the 5' carbon atom of the sugar, and the base is attached to the 1' carbon atom. In addition, a free hydroxyl (—OH) group is attached to the 3' carbon atom.

The 5' phosphate and 3' hydroxyl groups allow DNA and RNA to form long chains of nucleotides, because these two groups can react chemically with each other. The reaction between the phosphate group of one nucleotide and the hydroxyl group of another is a dehydration synthesis, eliminating a water molecule and forming a covalent bond that links the two groups (figure 14.8). The linkage is called a **phosphodiester bond** because

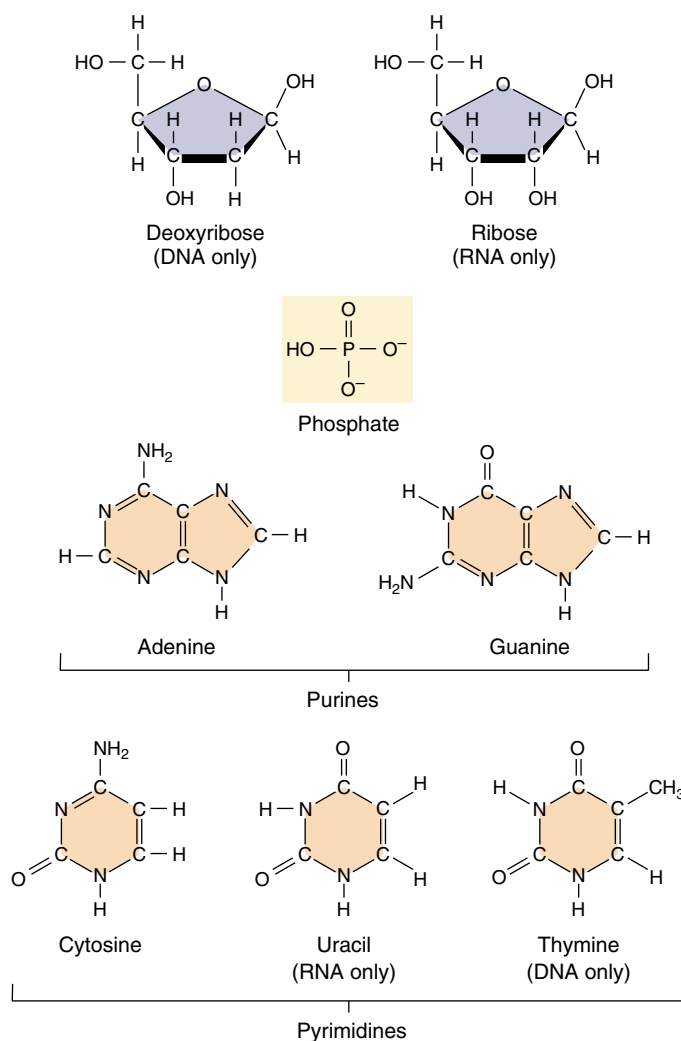


FIGURE 14.6
Nucleotide subunits of DNA and RNA. The nucleotide subunits of DNA and RNA are composed of three elements: a five-carbon sugar (deoxyribose in DNA and ribose in RNA), a phosphate group, and a nitrogenous base (either a purine or a pyrimidine).

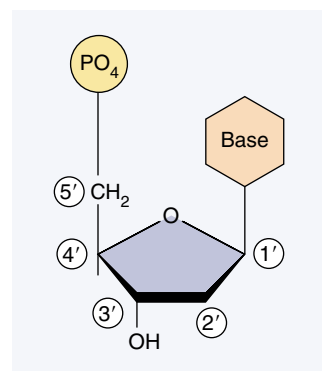


FIGURE 14.7
Numbering the carbon atoms in a nucleotide. The carbon atoms in the sugar of the nucleotide are numbered 1' to 5', proceeding clockwise from the oxygen atom. The "prime" symbol (') indicates that the carbon belongs to the sugar rather than the base.

Table 14.1 Chargaff's Analysis of DNA Nucleotide Base Compositions

Organism	Base Composition (Mole Percent)			
	A	T	G	C
<i>Escherichia coli</i> (K12)	26.0	23.9	24.9	25.2
<i>Mycobacterium tuberculosis</i>	15.1	14.6	34.9	35.4
Yeast	31.3	32.9	18.7	17.1
Herring	27.8	27.5	22.2	22.6
Rat	28.6	28.4	21.4	21.5
Human	30.9	29.4	19.9	19.8

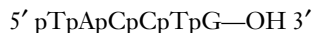
Source: Data from E. Chargaff and J. Davidson (editors), *The Nucleic Acids*, 1955, Academic Press, New York, NY.

the phosphate group is now linked to the two sugars by means of a pair of ester (P—O—C) bonds. The two-unit polymer resulting from this reaction still has a free 5' phosphate group at one end and a free 3' hydroxyl group at the other, so it can link to other nucleotides. In this way, many thousands of nucleotides can join together in long chains.

Linear strands of DNA or RNA, no matter how long, will almost always have a free 5' phosphate group at one end and a free 3' hydroxyl group at the other. Therefore, every DNA and RNA molecule has an intrinsic directionality, and we can refer unambiguously to each end of the molecule. By convention, the sequence of bases is usually expressed in the 5'-to-3' direction. Thus, the base sequence "GTCCAT" refers to the sequence,



where the phosphates are indicated by "p." Note that this is not the same molecule as that represented by the reverse sequence:



Levene's early studies indicated that all four types of DNA nucleotides were present in roughly equal amounts. This result, which later proved to be erroneous, led to the mistaken idea that DNA was a simple polymer in which the four nucleotides merely repeated (for instance, GCAT . . . GCAT . . . GCAT . . . GCAT . . .). If the sequence never varied, it was difficult to see how DNA might contain the hereditary information; this was why Avery's conclusion that DNA is the transforming principle was not readily accepted at first. It seemed more plausible that DNA was simply a structural element of the chromosomes, with proteins playing the central genetic role.

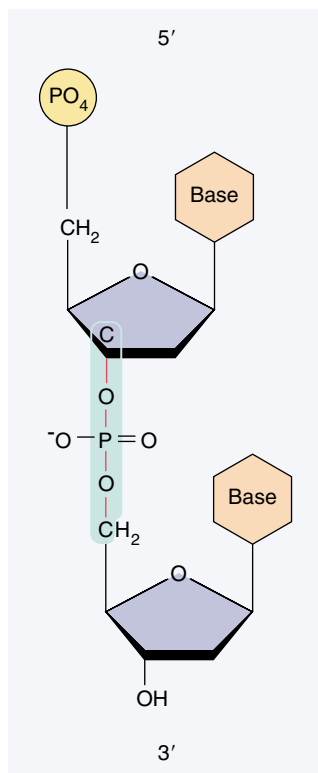


FIGURE 14.8
A phosphodiester bond.

Chargaff's Analysis: DNA Is Not a Simple Repeating Polymer

When Levene's chemical analysis of DNA was repeated using more sensitive techniques that became available after World War II, quite a different result was obtained. The four nucleotides were *not* present in equal proportions in DNA molecules after all. A careful study carried out by Erwin Chargaff showed that the nucleotide composition of DNA molecules varied in complex ways, depending on the source of the DNA (table 14.1). This strongly suggested that DNA was not a simple repeating polymer and might have the information-encoding properties genetic material must have. Despite DNA's complexity, however, Chargaff observed an important underlying regularity in double-stranded DNA: *the amount of adenine present in DNA always equals the amount of thymine, and the amount of guanine always equals the amount of cytosine*. These findings are commonly referred to as **Chargaff's rules**:

1. The proportion of A always equals that of T, and the proportion of G always equals that of C:
 $A = T$, and $G = C$.
2. It follows that there is always an equal proportion of purines (A and G) and pyrimidines (C and T).

A single strand of DNA or RNA consists of a series of nucleotides joined together in a long chain. In all natural double-stranded DNA molecules, the proportion of A equals that of T, and the proportion of G equals that of C.

The Three-Dimensional Structure of DNA

As it became clear that DNA was the molecule that stored the hereditary information, investigators began to puzzle over how such a seemingly simple molecule could carry out such a complex function.

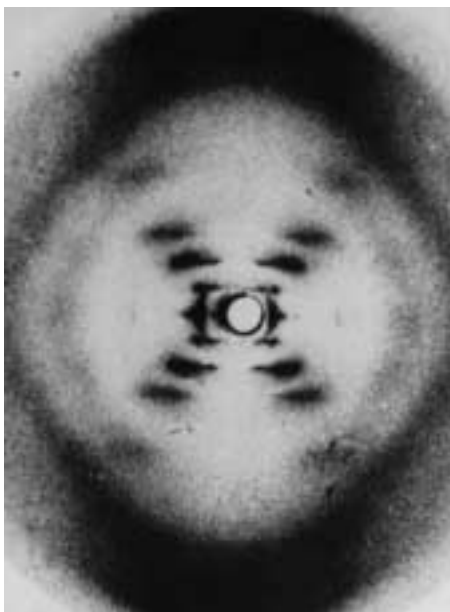
Franklin: X-ray Diffraction Patterns of DNA

The significance of the regularities pointed out by Chargaff were not immediately obvious, but they became clear when a British chemist, Rosalind Franklin (figure 14.9a), carried out an X-ray diffraction analysis of DNA. In X-ray diffraction, a molecule is bombarded with a beam of X rays. When individual rays encounter atoms, their path is bent or diffracted, and the diffraction pattern is recorded on photographic film. The patterns resemble the ripples created by tossing a rock into a smooth lake (figure 14.9b). When carefully analyzed, they yield information about the three-dimensional structure of a molecule.

X-ray diffraction works best on substances that can be prepared as perfectly regular crystalline arrays. However, it was impossible to obtain true crystals of natural DNA at the time Franklin conducted her analysis, so she had to use DNA in the form of fibers. Franklin worked in the laboratory of British biochemist Maurice Wilkins, who was able to prepare more uniformly oriented DNA fibers than anyone had previously. Using these fibers, Franklin succeeded in obtaining crude diffraction information on natural DNA. The diffraction patterns she obtained suggested that the DNA molecule had the shape of a helix, or corkscrew, with a diameter of about 2 nanometers and a complete helical turn every 3.4 nanometers (figure 14.9c).



(a)

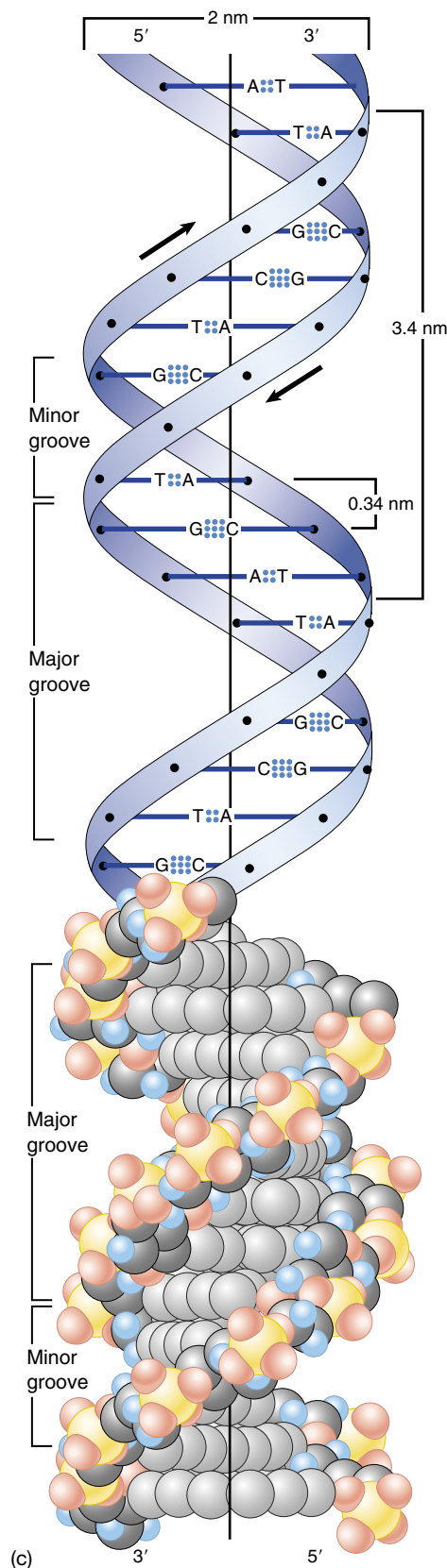


(b)

FIGURE 14.9

Rosalind Franklin's X-ray diffraction patterns suggested the shape of DNA.

(a) Rosalind Franklin developed techniques for taking X-ray diffraction pictures of fibers of DNA. (b) This is the telltale X-ray diffraction photograph of DNA fibers made in 1953 by Rosalind Franklin in the laboratory of Maurice Wilkins. (c) The X-ray diffraction studies of Rosalind Franklin suggested the dimensions of the double helix.



(c)

Watson and Crick: A Model of the Double Helix

Learning informally of Franklin's results before they were published in 1953, James Watson and Francis Crick, two young investigators at Cambridge University, quickly worked out a likely structure for the DNA molecule (figure 14.10), which we now know was substantially correct. They analyzed the problem deductively, first building models of the nucleotides, and then trying to assemble the nucleotides into a molecule that matched what was known about the structure of DNA. They tried various possibilities before they finally hit on the idea that the molecule might be a simple **double helix**, with the bases of two strands pointed inward toward each other, forming **base-pairs**. In their model, base-pairs always consist of purines, which are large, pointing toward pyrimidines, which are small, keeping the diameter of the molecule a constant 2 nanometers. Because hydrogen bonds can form between the bases in a base-pair, the double helix is stabilized as a duplex DNA molecule composed of two **antiparallel strands**, one chain running 3' to 5' and the other 5' to 3'. The base-pairs are planar (flat) and stack 0.34 nm apart as a result of hydrophobic interactions, contributing to the overall stability of the molecule.

The Watson-Crick model explained why Chargaff had obtained the results he had: in a double helix, adenine forms two hydrogen bonds with thymine, but it will not form hydrogen bonds properly with cytosine. Similarly, guanine forms three hydrogen bonds with cytosine, but it will not form hydrogen bonds properly with thymine. Consequently, adenine and thymine will always occur in the same proportions in any DNA molecule, as will guanine and cytosine, because of this base-pairing.

The DNA molecule is a double helix, the strands held together by base-pairing.

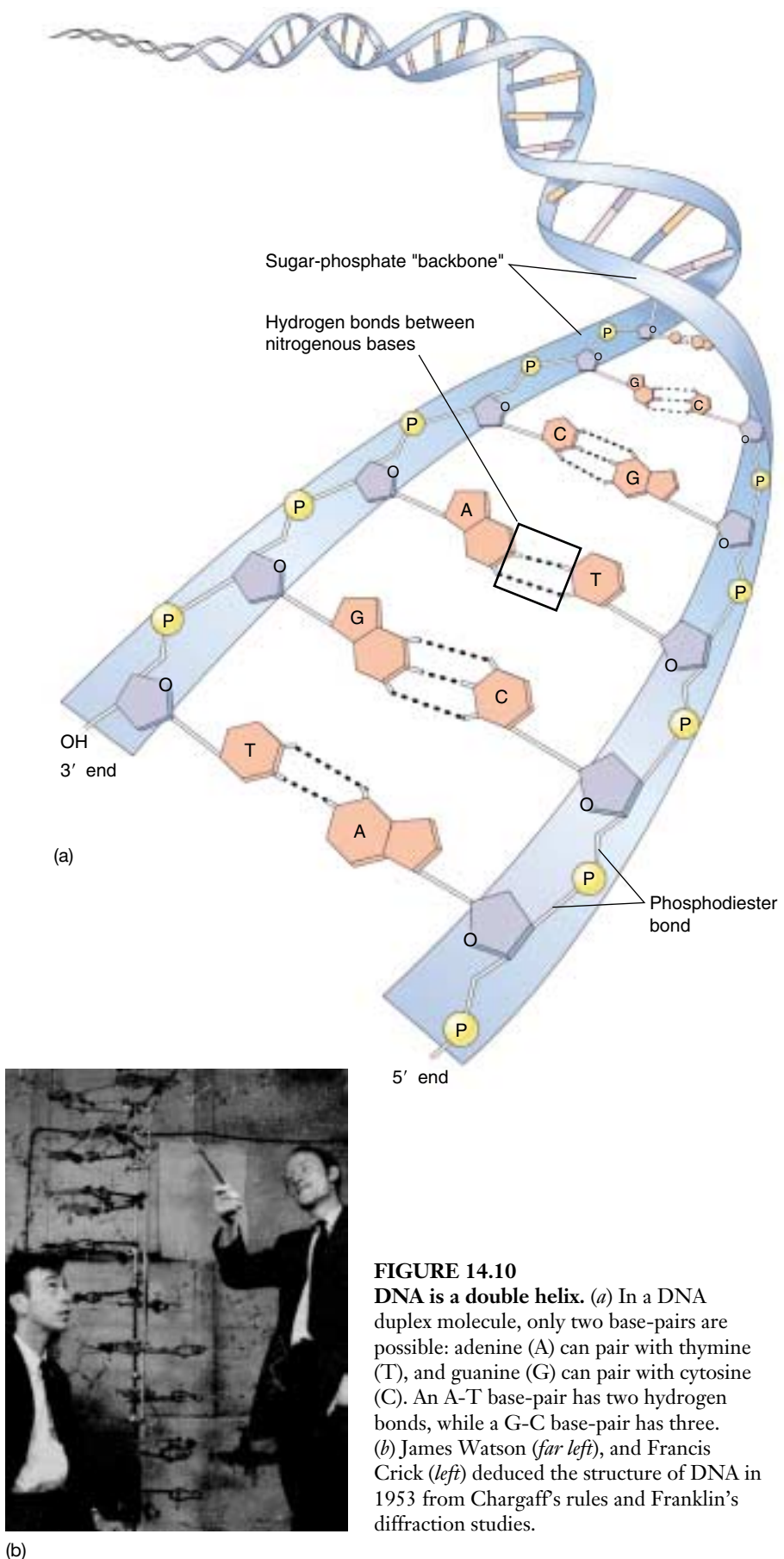


FIGURE 14.10

DNA is a double helix. (a) In a DNA duplex molecule, only two base-pairs are possible: adenine (A) can pair with thymine (T), and guanine (G) can pair with cytosine (C). An A-T base-pair has two hydrogen bonds, while a G-C base-pair has three. (b) James Watson (*far left*), and Francis Crick (*left*) deduced the structure of DNA in 1953 from Chargaff's rules and Franklin's diffraction studies.

14.3 How does DNA replicate?

The Meselson–Stahl Experiment: DNA Replication Is Semiconservative

The Watson–Crick model immediately suggested that the basis for copying the genetic information is **complementarity**. One chain of the DNA molecule may have any conceivable base sequence, but this sequence completely determines the sequence of its partner in the duplex. For example, if the sequence of one chain is 5'-ATTGCAT-3', the sequence of its partner *must* be 3'-TAACGTA-5'. Thus, each chain in the duplex is a complement of the other.

The complementarity of the DNA duplex provides a ready means of accurately duplicating the molecule. If one were to “unzip” the molecule, one would need only to assemble the appropriate complementary nucleotides on the exposed single strands to form two daughter duplexes with the same sequence. This form of DNA replication is called **semiconservative**, because while the sequence of the original duplex is conserved after one round of replication, the duplex itself is not. Instead, each strand of the duplex becomes part of another duplex.

Two other hypotheses of gene replication were also proposed. The conservative model stated that the parental double helix would remain intact and generate DNA copies consisting of entirely new molecules. The dispersive model predicted that parental DNA would become dispersed throughout the new copy so that each strand of all the daughter molecules would be a mixture of old and new DNA.

The three hypotheses of DNA replication were evaluated in 1958 by Matthew Meselson and Franklin Stahl of the California Institute of Technology. These two scientists grew bacteria in a medium containing the heavy isotope of nitrogen, ^{15}N , which became incorporated into the bases of the bacterial DNA. After several generations, the DNA of these bacteria was denser than that of bacteria grown in a medium containing the lighter isotope of nitrogen, ^{14}N . Meselson and Stahl then transferred the bacteria from the ^{15}N medium to the ^{14}N medium and collected the DNA at various intervals.

By dissolving the DNA they had collected in a heavy salt called cesium chloride and then spinning the solution at very high speeds in an ultracentrifuge, Meselson and Stahl were able to separate DNA strands of different densities. The enormous centrifugal forces generated by the ultracentrifuge caused the cesium ions to migrate toward the bottom of the centrifuge tube, creating a gradient of cesium concentration, and thus of density. Each DNA strand floats or sinks in the gradient until it reaches the position where its density exactly matches the density of the cesium there. Because ^{15}N strands are denser than ^{14}N

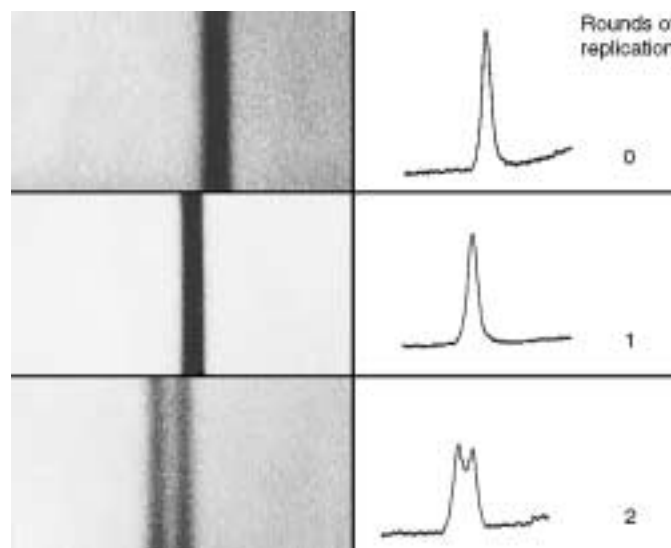


FIGURE 14.11

The key result of the Meselson and Stahl experiment. These bands of DNA, photographed on the left and scanned on the right, are from the density-gradient centrifugation experiment of Meselson and Stahl. At 0 generation, all DNA is heavy; after one replication all DNA has a hybrid density; after two replications, all DNA is hybrid or light.

strands, they migrate farther down the tube to a denser region of the cesium gradient.

The DNA collected immediately after the transfer was all dense. However, after the bacteria completed their first round of DNA replication in the ^{14}N medium, the density of their DNA had decreased to a value intermediate between ^{14}N -DNA and ^{15}N -DNA. After the second round of replication, two density classes of DNA were observed, one intermediate and one equal to that of ^{14}N -DNA (figure 14.11).

Meselson and Stahl interpreted their results as follows: after the first round of replication, each daughter DNA duplex was a hybrid possessing one of the heavy strands of the parent molecule and one light strand; when this hybrid duplex replicated, it contributed one heavy strand to form another hybrid duplex and one light strand to form a light duplex (figure 14.12). Thus, this experiment clearly confirmed the prediction of the Watson–Crick model that DNA replicates in a semiconservative manner.

The basis for the great accuracy of DNA replication is complementarity. A DNA molecule is a duplex, containing two strands that are complementary mirror images of each other, so either one can be used as a template to reconstruct the other.

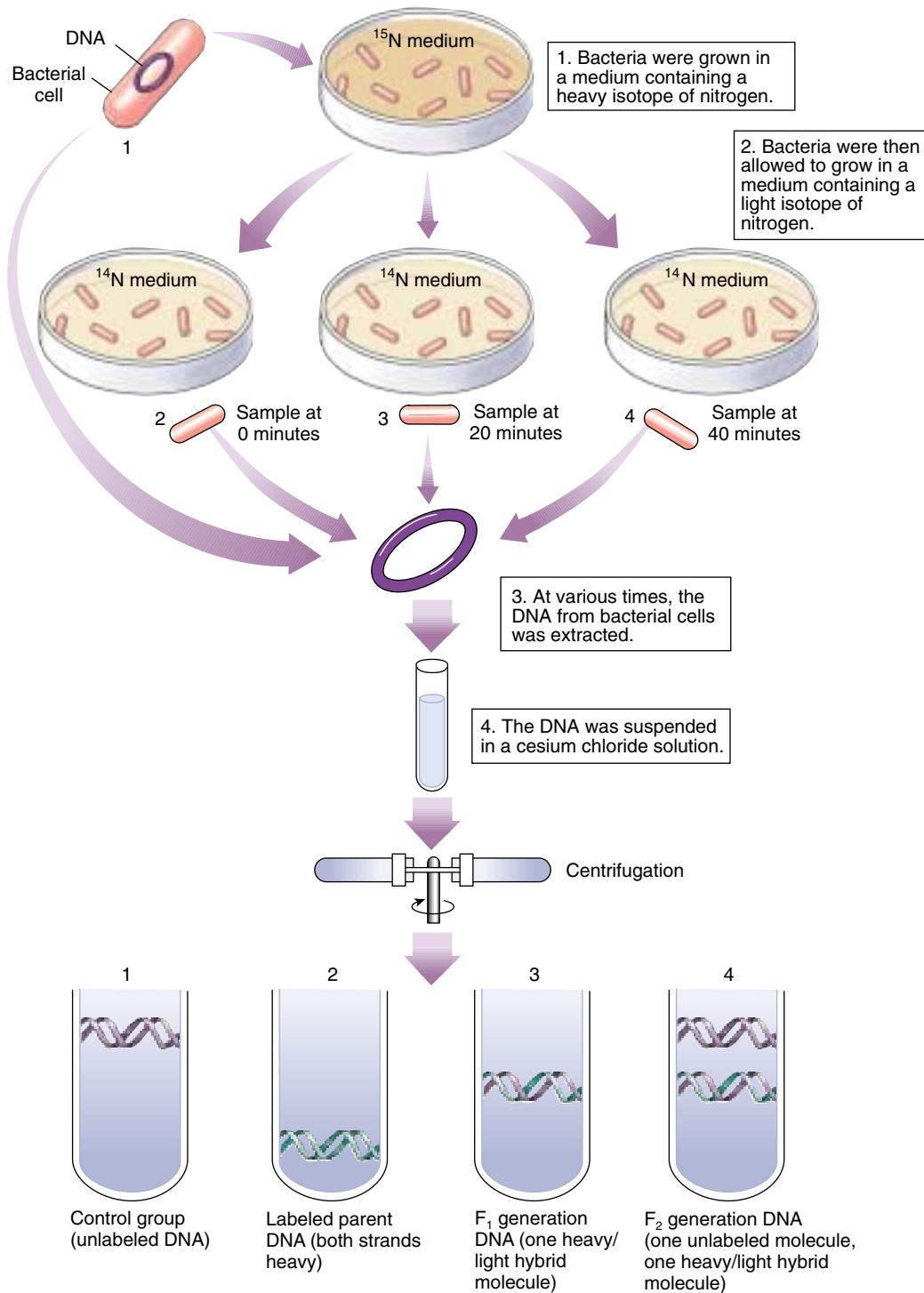


FIGURE 14.12

The Meselson and Stahl experiment: evidence demonstrating semiconservative replication. Bacterial cells were grown for several generations in a medium containing a heavy isotope of nitrogen (^{15}N) and then were transferred to a new medium containing the normal lighter isotope (^{14}N). At various times thereafter, samples of the bacteria were collected, and their DNA was dissolved in a solution of cesium chloride, which was spun rapidly in a centrifuge. Because the cesium ion is so massive, it tends to settle toward the bottom of the spinning tube, establishing a gradient of cesium density. DNA molecules sink in the gradient until they reach a place where their density equals that of the cesium; they then “float” at that position. DNA containing ^{15}N is denser than that containing ^{14}N , so it sinks to a lower position in the cesium gradient. After one generation in ^{14}N medium, the bacteria yielded a single band of DNA with a density between that of ^{14}N -DNA and ^{15}N -DNA, indicating that only one strand of each duplex contained ^{15}N . After two generations in ^{14}N medium, two bands were obtained; one of intermediate density (in which one of the strands contained ^{15}N) and one of low density (in which neither strand contained ^{15}N). Meselson and Stahl concluded that replication of the DNA duplex involves building new molecules by separating strands and assembling new partners on each of these templates.

The Replication Process

To be effective, DNA replication must be fast and accurate. The machinery responsible has been the subject of intensive study for 40 years, and we now know a great deal about it. The replication of DNA begins at one or more sites on the DNA molecule where there is a specific sequence of nucleotides called a **replication origin** (figure 14.13). There the DNA replicating enzyme **DNA polymerase III** and other enzymes begin a complex process that catalyzes the addition of nucleotides to the growing complementary strands of DNA (figure 14.14). Table 14.2 lists the proteins involved in DNA replication in bacteria. Before considering the replication process in detail, let's take a closer look at DNA polymerase III.

DNA Polymerase III

The first DNA polymerase enzyme to be characterized, DNA polymerase I of the bacterium *Escherichia coli*, is a relatively small enzyme that plays a key supporting role in

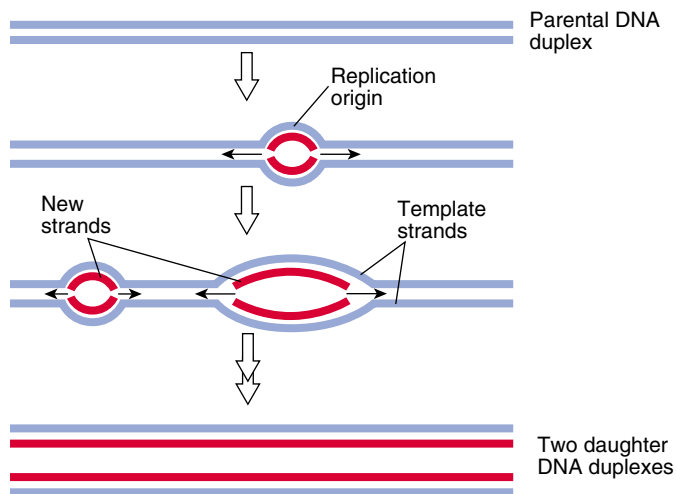


FIGURE 14.13

Origins of replication. At a site called the replication origin, the DNA duplex opens to create two separate strands, each of which can be used as a template for a new strand. Eukaryotic DNA has multiple origins of replication.

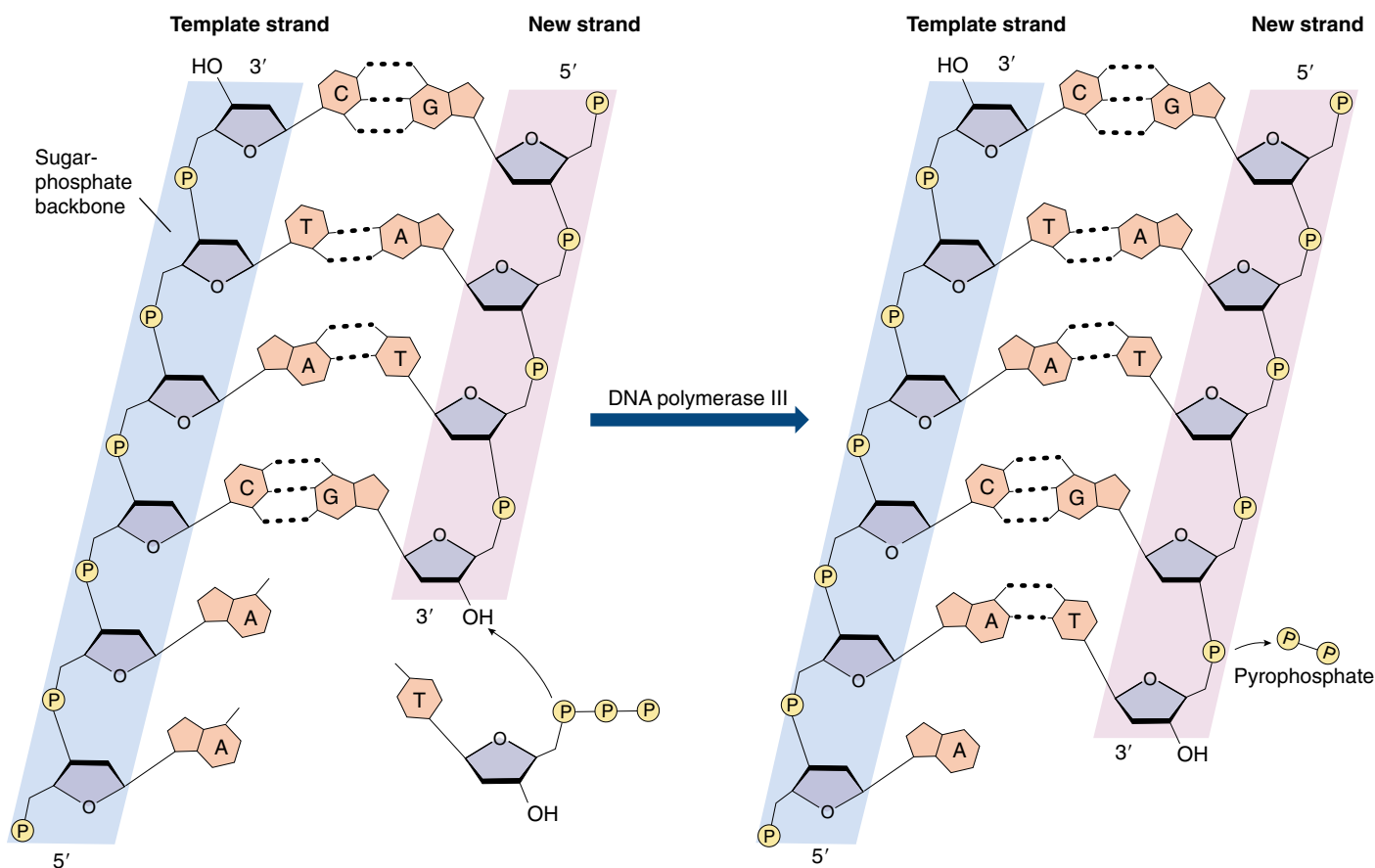


FIGURE 14.14

How nucleotides are added in DNA replication. DNA polymerase III, along with other enzymes, catalyzes the addition of nucleotides to the growing complementary strand of DNA. When a nucleotide is added, two of its phosphates are lost as pyrophosphate.

DNA replication. The true *E. coli* replicating enzyme, dubbed DNA polymerase III, is some 10 times larger and far more complex in structure. We know more about DNA polymerase III than any other organism's DNA polymerase, and so will describe it in detail here. Other DNA polymerases are thought to be broadly similar.

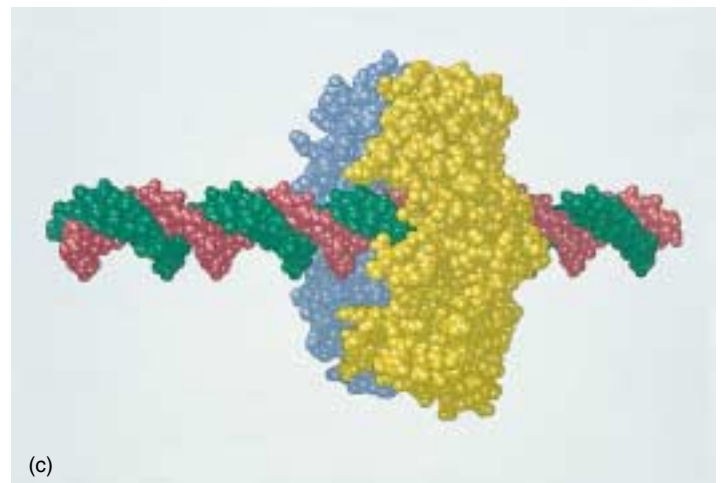
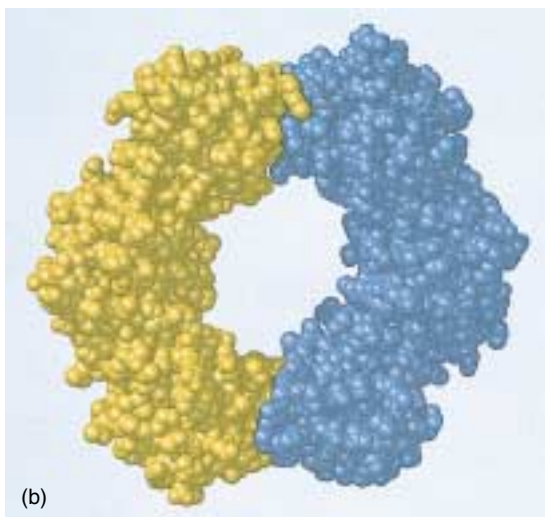
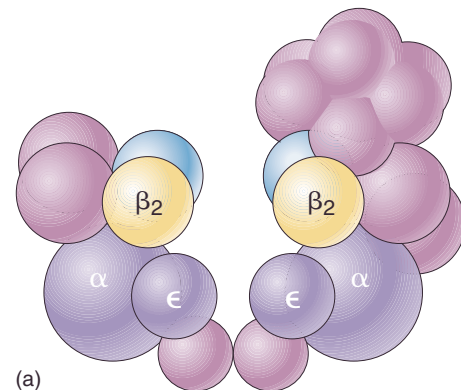
DNA polymerase III contains 10 different kinds of polypeptide chains, as illustrated in figure 14.15. The enzyme is a dimer, with two similar multisubunit complexes. Each complex catalyzes the replication of one DNA strand.

A variety of different proteins play key roles within each complex. The subunits include a single large catalytic α subunit that catalyzes 5' to 3' addition of nucleotides to a growing chain, a smaller ϵ subunit that proofreads 3' to 5' for mistakes, and a ring-shaped β_2 dimer subunit that clamps the polymerase III complex around the DNA double helix. Polymerase III progressively threads the DNA through the enzyme complex, moving it at a rapid rate, some 1000 nucleotides per second (100 full turns of the helix, 0.34 micrometers).

Protein	Role	Size (kd)	Molecules per Cell
Helicase	Unwinds the double helix	300	20
Primase	Synthesizes RNA primers	60	50
Single-strand binding protein	Stabilizes single-stranded regions	74	300
DNA gyrase	Relieves torque	400	250
DNA polymerase III	Synthesizes DNA	≈ 900	20
DNA polymerase I	Erases primer and fills gaps	103	300
DNA ligase	Joins the ends of DNA segments	74	300

FIGURE 14.15

The DNA polymerase III complex. (a) The complex contains 10 kinds of protein chains. The protein is a dimer because both strands of the DNA duplex must be replicated simultaneously. The catalytic (α) subunits, the proofreading (ϵ) subunits, and the "sliding clamp" (β_2) subunits (yellow and blue) are labeled. (b) The "sliding clamp" units encircle the DNA template and (c) move it through the catalytic subunit like a rope drawn through a ring.



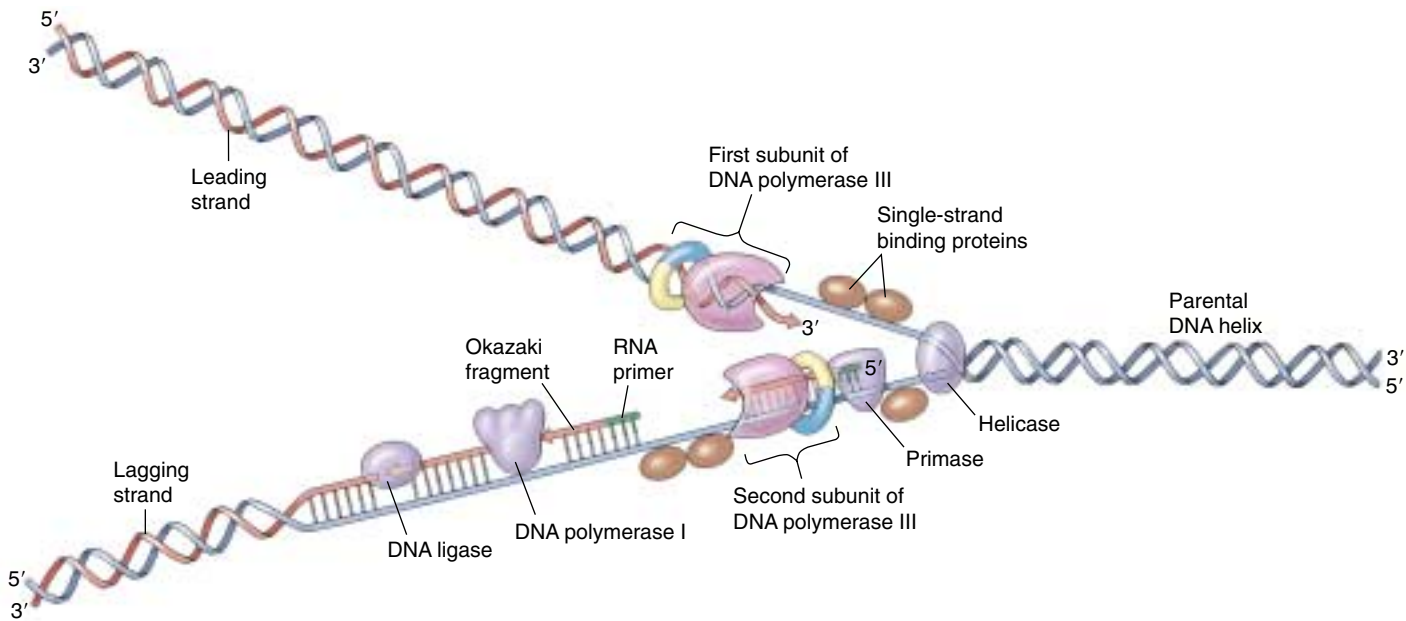


FIGURE 14.16

A DNA replication fork. Helicase enzymes separate the strands of the double helix, and single-strand binding proteins stabilize the single-stranded regions. Replication occurs by two mechanisms. (1) *Continuous synthesis*: After primase adds a short RNA primer, DNA polymerase III adds nucleotides to the 3' end of the leading strand. DNA polymerase I then replaces the RNA primer with DNA nucleotides. (2) *Discontinuous synthesis*: Primase adds a short RNA primer (green) ahead of the 5' end of the lagging strand. DNA polymerase III then adds nucleotides to the primer until the gap is filled in. DNA polymerase I replaces the primer with DNA nucleotides, and DNA ligase attaches the short segment of nucleotides to the lagging strand.

The Need for a Primer

One of the features of DNA polymerase III is that it can add nucleotides only to a chain of nucleotides that is already paired with the parent strands. Hence, DNA polymerase cannot link the first nucleotides in a newly synthesized strand. Instead, another enzyme, an RNA polymerase called **primase**, constructs an **RNA primer**, a sequence of about 10 RNA nucleotides complementary to the parent DNA template. DNA polymerase III recognizes the primer and adds DNA nucleotides to it to construct the new DNA strands. The RNA nucleotides in the primers are then replaced by DNA nucleotides.

The Two Strands of DNA Are Assembled in Different Ways

Another feature of DNA polymerase III is that it can add nucleotides only to the 3' end of a DNA strand (the end with an —OH group attached to a 3' carbon atom). This means that replication always proceeds in the 5' → 3' direction on a growing DNA strand. Because the two parent strands of a DNA molecule are antiparallel, *the new strands are oriented in opposite directions* along the parent templates at each replication fork (figure 14.16). Therefore, the new strands must be elongated by different mechanisms! The **leading strand**, which elongates *toward* the replication fork, is built up simply by adding nucleotides continuously

to its growing 3' end. In contrast, the **lagging strand**, which elongates *away from* the replication fork, is synthesized discontinuously as a series of short segments that are later connected. These segments, called **Okazaki fragments**, are about 100 to 200 nucleotides long in eukaryotes and 1000 to 2000 nucleotides long in prokaryotes. Each Okazaki fragment is synthesized by DNA polymerase III in the 5' → 3' direction, beginning at the replication fork and moving away from it. When the polymerase reaches the 5' end of the lagging strand, another enzyme, **DNA ligase**, attaches the fragment to the lagging strand. The DNA is further unwound, new RNA primers are constructed, and DNA polymerase III then jumps ahead 1000 to 2000 nucleotides (toward the replication fork) to begin constructing another Okazaki fragment. If one looks carefully at electron micrographs showing DNA replication in progress, one can sometimes see that one of the parent strands near the replication fork appears single-stranded over a distance of about 1000 nucleotides. Because the synthesis of the leading strand is continuous, while that of the lagging strand is discontinuous, the overall replication of DNA is said to be **semidiscontinuous**.

The Replication Process

The replication of the DNA double helix is a complex process that has taken decades of research to understand. It takes place in five interlocking steps:

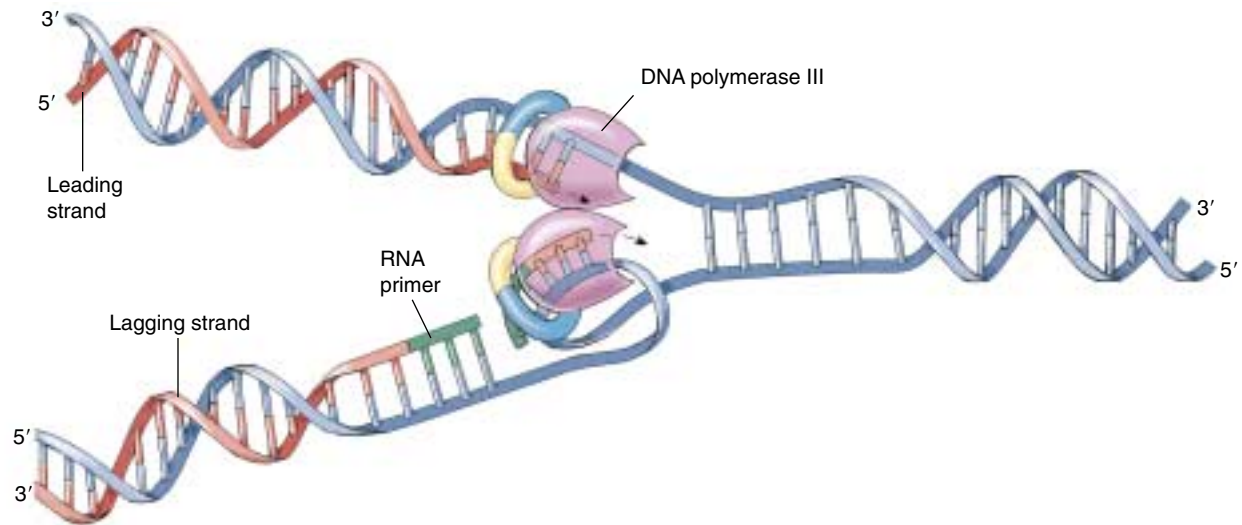


FIGURE 14.17

How DNA polymerase III works. This diagram presents a current view of how DNA polymerase III works. Note that the DNA on the lagging strand is folded to allow the dimeric DNA polymerase III molecule to replicate both strands of the parental DNA duplex simultaneously. This brings the 3' end of each completed Okazaki fragment close to the start site for the next fragment.

1. Opening up the DNA double helix. The very stable DNA double helix must be opened up and its strands separated from each other for semiconservative replication to occur.

Stage one: Initiating replication. The binding of **initiator proteins** to the replication origin starts an intricate series of interactions that opens the helix.

Stage two: Unwinding the duplex. After initiation, “unwinding” enzymes called **helicases** bind to and move along one strand, shouldering aside the other strand as they go.

Stage three: Stabilizing the single strands. The unwound portion of the DNA double helix is stabilized by **single-strand binding protein**, which binds to the exposed single strands, protecting them from cleavage and preventing them from rewinding.

Stage four: Relieving the torque generated by unwinding. For replication to proceed at 1000 nucleotides per second, the parental helix ahead of the replication fork must rotate 100 revolutions per second! To relieve the resulting twisting, called torque, enzymes known as topoisomerases—or, more informally, **gyrases**—cleave a strand of the helix, allow it to swivel around the intact strand, and then reseal the broken strand.

2. Building a primer. New DNA cannot be synthesized on the exposed templates until a primer is constructed, as DNA polymerases require 3' primers to initiate replication. The necessary primer is a short stretch of RNA, added by a specialized RNA polymerase called *primase* in a multisubunit complex in-

formally called a *primosome*. Why an RNA primer, rather than DNA? Starting chains on exposed templates introduces many errors; RNA marks this initial stretch as “temporary,” making this error-prone stretch easy to excise later.

3. Assembling complementary strands. Next, the dimeric DNA polymerase III then binds to the replication fork. While the leading strand complexes with one half of the polymerase dimer, the lagging strand is thought to loop around and complex with the other half of the polymerase dimer (figure 14.17). Moving in concert down the parental double helix, DNA polymerase III catalyzes the formation of complementary sequences on each of the two single strands at the same time.

4. Removing the primer. The enzyme DNA polymerase I now removes the RNA primer and fills in the gap, as well as any gaps between Okazaki fragments.

5. Joining the Okazaki fragments. After any gaps between Okazaki fragments are filled in, the enzyme DNA ligase joins the fragments to the lagging strand.

DNA replication involves many different proteins that open and unwind the DNA double helix, stabilize the single strands, synthesize RNA primers, assemble new complementary strands on each exposed parental strand—one of them discontinuously—remove the RNA primer, and join new discontinuous segments on the lagging strand.

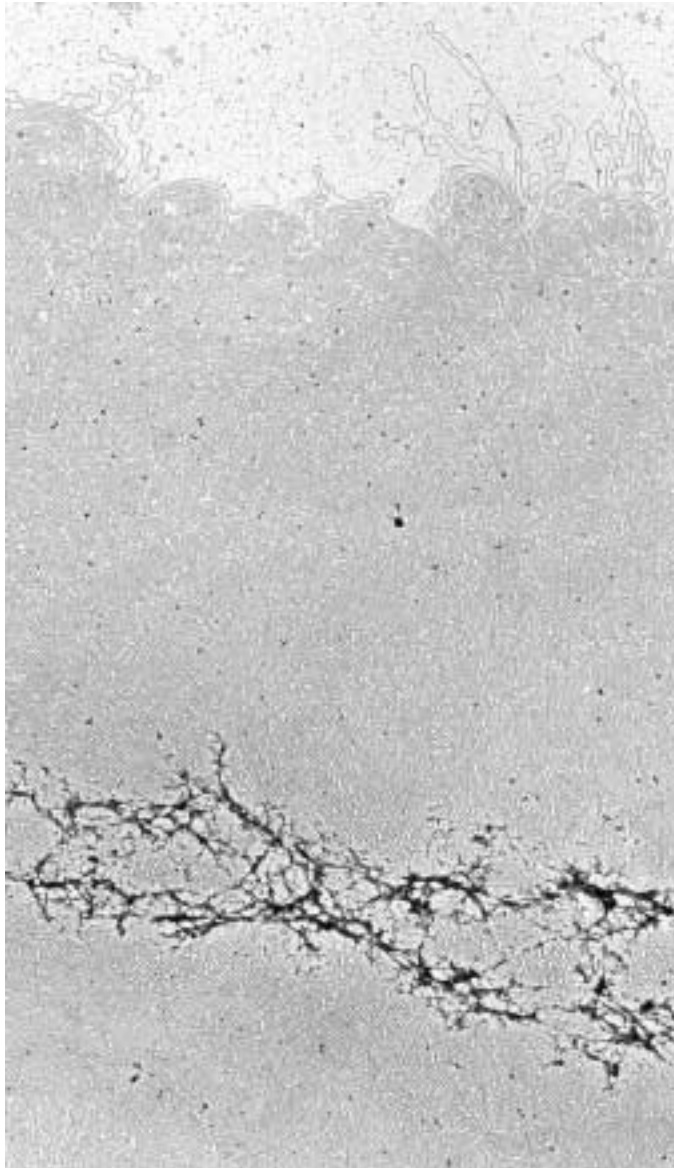


FIGURE 14.18
DNA of a single human chromosome. This chromosome has been “exploded,” or relieved, of most of its packaging proteins. The residual protein scaffolding appears as the dark material in the lower part of the micrograph.

Eukaryotic DNA Replication

In eukaryotic cells, the DNA is packaged in nucleosomes within chromosomes (figure 14.18). Each individual zone of a chromosome replicates as a discrete section called a **replication unit**, or **replicon**, which can vary in length from 10,000 to 1 million base-pairs; most are about 100,000 base-pairs long. Each replication unit has its own origin of replication, and multiple units may be undergoing replication at any given time, as can be seen in electron micrographs of replicating chromosomes (figure 14.19). Each unit replicates in a way fundamentally similar to prokaryotic DNA replication, using similar enzymes. The advantage of having multiple origins of replication in eukaryotes is speed: replication takes approximately eight hours in human cells, but if there were only one origin, it would take 100 times longer. Regulation of the replication process ensures that only one copy of the DNA is ultimately produced. How a cell achieves this regulation is not yet completely clear. It may involve periodic inhibitor or initiator proteins on the DNA molecule itself.

Eukaryotic chromosomes have multiple origins of replication.

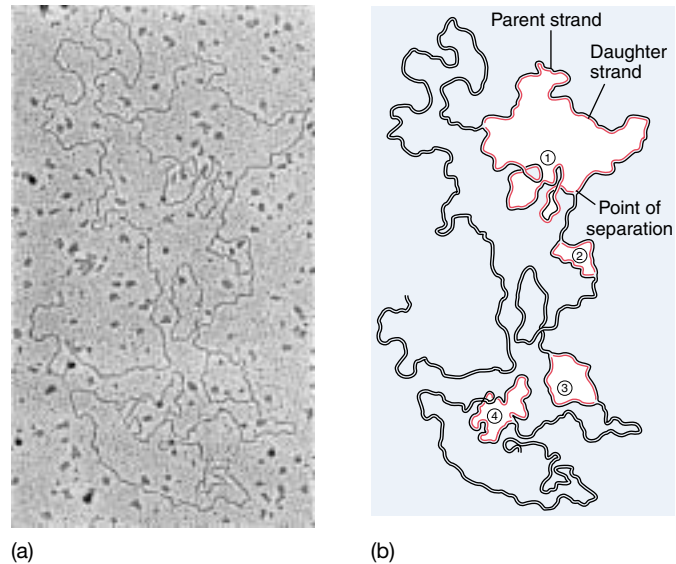


FIGURE 14.19
Eukaryotic chromosomes possess numerous replication forks spaced along their length. Four replication units (each with two replication forks) are producing daughter strands (*a*) in this electron micrograph, as indicated in *red* in the (*b*) corresponding drawing.

14.4 What is a gene?

The One-Gene/One-Polypeptide Hypothesis

As the structure of DNA was being solved, other biologists continued to puzzle over how the genes of Mendel were related to DNA.

Garrod: Inherited Disorders Can Involve Specific Enzymes

In 1902, a British physician, Archibald Garrod, was working with one of the early Mendelian geneticists, his countryman William Bateson, when he noted that certain diseases he encountered among his patients seemed to be more prevalent in particular families. By examining several generations of these families, he found that some of the diseases behaved as if they were the product of simple recessive alleles. Garrod concluded that these disorders were Mendelian traits and that they had resulted from changes in the hereditary information in an ancestor of the affected families.

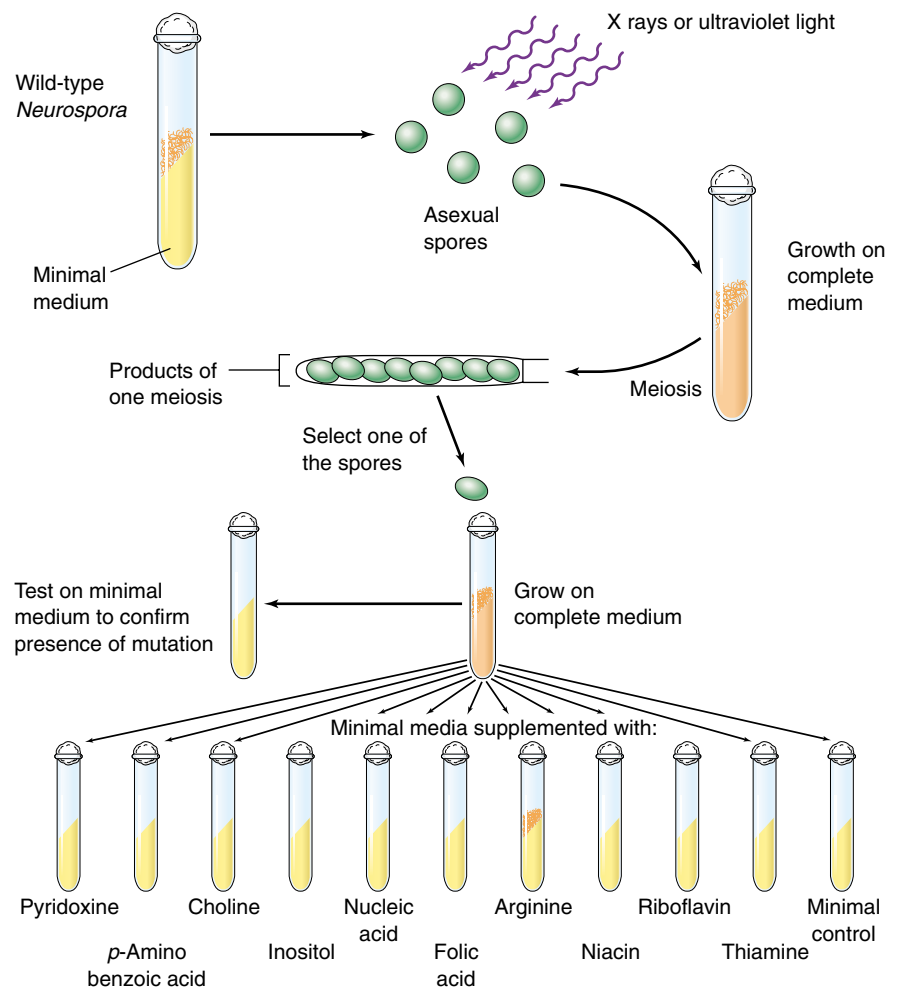
Garrod investigated several of these disorders in detail. In alkaptonuria the patients produced urine that contained homogentisic acid (alkapton). This substance

oxidized rapidly when exposed to air, turning the urine black. In normal individuals, homogentisic acid is broken down into simpler substances. With considerable insight, Garrod concluded that patients suffering from alkaptonuria lacked the enzyme necessary to catalyze this breakdown. He speculated that many other inherited diseases might also reflect enzyme deficiencies.

Beadle and Tatum: Genes Specify Enzymes

From Garrod's finding, it took but a short leap of intuition to surmise that the information encoded within the DNA of chromosomes acts to specify particular enzymes. This point was not actually established, however, until 1941, when a series of experiments by Stanford University geneticists George Beadle and Edward Tatum provided definitive evidence on this point. Beadle and Tatum deliberately set out to create Mendelian mutations in chromosomes and then studied the effects of these mutations on the organism (figure 14.20).

FIGURE 14.20
Beadle and Tatum's procedure for isolating nutritional mutants in *Neurospora*. This fungus grows easily on an artificial medium in test tubes. In this experiment, spores were irradiated to increase the frequency of mutation; they were then placed on a "complete" medium that contained all of the nutrients necessary for growth. Once the fungal colonies were established on the complete medium, individual spores were transferred to a "minimal" medium that lacked various substances the fungus could normally manufacture. Any spore that would not grow on the minimal medium but would grow on the complete medium contained one or more mutations in genes needed to produce the missing nutrients. To determine which gene had mutated, the minimal medium was supplemented with particular substances. The mutation illustrated here produced an arginine mutant, a collection of cells that lost the ability to manufacture arginine. These cells will not grow on minimal medium but will grow on minimal medium with only arginine added.



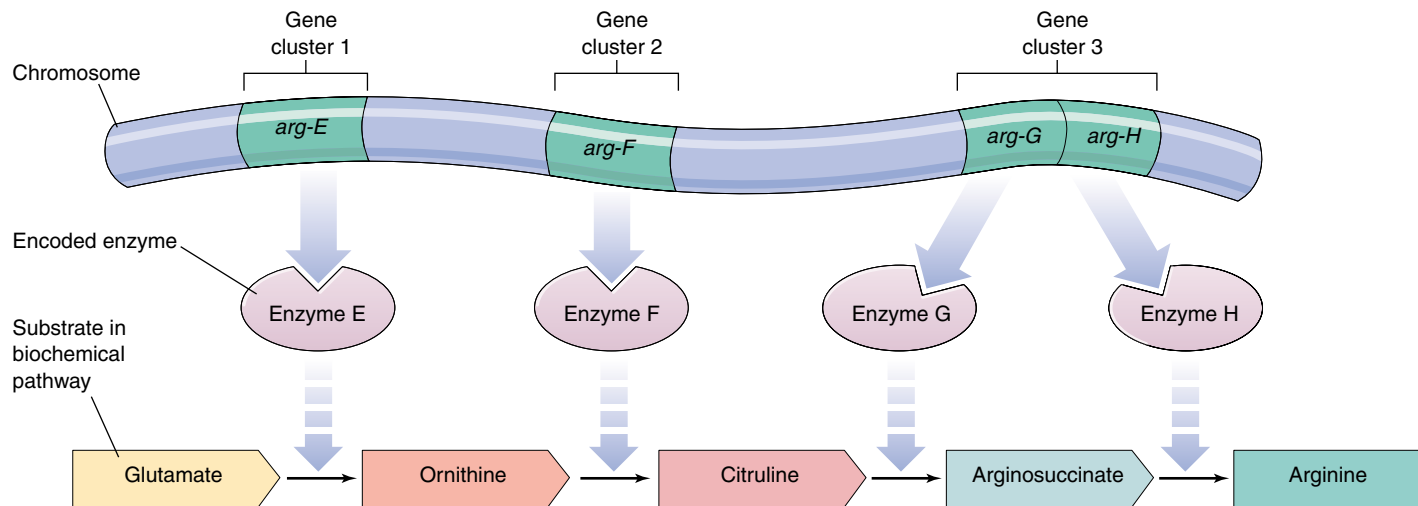


FIGURE 14.21

Evidence for the “one-gene/one-polypeptide” hypothesis. The chromosomal locations of the many arginine mutants isolated by Beadle and Tatum cluster around three locations. These locations correspond to the locations of the genes encoding the enzymes that carry out arginine biosynthesis.

A Defined System. One of the reasons Beadle and Tatum’s experiments produced clear-cut results is that the researchers made an excellent choice of experimental organism. They chose the bread mold *Neurospora*, a fungus that can be grown readily in the laboratory on a defined medium (a medium that contains only known substances such as glucose and sodium chloride, rather than some uncharacterized mixture of substances such as ground-up yeasts). Beadle and Tatum exposed *Neurospora* spores to X rays, expecting that the DNA in some of the spores would experience damage in regions encoding the ability to make compounds needed for normal growth (see figure 14.20). DNA changes of this kind are called mutations, and organisms that have undergone such changes (in this case losing the ability to synthesize one or more compounds) are called mutants. Initially, they allowed the progeny of the irradiated spores to grow on a defined medium containing all of the nutrients necessary for growth, so that any growth-deficient mutants resulting from the irradiation would be kept alive.

Isolating Growth-Deficient Mutants. To determine whether any of the progeny of the irradiated spores had mutations causing metabolic deficiencies, Beadle and Tatum placed subcultures of individual fungal cells on a “minimal” medium that contained only sugar, ammonia, salts, a few vitamins, and water. Cells that had lost the ability to make other compounds necessary for growth would not survive on such a medium. Using this approach, Beadle and Tatum succeeded in identifying and isolating many growth-deficient mutants.

Identifying the Deficiencies. Next the researchers added various chemicals to the minimal medium in an attempt to find one that would enable a given mutant strain

to grow. This procedure allowed them to pinpoint the nature of the biochemical deficiency that strain had. The addition of arginine, for example, permitted several mutant strains, dubbed *arg* mutants, to grow. When their chromosomal positions were located, the *arg* mutations were found to cluster in three areas (figure 14.21).

One-Gene/One-Polypeptide

For each enzyme in the arginine biosynthetic pathway, Beadle and Tatum were able to isolate a mutant strain with a defective form of that enzyme, and the mutation was always located at one of a few specific chromosomal sites. Most importantly, they found there was a different site for each enzyme. Thus, each of the mutants they examined had a defect in a single enzyme, caused by a mutation at a single site on one chromosome. Beadle and Tatum concluded that genes produce their effects by specifying the structure of enzymes and that each gene encodes the structure of one enzyme. They called this relationship the **one-gene/one-enzyme hypothesis**. Because many enzymes contain multiple protein or polypeptide subunits, each encoded by a separate gene, the relationship is today more commonly referred to as “**one-gene/one-polypeptide**.”

Enzymes are responsible for catalyzing the synthesis of all the parts of an organism. They mediate the assembly of nucleic acids, proteins, carbohydrates, and lipids. Therefore, by encoding the structure of enzymes and other proteins, DNA specifies the structure of the organism itself.

Genetic traits are expressed largely as a result of the activities of enzymes. Organisms store hereditary information by encoding the structures of enzymes and other proteins in their DNA.

How DNA Encodes Protein Structure

What kind of information must a gene encode to specify a protein? For some time, the answer to that question was not clear, as protein structure seemed impossibly complex.

Sanger: Proteins Consist of Defined Sequences of Amino Acids

The picture changed in 1953, the same year in which Watson and Crick unraveled the structure of DNA. That year, the English biochemist Frederick Sanger, after many years of work, announced the complete sequence of amino acids in the protein insulin. Insulin, a small protein hormone, was the first protein for which the amino acid sequence was determined. Sanger's achievement was extremely significant because it demonstrated for the first time that proteins consisted of definable sequences of amino acids—for any given form of insulin, every molecule has the same amino acid sequence. Sanger's work soon led to the sequencing of many other proteins, and it became clear that all enzymes and other proteins are strings of amino acids arranged in a certain definite order. The information needed to specify a protein such as an enzyme, therefore, is an ordered list of amino acids.

Ingram: Single Amino Acid Changes in a Protein Can Have Profound Effects

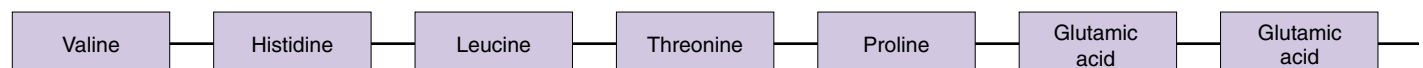
Following Sanger's pioneering work, Vernon Ingram in 1956 discovered the molecular basis of sickle cell anemia, a

protein defect inherited as a Mendelian disorder. By analyzing the structures of normal and sickle cell hemoglobin, Ingram, working at Cambridge University, showed that sickle cell anemia is caused by a change from glutamic acid to valine at a single position in the protein (figure 14.22). The alleles of the gene encoding hemoglobin differed only in their specification of this one amino acid in the hemoglobin amino acid chain.

These experiments and other related ones have finally brought us to a clear understanding of the unit of heredity. The characteristics of sickle cell anemia and most other hereditary traits are defined by changes in protein structure brought about by an alteration in the sequence of amino acids that make up the protein. This sequence in turn is dictated by the order of nucleotides in a particular region of the chromosome. For example, the critical change leading to sickle cell disease is a mutation that replaces a single thymine with an adenine at the position that codes for glutamic acid, converting the position to valine. The sequence of nucleotides that determines the amino acid sequence of a protein is called a **gene**. Although most genes encode proteins or subunits of proteins, some genes are devoted to the production of special forms of RNA, many of which play important roles in protein synthesis themselves.

A half-century of experimentation has made clear that DNA is the molecule responsible for the inheritance of traits, and that this molecule is divided into functional units called genes.

Normal hemoglobin β chain



Sickle cell anemia hemoglobin β chain

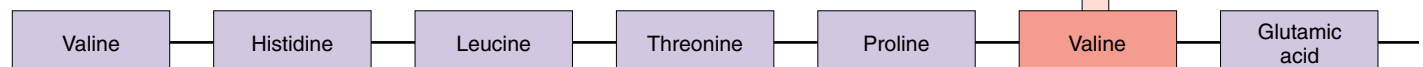


FIGURE 14.22

The molecular basis of a hereditary disease. Sickle cell anemia is produced by a recessive allele of the gene that encodes the hemoglobin β chains. It represents a change in a single amino acid, from glutamic acid to valine at the sixth position in the chains, which consequently alters the tertiary structure of the hemoglobin molecule, reducing its ability to carry oxygen.



Summary

Questions

Media Resources

14.1 What is the genetic material?

- Eukaryotic cells store hereditary information within the nucleus.
- In viruses, bacteria, and eukaryotes, the hereditary information resides in nucleic acids. The transfer of nucleic acids can lead to the transfer of hereditary traits.
- When radioactively labeled DNA viruses infect bacteria, the DNA but not the protein coat of the viruses enters the bacterial cells, indicating that the hereditary material is DNA rather than protein.

- In Hammerling's experiments on *Acetabularia*, what happened when a stalk from *A. crenulata* was grafted to a foot from *A. mediterranea*?
- How did Hershey and Chase determine which component of bacterial viruses contains the viruses' hereditary information?



- Experiment: Griffith/Hershey/Chase-DNA is the Genetic Material

14.2 What is the structure of DNA?

- Chargaff showed that the proportion of adenine in DNA always equals that of thymine, and the proportion of guanine always equals that of cytosine.
- DNA has the structure of a double helix, consisting of two chains of nucleotides held together by hydrogen bonds between adenines and thymines, and between guanines and cytosines.

- What is the three-dimensional shape of DNA, and how does this shape fit with Chargaff's observations on the proportions of purines and pyrimidines in DNA?
- How did Meselson and Stahl show that DNA replication is semiconservative?



- DNA Structure
- DNA Packaging



- Nucleic Acid
- DNA Structure



- Experiment: Kornbert-Isolating DNA Poly merase

14.3 How does DNA replicate?

- During the S phase of the cell cycle, the hereditary message in DNA is replicated with great accuracy.
- During replication, the DNA duplex is unwound, and two new strands are assembled in opposite directions along the original strands. One strand elongates by the continuous addition of nucleotides to its growing end; the other is constructed by the addition of segments containing 100 to 2000 nucleotides, which are then joined to the end of that strand.

- How is the leading strand of a DNA duplex replicated? How is the lagging strand replicated? What is the basis for the requirement that the leading and lagging strands be replicated by different mechanisms?



- DNA Replication



- DNA Replication



- Student Research: Microsatellites in Rabbits Experiment
- Meselson-Stahl—DNA Replication is Semiconservative
- Okazaki: DNA Synthesis is Discontinuous

14.4 What is a gene?

- Most hereditary traits reflect the actions of enzymes.
- The traits are hereditary because the information necessary to specify the structure of the enzymes is stored within the DNA.
- Each enzyme is encoded by a specific region of the DNA called a gene.

- What hypothesis did Beadle and Tatum test in their experiments on *Neurospora*? What did they do to change the DNA in individuals of this organism? How did they determine whether any of these changes affected enzymes in biosynthetic pathways?



- Scientists on Science: The Future of Molecular Biology
- Experiment: Ephrussi/Beadle/Tatum—Genes Encode Enzymes

15

Genes and How They Work

Concept Outline

15.1 The Central Dogma traces the flow of gene-encoded information.

Cells Use RNA to Make Protein. The information in genes is expressed in two steps, first being transcribed into RNA, and the RNA then being translated into protein.

15.2 Genes encode information in three-nucleotide code words.

The Genetic Code. The sequence of amino acids in a protein is encoded in the sequence of nucleotides in DNA, three nucleotides encoding an amino acid.

15.3 Genes are first transcribed, then translated.

Transcription. The enzyme RNA polymerase unwinds the DNA helix and synthesizes an RNA copy of one strand.

Translation. mRNA is translated by activating enzymes that select tRNAs to match amino acids. Proteins are synthesized on ribosomes, which provide a framework for the interaction of tRNA and mRNA.

15.4 Eukaryotic gene transcripts are spliced.

The Discovery of Introns. Eukaryotic genes contain extensive material that is not translated.

Differences between Bacterial and Eukaryotic Gene Expression. Gene expression is broadly similar in bacteria and eukaryotes, although it differs in some respects.



FIGURE 15.1

The unraveled chromosome of an *E. coli* bacterium. This complex tangle of DNA represents the full set of assembly instructions for the living organism *E. coli*.

Every cell in your body contains the hereditary instructions specifying that you will have arms rather than fins, hair rather than feathers, and two eyes rather than one. The color of your eyes, the texture of your fingernails, and all of the other traits you receive from your parents are recorded in the cells of your body. As we have seen, this information is contained in long molecules of DNA (figure 15.1). The essence of heredity is the ability of cells to use the information in their DNA to produce particular proteins, thereby affecting what the cells will be like. In that sense, proteins are the tools of heredity. In this chapter, we will examine how proteins are synthesized from the information in DNA, using both prokaryotes and eukaryotes as models.

15.1 The Central Dogma traces the flow of gene-encoded information.

Cells Use RNA to Make Protein

To find out how a eukaryotic cell uses its DNA to direct the production of particular proteins, you must first ask where in the cell the proteins are made. We can answer this question by placing cells in a medium containing radioactively labeled amino acids for a short time. The cells will take up the labeled amino acids and incorporate them into proteins. If we then look to see where in the cells radioactive proteins first appear, we will find that it is not in the nucleus, where the DNA is, but rather in the cytoplasm, on large RNA-protein aggregates called **ribosomes** (figure 15.2). These polypeptide-making factories are very complex, composed of several RNA molecules and over 50 different proteins (figure 15.3). Protein synthesis involves three different sites on the ribosome surface, called the P, A, and E sites, discussed later in this chapter.

Kinds of RNA

The class of RNA found in ribosomes is called **ribosomal RNA (rRNA)**. During polypeptide synthesis, rRNA provides the site where polypeptides are assembled. In addition to rRNA, there are two other major classes of RNA in cells. **Transfer RNA (tRNA)** molecules both transport the amino acids to the ribosome for use in building the polypeptides and position each amino acid at the correct place on the elongating polypeptide chain (figure 15.4). Human cells contain about 45 different kinds of tRNA molecules. **Messenger RNA (mRNA)** molecules are long strands of RNA that are transcribed from DNA and that travel to the ribosomes to direct precisely *which* amino acids are assembled into polypeptides.

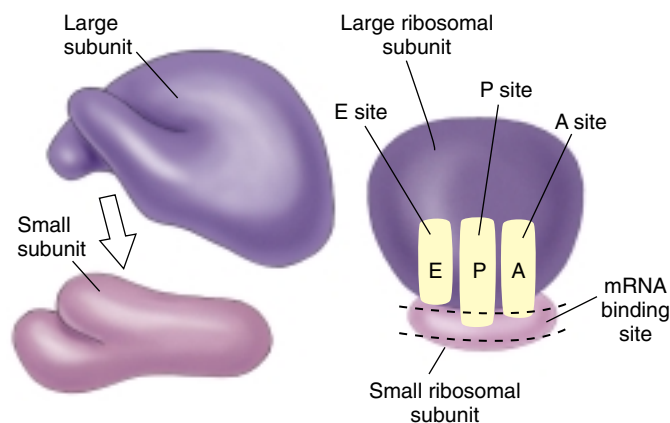


FIGURE 15.2

A ribosome is composed of two subunits. The smaller subunit fits into a depression on the surface of the larger one. The A, P, and E sites on the ribosome, discussed later in this chapter, play key roles in protein synthesis.

These RNA molecules, together with ribosomal proteins and certain enzymes, constitute a system that reads the genetic messages encoded by nucleotide sequences in the DNA and produces the polypeptides that those sequences specify. As we will see, biologists have also learned to read these messages. In so doing, they have learned a great deal about what genes are and how they are able to dictate what a protein will be like and when it will be made.

The Central Dogma

All organisms, from the simplest bacteria to ourselves, use the same basic mechanism of reading and expressing genes, so fundamental to life as we know it that it is often referred to as the “Central Dogma”: Information passes from the genes (DNA) to an RNA copy of the gene, and the RNA copy directs the sequential assembly of a chain of amino acids (figure 15.5). Said briefly,

DNA → RNA → protein

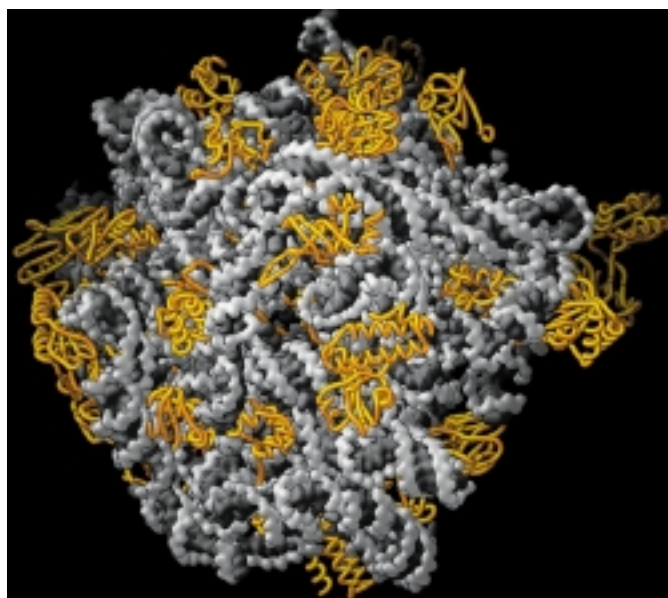


FIGURE 15.3

Ribosomes are very complex machines. The complete atomic structure of a bacterial large ribosomal subunit has been determined at 2.4 Å resolution. The RNA of the subunit is shown in gray and the proteins in gold. The subunit's RNA is twisted into irregular shapes that fit together like a three-dimensional jigsaw puzzle. Proteins are abundant everywhere on its surface except where peptide bonds form and where it contacts the small subunit. The proteins stabilize the structure by interacting with adjacent RNA strands, often with folded extensions that reach into the subunit's interior.

Transcription: An Overview

The first step of the Central Dogma is the transfer of information from DNA to RNA, which occurs when an mRNA copy of the gene is produced. Like all classes of RNA, mRNA is formed on a DNA template. Because the DNA sequence in the gene is transcribed into an RNA sequence, this stage is called **transcription**. Transcription is initiated when the enzyme **RNA polymerase** binds to a particular binding site called a **promoter** located at the beginning of a gene. Starting there, the RNA polymerase moves along the strand into the gene. As it encounters each DNA nucleotide, it adds the corresponding complementary RNA nucleotide to a growing mRNA strand. Thus, guanine (G), cytosine (C), thymine (T), and adenine (A) in the DNA would signal the addition of C, G, A, and uracil (U), respectively, to the mRNA.

When the RNA polymerase arrives at a transcriptional “stop” signal at the opposite end of the gene, it disengages from the DNA and releases the newly assembled RNA chain. This chain is a complementary transcript of the gene from which it was copied.

Translation: An Overview

The second step of the Central Dogma is the transfer of information from RNA to protein, which occurs when the information contained in the mRNA transcript is used to direct the sequence of amino acids during the synthesis of polypeptides by ribosomes. This process is called **translation** because the nucleotide sequence of the mRNA transcript is translated into an amino acid sequence in the polypeptide. Translation begins when an rRNA molecule within the ribosome recognizes and binds to a “start” sequence on the mRNA. The ribosome then moves along the mRNA molecule, three nucleotides at a time. Each group of three nucleotides is a codeword that specifies which amino acid will be added to the growing polypeptide chain. The ribosome continues in this fashion until it encounters a translational “stop” signal; then it disengages from the mRNA and releases the completed polypeptide.

The two steps of the Central Dogma, taken together, are a concise summary of the events involved in the expression of an active gene. Biologists refer to this process as **gene expression**.

The information encoded in genes is expressed in two phases: transcription, in which an RNA polymerase enzyme assembles an mRNA molecule whose nucleotide sequence is complementary to the DNA nucleotide sequence of the gene; and translation, in which a ribosome assembles a polypeptide, whose amino acid sequence is specified by the nucleotide sequence in the mRNA.

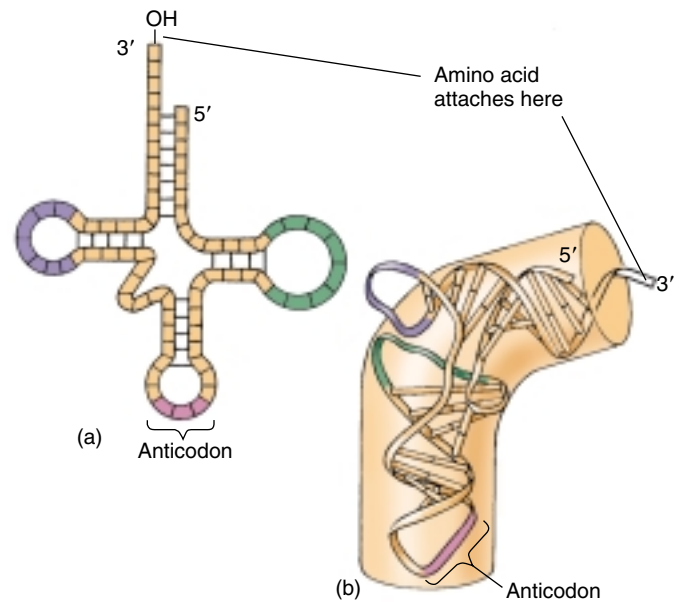


FIGURE 15.4

The structure of tRNA. (a) In the two-dimensional schematic, the three loops of tRNA are unfolded. Two of the loops bind to the ribosome during polypeptide synthesis, and the third loop contains the anticodon sequence, which is complementary to a three-base sequence on messenger RNA. Amino acids attach to the free, single-stranded —OH end. (b) In the three-dimensional structure, the loops of tRNA are folded.

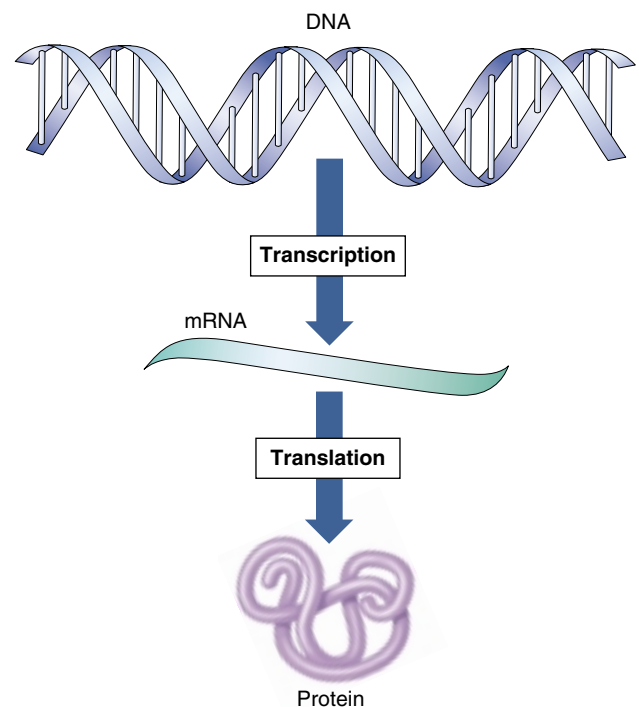


FIGURE 15.5

The Central Dogma of gene expression. DNA is transcribed to make mRNA, which is translated to make a protein.

15.2 Genes encode information in three-nucleotide code words.

The Genetic Code

The essential question of gene expression is, “How does the *order* of nucleotides in a DNA molecule encode the information that specifies the order of amino acids in a polypeptide?” The answer came in 1961, through an experiment led by Francis Crick. That experiment was so elegant and the result so critical to understanding the genetic code that we will describe it in detail.

Proving Code Words Have Only Three Letters

Crick and his colleagues reasoned that the genetic code most likely consisted of a series of blocks of information called **codons**, each corresponding to an amino acid in the encoded protein. They further hypothesized that the information within one codon was probably a sequence of three nucleotides specifying a particular amino acid. They arrived at the number three, because a two-nucleotide codon would not yield enough combinations to code for the 20 different amino acids that commonly occur in proteins. With four DNA nucleotides (G, C, T, and A), only 4^2 , or 16, different pairs of nucleotides could be formed. However, these same nucleotides can be arranged in 4^3 , or 64, different combinations of three, more than enough to code for the 20 amino acids.

In theory, the codons in a gene could lie immediately adjacent to each other, forming a continuous sequence of transcribed nucleotides. Alternatively, the sequence could be punctuated with untranscribed nucleotides between the codons, like the spaces that separate the words in this sentence. It was important to determine which method cells employ because these two ways of transcribing DNA imply different translating processes.

To choose between these alternative mechanisms, Crick and his colleagues used a chemical to delete one, two, or three nucleotides from a viral DNA molecule and then asked whether a gene downstream of the deletions was transcribed correctly. When they made a single deletion or two deletions near each other, the **reading frame** of the genetic message shifted, and the downstream gene was transcribed as nonsense. However, when they made three deletions, the correct reading frame was restored, and the sequences downstream were transcribed correctly. They obtained the same results when they made additions to the DNA consisting of one, two, or three nucleotides. As shown in figure 15.6, these results could not have been obtained if the codons were punctuated by untranscribed nucleotides. Thus, Crick and his colleagues concluded that the genetic code is read in increments consisting of three nucleotides (in other words, it is a **triplet code**) and that reading occurs continuously without punctuation between the three-nucleotide units.

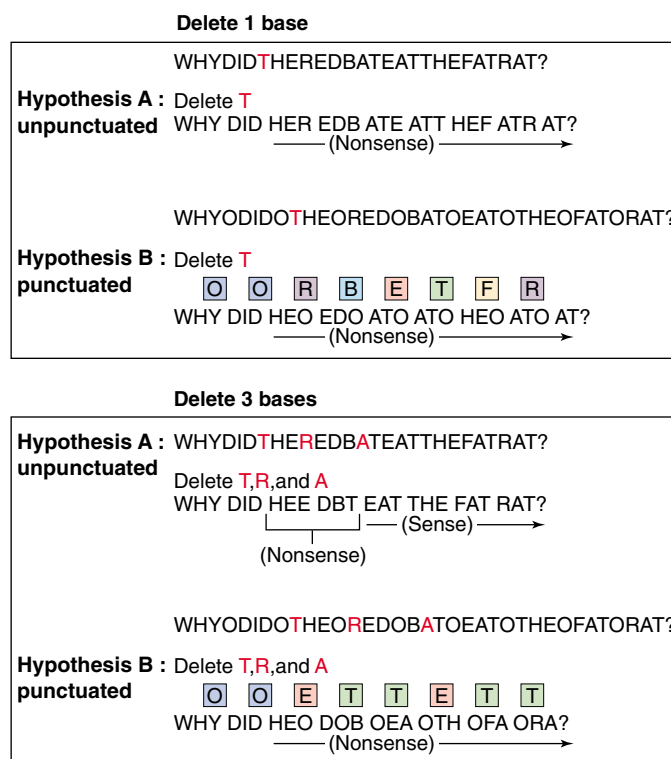


FIGURE 15.6

Using frame-shift alterations of DNA to determine if the genetic code is punctuated. The hypothetical genetic message presented here is “Why did the red bat eat the fat rat?” Under hypothesis B, which proposes that the message is punctuated, the three-letter words are separated by nucleotides that are not read (indicated by the letter “O”).

Breaking the Genetic Code

Within a year of Crick’s experiment, other researchers succeeded in determining the amino acids specified by particular three-nucleotide units. Marshall Nirenberg discovered in 1961 that adding the synthetic mRNA molecule polyU (an RNA molecule consisting of a string of uracil nucleotides) to cell-free systems resulted in the production of the polypeptide polyphenylalanine (a string of phenylalanine amino acids). Therefore, one of the three-nucleotide sequences specifying phenylalanine is UUU. In 1964, Nirenberg and Philip Leder developed a powerful **triplet binding assay** in which a specific triplet was tested to see which radioactive amino acid (complexed to tRNA) it would bind. Some 47 of the 64 possible triplets gave unambiguous results. Har Gobind Khorana decoded the remaining 17 triplets by constructing artificial mRNA molecules of defined sequence and examining what polypeptides they directed. In these ways, all 64 possible three-nucleotide sequences were tested, and the full genetic code was determined (table 15.1).

Table 15.1 The Genetic Code									
First Letter	Second Letter								Third Letter
	U		C		A		G		
U	UUU	Phenylalanine	UCU	Serine	UAU	Tyrosine	UGU	Cysteine	U
	UUC		UCC		UAC		UGC		C
	UUA	Leucine	UCA		UAA	Stop	UGA	Stop	A
	UUG		UCG		UAG	Stop	UGG	Tryptophan	G
C	CUU	Leucine	CCU	Proline	CAU	Histidine	CGU	Arginine	U
	CUC		CCC		CAC		CGC		C
	CUA		CCA		CAA	Glutamine	CGA		A
	CUG		CCG		CAG		CGG		G
A	AUU	Isoleucine	ACU	Threonine	AAU	Asparagine	AGU	Serine	U
	AUC		ACC		AAC		AGC		C
	AUA	Methionine; Start	ACA		AAA	Lysine	AGA	Arginine	A
	AUG		ACG		AAG		AGG		G
G	GUU	Valine	GCU	Alanine	GAU	Aspartate	GGU	Glycine	U
	GUC		GCC		GAC		GGC		C
	GUA		GCA		GAA	Glutamate	GGA		A
	GUG		GCG		GAG		GGG		G

A codon consists of three nucleotides read in the sequence shown. For example, ACU codes for threonine. The first letter, A, is in the First Letter column; the second letter, C, is in the Second Letter column; and the third letter, U, is in the Third Letter column. Each of the mRNA codons is recognized by a corresponding anticodon sequence on a tRNA molecule. Some tRNA molecules recognize more than one codon in mRNA, but they always code for the same amino acid. In fact, most amino acids are specified by more than one codon. For example, threonine is specified by four codons, which differ only in the third nucleotide (ACU, ACC, ACA, and ACG).

The Code Is Practically Universal

The genetic code is the same in almost all organisms. For example, the codon AGA specifies the amino acid arginine in bacteria, in humans, and in all other organisms whose genetic code has been studied. The universality of the genetic code is among the strongest evidence that all living things share a common evolutionary heritage. Because the code is universal, genes transcribed from one organism can be translated in another; the mRNA is fully able to dictate a functionally active protein. Similarly, genes can be transferred from one organism to another and be successfully transcribed and translated in their new host. This universality of gene expression is central to many of the advances of genetic engineering. Many commercial products such as the insulin used to treat diabetes are now manufactured by placing human genes into bacteria, which then serve as tiny factories to turn out prodigious quantities of insulin.

But Not Quite

In 1979, investigators began to determine the complete nucleotide sequences of the mitochondrial genomes in

humans, cattle, and mice. It came as something of a shock when these investigators learned that the genetic code used by these mammalian mitochondria was not quite the same as the “universal code” that has become so familiar to biologists. In the mitochondrial genomes, what should have been a “stop” codon, UGA, was instead read as the amino acid tryptophan; AUA was read as methionine rather than isoleucine; and AGA and AGG were read as “stop” rather than arginine. Furthermore, minor differences from the universal code have also been found in the genomes of chloroplasts and ciliates (certain types of protists).

Thus, it appears that the genetic code is not quite universal. Some time ago, presumably after they began their endosymbiotic existence, mitochondria and chloroplasts began to read the code differently, particularly the portion of the code associated with “stop” signals.

Within genes that encode proteins, the nucleotide sequence of DNA is read in blocks of three consecutive nucleotides, without punctuation between the blocks. Each block, or codon, codes for one amino acid.

15.3 Genes are first transcribed, then translated.

Transcription

The first step in gene expression is the production of an RNA copy of the DNA sequence encoding the gene, a process called **transcription**. To understand the mechanism behind the transcription process, it is useful to focus first on RNA polymerase, the remarkable enzyme responsible for carrying it out (figure 15.7).

RNA Polymerase

RNA polymerase is best understood in bacteria. Bacterial RNA polymerase is very large and complex, consisting of five subunits: two α subunits bind regulatory proteins, a β' subunit binds the DNA template, a β subunit binds RNA nucleoside subunits, and a σ subunit recognizes the promoter and initiates synthesis. Only one of the two strands of DNA, called the **template strand**, is transcribed. The RNA transcript's sequence is complementary to the template strand. The strand of DNA that is not transcribed is called the **coding strand**. It has the same sequence as the RNA transcript, except T takes the place of U. The coding strand is also known as the sense (+) strand, and the template strand as the antisense (–) strand.

In both bacteria and eukaryotes, the polymerase adds ribonucleotides to the growing 3' end of an RNA chain. No primer is needed, and synthesis proceeds in the 5' → 3' direction. Bacteria contain only one RNA polymerase enzyme, while eukaryotes have three different RNA polymerases: RNA polymerase I synthesizes rRNA in the nucleolus; RNA polymerase II synthesizes mRNA; and RNA polymerase III synthesizes tRNA.

Promoter

Transcription starts at RNA polymerase binding sites called **promoters** on the DNA template strand. A promoter is a short sequence that is not itself transcribed by the polymerase that binds to it. Striking similarities are evident in the sequences of different promoters. For example, two six-base sequences are common to many bacterial promoters, a TTGACA sequence called the **–35 sequence**, located 35 nucleotides upstream of the position where transcription actually starts, and a TATAAT sequence called the **–10 sequence**, located 10 nucleotides upstream of the start site. In eukaryotic DNA, the sequence TATAAA, called the **TATA box**, is located at –25 and is very similar to the prokaryotic –10 sequence but is farther from the start site.

Promoters differ widely in efficiency. Strong promoters cause frequent initiations of transcription, as often as every 2 seconds in some bacteria. Weak promoters may tran-

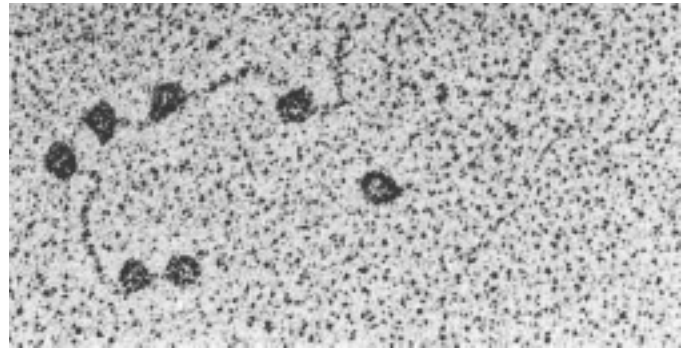


FIGURE 15.7

RNA polymerase. In this electron micrograph, the dark circles are RNA polymerase molecules bound to several promoter sites on bacterial virus DNA.

scribe only once every 10 minutes. Most strong promoters have unaltered –35 and –10 sequences, while weak promoters often have substitutions within these sites.

Initiation

The binding of RNA polymerase to the promoter is the first step in gene transcription. In bacteria, a subunit of RNA polymerase called σ (**sigma**) recognizes the –10 sequence in the promoter and binds RNA polymerase there. Importantly, this subunit can detect the –10 sequence without unwinding the DNA double helix. In eukaryotes, the –25 sequence plays a similar role in initiating transcription, as it is the binding site for a key protein factor. Other eukaryotic factors then bind one after another, assembling a large and complicated **transcription complex**. The eukaryotic transcription complex is described in detail in the following chapter.

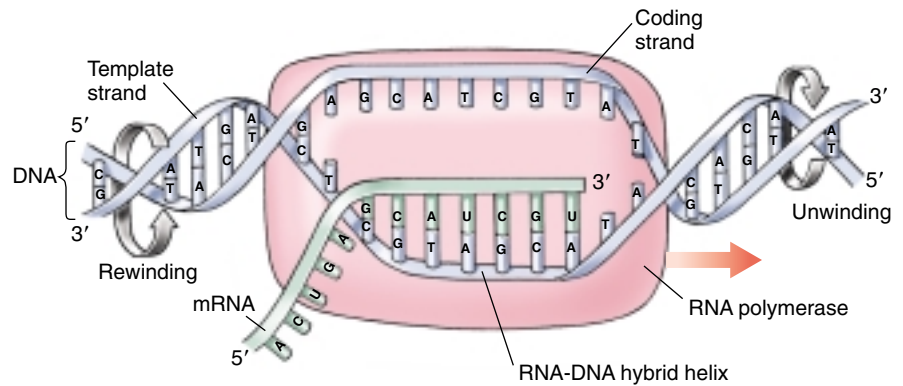
Once bound to the promoter, the RNA polymerase begins to unwind the DNA helix. Measurements indicate that bacterial RNA polymerase unwinds a segment approximately 17 base-pairs long, nearly two turns of the DNA double helix. This sets the stage for the assembly of the RNA chain.

Elongation

The transcription of the RNA chain usually starts with ATP or GTP. One of these forms the 5' end of the chain, which grows in the 5' → 3' direction as ribonucleotides are added. Unlike DNA synthesis, a primer is not required. The region containing the RNA polymerase, DNA, and growing RNA transcript is called the **transcription bubble** because it contains a locally unwound “bubble” of DNA (figure 15.8). Within the bubble, the first 12 bases of the

FIGURE 15.8

Model of a transcription bubble. The DNA duplex unwinds as it enters the RNA polymerase complex and rewinds as it leaves. One of the strands of DNA functions as a template, and nucleotide building blocks are assembled into RNA from this template.



newly synthesized RNA strand temporarily form a helix with the template DNA strand. Corresponding to not quite one turn of the helix, this stabilizes the positioning of the 3' end of the RNA so it can interact with an incoming ribonucleotide. The RNA-DNA hybrid helix rotates each time a nucleotide is added so that the 3' end of the RNA stays at the catalytic site.

The transcription bubble moves down the DNA at a constant rate, about 50 nucleotides per second, leaving the growing RNA strand protruding from the bubble. After the transcription bubble passes, the now transcribed DNA is rewound as it leaves the bubble.

Unlike DNA polymerase, RNA polymerase has no proofreading capability. Transcription thus produces many more copying errors than replication. These mistakes, however, are not transmitted to progeny. Most genes are transcribed many times, so a few faulty copies are not harmful.

Termination

At the end of a gene are “stop” sequences that cause the formation of phosphodiester bonds to cease, the RNA-DNA hybrid within the transcription bubble to dissociate, the RNA polymerase to release the DNA, and the DNA within the transcription bubble to rewind. The simplest stop signal is a series of GC base-pairs followed by a series of AT base-pairs. The RNA transcript of this stop region forms a GC hairpin (figure 15.9), followed by four or more U ribonucleotides. How does this structure terminate transcription? The hairpin causes the RNA polymerase to pause immediately after the polymerase has synthesized it, placing the polymerase directly over the run of four uracils. The pairing of U with DNA's A is the weakest of the four hybrid base-pairs and is not strong enough to hold the hybrid strands together during the long pause. Instead, the RNA strand dissociates from the DNA within the transcription bubble, and transcription stops. A variety of protein factors aid hairpin loops in terminating transcription of particular genes.

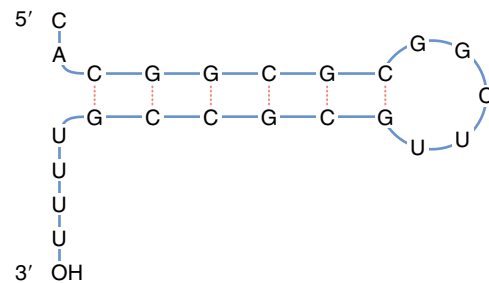


FIGURE 15.9

A GC hairpin. This structure stops gene transcription.

Posttranscriptional Modifications

In eukaryotes, every mRNA transcript must travel a long journey out from the nucleus into the cytoplasm before it can be translated. Eukaryotic mRNA transcripts are modified in several ways to aid this journey:

5' caps. Transcripts usually begin with A or G, and, in eukaryotes, the terminal phosphate of the 5' A or G is removed, and then a very unusual 5'-5' linkage forms with GTP. Called a **5' cap**, this structure protects the 5' end of the RNA template from nucleases and phosphatases during its long journey through the cytoplasm. Without these caps, RNA transcripts are rapidly degraded.

3' poly-A tails. The 3' end of eukaryotic transcript is cleaved off at a specific site, often containing the sequence AAUAAA. A special poly-A polymerase enzyme then adds about 250 A ribonucleotides to the 3' end of the transcript. Called a **3' poly-A tail**, this long string of As protects the transcript from degradation by nucleases. It also appears to make the transcript a better template for protein synthesis.

Transcription is carried out by the enzyme RNA polymerase, aided in eukaryotes by many other proteins.

Translation

In prokaryotes, translation begins when the initial portion of an mRNA molecule binds to an rRNA molecule in a ribosome. The mRNA lies on the ribosome in such a way that only one of its codons is exposed at the polypeptide-making site at any time. A tRNA molecule possessing the complementary three-nucleotide sequence, or anticodon, binds to the exposed codon on the mRNA.

Because this tRNA molecule carries a particular amino acid, that amino acid and no other is added to the polypeptide in that position. As the mRNA molecule moves through the ribosome, successive codons on the mRNA are exposed, and a series of tRNA molecules bind one after another to the exposed codons. Each of these tRNA molecules carries an attached amino acid, which it adds to the end of the growing polypeptide chain (figure 15.10).

There are about 45 different kinds of tRNA molecules. Why are there 45 and not 64 tRNAs (one for each codon)? Because the third base-pair of a tRNA anticodon allows some “wobble,” some tRNAs recognize more than one codon.

How do particular amino acids become associated with particular tRNA molecules? The key translation step, which pairs the three-nucleotide sequences with appropriate amino acids, is carried out by a remarkable set of enzymes called activating enzymes.

Activating Enzymes

Particular tRNA molecules become attached to specific amino acids through the action of activating enzymes called **aminoacyl-tRNA synthetases**, one of which exists for each of the 20 common amino acids (figure 15.11). Therefore, these enzymes must correspond to specific anticodon sequences on a tRNA molecule as well as particular amino acids. Some activating enzymes correspond to only one anticodon and thus only one tRNA molecule. Others recognize two, three, four, or six different tRNA molecules, each with a different anticodon but coding for the same amino acid (see table 15.1). If one considers the nucleotide sequence of mRNA a coded message, then the 20 activating enzymes are responsible for decoding that message.

“Start” and “Stop” Signals

There is no tRNA with an anticodon complementary to three of the 64 codons: UAA, UAG, and UGA. These codons, called **nonsense codons**, serve as “stop” signals in the mRNA message, marking the end of a polypeptide. The “start” signal that marks the beginning of a polypeptide within an mRNA message is the codon AUG, which also encodes the amino acid methionine. The ribosome will usually use the first AUG that it encounters in the mRNA to signal the start of translation.

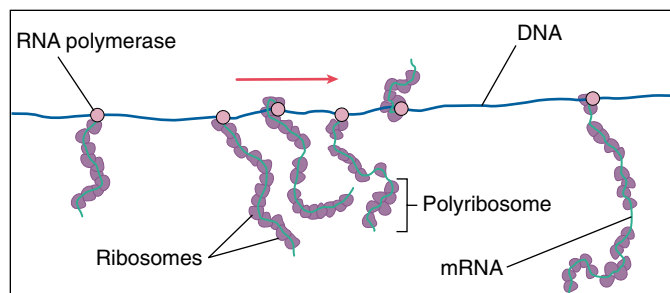
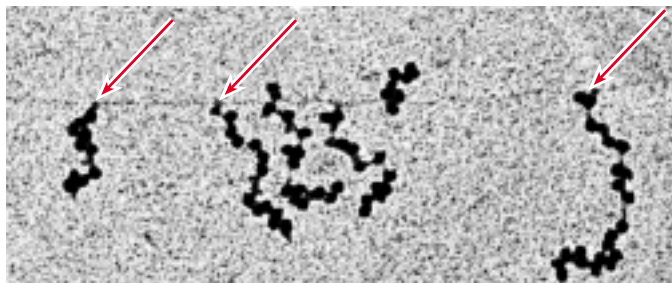


FIGURE 15.10

Translation in action. Bacteria have no nucleus and hence no membrane barrier between the DNA and the cytoplasm. In this electron micrograph of genes being transcribed in the bacterium *Escherichia coli*, you can see every stage of the process. The arrows point to RNA polymerase enzymes. From each mRNA molecule dangling from the DNA, a series of ribosomes is assembling polypeptides. These clumps of ribosomes are sometimes called “polyribosomes.”

Initiation

In prokaryotes, polypeptide synthesis begins with the formation of an **initiation complex**. First, a tRNA molecule carrying a chemically modified methionine called *N*-formylmethionine ($\text{tRNA}^{\text{fMet}}$) binds to the small ribosomal subunit. Proteins called **initiation factors** position the $\text{tRNA}^{\text{fMet}}$ on the ribosomal surface at the *P site* (for peptidyl), where peptide bonds will form. Nearby, two other sites will form: the *A site* (for aminoacyl), where successive amino acid-bearing tRNAs will bind, and the *E site* (for exit), where empty tRNAs will exit the ribosome (figure 15.12). This initiation complex, guided by another initiation factor, then binds to the anticodon AUG on the mRNA. Proper positioning of the mRNA is critical because it determines the reading frame—that is, which groups of three nucleotides will be read as codons. Moreover, the complex must bind to the beginning of the mRNA molecule, so that all of the transcribed gene will be translated. In bacteria, the beginning of each mRNA molecule is marked by a *leader sequence* complementary to one of the rRNA molecules on the ribosome. This complementarity ensures that the mRNA is read from the beginning. Bacteria often include several genes within a single mRNA transcript (polycistronic mRNA), while each eukaryotic gene is transcribed on a separate mRNA (monocistronic mRNA).

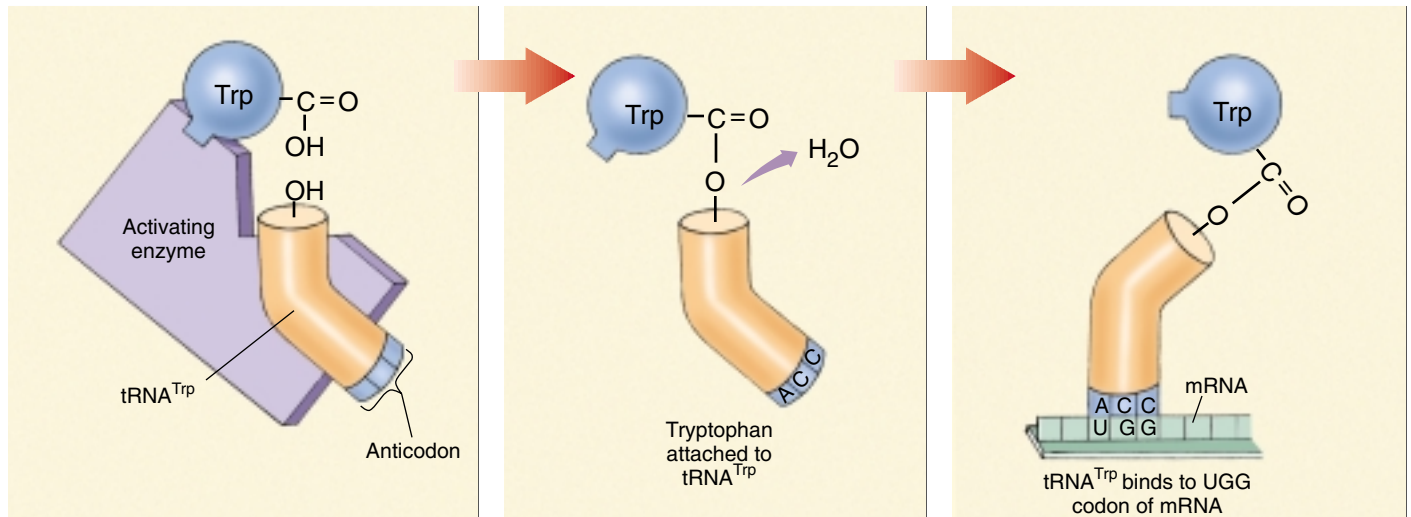


FIGURE 15.11

Activating enzymes “read” the genetic code. Each kind of activating enzyme recognizes and binds to a specific amino acid, such as tryptophan; it also recognizes and binds to the tRNA molecules with anticodons specifying that amino acid, such as ACC for tryptophan. In this way, activating enzymes link the tRNA molecules to specific amino acids.

Initiation in eukaryotes is similar, although it differs in two important ways. First, in eukaryotes, the initiating amino acid is methionine rather than *N*-formylmethionine. Second, the initiation complex is far more complicated than in bacteria, containing nine or more protein factors, many consisting of several subunits. Eukaryotic initiation complexes are discussed in detail in the following chapter.

Elongation

After the initiation complex has formed, the large ribosome subunit binds, exposing the mRNA codon adjacent to the

initiating AUG codon, and so positioning it for interaction with another amino acid-bearing tRNA molecule. When a tRNA molecule with the appropriate anticodon appears, proteins called elongation factors assist in binding it to the exposed mRNA codon at the A site. When the second tRNA binds to the ribosome, it places its amino acid directly adjacent to the initial methionine, which is still attached to its tRNA molecule, which in turn is still bound to the ribosome. The two amino acids undergo a chemical reaction, catalyzed by *peptidyl transferase*, which releases the initial methionine from its tRNA and attaches it instead by a peptide bond to the second amino acid.

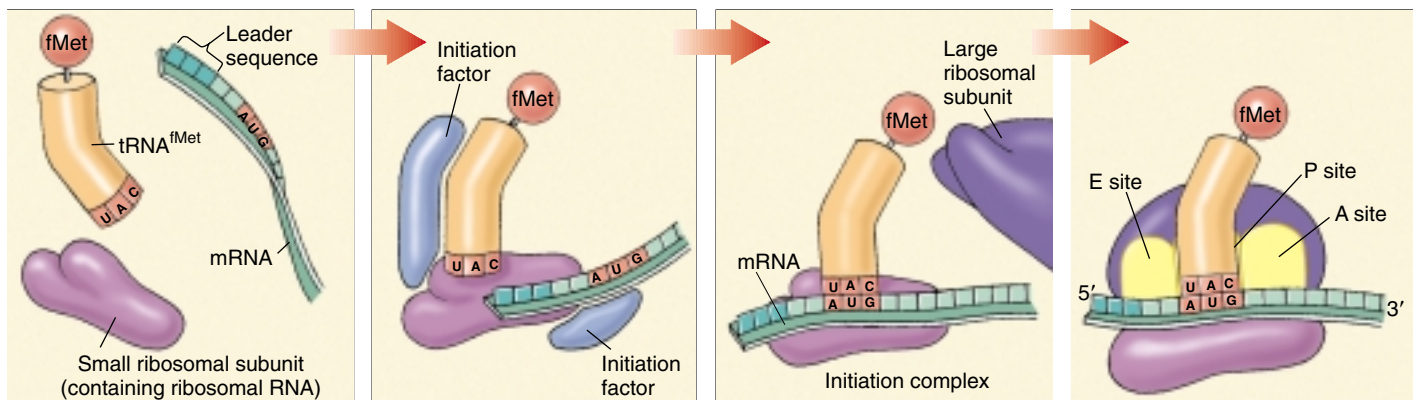


FIGURE 15.12

Formation of the initiation complex. In prokaryotes, proteins called initiation factors play key roles in positioning the small ribosomal subunit and the *N*-formylmethionine, or tRNA^{fMet}, molecule at the beginning of the mRNA. When the tRNA^{fMet} is positioned over the first AUG codon of the mRNA, the large ribosomal subunit binds, forming the P, A, and E sites where successive tRNA molecules bind to the ribosomes, and polypeptide synthesis begins.

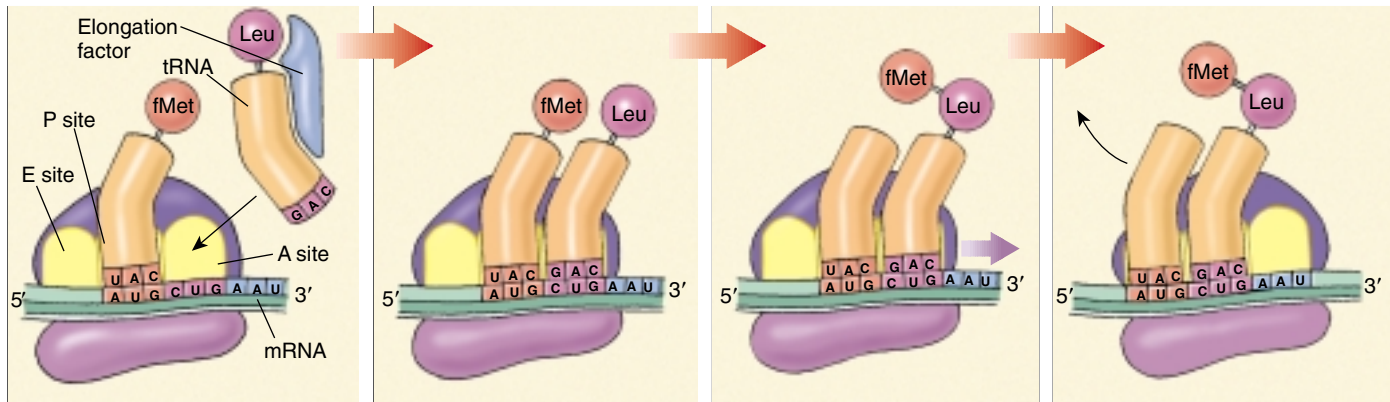


FIGURE 15.13

Translocation. The initiating $\text{tRNA}^{\text{fMet}}$ in prokaryotes ($\text{tRNA}^{\text{fMet}}$ in eukaryotes) occupies the P site, and a tRNA molecule with an anticodon complementary to the exposed mRNA codon binds at the A site. fMet is transferred to the incoming amino acid (Leu), as the ribosome moves three nucleotides to the right along the mRNA. The empty $\text{tRNA}^{\text{fMet}}$ moves to the E site to exit the ribosome, the growing polypeptide chain moves to the P site, and the A site is again exposed and ready to bind the next amino acid-laden tRNA.

Translocation

In a process called **translocation** (figure 15.13), the ribosome now moves (translocates) three more nucleotides along the mRNA molecule in the $5' \rightarrow 3'$ direction, guided by other elongation factors. This movement relocates the initial tRNA to the E site and ejects it from the ribosome, repositions the growing polypeptide chain (at this point containing two amino acids) to the P site, and exposes the next codon on the mRNA at the A site. When a tRNA molecule recognizing that codon appears, it binds to the codon at the A site, placing its amino acid adjacent to the growing chain. The chain then transfers to the new amino acid, and the entire process is repeated.

Termination

Elongation continues in this fashion until a chain-terminating nonsense codon is exposed (for example, UAA in figure 15.14). Nonsense codons do not bind to tRNA, but they are recognized by **release factors**, proteins that release the newly made polypeptide from the ribosome.

The first step in protein synthesis is the formation of an initiation complex. Each step of the ribosome's progress exposes a codon, to which a tRNA molecule with the complementary anticodon binds. The amino acid carried by each tRNA molecule is added to the end of the growing polypeptide chain.

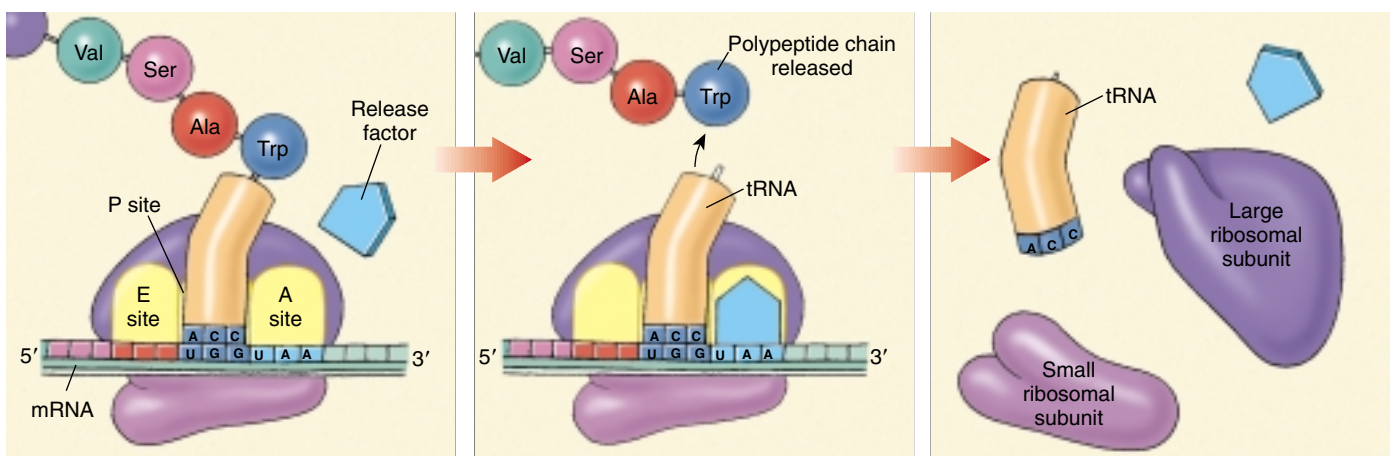


FIGURE 15.14

Termination of protein synthesis. There is no tRNA with an anticodon complementary to any of the three termination signal codons, such as the UAA nonsense codon illustrated here. When a ribosome encounters a termination codon, it therefore stops translocating. A specific release factor facilitates the release of the polypeptide chain by breaking the covalent bond that links the polypeptide to the P-site tRNA.

15.4 Eukaryotic gene transcripts are spliced.

The Discovery of Introns

While the mechanisms of protein synthesis are similar in bacteria and eukaryotes, they are not identical. One difference is of particular importance. Unlike bacterial genes, most eukaryotic genes are larger than they need to be to produce the polypeptides they code for. Such genes contain long sequences of nucleotides, known as **introns**, that do not code for any portion of the polypeptide specified by the gene. Introns are inserted between **exons**, much shorter sequences in the gene that do code for portions of the polypeptide.

In bacteria, virtually every nucleotide within a bacterial gene transcript is part of an amino acid-specifying codon. Scientists assumed for many years that this was true of all organisms. In the late 1970s, however, biologists were amazed to discover that many of the characteristics of prokaryotic gene expression did not apply to eukaryotes. In particular, they found that eukaryotic proteins are encoded by RNA segments that are excised from several locations along what is called the **primary RNA transcript** (or **primary transcript**) and then spliced together to form the mRNA that is eventually translated in the cytoplasm. The experiment that revealed this unexpected mode of gene expression consisted of several steps:

1. The mRNA transcribed from a particular gene was isolated and purified. For example, ovalbumin mRNA could be obtained fairly easily from unfertilized eggs.

2. Molecules of DNA complementary to the isolated mRNA were synthesized with the enzyme **reverse transcriptase**. These DNA molecules, which are called “copy” DNA (cDNA), had the same nucleotide sequence as the template strand of the gene that produced the mRNA.
3. With genetic engineering techniques (chapter 19), the portion of the nuclear DNA containing the gene that produced the mRNA was isolated. This procedure is referred to as *cloning* the gene in question.
4. Single-stranded forms of the cDNA and the nuclear DNA were mixed and allowed to pair with each other (to *hybridize*).

When the researchers examined the resulting hybrid DNA molecules with an electron microscope, they found that the DNA did not appear as a single duplex. Instead, they observed unpaired loops. In the case of the ovalbumin gene, they discovered seven loops, corresponding to sites where the nuclear DNA contained long nucleotide sequences not present in the cDNA. The conclusion was inescapable: nucleotide sequences must have been removed from the gene transcript before it appeared as cytoplasmic mRNA. These removed sequences are introns, and the remaining sequences are exons (figure 15.15). Because introns are excised from the RNA transcript before it is translated into protein, they do not affect the structure of the protein encoded by the gene in which they occur.

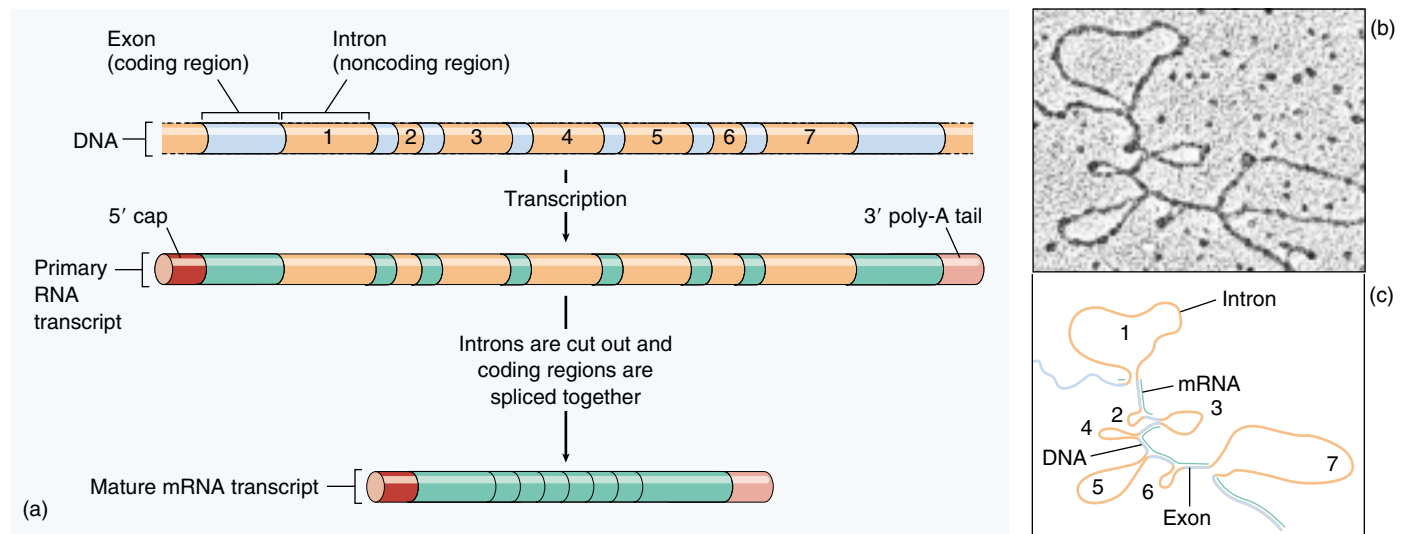


FIGURE 15.15

The eukaryotic gene that codes for ovalbumin in eggs contains introns. (a) The ovalbumin gene and its primary RNA transcript contain seven segments not present in the mRNA the ribosomes use to direct protein synthesis. Enzymes cut these segments (introns) out and splice together the remaining segments (exons). (b) The seven loops are the seven introns represented in the schematic drawing (c) of the mature mRNA transcript hybridized to DNA.

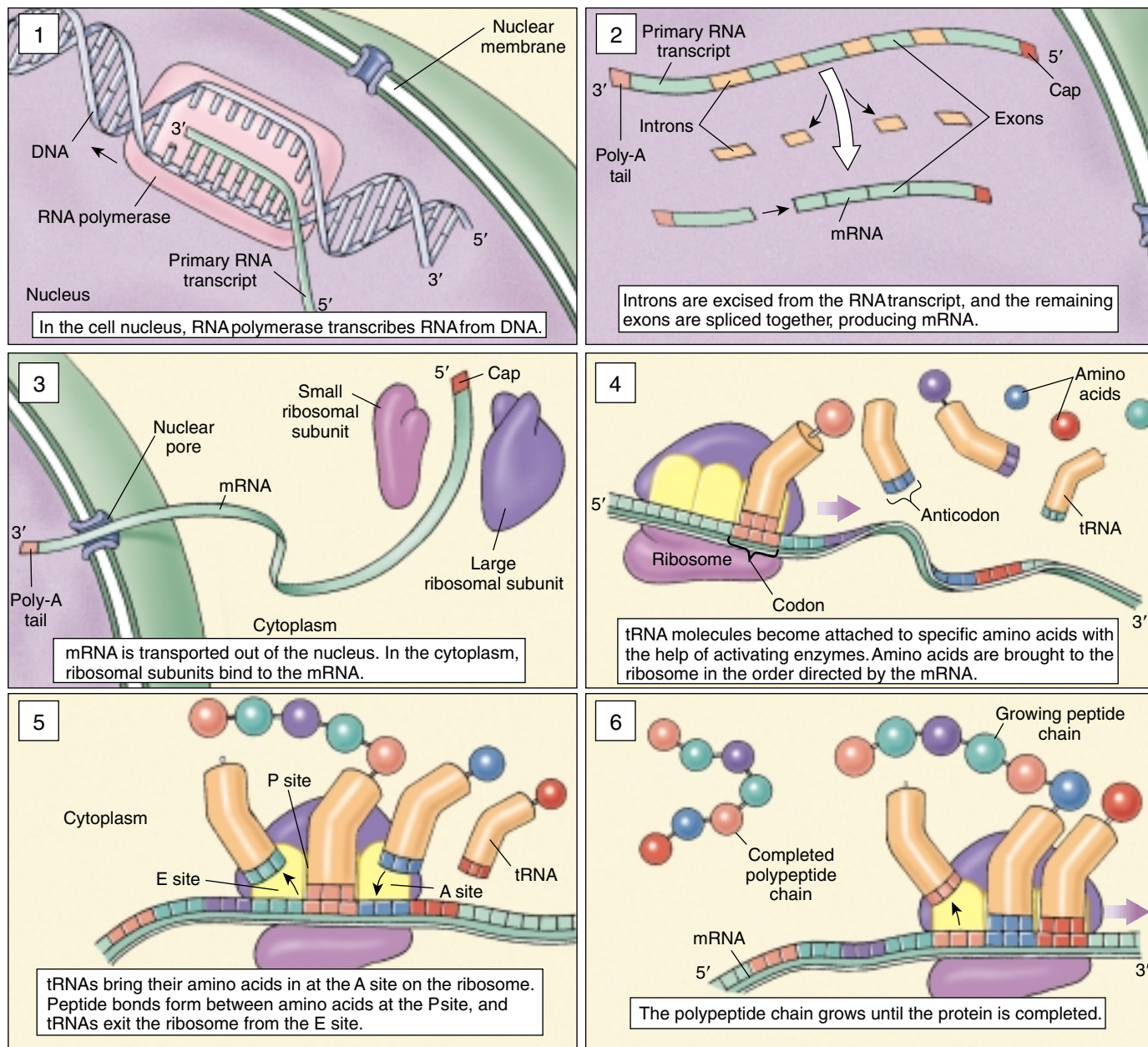


FIGURE 15.16
An overview of gene expression in eukaryotes.

RNA Splicing

When a gene is transcribed, the primary RNA transcript (that is, the gene copy as it is made by RNA polymerase, before any modification occurs) contains sequences complementary to the entire gene, including introns as well as exons. However, in a process called **RNA processing**, or **splicing**, the intron sequences are cut out of the primary transcript before it is used in polypeptide synthesis; therefore, those sequences are not translated. The remaining sequences, which correspond to the exons, are spliced together to form the final, “processed” mRNA molecule that

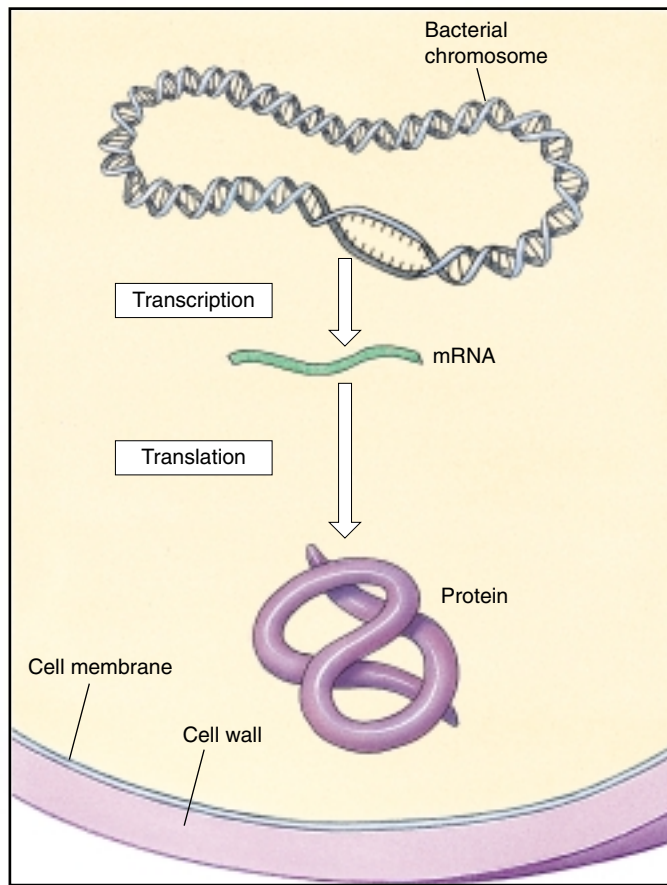
is translated. In a typical human gene, the introns can be 10 to 30 times larger than the exons. For example, even though only 432 nucleotides are required to encode the 144 amino acids of hemoglobin, there are actually 1356 nucleotides in the primary mRNA transcript of the hemoglobin gene. Figure 15.16 summarizes eukaryotic protein synthesis.

Much of a eukaryotic gene is not translated. Noncoding segments scattered throughout the gene are removed from the primary transcript before the mRNA is translated.

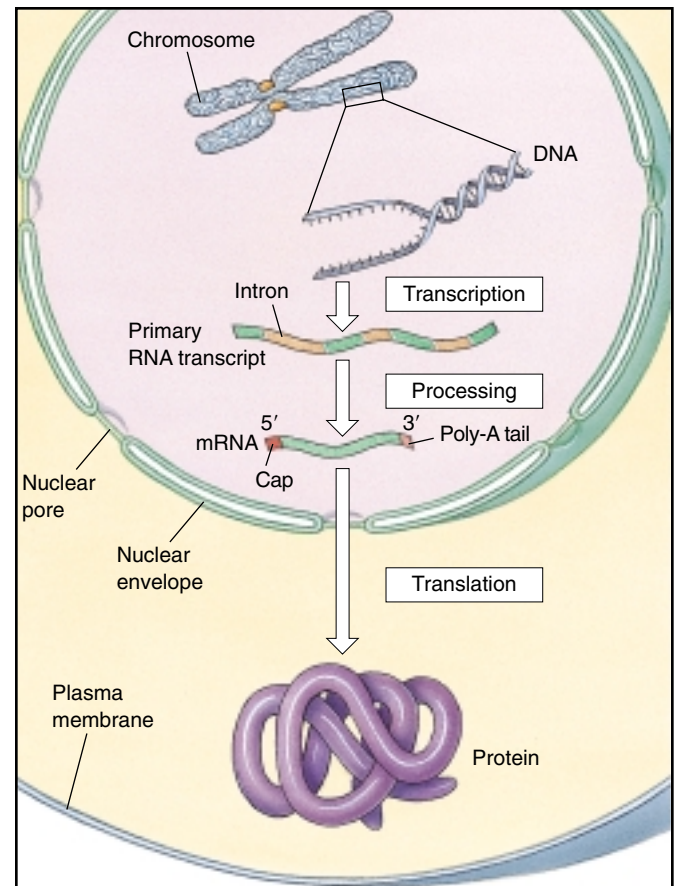
Differences between Bacterial and Eukaryotic Gene Expression

1. Most eukaryotic genes possess introns. With the exception of a few genes in the Archaeobacteria, prokaryotic genes lack introns (figure 15.17).
2. Individual bacterial mRNA molecules often contain transcripts of several genes. By placing genes with related functions on the same mRNA, bacteria coordinate the regulation of those functions. Eukaryotic mRNA molecules rarely contain transcripts of more than one gene. Regulation of eukaryotic gene expression is achieved in other ways.
3. Because eukaryotes possess a nucleus, their mRNA molecules must be completely formed and must pass across the nuclear membrane before they are translated. Bacteria, which lack nuclei, often begin translation of an mRNA molecule before its transcription is completed.
4. In bacteria, translation begins at an AUG codon preceded by a special nucleotide sequence. In eukaryotic cells, mRNA molecules are modified at the 5' leading end after transcription, adding a 5' cap, a methylated guanosine triphosphate. The cap initiates translation by binding the mRNA, usually at the first AUG, to the small ribosomal subunit.
5. Eukaryotic mRNA molecules are modified before they are translated: introns are cut out, and the remaining exons are spliced together; a 5' cap is added; and a 3' poly-A tail consisting of some 200 adenine (A) nucleotides is added. These modifications can delay the destruction of the mRNA by cellular enzymes.
6. The ribosomes of eukaryotes are a little larger than those of bacteria.

Gene expression is broadly similar in bacteria and eukaryotes, although it differs in some details.



(a)



(b)

FIGURE 15.17

Gene information is processed differently in prokaryotes and eukaryotes. (a) Bacterial genes are transcribed into mRNA, which is translated immediately. Hence, the sequence of DNA nucleotides corresponds exactly to the sequence of amino acids in the encoded polypeptide. (b) Eukaryotic genes are typically different, containing long stretches of nucleotides called introns that do not correspond to amino acids within the encoded polypeptide. Introns are removed from the primary RNA transcript of the gene and a 5' cap and 3' poly-A tail are added before the mRNA directs the synthesis of the polypeptide.



Summary

Questions

Media Resources

15.1 The Central Dogma traces the flow of gene-encoded information.

- There are three principal kinds of RNA: messenger RNA (mRNA), transcripts of genes used to direct the assembly of amino acids into proteins; ribosomal RNA (rRNA), which combines with proteins to make up the ribosomes that carry out the assembly process; and transfer RNA (tRNA), molecules that transport the amino acids to the ribosome for assembly into proteins.

- What are the three major classes of RNA? What is the function of each type?
- What is the function of RNA polymerase in transcription? What determines where RNA polymerase begins and ends its function?



- Experiment: Jacob/Meselson/Brenner-Discovery of Messenger RNA (mRNA)



- Gene Activity

15.2 Genes encode information in three-nucleotide code words.

- The sequence of nucleotides in DNA encodes the sequence of amino acids in proteins. The mRNA transcribed from the DNA is read by ribosomes in increments of three nucleotides called codons.

- How did Crick and his colleagues determine how many nucleotides are used to specify each amino acid? What is an anticodon?



- Experiment: Nirenberg/Khorana-Breaking the Genetic Code
- Experiment: The Genetic Code is Read in Three Bases at a Time

15.3 Genes are first transcribed, then translated.

- During transcription, the enzyme RNA polymerase manufactures mRNA molecules with nucleotide sequences complementary to particular segments of the DNA.
- During translation, the mRNA sequences direct the assembly of amino acids into proteins on cytoplasmic ribosomes.
- The information in a gene and in an mRNA molecule is read in three-nucleotide blocks called codons.
- On the ribosome, the mRNA molecule is positioned so that only one of its codons is exposed at any time.
- This exposure permits a tRNA molecule with the complementary base sequence (anticodon) to bind to it.
- Attached to the other end of the tRNA is an amino acid, which is added to the end of the growing polypeptide chain.

- During protein synthesis, what mechanism ensures that only one amino acid is added to the growing polypeptide at a time? What mechanism ensures the correct amino acid is added at each position in the polypeptide?

- How does an mRNA molecule specify where the polypeptide it encodes should begin? How does it specify where the polypeptide should end?

- What roles do elongation factors play in translation?



- Transcription
- Translation
- Polyribosomes



- Transcription
- Translation



- Experiment: Chapeville-Proving the tRNA Hypothesis

15.4 Eukaryotic gene transcripts are spliced.

- Most eukaryotic genes contain noncoding sequences (introns) interspersed randomly between coding sequences (exons).
- The portions of an mRNA molecule corresponding to the introns are removed from the primary RNA transcript before the remainder is translated.

- What is an intron? What is an exon? How is each involved in the mRNA molecule that is ultimately translated?



- Experiment: Chambon-Discovery of Introns

16

Control of Gene Expression

Concept Outline

16.1 Gene expression is controlled by regulating transcription.

An Overview of Transcriptional Control. In bacteria transcription is regulated by controlling access of RNA polymerase to the promoter in a flexible and reversible way; eukaryotes by contrast regulate many of their genes by turning them on and off in a more permanent fashion.

16.2 Regulatory proteins read DNA without unwinding it.

How to Read a Helix without Unwinding It. Regulatory proteins slide special segments called DNA-binding motifs along the major groove of the DNA helix, reading the sides of the bases.

Four Important DNA-Binding Motifs. DNA-binding proteins contain structural motifs such as the helix-turn-helix which fit into the major groove of the DNA helix.

16.3 Bacteria limit transcription by blocking RNA polymerase.

Controlling Transcription Initiation. Repressor proteins inhibit RNA polymerase's access to the promoter, while activators facilitate its binding.

16.4 Transcriptional control in eukaryotes operates at a distance.

Designing a Complex Gene Control System. Eukaryotic genes use a complex collection of transcription factors and enhancers to aid the polymerase in transcription.

The Effect of Chromosome Structure on Gene Regulation. The tight packaging of eukaryotic DNA into nucleosomes does not interfere with gene expression.

Posttranscriptional Control in Eukaryotes. Gene expression can be controlled at a variety of levels after transcription.

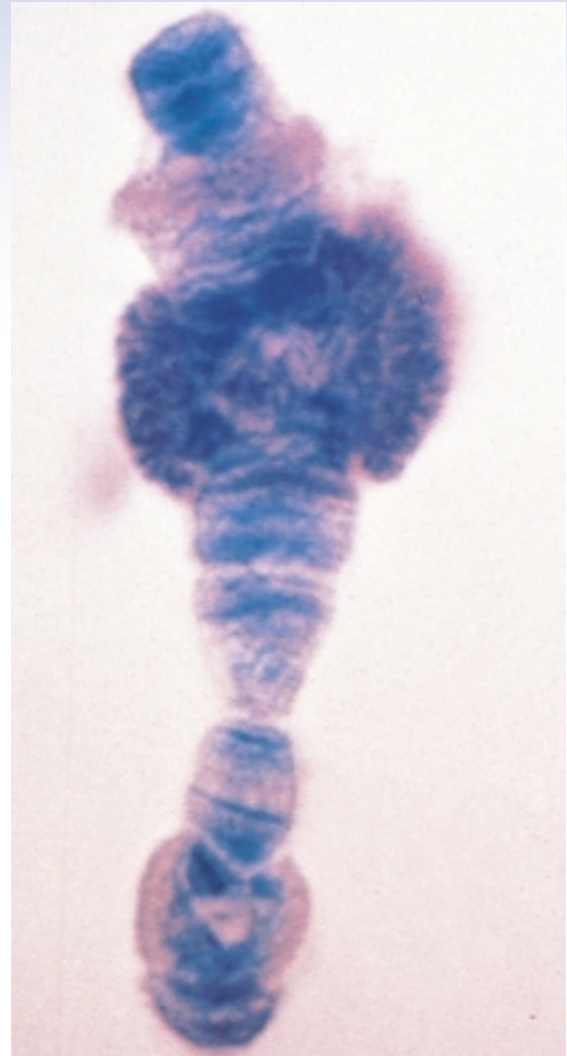


FIGURE 16.1

Chromosome puffs. In this chromosome of the fly *Drosophila melanogaster*, individual active genes can be visualized as “puffs” on the chromosomes. The RNA being transcribed from the DNA template has been radioactively labeled, and the dark specks indicate its position on the chromosome.

In an orchestra, all of the instruments do not play all the time; if they did, all they would produce is noise. Instead, a musical score determines which instruments in the orchestra play when. Similarly, all of the genes in an organism are not expressed at the same time, each gene producing the protein it encodes full tilt. Instead, different genes are expressed at different times, with a genetic score written in regulatory regions of the DNA determining which genes are active when (figure 16.1).

16.1 Gene expression is controlled by regulating transcription.

An Overview of Transcriptional Control

Control of gene expression is essential to all organisms. In bacteria, it allows the cell to take advantage of changing environmental conditions. In multicellular organisms, it is critical for directing development and maintaining homeostasis.

Regulating Promoter Access

One way to control transcription is to regulate the initiation of transcription. In order for a gene to be transcribed, RNA polymerase must have access to the DNA helix and must be capable of binding to the gene's **promoter**, a specific sequence of nucleotides at one end of the gene that tells the polymerase where to begin transcribing. How is the initiation of transcription regulated? Protein-binding nucleotide sequences on the DNA regulate the initiation of transcription by modulating the ability of RNA polymerase to bind to the promoter. These protein-binding sites are usually only 10 to 15 nucleotides in length (even a large regulatory protein has a “footprint,” or binding area, of only about 20 nucleotides). Hundreds of these regulatory sequences have been characterized, and each provides a binding site for a specific protein able to recognize the sequence. Binding the protein to the regulatory sequence either *blocks* transcription by getting in the way of RNA polymerase, or *stimulates* transcription by facilitating the binding of RNA polymerase to the promoter.

Transcriptional Control in Prokaryotes

Control of gene expression is accomplished very differently in bacteria than in the cells of complex multicellular organisms. Bacterial cells have been shaped by evolution to grow and divide as rapidly as possible, enabling them to exploit transient resources. In bacteria, the primary function of gene control is to adjust the cell's activities to its immediate environment. Changes in gene expression alter which enzymes are present in the cell in response to the quantity and type of available nutrients and the amount of oxygen present. Almost all of these changes are fully reversible, allowing the cell to adjust its enzyme levels up or down as the environment changes.

Transcriptional Control in Eukaryotes

The cells of multicellular organisms, on the other hand, have been shaped by evolution to be protected from transient changes in their immediate environment. Most of them experience fairly constant conditions. Indeed, **homeostasis**—

the maintenance of a constant internal environment—is considered by many to be the hallmark of multicellular organisms. Although cells in such organisms still respond to signals in their immediate environment (such as growth factors and hormones) by altering gene expression, in doing so they participate in regulating the body as a whole. In multicellular organisms with relatively constant internal environments, the primary function of gene control in a cell is not to respond to that cell's immediate environment, but rather to participate in regulating the body as a whole.

Some of these changes in gene expression compensate for changes in the physiological condition of the body. Others mediate the decisions that *produce* the body, ensuring that the right genes are expressed in the right cells at the right time during development. The growth and development of multicellular organisms entail a long series of biochemical reactions, each catalyzed by a specific enzyme. Once a particular developmental change has occurred, these enzymes cease to be active, lest they disrupt the events that must follow. To produce these enzymes, genes are transcribed in a carefully prescribed order, each for a specified period of time. In fact, many genes are activated only once, producing irreversible effects. In many animals, for example, **stem cells** develop into differentiated tissues like skin cells or red blood cells, following a fixed genetic program that often leads to programmed cell death. The one-time expression of the genes that guide this program is fundamentally different from the reversible metabolic adjustments bacterial cells make to the environment. In all multicellular organisms, changes in gene expression within particular cells serve the needs of the whole organism, rather than the survival of individual cells.

Posttranscriptional Control

Gene expression can be regulated at many levels. By far the most common form of regulation in both bacteria and eukaryotes is **transcriptional control**, that is, control of the transcription of particular genes by RNA polymerase. Other less common forms of control occur after transcription, influencing the mRNA that is produced from the genes or the activity of the proteins encoded by the mRNA. These controls, collectively referred to as **post-transcriptional controls**, will be discussed briefly later in this chapter.

Gene expression is controlled at the transcriptional and posttranscriptional levels. Transcriptional control, more common, is effected by the binding of proteins to regulatory sequences within the DNA.

16.2 Regulatory proteins read DNA without unwinding it.

How to Read a Helix without Unwinding It

It is the ability of certain proteins to bind to *specific* DNA regulatory sequences that provides the basic tool of gene regulation, the key ability that makes transcriptional control possible. To understand how cells control gene expression, it is first necessary to gain a clear picture of this molecular recognition process.

Looking into the Major Groove

Molecular biologists used to think that the DNA helix had to unwind before proteins could distinguish one DNA sequence from another; only in this way, they reasoned, could regulatory proteins gain access to the hydrogen bonds between base-pairs. We now know it is unnecessary for the helix to unwind because proteins can bind to its outside surface, where the edges of the base-pairs are exposed. Careful inspection of a DNA molecule reveals two helical grooves winding round the molecule, one deeper than the other. Within the deeper groove, called the **major**

groove, the nucleotides' hydrophobic methyl groups, hydrogen atoms, and hydrogen bond donors and acceptors protrude. The pattern created by these chemical groups is unique for each of the four possible base-pair arrangements, providing a ready way for a protein nestled in the groove to read the sequence of bases (figure 16.2).

DNA-Binding Motifs

Protein-DNA recognition is an area of active research; so far, the structures of over 30 regulatory proteins have been analyzed. Although each protein is unique in its fine details, the part of the protein that actually binds to the DNA is much less variable. Almost all of these proteins employ one of a small set of **structural**, or **DNA-binding motifs**, particular bends of the protein chain that permit it to interlock with the major groove of the DNA helix.

Regulatory proteins identify specific sequences on the DNA double helix, without unwinding it, by inserting DNA-binding motifs into the major groove of the double helix where the edges of the bases protrude.

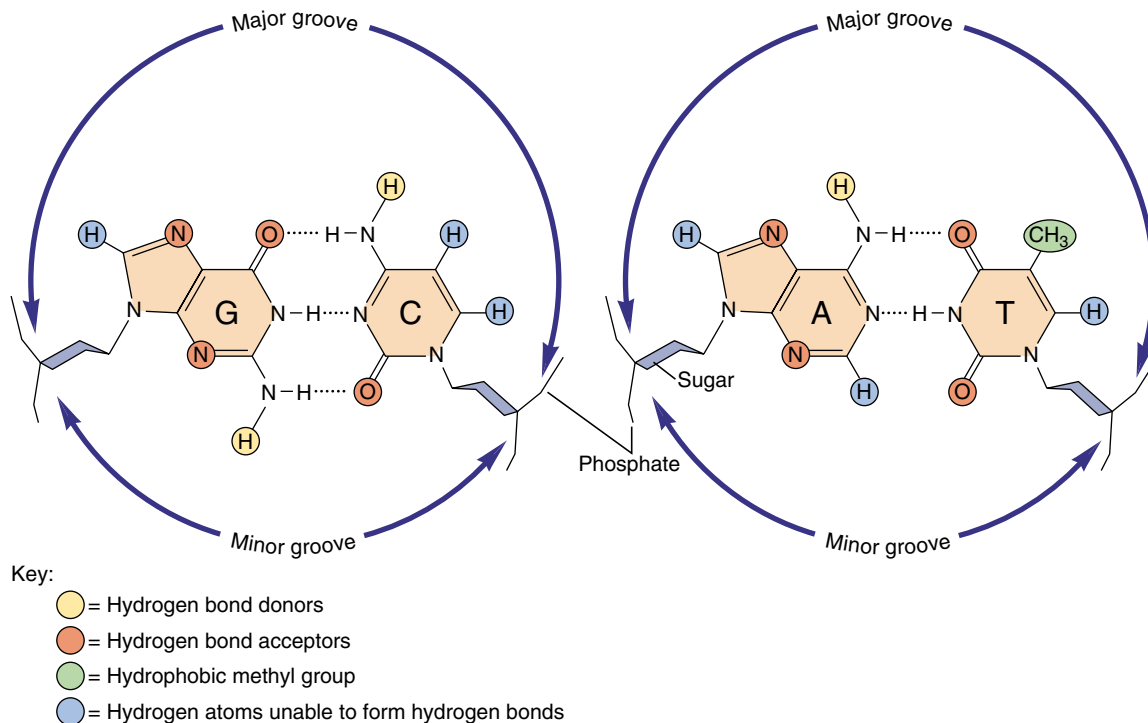


FIGURE 16.2

Reading the major groove of DNA. Looking down into the major groove of a DNA helix, we can see the edges of the bases protruding into the groove. Each of the four possible base-pair arrangements (two are shown here) extends a unique set of chemical groups into the groove, indicated in this diagram by differently colored balls. A regulatory protein can identify the base-pair arrangement by this characteristic signature.

Four Important DNA-Binding Motifs

The Helix-Turn-Helix Motif

The most common DNA-binding motif is the **helix-turn-helix**, constructed from two α -helical segments of the protein linked by a short nonhelical segment, the “turn” (figure 16.3). The first DNA-binding motif recognized, the helix-turn-helix motif has since been identified in hundreds of DNA-binding proteins.

A close look at the structure of a helix-turn-helix motif reveals how proteins containing such motifs are able to interact with the major groove of DNA. Interactions between the helical segments of the motif hold them at roughly right angles to each other. When this motif is pressed against DNA, one of the helical segments (called the recognition helix) fits snugly in the major groove of the DNA molecule, while the other butts up against the outside of the DNA molecule, helping to ensure the proper positioning of the recognition helix. Most DNA regulatory sequences recognized by helix-turn-helix motifs occur in symmetrical pairs. Such sequences are bound by proteins containing two helix-turn-helix motifs separated by 3.4 nm, the distance required for one turn of the DNA helix (figure 16.4). Having *two* protein/DNA-binding sites doubles the zone of contact between protein and DNA and so greatly strengthens the bond that forms between them.

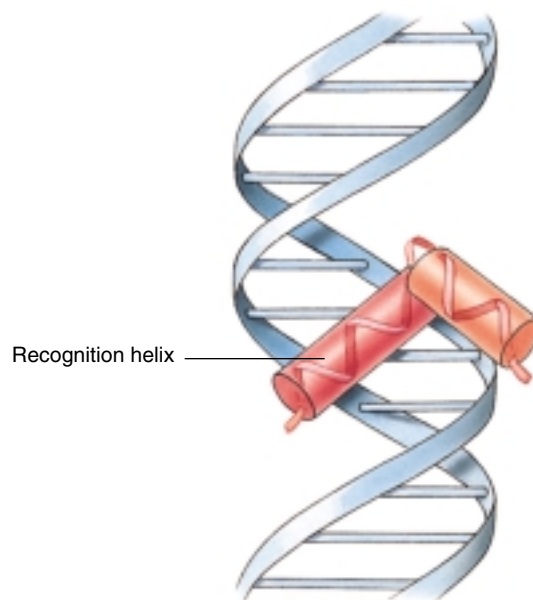


FIGURE 16.3

The helix-turn-helix motif. One helical region, called the recognition helix, actually fits into the major groove of DNA. There it contacts the edges of base-pairs, enabling it to recognize specific sequences of DNA bases.

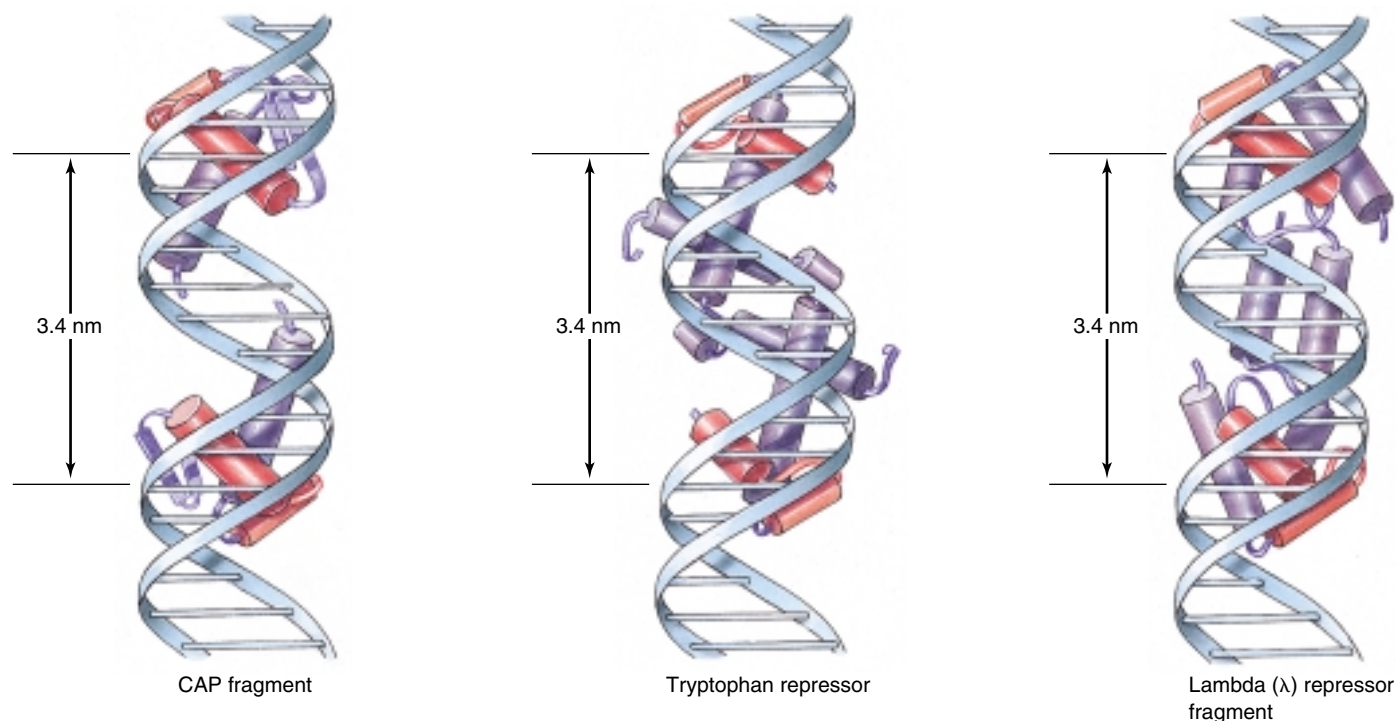


FIGURE 16.4

How the helix-turn-helix binding motif works. The three regulatory proteins illustrated here all bind to DNA using a pair of helix-turn-helix binding motifs. In each case, the two copies of the motif (*red*) are separated by 3.4 nm, precisely the spacing of one turn of the DNA helix. This allows the regulatory proteins to slip into two adjacent portions of the major groove in DNA, providing a strong attachment.

The Homeodomain Motif

A special class of helix-turn-helix motifs plays a critical role in development in a wide variety of eukaryotic organisms, including humans. These motifs were discovered when researchers began to characterize a set of homeotic mutations in *Drosophila* (mutations that alter how the parts of the body are assembled). They found that the mutant genes encoded regulatory proteins whose normal function was to initiate key stages of development by binding to developmental switch-point genes. More than 50 of these regulatory proteins have been analyzed, and they all contain a nearly identical sequence of 60 amino acids, the **homeodomain** (figure 16.5*b*). The center of the homeodomain is occupied by a helix-turn-helix motif that binds to the DNA. Surrounding this motif within the homeodomain is a region that always presents the motif to the DNA in the same way.

The Zinc Finger Motif

A different kind of DNA-binding motif uses one or more zinc atoms to coordinate its binding to DNA. Called **zinc fingers** (figure 16.5*c*), these motifs exist in several forms. In one form, a zinc atom links an α -helical segment to a β sheet segment so that the helical segment fits into the major groove of DNA. This sort of motif often occurs in clusters, the β sheets spacing the helical segments so that each helix contacts the major groove. The more zinc fingers in the cluster, the stronger the protein binds to the DNA. In other forms of the zinc finger motif, the β sheet's place is taken by another helical segment.

The Leucine Zipper Motif

In yet another DNA-binding motif, two different protein subunits cooperate to create a single DNA-binding site. This motif is created where a region on one of the subunits containing several hydrophobic amino acids (usually leucines) interacts with a similar region on the other subunit. This interaction holds the two subunits together at those regions, while the rest of the subunits are separated. This interaction holds the two subunits together at those regions, while the rest of the subunits are separated. Called a **leucine zipper**, this structure has the shape of a “Y,” with the two arms of the Y being helical regions that fit into the major groove of DNA (figure 16.5*d*). Because the two subunits can contribute quite different helical regions to the motif, leucine zippers allow for great flexibility in controlling gene expression.

Regulatory proteins bind to the edges of base-pairs exposed in the major groove of DNA. Most contain structural motifs such as the helix-turn-helix, homeodomain, zinc finger, or leucine zipper.

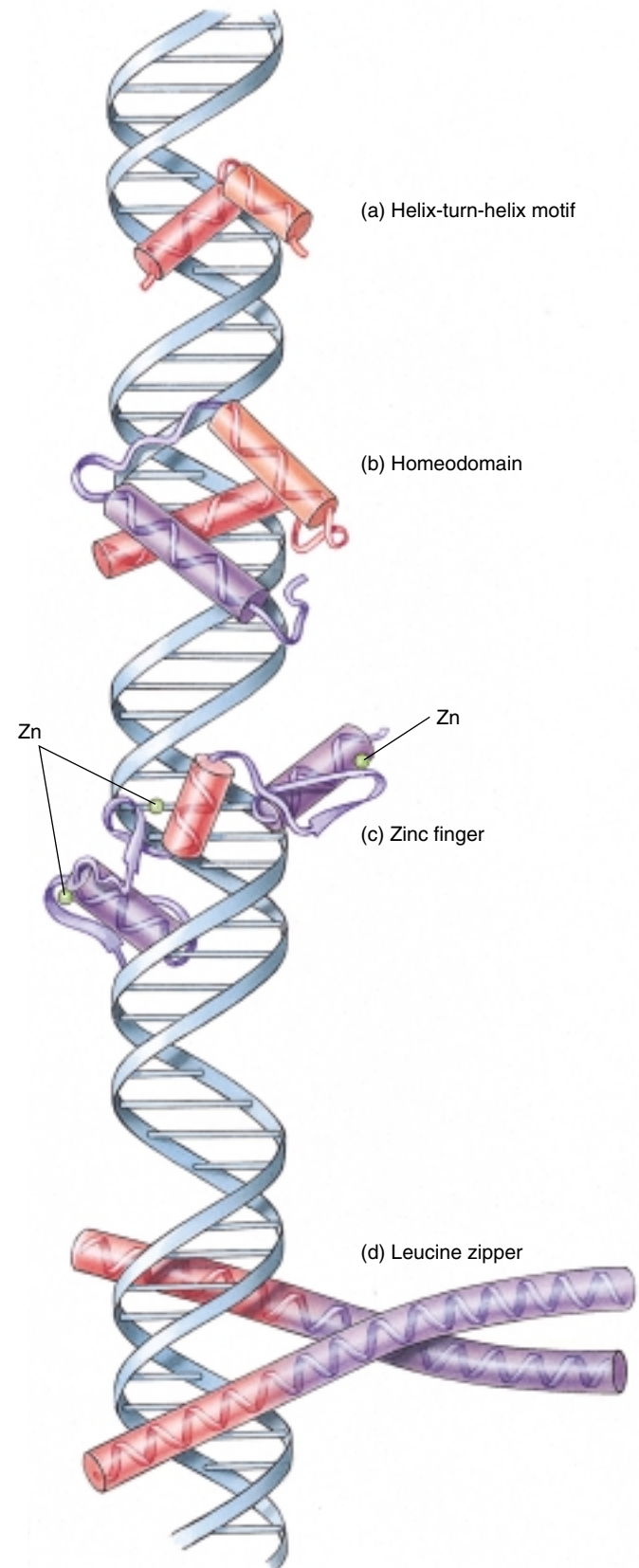


FIGURE 16.5
Major DNA-binding motifs.

16.3 Bacteria limit transcription by blocking RNA polymerase.

Controlling Transcription Initiation

How do organisms use regulatory DNA sequences and the proteins that bind them to control when genes are transcribed? The same basic controls are used in bacteria and eukaryotes, but eukaryotes employ several additional elements that reflect their more elaborate chromosomal structure. We will begin by discussing the relatively simple controls found in bacteria.

Repressors Are OFF Switches

A typical bacterium possesses genes encoding several thousand proteins, but only some are transcribed at any one time; the others are held in reserve until needed. When the cell encounters a potential food source, for example, it begins to manufacture the enzymes necessary to metabolize that food. Perhaps the best-understood example of this type of transcriptional control is the regulation of tryptophan-producing genes (*trp* genes), which was investigated in the pioneering work of Charles Yanofsky and his students at Stanford University.

Operons. The bacterium *Escherichia coli* uses proteins encoded by a cluster of five genes to manufacture the amino acid tryptophan. All five genes are transcribed together as a

unit called an **operon**, producing a single, long piece of mRNA. RNA polymerase binds to a promoter located at the beginning of the first gene, and then proceeds down the DNA, transcribing the genes one after another. Regulatory proteins shut off transcription by binding to an operator site immediately in front of the promoter and often overlapping it.

When tryptophan is present in the medium surrounding the bacterium, the cell shuts off transcription of the *trp* genes by means of a tryptophan **repressor**, a helix-turn-helix regulatory protein that binds to the operator site located within the *trp* promoter (figure 16.6). Binding of the repressor to the operator prevents RNA polymerase from binding to the promoter. The key to the functioning of this control mechanism is that the tryptophan repressor cannot bind to DNA unless it has first bound to two molecules of tryptophan. The binding of tryptophan to the repressor alters the orientation of a pair of helix-turn-helix motifs in the repressor, causing their recognition helices to fit into adjacent major grooves of the DNA (figure 16.7).

Thus, the bacterial cell's synthesis of tryptophan depends upon the absence of tryptophan in the environment. When the environment lacks tryptophan, there is nothing to activate the repressor, so the repressor cannot prevent

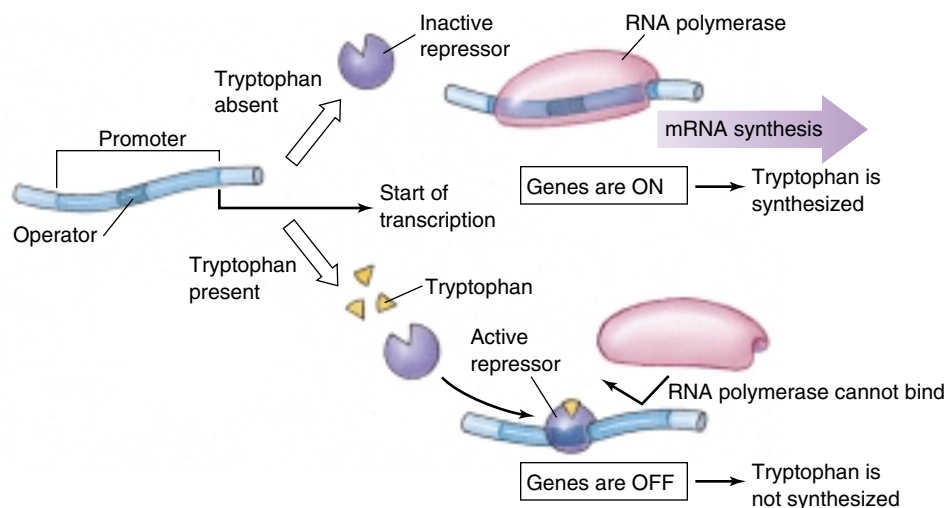


FIGURE 16.6

How the *trp* operon is controlled. The tryptophan repressor cannot bind the operator (which is located *within* the promoter) unless tryptophan first binds to the repressor. Therefore, in the absence of tryptophan, the promoter is free to function and RNA polymerase transcribes the operon. In the presence of tryptophan, the tryptophan-repressor complex binds tightly to the operator, preventing RNA polymerase from initiating transcription.

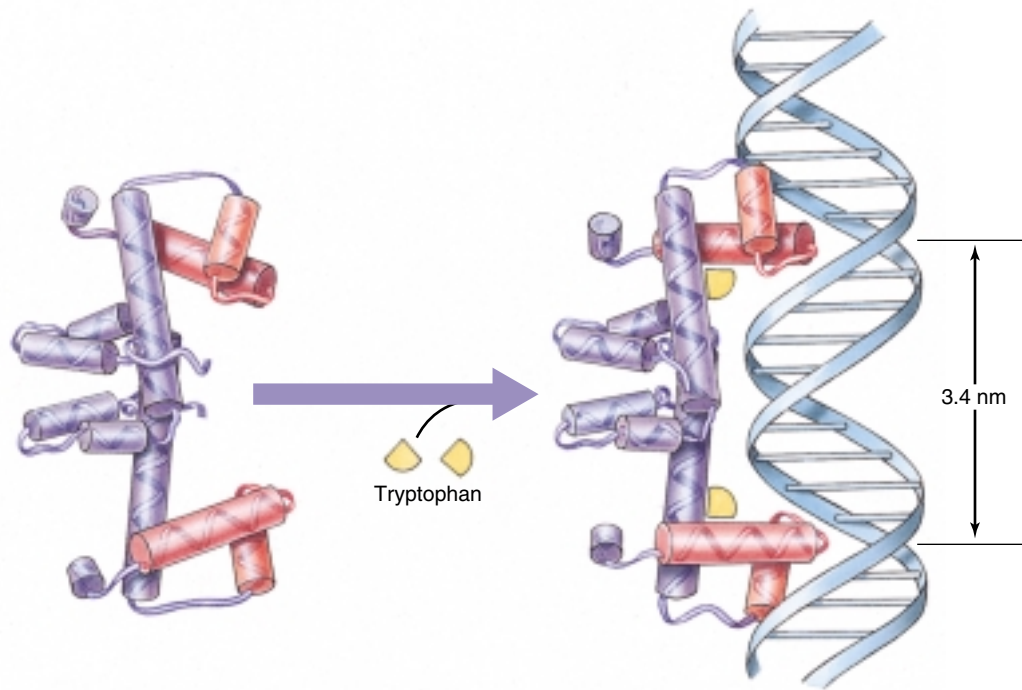


FIGURE 16.7

How the tryptophan repressor works. The binding of tryptophan to the repressor increases the distance between the two recognition helices in the repressor, allowing the repressor to fit snugly into two adjacent portions of the major groove in DNA.

RNA polymerase from binding to the *trp* promoter. The *trp* genes are transcribed, and the cell proceeds to manufacture tryptophan from other molecules. On the other hand, when tryptophan is present in the environment, it binds to the repressor, which is then able to bind to the *trp* promoter. This blocks transcription of the *trp* genes, and the cell's synthesis of tryptophan halts.

Activators Are ON Switches

Not all regulatory switches shut genes off—some turn them on. In these instances, bacterial promoters are deliberately constructed to be poor binding sites for RNA polymerase, and the genes these promoters govern are thus rarely transcribed—unless something happens to improve the promoter's ability to bind RNA polymerase. This can happen if a regulatory protein called a **transcriptional activator** binds to the DNA nearby. By contacting the polymerase protein itself, the activator protein helps hold the polymerase against the DNA promoter site so that transcription can begin.

A well-understood transcriptional activator is the catabolite activator protein (CAP) of *E. coli*, which initiates the transcription of genes that allow *E. coli* to use other molecules as food when glucose is not present. Falling levels of glucose lead to higher intracellular levels of the signaling molecule, cyclic AMP (cAMP), which binds to the

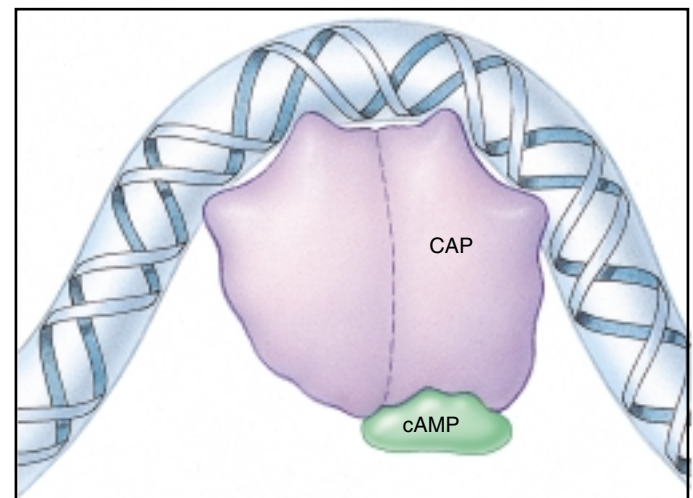


FIGURE 16.8

How CAP works. Binding of the catabolite activator protein (CAP) to DNA causes the DNA to bend around it. This increases the activity of RNA polymerase.

CAP protein. When cAMP binds to it, the CAP protein changes shape, enabling its helix-turn-helix motif to bind to the DNA near any of several promoters. Consequently, those promoters are activated and their genes can be transcribed (figure 16.8).

Combinations of Switches

By combining ON and OFF switches, bacteria can create sophisticated transcriptional control systems. A particularly well-studied example is the ***lac* operon** of *E. coli* (figure 16.9). This operon is responsible for producing three proteins that import the disaccharide lactose into the cell and break it down into two monosaccharides: glucose and galactose.

The Activator Switch. The *lac* operon possesses two regulatory sites. One is a CAP site located adjacent to the *lac* promoter. It ensures that the *lac* genes are not transcribed effectively when ample amounts of glucose are already present. In the absence of glucose, a high level of cAMP builds up in the cell. Consequently, cAMP is available to bind to CAP and allow it to change shape, bind to the DNA, and activate the *lac* promoter (figure 16.10). In the presence of glucose, cAMP levels are low, CAP is unable to bind to the DNA, and the *lac* promoter is not activated.

The Repressor Switch. Whether the *lac* genes are actually transcribed in the absence of glucose is determined by the second regulatory site, the **operator**, which is located adjacent to the promoter. A protein called the *lac* repressor is capable of binding to the operator, but only when lactose is absent. Because the operator and the promoter are close together, the repressor covers part of the promoter when it binds to the operator, preventing RNA polymerase from proceeding and so blocking transcription of the *lac* genes. These genes are then said to be “repressed” (figure 16.11). As a result, the cell does not transcribe genes whose products it has no use for. However, when lactose is present, a lactose isomer binds to the repressor, twisting its binding motif away from the major groove of the DNA. This prevents the repressor from binding to the operator and so allows RNA polymerase to bind to the promoter and transcribe the *lac* genes. Transcription of the *lac* operon is said to have been “induced” by lactose.

This two-switch control mechanism thus causes the cell to produce lactose-utilizing proteins whenever lactose is present but glucose is not, enabling it to make a metabolic decision to produce only what the cell needs, conserving its resources (figure 16.12).

Bacteria regulate gene expression transcriptionally through the use of repressor and activator “switches,” such as the *trp* repressor and the CAP activator. The transcription of some clusters of genes, such as the *lac* operon, is regulated by both repressors and activators.

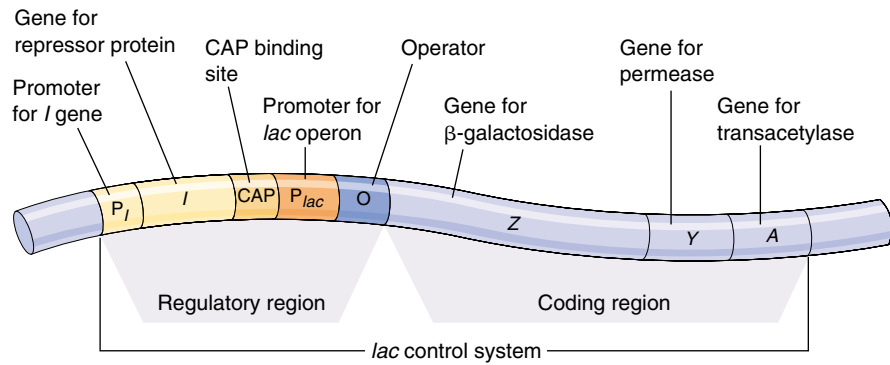


FIGURE 16.9

The *lac* region of the *Escherichia coli* chromosome. The *lac* operon consists of a promoter, an operator, and three genes that code for proteins required for the metabolism of lactose. In addition, there is a binding site for the catabolite activator protein (CAP), which affects whether or not RNA polymerase will bind to the promoter. Gene *I* codes for a repressor protein, which will bind to the operator and block transcription of the *lac* genes. The genes *Z*, *Y*, and *A* encode the two enzymes and the permease involved in the metabolism of lactose.

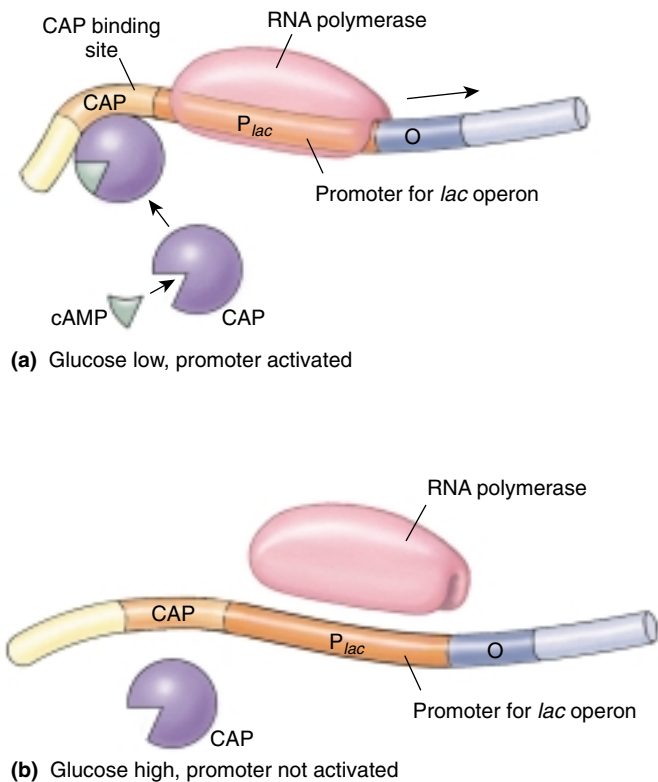


FIGURE 16.10

How the CAP site works. The CAP molecule can attach to the CAP binding site only when the molecule is bound to cAMP. (a) When glucose levels are low, cAMP is abundant and binds to CAP. The cAMP-CAP complex binds to the CAP site, bends in the DNA, and gives RNA polymerase access to the promoter. (b) When glucose levels are high, cAMP is scarce, and CAP is unable to activate the promoter.

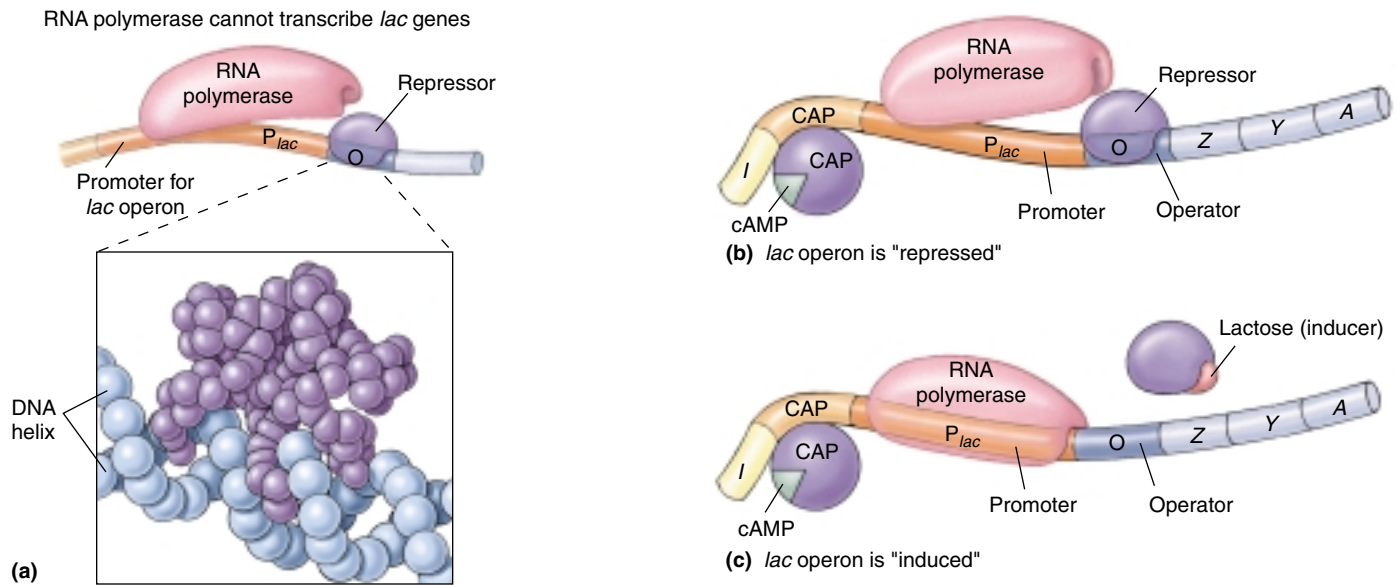


FIGURE 16.11

How the *lac* repressor works. (a) The *lac* repressor. Because the repressor fills the major groove of the DNA helix, RNA polymerase cannot fully attach to the promoter, and transcription is blocked. (b) The *lac* operon is shut down ("repressed") when the repressor protein is bound to the operator site. Because promoter and operator sites overlap, RNA polymerase and the repressor cannot functionally bind at the same time, any more than two people can sit in the same chair at once. (c) The *lac* operon is transcribed ("induced") when CAP is bound and when lactose binding to the repressor changes its shape so that it can no longer sit on the operator site and block RNA polymerase activity.

Glucose	Lactose	Diagram	Operon State
+	+		Operon OFF because CAP is not bound
+	-		Operon OFF both because <i>lac</i> repressor is bound and CAP is not
-	-		Operon OFF because <i>lac</i> repressor is bound
-	+		Operon ON because CAP is bound and <i>lac</i> repressor is not

FIGURE 16.12

Two regulatory proteins control the *lac* operon. Together, the *lac* repressor and CAP provide a very sensitive response to the cell's need to utilize lactose-metabolizing enzymes.

16.4 Transcriptional control in eukaryotes operates at a distance.

Designing a Complex Gene Control System

As we have seen, combinations of ON and OFF control switches allow bacteria to regulate the transcription of particular genes in response to the immediate metabolic demands of their environment. All of these switches work by interacting directly with RNA polymerase, either blocking or enhancing its binding to specific promoters. There is a limit to the complexity of this sort of regulation, however, because only a small number of switches can be squeezed into and around one promoter. In a eukaryotic organism that undergoes a complex development, many genes must interact with one another, requiring many more interacting elements than can fit around a single promoter (table 16.1).

In eukaryotes, this physical limitation is overcome by having distant sites on the chromosome exert control over the transcription of a gene (figure 16.13). In this way, many regulatory sequences scattered around the chromosomes can influence a particular gene's transcription. This "control-at-a-distance" mechanism includes two features: a set of proteins that help bind RNA polymerase to the promoter, and modular regulatory proteins that bind to distant sites. These two features produce a truly flexible control system.

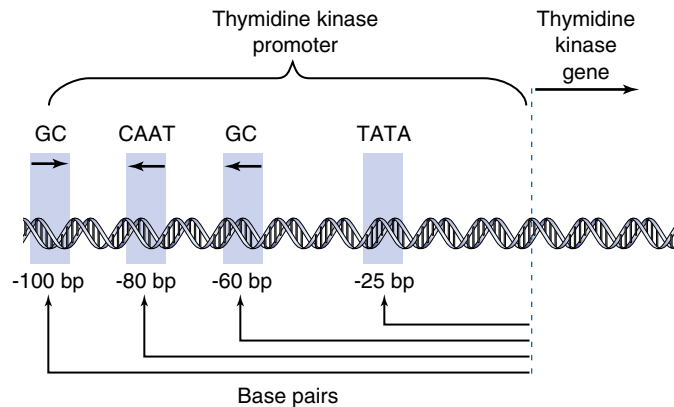


FIGURE 16.13

A eukaryotic promoter. This promoter for the gene encoding the enzyme thymidine kinase contains the TATA box that the initiation factor binds to, as well as three other DNA sequences that direct the binding of other elements of the transcription complex.

Table 16.1 Some Gene Regulatory Proteins and the DNA Sequences They Recognize

Regulatory Proteins of Species	DNA Sequence Recognized*	Regulatory Proteins of Species	DNA Sequence Recognized*
<i>ESCHERICHIA COLI</i>		<i>DROSOPHILA MELANOGASTER</i>	
<i>lac repressor</i>	AATTGTGAGCGGATAACAATT TTAACAACCTCGCCTATTGTTAA	Krüppel	AACGGGTAA TTGCCCAATT
CAP	TGTGAGTTAGCTCACT ACACTCAATCGAGTGA	bicoid	GGGATTAGA CCCTAATCT
λ repressor	TATCACCGCCAGAGGTA ATAGTGGCGGTCTCCAT		
YEAST		HUMAN	
GAL4	CGGAGGACTGTCCTCCG GCCTCCTGACAGGAGGC	Spl	GGGCGG CCCGCC
MAT α 2	CATGTAATT GTACATTAA	Oct-1	ATGCAAAT TACGTTTA
GCN4	ATGACTCAT TACTGAGTA	GATA-1	TGATAG ACTATC

*Each regulatory protein is able to recognize a family of closely related DNA sequences; only one member of each family is listed here.

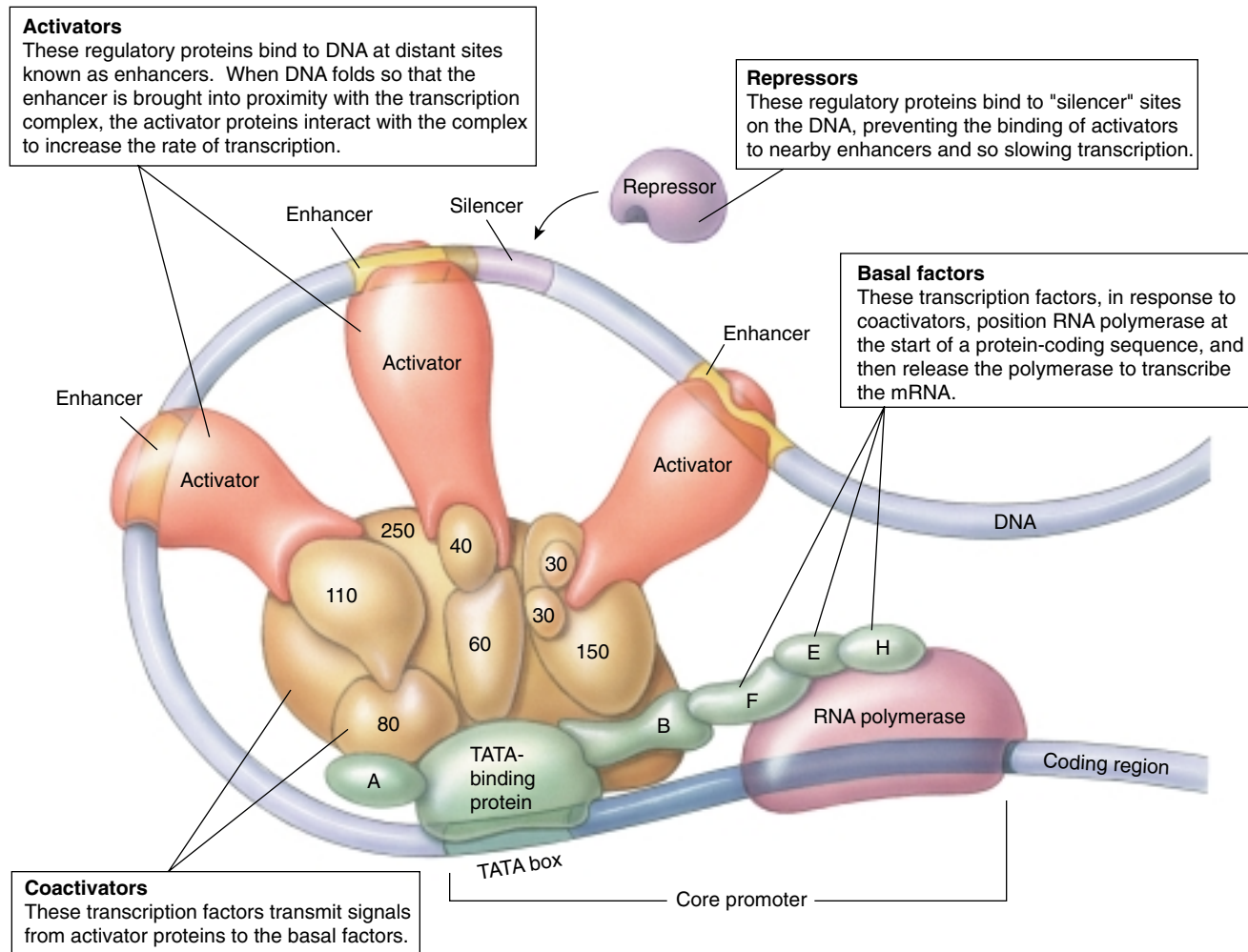


FIGURE 16.14

The structure of a human transcription complex. The transcription complex that positions RNA polymerase at the beginning of a human gene consists of four kinds of proteins. Basal factors (the green shapes at bottom of complex with letter names) are transcription factors that are essential for transcription but cannot by themselves increase or decrease its rate. They include the TATA-binding protein, the first of the basal factors to bind to the core promoter sequence. Coactivators (the tan shapes that form the bulk of the transcription complex, named according to their molecular weights) are transcription factors that link the basal factors with regulatory proteins called activators (the red shapes). The activators bind to enhancer sequences at other locations on the DNA. The interaction of individual basal factors with particular activator proteins is necessary for proper positioning of the polymerase, and the rate of transcription is regulated by the availability of these activators. When a second kind of regulatory protein called a repressor (the purple shape) binds to a so-called "silencer" sequence located adjacent to or overlapping an enhancer sequence, the corresponding activator that would normally have bound that enhancer is no longer able to do so. The activator is thus unavailable to interact with the transcription complex and initiate transcription.

Eukaryotic Transcription Factors

For RNA polymerase to successfully bind to a eukaryotic promoter and initiate transcription, a set of proteins called **transcription factors** must first assemble on the promoter, forming a complex that guides and stabilizes the binding of the polymerase (figure 16.14). The assembly process begins some 25 nucleotides upstream from the transcription start site, where a transcription factor composed of many subunits binds to a short TATA sequence (discussed in chapter 15). Other transcription factors then bind, eventually forming a full transcription factor com-

plex able to capture RNA polymerase. In many instances, the transcription factor complex then phosphorylates the bound polymerase, disengaging it from the complex so that it is free to begin transcription.

The binding of several different transcription factors provides numerous points where control over transcription may be exerted. Anything that reduces the availability of a particular factor (for example, by regulating the promoter that governs the expression and synthesis of that factor) or limits its ease of assembly into the transcription factor complex will inhibit transcription.

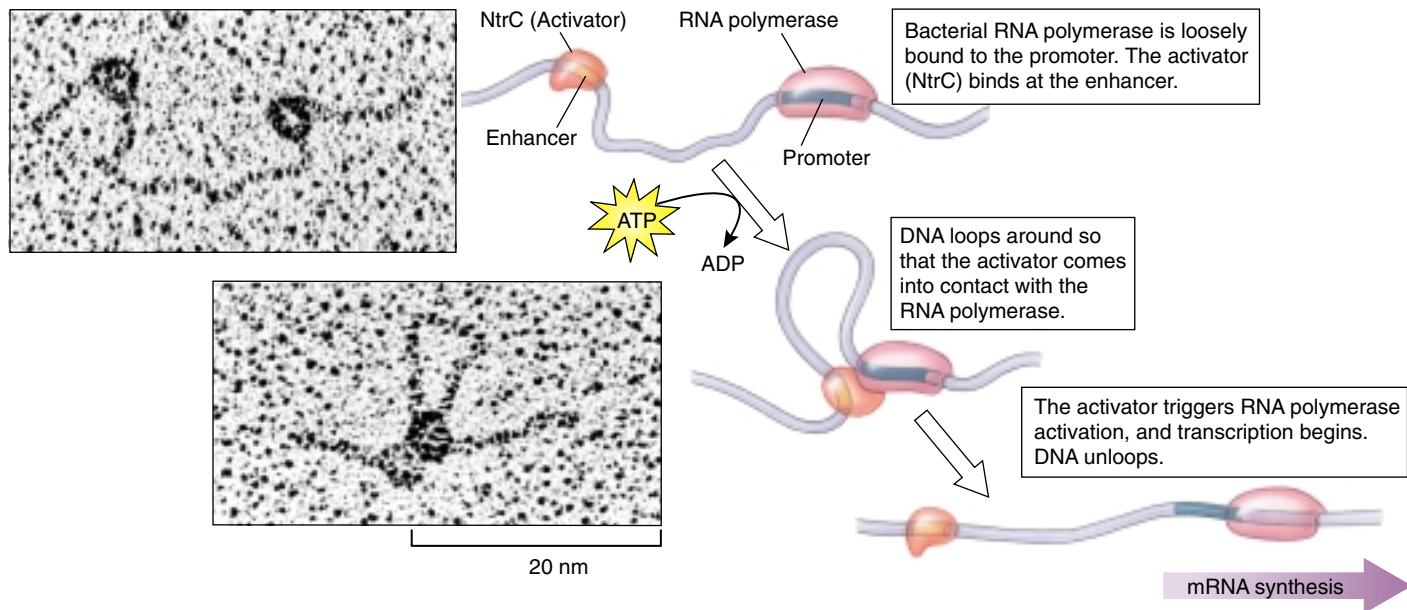


FIGURE 16.15

An enhancer in action. When the bacterial activator NtrC binds to an enhancer, it causes the DNA to loop over to a distant site where RNA polymerase is bound, activating transcription. While such enhancers are rare in bacteria, they are common in eukaryotes.

Enhancers

A key advance in the evolution of eukaryotic gene transcription was the advent of regulatory proteins composed of two distinct modules, or domains. The **DNA-binding domain** physically attaches the protein to the DNA at a specific site, using one of the structural motifs discussed earlier, while the **regulatory domain** interacts with other regulatory proteins.

The great advantage of this modular design is that it uncouples regulation from DNA binding, allowing a regulatory protein to bind to a specific DNA sequence at one site on a chromosome and exert its regulation over a promoter at another site, which may be thousands of nucleotides away. The distant sites where these regulatory proteins bind are called **enhancers**. Although enhancers also occur in exceptional instances in bacteria (figure 16.15), they are the rule rather than the exception in eukaryotes.

How can regulatory proteins affect a promoter when they bind to the DNA at enhancer sites located far from the promoter? Apparently the DNA loops around so that the enhancer is positioned near the promoter. This brings the regulatory domain of the protein attached to the enhancer into direct contact with the transcription factor complex attached to the promoter (figure 16.16).

The enhancer mode of transcriptional control that has evolved in eukaryotes adds a great deal of flexibility to the control process. The positioning of regulatory sites at a distance permits a large number of different regulatory sequences scattered about the DNA to influence a particular gene.

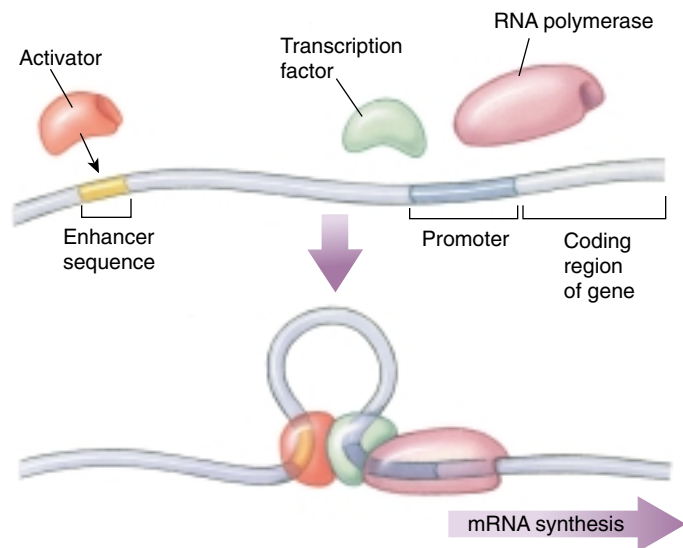


FIGURE 16.16

How enhancers work. The enhancer site is located far away from the gene being regulated. Binding of an activator (red) to the enhancer allows the activator to interact with the transcription factors (green) associated with RNA polymerase, activating transcription.

Transcription factors and enhancers confer great flexibility on the control of gene expression in eukaryotes.

The Effect of Chromosome Structure on Gene Regulation

The way DNA is packaged into chromosomes can have a profound effect on gene expression. As we saw in chapter 11, the DNA of eukaryotes is packaged in a highly compact form that enables it to fit into the cell nucleus. DNA is wrapped tightly around histone proteins to form nucleosomes (figure 16.17) and then the strand of nucleosomes is twisted into 30-nm filaments.

Promoter Blocking by Nucleosomes

Intensive study of eukaryotic chromosomes has shown that histones positioned over promoters block the assembly of transcription factor complexes. Therefore, transcription factors appear unable to bind to a promoter packaged in a nucleosome. In this way, nucleosomes may prevent continuous transcription initiation. On the other hand, nucleosomes do *not* inhibit activators and RNA polymerase. The regulatory domains of activators attached to enhancers apparently are able to displace the histones that block a promoter. In fact, this displacement of histones and the binding of activator to promoter are required for the assembly of the transcription factor complex. Once transcription has begun, RNA polymerase seems to push the histones aside as it traverses the nucleosome.

DNA Methylation

Chemical **methylation** of the DNA was once thought to play a major role in gene regulation in vertebrate cells. The addition of a methyl group to cytosine creates 5-methylcytosine but has no effect on base-pairing with guanine (figure 16.18), just as the addition of a methyl group to uracil produces thymine without affecting base-pairing with adenine. Many inactive mammalian genes are methylated, and it was tempting to conclude that methylation caused the inactivation. However, methylation is now viewed as having a less direct role, blocking accidental transcription of “turned-off” genes. Vertebrate cells apparently possess a protein that binds to clusters of 5-methylcytosine, preventing transcriptional activators from gaining access to the DNA. DNA methylation in vertebrates thus ensures that once a gene is turned off, it stays off.

Transcriptional control of gene expression occurs in eukaryotes despite the tight packaging of DNA into nucleosomes.

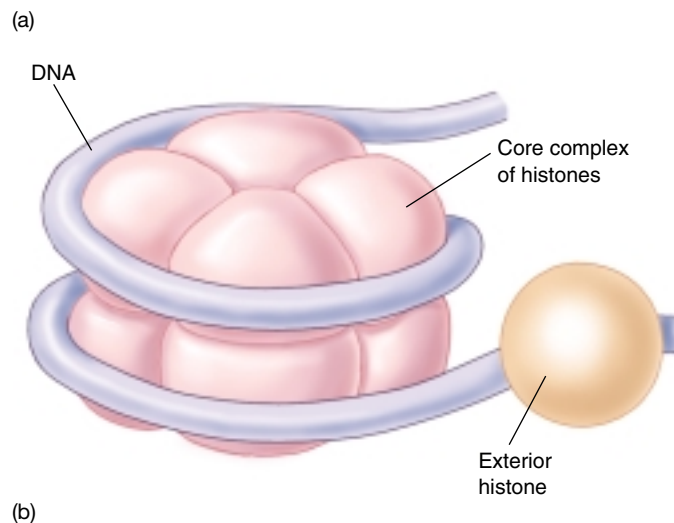
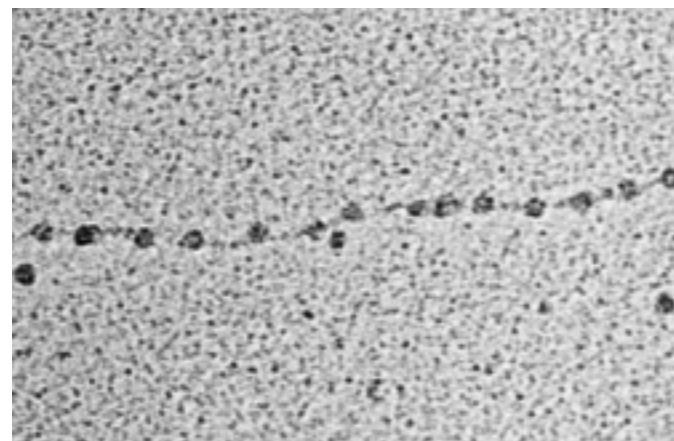


FIGURE 16.17
Nucleosomes. (a) In the electron micrograph, the individual nucleosomes have diameters of about 10 nm. (b) In the diagram of a nucleosome, the DNA double helix is wound around a core complex of eight histones; one additional histone binds to the outside of the nucleosome, exterior to the DNA.

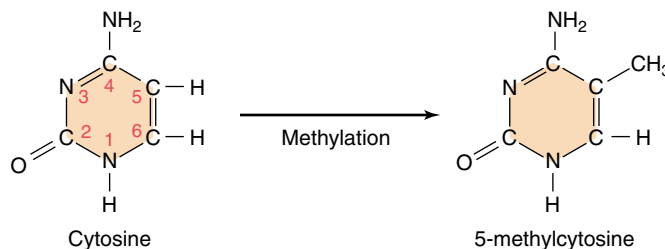


FIGURE 16.18
DNA methylation. Cytosine is methylated, creating 5-methylcytosine. Because the methyl group is positioned to the side, it does not interfere with the hydrogen bonds of a GC base-pair.

Posttranscriptional Control in Eukaryotes

Thus far we have discussed gene regulation entirely in terms of transcription initiation, that is, when and how often RNA polymerase starts “reading” a particular gene. Most gene regulation appears to occur at this point. However, there are many other points after transcription where gene expression could be regulated in principle, and all of them serve as control points for at least some eukaryotic genes. In general, these posttranscriptional control processes involve the recognition of specific sequences on the primary RNA transcript by regulatory proteins or other RNA molecules.

Processing of the Primary Transcript

As we learned in chapter 15, most eukaryotic genes have a patchwork structure, being composed of numerous short

coding sequences (exons) embedded within long stretches of noncoding sequences (introns). The initial mRNA molecule copied from a gene by RNA polymerase, the **primary transcript**, is a faithful copy of the entire gene, including introns as well as exons. Before the primary transcript is translated, the introns, which comprise on average 90% of the transcript, are removed in a process called **RNA processing**, or **RNA splicing**. Particles called *small nuclear ribonucleoproteins*, or *snRNPs* (more informally, **snurps**), are thought to play a role in RNA splicing. These particles reside in the nucleus of a cell and are composed of proteins and a special type of RNA called *small nuclear RNA*, or *snRNA*. One kind of snRNP contains snRNA that can bind to the 5' end of an intron by forming base-pairs with complementary sequences on the intron. When multiple snRNPs combine to form a larger complex called a **spliceosome**, the intron loops out and is excised (figure 16.19).

RNA splicing provides a potential point where the expression of a gene can be controlled, because exons can be

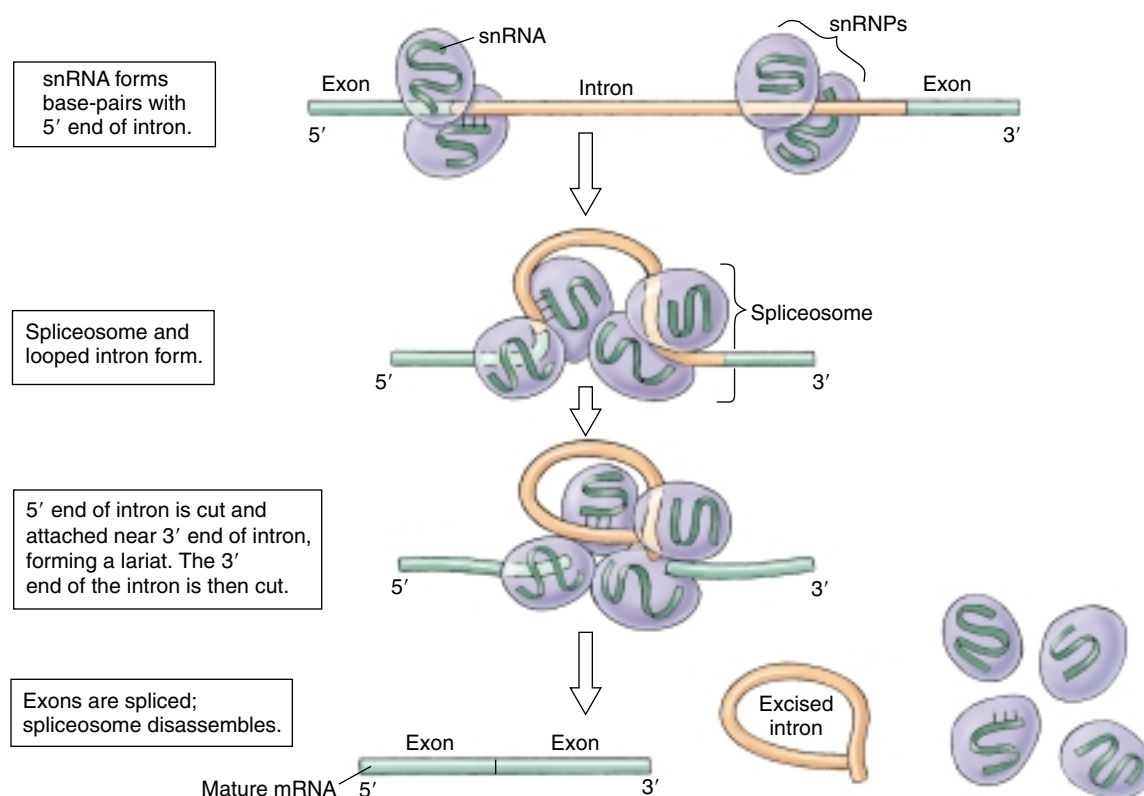


FIGURE 16.19

How spliceosomes process RNA. Particles called snRNPs contain snRNA that interacts with the 5' end of an intron. Several snRNPs come together and form a spliceosome. As the intron forms a loop, the 5' end is cut and linked to a site near the 3' end of the intron. The intron forms a lariat that is excised, and the exons are spliced together. The spliceosome then disassembles and releases the mature mRNA.

spliced together in different ways, allowing a variety of different polypeptides to be assembled from the same gene! Alternative splicing is common in insects and vertebrates, with two or three different proteins produced from one gene. In many cases, gene expression is regulated by changing which splicing event occurs during different stages of development or in different tissues.

An excellent example of alternative splicing in action is found in two different human organs, the thyroid and the hypothalamus. The thyroid gland (see chapter 56) is responsible for producing hormones that control processes such as metabolic rate. The hypothalamus, located in the brain, collects information from the body (for example, salt balance) and releases hormones that in turn regulate the release of hormones from other glands, such as the pituitary gland (see chapter 56). The two organs produce two distinct hormones, calcitonin and CGRP (calcitonin gene-related peptide) as part of their function. Calcitonin is responsible for controlling the amount of calcium we take up from our food and the balance of calcium in tissues like bone and teeth. CGRP is involved in a number of neural and endocrine functions. Although these two hormones are used for very different physiological purposes, the hormones are made using the same transcript (figure 16.20). The appearance of one product versus another is determined by tissue-specific factors that regulate the processing of the primary transcript. This ability offers another powerful way to control the expression of gene products, ranging from proteins with subtle differences to totally unrelated proteins.

Transport of the Processed Transcript Out of the Nucleus

Processed mRNA transcripts exit the nucleus through the nuclear pores described in chapter 5. The passage of a transcript across the nuclear membrane is an active process that requires that the transcript be recognized by receptors lining the interior of the pores. Specific portions of the transcript, such as the poly-A tail, appear to play a role in this recognition. The transcript cannot move through a pore as long as any of the splicing enzymes remain associated with the transcript, ensuring that partially processed transcripts are not exported into the cytoplasm.

There is little hard evidence that gene expression is regulated at this point, although it could be. On average, about 10% of transcribed genes are exon sequences, but only about 5% of the total mRNA produced as primary transcript ever reaches the cytoplasm. This suggests that about half of the exon primary transcripts never leave the nucleus, but it is not clear whether the disappearance of this mRNA is selective.

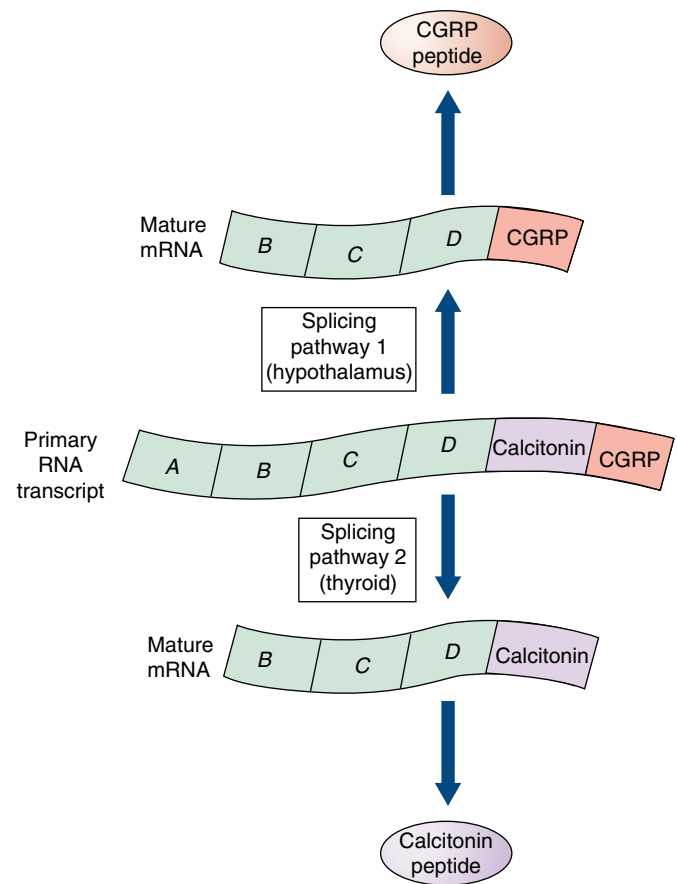


FIGURE 16.20
Alternative splicing products. The same transcript made from one gene can be spliced differently to give rise to two very distinct protein products, calcitonin and CGRP.

Selecting Which mRNAs Are Translated

The translation of a processed mRNA transcript by the ribosomes in the cytoplasm involves a complex of proteins called translation factors. In at least some cases, gene expression is regulated by modification of one or more of these factors. In other instances, **translation repressor proteins** shut down translation by binding to the beginning of the transcript, so that it cannot attach to the ribosome. In humans, the production of ferritin (an iron-storing protein) is normally shut off by a translation repressor protein called aconitase. Aconitase binds to a 30-nucleotide sequence at the beginning of the ferritin mRNA, forming a stable loop to which ribosomes cannot bind. When the cell encounters iron, the binding of iron to aconitase causes the aconitase to dissociate from the ferritin mRNA, freeing the mRNA to be translated and increasing ferritin production 100-fold.

A Vocabulary of Gene Expression

activator A regulatory protein that promotes gene transcription by binding to DNA sequences upstream of a promoter. Activator binding stimulates RNA polymerase activity.

anticodon The three-nucleotide sequence on one end of a tRNA molecule that is complementary to and base-pairs with an amino acid-specifying codon in mRNA.

codon The basic unit of the genetic code; a sequence of three adjacent nucleotides in DNA or mRNA that codes for one amino acid or for polypeptide termination.

exon A segment of eukaryotic DNA that is both transcribed into mRNA and translated into protein. Exons are typically scattered within much longer stretches of non-translated intron sequences.

intron A segment of eukaryotic DNA that is transcribed into mRNA but removed before translation.

nonsense codon A codon (UAA, UAG, or UGA) for which there is no tRNA with a complementary anticodon; a chain-terminating codon often called a “stop” codon.

operator A site of negative gene regulation; a sequence of nucleotides near or within the promoter that is recognized by a repressor. Binding of the repressor to the operator prevents the functional binding of RNA polymerase to the promoter and so blocks transcription.

operon A cluster of functionally related genes transcribed into a single mRNA molecule. A common mode of gene regulation in prokaryotes, it is rare in eukaryotes other than fungi.

promoter A site upstream from a gene to which RNA polymerase attaches to initiate transcription.

repressor A protein that regulates transcription by binding to the operator and so preventing RNA polymerase from initiating transcription from the promoter.

RNA polymerase The enzyme that transcribes DNA into RNA.

transcription The RNA polymerase-catalyzed assembly of an RNA molecule complementary to a strand of DNA.

translation The assembly of a polypeptide on the ribosomes, using mRNA to direct the sequence of amino acids.

Selectively Degrading mRNA Transcripts

Another aspect that affects gene expression is the stability of mRNA transcripts in the cell cytoplasm (figure 16.21). Unlike bacterial mRNA transcripts, which typically have a half-life of about 3 minutes, eukaryotic mRNA transcripts are very stable. For example, β -globin gene transcripts have a half-life of over 10 hours, an eternity in the fast-moving metabolic life of a cell. The transcripts encoding regulatory proteins and growth factors, however, are usually much less stable, with half-lives of less than 1 hour. What makes these particular transcripts so unstable? In many cases, they contain specific sequences near their 3' ends that make them attractive targets for enzymes that degrade mRNA. A sequence of A and U nucleotides near the 3' poly-A tail of a transcript promotes removal of the tail, which destabilizes the mRNA. Histone transcripts, for example, have a half-life of about 1 hour in cells that are actively synthesizing DNA; at other times during the cell cycle, the poly-A tail is lost and the transcripts are degraded within minutes. Other mRNA transcripts contain sequences near their 3' ends that are recognition sites for endonucleases, which causes these transcripts to be digested quickly. The short half-lives of the mRNA transcripts of many regulatory genes are critical to the function of those genes, as they enable the levels of regulatory proteins in the cell to be altered rapidly.

An Example of a Complex Gene Control System

Sunlight is an important gene-controlling signal for plants, from germination to seed formation. Plants must regulate their genes according to the presence of sunlight, the quality of the light source, the time of day, and many other environmental signals. The combination of these responses culminate in the way the genes are regulated, such as the genes *cab* (a chlorophyll-binding photosynthetic protein) and *rbcS* (a subunit of a carbon-fixing enzyme). For instance, photosynthesis-related genes tend to express early in the day, to carry out photosynthesis, and begin to shut down later in the day. Expression levels may also be regulated according to lighting conditions, such as cloudy days versus sunny days. When darkness arrives, the transcripts must be degraded in preparation for the next day. This is an example of how complex a gene control system can be, and scientists are just beginning to understand parts of such a complicated system.

Although less common than transcriptional control, posttranscriptional control of gene expression occurs in eukaryotes via RNA splicing, translation repression, and selective degradation of mRNA transcripts.

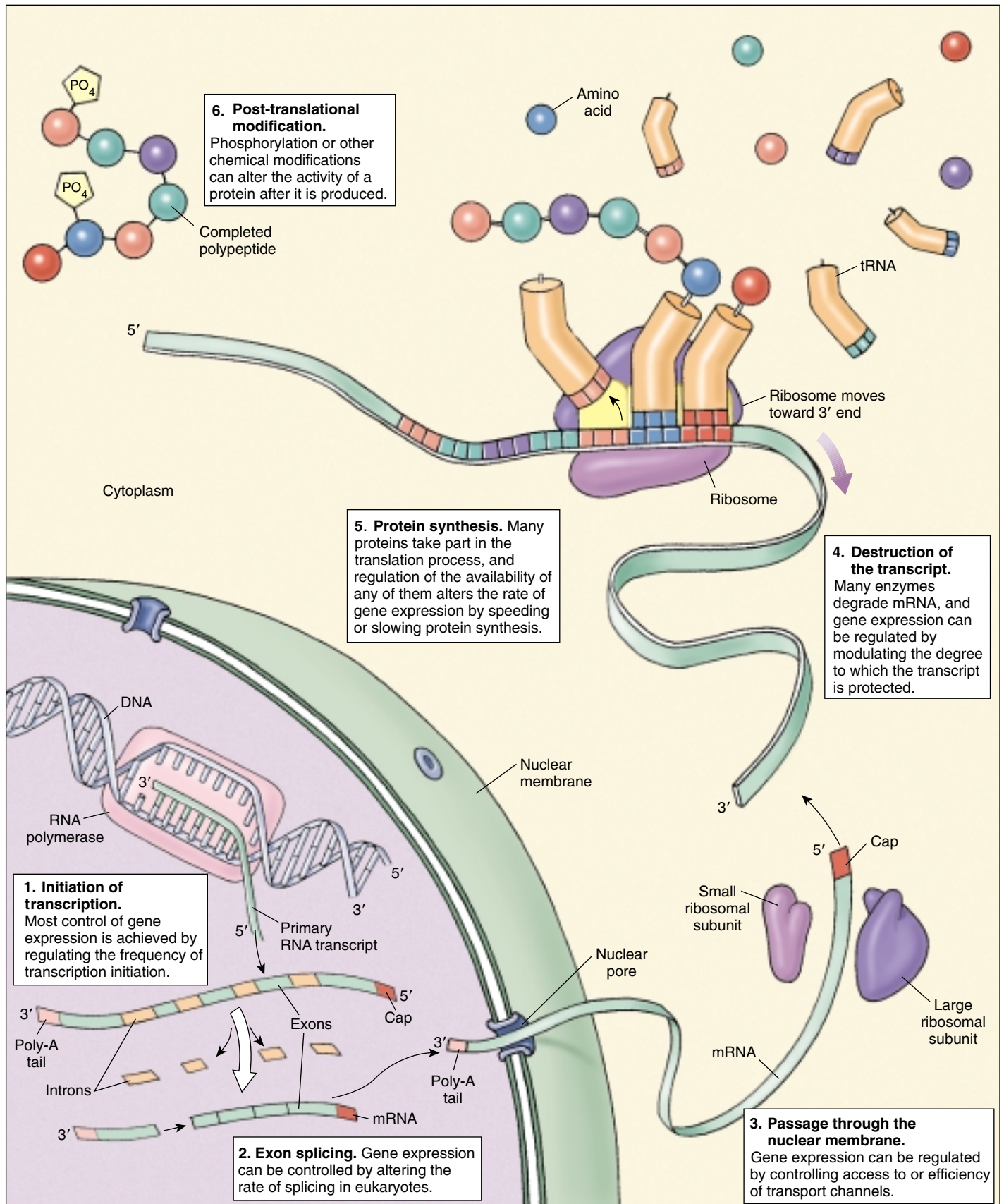


FIGURE 16.21
Six levels where gene expression can be controlled in eukaryotes.

Chapter 16



<http://www.mhhe.com/raven6e>



<http://www.biocourse.com>

Summary

Questions

Media Resources

16.1 Gene expression is controlled by regulating transcription.

- Regulatory sequences are short stretches of DNA that function in transcriptional control but are not transcribed themselves.
- Regulatory proteins recognize and bind to specific regulatory sequences on the DNA.

1. How do regulatory proteins identify specific nucleotide sequences without unwinding the DNA?



- Exploration: Gene regulation



- Student Research: Heat Shock Proteins

16.2 Regulatory proteins read DNA without unwinding it.

- Regulatory proteins possess structural motifs that allow them to fit snugly into the major groove of DNA, where the sides of the base-pairs are exposed.
- Common structural motifs include the helix-turn-helix, homeodomain, zinc finger, and leucine zipper.

2. What is a helix-turn-helix motif? What sort of developmental events are homeodomain motifs involved in?



- Exploration: Reading DNA

16.3 Bacteria limit transcription by blocking RNA polymerase.

- Many genes are transcriptionally regulated through repressors, proteins that bind to the DNA at or near the promoter and thereby inhibit transcription of the gene.
- Genes may also be transcriptionally regulated through activators, proteins that bind to the DNA and thereby stimulate the binding of RNA polymerase to the promoter.
- Transcription is often controlled by a *combination* of repressors and activators.

3. Describe the mechanism by which the transcription of *trp* genes is regulated in *Escherichia coli* when tryptophan is present in the environment.



- Art Activity: The lac operon

4. Describe the mechanism by which the transcription of *lac* genes is regulated in *E. coli* when glucose is absent but lactose is present in the environment.



- Regulation of *E. coli lac* operon
- Regulation of *E. coli trp* operon



- Gene Regulation

16.4 Transcriptional control in eukaryotes operates at a distance.

- In eukaryotes, RNA polymerase cannot bind to the promoter unless aided by a family of transcription factors.
- Anything that interferes with the activity of the transcription factors can block or alter gene expression.
- Eukaryotic DNA is packaged tightly in nucleosomes within chromosomes. This packaging appears to provide some inhibition of transcription, although regulatory proteins and RNA polymerase can still activate specific genes even when they are so packaged.
- Gene expression can also be regulated at the posttranscriptional level, through RNA splicing, translation repressor proteins, and the selective degradation of mRNA transcripts.

5. How do transcription factors promote transcription in eukaryotic cells? How do the enhancers of eukaryotic cells differ from most regulatory sites on bacterial DNA?

6. What role does the methylation of DNA likely play in transcriptional control?

7. How does the *primary* RNA transcript of a eukaryotic gene differ from the mRNA transcript of that gene as it is translated in the cytoplasm?

8. How can a eukaryotic cell control the translation of mRNA transcripts after they have been transported from the nucleus to the cytoplasm?

17

Cellular Mechanisms of Development

Concept Outline

17.1 Development is a regulated process.

Overview of Development. Studies of cellular mechanisms have focused on mice, fruit flies, nematodes, and flowering plants.

Vertebrate Development. Vertebrates develop in a highly orchestrated fashion.

Insect Development. Insect development is highly specialized, many key events occurring in a fused mass of cells.

Plant Development. Unlike animal development, which is buffered from the environment, plant development is sensitive to environmental influences.

17.2 Multicellular organisms employ the same basic mechanisms of development.

Cell Movement and Induction. Animal cells move by extending protein cables that they use to pull themselves past surrounding cells. Transcription within cells is influenced by signal molecules from other cells.

Determination. Cells become reversibly committed to particular developmental paths.

Pattern Formation. Diffusion of chemical inducers governs pattern formation in fly embryos.

Expression of Homeotic Genes. Master genes determine the form body segments will take.

Programmed Cell Death. Some genes, when activated, kill their cells.

17.3 Four model developmental systems have been extensively researched.

The Mouse. *Mus musculus*.

The Fruit Fly. *Drosophila melanogaster*.

The Nematode. *Caenorhabditis elegans*.

The Flowering Plant. *Arabidopsis thaliana*.

17.4 Aging can be considered a developmental process.

Theories of Aging. While there are many ideas about why cells age, no one theory of aging is widely accepted.



FIGURE 17.1

A collection of future fish undergo embryonic development. Inside a transparent fish egg, a single cell becomes millions of cells that form eyes, fins, gills, and other body parts.

In the previous chapter, we explored gene expression from the perspective of an individual cell, examining the diverse mechanisms that may be employed by a cell to control the transcription of particular genes. Now we will broaden our perspective and look at the unique challenge posed by the development of a cell into a multicellular organism (figure 17.1). In the course of this developmental journey, a pattern of decisions about transcription are made that cause particular lines of cells to proceed along different paths, spinning an incredibly complex web of cause and effect. Yet, for all its complexity, this developmental program works with impressive precision. In this chapter, we will explore the mechanisms used by multicellular organisms to control their development and achieve this precision.

17.1 Development is a regulated process.

Overview of Development

Organisms in all three multicellular kingdoms—fungi, plants, and animals—realize cell specialization by orchestrating gene expression. That is, different cells express different genes at different times. To understand development, we need to focus on how cells determine which genes to activate, and when.

Among the fungi, the specialized cells are largely limited to reproductive cells. In basidiomycetes and ascomycetes (the so-called higher fungi), certain cells produce hormones that influence other cells, but the basic design of all fungi is quite simple. For most of its life, a fungus has a two-dimensional body, consisting of long filaments of cells that are only imperfectly separated from each other. Fungal maturation is primarily a process of growth rather than specialization.

Development is far more complex in plants, where the adult individuals contain a variety of specialized cells organized into tissues and organs. A hallmark of plant development is flexibility; as a plant develops, the precise array of tissues it achieves is greatly influenced by its environment.

In animals, development is complex and rigidly controlled, producing a bewildering array of specialized cell types through mechanisms that are much less sensitive to the environment. The subject of intensive study, animal development has in the last decades become relatively well understood.

Here we will focus our attention on four developmental systems which researchers have studied intensively: (1) an animal with a very complexly arranged body, a mammal; (2) a less complex animal with an intricate developmental cycle, an insect; (3) a very simple animal, a nematode; and (4) a flowering plant (figure 17.2).

To begin our investigation of development, we will first examine the overall process of development in three quite different organisms, so we can sort through differences in the gross process to uncover basic similarities in underlying mechanisms. We will start by describing the overall process in vertebrates, because it is the best understood among the animals. Then we will examine the very different developmental process carried out by insects, in which genetics has allowed us to gain detailed knowledge of many aspects of the process. Finally we will look at development in a third very different organism, a flowering plant.

Almost all multicellular organisms undergo development. The process has been well studied in animals, especially in mammals, insects, nematodes, and flowering plants.

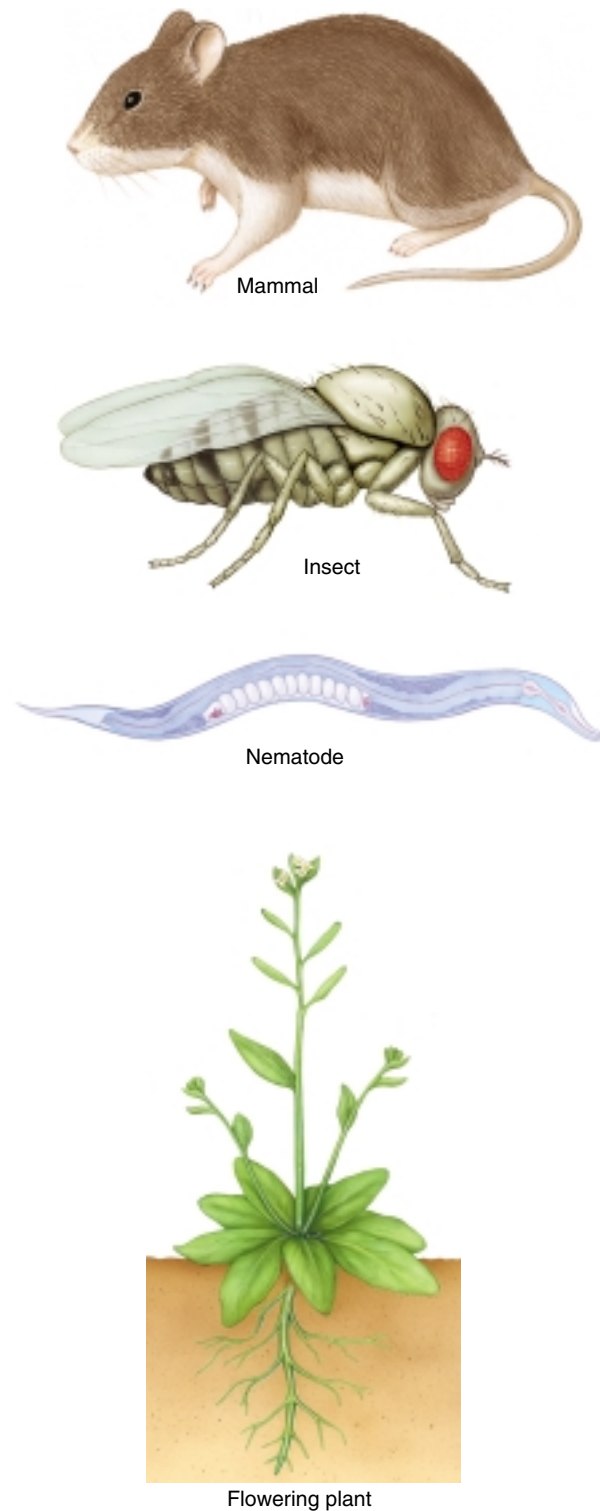


FIGURE 17.2
Four developmental systems. Researchers studying the cellular mechanisms of development have focused on these four organisms.

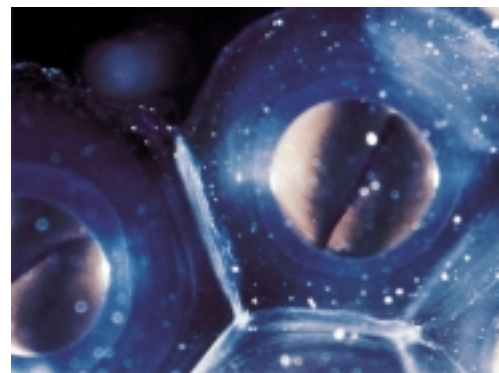
Vertebrate Development

Vertebrate development is a dynamic process in which cells divide rapidly and move over each other as they first establish the basic geometry of the body (figure 17.3). At different sites, particular cells then proceed to form the body's organs, and then the body grows to a size and shape that will allow it to survive after birth. The entire process, described more fully in chapter 60, is traditionally divided into phases. As in mitosis, however, the boundaries between phases are somewhat artificial, and the phases, in fact, grade into one another.

Cleavage

Vertebrates begin development as a single fertilized egg, the zygote. Within an hour after fertilization, the zygote begins to divide rapidly into a larger and larger number of smaller and smaller cells called **blastomeres**, until a solid ball of cells is produced (figure 17.4). This initial period of cell division, termed **cleavage**, is not accompanied by any increase in the overall size of the embryo; rather, the contents of the zygote are simply partitioned into the daughter cells. The two ends of the zygote are traditionally referred to as the **animal** and **vegetal poles**. In general, the blastomeres of the animal pole will go on to form the external tissues of the body, while those of the vegetal pole will form the internal tissues. The initial top-bottom (dorsal-ventral) orientation of the embryo is determined at fertilization by the location where the sperm nucleus enters the egg, a point that corresponds roughly to the future belly. After about 12 divisions, the burst of cleavage divisions slows, and transcription of key genes begins within the embryo cells.

FIGURE 17.4
Cleavage divisions producing a frog embryo. (a) The initial divisions are, in this case, on the side of the embryo facing you, producing (b) a cluster of cells on this side of the embryo, which soon expands to become a (c) compact mass of cells. (d) This mass eventually invaginates into the interior of the embryo, forming a gastrula, then a neurula.



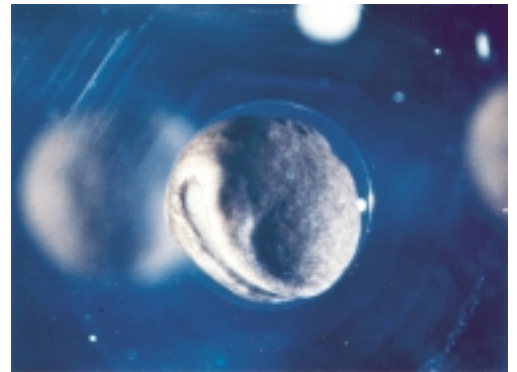
(a)



(b)



(c)



(d)



FIGURE 17.3
The miracle of development. This nine-week-old human fetus started out as a single cell: a fertilized egg, or zygote. The zygote's daughter cells have been repeatedly dividing and specializing to produce the distinguishable features of a fetus.

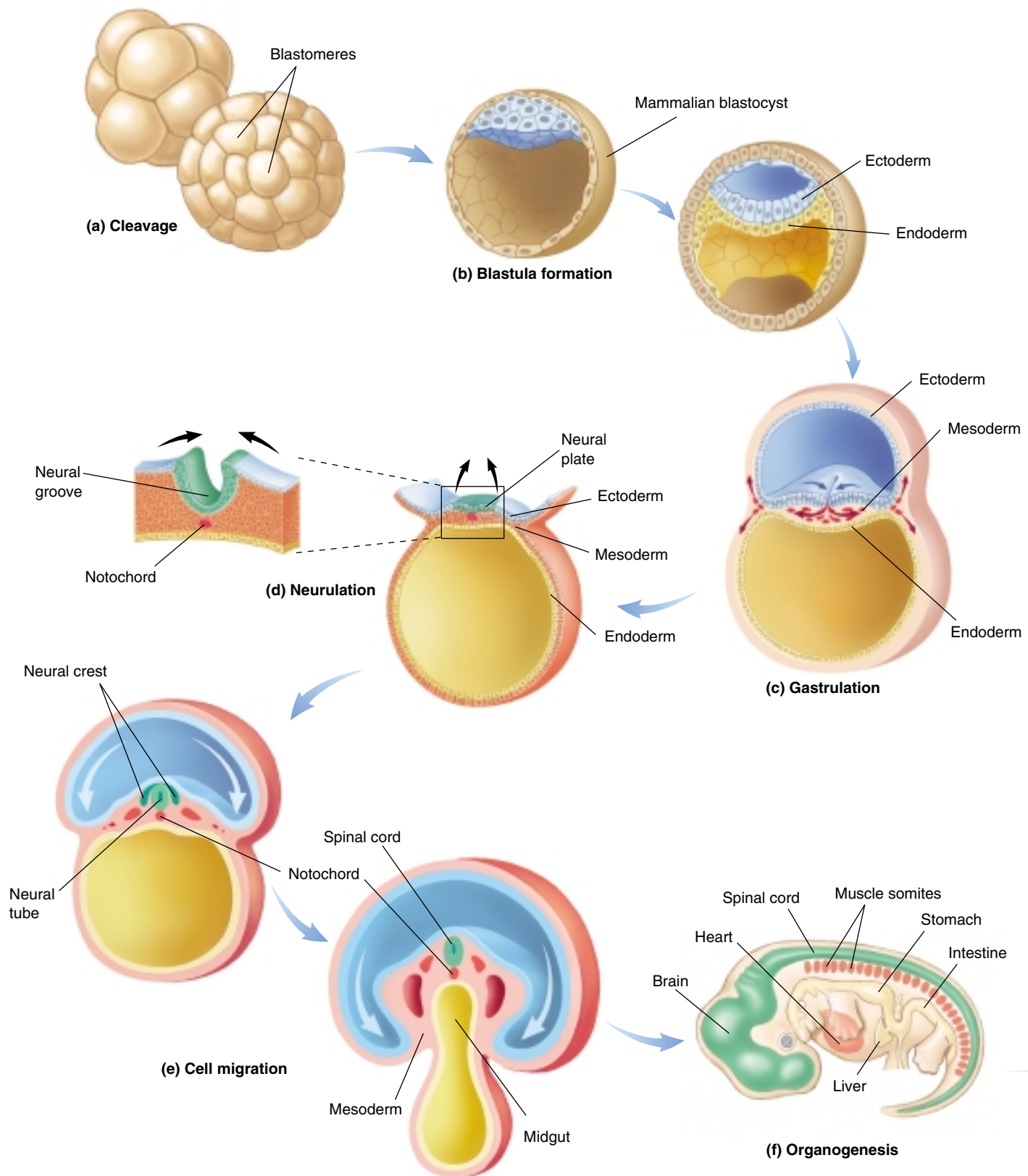


FIGURE 17.5
The path of vertebrate development. An illustration of the major events in the development of *Mus musculus*, the house mouse. (a) Cleavage. (b) Formation of blastula. (c) Gastrulation. (d) Neurulation. (e) Cell migration. (f) Organogenesis. (g) Growth.

Formation of the Blastula

The outermost blastomeres (figure 17.5a) in the ball of cells produced during cleavage are joined to one another by tight junctions, which, as you may recall from chapter 7, are belts of protein that encircle a cell and weld it firmly to its neighbors. These tight junctions create a seal that isolates the interior of the cell mass from the surrounding medium. At about the 16-cell stage, the cells in the interior of the mass begin to pump Na^+ from their cytoplasm into the spaces between cells. The resulting osmotic gradient causes water to be drawn into the center of the cell mass, enlarging the intercellular spaces. Eventually, the spaces coalesce to form a single large cavity within the cell mass. The resulting hollow ball of cells is called a **blastula**, or **blastocyst** in mammals (figure 17.5b).

Gastrulation

Some cells of the blastula then push inward, forming a **gastrula** that is invaginated. Cells move by using extensions called lamellipodia to crawl over neighboring cells, which respond by forming lamellipodia of their own. Soon a sheet of cells contracts on itself and shoves inward, starting the invagination. Called **gastrulation** (figure 17.5c), this process creates the main axis of the vertebrate body, converting the blastula into a bilaterally symmetrical embryo with a central gut. From this point on, the embryo has three **germ layers** whose organization foreshadows the future organization of the adult body. The cells that invaginate and form the tube of the primitive gut are endoderm; they give rise to the stomach, lungs,

liver, and most of the other internal organs. The cells that remain on the exterior are ectoderm, and their derivatives include the skin on the outside of the body and the nervous system. The cells that break away from the invaginating cells and invade the space between the gut and the exterior wall are mesoderm; they eventually form the notochord, bones, blood vessels, connective tissues, and muscles.

Neurulation

Soon after gastrulation is complete, a broad zone of ectoderm begins to thicken on the dorsal surface of the embryo, an event triggered by the presence of the notochord beneath it. The thickening is produced by the elongation of certain ectodermal cells. Those cells then assume a wedge shape by contracting bundles of actin filaments at one end. This change in shape causes the neural tissue to roll up into a tube, which eventually pinches off from the rest of the ectoderm and gives rise to the brain and spinal cord. This tube is called the **neural tube**, and the process by which it forms is termed **neurulation** (figure 17.5d).

Cell Migration

During the next stage of vertebrate development, a variety of cells migrate to form distant tissues, following specific paths through the embryo to particular locations (figure 17.5e). These migrating cells include those of the **neural crest**, which pinch off from the neural tube and form a number of structures, including some of the body's sense organs; cells that migrate from central blocks of muscle tissue called **somites** and form the skeletal muscles of the body; and the precursors of blood cells and gametes. When a migrating cell reaches its destination, receptor proteins on its surface interact with proteins on the surfaces of cells in the destination tissue, triggering changes in the cytoskeleton of the migrating cell that cause it to cease moving.

Organogenesis and Growth

At the end of this wave of cell migration and colonization, the basic vertebrate body plan has been established, although the embryo is only a few millimeters long and has only about 10^5 cells. Over the course of subsequent development, tissues will develop into organs (figure 17.5f), and the embryo will grow to be a hundred times larger, with a million times as many cells (figure 17.5g).



(g) Growth

Vertebrates develop in a highly orchestrated fashion. The zygote divides rapidly, forming a hollow ball of cells that then pushes inward, forming the main axis of an embryo that goes on to develop tissues, and after a process of cell migration, organs.

Insect Development

Like all animals, insects develop through an orchestrated series of cell changes, but the path of development is quite different from that of a vertebrate. Many insects produce two different kinds of bodies during their development, the first a tubular eating machine called a **larva**, and the second a flying machine with legs and wings. The passage from one body form to the other is called **metamorphosis** and involves a radical shift in development. Here we will describe development in the fruit fly *Drosophila* (figure 17.6), which is the subject of much genetic research.



FIGURE 17.6

The fruit fly, *Drosophila melanogaster*. A dorsal view of *Drosophila*, one of the most intensively studied animals in development.

Maternal Genes

The development of an insect like *Drosophila* begins before fertilization, with the construction of the egg. Specialized nurse cells that help the egg to grow move some of their own mRNA into the end of the egg nearest them (figure 17.7a). As a result, mRNAs produced by maternal genes are positioned in particular locations in the egg, so that after repeated divisions subdivide the fertilized egg, different daughter cells will contain different maternal products. Thus, the action of maternal (rather than zygotic) genes determines the initial course of development.

Syncytial Blastoderm

After fertilization, 12 rounds of nuclear division without cytokinesis produce about 6000 nuclei, all within a single cytoplasm. All of the nuclei within this **syncytial blastoderm** (figure 17.7b) can freely communicate with one another, but nuclei located in different sectors of the egg experience different maternal products. The nuclei then space themselves evenly along the surface of the blastoderm, and membranes grow between them. Folding of the embryo and primary tissue development soon follow, in a process fundamentally similar to that seen in vertebrate development. The tubular body that results within a day of fertilization is a larva.

Larval Instars

The larva begins to feed immediately, and as it does so, it grows. Its chitinous exoskeleton cannot stretch much, however, and within a day it sheds the exoskeleton. Before the new exoskeleton has had a chance to harden, the larva ex-

pands in size. A total of three larval stages, or **instars**, are produced over a period of four days (figure 17.7c).

Imaginal Discs

During embryonic growth, about a dozen groups of cells called **imaginal discs** are set aside in the body of the larva (figure 17.7d). Imaginal discs play no role in the life of the larva, but are committed to form key parts of the adult fly's body.

Metamorphosis

After the last larval stage, a hard outer shell forms, and the larva is transformed into a **pupa** (figure 17.7e). Within the pupa, the larval cells break down and release their nutrients, which are used in the growth and development of the various imaginal discs (eye discs, wing discs, leg discs, and so on). The imaginal discs then associate with one another, assembling themselves into the body of the adult fly (figure 17.7f). The metamorphosis of a *Drosophila* larva into a pupa and then into adult fly takes about four days, after which the pupal shell splits and the fly emerges.

Drosophila development proceeds through two discrete phases, the first a larval phase that gathers food, then an adult phase that is capable of flight and reproduction.

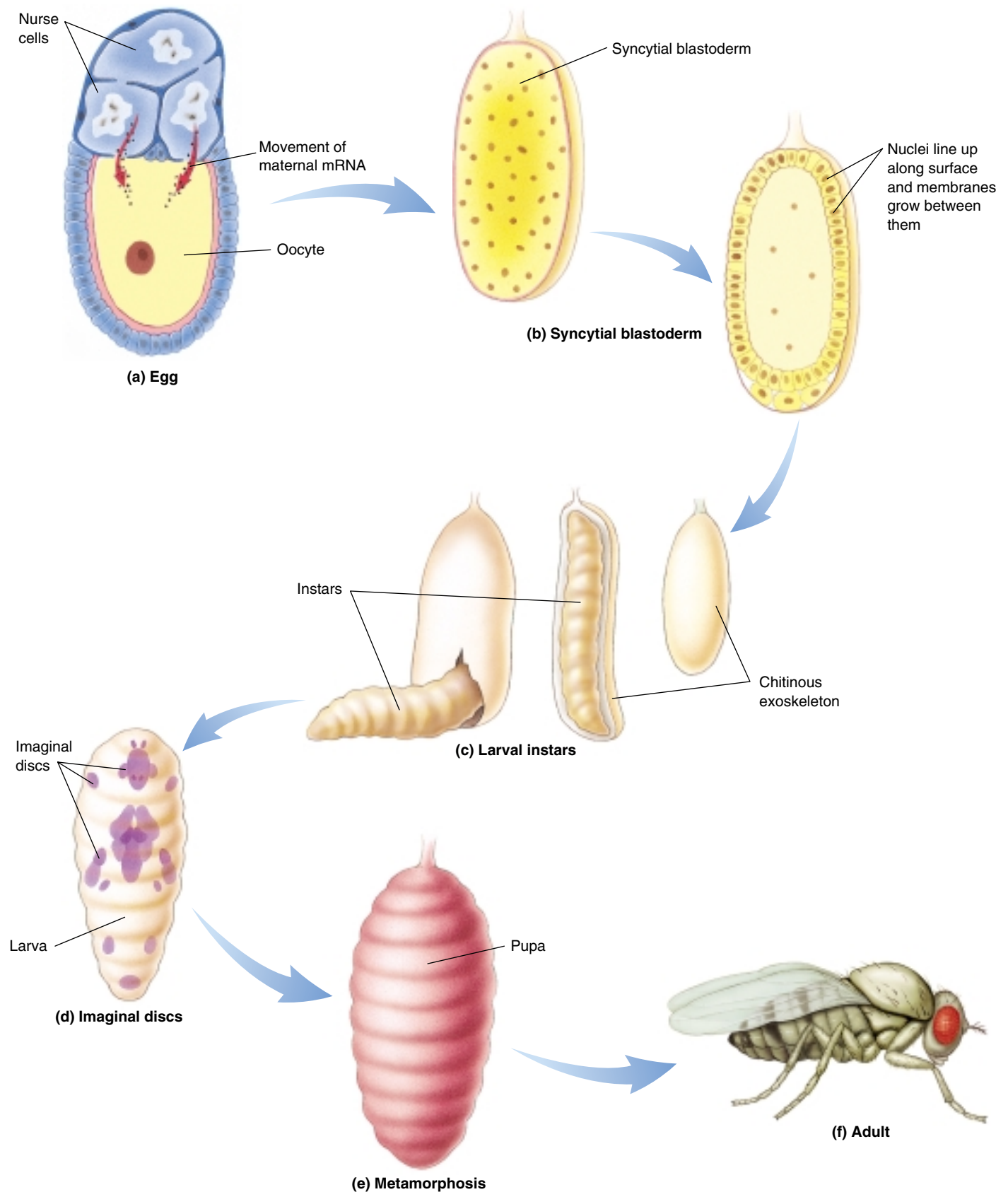


FIGURE 17.7
The path of insect development. An illustration of the major events in the development of *Drosophila melanogaster*. (a) Egg. (b) Syncytial blastoderm. (c) Larval instars. (d) Imaginal discs. (e) Metamorphosis. (f) Adult.

Plant Development

At the most basic level, the developmental paths of plants and animals share many key elements. However, the mechanisms used to achieve body form are quite different. While animal cells follow an orchestrated series of movements during development, plant cells are encased within stiff cellulose walls, and, therefore, cannot move. Each cell in a plant is fixed into position when it is created. Instead of using cell migration, plants develop by building their bodies outward, creating new parts from special groups of self-renewing cells called **meristems**. As meristem cells continually divide, they produce cells that can differentiate into the tissues of the plant.

Another major difference between animals and plants is that most animals are mobile and can move away from unfavorable circumstances, while plants are anchored in position and must simply endure whatever environment they experience. Plants compensate for this restriction by relaxing the rules of development to accommodate local circumstances. Instead of creating a body in which every part is specified to have a fixed size and location, a plant assembles its body from a few types of modules, such as leaves, roots, branch nodes, and flowers. Each module has a rigidly controlled structure and organization, but how the modules are utilized is quite flexible. As a plant develops, it simply adds more modules, with the environment having a major influence on the type, number, size, and location of what is added. In this way the plant is able to adjust the path of its development to local circumstances.

Early Cell Division

The first division of the fertilized egg in a flowering plant is off-center, so that one of the daughter cells is small, with dense cytoplasm (figure 17.8*a*). That cell, the future embryo, begins to divide repeatedly, forming a ball of cells. The other daughter cell also divides repeatedly, forming an elongated structure called a **suspensor**, which links the embryo to the nutrient tissue of the seed. The suspensor also provides a route for nutrients to reach the developing embryo. Just as the animal embryo acquires its initial axis as a cell mass formed during cleavage divisions, so the plant embryo forms its root-shoot axis at this time. Cells near the suspensor are destined to form a root, while those at the other end of the axis ultimately become a shoot.

Tissue Formation

Three basic tissues differentiate while the plant embryo is still a ball of cells (figure 17.8*b*), analogous to the formation of the three germ layers in animal embryos, although in plants, no cell movements are involved. The outermost cells in a plant embryo become **epidermal cells**. The bulk of the embryonic interior consists of **ground tissue** cells

that eventually function in food and water storage. Lastly, cells at the core of the embryo are destined to form the future **vascular tissue**.

Seed Formation

Soon after the three basic tissues form, a flowering plant embryo develops one or two seed leaves called **cotyledons**. At this point, development is arrested, and the embryo is now either surrounded by nutritive tissue or has amassed stored food in its cotyledons (figure 17.8*c*). The resulting package, known as a *seed*, is resistant to drought and other unfavorable conditions; in its dormant state, it is a vehicle for dispersing the embryo to distant sites and allows a plant embryo to survive in environments that might kill a mature plant.

Germination

A seed germinates in response to changes in its environment brought about by water, temperature, or other factors. The embryo within the seed resumes development and grows rapidly, its roots extending downward and its leaf-bearing shoots extending upward (figure 17.8*d*).

Meristematic Development

Plants development exhibits its great flexibility during the assembly of the modules that make up a plant body. Apical meristems at the root and shoot tips generate the large numbers of cells needed to form leaves, flowers, and all other components of the mature plant (figure 17.8*e*). At the same time, meristems ensheathing the stems and roots produce the wood and other tissues that allow growth in circumference. A variety of hormones produced by plant tissues influence meristem activity and, thus, the development of the plant body. Plant hormones (see chapter 41) are the tools that allow plant development to adjust to the environment.

Morphogenesis

The form of a plant body is largely determined by controlled changes in cell shape as they expand osmotically after they form (see figure 17.8*e*). Plant growth-regulating hormones and other factors influence the orientation of bundles of microtubules on the interior of the plasma membrane. These microtubules seem to guide cellulose deposition as the cell wall forms around the outside of a new cell. The orientation of the cellulose fibers, in turn, determines how the cell will elongate as it increases in volume, and so determines the cell's final shape.

In a developing plant, leaves, flowers, and branches are added to the growing body in ways that are strongly influenced by the environment.

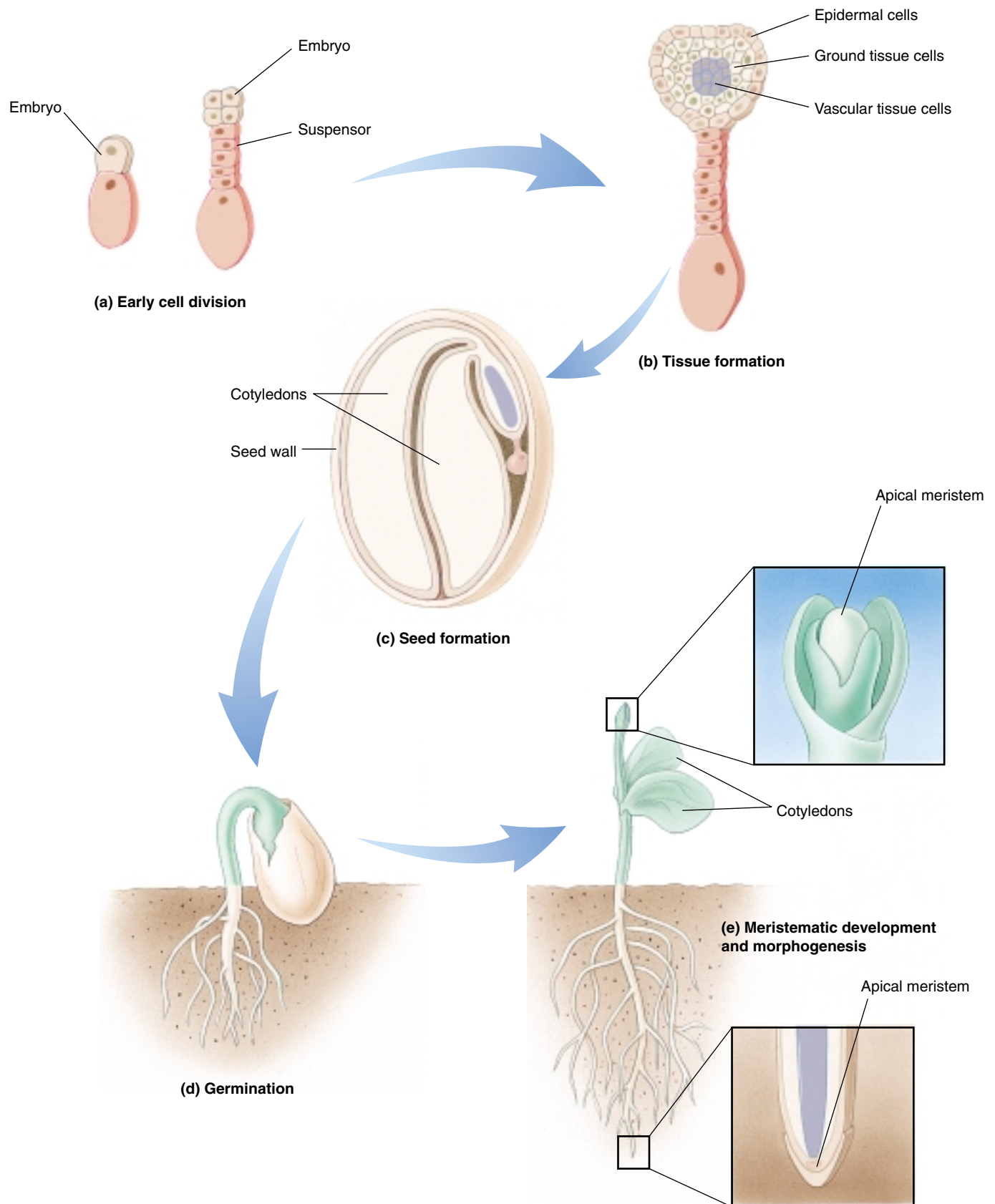


FIGURE 17.8

The path of plant development. An illustration of the developmental stages of *Arabidopsis thaliana*. (a) Early cell division. (b) Tissue formation. (c) Seed formation. (d) Germination. (e) Meristematic development and morphogenesis.

17.2 Multicellular organisms employ the same basic mechanisms of development.

Despite the many differences in the three developmental paths we have just discussed, it is becoming increasingly clear that most multicellular organisms develop according to molecular mechanisms that are fundamentally very similar. This observation suggests that these mechanisms evolved very early in the history of multicellular life. Here, we will focus on six mechanisms that seem to be of particular importance in the development of a wide variety of organisms. We will consider them in roughly the order in which they first become important during development.

Cell Movement and Induction

Cell Movement

Cells migrate during many stages in animal development, sometimes traveling great distances before reaching the site where they are destined to develop. By the time vertebrate development is complete, most tissues contain cells that originated from quite different parts of the early embryo. One way cells move is by pulling themselves along using cell adhesion molecules, such as the cadherin proteins you read about in chapter 7. Cadherins span the plasma membrane, protruding into the cytoplasm and extending out from the cell surface. The cytoplasmic portion of the molecule is attached to actin or intermediate filaments of the cytoskeleton, while the extracellular portion has five 100-amino acid segments linked end-to-end; three or more of these segments have Ca^{++} binding sites that play a critical role in the attachment of the cadherin to other cells. Over a dozen different cadherins have been discovered to date. Each type of cadherin attaches to others of its own type at its terminal segments, forming a two-cadherin link between the cytoskeletons of adjacent cells. As a cell migrates to a

different tissue, the nature of the cadherin it expresses changes, and if cells expressing two different cadherins are mixed, they quickly sort themselves out, aggregating into two separate masses. This is how the different imaginal discs of a *Drosophila* larva assemble into an adult. Other calcium-independent cell adhesion molecules, such as the neural cell adhesion molecules (N-CAMs) expressed by migrating nerve cells, reinforce the associations made by cadherins, but cadherins play the major role in holding aggregating cells together.

In some tissues, such as connective tissue, much of the volume of the tissue is taken up by the spaces *between* cells. These spaces are not vacant, however. Rather, they are filled with a network of molecules secreted by surrounding cells, principally, a matrix of long polysaccharide chains covalently linked to proteins (proteoglycans), within which are embedded strands of fibrous protein (collagen, elastin, and fibronectin). Migrating cells traverse this matrix by binding to it with cell surface proteins called integrins, which was also described in chapter 7. Integrins are attached to actin filaments of the cytoskeleton and protrude out from the cell surface in pairs, like two hands. The “hands” grasp a specific component of the matrix such as collagen or fibronectin, thus linking the cytoskeleton to the fibers of the matrix. In addition to providing an anchor, this binding can initiate changes within the cell, alter the growth of the cytoskeleton, and change the way in which the cell secretes materials into the matrix.

Thus, cell migration is largely a matter of changing patterns of cell adhesion. As a migrating cell travels, it continually extends projections that probe the nature of its environment. Tugged this way and that by different tentative attachments, the cell literally feels its way toward its ultimate target site.

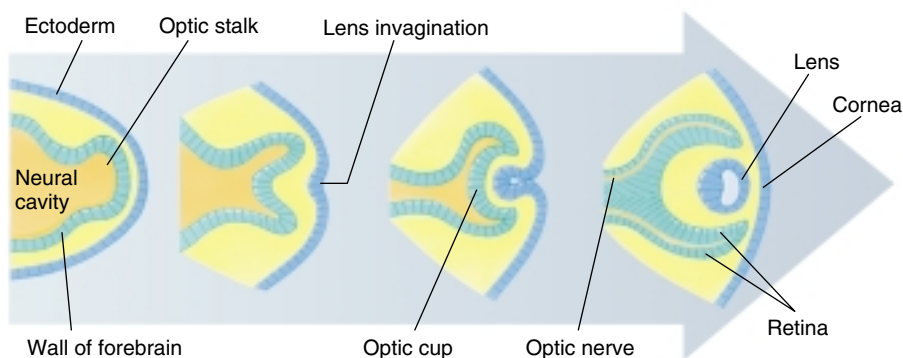


FIGURE 17.9

Development of the vertebrate eye proceeds by induction. The eye develops as an extension of the forebrain called the optic stalk that grows out until it contacts the ectoderm. This contact induces the formation of a lens from the ectoderm.

Induction

In *Drosophila* the initial cells created by cleavage divisions contain different developmental signals (called **determinants**) from the egg, setting individual cells off on different developmental paths. This pattern of development is called **mosaic development**. In mammals, by contrast, all of the blastomeres receive equivalent sets of determinants; body form is determined by cell-cell interactions, a pattern called **regulative development**.

We can demonstrate the importance of cell-cell interactions in development by separating the cells of an early blastula and allowing them to develop independently. Under these conditions, animal pole blastomeres develop features of ectoderm and vegetal pole blastomeres develop features of endoderm, but none of the cells ever develop features characteristic of mesoderm. However, if animal pole and vegetal pole cells are placed next to each other, some of the animal pole cells will develop as mesoderm. The interaction between the two cell types triggers a switch in the developmental path of the cells! When a cell switches from one path to another as a result of interaction with an adjacent cell, **induction** has taken place (figure 17.9).

How do cells induce developmental changes in neighboring cells? Apparently, the inducing cells secrete proteins that act as intercellular signals. Signal molecules, which we discussed in detail in chapter 7, are capable of producing abrupt changes in the patterns of gene transcription.

In some cases, particular groups of cells called **organizers** produce diffusible signal molecules that convey positional information to other cells. Organizers can have a profound influence on the development of surrounding tissues (see chapter 60). Working as signal beacons, they inform surrounding cells of their distance from the organizer. The closer a particular cell is to an organizer, the higher the concentration of the signal molecule, or **morphogen**, it experiences (figure 17.10). Although only a few morphogens have been isolated, they are thought to be part of a widespread mechanism for determining relative position during development.

A single morphogen can have different effects, depending upon how far away from the organizer the affected cell is located. Thus, low levels of the morphogen activin will cause cells of the animal pole of an early *Xenopus* embryo to develop into epidermis, while slightly higher levels will induce the cells to develop into muscles, and levels a little higher than that will induce them to form notochord (figure 17.11).

Cells migrate by extending probes to neighboring cells which they use to pull themselves along. Interactions between cells strongly influence the developmental paths they take. Signal molecules from an inducing cell alter patterns of transcription in cells which come in contact with it.

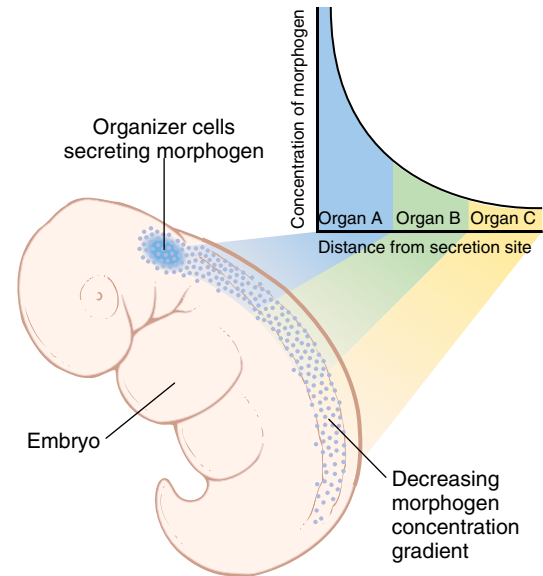


FIGURE 17.10

An organizer creates a morphogen gradient. As a morphogen diffuses from the organizer site, it becomes less concentrated. Different concentrations of the morphogen stimulate the development of different organs.

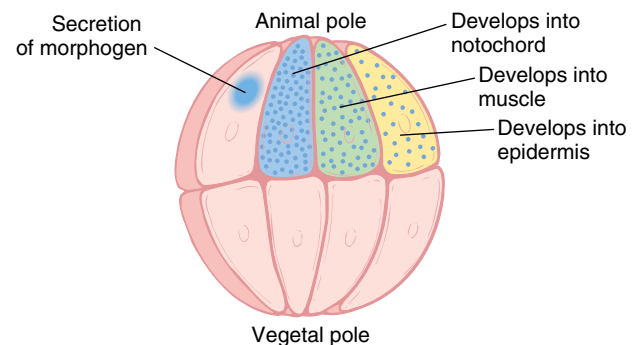


FIGURE 17.11

Fate of cells in an early *Xenopus* embryo. The fates of the individual cells are determined by the concentration of morphogen washing over them.

Determination

The mammalian egg is symmetrical in its contents as well as its shape, so that all of the cells of an early blastoderm are equivalent up to the eight-cell stage. The cells are said to be **totipotent**, meaning that they are potentially capable of expressing all of the genes of their genome. If they are separated from one another, any one of them can produce a completely normal individual. Indeed, just this sort of procedure has been used to produce sets of four or eight identical offspring in the commercial breeding of particularly valuable lines of cattle. The reverse process works, too; if cells from two different eight-cell-stage embryos are combined, a single normal individual results. Such an individual is called a **chimera**, because it contains cells from different genetic lines (figure 17.12).

Mammalian cells start to become different after the eight-cell stage as a result of cell-cell interactions like those we just discussed. At this point, the pathway that will influence the future developmental fate of the cells is determined. The commitment of a particular cell to a specialized developmental path is called **determination**. A cell in the prospective brain region of an amphibian embryo at the early gastrula stage has not yet been determined; if transplanted elsewhere in the embryo, it will develop like its new neighbors (see chapter 60). By the late gastrula stage, however, determination has taken place, and the cell will develop as neural tissue no matter where it is transplanted. Determination must be carefully distinguished from **differ-**

entiation, which is the cell specialization that occurs at the end of the developmental path. Cells may become *determined* to give rise to particular tissues long before they actually *differentiate* into those tissues. The cells of a *Drosophila* eye imaginal disc, for example, are fully determined to produce an eye, but they remain totally undifferentiated during most of the course of larval development.

The Mechanism of Determination

What is the molecular mechanism of determination? The gene regulatory proteins discussed in detail in chapter 16 are the tools used by cells to initiate developmental changes. When genes encoding these proteins are activated, one of their effects is to reinforce their own activation. This makes the developmental switch deterministic, initiating a chain of events that leads down a particular developmental pathway. Cells in which a set of regulatory genes have been activated may not actually undergo differentiation until some time later, when other factors interact with the regulatory protein and cause it to activate still other genes. Nevertheless, once the initial “switch” is thrown, the cell is fully committed to its future developmental path.

Often, before a cell becomes fully committed to a particular developmental path, it first becomes partially committed, acquiring **positional labels** that reflect its location in the embryo. These labels can have a great influence on how the pattern of the body subsequently develops. In a chicken

embryo, if tissue at the base of the leg bud (which would normally give rise to the thigh) is transplanted to the tip of the identical-looking wing bud (which would normally give rise to the wing tip), that tissue will develop into a toe rather than a thigh! The tissue has already been determined as leg but is not yet committed to being a particular part of the leg. Therefore, it can be influenced by the positional signaling at the tip of the wing bud to form a tip (in this case a tip of leg).

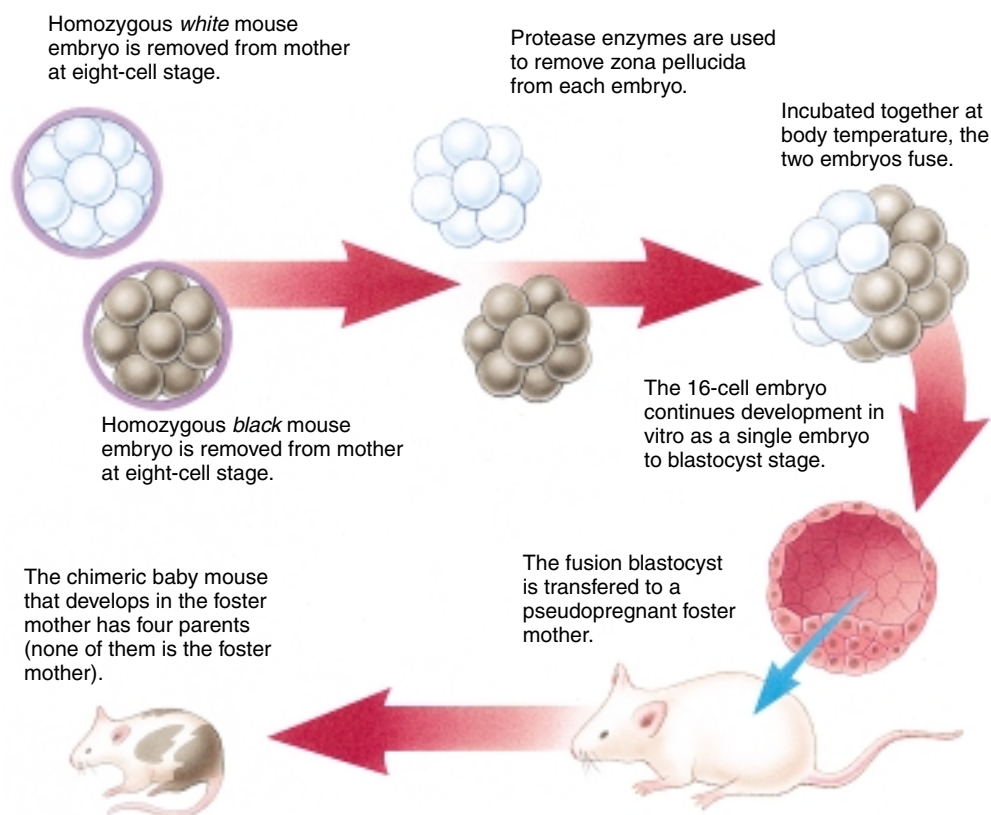


FIGURE 17.12
Constructing a chimeric mouse.
Cells from two eight-cell individuals fuse to form a single individual.

Is Determination Irreversible?

Until very recently, biologists thought determination was irreversible. Experiments carried out in the 1950s and 1960s by John Gurdon and others made what seemed a convincing case: using very fine pipettes (hollow glass tubes) to suck the nucleus out of a frog or toad egg, these researchers replaced the egg nucleus with a nucleus sucked out of a body cell taken from another individual (see figure 14.3). If the transplanted nucleus was obtained from an advanced embryo, the egg went on to develop into a tadpole, but died before becoming an adult.

Nuclear transplant experiments were attempted without success by many investigators, until finally, in 1984, Steen Willadsen, a Danish embryologist working in Texas, succeeded in cloning a sheep using the nucleus from a cell of an early embryo. The key to his success was in picking a cell very early in development. This exciting result was soon replicated by others in a host of other organisms, including pigs and monkeys.

Only early embryo cells seemed to work, however. Researchers became convinced, after many attempts to transfer older nuclei, that animal cells become irreversibly committed after the first few cell divisions of the developing embryo.

We now know this conclusion to have been unwarranted. The key advance unraveling this puzzle was made in Scotland by geneticists Keith Campbell and Ian Wilmut, who reasoned that perhaps the egg and the donated nucleus needed to be at the same stage in the cell cycle. They removed mammary cells from the udder of a six-year-old sheep. The origin of these cells gave the clone its name, “Dolly,” after the country singer Dolly Parton. The cells were grown in tissue culture; then, in preparation for cloning, the researchers substantially reduced for five days the concentration of serum nutrients on which the sheep mammary cells were subsisting. Starving the cells caused them to pause at the beginning of the cell cycle. In parallel preparation, eggs obtained from a ewe were enucleated (figure 17.13).

Mammary cells and egg cells were then surgically combined in January of 1996, inserting the mammary cells inside the covering around the egg cell. The researchers then applied a brief electrical shock. This caused the plasma membranes surrounding the two cells to become leaky, so

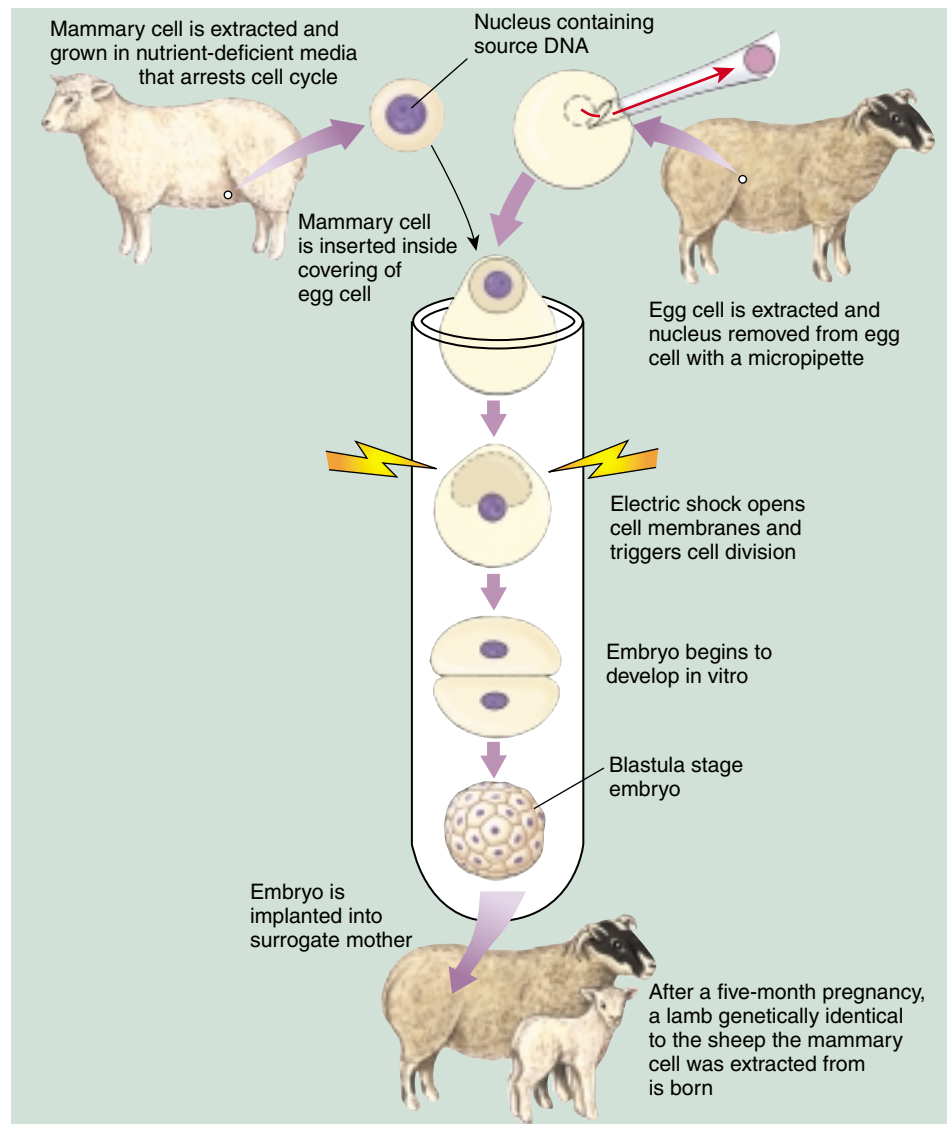


FIGURE 17.13

Proof that determination is reversible. This experiment by Campbell and Wilmut was the first successful cloning of an adult animal.

that the nucleus of the mammary cell passed into the egg cell—a neat trick. The shock also kick-started the cell cycle, causing the cell to begin to divide.

After six days, in 30 of 277 tries, the dividing embryo reached the hollow-ball “blastula” stage, and 29 of these were transplanted into surrogate mother sheep. A little over five months later, on July 5, 1996, one sheep gave birth to a lamb, Dolly, the first clone generated from a fully differentiated animal cell. Dolly established beyond all dispute that determination is reversible, that with the right techniques the fate of a fully differentiated cell *can* be altered.

The commitment of particular cells to certain developmental fates is fully reversible.

Pattern Formation

All animals seem to use positional information to determine the basic pattern of body compartments and, thus, the overall architecture of the adult body. How is positional information encoded in labels and read by cells? To answer this question, let us consider how positional labels are used in pattern formation in *Drosophila*. The Nobel Prize in Physiology or Medicine was awarded in 1995 for the unraveling of this puzzle.

As we noted previously, a *Drosophila* egg acquires an initial asymmetry long before fertilization as a result of mater-

nal mRNA molecules that are deposited in one end of the egg by nurse cells. Part of this maternal mRNA, from a gene called *bicoid*, remains near its point of entry, marking what will become the embryo's front end. Fertilization causes this mRNA to be translated into bicoid protein, which diffuses throughout the syncytial blastoderm, forming a morphogen gradient. Mothers unable to make bicoid protein produce embryos without a head or thorax (in effect, these embryos are two-tailed, or bicaudal—hence the name “bicoid”). Bicoid protein establishes the anterior (front) end of the embryo. If bicoid protein is injected into the anterior end of mutant embryos unable to make it, the

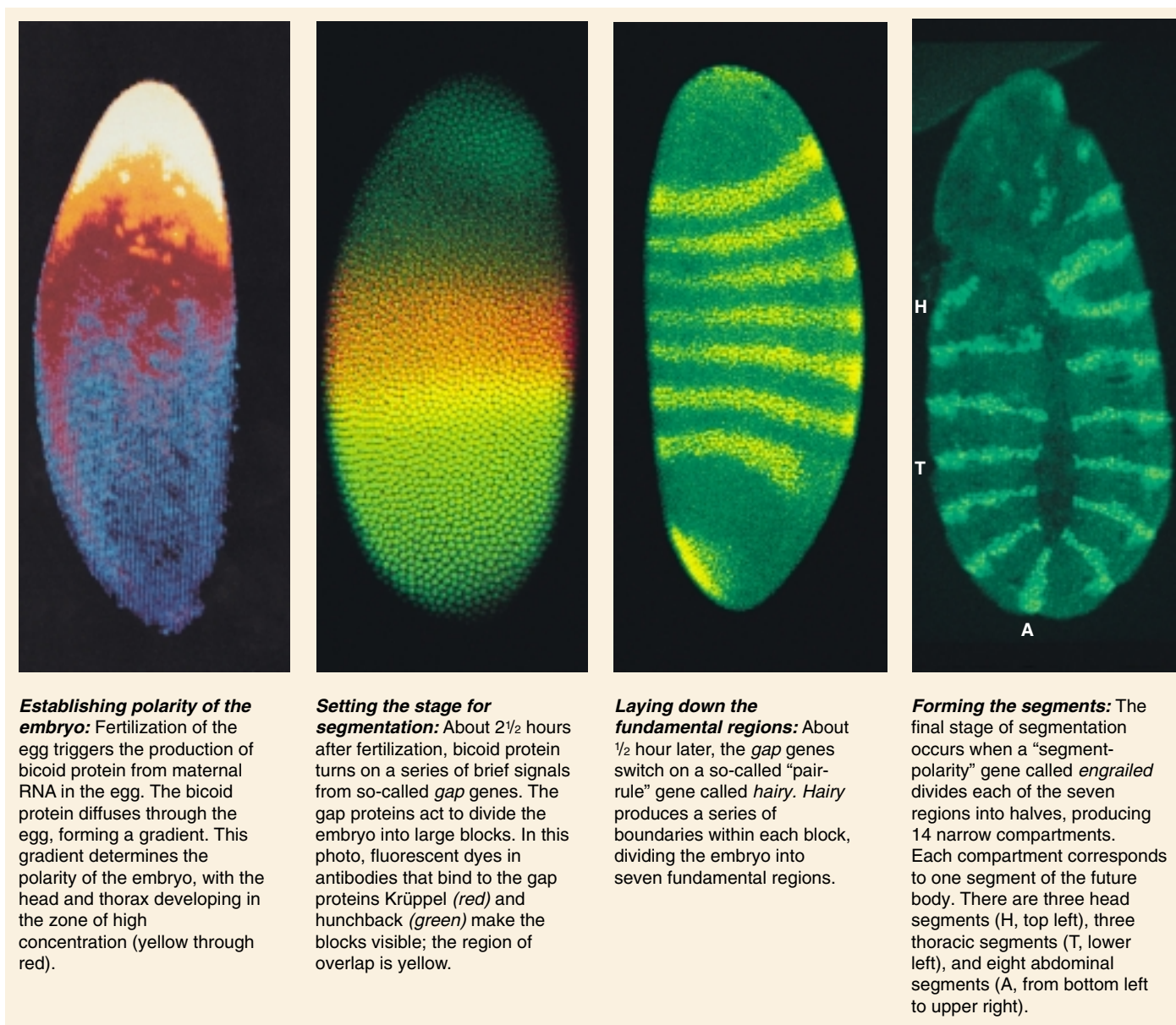


FIGURE 17.14

Body organization in an early *Drosophila* embryo. In these images by 1995 Nobel laureate, Christiane Nüsslein-Volhard, and Sean Carroll, we watch a *Drosophila* egg pass through the early stages of development, in which the basic segmentation pattern of the embryo is established.

embryos will develop normally. If it is injected into the opposite (posterior) end of normal embryos, a head and thorax will develop at that end.

Bicoid protein exerts this profound effect on the organization of the embryo by activating genes that encode the first mRNAs to be transcribed after fertilization. Within the first two hours, before cellularization of the syncytial blastoderm, a group of six genes called the **gap genes** begins to be transcribed. These genes map out the coarsest subdivision of the embryo (figure 17.14). One of them is a gene called *hunchback* (because an embryo without *hunchback* lacks a thorax and so, takes on a hunched shape). Although *hunchback* mRNA is distributed throughout the embryo, its translation is controlled by the protein product of another maternal mRNA called *nanos* (named after the Greek word for “dwarf,” as mutants without *nanos* genes lack abdominal segments and hence, are small). The **nanos** protein binds to *hunchback* mRNA, preventing it from being translated. The only place in the embryo where there is too little nanos protein to block translation of *hunchback* mRNA is the far anterior end. Consequently, hunchback protein is made primarily at the anterior end of the embryo. As it diffuses back toward the posterior end, it sets up a second morphogen gradient responsible for establishing the thoracic and abdominal segments.

Other gap genes act in more posterior regions of the embryo. They, in turn, activate 11 or more **pair-rule genes**. (When mutated, each of these genes alters every other body segment.) One of the pair-rule genes, named *hairy*, produces seven bands of protein, which look like stripes when visualized with fluorescent markers. These bands establish boundaries that divide the embryo into seven zones. Finally, a group of 16 or more **segment polarity genes** subdivide these zones. The *engrailed* gene, for example, divides each of the seven zones established by *hairy* into anterior and posterior compartments. The 14 compartments that result correspond to the three head segments, three thoracic segments, and eight abdominal segments of the embryo.

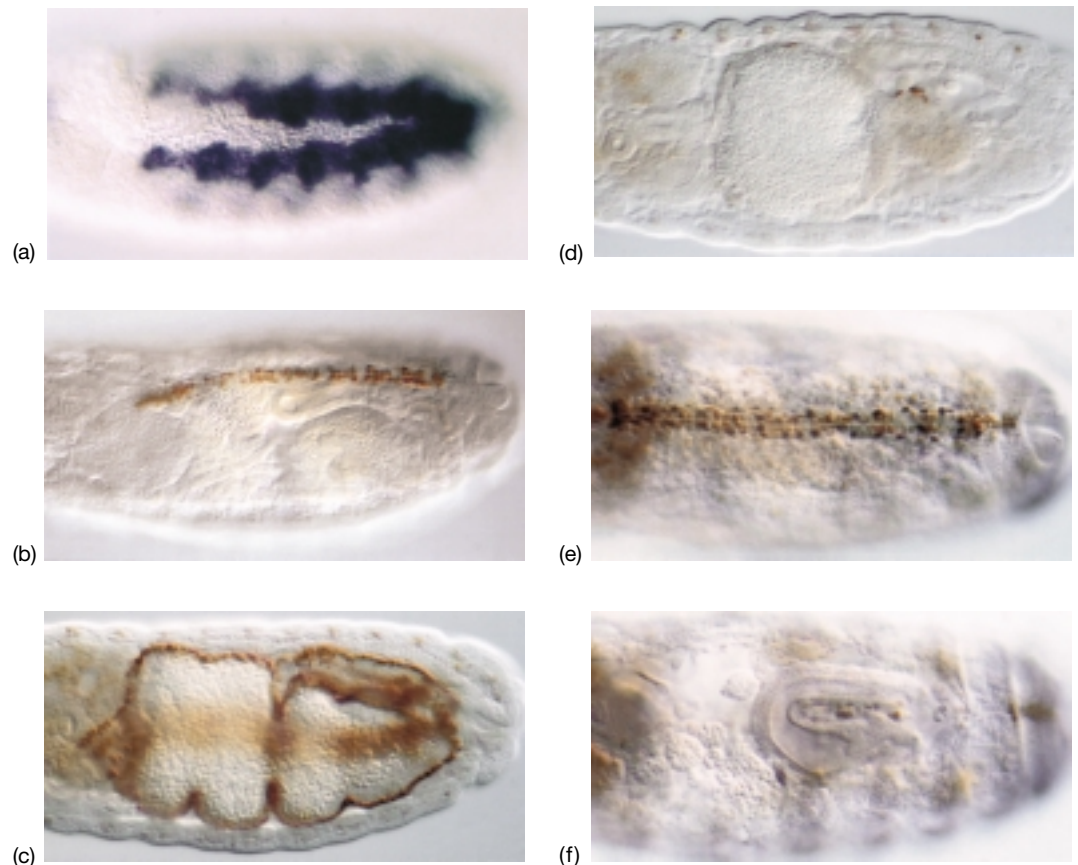
Thus, within three hours after fertilization, a highly orchestrated cascade of segmentation gene activity produces the fly embryo’s basic body plan. The activation of these and other developmentally important genes (figure 17.15) depends upon the free diffusion of morphogens that is possible within a syncytial blastoderm. In mammalian embryos with cell partitions, other mechanisms must operate.

In *Drosophila* diffusion of chemical inducers produces the embryo’s basic body plan, a cascade of genes dividing it into 14 compartments.

FIGURE 17.15

A gene controlling organ formation in *Drosophila*.

Called *tinman*, this gene is responsible for the formation of gut musculature and the heart. The dye shows expression of the *tinman* in five-hour (a) and seventeen-hour (b) *Drosophila* embryos. The gut musculature then appears along the edges of normal embryos (c) but is not present in embryos in which the gene has been mutated (d). The heart tissue develops along the center of normal embryos (e) but is missing in *tinman* mutant embryos (f).



Expression of Homeotic Genes

After pattern formation has successfully established the number of body segments in *Drosophila*, a series of **homeotic genes** act as master switches to determine the forms these segments will assume. Homeotic genes code for proteins that function as transcription factors. Each homeotic gene activates a particular module of the genetic program, initiating the production of specific body parts within each of the 14 compartments.

Homeotic Mutations

Mutations in homeotic genes lead to the appearance of perfectly normal body parts in unusual places. Mutations in *bithorax* (figure 17.16), for example, cause a fly to grow an extra pair of wings, as if it had a double thoracic segment, and mutations in *Antennapedia* cause legs to grow out of the head in place of antennae! In the early 1950s, geneticist Edward Lewis discovered that several homeotic genes, including *bithorax*, map together on the third chromosome of *Drosophila*, in a tight cluster called the **bithorax complex**.

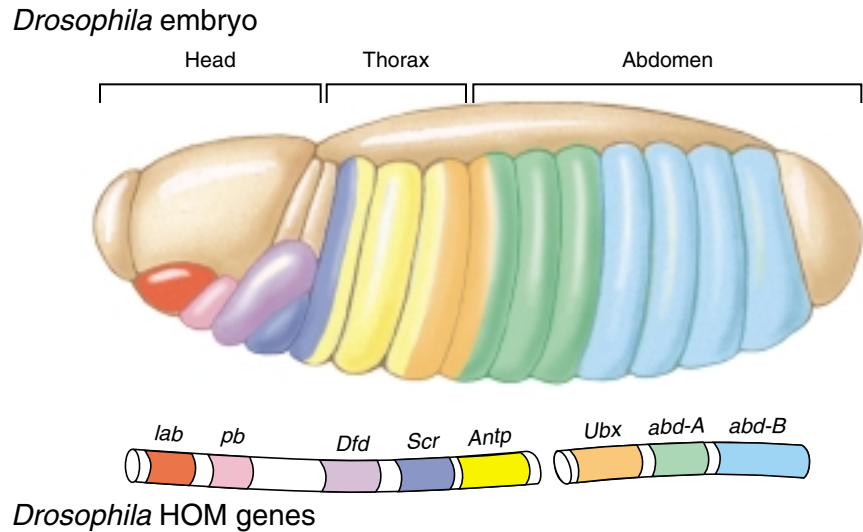


FIGURE 17.16

Mutations in homeotic genes. Three separate mutations in the *bithorax* gene caused this fruit fly to develop an extra thoracic segment, with accompanying wings. Compare this photograph with that of the normal fruit fly in figure 17.6.

FIGURE 17.17

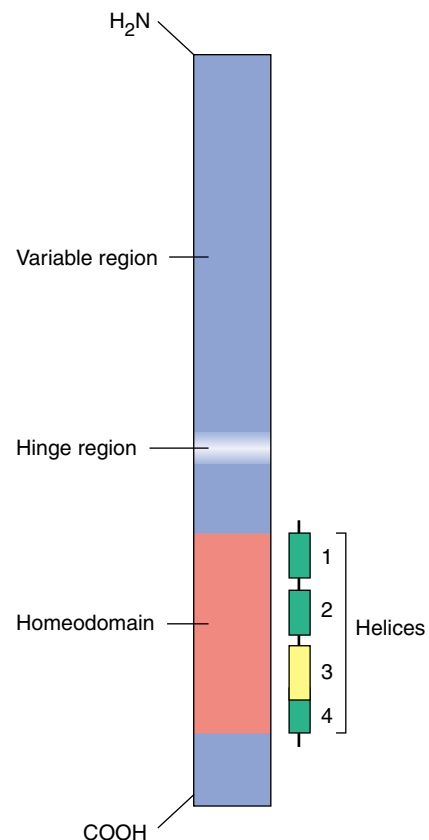
***Drosophila* homeotic genes.** Called the homeotic gene complex, or HOM complex, the genes are grouped into two clusters, the Antennapedia complex (anterior) and the bithorax complex (posterior).



Mutations in these genes all affect body parts of the thoracic and abdominal segments, and Lewis concluded that the genes of the bithorax complex control the development of body parts in the rear half of the thorax and all of the abdomen. Most interestingly, the order of the genes in the bithorax complex mirrors the order of the body parts they control, as if the genes are activated serially! Genes at the beginning of the cluster switch on development of the thorax, those in the middle control the anterior part of the abdomen, and those at the end affect the tip of the abdomen. A second cluster of homeotic genes, the **Antennapedia complex**, was discovered in 1980 by Thomas Kaufmann. The Antennapedia complex governs the anterior end of the fly, and the order of genes in it also corresponds to the order of segments they control (figure 17.17).

The Homeobox

Drosophila homeotic genes typically contain the **homeobox**, a sequence of 180 nucleotides that codes for a 60-amino acid DNA-binding peptide domain called the homeodomain (figure 17.18). As we saw in chapter 16, proteins that contain the homeodomain function as transcription factors, ensuring that developmentally related genes are transcribed at the appropriate time. Segmentation genes such as *bicoid* and *engrailed* also contain the homeobox sequence. Clearly, the homeobox distinguishes those portions of the genome devoted to pattern formation.

**FIGURE 17.18**

Homeodomain protein. This protein plays an important regulatory role when it binds to DNA and regulates expression of specific genes. The variable region of the protein determines the specific activity of the protein. Also included in this protein is a small hinge region and the homeodomain, a 60-amino-acid sequence common to all proteins of this type. The homeodomain region of the protein is coded for by the homeobox region of genes and is composed of four α helices. One of the helices recognizes and binds to a specific DNA sequence in target genes.

Evolution of Homeobox Genes

Since their initial discovery in *Drosophila*, homeotic genes have also been found in mice and humans, which are separated from insects by over 600 million years of evolution. Their presence in mammals and insects indicates that homeotic genes governing the positioning of body parts must have arisen very early in the evolutionary history of animals. Similar genes also appear to operate in flowering plants. Gene probes made using the homeobox sequence of *Drosophila* have been used to identify very similar sequences in a wide variety of other organisms, including frogs, mice, humans, cows, chickens, beetles, and even earthworms. Mice and humans have four clusters of homeobox-containing genes, called *Hox* genes in mice. Just as in flies, the homeotic genes of mammals appear to be lined up in the same order as the segments they control (figure 17.19). Thus, the ordered nature of homeotic gene clusters is highly conserved in evolution (figure 17.20). There is a total of 38 *Hox* genes in the four homeotic clusters of a mouse, and we are only beginning to understand how they interact.

Homeotic genes encode transcription factors that activate blocks of genes specifying particular body parts.

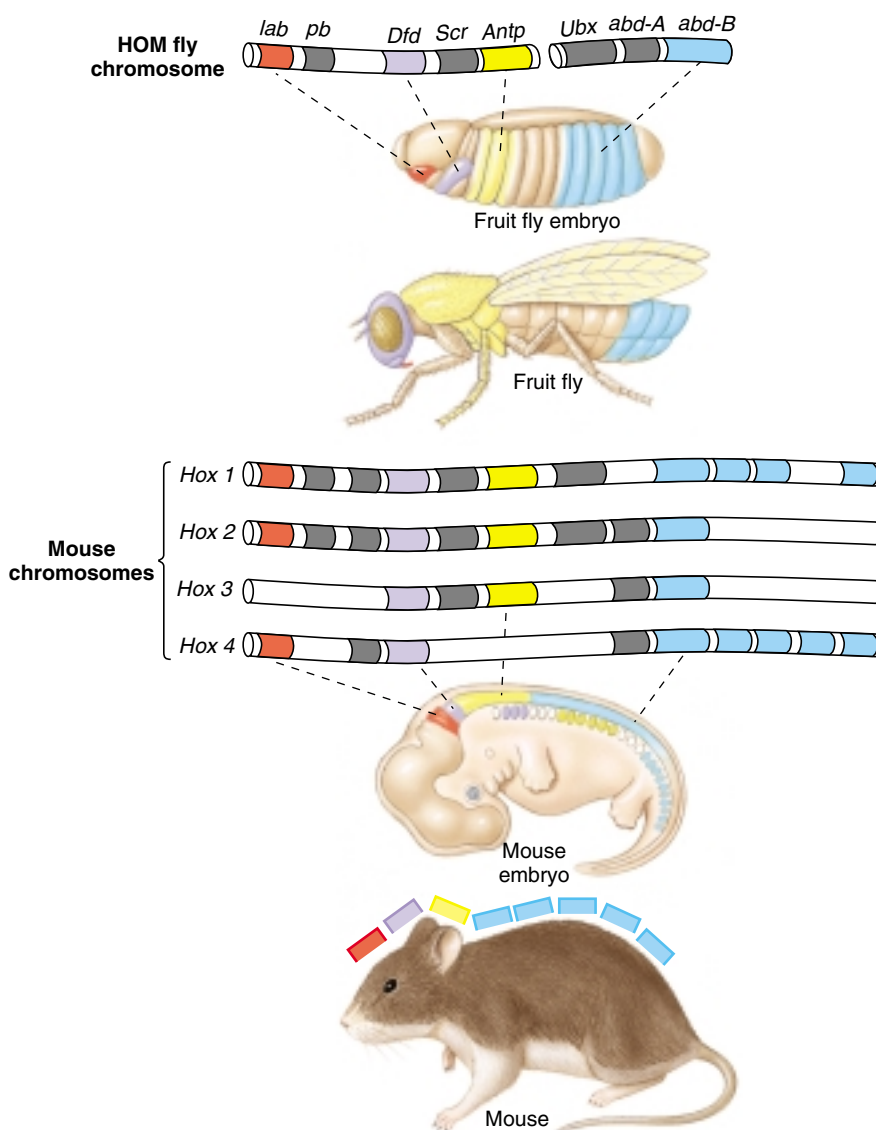
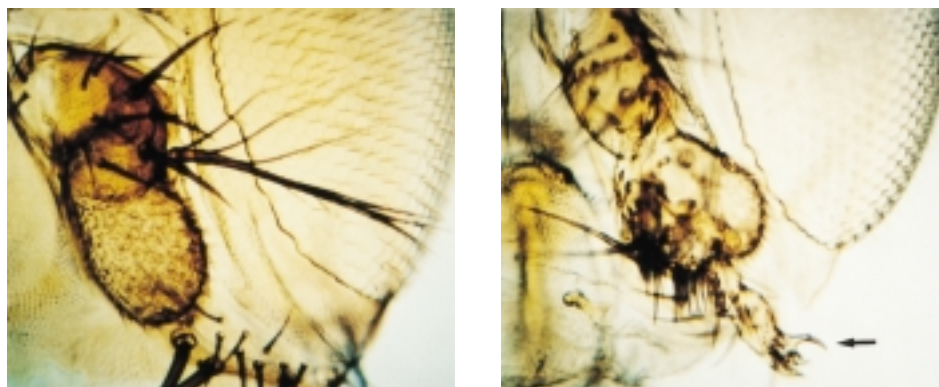


FIGURE 17.19

A comparison of homeotic gene clusters in the fruit fly *Drosophila melanogaster* and the mouse *Mus musculus*. Similar genes, the *Drosophila* HOM genes and the mouse *Hox* genes, control the development of front and back parts of the body. These genes are located on a single chromosome in the fly, and on four separate chromosomes in mammals. The genes are color-coded to match the parts of the body in which they are expressed.

FIGURE 17.20

The remarkably conserved homeobox series. By inserting a mouse homeobox-containing gene into a fruit fly, a mutant fly (*right*) can be manufactured with a leg (*arrow*) growing from where its antenna would be in a normal fly (*left*).



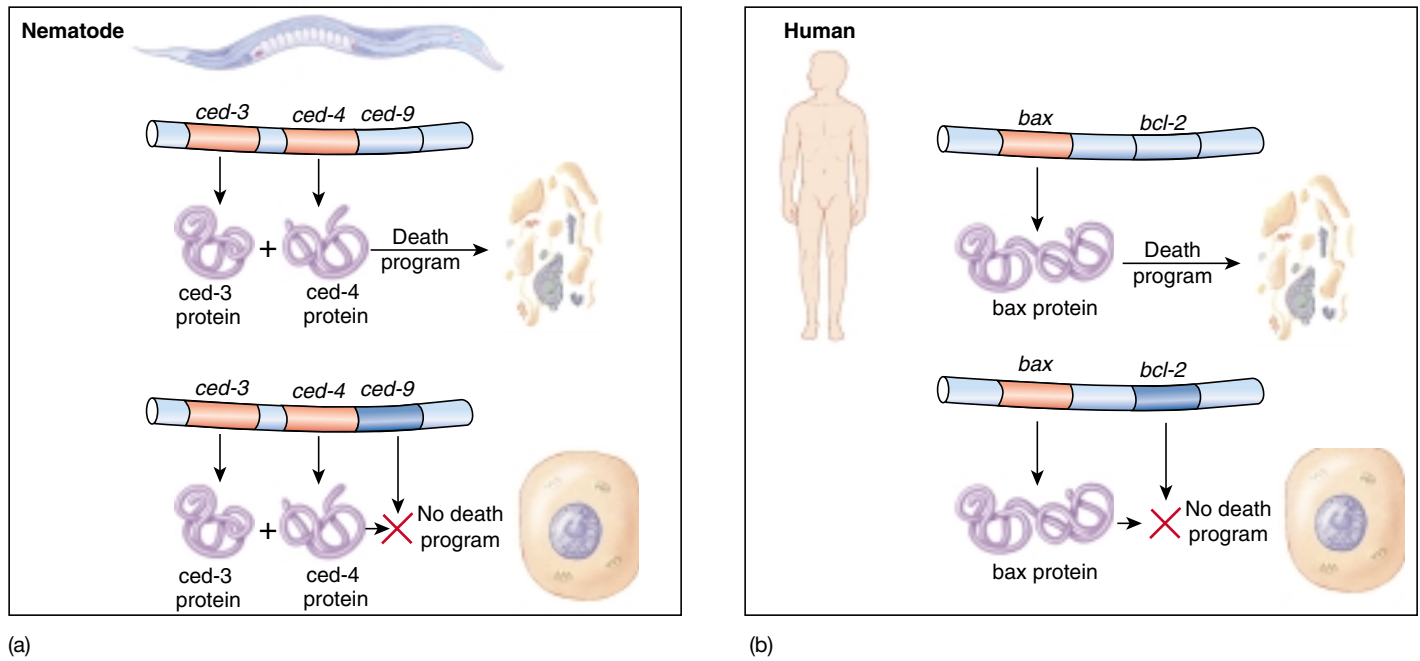


FIGURE 17.21

Programmed cell death. Apoptosis, or programmed cell death, is necessary for normal development of all animals. (a) In the developing nematode, for example, two genes, *ced-3* and *ced-4*, code for proteins that cause the programmed cell death of 131 specific cells. In the other cells of the developing nematode, the product of a third gene, *ced-9*, represses the death program encoded by *ced-3* and *ced-4*. (b) In developing humans, the product of a gene called *bax* causes a cell death program in some cells and is blocked by the *bcl-2* gene in other cells.

Programmed Cell Death

Not every cell that is produced during development is destined to survive. For example, the cells between your fingers and toes die; if they did not, you would have paddles rather than digits. Vertebrate embryos produce a very large number of neurons, ensuring that there are enough neurons to make all of the necessary synaptic connections, but over half of these neurons never make connections and die in an orderly way as the nervous system develops. Unlike accidental cell deaths due to injury, these cell deaths are planned for and indeed required for proper development. Cells that die due to injury typically swell and burst, releasing their contents into the extracellular fluid. This form of cell death is called **necrosis**. In contrast, cells programmed to die shrivel and shrink in a process called **apoptosis** (from the Greek word meaning shedding of leaves in autumn), and their remains are taken up by surrounding cells.

Gene Control of Apoptosis

This sort of developmentally regulated cell suicide occurs when a “death program” is activated. All animal cells appear to possess such programs. In the nematode worm, for example, the same 131 cells always die during development in a predictable and reproducible pattern of apoptosis. Three genes govern this process. Two (*ced-3* and *ced-4*)

constitute the death program itself; if either is mutant, those 131 cells do not die, and go on instead to form nervous and other tissue. The third gene (*ced-9*) represses the death program encoded by the other two (figure 17.21a). The same sorts of apoptosis programs occur in human cells: the *bax* gene encodes the cell death program, and another, an oncogene called *bcl-2*, represses it (figure 17.21b). The mechanism of apoptosis appears to have been highly conserved during the course of animal evolution. The protein made by *bcl-2* is 25% identical in amino acid sequence to that made by *ced-9*. If a copy of the human *bcl-2* gene is transferred into a nematode with a defective *ced-9* gene, *bcl-2* suppresses the cell death program of *ced-3* and *ced-4*!

How does *bax* kill a cell? The bax protein seems to induce apoptosis by binding to the permeability pore of the cell’s mitochondria, increasing its permeability and in doing so triggering cell death. How does *bcl-2* prevent cell death? One suggestion is that it prevents damage from free radicals, highly reactive fragments of atoms that can damage cells severely. Proteins or other molecules that destroy free radicals are called **antioxidants**. Antioxidants are almost as effective as *bcl-2* in blocking apoptosis.

Animal development involves programmed cell death (apoptosis), in which particular genes, when activated, kill their cells.

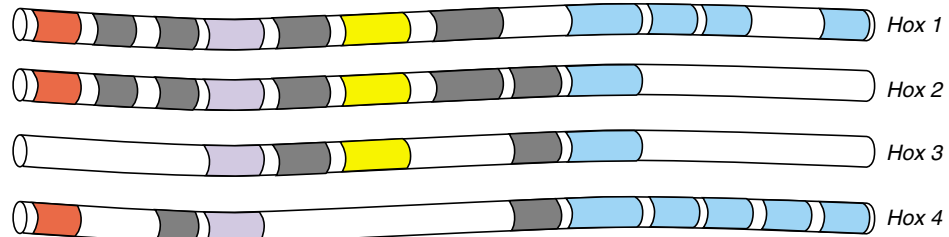
17.3 Four model developmental systems have been extensively researched.

The Mouse

Some of the most elegant investigations of the cellular mechanisms of development are being done with mammals, particularly the mouse *Mus musculus*. Mice have a battery of homeotic genes, the *Hox* genes (figure 17.22), which seem to be closely related to the homeotic genes of *Drosophila*. Very interestingly, not only do the same genes occur, but they also seem to operate in the same order! Clearly, the homeotic gene system has been highly conserved during the course of animal evolution.

What lends great power to this developmental model system is the ability to create chimeric mice containing cells from two different genetic lines. Mammalian embryos are unusual among vertebrates in that they arise from symmetrical eggs; there are no chemical gradients, and during the initial cleavage divisions, all of the daughter cells are identical. Up to the eight-cell stage, any one of the cells, if isolated, will form a normal adult. Moreover, two different eight-cell-stage embryos can be fused to form a single embryo that will go on to form a normal adult. The resulting adult is a chimera, containing cells from both embryos. In a very real sense, these chimeric mice each have four parents!

The *Hox* genes control body part development in mice.



Mouse chromosomes

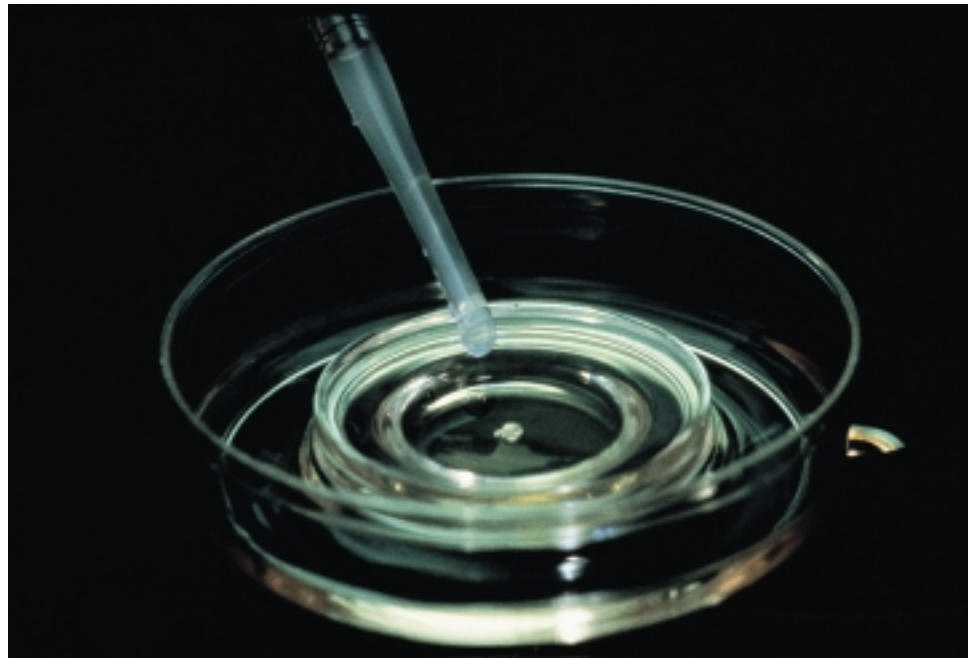
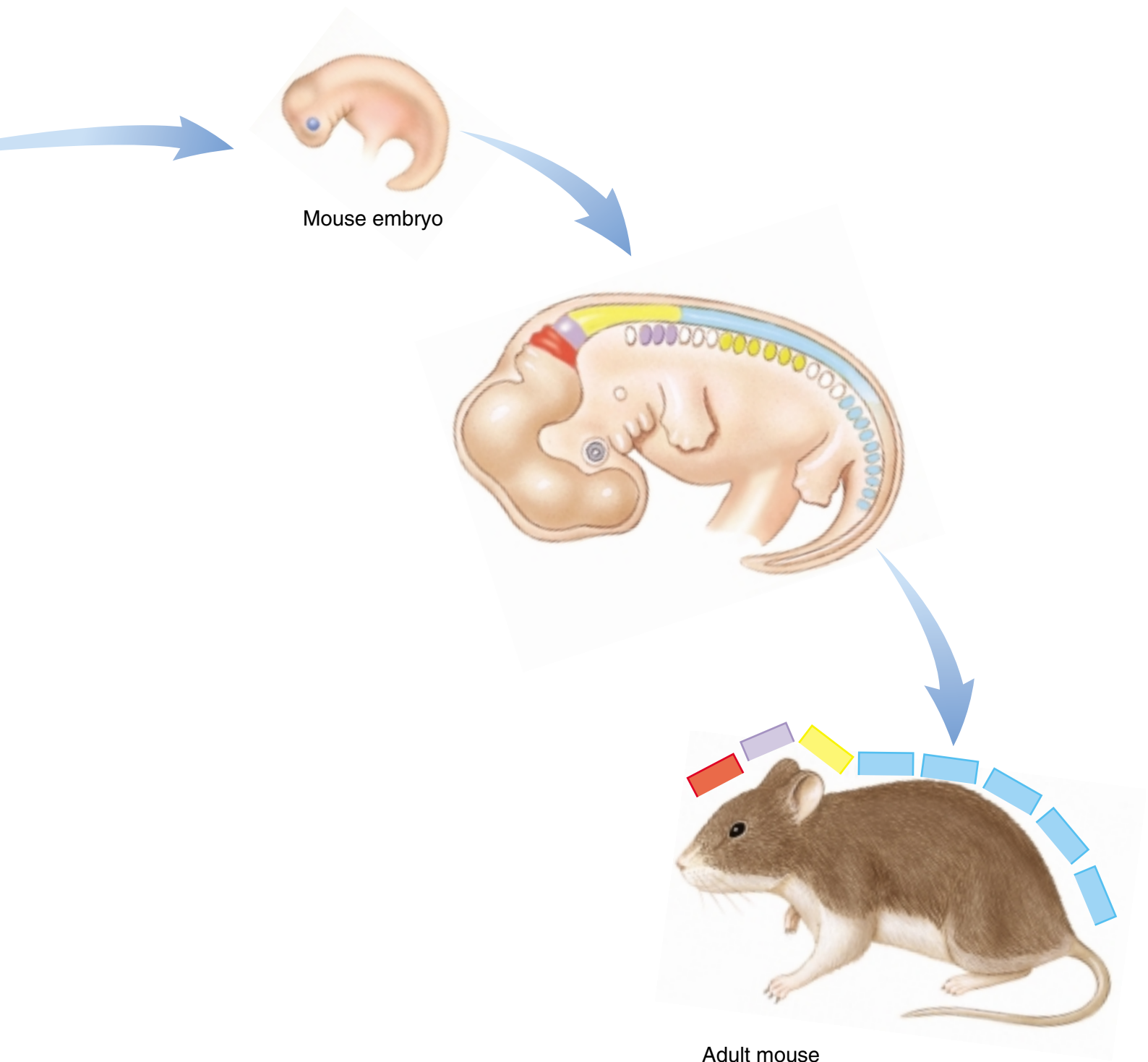


FIGURE 17.22
Studying development in the mouse.



The Fruit Fly

The tiny fruit fly *Drosophila melanogaster* has been a favorite of geneticists for over 90 years and is now playing a key role in our growing understanding of the cellular mechanisms of development. Over the last 10 years, researchers have pieced together a fairly complete picture of how genes expressed early in fruit fly development determine the pattern of the adult body (figure 17.23). The major parts of the adult body are determined as patches of tissue called imaginal discs that float within the body of the larva; during the pupal stage, these discs grow, develop, and associate to form the adult body.

The adult *Drosophila* body is divided into 17 segments, some bearing jointed appendages such as wings or legs. These segments are established during very early development, before the many nuclei of the blastoderm are fully separated from one another. Chemical gradients, established within the egg by material from the mother, create a polarity that directs embryonic development. Reacting to this gradient, a series of segmentation genes progressively subdivide the embryo, first into four broad stripes, and then into 7, 14, and finally 17 segments.

Within each segment, the development of key body parts is under the control of homeotic genes that determine where the body part will form. As we have seen, there are two clusters of homeotic genes, one called Antennapedia that governs the front (anterior) end of the body, and another called bithorax that governs the rear (posterior) end. The organization of genes within each cluster corresponds nicely with the order of the segments they affect. A very similar set of homeotic genes governs body architecture in mice and humans.

A series of segmentation genes divides a *Drosophila* embryo into parts; Antennapedia genes control anterior development, and bithorax genes control the development of the posterior.

Drosophila egg

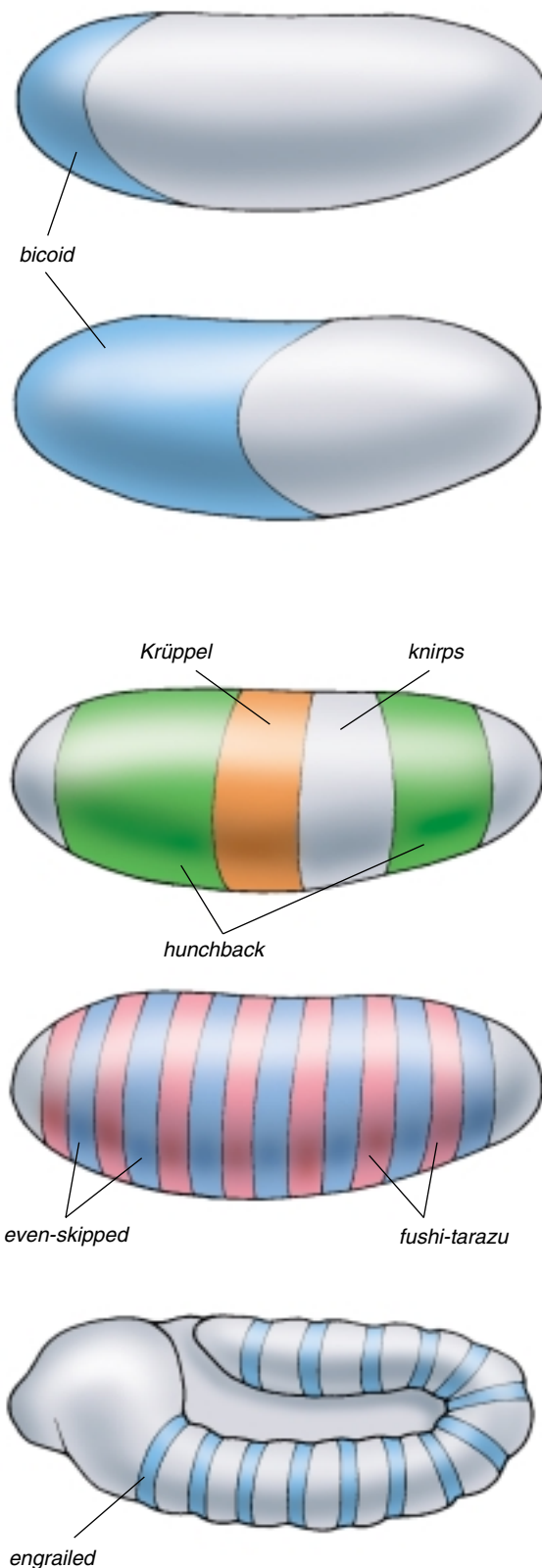
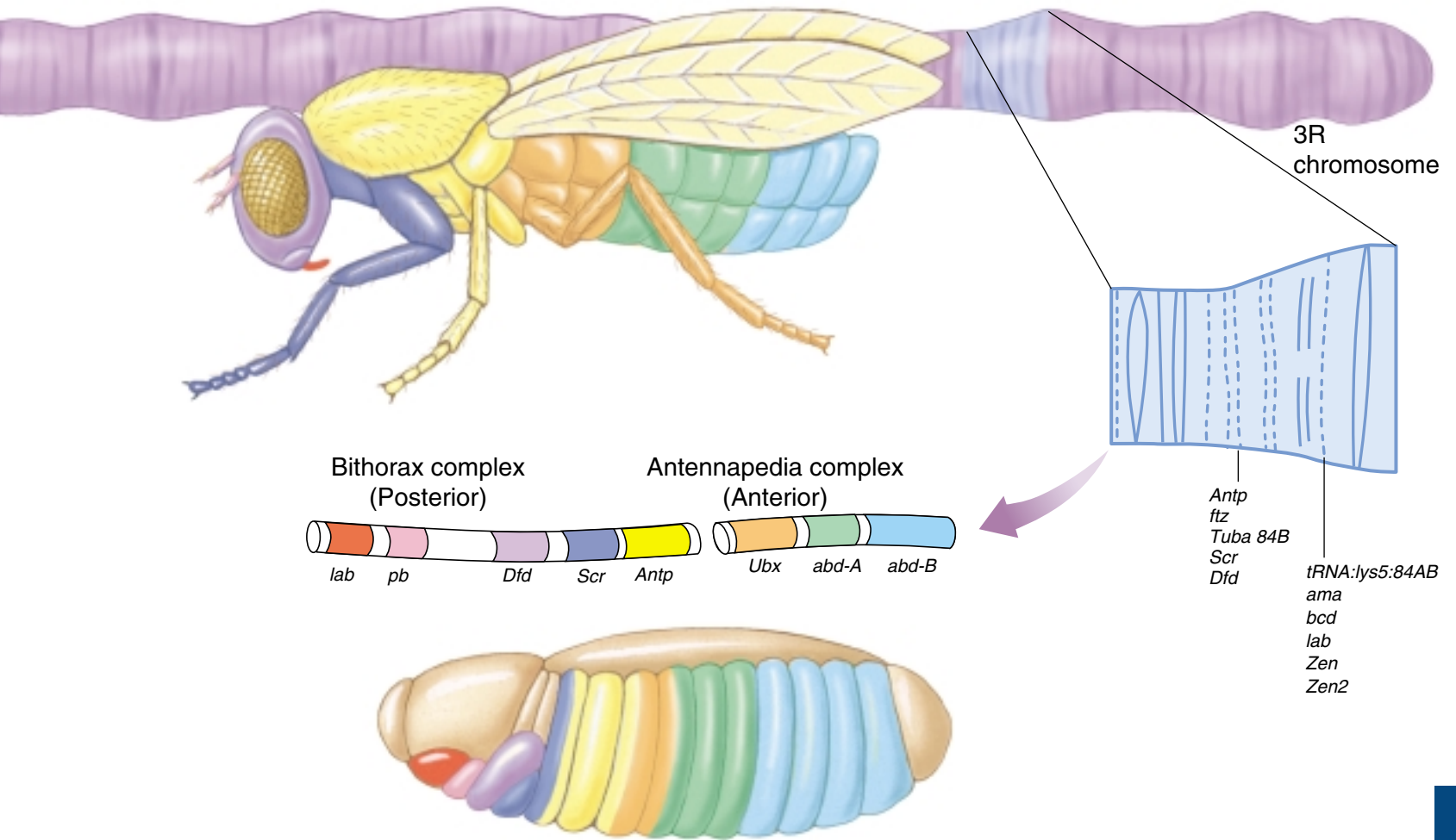
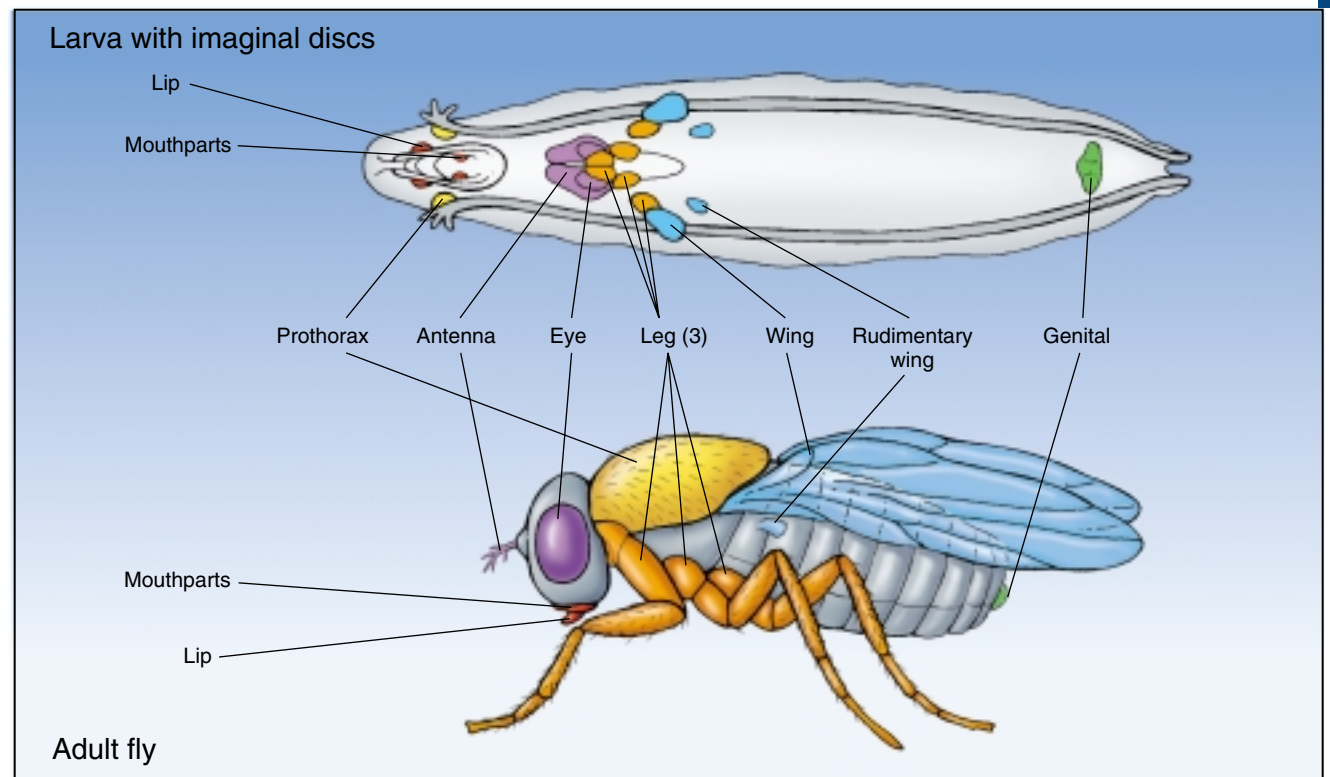
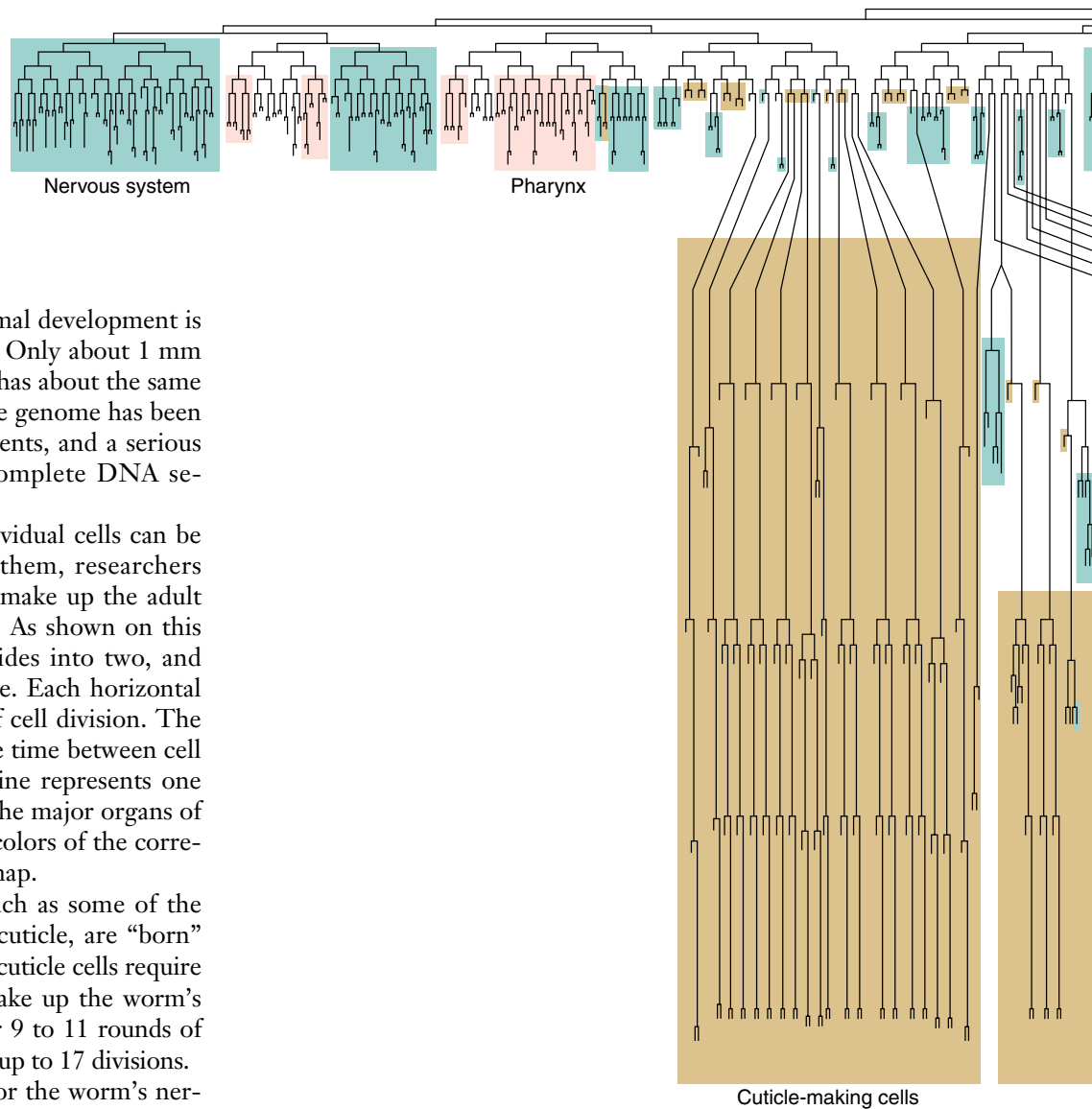


FIGURE 17.23
Studying development in the fruit fly.



Drosophila embryo





The Nematode

One of the most powerful models of animal development is the tiny nematode *Caenorhabditis elegans*. Only about 1 mm long, it consists of 959 somatic cells and has about the same amount of DNA as *Drosophila*. The entire genome has been mapped as a series of overlapping fragments, and a serious effort is underway to determine the complete DNA sequence of the genome.

Because *C. elegans* is transparent, individual cells can be followed as they divide. By observing them, researchers have learned how each of the cells that make up the adult worm is derived from the fertilized egg. As shown on this lineage map (figure 17.24), the egg divides into two, and then its daughter cells continue to divide. Each horizontal line on the map represents one round of cell division. The length of each vertical line represents the time between cell divisions, and the end of each vertical line represents one fully differentiated cell. In figure 17.24, the major organs of the worm are color-coded to match the colors of the corresponding groups of cells on the lineage map.

Some of these differentiated cells, such as some of the cells that generate the worm's external cuticle, are "born" after only 8 rounds of cell division; other cuticle cells require as many as 14 rounds. The cells that make up the worm's pharynx, or feeding organ, are born after 9 to 11 rounds of division, while cells in the gonads require up to 17 divisions.

Exactly 302 nerve cells are destined for the worm's nervous system. Exactly 131 cells are programmed to die, mostly within minutes of their birth. The fate of each cell is the same in every *C. elegans* individual, except for the cells that will become eggs and sperm.

The nematode develops 959 somatic cells from a single fertilized egg in a carefully orchestrated series of cell divisions which have been carefully mapped by researchers.

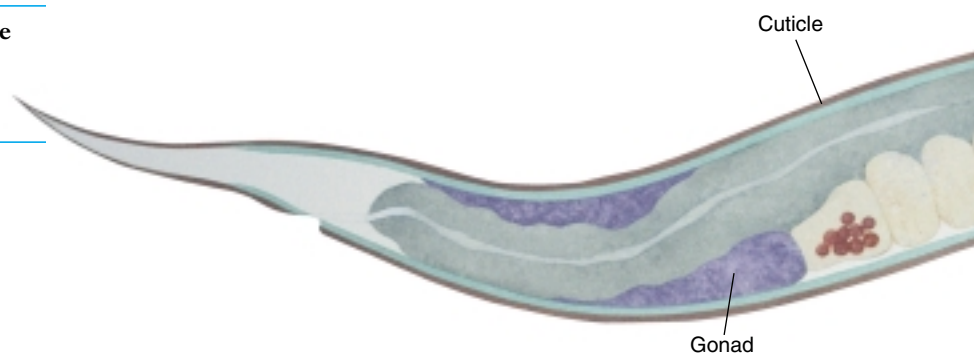
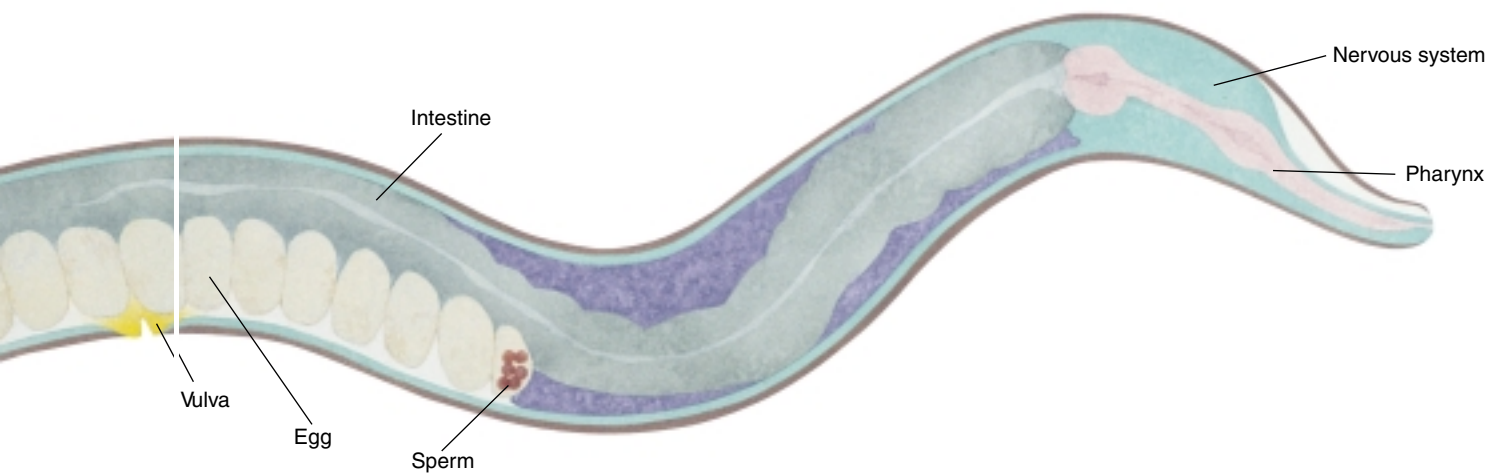
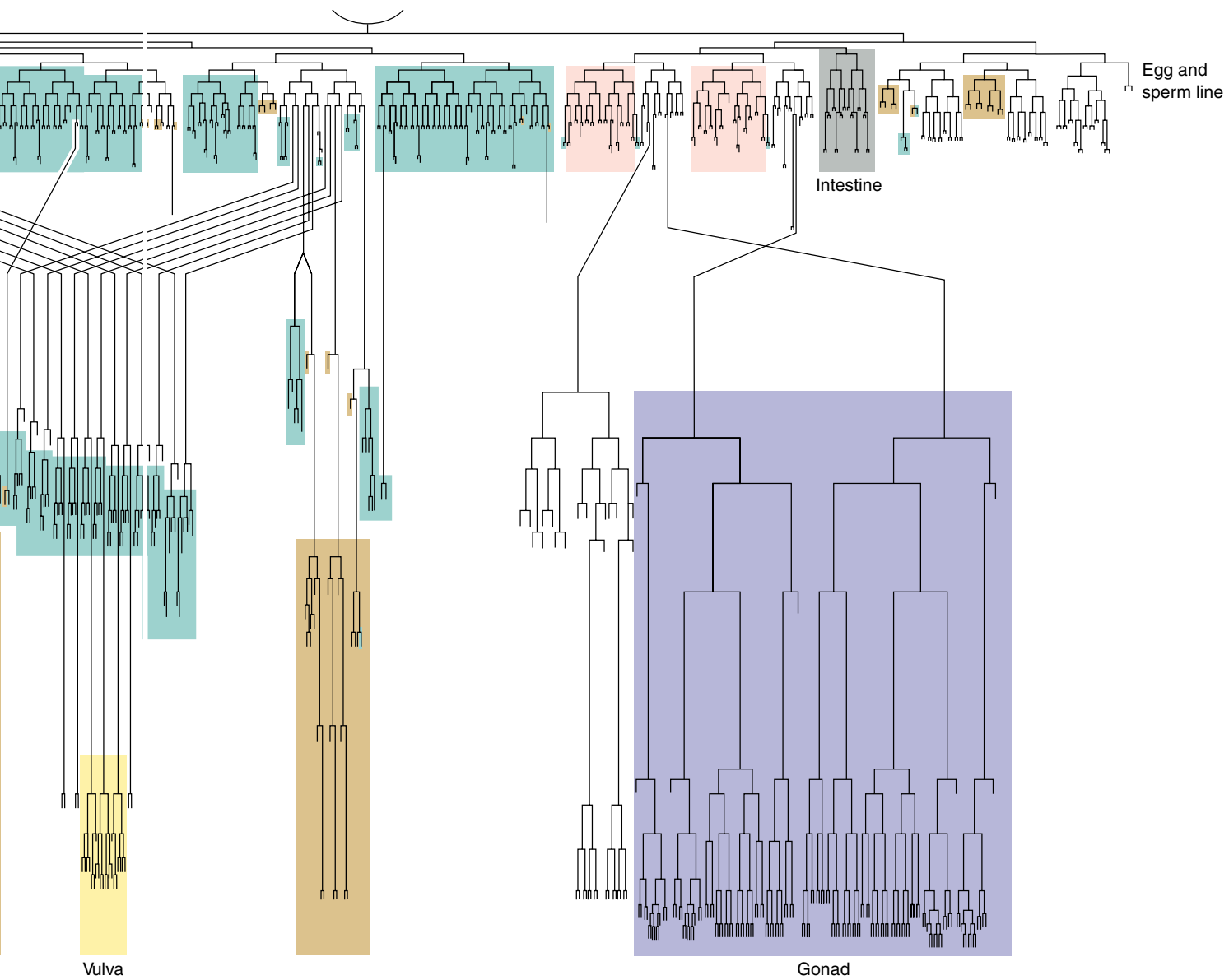


FIGURE 17.24
Studying development in the nematode.



The Flowering Plant

Scientists are only beginning to unravel the molecular biology of plant development, largely through intensive recent study of a small weedy relative of the mustard plant, the wall cress *Arabidopsis thaliana*. Easy to grow and cross, and with a short generation time, *Arabidopsis* makes an ideal model for investigating plant development. It is able to self-fertilize, like Mendel's pea plants, making genetic analysis convenient. *Arabidopsis* can be grown indoors in test tubes, a single plant producing thousands of offspring after only two months. Its genome is approximately the same size as those of the nematode *Caenorhabditis elegans* and the fruit fly *Drosophila melanogaster*. An ordered library of *Arabidopsis* gene clones was made available to researchers in 1997, and the full genome sequence was completed in 1999.

Pattern Formation

Much of the current work investigating *Arabidopsis* development has centered on obtaining and studying mutations that alter the plant's development. Many different sorts of mutations have been identified. Some of the most interesting of them alter the basic architecture of the embryo, the pattern of tissues laid down as the embryo first forms. Mutations in over 50 different genes that alter pattern formation in *Arabidopsis* embryos are now known, affecting every stage of development. While work in this area is still very preliminary, it appears that the mechanisms that establish patterns in the early *Arabidopsis* embryo are broadly similar to those known to function in animal development.

Organ Formation

Importantly, the subsequent development of organs in *Arabidopsis* also seems to parallel organ development in animals, and a similar set of regulatory genes control development in *Arabidopsis*, *Drosophila*, and mice. *Arabidopsis* flowers, for example, are modified leaves formed as four whorls in a specific order, and homeotic mutations have been identified that convert one part of the pattern to another, just as they do in the body segments of a fly (figure 17.25).

Scientists are only beginning to understand the molecular biology of plant development. In broad outline, it appears quite similar to the development in animals. The genes that determine pattern formation and organ development, for example, operate in the same way in plants and animals.

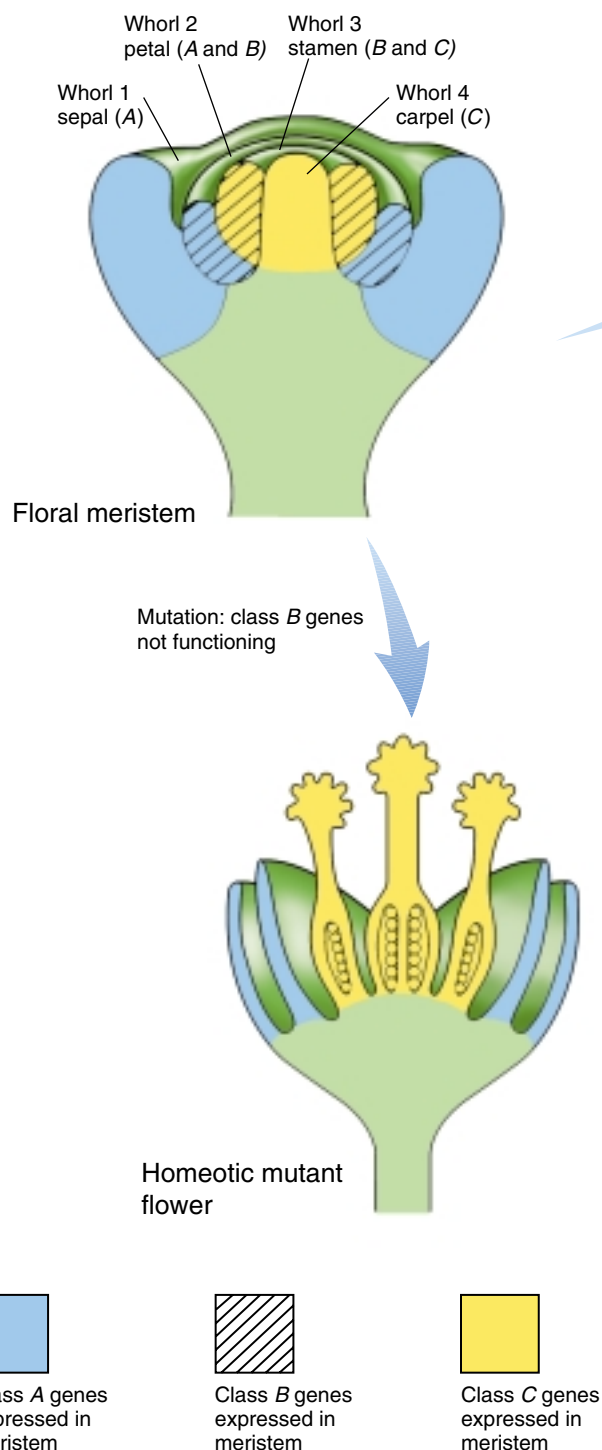
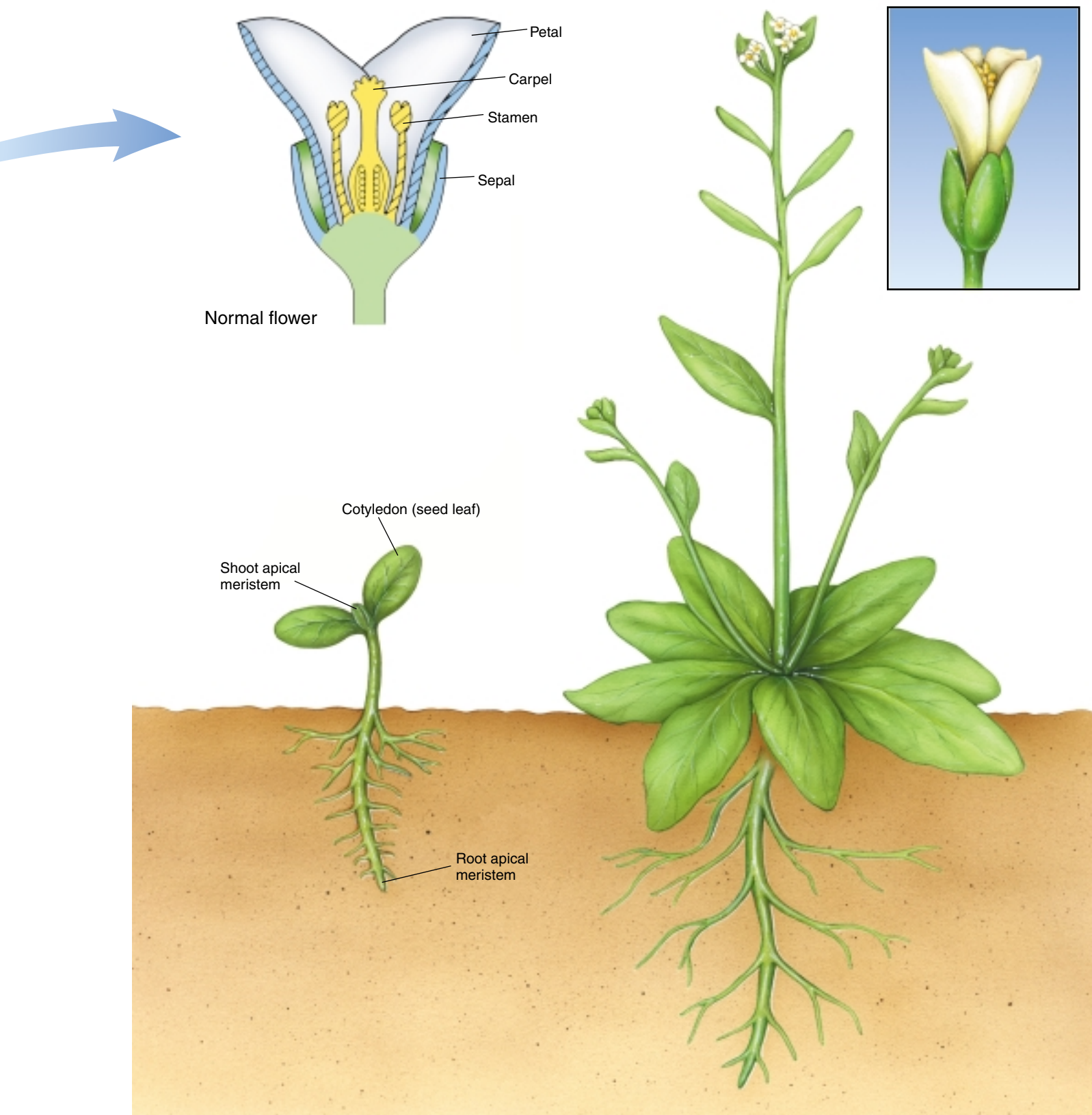


FIGURE 17.25
Studying development in a flowering plant.



17.4 Aging can be considered a developmental process.

Theories of Aging

All humans die. The oldest documented person, Jeanne Louise Calment of Arles, France, reached the age of 122 years before her death in 1997. The “safest” age is around puberty. As you can see in figure 17.26, 10- to 15-year-olds have the lowest risk of dying. The death rate begins to increase rapidly after puberty; the rate of mortality then begins to increase as an exponential function of increasing age. Plotted on a log scale as in figure 17.26 (in a so-called Gompertz plot), the mortality rate increases as a straight line from about 15 to 90 years, doubling about every eight years (the “Gompertz number”). By the time we reach 100, age has taken such a toll that the risk of dying reaches 50% per year.

A wide variety of theories have been advanced to explain why humans and other animals age. No one theory has gained general acceptance, but the following four are being intensively investigated:

Accumulated Mutation Hypothesis

The oldest general theory of aging is that cells accumulate mutations as they age, leading eventually to lethal damage. Careful studies have shown that somatic mutations do indeed accumulate during aging. As cells age, for example, they tend to accumulate the modified base 8-hydroxyguanine, in which an —OH group is added to the base guanine. There is little direct evidence, however, that these mutations *cause* aging. No acceleration in aging occurred among survivors of Hiroshima and Nagasaki despite their enormous added mutation load, arguing against any general relationship between mutation and aging.

Telomere Depletion Hypothesis

In a seminal experiment carried out in 1961, Leonard Hayflick demonstrated that fibroblast cells growing in tissue culture will divide only a certain number of times (figure 17.27). After about 50 population doublings, cell division stops, the cell cycle blocked just before DNA replication. If a cell sample is taken after 20 doublings and frozen, when thawed it resumes growth for 30 more doublings, then stops.

An explanation of the “Hayflick limit” was suggested in 1986 when Howard Cooke first glimpsed an extra length of DNA at the ends of chromosomes. These **telomeric regions**, repeats of the sequence TTAGGG, were found to be substantially shorter in older somatic tissue, and Cooke speculated that a 100 base-pair portion of the telomere cap was lost by a chromosome during each cycle of DNA replication. Eventually, after some 50 replication cycles, the protective telomeric cap would be used up, and

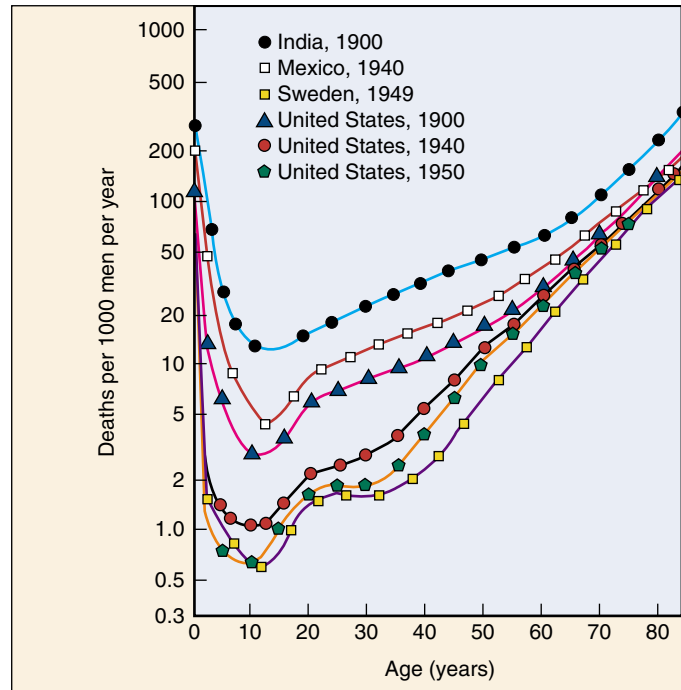


FIGURE 17.26

Gompertz curves. While human populations may differ 25-fold in their mortality rates before puberty, the slopes of their Gompertz curves are about the same in later years.

the cell line would then enter senescence, no longer able to proliferate. Cancer cells appear to avoid telomeric shortening.

Research reported in 1998 has confirmed Cooke’s hypothesis, providing direct evidence for a causal relation between telomeric shortening and cell senescence. Using genetic engineering, researchers transferred into human primary cell cultures a gene that leads to expression of telomerase, an enzyme that builds TTAGGG telomeric caps. The result was unequivocal. New telomeric caps were added to the chromosomes of the cells, and the cells with the artificially elongated telomeres did not senesce at the Hayflick limit, continuing to divide in a healthy and vigorous manner for more than 20 additional generations.

Wear-and-Tear Hypothesis

Numerous theories of aging focus in one way or another on the general idea that cells wear out over time, accumulating damage until they are no longer able to function. Loosely dubbed the “wear-and-tear” hypothesis, this idea implies that there is no inherent designed-in limit to aging, just a statistical one—over time, disruption, wear, and damage eventually erode a cell’s ability to function properly.

There is considerable evidence that aging cells do accumulate damage. Some of the most interesting evidence concerns free radicals, fragments of molecules or atoms that contain an unpaired electron. Free radicals are very reactive chemically and can be quite destructive in a cell. Free radicals are produced as natural by-products of oxidative metabolism, but most are mopped up by special enzymes that function to sweep the cell interior free of their destructive effects.

One of the most damaging free radical reactions that occurs in cells causes glucose to become linked to proteins, a nonenzymatic process called glycation. Two of the most commonly glycosylated proteins are collagen and elastin, key components of the connective tissues in our joints. Glycosylated collagen and elastin are not replaced, and individual molecules may be as old as the individual.

Glycation of collagen, elastin, and a diverse collection of other proteins within the cell produces a complex mixture of glucose-linked proteins called advanced glycosylation end products (AGEs). AGEs can cross-link to one another, reducing the flexibility of connective tissues in the joints and producing many of the other characteristic symptoms of aging.

Gene Clock Hypothesis

There is very little doubt that at least some aspects of aging are under the direct control of genes. Just as genes regulate the body's development, so they appear to regulate its rate of aging. Mutations in these genes can produce premature aging in the young. In the very rare recessive Hutchinson-Gilford syndrome, growth, sexual maturation, and skeletal development are retarded; atherosclerosis and strokes usually lead to death by age 12 years. Only some 20 cases have ever been described.

The similar Werner's syndrome is not as rare, affecting some 10 people per million worldwide. The syndrome is named after Otto Werner, who in Germany in 1904 reported a family affected by premature aging and said a genetic component was at work. Werner's syndrome makes its appearance in adolescence, usually producing death before age 50 of heart attack or one of a variety of rare connective tissue cancers. The gene responsible for Werner's syndrome was identified in 1996. Located on the short arm of chromosome 8, it seems to affect a helicase enzyme involved in the repair of DNA. The gene, which codes for a 1432-amino-acid protein, has been fully sequenced, and four mutant alleles identified. Helicase enzymes are needed to unwind the DNA double helix whenever DNA has to be replicated, repaired, or transcribed. The high incidence of certain cancers among Werner's syndrome patients leads investigators to speculate that the mutant helicase may fail to activate critical tumor suppressor genes. The potential role of helicases in aging is the subject of heated research.

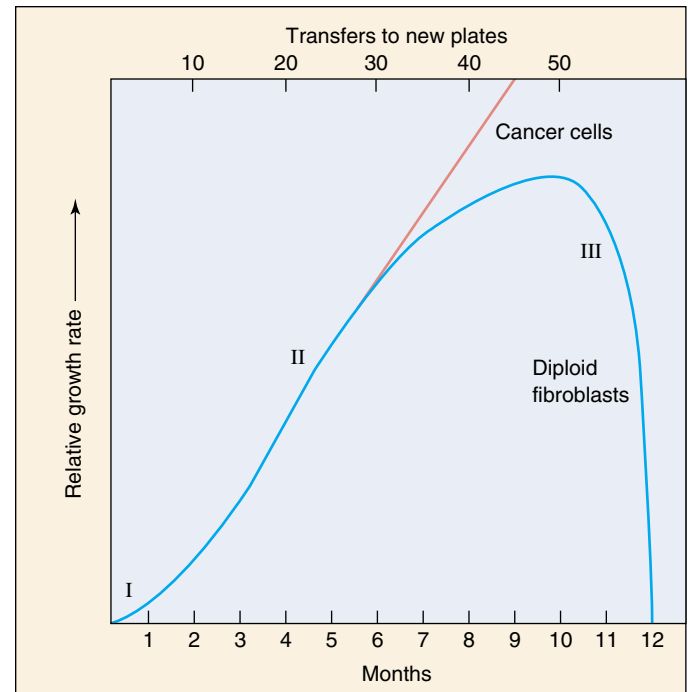


FIGURE 17.27

Hayflick's experiment. Fibroblast cells stop growing after about 50 doublings. Growth is rapid in phases I and II, but slows in phase III, as the culture becomes senescent, until the final doubling. Cancer cells, by contrast, do not "age."

Research on aging in other animals strongly supports the hypothesis that genes regulate the rate of aging. Particularly impressive results have been obtained in the nematode *Caenorhabditis elegans*, where genes discovered in 1996 seem to affect an intrinsic genetic clock. A combination of mutations can increase the worm's lifespan fivefold, the largest increase in lifespan seen in any organism! Mutations in the clock gene *clk-1* cause individual cells to divide more slowly, and the animal spends more time in each phase of its life cycle. Mutations in two other clock genes, *clk-2* and *clk-3*, have similar effects. Nematodes with mutations in two of the clock genes lived three to four times longer than normal. It seems that slowing life down in nematodes extends it. Perhaps, as the "wear-and-tear" theory suggests, aging results from damage to cells and their DNA by highly reactive oxidative by-products of metabolism. Living more slowly, destructive by-products may be produced less frequently, accumulate more slowly, and their damage be repaired more efficiently. Similar genes have been reported in yeasts, and attempts are now underway to isolate and clone these genes.

Among the many theories advanced to explain aging, many involve the progressive accumulation of damage to DNA. When genes affecting aging have been isolated, they affect DNA repair processes.

Chapter 17 Summary



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Summary

Questions

Media Resources

17.1 Development is a regulated process.

- Vertebrate development is initiated by a rapid cleavage of the fertilized egg into a hollow ball of cells, the blastula. Cell movements then form primary germ layers and organize the structure of the embryo.
- Cells determined in the insect embryo are carried within the body of larvae as imaginal discs, which are assembled into the adult body during pupation.
- Plant meristems continuously produce new tissues, which then differentiate into body parts. This differentiation is significantly influenced by the environment.

1. What is cleavage? How does the type of cleavage influence subsequent embryonic development?
2. What is a blastula? How does it form and what does it turn into?
3. What is a gastrula? Where are the germ layers in a gastrula?
4. What is neurulation? How and when does it occur?



- Introduction to Development



- Vertebrate Limb Formation

17.2 Multicellular organisms employ the same basic mechanisms of development.

- Cell movement in animal development is carried out by altering a cell's complement of surface adhesion molecules, which it uses to pull itself over other cells.
- A key to animal development is the ability of cells to alter the developmental paths of adjacent cells, a process called induction. Induction is achieved by diffusible chemicals called morphogens.
- Determination of a cell's ultimate developmental fate often involves the addition to it of positional labels that reflect its location in the embryo.
- The location of structures within body segments is dictated by a spatially organized assembly of homeotic genes, first discovered in *Drosophila* but now known to occur in all animals.
- Many cells are genetically programmed to die, usually soon after they are formed during development, in a process called apoptosis.

5. What role do cadherins and integrins play in cell movement?
6. What is the difference between mosaic development and regulative development?
7. How do organizers and morphogens participate in induction?
8. How is determination distinguished from differentiation?
9. What role does maternal mRNA play in the development of a *Drosophila* embryo?
10. What are homeotic genes and what do they do?



- Induction
- Pattern Formation

17.3 Four model developmental systems have been extensively researched.

- The four most intensively studied model systems of development are the mouse *Mus musculus*, the fruit fly *Drosophila melanogaster*, the nematode *Caenorhabditis elegans*, and the flowering plant *Arabidopsis thaliana*.
11. What are the major differences between vertebrate, insect, and plant developmental pathways? What are the similarities?

17.4 Aging can be considered a developmental process.

- Aging is not well understood, although not for want of theories, most of which involve progressive damage to DNA.
12. Cancer cell cultures never seem to grow old, dividing ceaselessly. What can you deduce about the state of their telomerase gene?

18

Altering the Genetic Message

Concept Outline

18.1 Mutations are changes in the genetic message.

Mutations Are Rare But Important. Changes in genes provide the raw material for evolution.

Kinds of Mutation. Some mutations alter genes themselves, others alter the positions of genes.

Point Mutations. Radiation damage or chemical modification can change one or a few nucleotides.

Changes in Gene Position. Chromosomal rearrangement and insertional inactivation reflect changes in gene position.

18.2 Cancer results from mutation of growth-regulating genes.

What Is Cancer? Cancer is a growth disorder of cells.

Kinds of Cancer. Cancer occurs in almost all tissues, but more in some than others.

Some Tumors Are Caused by Chemicals. Chemicals that mutate DNA cause cancer.

Other Tumors Result from Viral Infection. Viruses carrying growth-regulating genes can cause cancer.

Cancer and the Cell Cycle. Cancer results from mutation of genes regulating cell proliferation

Smoking and Cancer. Smoking causes lung cancer.

Curing Cancer. New approaches offer promise of a cure.

18.3 Recombination alters gene location.

An Overview of Recombination. Recombination is produced by gene transfer and by reciprocal recombination.

Gene Transfer. Many genes move within small circles of DNA called plasmids. Plasmids can move between bacterial cells and carry bacterial genes. Some gene sequences move from one location to another on a chromosome.

Reciprocal Recombination. Reciprocal recombination can alter genes in several ways.

Trinucleotide Repeats. Increases in the number of repeated triplets can produce gene disorders.

18.4 Genomes are continually evolving.

Classes of Eukaryotic DNA. Unequal crossing over expands eukaryotic genomes.

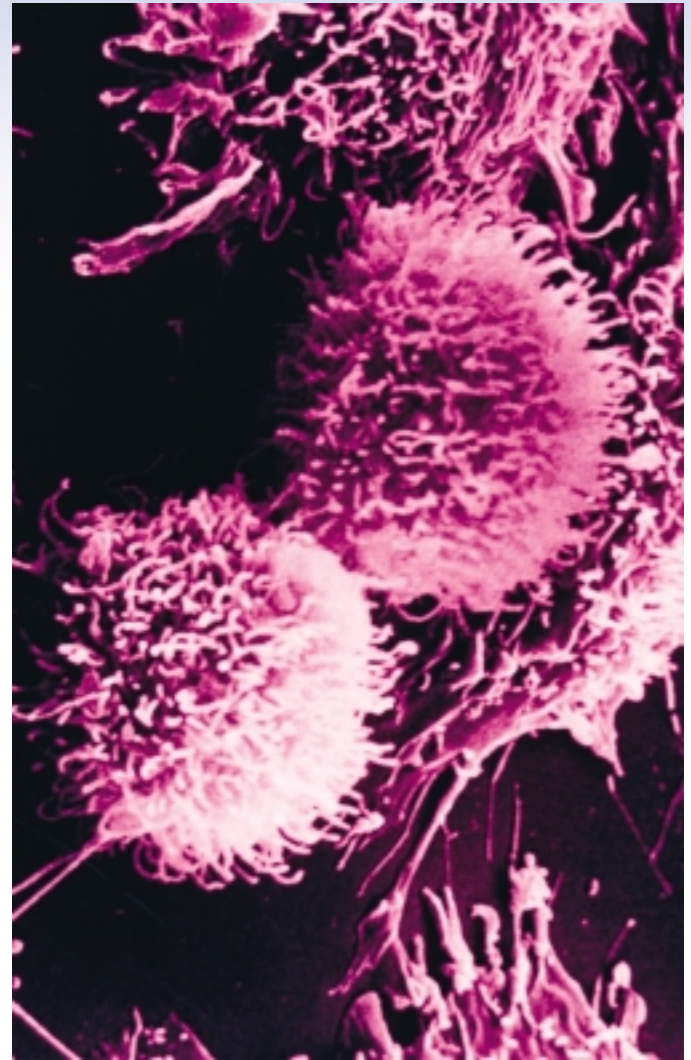


FIGURE 18.1

Cancer. A scanning electron micrograph of dead cancer cells (8000 $\times$).

In general, the genetic message can be altered in two broad ways: mutation and recombination. A change in the content of the genetic message—the base sequence of one or more genes—is referred to as a mutation. Some mutations alter the identity of a particular nucleotide, while others remove or add nucleotides to a gene. A change in the position of a portion of the genetic message is referred to as recombination. Some recombination events move a gene to a different chromosome; others alter the location of only part of a gene. In this chapter, we will first consider gene mutation, using cancer as a focus for our inquiry (figure 18.1). Then we will turn to recombination, focusing on how it has affected the organization of the eukaryotic genome.

18.1 Mutations are changes in the genetic message.

Mutations Are Rare But Important

The cells of eukaryotes contain an enormous amount of DNA. If the DNA in all of the cells of an adult human were lined up end-to-end, it would stretch nearly 100 billion kilometers—60 times the distance from Earth to Jupiter! The DNA in any multicellular organism is the final result of a long series of replications, starting with the DNA of a single cell, the fertilized egg. Organisms have evolved many different mechanisms to avoid errors during DNA replication and to preserve the DNA from damage. Some of these mechanisms “proof-read” the replicated DNA strands for accuracy and correct any mistakes. The proof-reading is not perfect, however. If it were, no variation in the nucleotide sequences of genes would be generated.

Mistakes Happen

In fact, cells do make mistakes during replication, and damage to the genetic message also occurs, causing mutation (figure 18.2). However, change is rare. Typically, a particular gene is altered in only one of a million gametes. If changes were common, the genetic instructions encoded in DNA would soon degrade into meaningless gibberish. Limited as it might seem, the steady trickle of change that does occur is the very stuff of evolution. Every difference in the genetic messages that specify different organisms arose as the result of genetic change.

The Importance of Genetic Change

All evolution begins with alterations in the genetic message: mutation creates new alleles, gene transfer and transposition alter gene location, reciprocal recombination shuffles and sorts these changes, and chromosomal rearrangement alters the organization of entire chromosomes. Some changes in germ-line tissue produce alterations that enable an organism to leave more offspring, and those changes tend to be preserved as the genetic endowment of future generations. Other changes reduce the ability of an organism to leave offspring. Those changes tend to be lost, as the organisms that carry them contribute fewer members to future generations.



FIGURE 18.2

Mutation. Normal fruit flies have one pair of wings extending from the thorax. This fly is a mutant because of changes in *bithorax*, a gene regulating a critical stage of development; it possesses two thoracic segments and thus two sets of wings.

Evolution can be viewed as the selection of particular combinations of alleles from a pool of alternatives. The rate of evolution is ultimately limited by the rate at which these alternatives are generated. Genetic change through mutation and recombination provides the raw material for evolution.

Genetic changes in somatic cells do not pass on to offspring, and so have less evolutionary consequence than germ-line change. However, changes in the genes of somatic cells can have an important immediate impact, particularly if the gene affects development or is involved with regulation of cell proliferation.

Rare changes in genes, called mutations, can have significant effects on the individual when they occur in somatic tissue, but are only inherited if they occur in germ-line tissue. Inherited changes provide the raw material for evolution.

Kinds of Mutation

Because mutations can occur randomly anywhere in a cell's DNA, mutations can be detrimental, just as making a random change in a computer program or a musical score usually worsens performance. The consequences of a detrimental mutation may be minor or catastrophic, depending on the function of the altered gene.

Mutations in Germ-Line Tissues

The effect of a mutation depends critically on the identity of the cell in which the mutation occurs. During the embryonic development of all multicellular organisms, there comes a point when cells destined to form gametes (**germ-line cells**) are segregated from those that will form the other cells of the body (**somatic cells**). Only when a mutation occurs within a germ-line cell is it passed to subsequent generations as part of the hereditary endowment of the gametes derived from that cell.

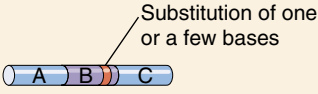
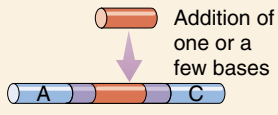
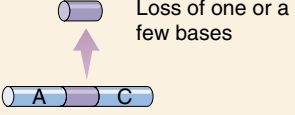
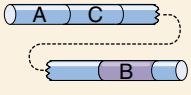
Mutations in Somatic Tissues

Mutations in germ-line tissue are of enormous biological importance because they provide the raw material from which natural selection produces evolutionary change. Change can occur only if there are new, different allele combinations available to replace the old. Mutation produces new alleles, and recombination puts the alleles together in different combinations. In animals, it is the occurrence of these two processes in germ-line tissue that is important to evolution, as mutations in somatic cells (**somatic mutations**) are not passed from one generation to the next. However, a somatic mutation may have drastic effects on the individual organism in which it occurs, as it is passed on to all of the cells that are descended from the original mutant cell. Thus, if a mutant lung cell divides, all cells derived from it will carry the mutation. Somatic mutations of lung cells are, as we shall see, the principal cause of lung cancer in humans.

Point Mutations

One category of mutational changes affects the message itself, producing alterations in the sequence of DNA nucleotides (table 18.1 summarizes the sources and types of mutations). If alterations involve only one or a few base-pairs in the coding sequence, they are called **point mutations**. While some point mutations arise due to spontaneous pairing errors that occur during DNA replication, others result from damage to the DNA caused by **mutagens**, usually radiation or chemicals. The latter class of mutations is of particular practical importance because modern industrial societies often release many chemical mutagens into the environment.

Table 18.1 Types of Mutation

Mutation	Example result
NO MUTATION	
	Normal B protein is produced by the B gene.
POINT MUTATION	
Base substitution	B protein is inactive because changed amino acid disrupts function.
	
Insertion	B protein is inactive because inserted material disrupts proper shape.
	
Deletion	B protein is inactive because portion of protein is missing.
	
CHANGES IN GENE POSITION	
Transposition	B gene or B protein may be regulated differently because of change in gene position.
	
Chromosomal rearrangement	B gene may be inactivated or regulated differently in its new location on chromosome.
	

Changes in Gene Position

Another category of mutations affects the way the genetic message is organized. In both bacteria and eukaryotes, individual genes may move from one place in the genome to another by **transposition**. When a particular gene moves to a different location, its expression or the expression of neighboring genes may be altered. In addition, large segments of chromosomes in eukaryotes may change their relative locations or undergo duplication. Such **chromosomal rearrangements** often have drastic effects on the expression of the genetic message.

Point mutations are changes in the hereditary message of an organism. They may result from spontaneous errors during DNA replication or from damage to the DNA due to radiation or chemicals.

Point Mutations

Physical Damage to DNA

Ionizing Radiation. High-energy forms of radiation, such as X rays and gamma rays, are highly mutagenic. When such radiation reaches a cell, it is absorbed by the atoms it encounters, imparting energy to the electrons in their outer shells. These energized electrons are ejected from the atoms, leaving behind free radicals, ionized atoms with unpaired electrons. Free radicals react violently with other molecules, including DNA.

When a free radical breaks *both* phosphodiester bonds of a DNA helix, causing a **double-strand break**, the cell's usual mutational repair enzymes cannot fix the damage. The two fragments created by the break must be aligned while the phosphodiester bonds between them form again. Bacteria have no mechanism to achieve this alignment, and double-strand breaks are lethal to their descendants. In eukaryotes, which almost all possess multiple copies of their chromosomes, the synaptonemal complex assembled in meiosis is used to pair the fragmented chromosome with its homologue. In fact, it is speculated that meiosis may have evolved initially as a mechanism to repair double-strand breaks in DNA (see chapter 12).

Ultraviolet Radiation. Ultraviolet (UV) radiation, the component of sunlight that tans (and burns), contains much less energy than ionizing radiation. It does not induce atoms to eject electrons, and thus it does not produce free radicals. The only molecules capable of absorbing UV radiation are certain organic ring compounds, whose outer-shell electrons become reactive when they absorb UV energy.

DNA strongly absorbs UV radiation in the pyrimidine bases, thymine and cytosine. If one of the nucleotides on either side of the absorbing pyrimidine is also a pyrimidine, a double covalent bond forms between them. The resulting cross-link between adjacent pyrimidines is called a **pyrimidine dimer** (figure 18.3). In most cases, cellular UV repair systems either cleave the bonds that link the adjacent pyrimidines or excise the entire pyrimidine dimer from the strand and fill in the gap, using the other strand as a template (figure 18.4). In those rare instances in which a pyrimidine dimer goes unrepaired, DNA polymerase may fail to replicate the portion of the strand that includes the dimer, skipping ahead and leaving the problem area to be filled in later. This filling-in process is often error-prone, however, and it may create mutational changes in the base sequence of the gap region. Some unrepaired pyrimidine dimers block DNA replication altogether, which is lethal to the cell.

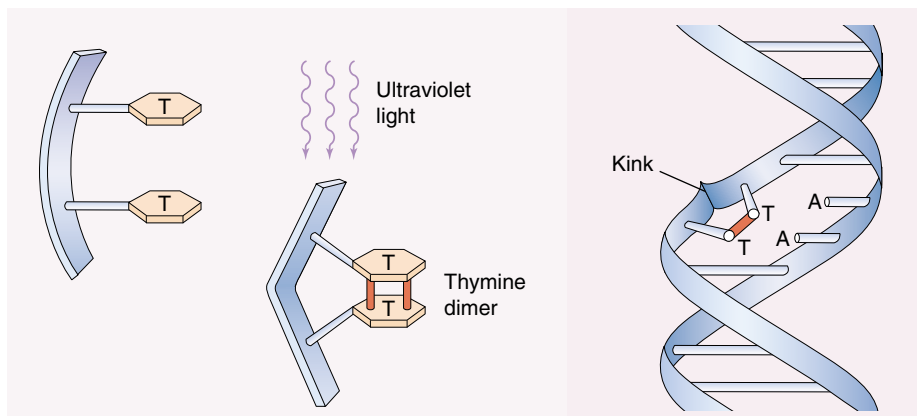


FIGURE 18.3

Making a pyrimidine dimer. When two pyrimidines, such as two thymines, are adjacent to each other in a DNA strand, the absorption of UV radiation can cause covalent bonds to form between them—creating a pyrimidine dimer. The dimer introduces a “kink” into the double helix that prevents replication of the duplex by DNA polymerase.

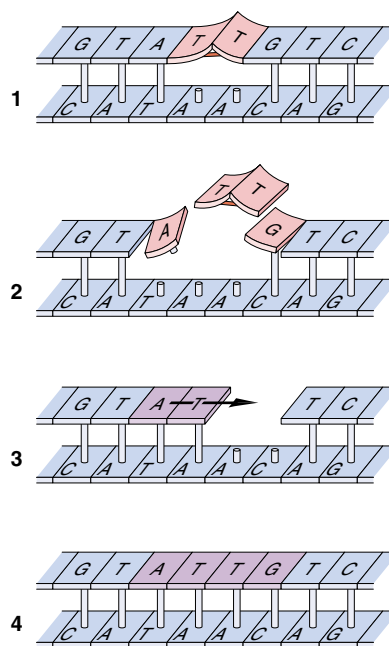


FIGURE 18.4

Repair of a pyrimidine dimer. Some pyrimidine dimers are repaired by excising the dimer, as well as a short run of nucleotides on either side of it, and then filling in the gap using the other strand as a template.

Sunlight can wreak havoc on the cells of the skin because its UV light causes mutations. Indeed, a strong and direct association exists between exposure to bright sunlight, UV-induced DNA damage, and skin cancer. A deep tan is *not* healthy! A rare hereditary disorder among humans called **xeroderma pigmentosum** causes these problems after a lesser exposure to UV. Individuals with this disorder develop extensive skin tumors after exposure to sunlight because they lack a mechanism for repairing the DNA damage UV radiation causes. Because of the many different proteins involved in excision and repair of pyrimidine dimers, mutations in as many as eight different genes cause the disease.

Chemical Modification of DNA

Many mutations result from direct chemical modification of the DNA. The chemicals that act on DNA fall into three classes: (1) chemicals that resemble DNA nucleotides but pair incorrectly when they are incorporated into DNA (figure 18.5). Some of the new AIDS chemotherapeutic drugs are analogues of nitrogenous bases that are inserted into the viral or infected cell DNA. This DNA cannot be properly transcribed, so viral growth slows; (2) chemicals that remove the amino group from adenine or cytosine, causing them to mispair; and (3) chemicals that add hydrocarbon groups to nucleotide bases, also causing them to mispair. This last group includes many particularly potent mutagens commonly used in laboratories, as well as compounds sometimes released into the environment, such as mustard gas.

Spontaneous Mutations

Many point mutations occur spontaneously, without exposure to radiation or mutagenic chemicals. Sometimes nucleotide bases spontaneously shift to alternative conformations, or isomers, which form different kinds of hydrogen bonds than the normal conformations. During replication, DNA polymerase pairs a different nucleotide with the isomer than it would have otherwise selected. Unrepaired spontaneous errors occur in fewer than one in a billion nucleotides per generation, but they are still an important source of mutation.

Sequences sometimes misalign when homologous chromosomes pair, causing a portion of one strand to loop out. These misalignments, called **slipped mispairing**, are usually only transitory, and the chromosomes quickly revert to the normal arrangement (figure 18.6). If the error-correcting system of the cell encounters a slipped mispairing before it reverts, however, the system will attempt to “correct” it, usually by excising the loop. This may result in a **deletion** of several hundred nucleotides from one of the chromosomes. Many of these deletions start or end in the middle of a codon, thereby shifting the reading frame by one or two bases. These so-called **frame-shift mutations** cause the gene to be read in the wrong three-base groupings, distorting the genetic message, just as the deletion of the letter F from the sentence, THE FAT CAT ATE THE RAT shifts the reading frame of the sentence, producing the meaningless message, THE ATC ATA TET HER AT. Some chemicals specifically promote deletions and frame-shift mutations by stabilizing the loops produced during slipped mispairing, thus increasing the time the loops are vulnerable to excision.

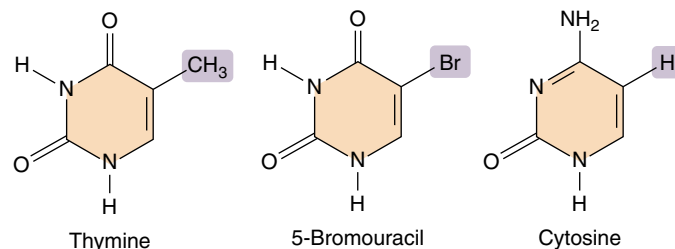


FIGURE 18.5

Chemicals that resemble DNA bases can cause mutations. For example, DNA polymerase cannot distinguish between thymine and 5-bromouracil, which are similar in shape. Once incorporated into a DNA molecule, however, 5-bromouracil tends to rearrange to a form that resembles cytosine and pairs with guanine. When this happens, what was originally an A-T base-pair becomes a G-C base-pair.

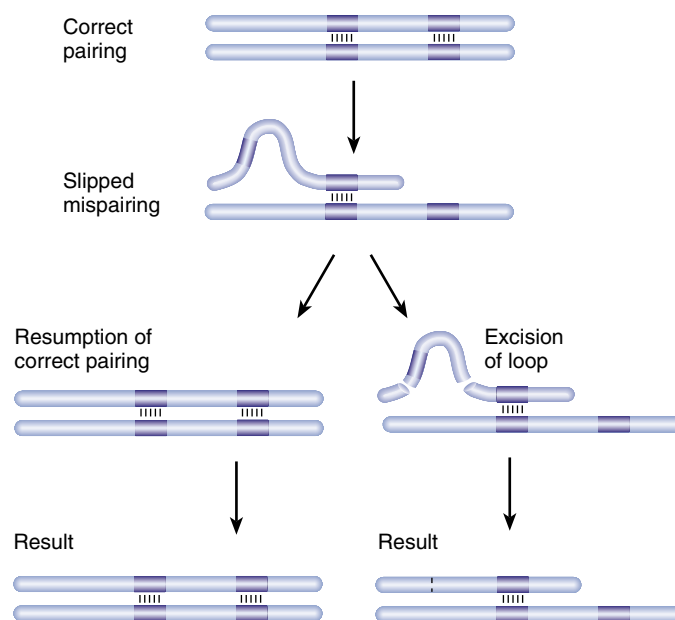


FIGURE 18.6

Slipped mispairing. Slipped mispairing occurs when a sequence is present in more than one copy on a chromosome and the copies on homologous chromosomes pair out of register, like a shirt buttoned wrong. The loop this mistake produces is sometimes excised by the cell's repair enzymes, producing a short deletion and often altering the reading frame. Any chemical that stabilizes the loop increases the chance it will be excised.

The major sources of physical damage to DNA are ionizing radiation, which breaks the DNA strands; ultraviolet radiation, which creates nucleotide cross-links whose removal often leads to errors in base selection; and chemicals that modify DNA bases and alter their base-pairing behavior. Unrepaired spontaneous errors in DNA replication occur rarely.

Changes in Gene Position

Chromosome location is an important factor in determining whether genes are transcribed. Some genes cannot be transcribed if they are adjacent to a tightly coiled region of the chromosome, even though the same gene can be transcribed normally in any other location. Transcription of many chromosomal regions appears to be regulated in this manner; the binding of specific proteins regulates the degree of coiling in local regions of the chromosome, determining the accessibility RNA polymerase has to genes located within those regions.

Chromosomal Rearrangements

Chromosomes undergo several different kinds of gross physical alterations that have significant effects on the locations of their genes. The two most important are **translocations**, in which a segment of one chromosome becomes part of another chromosome, and **inversions**, in which the orientation of a portion of a chromosome is reversed. Translocations often have significant effects on gene expression. Inversions, on the other hand, usually do not alter gene expression, but they are nonetheless important. Recombination within a region that is inverted on one homologue but not the other (figure 18.7) leads to serious problems: none of the gametes that contain chromatids produced following such a crossover event will have a complete set of genes.

Other chromosomal alterations change the number of gene copies an individual possesses. Particular genes or segments of chromosomes may be deleted or duplicated, whole chromosomes may be lost or gained (*aneuploidy*), and entire sets of chromosomes may be added (*polyploidy*). Most deletions are harmful because they halve the number of gene copies within a diploid genome and thus seriously affect the level of transcription. Duplications cause gene imbalance and are also usually harmful.

Insertional Inactivation

Many small segments of DNA are capable of moving from one location to another in the genome, using an enzyme to cut and paste themselves into new genetic neighborhoods. We call these mobile bits of DNA transposable elements, or **transposons**. Transposons select their new locations at random, and are as likely to enter one segment of a chromosome as another. Inevitably, some transposons end up inserted into genes, and this almost always inactivates the gene. The encoded protein now has a large meaningless chunk inserted within it, disrupting its structure. This form of mutation, called **insertional inactivation**, is common in nature. Indeed, it seems to be one of the most significant causes of mutation. The original white-eye mutant of *Drosophila* discovered by Morgan (see chapter 13) is the result of a transposition event, a transposon nested within a gene encoding a pigment-producing enzyme.

As you might expect, a variety of human gene disorders are the result of transposition. The human transposon called *Alu*, for example, is responsible for an X-linked hemophilia, inserting into clotting factor IX and placing a premature stop codon there. It also causes inherited high levels of cholesterol (hypercholesterolemia), *Alu* elements inserting into the gene encoding the low density lipoprotein (LDL) receptor. In one very interesting case, a *Drosophila* transposon called *Mariner* proves responsible for a rare human neurological disorder called Charcot-Marie-Tooth disease, in which the muscles and nerves of the legs and feet gradually wither away. The Mariner transposon is inserted into a key gene called *CMT* on chromosome 17, creating a weak site where the chromosome can break. No one knows how the *Drosophila* transposon got into the human genome.

Many mutations result from changes in gene location or from insertional inactivation.

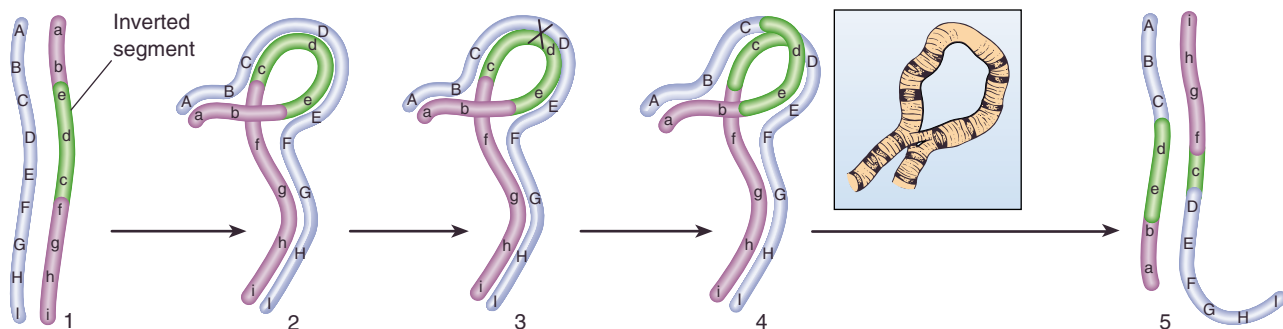


FIGURE 18.7

The consequence of inversion. (1) When a segment of a chromosome is inverted, (2) it can pair in meiosis only by forming an internal loop. (3) Any crossing over that occurs within the inverted segment during meiosis will result in nonviable gametes; some genes are lost from each chromosome, while others are duplicated (4 and 5). For clarity, only two strands are shown, although crossing over occurs in the four-strand stage. The pairing that occurs between inverted segments is sometimes visible under the microscope as a characteristic loop (*inset*).

18.2 Cancer results from mutation of growth-regulating genes.

What Is Cancer?

Cancer is a growth disorder of cells. It starts when an apparently normal cell begins to grow in an uncontrolled and invasive way (figure 18.8). The result is a cluster of cells, called a **tumor**, that constantly expands in size. Cells that leave the tumor and spread throughout the body, forming new tumors at distant sites, are called **metastases** (figure 18.9). Cancer is perhaps the most pernicious disease. Of the children born in 1999, one-third will contract cancer at some time during their lives; one-fourth of the male children and one-third of the female children will someday die of cancer. Most of us have had family or friends affected by the disease. In 1997, 560,000 Americans died of cancer.

Not surprisingly, researchers are expending a great deal of effort to learn the cause of this disease. Scientists have made a great deal of progress in the last 20 years using molecular biological techniques, and the rough outlines of understanding are now emerging. We now know that cancer is a gene disorder of somatic tissue, in which damaged genes fail to properly control cell proliferation. The cell division cycle is regulated by a sophisticated group of proteins described in chapter 11. Cancer results from the mutation of the genes encoding these proteins.

Cancer can be caused by chemicals that mutate DNA or in some instances by viruses that circumvent the cell's normal proliferation controls. Whatever the immediate cause, however, all cancers are characterized by unrestrained growth and division. Cell division never stops in a cancerous line of cells. Cancer cells are virtually immortal—until the body in which they reside dies.

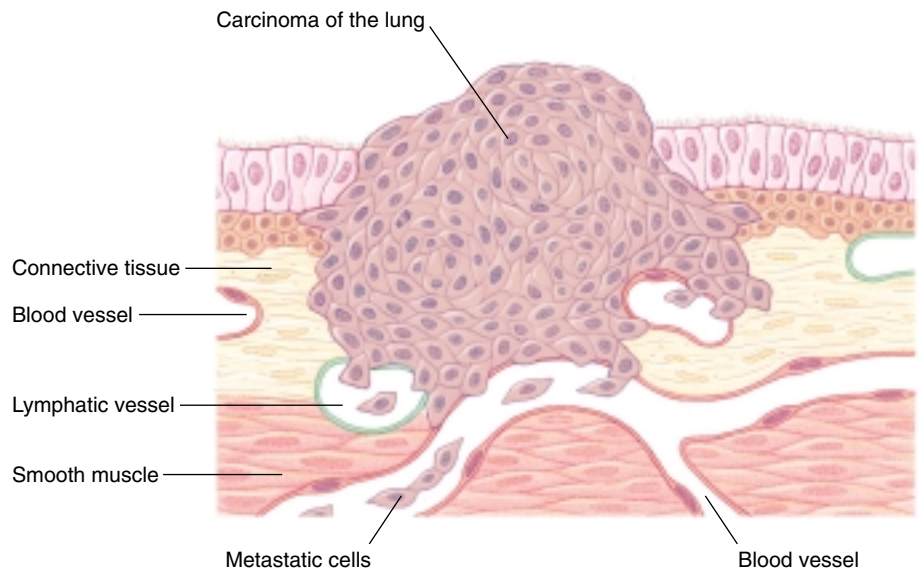
Cancer is unrestrained cell proliferation caused by damage to genes regulating the cell division cycle.



FIGURE 18.8
Lung cancer cells (530 $\times$). These cells are from a tumor located in the alveolus (air sac) of a lung.

FIGURE 18.9

Portrait of a cancer. This ball of cells is a carcinoma (cancer tumor) developing from epithelial cells that line the interior surface of a human lung. As the mass of cells grows, it invades surrounding tissues, eventually penetrating lymphatic and blood vessels, both plentiful within the lung. These vessels carry metastatic cancer cells throughout the body, where they lodge and grow, forming new masses of cancerous tissue.



Kinds of Cancer

Cancer can occur in almost any tissue, so a bewildering number of different cancers occur. Tumors arising from cells in connective tissue, bone, or muscle are known as **sarcomas**, while those that originate in epithelial tissue such as skin are called **carcinomas**. In the United States, the three deadliest human cancers are lung cancer, cancer of the colon and rectum, and breast cancer (table 18.2). Lung cancer, responsible for the most cancer deaths, is largely preventable; most cases result from smoking cigarettes. Colorectal cancers appear to be fostered by the high-meat diets so favored in the United States. The cause of breast cancer is still a mystery, although in 1994 and 1995 researchers isolated two genes responsible for hereditary susceptibility to breast cancer, *BRCA1* and *BRCA2* (Breast Cancer genes #1 and #2 located on human chromosomes 17 and 13); their discovery offers hope that researchers will soon be able to unravel the fundamental mechanism leading to hereditary breast cancer, about one-third of all breast cancers.

The association of particular chemicals with cancer, particularly chemicals that are potent mutagens, led researchers early on to the suspicion that cancer might be caused, at least in part, by chemicals, the so-called **chemical carcinogenesis theory**. Agents thought to cause cancer are called **carcinogens**. A simple and effective way to test if a chemical is mutagenic is the Ames test (figure 18.10), named for its developer, Bruce Ames. The test uses a strain of *Salmonella* bacteria that has a defective histidine-synthesizing gene. Because these bacteria cannot make histidine, they cannot grow on media without it. Only a back-mutation that restores the ability to manufacture histidine will permit growth. Thus the number of colonies of these bacteria that grow on histidine-free medium is a measure of the frequency of back-mutation. A majority of chemicals that cause back-mutations in this test are carcinogenic, and vice versa. To increase the sensitivity of the test, the strains of bacteria are altered to disable their DNA repair machinery. The search for the cause of cancer has focused in part on chemical carcinogens and other environmental factors, including ionizing radiation such as X rays (figure 18.11).

Cancers occur in all tissues, but are more common in some than others.

Table 18.2 Incidence of Cancer in the United States in 1999

Type of Cancer	New Cases	Deaths	% of Cancer Deaths
Lung	171,600	158,900	28
Colon and rectum	129,400	56,600	10
Leukemia/lymphoma	94,200	49,100	9
Breast	176,300	43,700	8
Prostate	179,300	37,000	7
Pancreas	28,600	28,600	5
Ovary	25,200	14,500	3
Stomach	21,900	13,500	2
Liver	14,500	13,600	2
Nervous system/eye	19,000	13,300	2
Bladder	54,200	12,100	2
Oral cavity	29,800	8,100	2
Kidney	30,000	11,900	2
Cervix/uterus	50,200	11,200	2
Malignant melanoma	44,200	7,300	1
Sarcoma (connective tissue)	10,400	5,800	1
All other cancers	143,000	77,900	14

In the United States in 1999 there were 1,221,800 reported cases of new cancers and 563,100 cancer deaths, indicating that roughly half the people who develop cancer die from it.

Source: Data from the American Cancer Society, Inc., 1999.

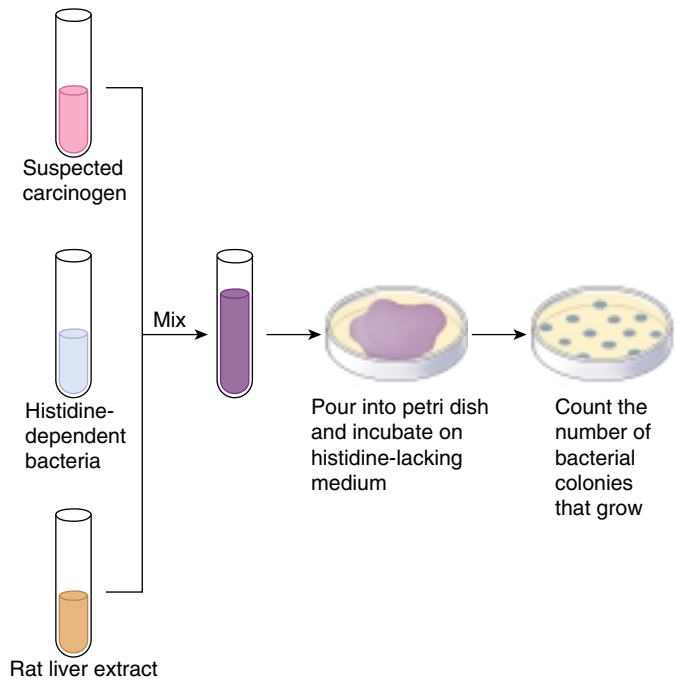


FIGURE 18.10
The Ames test. This test is uses a strain of *Salmonella* bacteria that requires histidine in the growth medium due to a mutated gene. If a suspected carcinogen is mutagenic, it can reverse this mutation. Rat liver extract is added because it contains enzymes that can convert carcinogens into mutagens. The mutagenicity of the carcinogen can be quantified by counting the number of bacterial colonies that grow on a medium lacking histidine.

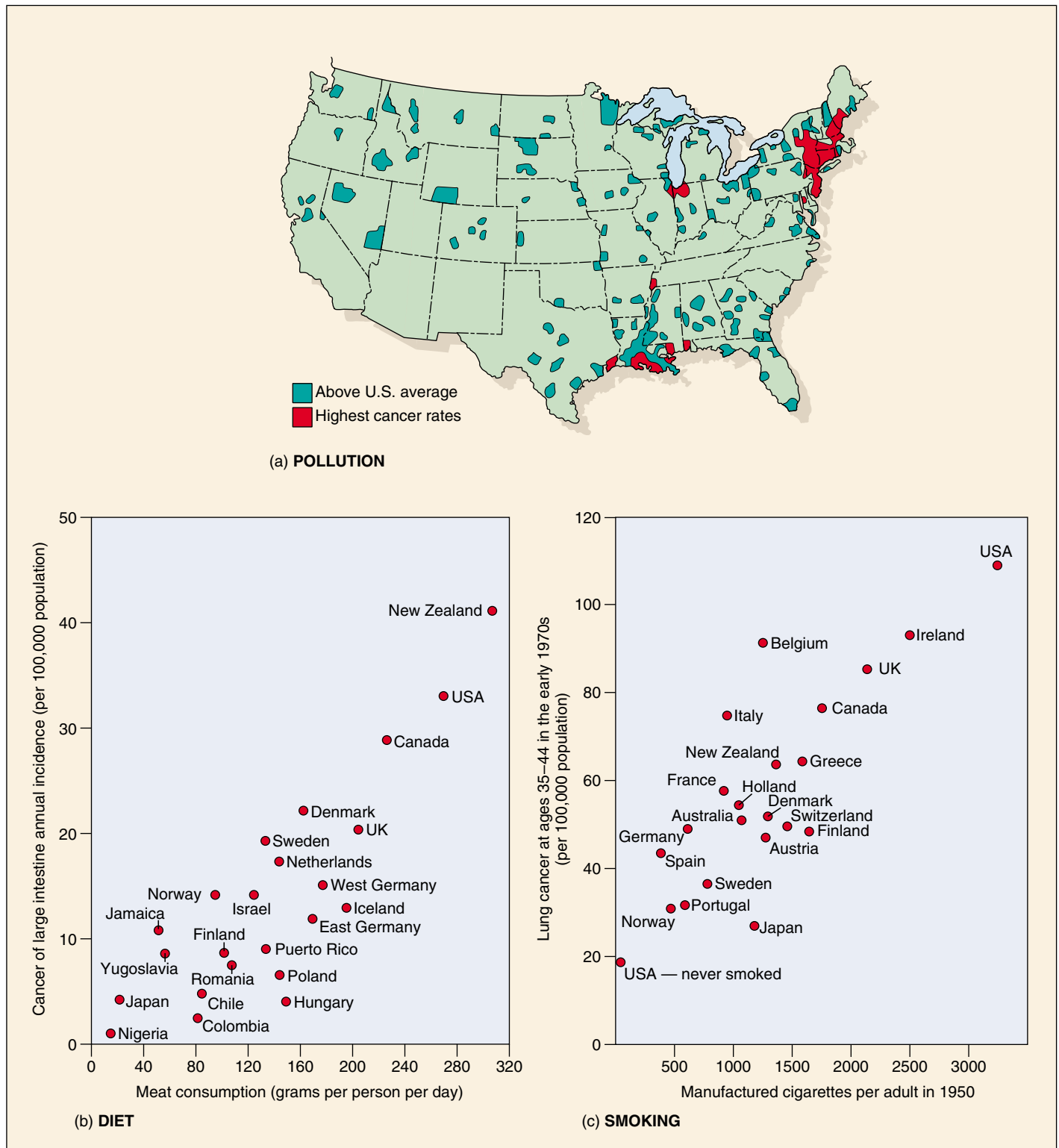


FIGURE 18.11

Potential cancer-causing agents. (a) The incidence of cancer per 1000 people is not uniform throughout the United States. The incidence is higher in cities and in the Mississippi Delta, suggesting that pollution and pesticide runoff may contribute to the development of cancer. (b) One of the deadliest cancers in the United States, cancer of the large intestine, is uncommon in many other countries. Its incidence appears to be related to the amount of meat a person consumes: a high-meat diet slows the passage of food through the intestine, prolonging exposure of the intestinal wall to digestive waste. (c) The biggest killer among cancers is lung cancer, and the most deadly environmental agent producing lung cancer is cigarette smoke. The incidence of lung cancer among men 35 to 44 years of age in various countries strongly correlates with the cigarette consumption in that country 20 years earlier.

Some Tumors Are Caused by Chemicals

Early Ideas

The chemical carcinogenesis theory was first advanced over 200 years ago in 1761 by Dr. John Hill, an English physician, who noted unusual tumors of the nose in heavy snuff users and suggested tobacco had produced these cancers. In 1775, a London surgeon, Sir Percivall Pott, made a similar observation, noting that men who had been chimney sweeps exhibited frequent cancer of the scrotum, and suggesting that soot and tars might be responsible. British sweeps washed themselves infrequently and always seemed covered with soot. Chimney sweeps on the continent, who washed daily, had much less of this scrotal cancer. These and many other observations led to the hypothesis that cancer results from the action of chemicals on the body.

Demonstrating That Chemicals Can Cause Cancer

It was over a century before this hypothesis was directly tested. In 1915, Japanese doctor Katsusaburo Yamagiwa applied extracts of coal tar to the skin of 137 rabbits every 2 or 3 days for 3 months. Then he waited to see what would happen. After a year, cancers appeared at the site of application in seven of the rabbits. Yamagiwa had induced cancer with the coal tar, the first direct demonstration of chemical carcinogenesis. In the decades that followed, this approach demonstrated that many chemicals were capable of causing cancer. Importantly, most of them were potent mutagens.

Because these were lab studies, many people did not accept that the results applied to real people. Do tars in fact induce cancer in humans? In 1949, the American physician Ernst Winder and the British epidemiologist Richard Doll independently reported that lung cancer showed a strong link to the smoking of cigarettes, which introduces tars into the lungs. Winder interviewed 684 lung cancer patients and 600 normal controls, asking whether each had ever smoked. Cancer rates were 40 times higher in heavy smokers than in nonsmokers. Doll's study was even more convincing. He interviewed a large number of British physicians, noting which ones smoked, then waited to see which would develop lung cancer. Many did. Overwhelmingly, those who did were smokers. From these studies, it seemed likely as long as 50 years ago that tars and other chemicals in cigarette smoke induce cancer in the lungs of persistent smokers. While this suggestion was (and is) resisted by the tobacco industry, the evidence that has accumulated since these pioneering studies makes a clear case, and there is no longer any real doubt. Chemicals in cigarette smoke cause cancer.

Carcinogens Are Common

In ongoing investigations over the last 50 years, many hundreds of synthetic chemicals have been shown capable

Table 18.3 Chemical Carcinogens in the Workplace

Chemical	Cancer	Workers at Risk for Exposure
COMMON EXPOSURE		
Benzene	Myelogenous leukemia	Painters; dye users; furniture finishers
Diesel exhaust	Lung	Railroad and bus-garage workers; truckers; miners
Mineral oils	Skin	Metal machinists
Pesticides	Lung	Sprayers
Cigarette tar	Lung	Smokers
UNCOMMON EXPOSURE		
Asbestos	Mesothelioma, lung	Brake-lining, insulation workers
Synthetic mineral fibers	Lung	Wall and pipe insulation and duct wrapping users
Hair dyes	Bladder	Hairdressers and barbers
Paint	Lung	Painters
Polychlorinated biphenyls	Liver, skin	Users of hydraulic fluids and lubricants, inks, adhesives, insecticides
Soot	Skin	Chimney sweeps; bricklayers; firefighters; heating-unit service workers
RARE EXPOSURE		
Arsenic	Lung, skin	Insecticide/herbicide sprayers; tanners; oil refiners
Formaldehyde	Nose	Wood product, paper, textiles, and metal product workers

of causing cancer in laboratory animals. Among them are trichloroethylene, asbestos, benzene, vinyl chloride, arsenic, arylamide, and a host of complex petroleum products with chemical structures resembling chicken wire. People in the workplace encounter chemicals daily (table 18.3).

In addition to identifying potentially dangerous substances, what have the studies of potential carcinogens told us about the nature of cancer? What do these cancer-causing chemicals have in common? *They are all mutagens, each capable of inducing changes in DNA.*

Chemicals that produce mutations in DNA are often potent carcinogens. Tars in cigarette smoke, for example, are the direct cause of most lung cancers.

Other Tumors Result from Viral Infection

Chemical mutagens are not the only carcinogens, however. Some tumors seem almost certainly to result from viral infection. Viruses can be isolated from certain tumors, and these viruses cause virus-containing tumors to develop in other individuals. About 15% of human cancers are associated with viruses.

A Virus That Causes Cancer

In 1911, American medical researcher Peyton Rous reported that a virus, subsequently named **Rous avian sarcoma virus (RSV)**, was associated with chicken sarcomas. He found that RSV could infect and initiate cancer in chicken fibroblast (connective tissue) cells growing in culture; from those cancerous cells, more viruses could be isolated. Rous was awarded the 1966 Nobel Prize in Physiology or Medicine for this discovery. RSV proved to be a kind of RNA virus called a **retrovirus**. When retroviruses infect a cell, they make a DNA copy of their RNA genome and insert that copy into the host cell's DNA.

How RSV Causes Cancer

How does RSV initiate cancer? When RSV was compared to a closely related virus, RAV-O, which is unable to transform normal chicken cells into cancerous cells, the two viruses proved to be identical except for one gene that was present in RSV but absent from RAV-O. That gene was called the *src* gene, short for sarcoma.

How do viral genes cause cancer? An essential clue came in 1970, when temperature-sensitive RSV mutants were isolated. These mutants would transform tissue culture cells into cancer cells at 35°C, but not at 41°C. Temperature sensitivity of this kind is almost always associated with proteins. It seemed likely, therefore, that the *src* gene was actively transcribed by the cell, rather than serving as a recognition site for some sort of regulatory protein. This was an exciting result, suggesting that the protein specified by this cancer-causing gene, or **oncogene**, could be isolated and its properties studied.

The *src* protein was first isolated in 1977 by J. Michael Bishop and Harold Varmus, who won the Nobel Prize for their efforts. It turned out to be an enzyme of moderate size that phosphorylates (adds a phosphate group to) the tyrosine amino acids of proteins. Such enzymes, called **tyrosine kinases**, are quite common in animal cells. One example is an enzyme that also serves as a plasma membrane receptor for **epidermal growth factor**, a protein that signals the initiation of cell division. This finding raised the exciting possibility that RSV may cause cancer by introducing into cells an active form of a normally quiescent growth-promoting enzyme. Later experiments showed this is indeed the case.

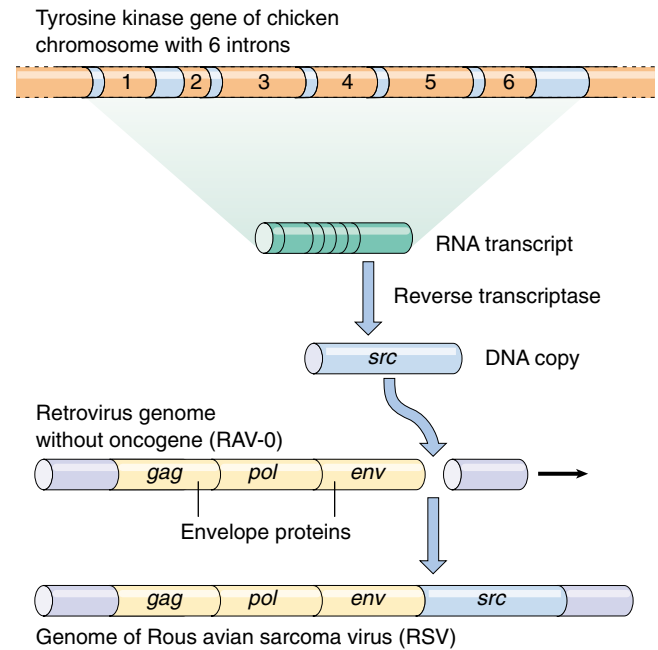


FIGURE 18.12

How a chicken gene got into the RSV genome. RSV contains only a few genes: *gag* and *env*, which encode the viral protein coat and envelope proteins, and *pol*, which encodes reverse transcriptase. It also contains the *src* gene that causes sarcomas, which the RAV-O virus lacks. RSV originally obtained its *src* gene from chickens, where a copy of the gene occurs normally and is controlled by the chicken's regulatory genes.

Origin of the *src* Gene

Does the *src* gene actually integrate into the host cell's chromosome along with the rest of the RSV genome? One way to answer this question is to prepare a radioactive version of the gene, allow it to bind to complementary sequences on the chicken chromosomes, and examine where the chromosomes become radioactive. The result of this experiment is that radioactive *src* DNA does in fact bind to the site where RSV DNA is inserted into the chicken genome—but it also binds to a second site where there is no part of the RSV genome!

The explanation for the second binding site is that the *src* gene is not exclusively a viral gene. It is also a growth-promoting gene that evolved in and occurs normally in chickens. This normal chicken gene is the second site where *src* binds to chicken DNA. Somehow, an ancestor of RSV picked up a copy of the normal chicken gene in some past infection. Now part of the virus, the gene is transcribed under the control of viral promoters rather than under the regulatory system of the chicken genome (figure 18.12).

Studies of RSV reveal that cancer results from the inappropriate activity of growth-promoting genes that are less active or completely inactive in normal cells.

Cancer and the Cell Cycle

An important technique used to study tumors is called **transfection**. In this procedure, the nuclear DNA from tumor cells is isolated and cleaved into random fragments. Each fragment is then tested individually for its ability to induce cancer in the cells that assimilate it.

Using transfection, researchers have discovered that most human tumors appear to result from the mutation of genes that regulate the cell cycle. Sometimes the mutation of only one or two gene is all that is needed to transform normally dividing cells into cancerous cells in tissue culture (table 18.4).

Point Mutations Can Lead to Cancer

The difference between a normal gene encoding a protein that regulates the cell cycle and a cancer-inducing version can be a single point mutation in the DNA. In one case of *ras*-induced bladder cancer, for example, a single DNA base change from guanine to thymine converts a glycine in the normal *ras* protein into a valine in the cancer-causing version. Several other *ras*-induced human carcinomas have been shown to also involve single nucleotide substitutions.

Telomerase and Cancer

Telomeres are short sequences of nucleotides repeated thousands of times on the ends of chromosomes. Because DNA polymerase is unable to copy chromosomes all the way to the tip (there is no place for the primer necessary to copy the last Okazaki fragment), telomeric segments are lost every time a cell divides.

In healthy cells a tumor suppressor inhibits production of a special enzyme called telomerase that adds the lost telomere material back to the tip. Without this enzyme, a cell's chromosomes lose material from their telomeres with each replication. Every time a chromosome is copied as the cell prepares to divide, more of the tip is lost. After some 30 divisions, so much is lost that copying is no longer possible. Cells in the tissues of an adult human have typically undergone 25 or more divisions. Cancer can't get very far with only the 5 remaining cell divisions. Were cancer to start, it would grind to a halt after only a few divisions for lack of telomere.

Thus, we see that the cell's inhibition of telomerase in somatic cells is a very effective natural brake on the cancer process. Any mutation that destroys the telomerase inhibitor releases that brake, making cancer possible. Thus, when researchers looked for telomerase in human ovarian tumor cells, they found it. These cells contained mutations that had inactivated the cell control that blocks the transcription of the telomerase gene. Telomerase produced in these cells reversed normal telomere shortening, allowing the cells to proliferate and gain the immortality of cancer cells.

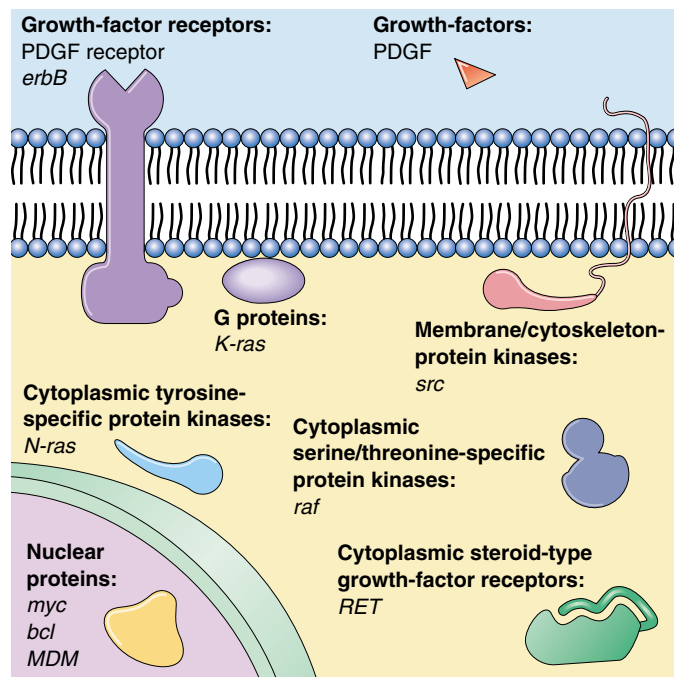


FIGURE 18.13

The main classes of oncogenes. Before they are altered by mutation to their cancer-causing condition, oncogenes are called proto-oncogenes (that is, genes able to become oncogenes). Illustrated here are the principal classes of proto-oncogenes, with some typical representatives indicated.

Mutations in Proto-Oncogenes: Accelerating the Cell Cycle

Most cancers are the direct result of mutations in growth-regulating genes. There are two general classes of cancer-inducing mutations: mutations of proto-oncogenes and mutations of tumor-suppressor genes.

Genes known as **proto-oncogenes** encode proteins that stimulate cell division. Mutations that overactivate these stimulatory proteins cause the cells that contain them to proliferate excessively. Mutated proto-oncogenes become cancer-causing genes called **oncogenes** (Greek *onco*-, "tumor") (figure 18.13). Often the induction of these cancers involves changes in the activity of intracellular signalling molecules associated with receptors on the surface of the plasma membrane. In a normal cell, the signalling pathways activated by these receptors trigger passage of the G₁ checkpoint of cell proliferation (see figure 11.17).

The mutated alleles of these oncogenes are genetically dominant. Among the most widely studied are *myc* and *ras*. Expression of *myc* stimulates the production of cyclins and cyclin-dependent protein kinases (Cdks), key elements in regulating the checkpoints of cell division.

Table 18.4 Some Genes Implicated in Human Cancers

Gene	Product	Cancer
ONCOGENES		
Genes Encoding Growth Factors or Their Receptors		
<i>erb-B</i>	Receptor for epidermal growth factor	Glioblastoma (a brain cancer); breast cancer
<i>erb-B2</i>	A growth factor receptor (gene also called <i>neu</i>)	Breast cancer; ovarian cancer; salivary gland cancer
<i>PDGF</i>	Platelet-derived growth factor	Glioma (a brain cancer)
<i>RET</i>	A growth factor receptor	Thyroid cancer
Genes Encoding Cytoplasmic Relays in Intracellular Signaling Pathways		
<i>K-ras</i>	Protein kinase	Lung cancer; colon cancer; ovarian cancer; pancreatic cancer
<i>N-ras</i>	Protein kinase	Leukemias
Genes Encoding Transcription Factors That Activate Transcription of Growth-Promoting Genes		
<i>c-myc</i>	Transcription factor	Lung cancer; breast cancer; stomach cancer; leukemias
<i>L-myc</i>	Transcription factor	Lung cancer
<i>N-myc</i>	Transcription factor	Neuroblastoma (a nerve cell cancer)
Genes Encoding Other Kinds of Proteins		
<i>bcl-2</i>	Protein that blocks cell suicide	Follicular B cell lymphoma
<i>bcl-1</i>	Cyclin D1, which stimulates the cell cycle clock (gene also called <i>PRAD1</i>)	Breast cancer; head and neck cancers
<i>MDM2</i>	Protein antagonist of p53 tumor-suppressor protein	Wide variety of sarcomas (connective tissue cancers)
TUMOR-SUPPRESSOR GENES		
Genes Encoding Cytoplasmic Proteins		
<i>APC</i>	Step in a signaling pathway	Colon cancer; stomach cancer
<i>DPC4</i>	A relay in signaling pathway that inhibits cell division	Pancreatic cancer
<i>NF-1</i>	Inhibitor of ras, a protein that stimulates cell division	Neurofibroma; myeloid leukemia
<i>NF-2</i>	Inhibitor of ras	Meningioma (brain cancer); schwannoma (cancer of cells supporting peripheral nerves)
Genes Encoding Nuclear Proteins		
<i>MTS1</i>	p16 protein, which slows the cell cycle clock	A wide range of cancers
<i>p53</i>	p53 protein, which halts cell division at the G ₁ checkpoint	A wide range of cancers
<i>Rb</i>	Rb protein, which acts as a master brake of the cell cycle	Retinoblastoma; breast cancer; bone cancer; bladder cancer
Genes Encoding Proteins of Unknown Cellular Locations		
<i>BRCA1</i>	?	Breast cancer; ovarian cancer
<i>BRCA2</i>	?	Breast cancer
<i>VHL</i>	?	Renal cell cancer

The *ras* gene product is involved in the cellular response to a variety of growth factors such as EGF, an intercellular signal that normally initiates cell proliferation. When EGF binds to a specific receptor protein on the plasma membrane of epithelial cells, the portion of the receptor that protrudes into the cytoplasm stimulates the *ras* protein to bind to GTP. The *ras* protein/GTP complex in turn re-

cruits and activates a protein called Raf to the inner surface of the plasma membrane, which in turn activates cytoplasmic kinases and so triggers an intracellular signaling system (see chapter 7). The final step in the pathway is the activation of transcription factors that trigger cell proliferation. Cancer-causing mutations in *ras* greatly reduce the amount of EGF necessary to initiate cell proliferation.

Mutations in Tumor-Suppressor Genes: Inactivating the Cell's Inhibitors of Proliferation

If the first class of cancer-inducing mutations “steps on the accelerator” of cell division, the second class of cancer-inducing mutations “removes the brakes.” Cell division is normally turned off in healthy cells by proteins that prevent cyclins from binding to Cdks. The genes that encode these proteins are called **tumor-suppressor genes**. Their mutant alleles are genetically recessive.

Among the most widely studied tumor-suppressor genes are *Rb*, *p16*, *p21*, and *p53*. The unphosphorylated product of the *Rb* gene ties up transcription factor E2F, which transcribes several genes required for passage

through the G₁ checkpoint into S phase of the cell cycle (figure 18.14). The proteins encoded by *p16* and *p21* reinforce the tumor-suppressing role of the Rb protein, preventing its phosphorylation by binding to the appropriate Cdk/cyclin complex and inhibiting its kinase activity. The p53 protein senses the integrity of the DNA and is activated if the DNA is damaged (figure 18.15). It appears to act by inducing the transcription of *p21*, which binds to cyclins and Cdk and prevents them from interacting. One of the reasons repeated smoking leads inexorably to lung cancer is that it induces *p53* mutations. Indeed, almost half of all cancers involve mutations of the *p53* gene.

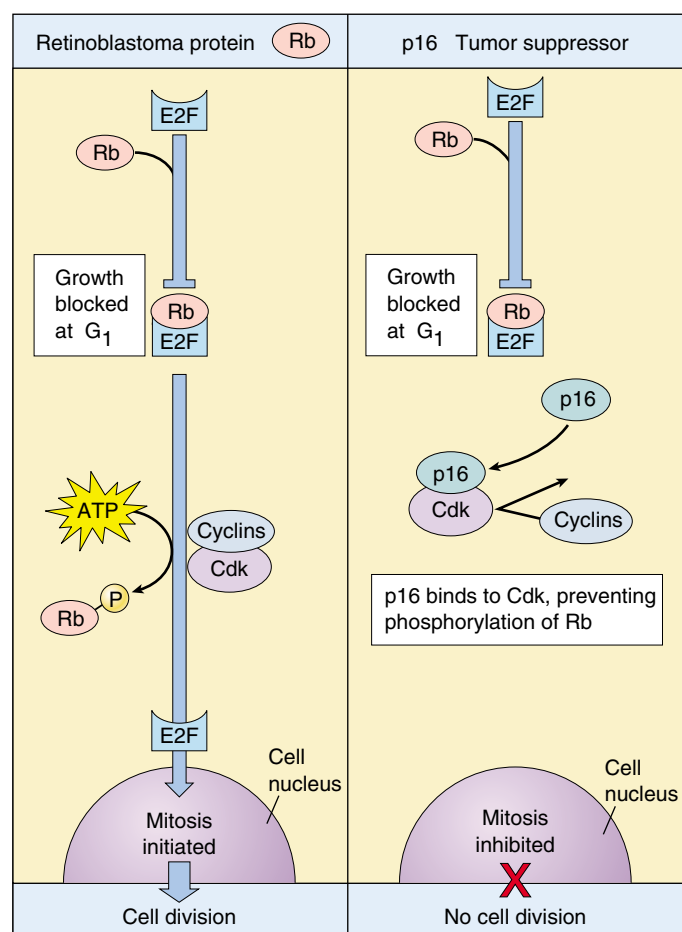


FIGURE 18.14

How the tumor-suppressor genes *Rb* and *p16* interact to block cell division. The retinoblastoma protein (Rb) binds to the transcription factor (E2F) that activates genes in the nucleus, preventing this factor from initiating mitosis. The G₁ checkpoint is passed when Cdk interacts with cyclins to phosphorylate Rb, releasing E2F. The p16 tumor-suppressor protein reinforces Rb's inhibitory action by binding to Cdk so that Cdk is not available to phosphorylate Rb.

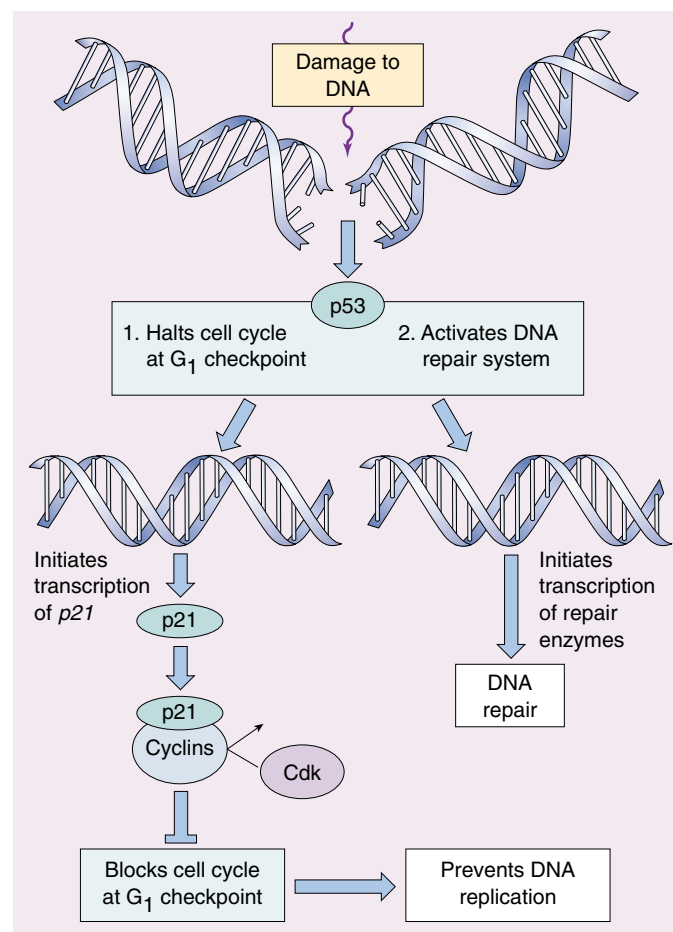


FIGURE 18.15

The role of tumor-suppressor *p53* in regulating the cell cycle. The p53 protein works at the G₁ checkpoint to check for DNA damage. If the DNA is damaged, p53 activates the DNA repair system and stops the cell cycle at the G₁ checkpoint (before DNA replication). This allows time for the damage to be repaired. p53 stops the cell cycle by inducing the transcription of *p21*. The p21 protein then binds to cyclins and prevents them from complexing with Cdk.

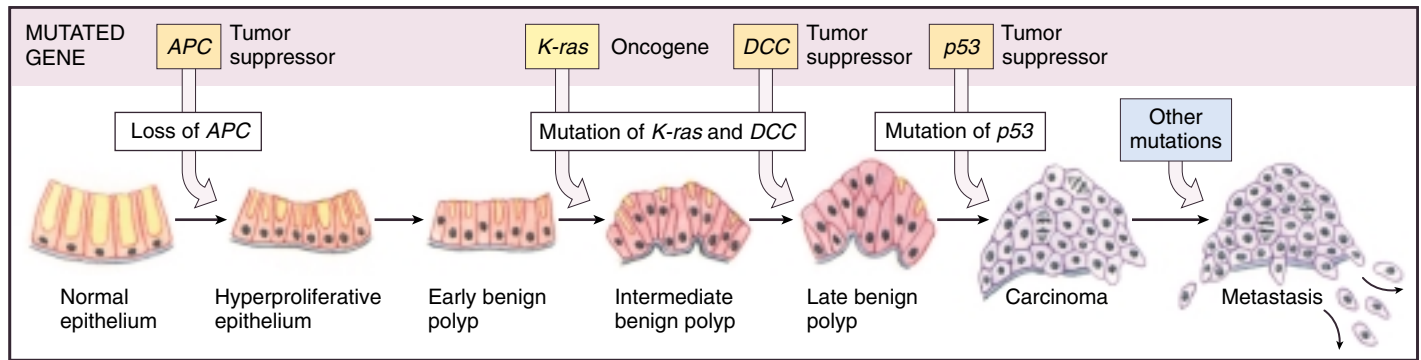


FIGURE 18.16

The progression of mutations that commonly lead to colorectal cancer. The fatal metastasis is the last of six serial changes that the epithelial cells lining the rectum undergo. One of these changes is brought about by mutation of a proto-oncogene, and three of them involve mutations that inactivate tumor-suppressor genes.

Cancer-Causing Mutations Accumulate over Time

Cells control proliferation at several checkpoints, and all of these controls must be inactivated for cancer to be initiated. Therefore, the induction of most cancers involves the mutation of multiple genes; four is a typical number (figure 18.16). In many of the tissue culture cell lines used to study cancer, most of the controls are already inactivated, so that mutations in only one or a few genes transform the line into cancerous growth. The need to inactivate several regulatory genes almost certainly explains why most cancers occur in people over 40 years old (figure 18.17); in older persons, there has been more time for individual cells to accumulate multiple mutations. It is now clear that mutations, including those in potentially cancer-causing genes, do accumulate over time. Using the polymerase chain reaction (PCR), researchers in 1994 searched for a certain cancer-associated gene mutation in the blood cells of 63 cancer-free people. They found that the mutation occurred 13 times more often in people over 60 years old than in people under 20.

Cancer is a disease in which the controls that normally restrict cell proliferation do not operate. In some cases, cancerous growth is initiated by the inappropriate activation of proteins that regulate the cell cycle; in other cases, it is initiated by the inactivation of proteins that normally suppress cell division.

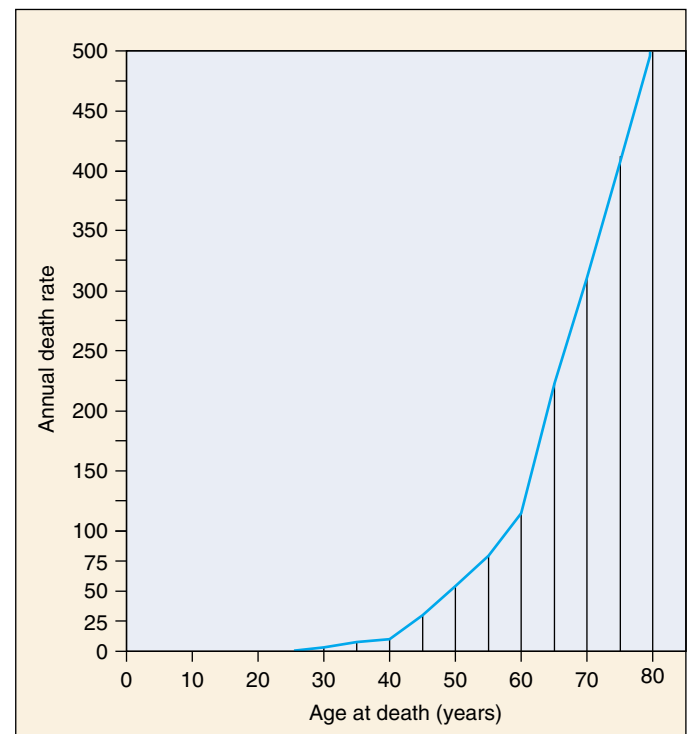


FIGURE 18.17

The annual death rate from cancer climbs with age. The rate of cancer deaths increases steeply after age 40 and even more steeply after age 60, suggesting that several independent mutations must accumulate to give rise to cancer.

Smoking and Cancer

How can we prevent cancer? The most obvious strategy is to minimize mutational insult. Anything that decreases exposure to mutagens can decrease the incidence of cancer because exposure has the potential to mutate a normal gene into an oncogene. It is no accident that the most reliable tests for the carcinogenicity of a substance are tests that measure the substance's mutagenicity.

The Association between Smoking and Cancer

About a third of all cases of cancer in the United States are directly attributable to cigarette smoking. The association between smoking and cancer is particularly striking for lung cancer (figure 18.18). Studies of male smokers show a highly positive correlation between the number of cigarettes smoked per day and the incidence of lung cancer (figure 18.19). For individuals who smoke two or more packs a day, the risk of contracting lung cancer is at least 40 times greater than it is for nonsmokers, whose risk level approaches zero. Clearly, an effective way to avoid lung cancer is not to smoke. Other studies have shown a clear relationship between cigarette smoking and reduced life expectancy (figure 18.20). Life insurance companies have calculated that smoking a single cigarette lowers one's life expectancy by 10.7 minutes (longer than it takes to smoke the cigarette)! Every pack of 20 cigarettes bears an unwritten label:

"The price of smoking this pack of cigarettes is 3½ hours of your life."

Smoking Introduces Mutagens to the Lungs

Over half a million people died of cancer in the United States in 1999; about 28% of them died of lung cancer. About 140,000 persons were diagnosed with lung cancer each year in the 1980s. Around 90% of them died within three years after diagnosis; 96% of them were cigarette smokers.

Smoking is a popular pastime. In the United States, 24% of the population smokes, and U.S. smokers consumed over 450 billion cigarettes in 1999. The smoke emitted from these cigarettes contains some 3000 chemical components, including vinyl chloride, benzo[*a*]pyrenes, and nitroso-*nor*-nicotine, all potent mutagens. Smoking places these mutagens into direct contact with the tissues of the lungs.

Mutagens in the Lung Cause Cancer

Introducing powerful mutagens to the lungs causes considerable damage to the genes of the epithelial cells that line the lungs and are directly exposed to the chemicals. Among the genes that are mutated as a result are some whose normal function is to regulate cell proliferation. When these genes are damaged, lung cancer results.

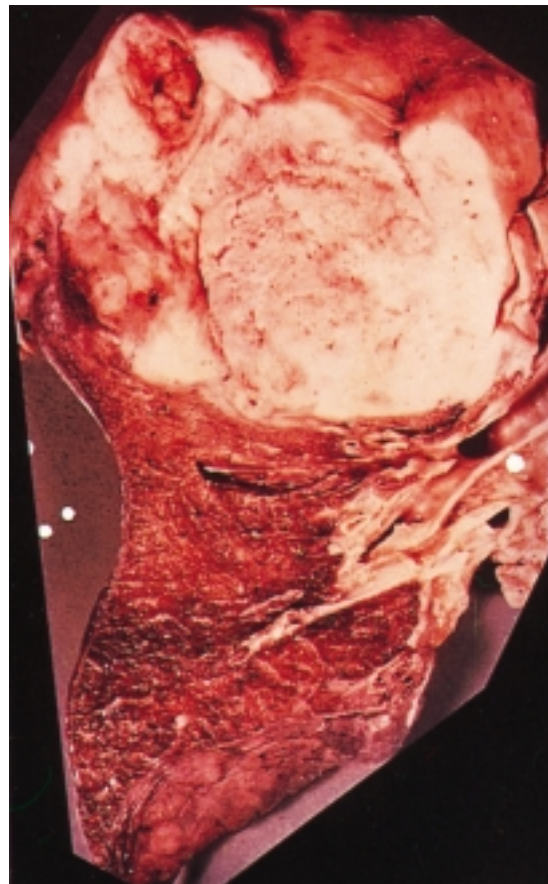


FIGURE 18.18

Photo of a cancerous human lung. The bottom half of the lung is normal, while a cancerous tumor has completely taken over the top half. The cancer cells will eventually break through into the lymph and blood vessels and spread through the body.

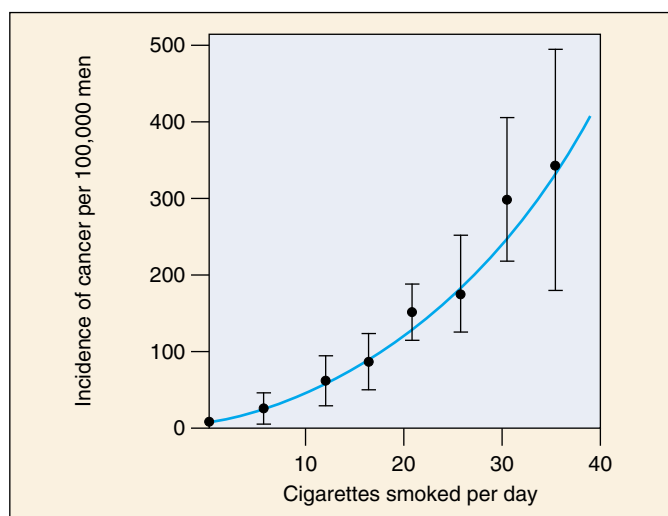


FIGURE 18.19

Smoking causes cancer. The annual incidence of lung cancer per 100,000 men clearly increases with the number of cigarettes smoked per day.

This process has been clearly demonstrated for benzo[*a*]pyrene (BP), one of the potent mutagens released into cigarette smoke from tars in the tobacco. The epithelial cells of the lung absorb BP from tobacco smoke and chemically alter it to a derivative form. This derivative form, benzo[*a*]pyrene-diolepoxide (BPDE), binds directly to the tumor-suppressor gene *p53* and mutates it to an inactive form. The protein encoded by *p53* oversees the G₁ cell cycle checkpoint described in chapter 11 and is one of the body's key mechanisms for preventing uncontrolled cell proliferation. The destruction of *p53* in lung epithelial cells greatly hastens the onset of lung cancer—*p53* is mutated to an inactive form in over 70% of lung cancers. When examined, the *p53* mutations in cancer cells almost all occur at one of three “hot spots.” The key evidence linking smoking and cancer is that when the mutations of *p53* caused by BPDE from cigarettes are examined, they occur at the same three specific “hot spots!”

The Incidence of Cancer Reflects Smoking

Cigarette manufacturers argue that the causal connection between smoking and cancer has not been proved, and that somehow the relationship is coincidental. Look carefully at the data presented in figure 18.21 and see if you agree. The upper graph, compiled from data on American men, shows the incidence of smoking from 1900 to 1990 and the incidence of lung cancer over the same period. Note that as late as 1920, lung cancer was a rare disease. About 20 years after the incidence of smoking began to increase among men, lung cancer also started to become more common.

Now look at the lower graph, which presents data on American women. Because of social mores, significant numbers of American women did not smoke until after World War II, when many social conventions changed. As late as 1963, when lung cancer among males was near current levels, this disease was still rare in women. In the United States that year, only 6588 women died of lung cancer. But as more women smoked, more developed lung cancer, again with a lag of about 20 years. American women today have achieved equality with men in the numbers of cigarettes they smoke, and their lung cancer death rates are today approaching those for men. In 1990, more than 49,000 women died of lung cancer in the United States. The current annual rate of deaths from lung cancer in male and female smokers is 180 per 100,000, or about 2 out of every 1000 smokers *each year*.

The easiest way to avoid cancer is to avoid exposure to mutagens. The single greatest contribution one can make to a longer life is not to smoke.

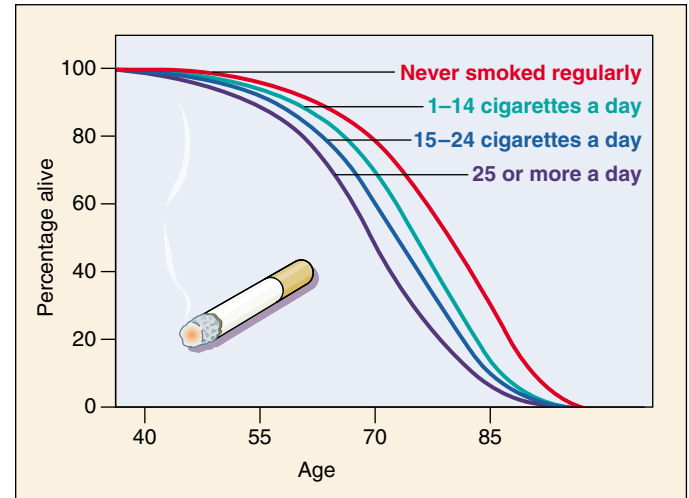


FIGURE 18.20

Tobacco reduces life expectancy. The world's longest-running survey of smoking, begun in 1951 in Britain, revealed that by 1994 the death rate for smokers had climbed to three times the rate for nonsmokers among men 35 to 69 years of age.

Source: Data from *New Scientist*, October 15, 1994.

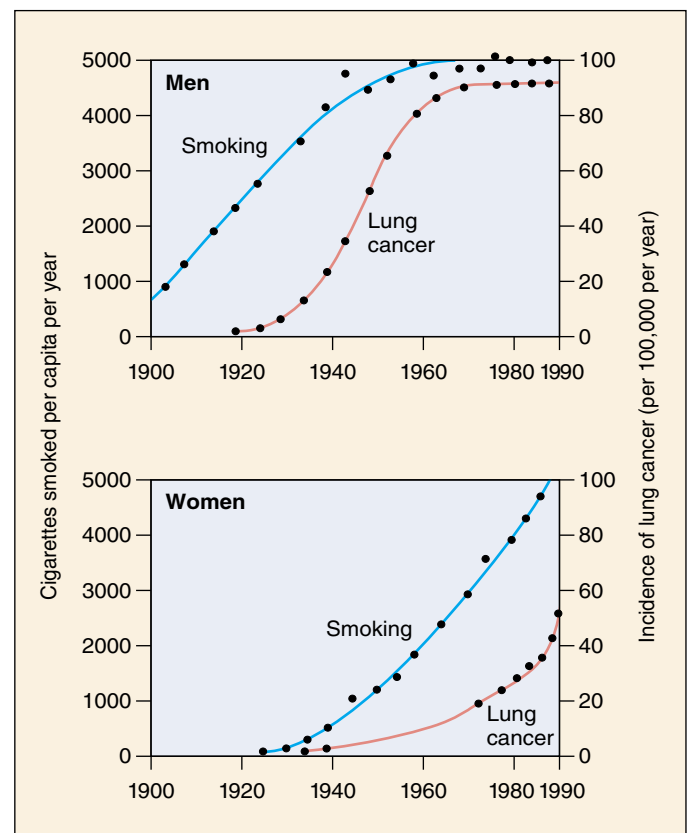


FIGURE 18.21

The incidence of lung cancer in men and women. What do these graphs indicate about the connection between smoking and lung cancer?

Curing Cancer

Potential cancer therapies are being developed on many fronts (figure 18.22). Some act to prevent the start of cancer within cells. Others act outside cancer cells, preventing tumors from growing and spreading.

Preventing the Start of Cancer

Many promising cancer therapies act within potential cancer cells, focusing on different stages of the cell's "Shall I divide?" decision-making process.

1. Receiving the Signal to Divide. The first step in the decision process is the reception of a "divide" signal, usually a small protein called a growth factor released from a neighboring cell. The growth factor is received by a protein receptor on the cell surface. Mutations that increase the number of receptors on the cell surface amplify the division signal and so lead to cancer. Over 20% of breast cancer tumors prove to overproduce a protein called HER2 associated with the receptor for epidermal growth factor.

Therapies directed at this stage of the decision process utilize the human immune system to attack cancer cells. Special protein molecules called "monoclonal antibodies," created by genetic engineering, are the therapeutic agents. These monoclonal antibodies are designed to seek out and stick to HER2. Like waving a red flag, the presence of the monoclonal antibody calls down attack by the immune system on the HER2 cell. Because breast cancer cells overproduce HER2, they are killed preferentially. Genentech's recently approved monoclonal antibody, called "herceptin," has given promising results in clinical tests. In other tests, the monoclonal antibody C225, directed against epidermal growth factor receptors, has succeeded in curing advanced colon cancer. Clinical trials of C225 have begun.

2. The Relay Switch. The second step in the decision process is the passage of the signal into the cell's interior, the cytoplasm. This is carried out in normal cells by a protein called Ras that acts as a relay switch. When growth factor binds to a receptor like EGF, the adjacent Ras protein acts like it has been "goosed," contorting into a new shape. This new shape is chemically active, and initiates a chain of reactions that passes the "divide" signal inward toward the nucleus. Mutated forms of the Ras protein behave like a relay switch stuck in the "ON" position, continually instructing the cell to divide when it should not. 30% of all cancers have a mutant form of Ras.

Therapies directed at this stage of the decision process take advantage of the fact that normal Ras proteins are inactive when made. Only after it has been modified by the special enzyme *farnesyl transferase* does Ras protein become able to function as a relay switch. In tests on animals, farnesyl transferase inhibitors induce the regression of tumors and prevent the formation of new ones.

3. Amplifying the Signal. The third step in the decision process is the amplification of the signal within the cytoplasm. Just as a TV signal needs to be amplified in order to be received at a distance, so a "divide" signal must be amplified if it is to reach the nucleus at the interior of the cell, a very long journey at a molecular scale. Cells use an ingenious trick to amplify the signal. Ras, when "ON," activates an enzyme, a protein kinase. This protein kinase activates other protein kinases that in their turn activate still others. The trick is that once a protein kinase enzyme is activated, it goes to work like a demon, activating hoards of others every second! And each and every one it activates behaves the same way too, activating still more, in a cascade of ever-widening effect. At each stage of the relay, the signal is amplified a thousand-fold. Mutations stimulating any of the protein kinases can dangerously increase the already amplified signal and lead to cancer. Five percent of all cancers, for example, have a mutant hyperactive form of the protein kinase Src.

Therapies directed at this stage of the decision process employ so-called "anti-sense RNA" directed specifically against Src or other cancer-inducing kinase mutations. The idea is that the *src* gene uses a complementary copy of itself to manufacture the Src protein (the "sense" RNA or messenger RNA), and a mirror image complementary copy of the sense RNA ("anti-sense RNA") will stick to it, gumming it up so it can't be used to make Src protein. The approach appears promising. In tissue culture, anti-sense RNAs inhibit the growth of cancer cells, and some also appear to block the growth of human tumors implanted in laboratory animals. Human clinical trials are underway.

4. Releasing the Brake. The fourth step in the decision process is the removal of the "brake" the cell uses to restrain cell division. In healthy cells this brake, a tumor suppressor protein called Rb, blocks the activity of a transcription factor protein called E2F. When free, E2F enables the cell to copy its DNA. Normal cell division is triggered to begin when Rb is inhibited, unleashing E2F. Mutations which destroy Rb release E2F from its control completely, leading to ceaseless cell division. Forty percent of all cancers have a defective form of Rb.

Therapies directed at this stage of the decision process are only now being attempted. They focus on drugs able to inhibit E2F, which should halt the growth of tumors arising from inactive Rb. Experiments in mice in which the E2F genes have been destroyed provide a model system to study such drugs, which are being actively investigated.

5. Checking That Everything Is Ready. The fifth step in the decision process is the mechanism used by the cell to ensure that its DNA is undamaged and ready to divide. This job is carried out in healthy cells by the tumor-suppressor protein p53, which inspects the integrity of the DNA. When it detects damaged or foreign DNA, p53 stops cell division and activates the cell's DNA repair systems. If the damage doesn't

get repaired in a reasonable time, p53 pulls the plug, triggering events that kill the cell. In this way, mutations such as those that cause cancer are either repaired or the cells containing them eliminated. If p53 is itself destroyed by mutation, future damage accumulates unrepaired. Among this damage are mutations that lead to cancer. Fifty percent of all cancers have a disabled p53. Fully 70 to 80% of lung cancers have a mutant inactive p53—the chemical benzo[*a*]pyrene in cigarette smoke is a potent mutagen of p53.

A promising new therapy using adenovirus (responsible for mild colds) is being targeted at cancers with a mutant p53. To grow in a host cell, adenovirus must use the product of its gene *E1B* to block the host cell's p53, thereby enabling replication of the adenovirus DNA. This means that while mutant adenovirus without *E1B* cannot grow in healthy cells, the mutants should be able to grow in, and destroy, cancer cells with defective p53. When human colon and lung cancer cells are introduced into mice lacking an immune system and allowed to produce substantial tumors, 60% of the tumors simply disappear when treated with E1B-deficient adenovirus, and do not reappear later. Initial clinical trials are less encouraging, as many people possess antibodies to adenovirus.

6. Stepping on the Gas. Cell division starts with replication of the DNA. In healthy cells, another tumor suppressor “keeps the gas tank nearly empty” for the DNA replication process by inhibiting production of an enzyme called telomerase. Without this enzyme, a cell's chromosomes lose material from their tips, called telomeres. Every time a chromosome is copied, more tip material is lost. After some thirty divisions, so much is lost that copying is no longer possible. Cells in the tissues of an adult human have typically undergone twenty five or more divisions. Cancer can't get very far with only the five remaining cell divisions, so inhibiting telomerase is a very effective natural break on the cancer process. It is thought that almost all cancers involve a mutation that destroys the telomerase inhibitor, releasing this break and making cancer possible. It should be possible to block cancer by reapplying this inhibition. Cancer therapies that inhibit telomerase are just beginning clinical trials.

Preventing the Spread of Cancer

7. Tumor Growth. Once a cell begins cancerous growth, it forms an expanding tumor. As the tumor grows ever-larger, it requires an increasing supply of food and nutrients, obtained from the body's blood supply. To facilitate this necessary grocery shopping, tumors leak out substances into the surrounding tissues that encourage angiogenesis, the formation of small blood vessels. Chemicals that inhibit this process are called angiogenesis inhibitors. In mice, two such angiogenesis inhibitors, angiostatin and endostatin, caused tumors to regress to microscopic size. This very exciting result has proven controversial, but initial human trials seem promising.

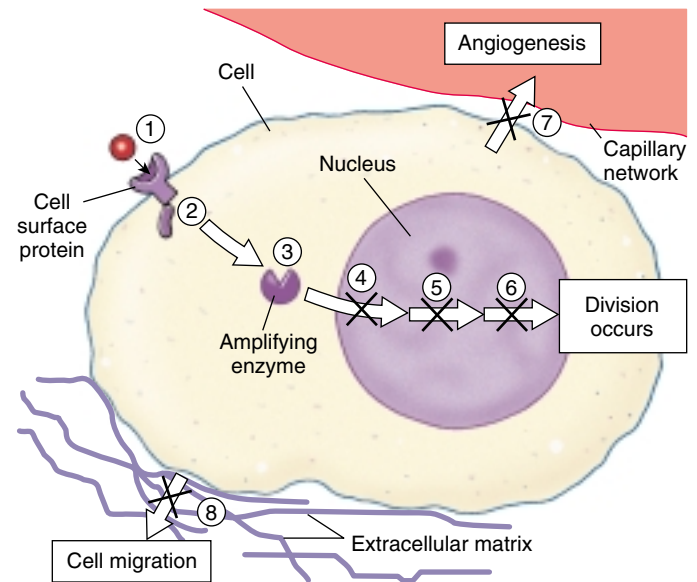


FIGURE 18.22

New molecular therapies for cancer target eight different stages in the cancer process. (1) On the cell surface, a growth factor signals the cell to divide. (2) Just inside the cell, a protein relay switch passes on the divide signal. (3) In the cytoplasm, enzymes amplify the signal. In the nucleus, (4) a “brake” preventing DNA replication is released, (5) proteins check that the replicated DNA is not damaged, and (6) other proteins rebuild chromosome tips so DNA can replicate. (7) The new tumor promotes angiogenesis, the formation of growth-promoting blood vessels. (8) Some cancer cells break away from the extracellular matrix and invade other parts of the body.

8. Metastasis. If cancerous tumors simply continued to grow where they form, many could be surgically removed, and far fewer would prove fatal. Unfortunately, many cancerous tumors eventually metastasize, individual cancer cells breaking their moorings to the extracellular matrix and spreading to other locations in the body where they initiate formation of secondary tumors. This process involves metal-requiring protease enzymes that cleave the cell-matrix linkage, components of the extracellular matrix such as fibronectin that also promote the migration of several non-cancerous cell types, and RhoC, a GTP-hydrolyzing enzyme that promotes cell migration by providing needed GTP. All of these components offer promising targets for future anti-cancer therapy.

Therapies such as those described here are only part of a wave of potential treatments under development and clinical trial. The clinical trials will take years to complete, but in the coming decade we can expect cancer to become a curable disease.

Understanding of how mutations produce cancer has progressed to the point where promising potential therapies can be tested.

18.3 Recombination alters gene location.

An Overview of Recombination

Mutation is a change in the *content* of an organism's genetic message, but it is not the only source of genetic diversity. Diversity is also generated when existing elements of the genetic message move around within the genome. As an analogy, consider the pages of this book. A point mutation would correspond to a change in one or more of the letters on the pages. For example, “. . . in one or more of the letters of the pages” is a mutation of the previous sentence, in which an “n” is changed to an “f.” A significant alteration is also achieved, however, when we move the position of words, as in “. . . in one or more of the pages on the letters.” The change alters (and destroys) the meaning of the sentence by exchanging the position of the words “letters” and “pages.” This second kind of change, which represents an alteration in the genomic *location* of a gene or a fragment of a gene, demonstrates **genetic recombination**.

Gene Transfer

Viewed broadly, genetic recombination can occur by two mechanisms (table 18.5). In **gene transfer**, one chromosome or genome donates a segment to another chromosome or genome. The transfer of genes from the human immunodeficiency virus (HIV) to a human chromosome is an example of gene transfer. Because gene transfer occurs in both prokaryotes and eukaryotes, it is thought to be the more primitive of the two mechanisms.

Reciprocal Recombination

Reciprocal recombination is when two chromosomes trade segments. It is exemplified by the crossing over that occurs between homologous chromosomes during meiosis. Independent assortment during meiosis is another form of reciprocal recombination. Discussed in chapters 12 and 13, it is responsible for the 9:3:3:1 ratio of phenotypes in a dihybrid cross and occurs only in eukaryotes.

Genetic recombination is a change in the genomic association among genes. It often involves a change in the position of a gene or portion of a gene. Recombination of this sort may result from one-way gene transfer or reciprocal gene exchange.



FIGURE 18.23
A Nobel Prize for discovering gene transfer by transposition.
Barbara McClintock receiving her Nobel Prize in 1983.

Table 18.5 Classes of Genetic Recombination

Class	Occurrence
GENE TRANSFERS	
Conjugation	Occurs predominantly but not exclusively in bacteria and is targeted to specific locations in the genome
Transposition	Common in both bacteria and eukaryotes; genes move to new genomic locations, apparently at random
RECIPROCAL RECOMBINATIONS	
Crossing over	Requires the pairing of homologous chromosomes and may occur anywhere along their length
Unequal crossing over	The result of crossing over between mismatched segments; leads to gene duplication and deletion
Gene conversion	Occurs when homologous chromosomes pair and one is “corrected” to resemble the other
Independent assortment	Haploid cells produced by meiosis contain only one randomly selected member of each pair of homologous chromosomes

Gene Transfer

Genes are not fixed in their locations on chromosomes or the circular DNA molecules of bacteria; they can move around. Some genes move because they are part of small, circular, extrachromosomal DNA segments called **plasmids**. Plasmids enter and leave the main genome at specific places where a nucleotide sequence matches one present on the plasmid. Plasmids occur primarily in bacteria, in which the main genomic DNA can interact readily with other DNA fragments. About 5% of the DNA that occurs in a bacterium is plasmid DNA. Some plasmids are very small, containing only one or a few genes, while others are quite complex and contain many genes. Other genes move within **transposons**, which jump from one genomic position to another at random in both bacteria and eukaryotes.

Gene transfer by plasmid movement was discovered by Joshua Lederberg and Edward Tatum in 1947. Three years later, transposons were discovered by Barbara McClintock. However, her work implied that the position of genes in a genome need not be constant. Researchers accustomed to viewing genes as fixed entities, like beads on a string, did not readily accept the idea of transposons. Therefore, while Lederberg and Tatum were awarded a Nobel Prize for their discovery in 1958, McClintock did not receive the same recognition for hers until 1983 (figure 18.23).

Plasmid Creation

To understand how plasmids arise, consider a hypothetical stretch of bacterial DNA that contains two copies of the same nucleotide sequence. It is possible for the two copies to base-pair with each other and create a transient “loop,” or double duplex. All cells have recombination enzymes that can cause such double duplexes to undergo a **reciprocal exchange**, in which they exchange strands. As a result of the exchange, the loop is freed from the rest of the DNA molecule and becomes a plasmid (figure 18.24, steps 1–3). Any genes between the duplicated sequences (such as gene A in figure 18.24) are transferred to the plasmid.

Once a plasmid has been created by reciprocal exchange, DNA polymerase will replicate it if it contains a replication origin, often without the controls that restrict the main genome to one replication per cell division. Consequently, some plasmids may be present in multiple copies, others in just a few copies, in a given cell.

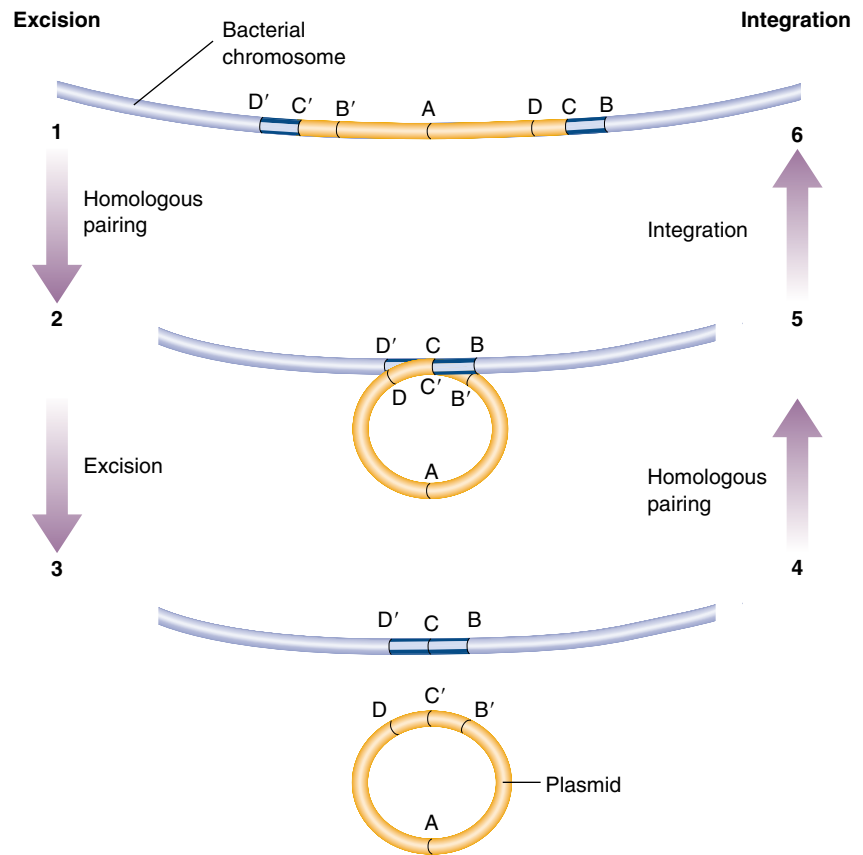


FIGURE 18.24

Integration and excision of a plasmid. Because the ends of the two sequences in the bacterial genome are the same (D', C', B', and D, C, B), it is possible for the two ends to pair. Steps 1–3 show the sequence of events if the strands exchange during the pairing. The result is excision of the loop and a free circle of DNA—a plasmid. Steps 4–6 show the sequence when a plasmid integrates itself into a bacterial genome.

Integration

A plasmid created by recombination can reenter the main genome the same way it left. Sometimes the region of the plasmid DNA that was involved in the original exchange, called the **recognition site**, aligns with a matching sequence on the main genome. If a recombination event occurs anywhere in the region of alignment, the plasmid will integrate into the genome (figure 18.24, steps 4–6). Integration can occur wherever any shared sequences exist, so plasmids may be integrated into the main genome at positions other than the one from which they arose. If a plasmid is integrated at a new position, it transfers its genes to that new position.

Transposons and plasmids transfer genes to new locations on chromosomes. Plasmids can arise from and integrate back into a genome wherever DNA sequences in the genome and in the plasmid match.

Gene Transfer by Conjugation

One of the startling discoveries Lederberg and Tatum made was that plasmids can pass from one bacterium to another. The plasmid they studied was part of the genome of *Escherichia coli*. It was given the name F for fertility factor because only cells which had that plasmid integrated into their DNA could act as plasmid donors. These cells are called Hfr cells (for “high-frequency recombination”). The F plasmid contains a DNA replication origin and several genes that promote its transfer to other cells. These genes encode protein subunits that assemble on the surface of the bacterial cell, forming a hollow tube called a **pilus**.

When the pilus of one cell (F⁺) contacts the surface of another cell that lacks a pilus, and therefore does not contain an F plasmid (F⁻), the pilus draws the two cells close together so that DNA can be exchanged (figure 18.25). First, the F plasmid binds to a site on the interior of the F⁺ cell just beneath the pilus. Then, by a process called **rolling-circle replication**, the F plasmid begins to copy its DNA at the binding point. As it is replicated, the single-stranded copy of the plasmid passes into the other cell. There a complementary strand is added, creating a new, stable F plasmid (figure 18.26). In this way, genes are passed from one bacterium to another. This transfer of genes between bacteria is called **conjugation**.

In an Hfr cell, with the F plasmid integrated into the main bacterial genome rather than free in the cytoplasm, the F plasmid can still organize the transfer of genes. In this case, the integrated F region binds beneath the pilus and initiates the *replication of the bacterial genome*, transferring the newly replicated portion to the recipient cell. Transfer proceeds as if the bacterial genome were simply a part of the F plasmid. By studying this phenomenon, researchers have been able to locate the positions of different genes in bacterial genomes (figure 18.27).

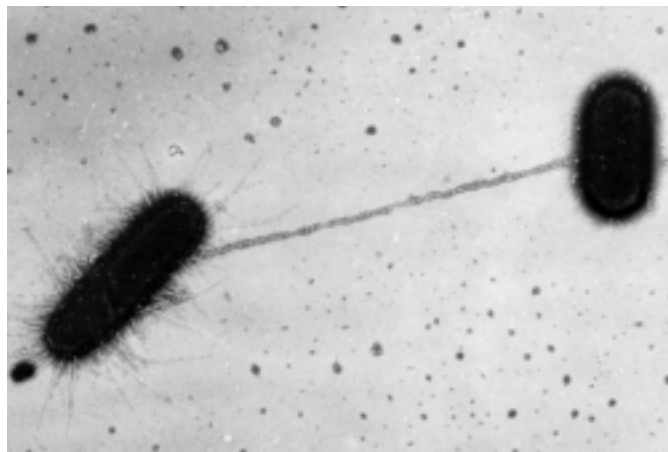


FIGURE 18.25

Contact by a pilus. The pilus of an F⁺ cell connects to an F⁻ cell and draws the two cells close together so that DNA transfer can occur.

Gene Transfer by Transposition

Like plasmids, transposons (figure 18.28) move from one genomic location to another. After spending many generations in one position, a transposon may abruptly move to a new position in the genome, carrying various genes along with it. Transposons encode an enzyme called **transposase**, that inserts the transposon into the genome (figure 18.29). Because this enzyme usually does not recognize any particular sequence on the genome, transposons appear to move to random destinations.

The movement of any given transposon is relatively rare: it may occur perhaps once in 100,000 cell generations. Although low, this rate is still about 10 times as frequent as

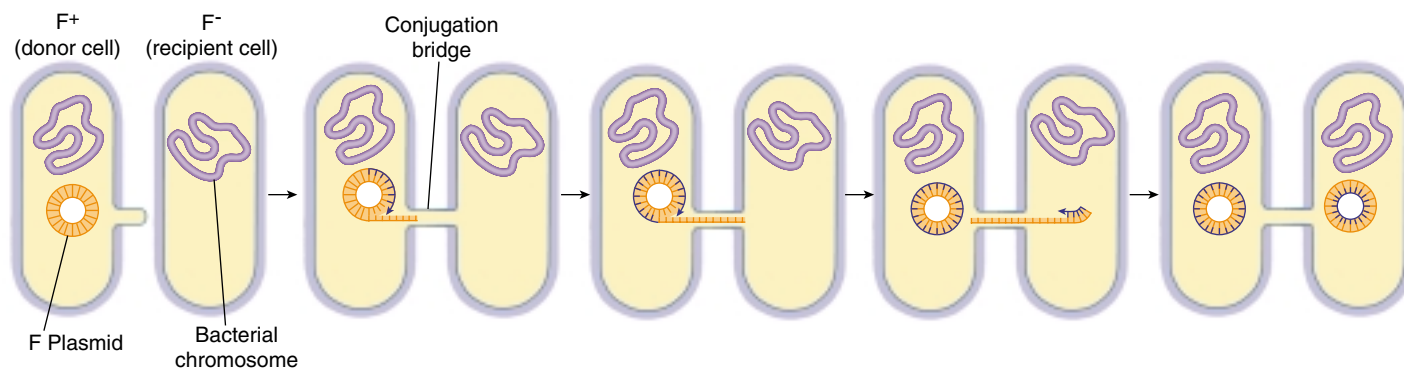


FIGURE 18.26

Gene transfer between bacteria. Donor cells (F⁺) contain an F plasmid that recipient cells (F⁻) lack. The F plasmid replicates itself and transfers the copy across a conjugation bridge. The remaining strand of the plasmid serves as a template to build a replacement. When the single strand enters the recipient cell, it serves as a template to assemble a double-stranded plasmid. When the process is complete, both cells contain a complete copy of the plasmid.

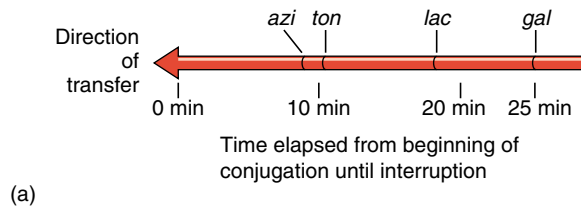
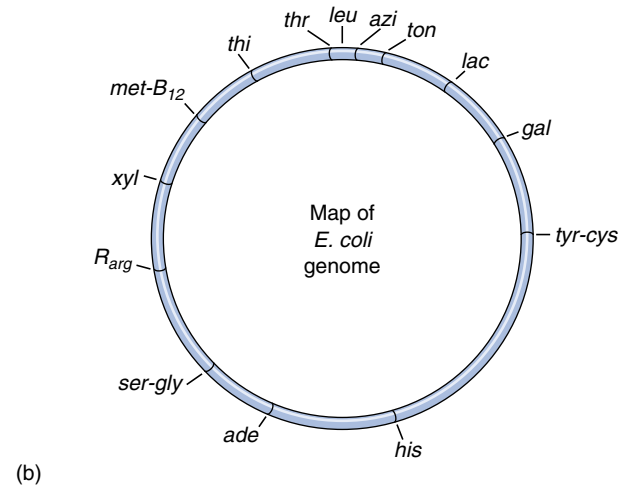


FIGURE 18.27

A conjugation map of the *E. coli* chromosome. Scientists have been able to break the *Escherichia coli* conjugation bridges by agitating the cell suspension rapidly in a blender. By agitating at different intervals after the start of conjugation, investigators can locate the positions of various genes along the bacterial genome. (a) The closer the genes are to the origin of replication, the sooner one has to turn on the blender to block their transfer. (b) Map of the *E. coli* genome developed using this method.



the rate at which random mutational changes occur. Furthermore, there are many transposons in most cells. Hence, over long periods of time, transposition can have an enormous evolutionary impact.

One way this impact can be felt is through mutation. The insertion of a transposon into a gene often destroys the gene's function, resulting in what is termed **insertional inactivation**. This phenomenon is thought to be the cause of a significant number of the spontaneous mutations observed in nature.

Transposition can also facilitate **gene mobilization**, the bringing together in one place of genes that are usually located at different positions in the genome. In bacteria, for example, a number of genes encode enzymes that make the bacteria resistant to antibiotics such as penicillin, and many of these genes are located on plasmids. The simultaneous exposure of bacteria to multiple antibiotics, a common medical practice some years ago, favors the persistence of plasmids that have managed to acquire several resistance genes. Transposition can rapidly generate such composite plasmids, called **resistance transfer factors** (RTFs), by moving antibiotic resistance genes from several plasmids to one. Bacteria possessing RTFs are thus able to survive treatment with a wide variety of antibiotics. RTFs are thought to be responsible for much of the recent difficulty in treating hospital-engendered *Staphylococcus aureus* infections and the new drug-resistant strains of tuberculosis.

Plasmids transfer copies of bacterial genes (and even entire genomes) from one bacterium to another. Transposition is the one-way transfer of genes to a randomly selected location in the genome. The genes move because they are associated with mobile genetic elements called transposons.

FIGURE 18.28

Transposon. Transposons form characteristic stem-and-loop structures called “lollipops” because their two ends have the same nucleotide sequence as inverted repeats. These ends pair together to form the stem of the lollipop.

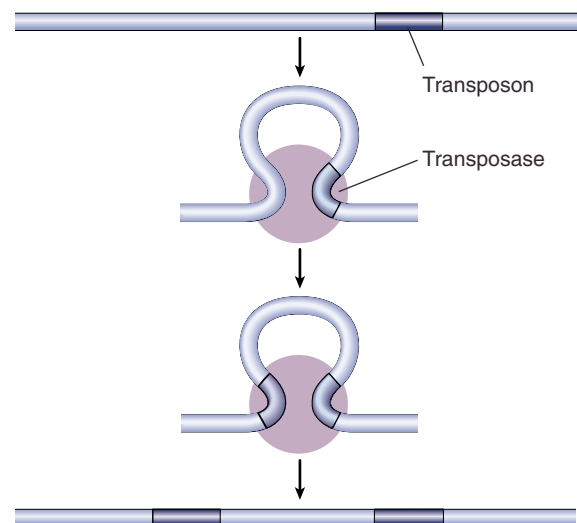
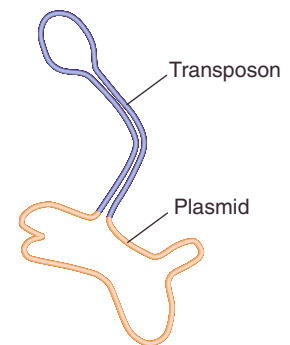


FIGURE 18.29

Transposition. Transposase does not recognize any particular DNA sequence; rather, it selects one at random, moving the transposon to a random location. Some transposons leave a copy of themselves behind when they move.

Reciprocal Recombination

In the second major mechanism for producing genetic recombination, reciprocal recombination, two homologous chromosomes exchange all or part of themselves during the process of meiosis.

Crossing Over

As we saw in chapter 12, crossing over occurs in the first prophase of meiosis, when two homologous chromosomes line up side by side within the synaptonemal complex. At this point, the homologues exchange DNA strands at one or more locations. This exchange of strands can produce chromosomes with new combinations of alleles.

Imagine, for example, that a giraffe has genes encoding neck length and leg length at two different loci on one of its chromosomes. Imagine further that a recessive mutation occurs at the neck length locus, leading after several rounds of independent assortment to some individuals that are homozygous for a variant “long-neck” allele. Similarly, a recessive mutation at the leg length locus leads to homozygous “long-leg” individuals.

It is very unlikely that these two mutations would arise at the same time in the same individual because the probability of two independent events occurring together is the product of their individual probabilities. If the spontaneous occurrence of both mutations in a single individual were the only way to produce a giraffe with both a long neck and long legs, it would be extremely unlikely that such an individual would ever occur. Because of recombination, however, a crossover in the interval between the two genes could in one meiosis produce a chromosome bearing both variant alleles. This ability to reshuffle gene combinations rapidly is what makes recombination so important to the production of natural variation.

Unequal Crossing Over

Reciprocal recombination can occur in any region along two homologous chromosomes with sequences similar enough to permit close pairing. Mistakes in pairing occasionally happen when several copies of a sequence exist in different locations on a chromosome. In such cases, one copy of a sequence may line up with one of the duplicate copies instead of with its homologous copy. Such misalignment causes slipped mispairing, which, as we discussed earlier, can lead to small deletions and frame-shift mutations. If a crossover occurs in the pairing region, it will result in unequal crossing over because the two homologues will exchange segments of unequal length.

In unequal crossing over, one chromosome gains extra copies of the multicopy sequences, while the other chromosome loses them (figure 18.30). This process can gener-

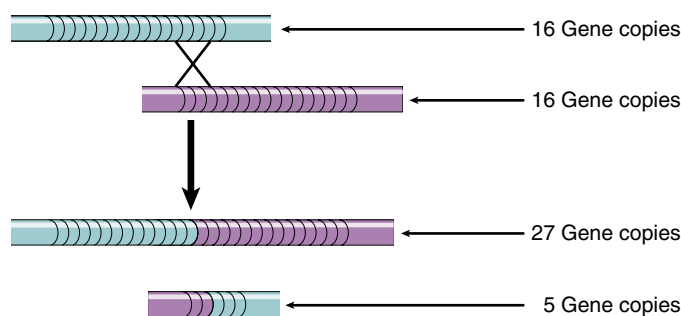


FIGURE 18.30

Unequal crossing over. When a repeated sequence pairs out of register, a crossover within the region will produce one chromosome with fewer gene copies and one with more. Much of the gene duplication that has occurred in eukaryotic evolution may well be the result of unequal crossing over.

ate a chromosome with hundreds of copies of a particular gene, lined up side by side in tandem array.

Because the genomes of most eukaryotes possess multiple copies of transposons scattered throughout the chromosomes, unequal crossing over between copies of transposons located in different positions has had a profound influence on gene organization in eukaryotes. As we shall see later, most of the genes of eukaryotes appear to have been duplicated one or more times during their evolution.

Gene Conversion

Because the two homologues that pair within a synaptonemal complex are not identical, some nucleotides in one homologue are not complementary to their counterpart in the other homologue with which it is paired. These occasional nonmatching pairs of nucleotides are called **mismatch pairs**.

As you might expect, the cell’s error-correcting machinery is able to detect mismatch pairs. If a mismatch is detected during meiosis, the enzymes that “proofread” new DNA strands during DNA replication correct it. The mismatched nucleotide in one of the homologues is excised and replaced with a nucleotide complementary to the one in the other homologue. Its base-pairing partner in the first homologue is then replaced, producing two chromosomes with the same sequence. This error correction causes one of the mismatched sequences to convert into the other, a process called **gene conversion**.

Unequal crossing over is a crossover between chromosomal regions that are similar in nucleotide sequence but are not homologous. Gene conversion is the alteration of one homologue by the cell’s error-detection and repair system to make it resemble the other homologue.

Trinucleotide Repeats

In 1991, a new kind of change in the genetic material was reported, one that involved neither changes in the identity of nucleotides (mutation) nor changes in the position of nucleotide sequences (recombination), but rather an increase in the number of copies of repeated trinucleotide sequences. Called **trinucleotide repeats**, these changes appear to be the root cause of a surprisingly large number of inherited human disorders.

The first examples of disorders resulting from the expansion of trinucleotide repeat sequences were reported in individuals with *fragile X syndrome* (the most common form of developmental disorder) and *spinal muscular atrophy*. In both disorders, genes containing runs of repeated nucleotide triplets (CGG in fragile X syndrome and CAG in spinal muscular atrophy) exhibit large increases in copy number. In individuals with fragile X syndrome, for example, the CGG sequence is repeated hundreds of times (figure 18.31), whereas in normal individuals it repeats only about 30 times.

Ten additional human genes are now known to have alleles with expanded trinucleotide repeats (figure 18.32). Many (but not all) of these alleles are GC-rich. A few of the alleles appear benign, but most are associated with heritable disorders, including Huntington's disease, myotonic dystrophy, and a variety of neurological ataxias. In each case, the expansion transmits as a dominant trait. Often the repeats are found within the exons of their genes, but sometimes, as in the case of fragile X syndrome, they are located outside the coding segment. Furthermore, although the repeat number is stably transmitted in normal families, it shows marked instability once it has abnormally expanded. Siblings often exhibit unique repeat lengths.

As the repeat number increases, disease severity tends to increase in step. In fragile X syndrome, the CGG triplet number first increases from the normal stable range of 5 to 55 times (the most common allele has 29 repeats) to an unstable number of repeats ranging from 50 to 200, with no

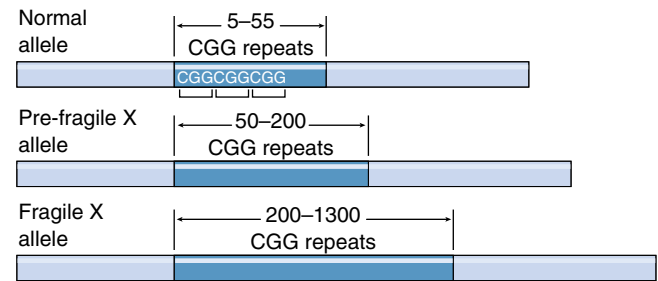


FIGURE 18.31

CGG repeats in fragile X alleles. The CGG triplet is repeated approximately 30 times in normal alleles. Individuals with pre-fragile X alleles show no detectable signs of the syndrome but do have increased numbers of CGG repeats. In fragile X alleles, the CGG triplet repeats hundreds of times.

detectable effect. In offspring, the number increases markedly, with copy numbers ranging from 200 to 1300, with significant mental retardation (see figure 18.31). Similarly, the normal allele for myotonic dystrophy has 5 GTC repeats. Mildly affected individuals have about 50, and severely affected individuals have up to 1000.

Trinucleotide repeats appear common in human genes, but their function is unknown. Nor do we know the mechanism behind trinucleotide repeat expansion. It may involve unequal crossing over, which can readily produce copy-number expansion, or perhaps some sort of stutter in the DNA polymerase when it encounters a run of triplets. The fact that di- and tetranucleotide repeat expansions are not found seems an important clue. Undoubtedly, further examples of this remarkable class of genetic change will be reported in the future. Considerable research is currently focused on this extremely interesting area.

Many human genes contain runs of a trinucleotide sequence. Their function is unknown, but if the copy number expands, hereditary disorders often result.

	CGG	GAA	CAG	CTG
Repeated trinucleotide	Exon 1	Intron 1	Exon 2	Intron 2
Condition	Fragile X syndrome Fragile site 11B Fragile XE syndrome	Friedreich's ataxia	Spinal and bulbar muscular atrophy Spinocerebellar ataxia type 1 Huntington's disease Dentatorubral-pallidoluysian atrophy Machado-Joseph disease	Myotonic dystrophy

FIGURE 18.32

A hypothetical gene showing the locations and types of trinucleotide repeats associated with various human diseases. The CGG repeats of fragile X syndrome, fragile XE syndrome, and fragile site 11B occur in the first exon of their respective genes. GAA repeats characteristic of Friedreich's ataxia exist in the first intron of its gene. The genes for five different diseases, including Huntington's disease, have CAG repeats within their second exons. Lastly, the myotonic dystrophy gene contains CTG repeats within the third exon.

18.4 Genomes are continually evolving.

Classes of Eukaryotic DNA

The two main mechanisms of genetic recombination, gene transfer and reciprocal recombination, are directly responsible for the architecture of the eukaryotic chromosome. They determine where genes are located and how many copies of each exist. To understand how recombination shapes the genome, it is instructive to compare the effects of recombination in bacteria and eukaryotes.

Comparing Bacterial and Eukaryotic DNA Sequences

Bacterial genomes are relatively simple, containing genes that almost always occur as single copies. Unequal crossing over between repeated transposition elements in their circular DNA molecules tends to *delete* material, fostering the maintenance of a minimum genome size (figure 18.33a). For this reason, these genomes are very tightly packed, with few or no noncoding nucleotides. Recall the efficient use of space in the organization of the *lac* genes described in chapter 16.

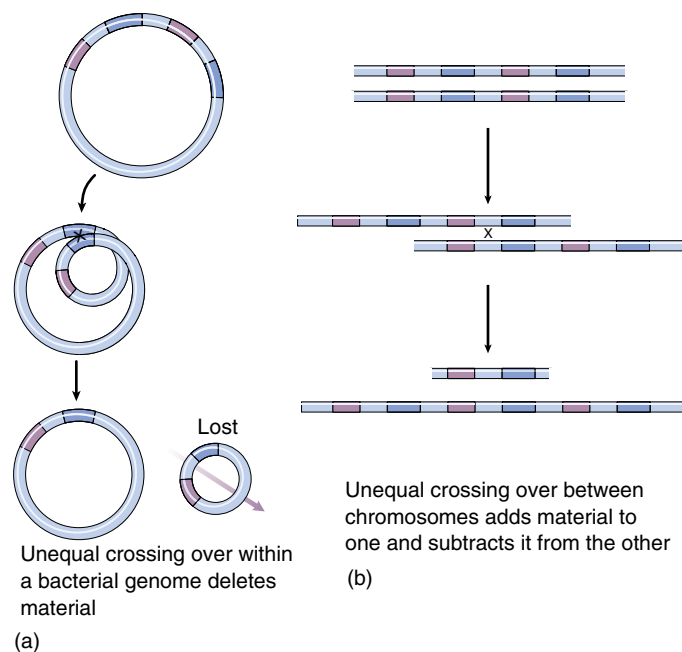


FIGURE 18.33

Unequal crossing over has different consequences in bacteria and eukaryotes. (a) Bacteria have a circular DNA molecule, and a crossover between duplicate regions within the molecule deletes the intervening material. (b) In eukaryotes, with two versions of each chromosome, crossing over adds material to one chromosome; thus, gene amplification occurs in that chromosome.

In eukaryotes, by contrast, the introduction of *pairs* of homologous chromosomes (presumably because of their importance in repairing breaks in double-stranded DNA) has led to a radically different situation. Unequal crossing over between homologous chromosomes tends to promote the *duplication* of material rather than its reduction (figure 18.33b). Consequently, eukaryotic genomes have been in a constant state of flux during the course of their evolution. Multiple copies of genes have evolved, some of them subsequently diverging in sequence to become different genes, which in turn have duplicated and diverged.

Six different classes of eukaryotic DNA sequences are commonly recognized, based on the number of copies of each (table 18.6).

Transposons

Transposons exist in multiple copies scattered about the genome. In *Drosophila*, for example, more than 30 different transposons are known, most of them present at 20 to 40 different sites throughout the genome. In all, the known transposons of *Drosophila* account for perhaps 5% of its DNA. Mammalian genomes contain fewer kinds of transposons than the genomes of many other organisms, although the transposons in mammals are repeated more often. The family of human transposons called *ALU* elements, for example, typically occurs about 300,000 times in each cell. Transposons are transcribed but appear to play no functional role in the life of the cell. As noted earlier in this chapter, many transposition events carry transposons into the exon portions of genes, disrupting the function of the protein specified by the gene transcript. These insertional inactivations are thought to be responsible for many naturally occurring mutations.

Tandem Clusters

A second class consists of DNA sequences that are repeated many times, one copy following another in tandem array. By transcribing all of the copies in these **tandem clusters** simultaneously, a cell can rapidly obtain large amounts of the product they encode. For example, the genes encoding rRNA are present in several hundred copies in most eukaryotic cells. Because these clusters are active sites of rRNA synthesis, they are readily visible in cytological preparations, where they are called **nucleolar organizer regions**. When transcription of the rRNA gene clusters ceases during cell division, the nucleolus disappears from view under the microscope, but it reappears when transcription begins again.

The genes present in a tandem cluster are very similar in sequence but not always identical; some may differ by one

Table 18.6 Classes of DNA Sequences Found in Eukaryotes	
Class	Description
Transposons	Thousands of copies scattered around the genome
Tandem clusters	Clusters containing hundreds of nearly identical copies of a gene
Multigene families	Clusters of a few to several hundred copies of related but distinctly different genes
Satellite DNA	Short sequences present in millions of copies per genome
Dispersed pseudogenes	Inactive members of a multigene family separated from other members of the family
Single-copy genes	Genes that exist in only one copy in the genome

or a few nucleotides. Each gene in the cluster is separated from its neighbors by a short “spacer” sequence that is not transcribed. Unlike the genes, the spacers in a cluster vary considerably in sequence and in length.

Multigene Families

As we have learned more about the nucleotide sequences of eukaryotic genomes, it has become apparent that many genes exist as parts of **multigene families**, groups of related but distinctly different genes that often occur together in a cluster. Multigene families differ from tandem clusters in that they contain far fewer genes (from three to several hundred), and those genes differ much more from one another than the genes in tandem clusters. Despite their differences, the genes in a multigene family are clearly related in their sequences, making it likely that they arose from a single ancestral sequence through a series of unequal crossing over events. For example, studies of the evolution of the hemoglobin multigene family indicate that the ancestral globin gene is at least 800 million years old. By the time modern fishes evolved, this ancestral gene had already duplicated, forming the α and β forms. Later, after the evolutionary divergence of amphibians and reptiles, these two globin gene forms moved apart on the chromosome; the mechanism of this movement is not known, but it may have involved transposition. In mammals, two more waves of duplication occurred to produce the array of 11 globin genes found in the human genome. Three of these genes are silent, encoding nonfunctional proteins. Other genes are expressed only during embryonic (ζ and ϵ) or fetal (γ) development. Only four (δ , β , α_1 , and α_2) encode the polypeptides that make up adult human hemoglobin.

Satellite DNA

Some short nucleotide sequences are repeated several million times in eukaryotic genomes. These sequences are collectively called **satellite DNA** and occur outside the main body of DNA. Almost all satellite DNA is either clustered around the centromere or located near the ends of the chromosomes, at the telomeres. These regions of the chromosomes remain highly condensed, tightly coiled, and un-

transcribed throughout the cell cycle; this suggests that satellite DNA may serve some sort of structural function, such as initiating the pairing of homologous chromosomes in meiosis. About 4% of the human genome consists of satellite DNA.

Dispersed Pseudogenes

Silent copies of a gene, inactivated by mutation, are called **pseudogenes**. Such mutations may affect the gene’s promoter (see chapter 16), shift the reading frame of the gene, or produce a small deletion. While some pseudogenes occur within a multigene family cluster, others are widely separated. The latter are called **dispersed pseudogenes** because they are believed to have been dispersed from their original position within a multigene family cluster. No one suspected the existence of dispersed pseudogenes until a few years ago, but they are now thought to be of major evolutionary significance in eukaryotes.

Single-Copy Genes

Ever since eukaryotes appeared, processes such as unequal crossing over between different copies of transposons have repeatedly caused segments of chromosomes to duplicate, and it appears that no portion of the genome has escaped this phenomenon. The duplication of genes, followed by the conversion of some of the copies into pseudogenes, has probably been the major source of “new” genes during the evolution of eukaryotes. As pseudogenes accumulate mutational changes, a fortuitous combination of changes may eventually result in an active gene encoding a protein with different properties. When that new gene first arises, it is a **single-copy gene**, but in time it, too, will be duplicated. Thus, a single-copy gene is but one stage in the cycle of duplication and divergence that has characterized the evolution of the eukaryotic genome.

Gene sequences in eukaryotes vary greatly in copy number, some occurring many thousands of times, others only once. Many protein-encoding eukaryotic genes occur in several nonidentical copies, some of them not transcribed.



Summary

Questions

Media Resources

18.1 Mutations are changes in the genetic message.

- A mutation is any change in the hereditary message.
- Mutations that change one or a few nucleotides are called point mutations. They may arise as a result of damage from ionizing or ultraviolet radiation, chemical mutagens, or errors in pairing during DNA replication.

1. What are pyrimidine dimers? How do they form? How are they repaired? What may happen if they are not repaired?
2. Explain how slipped mispairing can cause deletions and frame-shift mutations.



- Mutations
- DNA repair



- Experiment: Luria/Delbrück-Mutations Occur in Random

18.2 Cancer results from mutation of growth-regulating genes.

- Cancer is a disease in which the regulatory controls that normally restrain cell division are disrupted.
- A variety of environmental factors, including ionizing radiation, chemical mutagens, and viruses, have been implicated in causing cancer.
- The best way to avoid getting cancer is to avoid exposure to mutagens, especially those in cigarette smoke.

3. What is transfection? What has it revealed about the genetic basis of cancer?
4. About how many genes can be mutated to cause cancer? Why do most cancers require mutations in multiple genes?



- Polymerase Chain Reaction
- Student Research: Age and Breast Cancer
- On Science Articles: Understanding Cancer
- Evidence Links Cigarette Smoking to Lung Cancer
- Deadly Cancer is Becoming More Common

18.3 Recombination alters gene location.

- Recombination is the creation of new gene combinations. It includes changes in the position of genes or fragments of genes as well as the exchange of entire chromosomes during meiosis.
- Genes may be transferred between bacteria when they are included within small circles of DNA called plasmids.
- Transposition is the random movement of genes within transposons to new locations in the genome. It is responsible for many naturally occurring mutations, as the insertion of a transposon into a gene often inactivates the gene.
- Crossing over involves a physical exchange of genetic material between homologous chromosomes during the close pairing that occurs in meiosis. It may produce chromosomes that have different combinations of alleles.

5. What is genetic recombination? What mechanisms produce it? Which of these mechanisms occurs in prokaryotes, and which occurs in eukaryotes?
6. What is a plasmid? What is a transposon? How are plasmids and transposons similar, and how are they different?
7. What are mismatched pairs? How are they corrected? What effect does this correction have on the genetic message?



- Recombinant DNA/Technology
- Experiments: McClintock/Stern

18.4 Genomes are continually evolving.

- Satellite sequences are short sequences of nucleotides repeated millions of times.
- Tandem clusters are genes that occur in thousands of copies grouped together at one or a few sites on a chromosome. These genes encode products that are required by the cell in large amounts.
- Multigene families consist of copies of genes clustered at one site on a chromosome that diverge in sequence more than the genes in a tandem cluster.

8. What kinds of genes exist in multigene families? How are these families thought to have evolved?
9. What are pseudogenes? How might they have been involved in the evolution of single-copy genes?



- Student research: DNA repair in fish

19

Gene Technology

Concept Outline

19.1 The ability to manipulate DNA has led to a new genetics.

Restriction Endonucleases. Enzymes that cleave DNA at specific sites allow DNA segments from different sources to be spliced together.

Using Restriction Endonucleases to Manipulate Genes. Fragments produced by cleaving DNA with restriction endonucleases can be spliced into plasmids, which can be used to insert the DNA into host cells.

19.2 Genetic engineering involves easily understood procedures.

The Four Stages of a Genetic Engineering Experiment. Gene engineers cut DNA into fragments that they splice into vectors that carry the fragments into cells.

Working with Gene Clones. Gene technology is used in a variety of procedures involving DNA manipulation.

19.3 Biotechnology is producing a scientific revolution.

DNA Sequence Technology. The complete nucleotide sequence of the genomes of many organisms are now known. The unique DNA of every individual can be used to identify sperm, blood, or other tissues.

Biochips. Biochips are squares of glass etched with DNA strands and can be used for genetic screening.

Medical Applications. Many drugs and vaccines are now produced with gene technology.

Agricultural Applications. Gene engineers have developed crops resistant to pesticides and pests, as well as commercially superior animals.

Cloning. Recent experiments show it is possible to clone agricultural animals, a result with many implications for both agriculture and society.

Stem Cells. Both embryonic stem cells and tissue-specific stem cells can potentially be used to repair or replace damaged tissue.

Ethics and Regulation. Genetic engineering raises important questions about danger and privacy.



FIGURE 19.1

A famous plasmid. The circular molecule in this electron micrograph is pSC101, the first plasmid used successfully to clone a vertebrate gene. Its name comes from the fact that it was the one-hundred-and-first plasmid isolated by Stanley Cohen.

Over the past decades, the development of new and powerful techniques for studying and manipulating DNA has revolutionized genetics (figure 19.1). These techniques have allowed biologists to intervene directly in the genetic fate of organisms for the first time. In this chapter, we will explore these technologies and consider how they apply to specific problems of great practical importance. Few areas of biology will have as great an impact on our future lives.

19.1 The ability to manipulate DNA has led to a new genetics.

Restriction Endonucleases

In 1980, geneticists used the relatively new technique of gene splicing, which we will describe in this chapter, to introduce the human gene that encodes **interferon** into a bacterial cell's genome. Interferon is a rare blood protein that increases human resistance to viral infection, and medical scientists have been interested in its possible usefulness in cancer therapy. This possibility was difficult to investigate before 1980, however, because purification of the large amounts of interferon required for clinical testing would have been prohibitively expensive, given interferon's scarcity in the blood. An inexpensive way to produce interferon was needed, and introducing the gene responsible for its production into a bacterial cell made that possible. The cell that had acquired the human interferon gene proceeded to produce interferon at a rapid rate, and to grow and divide. Soon there were millions of interferon-producing bacteria in the culture, all of them descendants of the cell that had originally received the human interferon gene.

The Advent of Genetic Engineering

This procedure of producing a line of genetically identical cells from a single altered cell, called **cloning**, made every cell in the culture a miniature factory for producing interferon. The human insulin gene has also been cloned in bacteria, and now large amounts of insulin, a hormone essential for treating some forms of diabetes, can be manufactured at relatively little expense. Beyond these clinical applications, cloning and related molecular techniques are used to obtain basic information about how genes are put together and regulated. The interferon experiment and others like it marked the beginning of a new genetics, **genetic engineering**.

The essence of genetic engineering is the ability to cut DNA into recognizable pieces and rearrange those pieces in different ways. In the interferon experiment, a piece of DNA carrying the interferon gene was inserted into a plasmid, which then carried the gene into a bacterial cell. Most other genetic engineering approaches have used the same general strategy, bringing the gene of interest into the target cell by first incorporating it into a plasmid or an infective virus. To make these experiments work, one must be able to cut the source DNA (human DNA in the interferon experiment, for example) and the plasmid DNA in such a way that the desired fragment of source DNA can be spliced permanently into the plasmid. This cutting is performed by enzymes that recognize and cleave specific se-

quences of nucleotides in DNA. These enzymes are the basic tools of genetic engineering.

Discovery of Restriction Endonucleases

Scientific discoveries often have their origins in seemingly unimportant observations that receive little attention by researchers before their general significance is appreciated. In the case of genetic engineering, the original observation was that bacteria use enzymes to defend themselves against viruses.

Most organisms eventually evolve means of defending themselves from predators and parasites, and bacteria are no exception. Among the natural enemies of bacteria are bacteriophages, viruses that infect bacteria and multiply within them. At some point, they cause the bacterial cells to burst, releasing thousands more viruses. Through natural selection, some types of bacteria have acquired powerful weapons against these viruses: they contain enzymes called **restriction endonucleases** that fragment the viral DNA as soon as it enters the bacterial cell. Many restriction endonucleases recognize specific nucleotide sequences in a DNA strand, bind to the DNA at those sequences, and cleave the DNA at a particular place within the recognition sequence.

Why don't restriction endonucleases cleave the bacterial cells' own DNA as well as that of the viruses? The answer to this question is that bacteria modify their own DNA, using other enzymes known as **methylases** to add methyl ($-\text{CH}_3$) groups to some of the nucleotides in the bacterial DNA. When nucleotides within a restriction endonuclease's recognition sequence have been methylated, the endonuclease cannot bind to that sequence. Consequently, the bacterial DNA is protected from being degraded at that site. Viral DNA, on the other hand, has not been methylated and therefore is not protected from enzymatic cleavage.

How Restriction Endonucleases Cut DNA

The sequences recognized by restriction endonucleases are typically four to six nucleotides long, and they are often palindromes. This means the nucleotides at one end of the recognition sequence are complementary to those at the other end, so that the two strands of the DNA duplex have the same nucleotide sequence running in opposite directions for the length of the recognition sequence. Two important consequences arise from this arrangement of nucleotides.

First, because the same recognition sequence occurs on both strands of the DNA duplex, the restriction endonuclease can bind to and cleave both strands, effectively cutting the DNA in half. This ability to cut across both strands is almost certainly the reason that restriction endonucleases have evolved to recognize nucleotide sequences with twofold rotational symmetry.

Second, because the bond cleaved by a restriction endonuclease is typically not positioned in the center of the recognition sequence to which it binds, and because the DNA strands are antiparallel, the cut sites for the two strands of a duplex are offset from each other (figure 19.2). After cleavage, each DNA fragment has a single-stranded end a few nucleotides long. The single-stranded ends of the two fragments are complementary to each other.

Why Restriction Endonucleases Are So Useful

There are hundreds of bacterial restriction endonucleases, and each one has a specific recognition sequence. By chance, a particular endonuclease's recognition sequence is likely to occur somewhere in any given sample of DNA; the shorter the sequence, the more often it will arise by chance within a sample. Therefore, a given restriction endonuclease can probably cut DNA from any source into fragments. Each fragment will have complementary single-stranded ends characteristic of that endonuclease. Because of their complementarity, these single-stranded ends can pair with each other (consequently, they are sometimes called "sticky ends"). Once their ends have paired, two fragments can then be joined together with the aid of the enzyme **DNA ligase**, which re-forms the phosphodiester bonds of DNA. What makes restriction endonucleases so valuable for genetic engineering is the fact that *any* two fragments produced by the same restriction endonuclease can be joined together. Fragments of elephant and ostrich DNA cleaved by the same endonuclease can be joined to one another as readily as two bacterial DNA fragments.

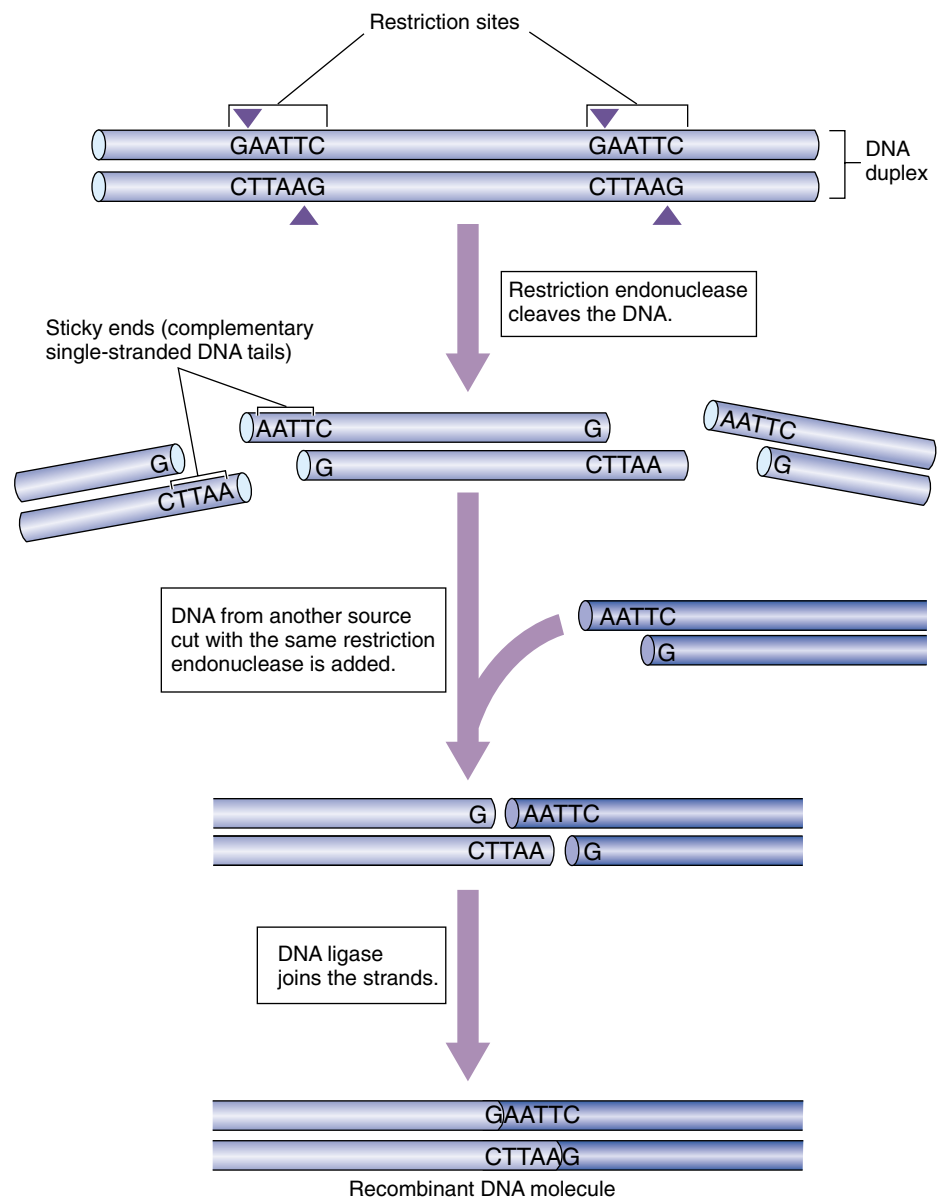


FIGURE 19.2

Many restriction endonucleases produce DNA fragments with "sticky ends." The restriction endonuclease *EcoRI* always cleaves the sequence GAATTC between G and A. Because the same sequence occurs on both strands, both are cut. However, the two sequences run in opposite directions on the two strands. As a result, single-stranded tails are produced that are complementary to each other, or "sticky."

Genetic engineering involves manipulating specific genes by cutting and rearranging DNA. A restriction endonuclease cleaves DNA at a specific site, generating in most cases two fragments with short single-stranded ends. Because these ends are complementary to each other, any pair of fragments produced by the same endonuclease, from any DNA source, can be joined together.

Using Restriction Endonucleases to Manipulate Genes

A **chimera** is a mythical creature with the head of a lion, body of a goat, and tail of a serpent. Although no such creatures existed in nature, biologists have made chimeras of a more modest kind through genetic engineering.

Constructing pSC101

One of the first chimeras was manufactured from a bacterial plasmid called a resistance transfer factor by American geneticists Stanley Cohen and Herbert Boyer in 1973. Cohen and Boyer used a restriction endonuclease called *EcoRI*, which is obtained from *Escherichia coli*, to cut the plasmid into fragments. One fragment, 9000 nucleotides in length, contained both the origin of replication necessary for replicating the plasmid and a gene that conferred resistance to the antibiotic tetracycline (*tet^r*). Because both ends of this fragment were cut by the same restriction endonuclease, they could be ligated to form a circle, a smaller plasmid Cohen dubbed pSC101 (figure 19.3).

Using pSC101 to Make Recombinant DNA

Cohen and Boyer also used *EcoRI* to cleave DNA that coded for rRNA that they had isolated from an adult amphibian, the African clawed frog, *Xenopus laevis*. They then mixed the fragments of *Xenopus* DNA with pSC101 plasmids that had been “reopened” by *EcoRI* and allowed bacterial cells to take up DNA from the mixture. Some of the bacterial cells immediately became resistant to tetracycline, indicating that they had incorporated the pSC101 plasmid with its antibiotic-resistance gene. Furthermore, some of these pSC101-containing bacteria also began to produce frog ribosomal RNA! Cohen and Boyer concluded that the frog rRNA gene must have been inserted into the pSC101 plasmids in those bacteria. In other words, the two ends of the pSC101 plasmid, produced by cleavage with *EcoRI*, had joined to the two ends of a frog DNA fragment that contained the rRNA gene, also cleaved with *EcoRI*.

The pSC101 plasmid containing the frog rRNA gene is a true chimera, an entirely new genome that never existed in nature and never would have evolved by natural means. It is a form of **recombinant DNA**—that is, DNA created

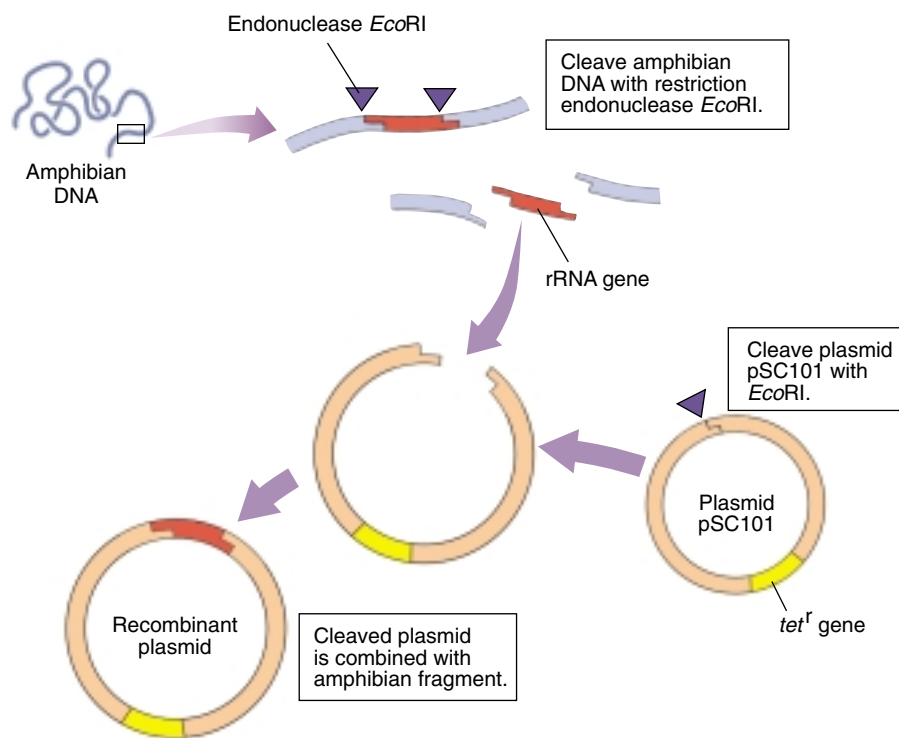


FIGURE 19.3

One of the first genetic engineering experiments. This diagram illustrates how Cohen and Boyer inserted an amphibian gene encoding rRNA into pSC101. The plasmid contains a single site cleaved by the restriction endonuclease *EcoRI*; it also contains *tet^r*, a gene which confers resistance to the antibiotic tetracycline. The rRNA-encoding gene was inserted into pSC101 by cleaving the amphibian DNA and the plasmid with *EcoRI* and allowing the complementary sequences to pair.

in the laboratory by joining together pieces of different genomes to form a novel combination.

Other Vectors

The introduction of foreign DNA fragments into host cells has become common in molecular genetics. The genome that carries the foreign DNA into the host cell is called a **vector**. Plasmids, with names like pUC18 can be induced to make hundreds of copies of themselves and thus of the foreign genes they contain. Much larger pieces of DNA can be introduced using YAKs (yeast artificial chromosomes) as a vector instead of a plasmid. Not all vectors have bacterial targets. Animal viruses such as the human cold virus adenovirus, for example, are serving as vectors to carry genes into monkey and human cells, and animal genes have even been introduced into plant cells.

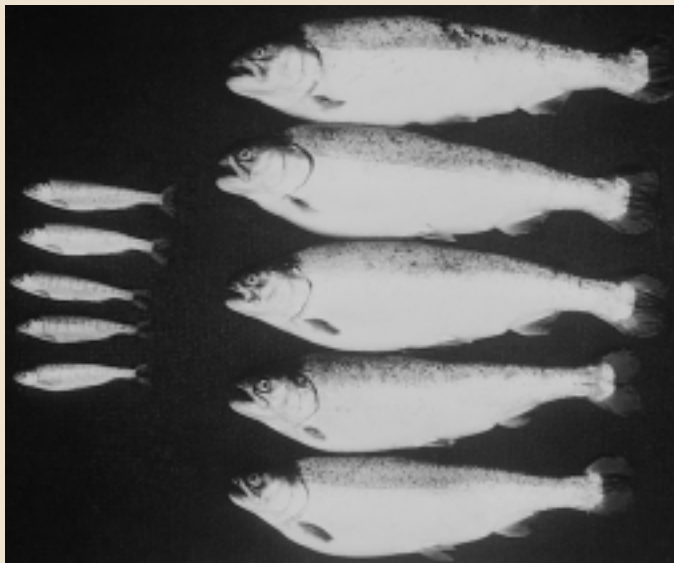
One of the first recombinant genomes produced by genetic engineering was a bacterial plasmid into which an amphibian ribosomal RNA gene was inserted. Viruses can also be used as vectors to insert foreign DNA into host cells and create recombinant genomes.

Examples of Gene Manipulation



HERMAN THE WONDER BULL

GenPharm, a California biotechnology company, engineered Herman, a bull that possesses the gene for human lactoferrin (HLF). HLF confers antibacterial and iron transport properties to humans. Many of Herman's female offspring now produce milk containing HLF, and GenPharm intends to build a herd of transgenic cows for the large-scale commercial production of HLF.



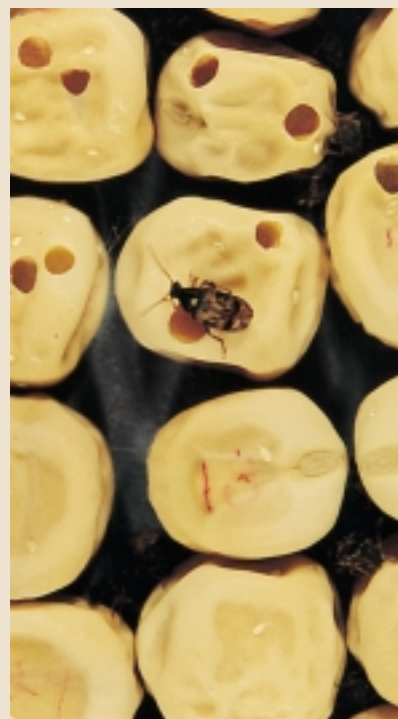
SUPER SALMON!

Canadian fisheries scientists have inserted recombinant growth hormone genes into developing salmon embryos, creating the first transgenic salmon. Not only do these transgenic fish have shortened production cycles, they are, on an average, 11 times heavier than nontransgenic salmon! The implications for the fisheries industry and for worldwide food production are obvious.



WILT-PROOF FLOWERS

Ethylene, the plant hormone that causes fruit to ripen, also causes flowers to wilt. Researchers at Purdue have found the gene that makes flower petals respond to ethylene by wilting and replaced it with a gene insensitive to ethylene. The transgenic carnations they produced lasted for 3 weeks after cutting, while normal carnations last only 3 days.



WEEVIL-PROOF PEAS

Not only has gene technology afforded agriculture viral and pest control in the field, it has also provided a pest control technique for the storage bin. A team of U.S. and Australian scientists have engineered a gene that is expressed only in the seed of the pea plant. The enzyme inhibitor encoded by this gene inhibits feeding by weevils, one of the most notorious pests affecting stored crops. The worldwide ramifications are significant as up to 40% of stored grains are lost to pests.

19.2 Genetic engineering involves easily understood procedures.

The Four Stages of a Genetic Engineering Experiment

Like the experiment of Cohen and Boyer, most genetic engineering experiments consist of four stages: DNA cleavage, production of recombinant DNA, cloning, and screening.

Stage 1: DNA Cleavage

A restriction endonuclease is used to cleave the source DNA into fragments. Because the endonuclease's recognition sequence is likely to occur many times within the source DNA, cleavage will produce a large number of different fragments. A different set of fragments will be

obtained by employing endonucleases that recognize different sequences. The fragments can be separated from one another according to their size by electrophoresis (figure 19.4).

Stage 2: Production of Recombinant DNA

The fragments of DNA are inserted into plasmids or viral vectors, which have been cleaved with the same restriction endonuclease as the source DNA.

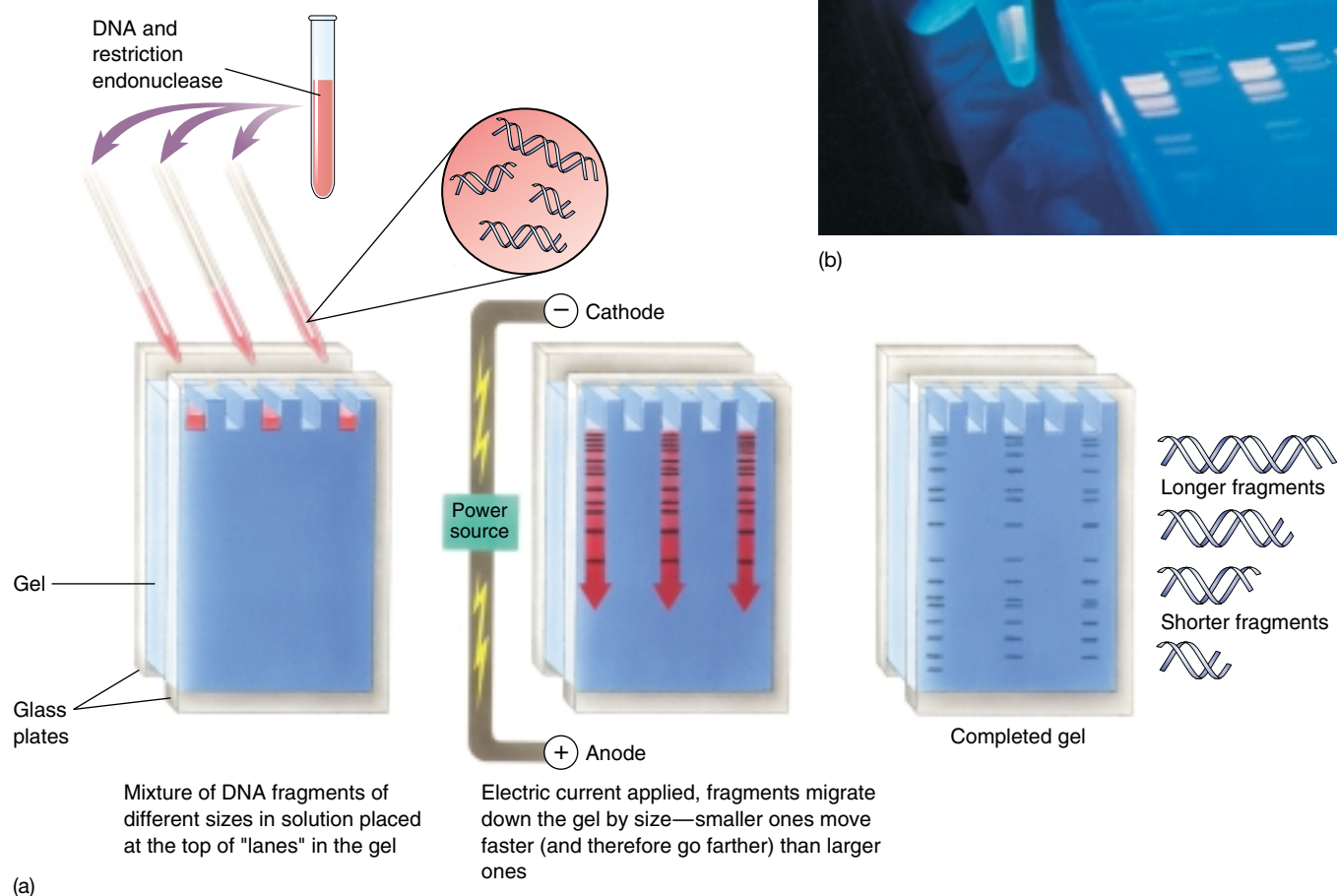


FIGURE 19.4

Gel electrophoresis. (a) After restriction endonucleases have cleaved the DNA, the fragments are loaded on a gel, and an electric current is applied. The DNA fragments migrate through the gel, with bigger ones moving more slowly. The fragments can be visualized easily, as the migrating bands fluoresce in UV light when stained with ethidium bromide. (b) In the photograph, one band of DNA has been excised from the gel for further analysis and can be seen glowing in the tube the technician holds.

Stage 3: Cloning

The plasmids or viruses serve as vectors that can introduce the DNA fragments into cells—usually, but not always, bacteria (figure 19.5). As each cell reproduces, it

forms a clone of cells that all contain the fragment-bearing vector. Each clone is maintained separately, and all of them together constitute a clone library of the original source DNA.

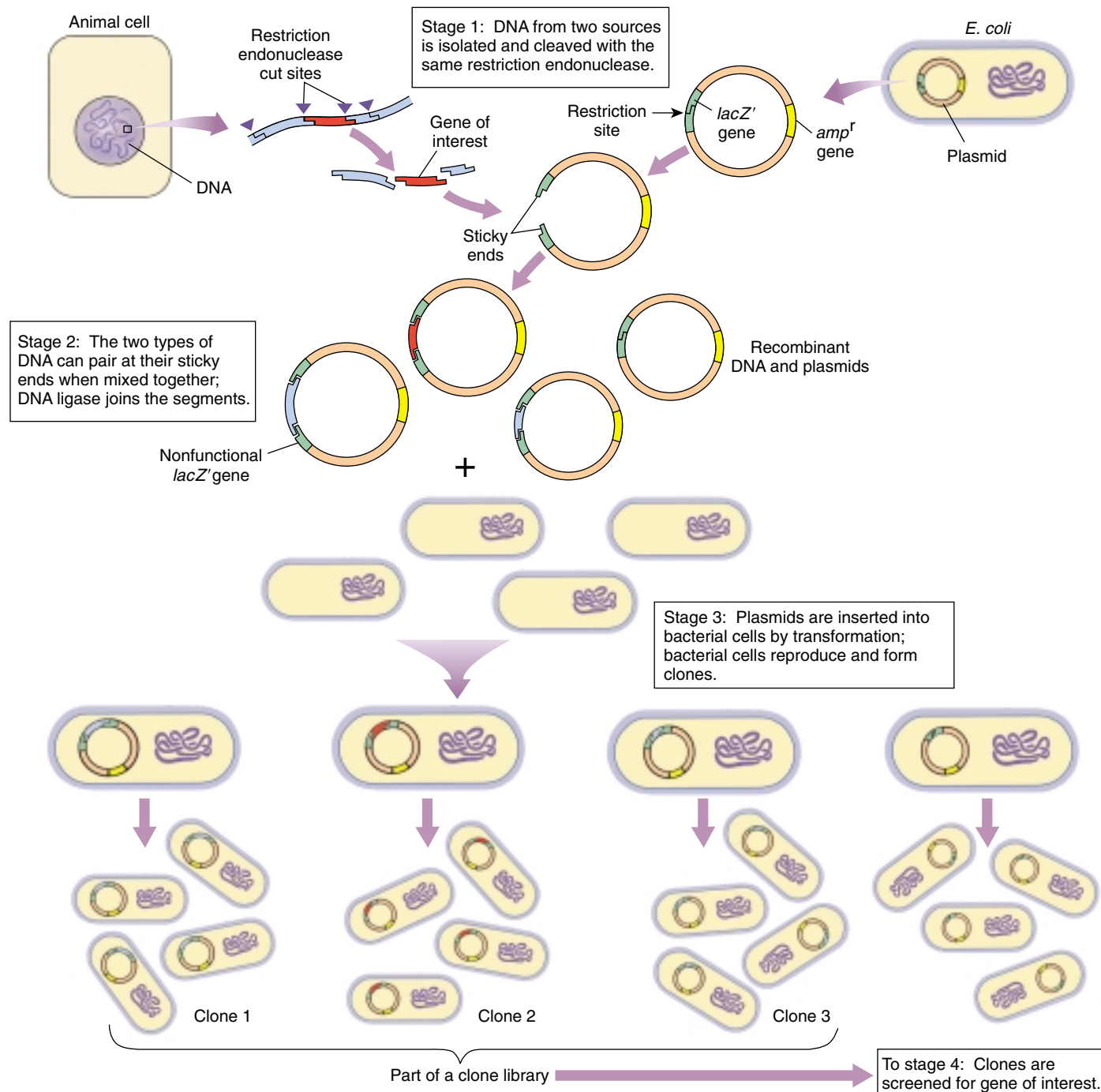


FIGURE 19.5

Stages in a genetic engineering experiment. In stage 1, DNA containing the gene of interest (in this case, from an animal cell) and DNA from a plasmid are cleaved with the same restriction endonuclease. The genes *amp^r* and *lacZ'* are contained within the plasmid and used for screening a clone (stage 4). In stage 2, the two cleaved sources of DNA are mixed together and pair at their sticky ends. In stage 3, the recombinant DNA is inserted into a bacterial cell, which reproduces and forms clones. In stage 4, the bacterial clones will be screened for the gene of interest.

Stage 4: Screening

The clones containing a specific DNA fragment of interest, often a fragment that includes a particular gene, are identified from the clone library. Let's examine this stage in more detail, as it is generally the most challenging in any genetic engineering experiment.

4-I: The Preliminary Screening of Clones. Investigators initially try to eliminate from the library any clones that do not contain vectors, as well as clones whose vectors do not contain fragments of the source DNA. The first category of clones can be eliminated by employing a vector with a gene that confers resistance to a specific antibiotic, such as tetracycline, penicillin, or ampicillin. In figure 19.6a, the gene *amp^r* is incorporated into the plasmid and confers resistance to the antibiotic ampicillin. When the clones are exposed to a medium containing that antibiotic, only clones that contain the vector will be resistant to the antibiotic and able to grow.

One way to eliminate clones with vectors that do not have an inserted DNA fragment is to use a vector that, in addition to containing antibiotic resistance genes, contains the *lacZ'* gene which is required to produce β -galactosidase, an enzyme that enables the cells to metabolize the sugar, X-gal. Metabolism of X-gal results in the formation of a blue reaction product, so any cells whose vectors contain a functional version of this gene will turn blue in the presence of X-gal (figure 19.6b). However, if one uses a restriction endonuclease whose recognition sequence lies within the *lacZ'* gene, the gene will be interrupted when recombinants are formed, and the cell will be unable to metabolize X-gal. Therefore, cells with vectors that contain a fragment of source DNA should remain colorless in the presence of X-gal.

Any cells that are able to grow in a medium containing the antibiotic but don't turn blue in the medium with X-gal must have incorporated a vector with a fragment of source DNA. Identifying cells that have a *specific* fragment of the source DNA is the next step in screening clones.

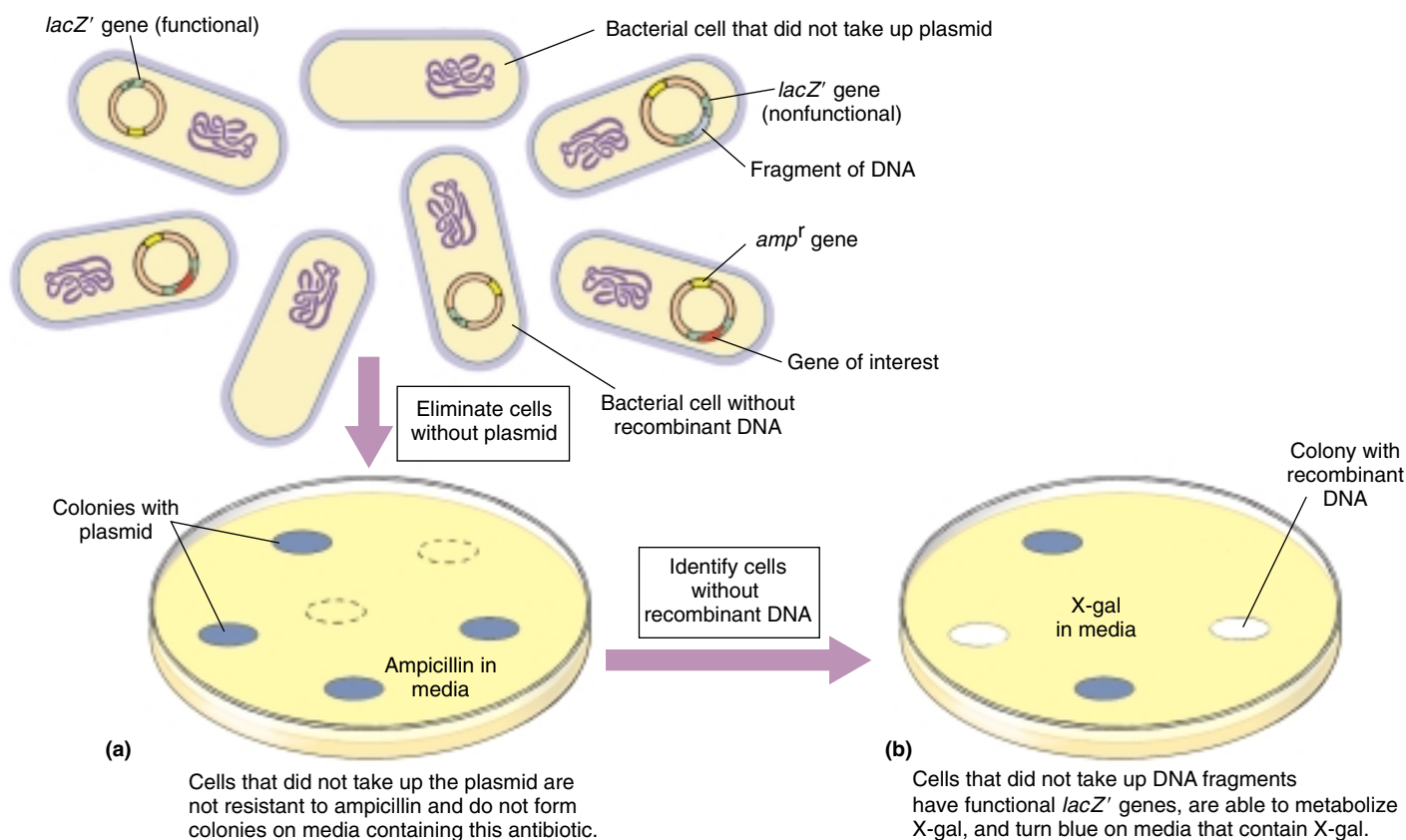


FIGURE 19.6

Stage 4-I: Using antibiotic resistance and X-gal as preliminary screens of restriction fragment clones. Bacteria are transformed with recombinant plasmids that contain a gene (*amp^r*) that confers resistance to the antibiotic ampicillin and a gene (*lacZ'*) that is required to produce β -galactosidase, the enzyme which enables the cells to metabolize the sugar X-gal. (a) Only those bacteria that have incorporated a plasmid will be resistant to ampicillin and will grow on a medium that contains the antibiotic. (b) Ampicillin-resistant bacteria will be able to metabolize X-gal if their plasmid does *not* contain a DNA fragment inserted in the *lacZ'* gene; such bacteria will turn blue when grown on a medium containing X-gal. Bacteria with a plasmid that has a DNA fragment inserted within the *lacZ'* gene will not be able to metabolize X-gal and, therefore, will remain colorless in the presence of X-gal.

4-II: Finding the Gene of Interest. A clone library may contain anywhere from a few dozen to many thousand individual fragments of source DNA. Many of those fragments will be identical, so to assemble a complete library of the entire source genome, several hundred thousand clones could be required. A complete *Drosophila* (fruit fly) library, for example, contains more than 40,000 different clones; a complete human library consisting of fragments 20 kilobases long would require close to a million clones. To search such an immense library for a clone that contains a fragment corresponding to a particular gene requires ingenuity, but many different approaches have been successful.

The most general procedure for screening clone libraries to find a particular gene is **hybridization** (figure 19.7). In this method, the cloned genes form base-pairs with complementary sequences on another nucleic acid. The complementary nucleic acid is called a **probe** because it is used to probe for the presence of the gene of interest. At least part of the nucleotide sequence of the gene of interest must be known to be able to construct the probe.

In this method of screening, bacterial colonies containing an inserted gene are grown on agar. Some cells are

transferred to a filter pressed onto the colonies, forming a replica of the plate. The filter is then treated with a solution that denatures the bacterial DNA and that contains a radioactively labeled probe. The probe hybridizes with complementary single-stranded sequences on the bacterial DNA.

When the filter is laid over photographic film, areas that contain radioactivity will expose the film (autoradiography). Only colonies which contain the gene of interest hybridize with the radioactive probe and emit radioactivity onto the film. The pattern on the film is then compared to the original master plate, and the gene-containing colonies may be identified.

Genetic engineering generally involves four stages: cleaving the source DNA; making recombinants; cloning copies of the recombinants; and screening the cloned copies for the desired gene. Screening can be achieved by making the desired clones resistant to certain antibiotics and giving them other properties that make them readily identifiable.

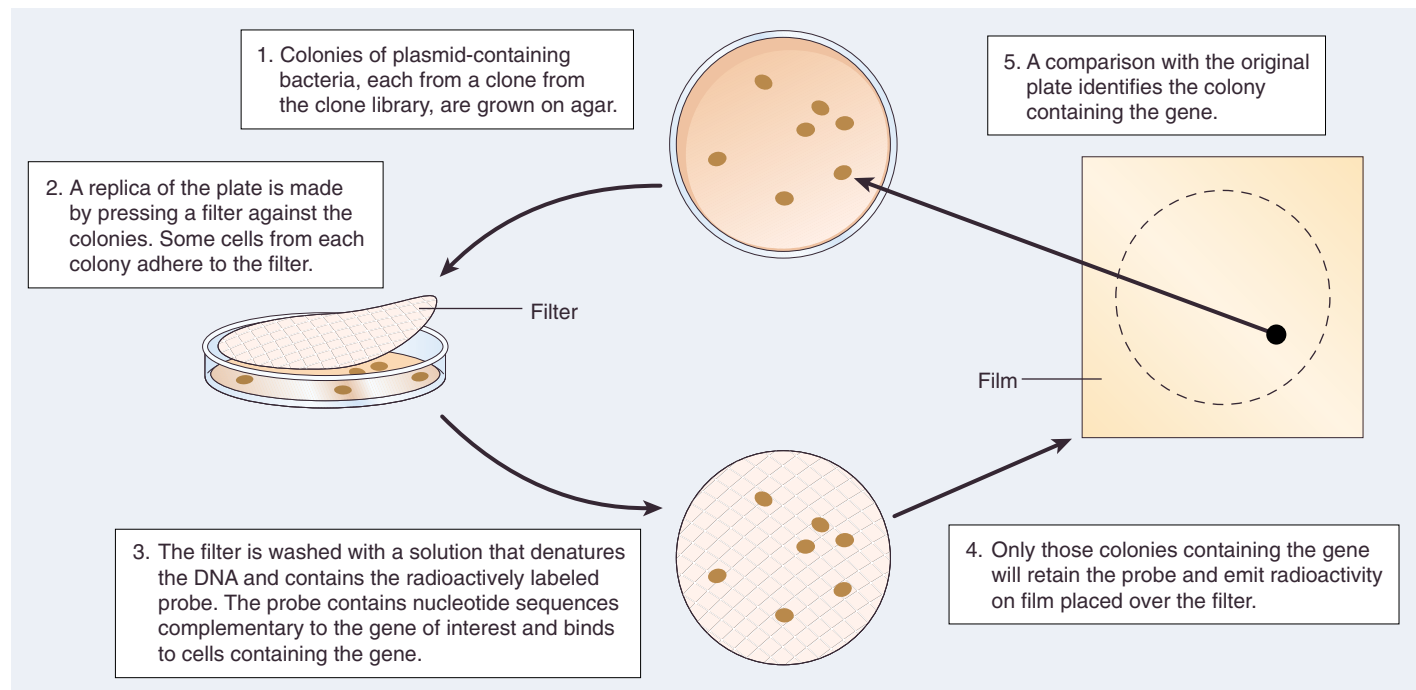


FIGURE 19.7

Stage 4-II: Using hybridization to identify the gene of interest. (1) Each of the colonies on these bacterial culture plates represents millions of clones descended from a single cell. To test whether a certain gene is present in any particular clone, it is necessary to identify colonies whose cells contain DNA that hybridizes with a probe containing DNA sequences complementary to the gene. (2) Pressing a filter against the master plate causes some cells from each colony to adhere to the filter. (3) The filter is then washed with a solution that denatures the DNA and contains the radioactively labeled probe. (4) Only those colonies that contain DNA that hybridizes with the probe, and thus contain the gene of interest, will expose film in autoradiography. (5) The film is then compared to the master plate to identify the gene-containing colony.

Working with Gene Clones

Once a gene has been successfully cloned, a variety of procedures are available to characterize it.

Getting Enough DNA to Work with: The Polymerase Chain Reaction

Once a particular gene is identified within the library of DNA fragments, the final requirement is to make multiple copies of it. One way to do this is to insert the identified fragment into a bacterium; after repeated cell divisions, millions of cells will contain copies of the fragment. A far more direct approach, however, is to use DNA polymerase to copy the gene sequence of interest through the **polymerase chain reaction (PCR)** (figure 19.8). Kary Mullis developed PCR in 1983 while he was a staff chemist at the Cetus Corporation; in 1993, it won him the Nobel Prize in Chemistry. PCR can amplify specific sequences or add sequences (such as endonuclease recognition sequences) as primers to cloned DNA. There are three steps in PCR:

Step 1: Denaturation. First, an excess of primer (typically a synthetic sequence of 20 to 30 nucleotides) is mixed with the DNA fragment to be amplified. This mixture of primer and fragment is heated to about 98° C. At this temperature, the double-stranded DNA fragment dissociates into single strands.

Step 2: Annealing of Primers. Next, the solution is allowed to cool to about 60°C. As it cools, the single strands of DNA reassociate into double strands. However, because of the large excess of primer, each strand of the fragment base-pairs with a complementary primer flanking the region to be amplified, leaving the rest of the fragment single-stranded.

Step 3: Primer Extension. Now a very heat-stable type of DNA polymerase, called Taq polymerase (after the thermophilic bacterium *Thermus aquaticus*, from which Taq is extracted) is added, along with a supply of all four nucleotides. Using the primer, the polymerase copies the rest of the fragment as if it were replicating DNA. When it is done, the primer has been lengthened into a complementary copy of the entire single-stranded fragment. Because *both* DNA strands are replicated, there are now two copies of the original fragment.

Steps 1 to 3 are now repeated, and the two copies become four. It is not necessary to add any more polymerase, as the heating step does not harm this particular enzyme. Each heating and cooling cycle, which can be as short as 1 or 2 minutes, doubles the number of DNA molecules. After 20 cycles, a single fragment produces more than one million (2^{20}) copies! In a few hours, 100 billion copies of the fragment can be manufactured.

PCR, now fully automated, has revolutionized many aspects of science and medicine because it allows the investigation of minute samples of DNA. In criminal investiga-

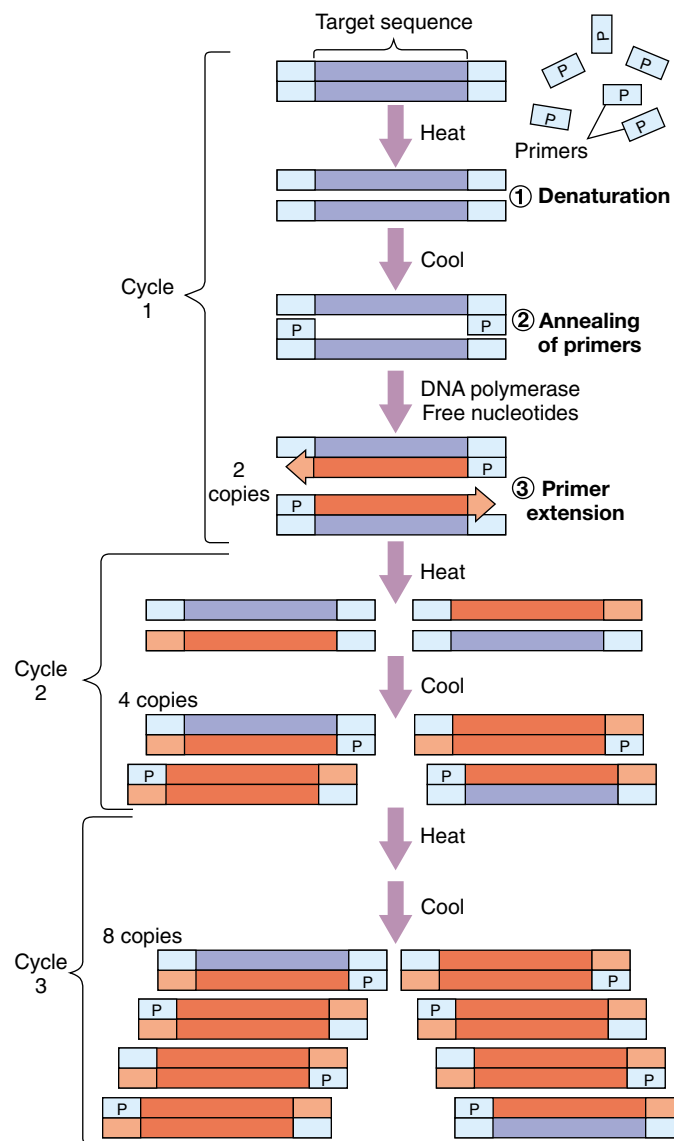


FIGURE 19.8

The polymerase chain reaction. (1) *Denaturation.* A solution containing primers and the DNA fragment to be amplified is heated so that the DNA dissociates into single strands. (2) *Annealing of primers.* The solution is cooled, and the primers bind to complementary sequences on the DNA flanking the region to be amplified. (3) *Primer extension.* DNA polymerase then copies the remainder of each strand, beginning at the primer. Steps 1–3 are then repeated with the replicated strands. This process is repeated many times, each time doubling the number of copies, until enough copies of the DNA fragment exist for analysis.

tions, “DNA fingerprints” are prepared from the cells in a tiny speck of dried blood or at the base of a single human hair. Physicians can detect genetic defects in very early embryos by collecting a few sloughed-off cells and amplifying their DNA. PCR could also be used to examine the DNA of historical figures such as Abraham Lincoln and of now-extinct species, as long as even a minuscule amount of their DNA remains intact.

Identifying DNA: Southern Blotting

Once a gene has been cloned, it may be used as a probe to identify the same or a similar gene in another sample (figure 19.9). In this procedure, called a **Southern blot**, DNA from the sample is cleaved into restriction fragments with a restriction endonuclease, and the fragments are spread apart by gel electrophoresis. The double-stranded helix of each DNA fragment is then denatured into single strands by making the pH of the gel basic, and the gel is “blotted”

with a sheet of nitrocellulose, transferring some of the DNA strands to the sheet. Next, a probe consisting of purified, single-stranded DNA corresponding to a specific gene (or mRNA transcribed from that gene) is poured over the sheet. Any fragment that has a nucleotide sequence complementary to the probe’s sequence will hybridize (base-pair) with the probe. If the probe has been labeled with ^{32}P , it will be radioactive, and the sheet will show a band of radioactivity where the probe hybridized with the complementary fragment.

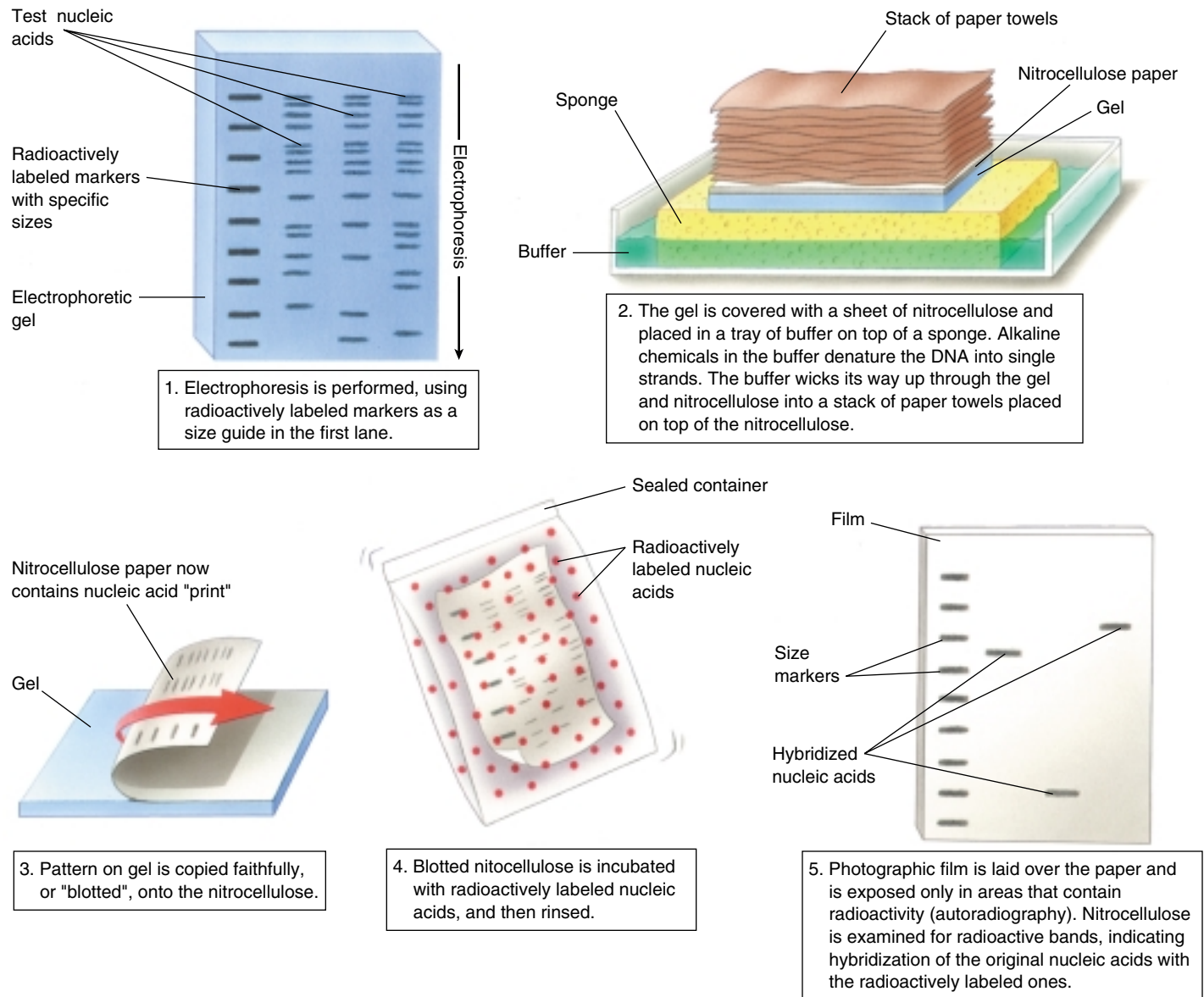


FIGURE 19.9

The Southern blot procedure. E. M. Southern developed this procedure in 1975 to enable DNA fragments of interest to be visualized in a complex sample containing many other fragments of similar size. The DNA is separated on a gel, then transferred (“blotted”) onto a solid support medium such as nitrocellulose paper or a nylon membrane. It is then incubated with a radioactive single-strand copy of the gene of interest, which hybridizes to the blot at the location(s) where there is a fragment with a complementary sequence. The positions of radioactive bands on the blot identify the fragments of interest.

Distinguishing Differences in DNA: RFLP Analysis

Often a researcher wishes not to find a specific gene, but rather to identify a particular *individual* using a specific gene as a marker. One powerful way to do this is to analyze **restriction fragment length polymorphisms**, or **RFLPs** (figure 19.10). Point mutations, sequence repetitions, and transposons (see chapter 18) that occur within or between the restriction endonuclease recognition sites will alter the length of the DNA fragments (restriction fragments) the restriction endonucleases produce. DNA from different individuals rarely has exactly the same array of restriction sites and distances between sites, so the population is said to be polymorphic (having many forms) for their restriction fragment patterns. By cutting a DNA sample with a particular restriction endonuclease, separating the fragments according to length on an electrophoretic gel, and then using a radioactive probe to identify the fragments on the gel, one can obtain a pattern of bands often unique for each region of DNA analyzed. These “DNA fingerprints” are used in forensic analysis during criminal investigations. RFLPs are also useful as markers to identify particular groups of people at risk for some genetic disorders.

Making an Intron-Free Copy of a Eukaryotic Gene

Recall from chapter 15 that eukaryotic genes are encoded in exons separated by numerous nontranslated introns. When the gene is transcribed to produce the primary transcript, the introns are cut out during RNA processing to produce the mature mRNA transcript. When transferring eukaryotic genes into bacteria, it is desirable to transfer DNA already processed this way, instead of the raw eukaryotic DNA, because bacteria lack the enzymes to carry out the processing. To do this, genetic engineers first isolate from the cytoplasm the mature mRNA corresponding to a particular gene. They then use an enzyme called reverse transcriptase to make a DNA version of the mature mRNA transcript (figure 19.11). The single strand of DNA can then serve as a template for the synthesis of a complementary strand. In this way, one can produce a double-stranded molecule of DNA that contains a gene lacking introns. This molecule is called **complementary DNA**, or **cDNA**.

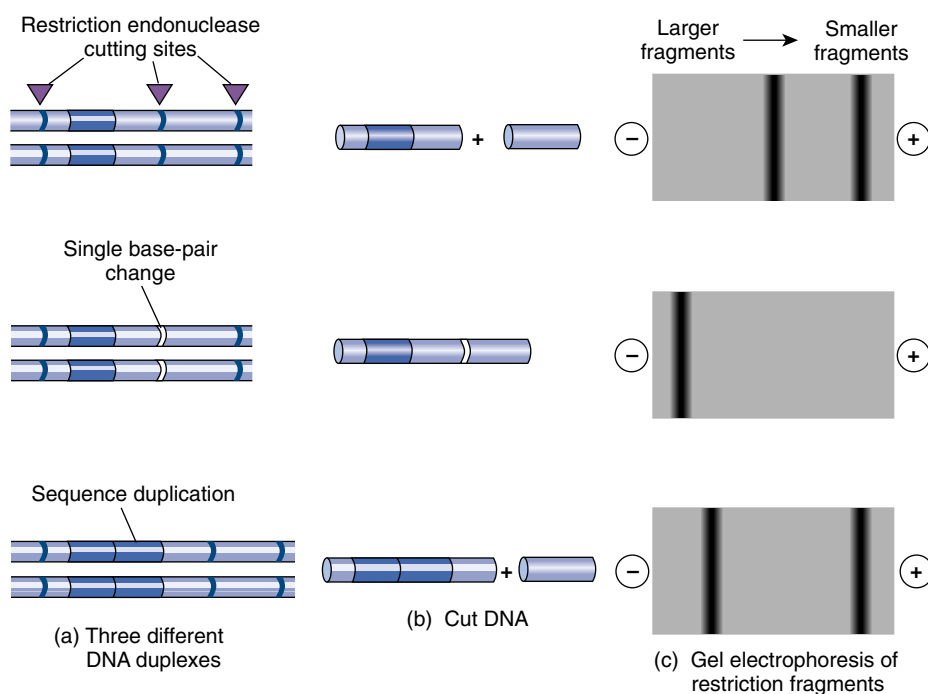


FIGURE 19.10

Restriction fragment length polymorphism (RFLP) analysis. (a) Three samples of DNA differ in their restriction sites due to a single base-pair substitution in one case and a sequence duplication in another case. (b) When the samples are cut with a restriction endonuclease, different numbers and sizes of fragments are produced. (c) Gel electrophoresis separates the fragments, and different banding patterns result.

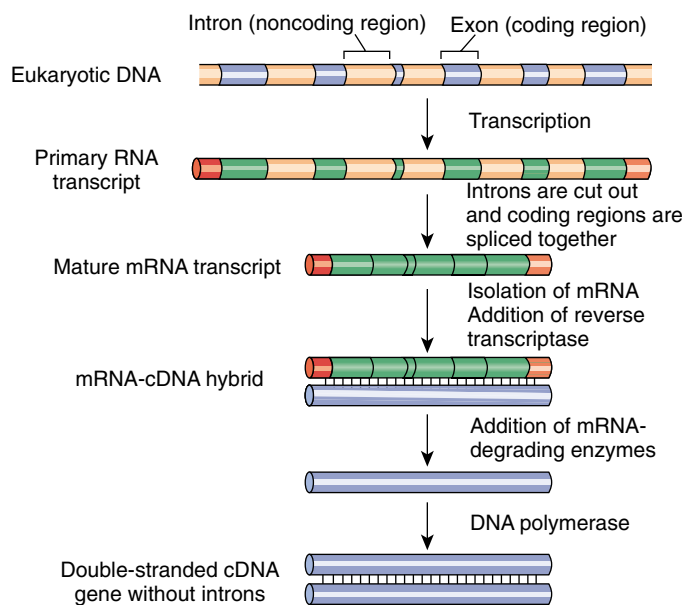


FIGURE 19.11

The formation of cDNA. A mature mRNA transcript is isolated from the cytoplasm of a cell. The enzyme reverse transcriptase is then used to make a DNA strand complementary to the processed mRNA. That newly made strand of DNA is the template for the enzyme DNA polymerase, which assembles a complementary DNA strand along it, producing cDNA, a double-stranded DNA version of the intron-free mRNA.

Sequencing DNA: The Sanger Method

Most DNA sequencing is currently done using the “chain termination” technique developed initially by Frederick Sanger, for which he earned his *second* Nobel Prize (figure 19.12). (1) A short single-stranded primer is added to the end of a single-stranded DNA fragment of unknown sequence. The primer provides a 3′ end for DNA polymerase. (2) The primed fragment is added, along with DNA polymerase and a supply of all four deoxynucleotides (d-nucleotides), to four synthesis tubes. Each contains a different *dideoxynucleotide* (dd-nucleotide); such nucleotides lack both the 2′ and the 3′ —OH groups and are thus chain-terminating. The first tube, for example, contains ddATP and stops synthesis whenever ddA is incorporated into DNA instead of dATP. Because of the relatively low concentration of ddATP compared to dATP, ddA will

not necessarily be added to the first A site; this tube will contain a series of fragments of different lengths, corresponding to the different distances the polymerase traveled from the primer before a ddA was incorporated. (3) These fragments can be separated according to size by electrophoresis. (4) A radioactive label (here dATP*) allows the fragments to be visualized on X-ray film, and the newly made sequence can be read directly from the film. Try it. (5) The original fragment has the complementary sequence.

Techniques such as Southern blotting and PCR enable investigators to identify specific genes and produce them in large quantities, while RFLP analysis and the Sanger method identify individuals and unknown gene sequences.

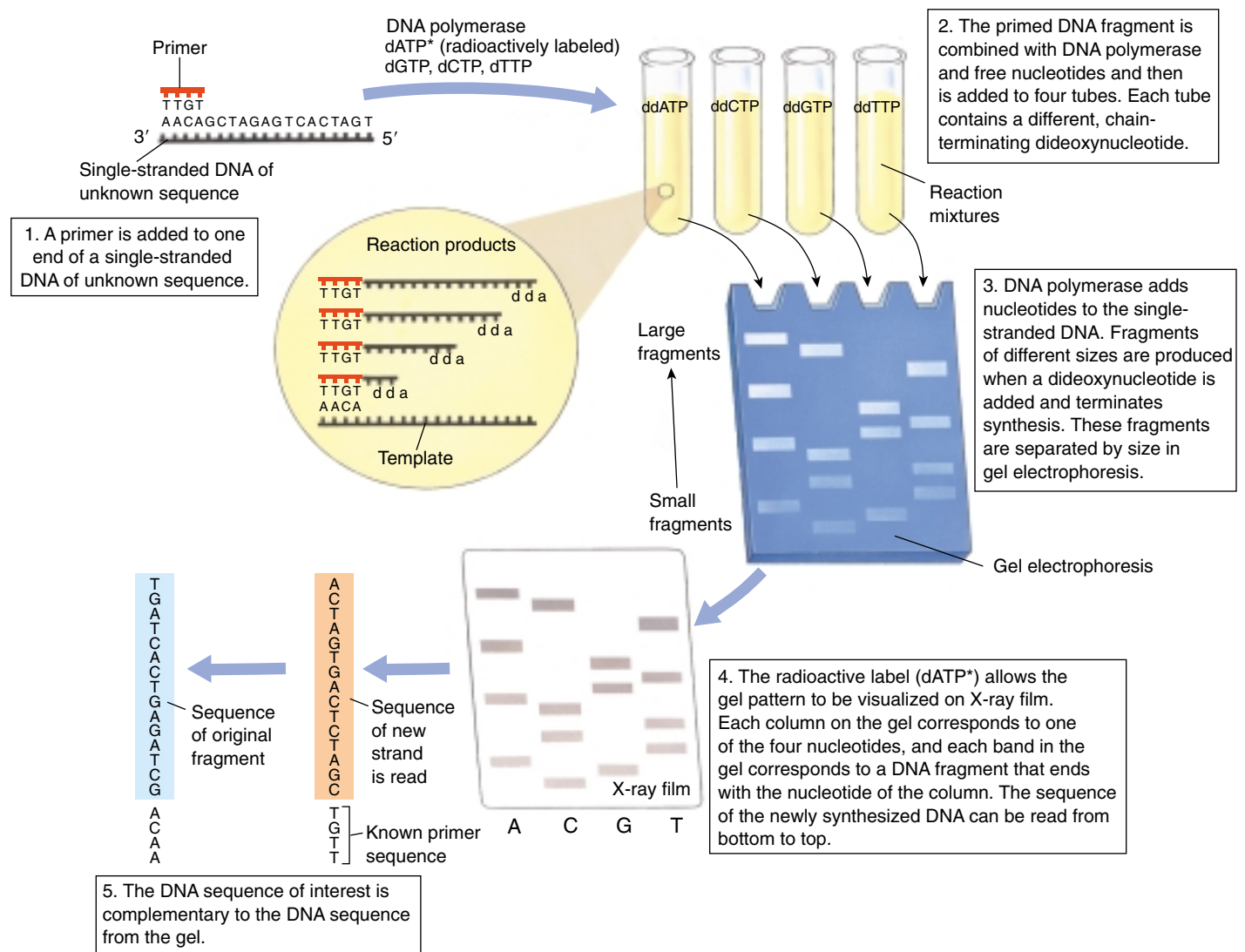


FIGURE 19.12
The Sanger dideoxynucleotide sequencing method.

19.3 Biotechnology is producing a scientific revolution.

DNA Sequence Technology

The 1980s saw an explosion of interest in **biotechnology**, the application of genetic engineering to practical human problems. Let us examine some of the major areas where these techniques have been put to use.

Genome Sequencing

Genetic engineering techniques are enabling us to learn a great deal more about the human genome. Several clonal libraries of the human genome have been assembled, using large-size restriction fragments. Any cloned gene can now be localized to a specific chromosomal site by using probes to detect in situ hybridization (that is, binding between the probe and a complementary sequence on the chromosome). Genes are now being mapped at an astonishing rate: genes that contribute to dyslexia, obesity, and cholesterol-proof blood are some of the important ones that were mapped in 1994 and 1995 alone! With an understanding of where specific genes are located in the human genome and how they work, it is not difficult to imagine a future in which virtually any genetic disease could be treated or perhaps even cured with gene therapy. As we mentioned in chapter 13, some success has already been reported in treating patients who have cystic fibrosis with a genetically corrected version of the cystic fibrosis gene.

An exciting scientific by-product of the human genome project has been the complete genome sequencing of many microorganisms with smaller genomes, on the order of a few Mb (table 19.1). In general, about half of the genes prove to have a known function; what the other half of the genes are doing is a complete mystery. The first eukaryotic genome to be sequenced in its entirety was that of brewer's yeast *Saccharomyces cerevisiae*; many of its approximately 6000 genes have a similar structure to some human genes. The complete sequences of many much larger genomes have recently been completed, including the malarial *Plasmodium* parasite (30 Mb), the nematode (100 Mb), the plant *Arabidopsis* (100 Mb) (figure 19.13), the fruit fly *Drosophila* (120 Mb), and the mouse (300 Mb).

The international scientific community has over the last several years mounted a major effort to sequence the entire human genome. Because the human genome contains some 3000 Mb (million nucleotide base-pairs), this task has presented no small challenge. Rapid progress was made possible by the use of so-called shotgun cloning techniques, in which the entire genome is first fragmented, then each of the fragments is sequenced by automated machines, and finally computers use overlaps to order the fragments. All but a small portion of the sequence was completed by the beginning of the year 2000.

Table 19.1 Genome Sequencing Projects

Organism	Genome Size (Mb)	Description
ARCHAEBACTERIA		
<i>Methanococcus jannaschi</i>	1.7	Extreme thermophile
EUBACTERIA		
<i>Escherichia coli</i>	4.6	Laboratory standard
FUNGI		
<i>Saccharomyces cerevisiae</i>	13	Baker's yeast
PROTIST		
<i>Plasmodium</i>	30	Malarial parasite
PLANT		
<i>Arabidopsis thaliana</i>	100	Relative of mustard plant
ANIMAL		
<i>Caenorhabditis elegans</i>	100	Nematode
<i>Drosophila melanogaster</i>	120	Fruit fly
<i>Mus musculus</i>	300	Mouse
<i>Homo sapiens</i>	3000	Human

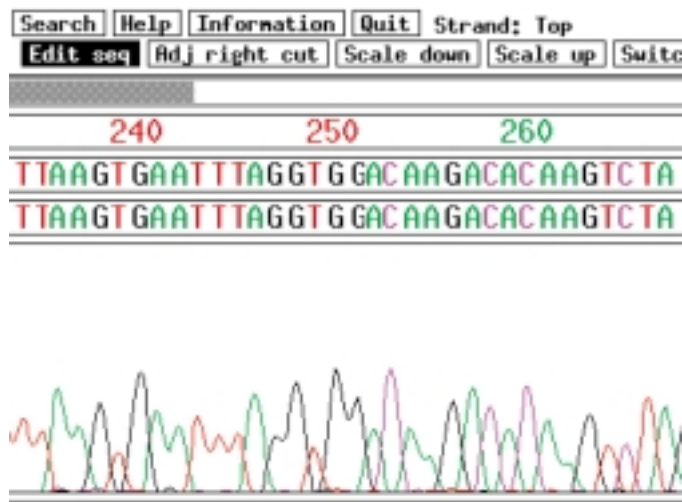


FIGURE 19.13
Part of the genome sequence of the plant *Arabidopsis*. Data from an automated DNA-sequencing run shows the nucleotide sequence for a small section of the *Arabidopsis* genome. Automated DNA sequencing has greatly increased the speed at which genomes can be sequenced.

DNA Fingerprinting

Figure 19.14 shows the DNA fingerprints a prosecuting attorney presented in a rape trial in 1987. They consisted of autoradiographs, parallel bars on X-ray film resembling the line patterns of the universal price code found on groceries. Each bar represents the position of a DNA restriction endonuclease fragment produced by techniques similar to those described in figures 19.4 and 19.10. The lane with many bars represents a standardized control. Two different probes were used to identify the restriction fragments. A vaginal swab had been taken from the victim within hours of her attack; from it semen was collected and the semen DNA analyzed for its restriction endonuclease patterns.

Compare the restriction endonuclease patterns of the semen to that of the suspect Andrews. You can see that the suspect's two patterns match that of the rapist (and are not at all like those of the victim). Clearly the semen collected from the rape victim and the blood sample from the suspect came from the same person. The suspect was Tommie Lee Andrews, and on November 6, 1987, the jury returned a verdict of guilty. Andrews became the first person in the United States to be convicted of a crime based on DNA evidence.

Since the Andrews verdict, DNA fingerprinting has been admitted as evidence in more than 2000 court cases (figure 19.15). While some probes highlight profiles shared by many people, others are quite rare. Using several probes, identity can be clearly established or ruled out.

Just as fingerprinting revolutionized forensic evidence in the early 1900s, so DNA fingerprinting is revolutionizing it today. A hair, a minute speck of blood, a drop of semen can all serve as sources of DNA to damn or clear a suspect. As the man who analyzed Andrews' DNA says: "It's like leaving your name, address, and social security number at the scene of the crime. It's that precise." Of course, laboratory analyses of DNA samples must be carried out properly—sloppy procedures could lead to a wrongful conviction. After widely publicized instances of questionable lab procedures, national standards are being developed.

The genomes of several organisms have been completely sequenced. When DNA is digested with restriction endonucleases, distinctive profiles on electrophoresis gels can be used to identify the individual that was the source of the tissue.

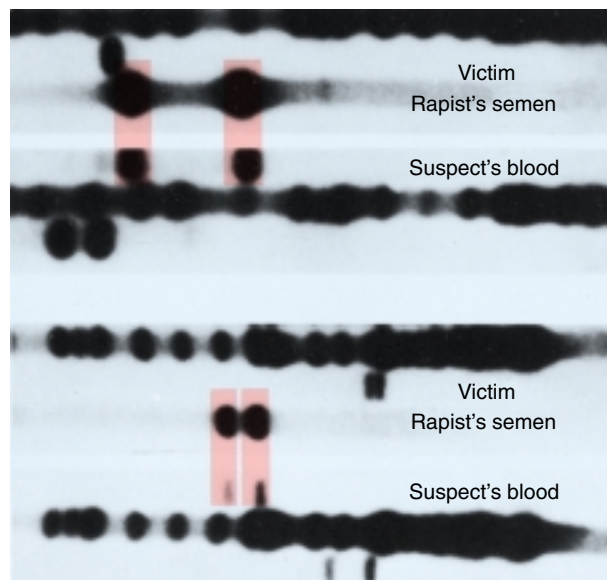


FIGURE 19.14
Two of the DNA profiles that led to the conviction of Tommie Lee Andrews for rape in 1987. The two DNA probes seen here were used to characterize DNA isolated from the victim, the semen left by the rapist, and the suspect. The dark channels are multiband controls. There is a clear match between the suspect's DNA and the DNA of the rapist's semen in these.

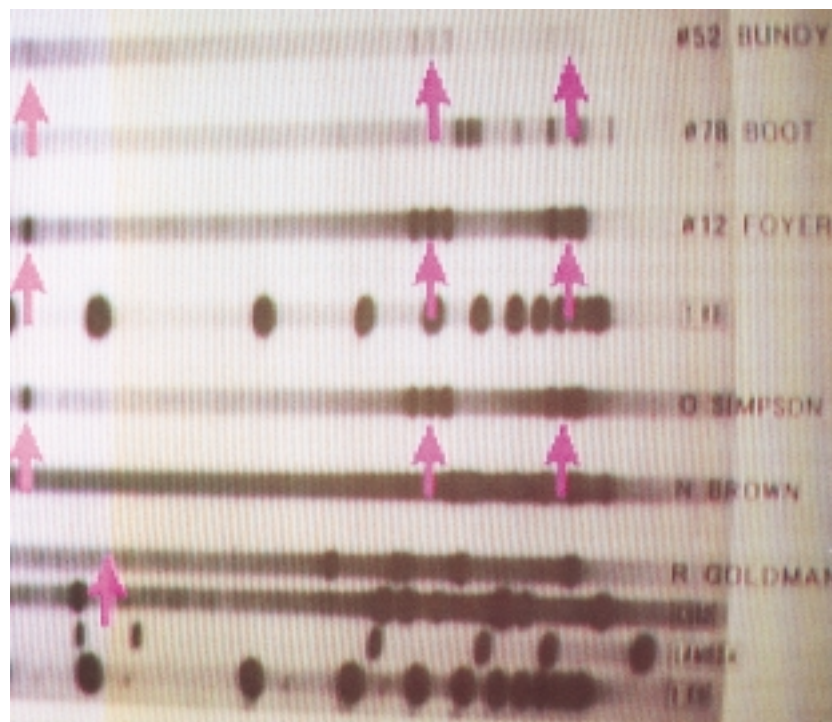


FIGURE 19.15
The DNA profiles of O. J. Simpson and blood samples from the murder scene of his former wife from his highly publicized and controversial murder trial in 1995.

Biochips

A biochip, also called a gene microarray, is a square of glass smaller than a postage stamp, covered with millions of strands of DNA like blades of grass. Biochips were invented nine years ago by gene scientist Stephen Fodor. In a flash of insight, he saw that photolithography, the process used to etch semiconductor circuits into silicon, could also be used to assemble particular DNA molecules on a chip—a biochip.

Think of the chip surface as a field of assembly sites, much as a TV screen is a field of colored dots. Just as a scanning beam moves over each individual TV dot instructing it to be red, green, or blue (the three components of color), so a scanning beam moves over each biochip spot, commanding the addition there of a base to a growing strand of DNA. A computer, by varying the wavelength of the scanning beam, determines which of four possible nucleotides is added to the growing DNA strand anchored to each spot. When the entire chip has been scanned, each DNA strand has been lengthened one nucleotide unit. The computer repeats the process, layer by layer, until each DNA strand is an entire gene or gene fragment. One biochip made in this way contains hundreds of thousands of specific gene sequences.

How could you use such a biochip to delve into a person's genes? All you would have to do is to obtain a little of the person's DNA, say from a blood sample or even a bit of hair. Flush fluid containing the DNA over the biochip surface. Every place that the DNA has a gene matching one of the biochip strands, it will stick to it in a way the computer can detect.

Now here is where it gets interesting. The mad rush to sequence the human genome is over. The gene research firm Celera has recently announced it has essentially completed the sequence, with over 90% of genes done. Already the researchers are busily comparing their consensus "reference sequence" to the DNA of individual people, and noting any differences they detect. Called single nucleotide polymorphisms, or SNPs (pronounced "snips"), these spot differences in the identity of particular nucleotides collectively record every way in which a particular individual differs from the reference sequence. Some SNPs cause diseases like cystic fibrosis or sickle cell anemia. Others may give you red hair or elevated cholesterol in your blood. As the human genome project charges toward completion, its researchers are excitedly assembling a huge database of SNPs. Research indicates that SNPs can be expected to occur at a frequency of about one per thousand nucleotides, scattered about randomly over the chromosomes. Each of us thus differs from the standard "type sequence" in several thousand nucleotide SNPs. Everything genetic about you that is different from a stranger you meet is caused by a few thousand SNPs; otherwise you and that stranger are identical.

How Biochips Can Be Used to Screen for Cancer

One of the biggest decisions facing an oncologist (cancer doctor) treating a tumor is to select the proper treatment. Most cancer cells look alike, although the tumors may in fact be caused by quite different forms of cancer. If the oncologist could clearly identify the cancer, very targeted therapies might be possible. Unable to tell the difference for sure, however, oncologists take no chances. Tumors are treated with therapy that attacks all cancers, usually with severe side effects.

This year Boston researchers Todd Golub and Eric Lander took a vital step towards treating cancer, using new DNA technology to sniff out the differences between different forms of a deadly cancer of the immune system. Golub and Lander worked with biochips.

The way to tell the difference between two kinds of cancer is to compare the mutations that led to the cancer in the first place. Biologists call such gene changes mutations. The mutations that cause many lung cancers are caused by a tobacco-induced alteration of a single DNA nucleotide in one gene. Such spot differences between the version of a gene one person has and another person has, or a cancer patient has, are examples of SNPs.

Golub and Lander obtained bone marrow cells from patients with two types of leukemia (cancer of white blood cells), and exposed DNA from each to biochips containing all known human genes, 6817 in all (figure 19.16). Using high-speed computer programs, Golub and Lander examined each of the 6817 positions on the chip. The two forms of leukemia each showed gene changes from normal, but, importantly, the changes were different in each case! Each had their own characteristic SNP.

Biochips thus may offer a quick and reliable way to identify any type of cancer. Just look and see what SNP is present.

The Use of Gene Chips Will Soon Be Widespread

Biochip technology is likely to dominate medicine in the coming millennium, a prospect both exciting and scary. Researchers have announced plans to compile a database of hundreds of thousands of SNPs over the next two years. Screening SNPs and comparing them to known SNP databases will soon allow doctors to screen each of us for copies of genes leading to genetic diseases. Many genetic diseases are associated with SNPs, including cystic fibrosis and muscular dystrophy.

Biochips Raise Critical Issues of Personal Privacy

The scary part is SNPs on chips. Researchers plan to have identified some 300,000 different SNPs by 2001, all of which could reside on a single biochip. When your DNA is flushed over a SNP biochip, the sequences that light up

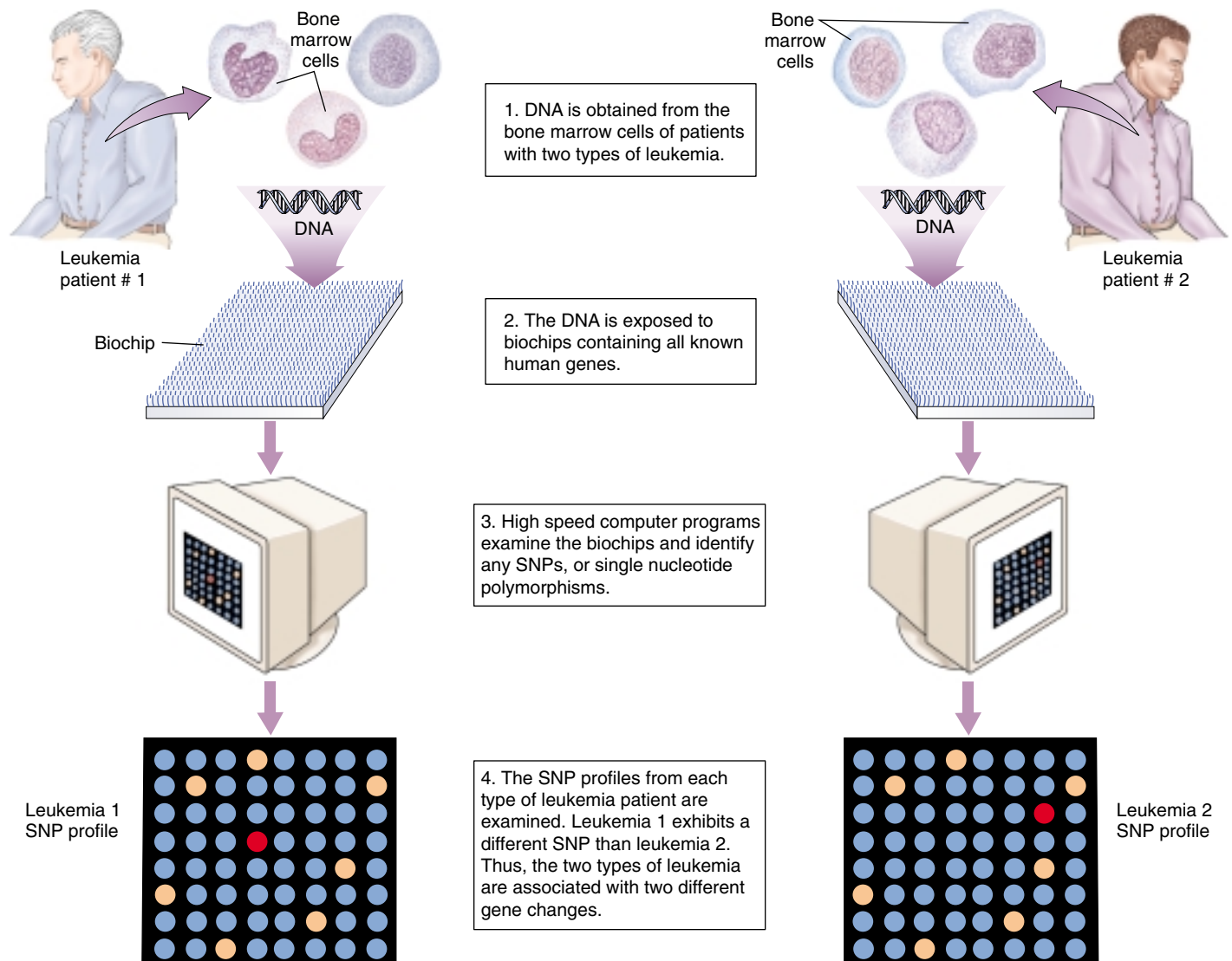


FIGURE 19.16
Biochips can help in identifying precise forms of cancer.

will instantly reveal your SNP profile. The genetic characteristics that make you you, genes that might affect your health, your behavior, your future potential—all are there to be read by anyone clever enough to interpret the profile.

To what extent are you your genes? Scientists fight about this question, and no one really knows the answer. It is clear that much of what each of us is like is strongly affected by our genetic makeup. Researchers have proven beyond any real dispute that intelligence and major personality traits like aggressiveness and inquisitiveness are about 80% heritable (that is, 80% of the variation in these traits reflects variation in genes).

Your SNP profile will reflect all of this variation, a table of contents of your chromosomes, a molecular window to who you are. When millions of such SNP profiles have been

gathered over the coming years, computers will be able to identify other individuals with profiles like yours, and, by examining health records, standard personality tests, and the like, correlate parts of your profile with particular traits. Even behavioral characteristics involving many genes, which until now have been thought too complex to ever analyze, cannot resist a determined assault by a computer comparing SNP profiles.

A biochip is a discrete collection of gene fragments on a stamp-sized chip that can be used to screen for the presence of particular gene variants. Biochips allow rapid screening of gene profiles, a tool that promises to have a revolutionary impact on medicine and society.

Medical Applications

Pharmaceuticals

The first and perhaps most obvious commercial application of genetic engineering was the introduction of genes that encode clinically important proteins into bacteria. Because bacterial cells can be grown cheaply in bulk (fermented in giant vats, like the yeasts that make beer), bacteria that incorporate recombinant genes can synthesize large amounts of the proteins those genes specify. This method has been used to produce several forms of human insulin and interferon, as well as other commercially valuable proteins such as growth hormone (figure 19.17) and erythropoietin, which stimulates red blood cell production.

Among the medically important proteins now manufactured by these approaches are **atrial peptides**, small proteins that may provide a new way to treat high blood pressure and kidney failure. Another is **tissue plasminogen activator**, a human protein synthesized in minute amounts that causes blood clots to dissolve and may be effective in preventing and treating heart attacks and strokes.

A problem with this general approach has been the difficulty of separating the desired protein from the others the bacteria make. The purification of proteins from such complex mixtures is both time-consuming and expensive, but it is still easier than isolating the proteins from the tissues of animals (for example, insulin from hog pancreases), which is how such proteins used to be obtained. Recently, however, researchers have succeeded in producing RNA transcripts of cloned genes; they can then use the transcripts to produce only these proteins in a test tube containing the transcribed RNA, ribosomes, cofactors, amino acids, tRNA, and ATP.

Gene Therapy

In 1990, researchers first attempted to combat genetic defects by the transfer of human genes. When a hereditary disorder is the result of a single defective gene, an obvious way to cure the disorder is to add a working copy of the gene. This approach is being used in an attempt to combat cystic fibrosis, and it offers potential for treating muscular dystrophy and a variety of other disorders (table 19.2). One of the first successful attempts was the transfer of a gene encoding the enzyme adenosine deaminase into the bone marrow of two girls suffering from a rare blood disorder caused by the lack of this enzyme. However, while many clinical trials are underway, no others have yet proven successful. This extremely promising approach will require a lot of additional effort.

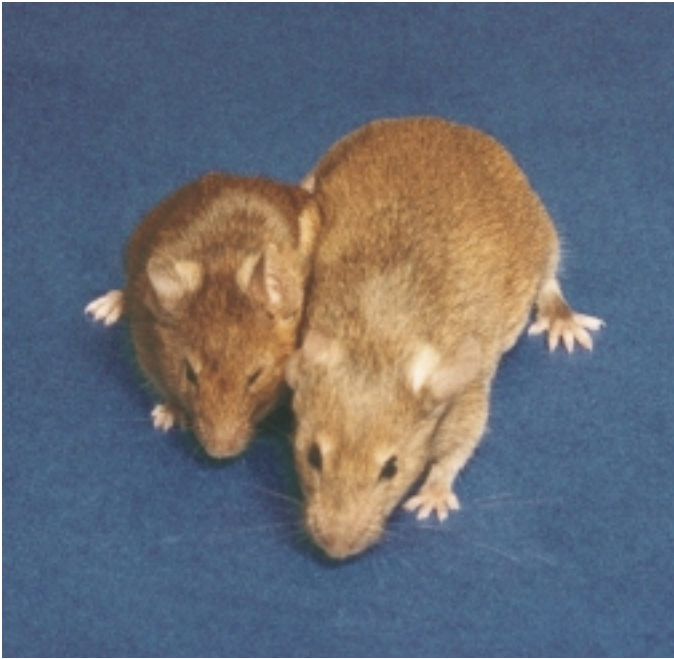


FIGURE 19.17
Genetically engineered human growth hormone. These two mice are genetically identical, but the large one has one extra gene: the gene encoding human growth hormone. The gene was added to the mouse’s genome by genetic engineers and is now a stable part of the mouse’s genetic endowment.

Table 19.2 Diseases Being Treated in Clinical Trials of Gene Therapy

Disease
Cancer (melanoma, renal cell, ovarian, neuroblastoma, brain, head and neck, lung, liver, breast, colon, prostate, mesothelioma, leukemia, lymphoma, multiple myeloma)
SCID (severe combined immunodeficiency)
Cystic fibrosis
Gaucher’s disease
Familial hypercholesterolemia
Hemophilia
Purine nucleoside phosphorylase deficiency
Alpha-1 antitrypsin deficiency
Fanconi’s anemia
Hunter’s syndrome
Chronic granulomatous disease
Rheumatoid arthritis
Peripheral vascular disease
AIDS

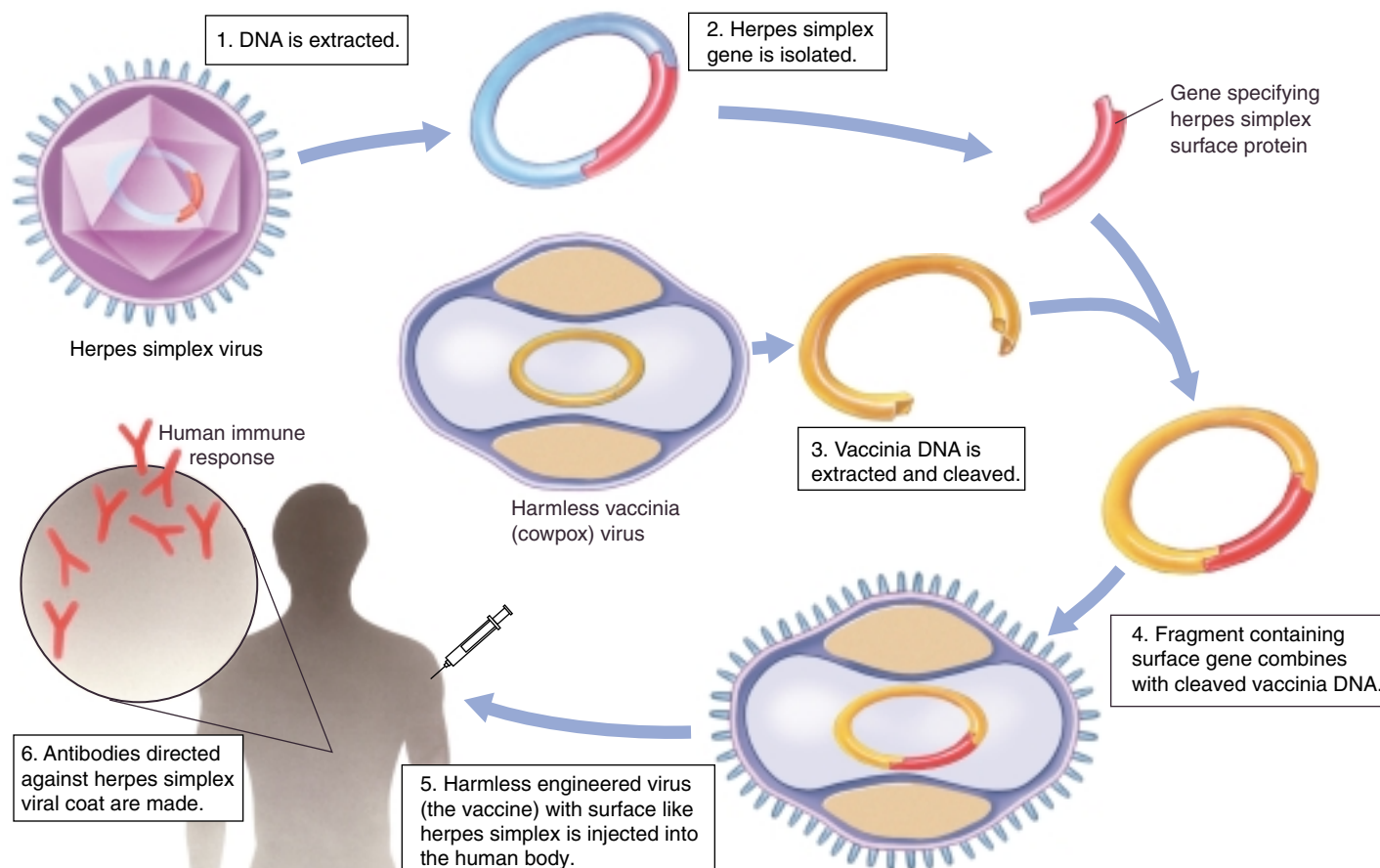


FIGURE 19.18
Strategy for constructing a subunit vaccine for herpes simplex.

Piggyback Vaccines

Another area of potential significance involves the use of genetic engineering to produce **subunit vaccines** against viruses such as those that cause herpes and hepatitis. Genes encoding part of the protein-polysaccharide coat of the herpes simplex virus or hepatitis B virus are spliced into a fragment of the vaccinia (cowpox) virus genome (figure 19.18). The vaccinia virus, which British physician Edward Jenner used almost 200 years ago in his pioneering vaccinations against smallpox, is now used as a vector to carry the herpes or hepatitis viral coat gene into cultured mammalian cells. These cells produce many copies of the recombinant virus, which has the outside coat of a herpes or hepatitis virus. When this recombinant virus is injected into a mouse or rabbit, the immune system of the infected animal produces antibodies directed against the coat of the recombinant virus. It therefore develops an immunity to herpes or hepatitis virus. Vaccines produced in this way are harmless because the vaccinia virus is benign and only a small fragment of the DNA from the disease-causing virus is introduced via the recombinant virus.

The great attraction of this approach is that it does not

depend upon the nature of the viral disease. In the future, similar recombinant viruses may be injected into humans to confer resistance to a wide variety of viral diseases.

In 1995, the first clinical trials began of a novel new kind of **DNA vaccine**, one that depends not on antibodies but rather on the second arm of the body's immune defense, the so-called cellular immune response, in which blood cells known as killer T cells attack infected cells. The infected cells are attacked and destroyed when they stick fragments of foreign proteins onto their outer surfaces that the T cells detect (the discovery by Peter Doherty and Rolf Zinkernagel that infected cells do so led to their receiving the Nobel Prize in Physiology or Medicine in 1996). The first DNA vaccines spliced an influenza virus gene encoding an internal nucleoprotein into a plasmid, which was then injected into mice. The mice developed strong cellular immune responses to influenza. New and controversial, the approach offers great promise.

Genetic engineering has produced commercially valuable proteins, gene therapies, and, possibly, new and powerful vaccines.

Agricultural Applications

Another major area of genetic engineering activity is manipulation of the genes of key crop plants. In plants the primary experimental difficulty has been identifying a suitable vector for introducing recombinant DNA. Plant cells do not possess the many plasmids that bacteria do, so the choice of potential vectors is limited. The most successful results thus far have been obtained with the **Ti** (tumor-inducing) **plasmid** of the plant bacterium *Agrobacterium tumefaciens*, which infects broadleaf plants such as tomato, tobacco, and soybean. Part of the Ti plasmid integrates into the plant DNA, and researchers have succeeded in attaching other genes to this portion of the plasmid (figure 19.19). The characteristics of a number of plants have been altered using this technique, which should be valuable in improving crops and forests. Among the features scientists would like to affect are resistance to disease, frost, and other forms of stress; nutritional balance and protein content; and herbicide resistance. Unfortunately, *Agrobacterium* generally does not infect cereals such as corn, rice, and wheat, but alternative methods can be used to introduce new genes into them.

A recent advance in genetically manipulated fruit is Calgene's "Flavr Savr" tomato, which has been approved for

sale by the USDA. The tomato has been engineered to inhibit genes that cause cells to produce ethylene. In tomatoes and other plants, ethylene acts as a hormone to speed fruit ripening. In Flavr Savr tomatoes, inhibition of ethylene production delays ripening. The result is a tomato that can stay on the vine longer and that resists overripening and rotting during transport to market.

Herbicide Resistance

Recently, broadleaf plants have been genetically engineered to be resistant to **glyphosate**, the active ingredient in Roundup, a powerful, biodegradable herbicide that kills most actively growing plants (figure 19.20). Glyphosate works by inhibiting an enzyme called EPSP synthetase, which plants require to produce aromatic amino acids. Humans do not make aromatic amino acids; they get them from their diet, so they are unaffected by glyphosate. To make glyphosate-resistant plants, agricultural scientists used a Ti plasmid to insert extra copies of the EPSP synthetase genes into plants. These engineered plants produce 20 times the normal level of EPSP synthetase, enabling them to synthesize proteins and grow despite glyphosate's suppression of the enzyme. In later experiments, a bacterial form of the EPSP synthetase gene that differs from the

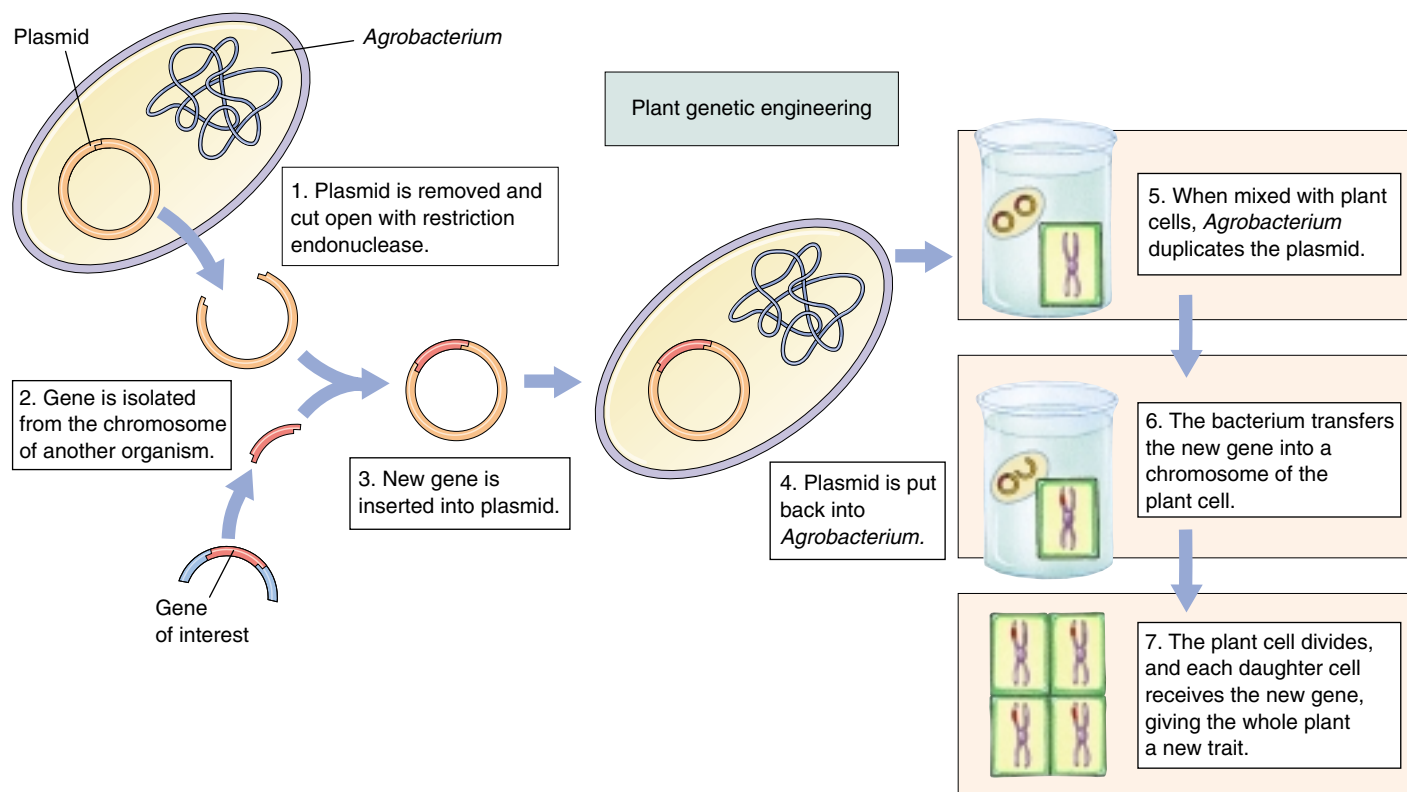


FIGURE 19.19

The **Ti plasmid**. This *Agrobacterium tumefaciens* plasmid is used in plant genetic engineering.

plant form by a single nucleotide was introduced into plants via Ti plasmids; the bacterial enzyme in these plants is not inhibited by glyphosate.

These advances are of great interest to farmers because a crop resistant to Roundup would never have to be weeded if the field were simply treated with the herbicide. Because Roundup is a broad-spectrum herbicide, farmers would no longer need to employ a variety of different herbicides, most of which kill only a few kinds of weeds. Furthermore, glyphosate breaks down readily in the environment, unlike many other herbicides commonly used in agriculture. A plasmid is actively being sought for the introduction of the EPSP synthetase gene into cereal plants, making them also glyphosate-resistant.

Nitrogen Fixation

A long-range goal of agricultural genetic engineering is to introduce the genes that allow soybeans and other legume plants to “fix” nitrogen into key crop plants. These so-called ***nif* genes** are found in certain symbiotic root-colonizing bacteria. Living in the root nodules of legumes, these bacteria break the powerful triple bond of atmospheric nitrogen gas, converting N_2 into NH_3 (ammonia). The plants then use the ammonia to make amino acids and other nitrogen-containing molecules. Other plants lack these bacteria and cannot fix nitrogen, so they must obtain their nitrogen from the soil. Farmland where these crops are grown soon becomes depleted of nitrogen, unless nitrogenous fertilizers are applied. Worldwide, farmers applied over 60 million metric tons of such fertilizers in 1987, an expensive undertaking. Farming costs would be much lower if major crops like wheat and corn could be engineered to carry out biological nitrogen fixation. However, introducing the nitrogen-fixing genes from bacteria into plants has proved difficult because these genes do not seem to function properly in eukaryotic cells. Researchers are actively experimenting with other species of nitrogen-fixing bacteria whose genes might function better in plant cells.

Insect Resistance

Many commercially important plants are attacked by insects, and the traditional defense against such attacks is to apply insecticides. Over 40% of the chemical insecticides used today are targeted against boll weevils, bollworms, and other insects that eat cotton plants. Genetic engineers are now attempting to produce plants that are resistant to insect pests, removing the need to use many externally applied insecticides.

The approach is to insert into crop plants genes encoding proteins that are harmful to the insects that feed on the plants but harmless to other organisms. One such insecticidal protein has been identified in *Bacillus thuringiensis*, a soil bacterium. When the tomato hornworm caterpillar ingests this protein, enzymes in the caterpillar’s stomach convert it



FIGURE 19.20

Genetically engineered herbicide resistance. All four of these petunia plants were exposed to equal doses of the herbicide Roundup. The two on top were genetically engineered to be resistant to glyphosate, the active ingredient of Roundup, while the two on the bottom were not.

into an insect-specific toxin, causing paralysis and death. Because these enzymes are not found in other animals, the protein is harmless to them. Using the Ti plasmid, scientists have transferred the gene encoding this protein into tomato and tobacco plants. They have found that these **transgenic** plants are indeed protected from attack by the insects that would normally feed on them. In 1995, the EPA approved altered forms of potato, cotton, and corn. The genetically altered potato can kill the Colorado potato beetle, a common pest. The altered cotton is resistant to cotton bollworm, budworm, and pink bollworm. The corn has been altered to resist the European corn borer and other mothlike insects.

Monsanto scientists screening natural compounds extracted from plant and soil samples have recently isolated a new insect-killing compound from a fungus, the enzyme cholesterol oxidase. Apparently, the enzyme disrupts membranes in the insect gut. The fungus gene, called the Bollgard gene after its discoverer, has been successfully inserted into a variety of crops. It kills a wide range of insects, including the cotton boll weevil and the Colorado potato beetle, both serious agricultural pests. Field tests began in 1996.

Some insect pests attack plant roots, and *B. thuringiensis* is being employed to counter that threat as well. This bacterium does not normally colonize plant roots, so biologists have introduced the *B. thuringiensis* insecticidal protein gene into root-colonizing bacteria, especially strains of *Pseudomonas*. Field testing of this promising procedure has been approved by the Environmental Protection Agency.

The Real Promise of Plant Genetic Engineering

In the last decade the cultivation of genetically modified crops of corn, cotton, and soybeans has become commonplace in the United States—in 1999, over half of the 72 million acres planted with soybeans in the United States were planted with seeds genetically modified to be herbicide resistant, with the result that less tillage has been needed, and as a consequence soil erosion has been greatly lessened. These benefits, while significant, have been largely confined to farmers, making their cultivation of crops cheaper and more efficient. The food that the public gets is the same, it just costs less to get it to the table.

Like the first act of a play, these developments have served mainly to set the stage for the real action, which is only now beginning to happen. The real promise of plant genetic engineering is to produce genetically modified plants with desirable traits that directly benefit the consumer.

One recent advance, nutritionally improved rice, gives us a hint of what is to come. In developing countries large numbers of people live on simple diets that are poor sources of vitamins and minerals (what botanists called "micronutrients"). Worldwide, the two major micronutrient deficiencies are iron, which affects 1.4 billion women, 24% of the world population, and vitamin A, affecting 40 million children, 7% of the world population. The deficiencies are especially severe in developing countries where the major staple food is rice. In recent research, Swiss bioengineer Ingo Potrykus and his team at the Institute of Plant Sciences, Zurich, have gone a long way towards solving this problem. Supported by the Rockefeller Foundation and with results to be made free to developing countries, the work is a model of what plant genetic engineering can achieve.

To solve the problem of dietary iron deficiency among rice eaters, Potrykus first asked why rice is such a poor source of dietary iron. The problem, and the answer, proved to have three parts:

1. *Too little iron.* The proteins of rice endosperm have unusually low amounts of iron. To solve this problem, a ferritin gene was transferred into rice from beans (figure 19.21). Ferritin is a protein with an extraordinarily high iron content, and so greatly increased the iron content of the rice.
2. *Inhibition of iron absorption by the intestine.* Rice contains an unusually high concentration of a chemical called phytate, which inhibits iron reabsorption in the intestine—it stops your body from taking up the iron in the rice. To solve this problem, a gene encoding an enzyme that destroys phytate was transferred into rice from a fungus.
3. *Too little sulfur for efficient iron absorption.* Sulfur is required for iron uptake, and rice has very little of it. To solve this problem, a gene encoding a particularly sulfur-rich metallothionin protein was transferred into rice from wild rice.

To solve the problem of vitamin A deficiency, the same approach was taken. First, the problem was identified. It turns out rice only goes part way toward making beta-carotene (provitamin A); there are no enzymes in rice to catalyze the last four steps. To solve the problem, genes encoding these four enzymes were added to rice from a familiar flower, the daffodil.

Potrykus's development of transgenic rice to combat dietary deficiencies involved no subtle tricks, just straightforward bioengineering and the will to get the job done. The transgenic rice he has developed will directly improve the lives of millions of people. His work is rep-

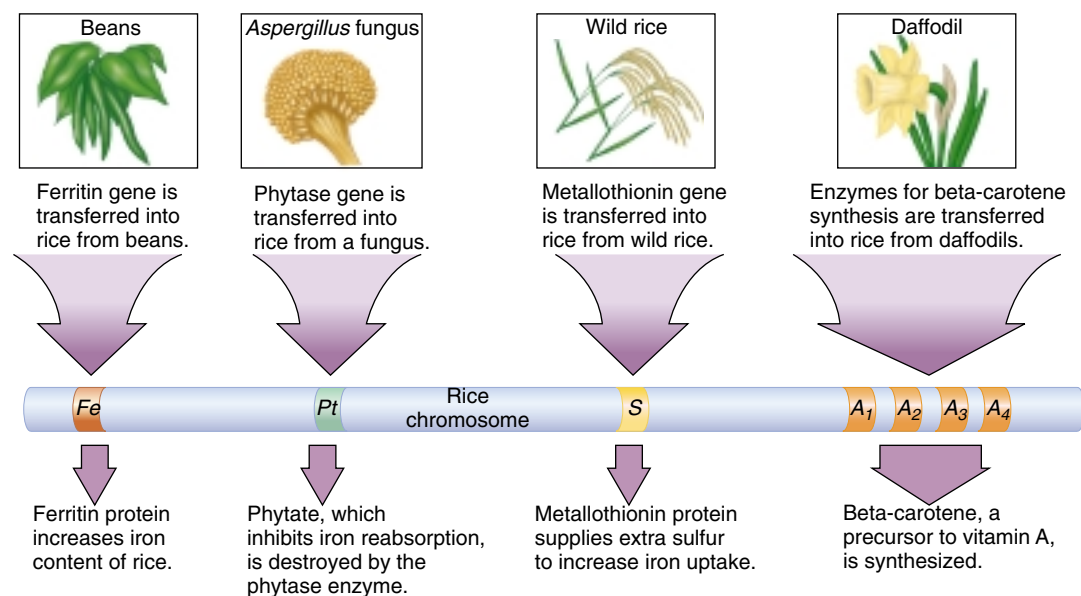


FIGURE 19.21

Transgenic rice. Developed by Swiss bioengineer Ingo Potrykus, transgenic rice offers the promise of improving the diets of people in rice-consuming developing countries, where iron and vitamin A deficiencies are a serious problem.

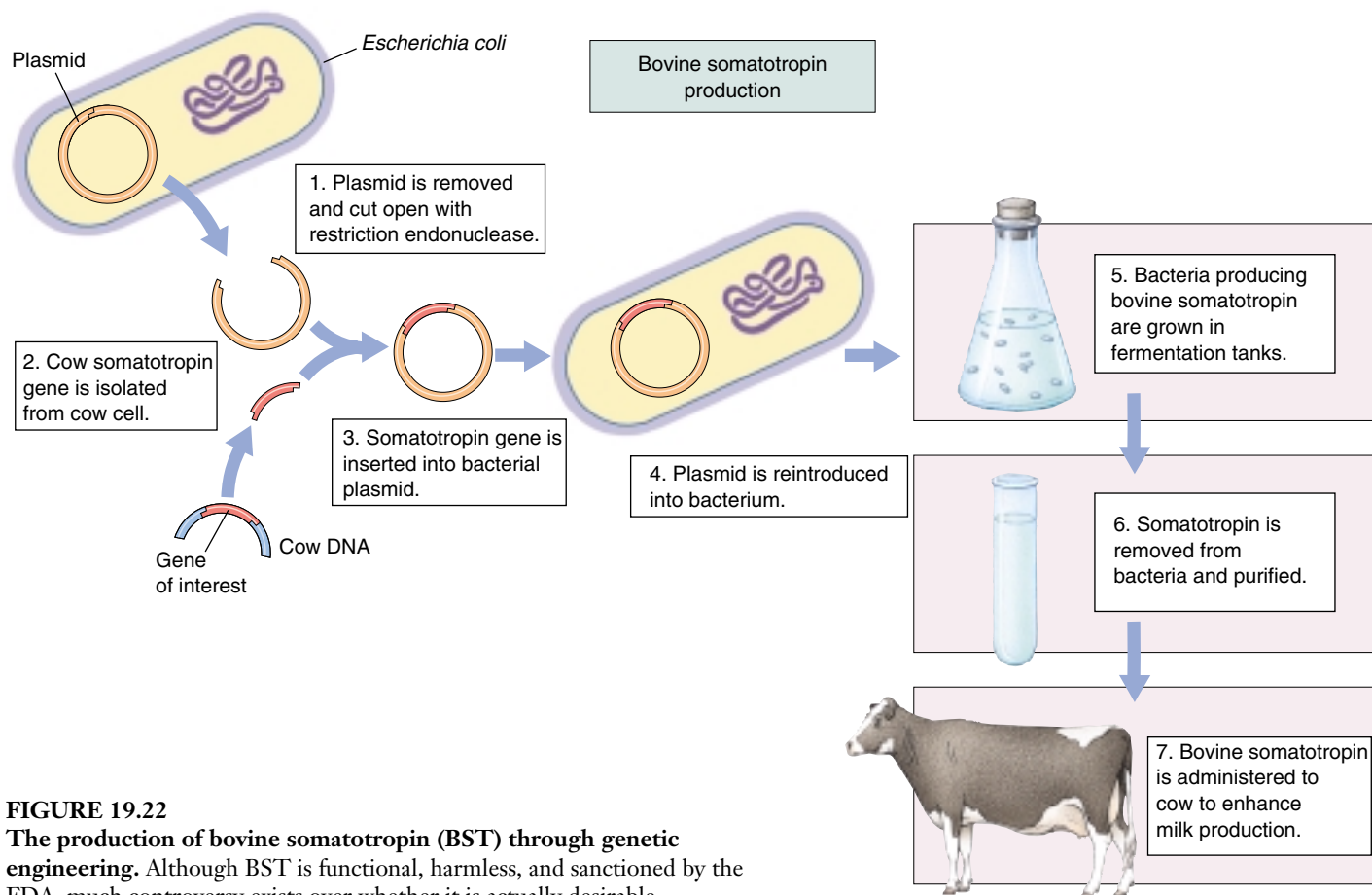


FIGURE 19.22
The production of bovine somatotropin (BST) through genetic engineering. Although BST is functional, harmless, and sanctioned by the FDA, much controversy exists over whether it is actually desirable.

representative of the very real promise of genetic engineering to help meet the challenges of the coming new millennium.

The list of gene modifications that directly aid consumers will only grow. In Holland, Dutch bioengineers announced last month that they are genetically engineering plants to act as vaccine-producing factories! To petunias they have added a gene for a vaccine against dog parvovirus, hiding the gene within the petunia genes that direct nectar production. The drug is produced in the nectar, collected by bees, and extracted from the honey. It is hard to believe this isn't science fiction. Clearly, the real promise of plant genetic engineering lies ahead, and not very far.

Farm Animals

The gene encoding the growth hormone somatotropin was one of the first to be cloned successfully. In 1994, Monsanto received federal approval to make its recombinant bovine somatotropin (BST) commercially available, and dairy farmers worldwide began to add the hormone as a supplement to their cows' diets, increasing the animals' milk production (figure 19.22). Genetically engi-

neered somatotropin is also being tested to see if it increases the muscle weight of cattle and pigs, and as a treatment for human disorders in which the pituitary gland fails to make adequate levels of somatotropin, producing dwarfism. BST ingested in milk or meat has no effect on humans, because it is a protein and is digested in the stomach. Nevertheless, BST has met with some public resistance, due primarily to generalized fears of gene technology. Some people mistrust milk produced through genetic engineering, even though the milk itself is identical to other milk. Problems concerning public perception are not uncommon as gene technology makes an even greater impact on our lives.

Transgenic animals engineered to have specific desirable genes are becoming increasingly available to breeders. Now, instead of selectively breeding for several generations to produce a racehorse or a stud bull with desirable qualities, the process can be shortened by simply engineering such an animal right at the start.

Gene technology is revolutionizing agriculture, increasing yields and resistance to pests, and producing animals with desirable traits.

Cloning

The difficulty in using transgenic animals to improve livestock is in getting enough of them. Breeding produces offspring only slowly, and recombination acts to undo the painstaking work of the genetic engineer. Ideally, one would like to “Xerox” many exact genetic copies of the transgenic strain—but until 1997 it was commonly accepted that adult animals can’t be cloned. Now the holy grail of agricultural genetic engineers seems within reach. In 1997, scientists announced the first successful cloning of differentiated vertebrate tissue, a lamb grown from a cell taken from an adult sheep. This startling result promises to revolutionize agricultural science.

Spemann’s “Fantastical Experiment”

The idea of cloning animals was first suggested in 1938 by German embryologist Hans Spemann (called the “father of modern embryology”), who proposed what he called a “fantastical experiment”: remove the nucleus from an egg cell, and put in its place a nucleus from another cell.

It was 14 years before technology advanced far enough for anyone to take up Spemann’s challenge. In 1952, two American scientists, Robert Briggs and T. J. King, used very fine pipettes to suck the nucleus from a frog egg (frog eggs are unusually large, making the experiment feasible) and transfer a nucleus sucked from a body cell of an adult frog into its place. The experiment did not work when done this way, but partial success was achieved 18 years later by the British developmental biologist John Gurdon, who in 1970 inserted nuclei from advanced frog embryos rather than adult tissue. The frog eggs developed into tadpoles, but died before becoming adults.

The Path to Success

For 14 years, nuclear transplant experiments were attempted without success. Technology continued to advance however, until finally in 1984, Steen Willadsen, a Danish embryologist working in Texas, succeeded in cloning a sheep using a nucleus from a cell of an early embryo. This exciting result was soon replicated by others in a host of other organisms, including cattle, pigs, and monkeys.

Only early embryo cells seemed to work, however. Researchers became convinced that animal embryo cells become irreversibly “committed” after the first few cell divisions. After that, nuclei from differentiated animal cells cannot be used to clone entire organisms.

We now know this conclusion to have been unwarranted. The key advance for unraveling this puzzle was made in Scotland by geneticist Keith Campbell, a specialist in studying the cell cycle of agricultural animals. By the early 1990s, knowledge of how the cell cycle is controlled, advanced by cancer research, had led to an understanding that cells don’t divide until conditions are appropriate. Just as a washing machine checks that the water has completely emptied before initiating the spin cycle, so the cell checks that everything needed is on hand before initiating cell division. Campbell reasoned: “Maybe the egg and the donated nucleus need to be at the same stage in the cell cycle.”

This proved to be a key insight. In 1994 researcher Neil First, and in 1995 Campbell himself working with reproductive biologist Ian Wilmut, succeeded in cloning farm animals from advanced embryos by first starving the cells, so that they paused at the beginning of the cell cycle at the G₁ checkpoint. Two starved cells are thus synchronized at the same point in the cell cycle.

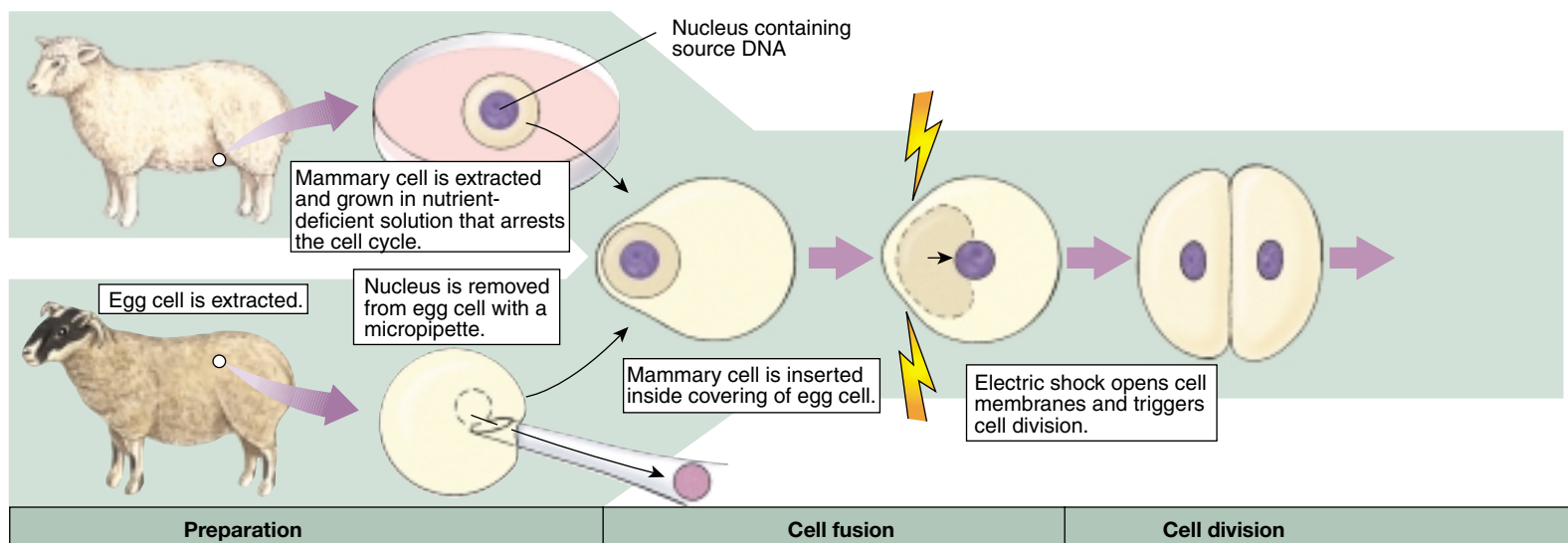


FIGURE 19.23

Wilmut’s animal cloning experiment. Wilmut combined a nucleus from a mammary cell and an egg cell (with its nucleus removed) to successfully clone a sheep.

Wilmut's Lamb

Wilmut then set out to attempt the key breakthrough, the experiment that had eluded researchers since Spemann proposed it 59 years before: to transfer the nucleus from an adult differentiated cell into an enucleated egg, and allow the resulting embryo to grow and develop in a surrogate mother, hopefully producing a healthy animal.

Wilmut removed mammary cells from the udder of a six-year-old sheep (figure 19.23). The origin of these cells, gave the clone its name, “Dolly” after the country singer Dolly Parton. The cells were grown in tissue culture, and some frozen so that in the future it would be possible with genetic fingerprinting to prove that a clone was indeed genetically identical with the six-year-old sheep.

In preparation for cloning, Wilmut's team reduced for five days the concentration of serum on which the sheep mammary cells were subsisting. In parallel preparation, eggs obtained from a ewe were enucleated, the nucleus of each egg carefully removed with a micropipette.

Mammary cells and egg cells were then surgically combined in January of 1996, the mammary cells inserted inside the covering around the egg cell. Wilmut then applied a brief electrical shock. A neat trick, this causes the plasma membranes surrounding the two cells to become leaky, so that the contents of the mammary cell passes into the egg cell. The shock also kick-starts the cell cycle, causing the cell to begin to divide.

After six days, in 30 of 277 tries, the dividing embryo reached the hollow-ball “blastula” stage, and 29 of these were transplanted into surrogate mother sheep. A little over five months later, on July 5, 1997, one sheep gave birth to a lamb. This lamb, “Dolly,” was the first successful clone generated from a differentiated animal cell.

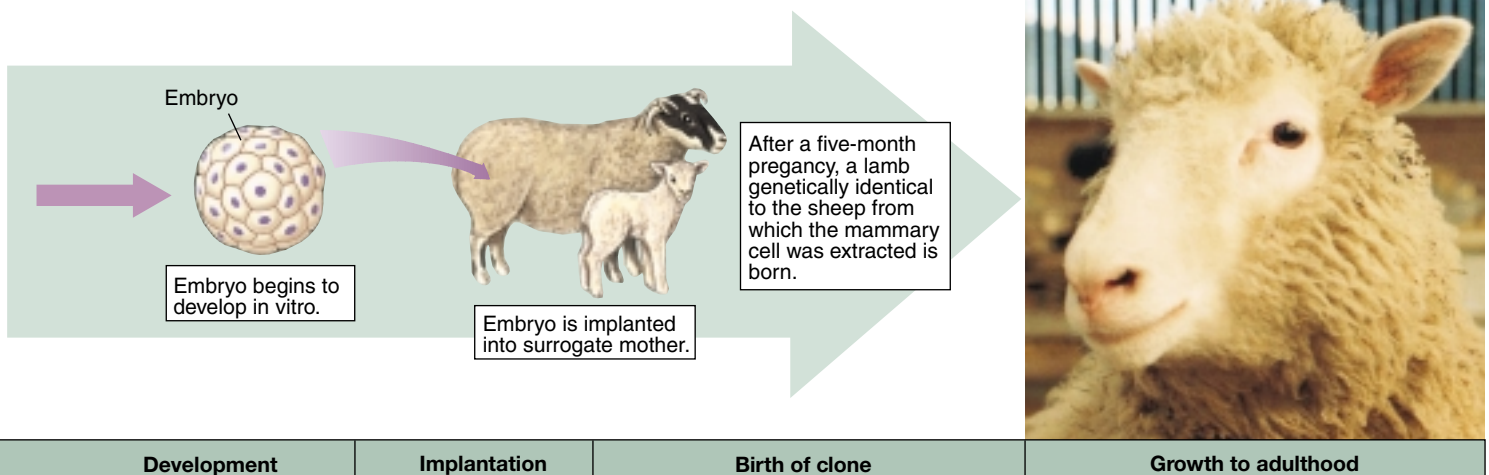
The Future of Cloning

Wilmut's successful cloning of fully differentiated sheep cells is a milestone event in gene technology. Even though his procedure proved inefficient (only one of 277 trials succeeded), it established the point beyond all doubt that cloning of adult animal cells *can* be done. In the following four years researchers succeeded in greatly improving the efficiency of cloning. Seizing upon the key idea in Wilmut's experiment, to clone a resting-stage cell, they have returned to the nuclear transplant procedure pioneered by Briggs and King. It works well. Many different mammals have been successfully cloned including mice, pigs, and cattle.

Transgenic cloning can be expected to have a major impact on medicine as well as agriculture. Animals with human genes can be used to produce rare hormones. For example, sheep that have recently been genetically engineered to secrete a protein called alpha-1 antitrypsin (helpful in relieving the symptoms of cystic fibrosis) into their milk may be cloned, greatly cheapening the production of this expensive drug.

It is impossible not to speculate on the possibility of cloning a human. There is no reason to believe such an experiment would not work, but many reasons to question whether it should be done. Because much of Western thought is based on the concept of human individuality, we can expect the possibility of human cloning to engender considerable controversy.

Recent experiments have demonstrated the possibility of cloning differentiated mammalian tissue, opening the door for the first time to practical transgenic cloning of farm animals.



Stem Cells

Since the isolation of embryonic stem cells in 1998, labs all over the world have been exploring the possibility of using stem cells to restore damaged or lost tissue. Exciting results are now starting to come in.

What is a stem cell? At the dawn of a human life, a sperm fertilizes an egg to create a single cell destined to become a child. As development commences, that cell begins to divide, producing a small ball of a few dozen cells. At this very early point, each of these cells is identical. We call these cells *embryonic stem cells*. Each one of them is capable by itself of developing into a healthy individual. In cattle breeding, for example, these cells are frequently separated by the breeder and used to produce multiple clones of valuable offspring.

The exciting promise of these embryonic stem cells is that, because they can develop into any tissue, they may give us the ability to restore damaged heart or spine tissue (figure 19.24). Experiments have already been tried successfully in mice. Heart muscle cells have been grown from mouse embryonic stem cells and successfully integrated with the heart tissue of a living mouse. This suggests that the damaged heart muscle of heart attack victims might be repairable with stem cells, and that injured spinal cords might be repairable. These very promising experiments are being pursued aggressively. They are, however, quite controversial, as embryonic stem cells are typically isolated from tissue of discarded or aborted embryos, raising serious ethical issues.

Tissue-Specific Stem Cells

New results promise a neat way around the ethical maze presented by stem cells derived from embryos. Go back for

a moment to what we were saying about how a human child develops. What happens next to the embryonic stem cells? They start to take different developmental paths. Some become destined to form nerve tissue and, after this decision is taken, cannot ever produce any other kind of cell. They are then called nerve stem cells. Others become specialized to produce blood, still others muscle. Each major tissue is represented by its own kind of *tissue-specific stem cell*. Now here's the key point: as development proceeds, these tissue-specific stem cells persist. Even in adults. So why not use these adult cells, rather than embryonic stem cells?

Transplanted Tissue-Specific Stem Cells Cure MS in Mice

In pathfinding 1999 laboratory experiments by Dr. Evan Snyder of Harvard Medical School, tissue-specific stem cells were able to restore lost brain tissue. He and his co-workers injected neural stem cells (immediate descendants of embryonic stem cells able to become any kind of neural cell) into the brains of newborn mice with a disease resembling multiple sclerosis (MS). These mice lacked the cells that maintain the layers of myelin insulation around signal-conducting nerves. The injected stem cells migrated all over the brain, and were able to convert themselves into the missing type of cell. The new cells then proceeded to repair the ravages of the disease by replacing the lost insulation of signal-conducting nerve cells. Many of the treated mice fully recovered. In mice at least, tissue-specific stem cells offer a treatment for MS.

The approach seems very straightforward, and should apply to humans. Indeed, blood stem cells are already routinely used in humans to replenish the bone marrow of cancer patients after marrow-destroying therapy. The problem

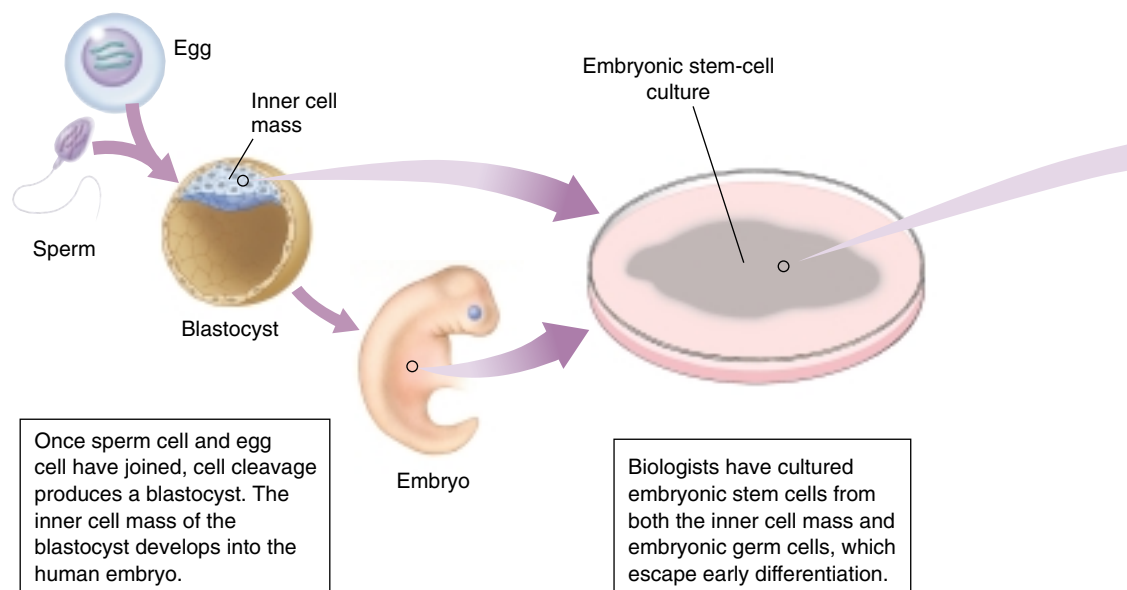


FIGURE 19.24

Using embryonic stem cells to restore damaged tissue. Embryonic stem cells can develop into any body tissue. Methods for growing the tissue and using it to repair damaged tissue in adults, such as the brain cells of multiple sclerosis patients, heart muscle, and spinal nerves, are being developed.

with extending the approach to other kinds of tissue-specific stem cells is that it has not always been easy to find the kind of tissue-specific stem cell you want.

Transplanted Stem Cells Reverse Juvenile Diabetes in Mice

Very promising experiments carried out in 2000 by Dr. Ammon Peck and a team of researchers at the University of Florida concern a particularly vexing problem, that of type 1 or juvenile diabetes. A person with juvenile diabetes lacks insulin-producing pancreas cells, because their immune system has mistakenly turned against them and destroyed them. They are no longer able to produce enough insulin to control their blood sugar levels and must take insulin daily. Adding back new insulin-producing cells called islet cells has been tried many times, but doesn't work well. Immune cells continue to destroy them.

Peck and his team reasoned, why not add instead the stem cells that produce islet cells? They would be able to produce a continuous supply of new islet cells, replacing those lost to immune attack. Because there would always be cells to make insulin, the diabetes would be cured.

No one knew just what such a stem cell looked like, but the researchers knew they come from the epithelial cells that line the pancreas ducts. Surely some must still lurk there unseen. So the research team took a bunch of these epithelial cells from mice and grew them in tissue culture until they had lots of them.

Were the stem cells they sought present in the cell culture they had prepared? Yes. In laboratory dishes the cell culture produced insulin in response to sugar, indicating islet cells had developed in the growing culture, islet cells that must have been produced from stem cells.

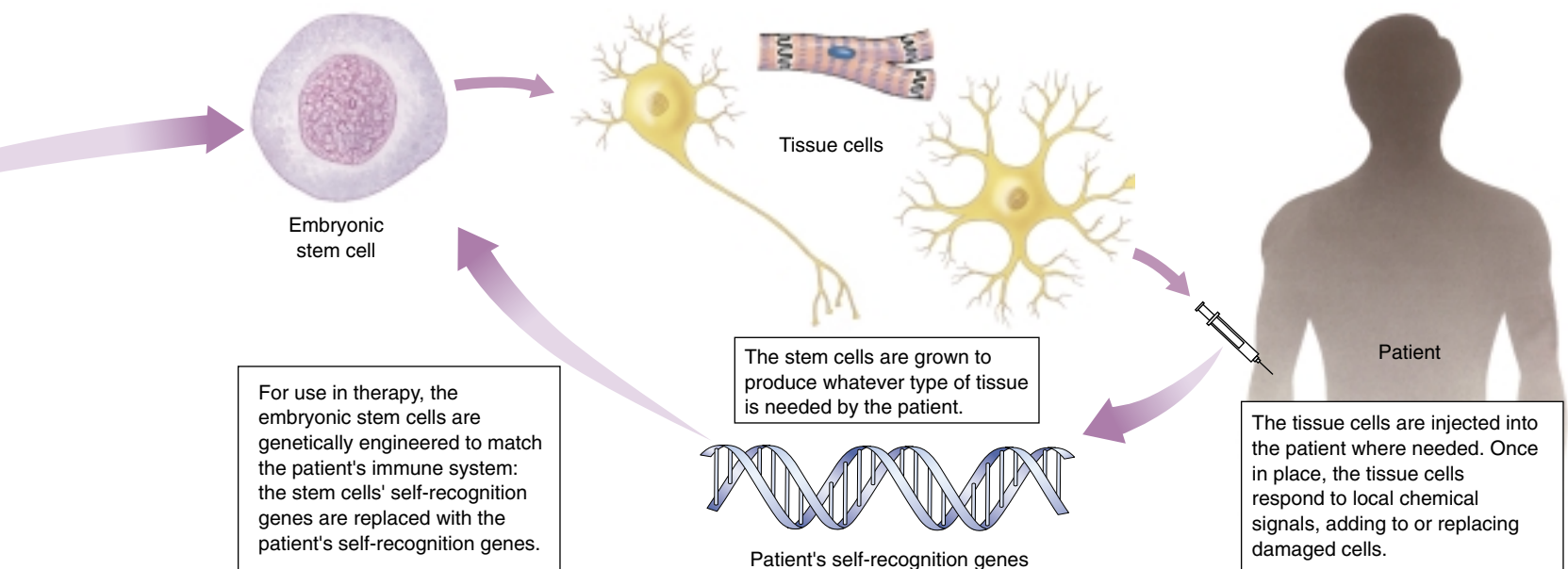
Now on to juvenile diabetes. The scientists injected their cell culture into the pancreas of mice specially bred to develop juvenile diabetes. Unable to manufacture their own insulin because they had no islet cells, these diabetic mice could not survive without daily insulin. What happened? The diabetes was reversed! The mice no longer required insulin.

Impatient to see in more detail what had happened, the researchers sacrificed the mice and examined the cells of their pancreas. The mice appeared to have perfectly normal islet cells.

One might have wished the researchers waited a little longer before terminating the experiment. It is not clear whether the cure was transitory or long term. Still, there is no escaping the conclusion that injection of a culture of adult stem cells cured their juvenile diabetes.

While certainly encouraging, a mouse is not a human, and there is no guarantee the approach will work in humans. But there is every reason to believe it might. The experiment is being repeated now with humans. People suffering from juvenile diabetes are being treated with human pancreatic duct cells obtained from people who have died and donated their organs for research. No ethical issues arise from using cells of adult organ donors, and initial results look promising.

Transplanted stem cells may allow us to replace damaged or lost tissue, offering cures for many disorders that cannot now be treated. Current work focuses on tissue-specific stem cells, which do not present the ethical problems that embryonic stem cells do.



Ethics and Regulation

The advantages afforded by genetic engineering are revolutionizing our lives. But what are the disadvantages, the potential costs and dangers of genetic engineering? Many people, including influential activists and members of the scientific community, have expressed concern that genetic engineers are “playing God” by tampering with genetic material. For instance, what would happen if one fragmented the DNA of a cancer cell, and then incorporated the fragments at random into vectors that were propagated within bacterial cells? Might there not be a danger that some of the resulting bacteria would transmit an infective form of cancer? Could genetically engineered products administered to plants or animals turn out to be dangerous for consumers after several generations? What kind of unforeseen impact on the ecosystem might “improved” crops have? Is it ethical to create “genetically superior” organisms, including humans?

How Do We Measure the Potential Risks of Genetically Modified Crops?

While the promise of genetic engineering is very much in evidence, this same genetic engineering has this summer been the cause of outright war between researchers and protesters in England. In June 1999, British protesters attacked an experimental plot of genetically modified (GM) sugar beets; the following August they destroyed a test field of GM canola (used for cooking oil and animal feed). The contrast could not be more marked between American acceptance of genetically modified crops on the one hand, and European distrust of genetically modified foods, on the other. The intense feelings generated by this dispute point to the need to understand how we measure the risks associated with the genetic engineering of plants.

Two sets of risks need to be considered. The first stems from eating genetically modified foods, the other concerns potential ecological effects.

Is Eating Genetically Modified Food Dangerous? Protesters worry that genetically modified food may have been rendered somehow dangerous. To sort this out, it is useful to bear in mind that bioengineers modify crops in two quite different ways. One class of gene modification makes the crop easier to grow; a second class of modification is intended to improve the food itself.

The introduction of Roundup-resistant soybeans to Europe is an example of the first class of modification. This modification has been very popular with farmers in the United States, who planted half their crop with these soybeans in 1999. They like GM soybeans because the beans can be raised without intense cultivation (weeds are killed with Roundup herbicide instead), which both saves money and lessens soil erosion. But is the soybean that results nutritionally different? No. The gene that confers Roundup resistance in soybeans does so by protecting the plant's

ability to manufacture so-called “aromatic” amino acids. In unprotected weeds, by contrast, Roundup blocks this manufacturing process, killing the weed. Because humans don't make any aromatic amino acids anyway (we get them in our diets), Roundup doesn't hurt us. The GM soybean we eat is nutritionally the same as an “organic” one, just cheaper to produce.

In the second class of modification, where a gene is added to improve the nutritional character of some food, the food will be nutritionally different. In each of these instances, it is necessary to examine the possibility that consumers may prove allergic to the product of the introduced gene. In one instance, for example, addition of a methionine-enhancing gene from Brazil nut into soybeans (which are deficient in this amino acid) was discontinued when six of eight individuals allergic to Brazil nuts produced antibodies to the GM soybeans, suggesting the possibility of a reverse reaction. Instead, methionine levels in GM crops are being increased with genes from sunflowers. Screening for allergy problems is now routine.

On both scores, then, the risk of bioengineering to the food supply seems to be very slight. GM foods to date seem completely safe.

Are GM Crops Harmful to the Environment? What are we to make of the much-publicized report that Monarch butterflies might be killed by eating pollen blowing out of fields planted with GM corn? First, it should come as no surprise. The GM corn (so-called Bt corn) was engineered to contain an insect-killing toxin (harmless to people) in order to combat corn borer pests. Of course it will kill any butterflies or other insects in the immediate vicinity of the field. However, focus on the fact that the GM corn fields do not need to be sprayed with pesticide to control the corn borer. An estimated \$9 billion in damage is caused annually by the application of pesticides in the United States, and billions of insects and other animals, including an estimated 67 million birds, are killed each year. This pesticide-induced murder of wildlife is far more damaging ecologically than any possible effects of GM crops on butterflies.

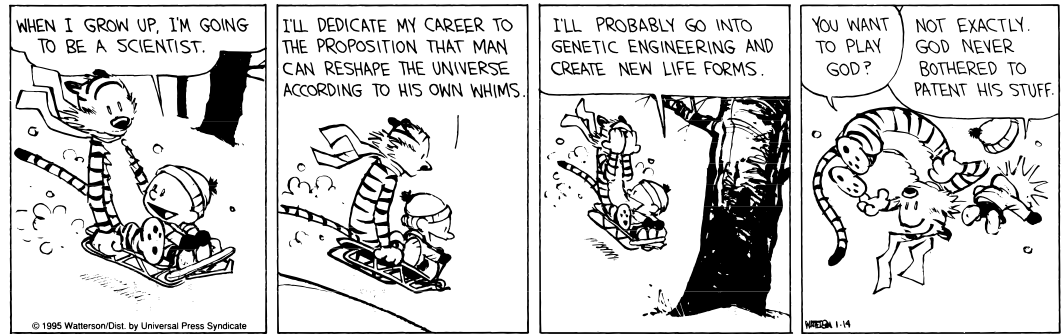
Will pests become resistant to the GM toxin? Not nearly as fast as they now become resistant to the far higher levels of chemical pesticide we spray on crops.

How about the possibility that introduced genes will pass from GM crops to their wild or weedy relatives? This sort of gene flow happens naturally all the time, and so this is a legitimate question. But so what if genes for resistance to Roundup herbicide spread from cultivated sugar beets to wild populations of sugar beets in Europe? Why would that be a problem? Besides, there is almost never a potential relative around to receive the modified gene from the GM crop. There are no wild relatives of soybeans in Europe, for example. Thus, there can be no gene escape from GM soybeans in Europe, any more than genes can flow from you to other kinds of animals.

Calvin and Hobbes

by Bill Watterson

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On either score, then, the risk of bioengineering to the environment seems to be very slight. Indeed, in some cases it lessens the serious environmental damage produced by cultivation and agricultural pesticides.

Should We Label Genetically Modified Foods?

While there seems little tangible risk in the genetic modification of crops, public assurance that these risks are being carefully assessed is important. Few issues manage to raise the temperature of discussions about plant genetic engineering more than labeling of genetically modified (GM) crops. Agricultural producers have argued that there are no demonstrable risks, so that a GM label can only have the function of scaring off wary consumers. Consumer advocates respond that consumers have every right to make that decision, and to the information necessary to make it.

In considering this matter, it is important to separate two quite different issues, the *need* for a label, and the *right* of the public to have one. Every serious scientific investigation of the risks of GM foods has concluded that they are safe—indeed, in the case of soybeans and many other crops modified to improve cultivation, the foods themselves are not altered in any detectable way, and no nutritional test could distinguish them from "organic" varieties. So there seems to be little if any health need for a GM label for genetically engineered foods.

The right of the public to know what it is eating is a very different issue. There is widespread fear of genetic manipulation in Europe, because it is unfamiliar. People there don't trust their regulatory agencies as we do here, because their agencies have a poor track record of protecting them. When they look at genetically modified foods, they are haunted by past experiences of regulatory ineptitude. In England they remember British regulators' failure to protect consumers from meat infected with mad cow disease.

It does no good whatsoever to tell a fearful European that there is no evidence to warrant fear, no trace of data supporting danger from GM crops. A European consumer will simply respond that the harm is not yet evident, that

we don't know enough to see the danger lurking around the corner. "Slow down," the European consumers say. "Give research a chance to look around all the corners. Lets be sure." No one can argue against caution, but it is difficult to imagine what else researchers can look into—safety has been explored very thoroughly. The fear remains, though, for the simple reason that no amount of information can remove it. Like a child scared of a monster under the bed, looking under the bed again doesn't help—the monster still might be there next time. And that means we are going to have to have GM labels, for people have every right to be informed about something they fear.

What should these labels be like? A label that only says "GM FOOD" simply acts as a brand—like a POISON label, it shouts a warning to the public of lurking danger. Why not instead have a GM label that provides information to the consumer, that tells the consumer what regulators know about that product?

(For *Bt corn*): The production of this food was made more efficient by the addition of genes that made plants resistant to pests so that less pesticides were required to grow the crop.

(For *Roundup-ready soybeans*): Genes have been added to this crop to render it resistant to herbicides—this reduces soil erosion by lessening the need for weed-removing cultivation.

(For *high beta-carotene rice*): Genes have been added to this food to enhance its beta-carotene content and so combat vitamin A deficiency.

GM food labels that in each instance actually tell consumers what has been done to the gene-modified crop would go a long way toward hastening public acceptance of gene technology in the kitchen.

Genetic engineering affords great opportunities for progress in medicine and food production, although many are concerned about possible risks. On balance, the risks appear slight, and the potential benefits substantial.



Summary

Questions

Media Resources

19.1 The ability to manipulate DNA has led to a new genetics.

- Genetic engineering involves the isolation of specific genes and their transfer to new genomes.
- An important component of genetic engineering technology is a special class of enzymes called restriction endonucleases, which cleave DNA molecules into fragments.
- The first such recombinant DNA was made by Cohen and Boyer in 1973, when they inserted a frog ribosomal RNA gene into a bacterial plasmid.

1. Why do the ends of the DNA fragments created by restriction endonucleases enable fragments from different genomes to be spliced together?



- Experiment: Cohen/Boyer/Berg- The first Genetically Engineered Organism
- Student Research: Homeobox Genes in the Medicinal Leech

19.2 Genetic engineering involves easily understood procedures.

- Genetic engineering experiments consist of four stages: isolation of DNA, production of recombinant DNA, cloning, and screening for the gene(s) of interest.
- Preliminary screening can be accomplished by making the desired clones resistant to an antibiotic; hybridization can then be employed to identify the gene of interest.
- Gene technologies, including PCR, Southern blotting, RFLP analysis, and the Sanger method, enable researchers to isolate genes and produce them in large quantities.

2. Describe the procedure used to eliminate clones that have not incorporated a vector in a genetic engineering experiment.



- Polymerase Chain Reaction

3. What is used as a probe in a Southern blot? With what does the probe hybridize? How are the regions of hybridization visualized?



- Recombinant



- On Science Article: How Genetic Engineering is Done

19.3 Biotechnology is producing a scientific revolution.

- Extensive research on the human genome has yielded important information about the location of genes, such as those that may be involved in dyslexia, obesity, and resistance to high blood cholesterol levels.
- Gene splicing holds great promise as a clinical tool, particularly in the prevention of disease with bioengineered vaccines.
- A major focus of genetic engineering activity has been agriculture, where genes conferring resistance to herbicides or insect pests have been incorporated into crop plants.
- Recent experiments open the way for cloning of genetically altered animals and suggest that human cloning is feasible.
- The impact of genetic engineering has skyrocketed over the past decade, providing many useful innovations for society; its moral and ethical aspects still provide a topic for heated debates.

4. What is the primary vector used to introduce genes into plant cells? What types of plants are generally infected by this vector? Describe three examples of how this vector has been used for genetic engineering, and explain the agricultural significance of each example.



- Exploration: DNA from Real Court Cases

5. How is the genetic engineering of bovine somatotropin (BST) used to increase milk production in the dairy industry? What effect would BST in milk have on persons who drink it?



- On Science Articles:
 - The Real Promise of Plant Genetic Engineering
 - Should We Label Genetically Modified Foods?
 - Measuring Risks of Genetically Modified Crops
 - The Search for the Natural Relatives of Cassava
 - Renouncing the Terminator
 - Frankenstein Grass is Poised to Invade my Back Yard
 - The Road to Dolly
 - Should a Clone Have Rights?
 - Who Should Own the Secrets of Your Genes?

Evolution

Catching evolution in action

A hundred years ago Charles Darwin's theory of evolution by natural selection was taught as the foundation of biology in public schools throughout the United States. Then something happened. In the 1920s, conservative religious groups began to argue against the teaching of evolution in our nation's schools. Darwinism, they said, contradicted the revealed word of God in the Bible and thus was a direct attack on their religious beliefs. Many of you will have read about the 1925 Scopes "monkey trial" or seen the movie about it, *Inherit the Wind*. In the backwash of this controversy, evolution for the first time in this century disappeared from the schools. Textbook publishers and local school boards, in a wish to avoid the dispute, simply chose not to teach evolution. By 1959, 100 years after Darwin's book, a famous American geneticist cried in anguish, "A hundred years without Darwin is enough!" What he meant was that the theory of evolution by natural selection has become the central operating concept of the science of biology, organic evolution being one of the most solidly validated facts of science. How could we continue to hide this truth from our children, crippling their understanding of science?

In the 1970s, Darwin reappeared in our nation's schools, part of the wave of concern about science that followed *Sputnik*. Not for long, however. Cries from creationists for equal time in the classroom soon had evolution out of our classrooms again. Only in recent years, amid considerable uproar, have states like California succeeded in reforming their school curriculums, focusing on evolution as the central principle of biology. In other states, teaching Darwin remains controversial.

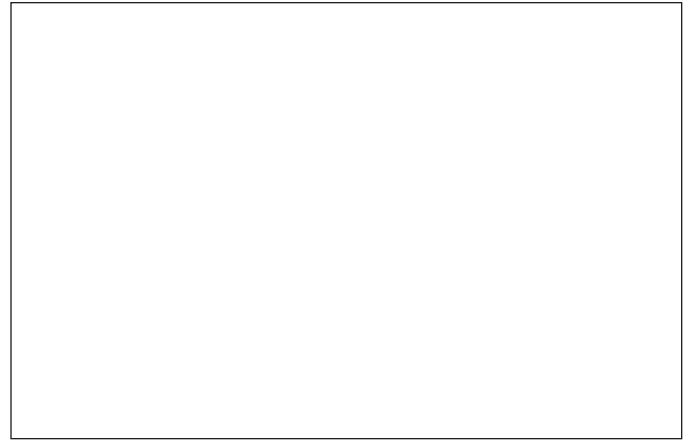
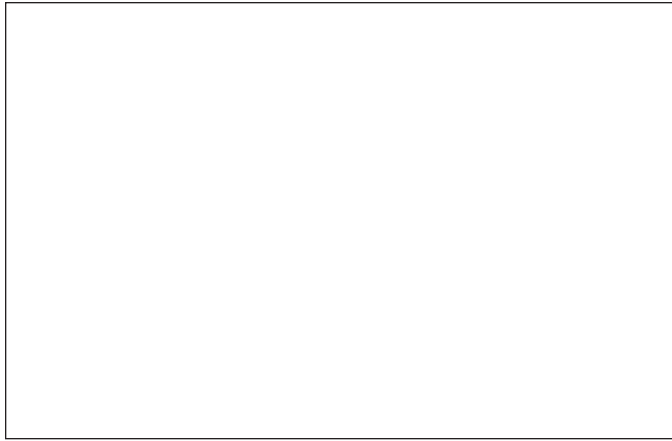
While Darwin's proposal that evolution occurs as the result of natural selection remains controversial in many local school boards, it is accepted by practically every biologist who has examined it seriously. In this section, we will review the evidence supporting Darwin's theory. Evolutionary biology is unlike most other fields of biology in which hypotheses are tested directly with experimental methods. To study evolution, we need to investigate what happened in the past, sometimes many millions of years ago. In this way, evolutionary biology is similar to astron-

The evolution of protective coloration in guppies. In pools below waterfalls where predation is high, guppies are drab colored. In the absence of the highly predatory pike cichlid, guppies in pools above waterfalls are much more colorful and attractive to females. The evolution of these differences can be experimentally tested.

omy and history, relying on observation and deduction rather than experiment and induction to examine ideas about past events.

Nonetheless, evolutionary biology is not entirely an observational science. Darwin was right about many things, but one area in which he was mistaken concerns the pace at which evolution occurs. Darwin thought that evolution occurred at a very slow, almost imperceptible, pace. However, in recent years many case studies of natural populations have demonstrated that in some circumstances evolutionary change can occur rapidly. In these instances, it is possible to establish experimental studies to directly test evolutionary hypotheses. Although laboratory studies on fruit flies and other organisms have been common for more than 50 years, it has only been in recent years that scientists have started conducting experimental studies of evolution in nature.

To conduct experimental tests of evolution, it is first necessary to identify a population in nature upon which strong selection might be operating (see above). Then, by manipulating the strength of the selection, an investigator can predict what outcome selection might produce, then look and see the actual effect on the population.



Evolutionary change in spot number. Populations transported to the low-predation environment quickly increased in number of spots, whereas selection in more dangerous environments, like the predator-filled pool *above right*, led to less conspicuous fish.

The Experiment

Guppies offer an excellent experimental opportunity. The guppy, *Poecilia reticulata*, is found in small streams in north-eastern South America and the nearby island of Trinidad. In Trinidad, guppies are found in many mountain streams. One interesting feature of several streams is that they have waterfalls. Amazingly, guppies are capable of colonizing portions of the stream above the waterfall. During flood seasons, rivers sometimes overflow their banks, creating secondary channels that move through the forest. During these occasions, guppies may be able to move upstream and invade pools above waterfalls. By contrast, not all species are capable of such dispersal and thus are only found in these streams below the first waterfall. One species whose distribution is restricted by waterfalls is the pike cichlid, *Crenicichla alta*, a voracious predator that feeds on other fish, including guppies.

Because of these barriers to dispersal, guppies can be found in two very different environments. In pools just below the waterfalls, predation is a substantial risk and rates of survival are relatively low. By contrast, in similar pools just above the waterfall, few predators prey on guppies. As a result, guppy populations above and below waterfalls have evolved many differences. In the high-predation pools, guppies exhibit drab coloration. Moreover, they tend to reproduce at a younger age.

The differences suggest the action of natural selection. Perhaps as a result of shunting energy to reproduction rather than growth, the fish in high-predation pools attain relatively smaller adult sizes. By contrast, male fish above the waterfall display gaudy colors that they use to court females. Adults there mature later and grow to larger sizes.

Although the differences between guppies living above and below the waterfalls are consistent with the hypothesis that they represent evolutionary responses to differences in the strength of predation, alternative explanations are possible. Perhaps, for example, only very large fish are capable of moving past the waterfall to colonize pools. If this were

the case, then a founder effect would occur in which the new population was established solely by individuals with genes for large size.

The only way to rule out such alternative possibilities is to conduct a controlled experiment. The first experiments were conducted in large pools in laboratory greenhouses. At the start of the experiment, a group of 2000 guppies were divided equally among 10 large pools. Six months later, pike cichlids were added to four of the pools and killifish (which rarely prey on guppies) to another four, with the remaining pools left as “no predator” controls.

The Results

Fourteen months later (which corresponds to 10 guppy generations), the scientists compared the populations. The guppies in the killifish and control pools were indistinguishable, brightly colored and large. In contrast, the guppies in the pike cichlid pools were smaller and drab in coloration. These results established that predation can lead to rapid evolutionary change, but does this laboratory experiments reflect what occurs in nature?

To find out, scientists located two streams that had guppies in pools below a waterfall, but not above it. As in other Trinidadian streams, the pike cichlid was present in the lower pools, but only the killifish was found above the waterfalls. The scientists then transplanted guppies to the upper pools and returned at several-year intervals to monitor the populations. Despite originating from populations in which predation levels were high, the transplanted populations rapidly evolved the traits characteristic of low-predation guppies: they matured late, attained greater size and brighter colors. Control populations in the lower pools, by contrast, continued to mature early and at smaller size. These results demonstrate that substantial evolutionary change can occur in less than 12 years.

To explore this concept further go to our interactive lab at www.mhhe.com/raven6e

20

Genes within Populations

Concept Outline

20.1 Genes vary in natural populations.

Gene Variation Is the Raw Material of Evolution.

Selection acts on the genetic variation present in populations, favoring variants that increase the likelihood of survival and reproduction.

Gene Variation in Nature. Natural populations contain considerable amounts of variation, present at the DNA level and expressed in proteins.

20.2 Why do allele frequencies change in populations?

The Hardy–Weinberg Principle. The proportion of homozygotes and heterozygotes in a population is not altered by meiosis or sexual reproduction.

Five Agents of Evolutionary Change. The frequency of alleles in a population can be changed by evolutionary forces like gene flow and selection.

Identifying the Evolutionary Forces Maintaining Polymorphism. A number of processes can influence allele frequencies in natural populations, but it is difficult to ascertain their relative importance.

Heterozygote Advantage.—In some cases, heterozygotes are superior to either type of homozygote. The gene for sickle cell anemia is one particularly well-understood example.

20.3 Selection can act on traits affected by many genes.

Forms of Selection. Selection can act on traits like height or weight to stabilize or change the level at which the trait is expressed.

Limits to What Selection Can Accomplish. Selection cannot act on traits with little or no genetic variation.



FIGURE 20.1

Genetic variation. The range of genetic material in a population is expressed in a variety of ways—including color.

No other human being is exactly like you (unless you have an identical twin). Often the particular characteristics of an individual have an important bearing on its survival, on its chances to reproduce, and on the success of its offspring. Evolution is driven by such consequences. Genetic variation that influences these characteristics provides the raw material for natural selection, and natural populations contain a wealth of such variation. In plants (figure 20.1), insects, and vertebrates, practically every gene exhibits some level of variation. In this chapter, we will explore genetic variation in natural populations and consider the evolutionary forces that cause allele frequencies in natural populations to change. These deceptively simple matters lie at the core of evolutionary biology.

20.1 Genes vary in natural populations.

Gene Variation Is the Raw Material of Evolution

Evolution Is Descent with Modification

The word “*evolution*” is widely used in the natural and social sciences. It refers to how an entity—be it a social system, a gas, or a planet—changes through time. Although development of the modern concept of evolution in biology can be traced to Darwin’s *On the Origin of Species*, the first five editions of this book never actually used the term! Rather, Darwin used the phrase “descent with modification.” Although many more complicated definitions have been proposed, Darwin’s phrase probably best captures the essence of biological evolution: all species arise from other, pre-existing species. However, through time, they accumulate differences such that ancestral and descendant species are not identical.

Natural Selection Is an Important Mechanism of Evolutionary Change

Darwin was not the first to propose a theory of evolution. Rather, he followed a long line of earlier philosophers and naturalists who deduced that the many kinds of organisms around us were produced by a process of evolution. Unlike his predecessors, however, Darwin proposed **natural selection** as the mechanism of evolution. Natural selection produces evolutionary change when in a population some individuals, which possess certain inherited characteristics, produce more surviving offspring than individuals lacking these characteristics. As a result, the population will gradually come to include more and more individuals with the advantageous characteristics. In this way, the population evolves and becomes better adapted to its local circumstances.

Natural selection was by no means the only evolutionary mechanism proposed. A rival theory, championed by the prominent biologist Jean-Baptiste Lamarck, was that evolution occurred by **the inheritance of acquired characteristics**. According to Lamarck, individuals passed on to offspring body and behavior changes acquired during their lives. Thus, Lamarck proposed that ancestral giraffes with short necks tended to stretch their necks to feed on tree leaves, and this extension of the neck was passed on to subsequent generations, leading to the long-necked giraffe (figure 20.2a). In Darwin’s theory, by contrast, the variation is not created by experience, but is the result of preexisting genetic differences among individuals (figure 20.2b).

Although the efficacy of natural selection is now widely accepted, it is not the only process that can lead to changes

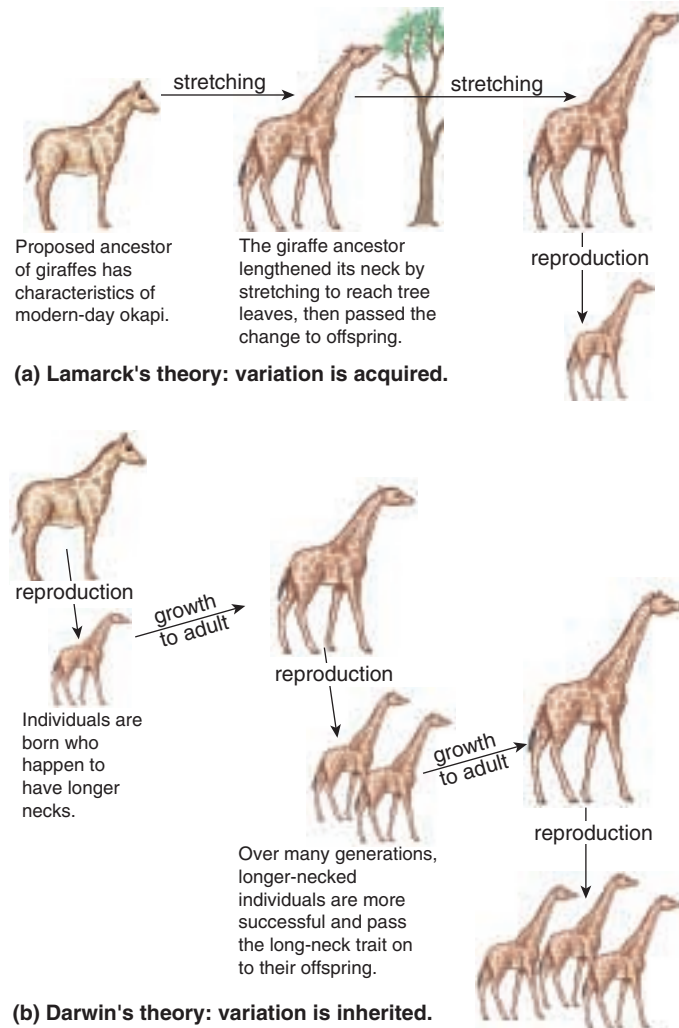


FIGURE 20.2
How did giraffes evolve a long neck?

in the genetic makeup of populations. Allele frequencies can also change as the result of repeated mutations from one allele to another and from migrants bringing alleles into a population. In addition, when populations are small, the frequencies of alleles can change randomly as the result of chance events. Evolutionary biologists debate the relative strengths of these processes. Although no one denies that natural selection is a powerful force leading to adaptive change, the importance of other processes is less certain.

Darwin proposed that natural selection on variants within populations leads to the evolution of different species.

Gene Variation in Nature

Evolution within a species may result from any process that causes a change in the genetic composition of a population. In considering this theory of population genetics, it is best to start by looking at the genetic variation present among individuals within a species. This is the raw material available for the selective process.

Measuring Levels of Genetic Variation

As we saw in chapter 13, a natural population can contain a great deal of genetic variation. This is true not only of humans, but of all organisms. How much variation usually occurs? Biologists have looked at many different genes in an effort to answer this question:

1. **Blood groups.** Chemical analysis has revealed the existence of more than 30 blood group genes in humans, in addition to the ABO locus. At least a third of these genes are routinely found in several alternative allelic forms in human populations. In addition to these, there are more than 45 variable genes encoding other proteins in human blood cells and plasma which are not considered blood groups. Thus, there are more than 75 genetically variable genes in this one system alone.
2. **Enzymes.** Alternative alleles of genes specifying particular enzymes are easy to distinguish by measuring how fast the alternative proteins migrate in an electric field (a process called **electrophoresis**). A great deal of variation exists at enzyme-specifying loci. About 5% of the enzyme loci of a typical human are heterozygous: if you picked an individual at random, and in turn selected one of the enzyme-encoding genes of that individual at random, the chances are 1 in 20 (5%) that the gene you selected would be heterozygous in that individual.

Considering the entire human genome, it is fair to say that almost all people are different from one another. This is also true of other organisms, except for those that reproduce asexually. In nature, genetic variation is the rule.

Enzyme Polymorphism

Many loci in a given population have more than one allele at frequencies significantly greater than would occur from mutation alone. Researchers refer to a locus with more variation than can be explained by mutation as **polymorphic** (*poly*, “many,” *morphic*, “forms”) (figure 20.3). The extent of such variation within natural populations was not even suspected a few decades ago, until modern techniques such as gel electrophoresis made it possible to examine enzymes and other proteins directly. We now know that most populations of insects and plants are polymorphic (that is, have more than one allele occurring at a frequency greater



FIGURE 20.3

Polymorphic variation. These Australian snails, all of the species *Bankivia fasciata*, exhibit considerable variation in pattern and color. Individual variations are heritable and passed on to offspring.

than 5%) at more than half of their enzyme-encoding loci, although vertebrates are somewhat less polymorphic. **Heterozygosity** (that is, the probability that a randomly selected gene will be heterozygous for a randomly selected individual) is about 15% in *Drosophila* and other invertebrates, between 5% and 8% in vertebrates, and around 8% in outcrossing plants. These high levels of genetic variability provide ample supplies of raw material for evolution.

DNA Sequence Polymorphism

With the advent of gene technology, it has become possible to assess genetic variation even more directly by sequencing the DNA itself. In a pioneering study in 1989, Martin Kreitman sequenced ADH genes isolated from 11 individuals of the fruit fly *Drosophila melanogaster*. He found 43 variable sites, only one of which had been detected by protein electrophoresis! In the following decade, numerous other studies of variation at the DNA level have confirmed these findings: abundant variation exists in both the coding regions of genes and in their nontranslated introns—considerably more variation than we can detect examining enzymes with electrophoresis.

Natural populations contain considerable amounts of genetic variation—more than can be accounted for by mutation alone.

20.2 Why do allele frequencies change in populations?

Population genetics is the study of the properties of genes in populations. Genetic variation within natural populations was a puzzle to Darwin and his contemporaries. The way in which meiosis produces genetic segregation among the progeny of a hybrid had not yet been discovered. Selection, scientists then thought, should always favor an optimal form, and so tend to eliminate variation. Moreover, the theory of **blending inheritance**—in which offspring were expected to be phenotypically intermediate relative to their parents—was widely accepted. If blending inheritance were correct, then the effect of any new genetic variant would quickly be diluted to the point of disappearance in subsequent generations.

The Hardy–Weinberg Principle

Following the rediscovery of Mendel’s research, two people in 1908 independently solved the puzzle of why genetic variation persists—G. H. Hardy, an English mathematician, and G. Weinberg, a German physician. They pointed out that the original proportions of the genotypes in a population will remain constant from generation to generation, as long as the following assumptions are met:

1. The population size is very large.
2. Random mating is occurring.
3. No mutation takes place.
4. No genes are input from other sources (no immigration takes place).
5. No selection occurs.

Dominant alleles do not, in fact, replace recessive ones. Because their proportions do not change, the genotypes are said to be in **Hardy–Weinberg equilibrium**.

In algebraic terms, the Hardy–Weinberg principle is written as an equation. Consider a population of 100 cats, with 84 black and 16 white cats. In statistics, **frequency** is defined as the proportion of individuals falling within a certain category in relation to the total number of individuals under consideration. In this case, the respective frequencies would be 0.84 (or 84%) and 0.16 (or 16%). Based on these phenotypic frequencies, can we deduce the underlying frequency of genotypes? If we assume that the white cats are homozygous recessive for an allele we designate b , and the black cats are therefore either homozygous dominant BB or heterozygous Bb , we can calculate the **allele frequencies** of the two alleles in the population from the proportion of black and white individuals. Let the letter p designate the frequency of one allele and the letter q the frequency of the alternative allele. Because there are only two alleles, p plus q must always equal 1.

The Hardy–Weinberg equation can now be expressed in the form of what is known as a binomial expansion:

$$(p + q)^2 = p^2 + 2pq + q^2$$

(Individuals homozygous for allele B)	(Individuals heterozygous with alleles $B + b$)	(Individuals homozygous for allele b)
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If $q^2 = 0.16$ (the frequency of white cats), then $q = 0.4$. Therefore, p , the frequency of allele B , would be 0.6 ($1.0 - 0.4 = 0.6$). We can now easily calculate the **genotype frequencies**: there are $p^2 = (0.6)^2 \times 100$ (the number of cats in the total population), or 36 homozygous dominant BB individuals. The heterozygous individuals have the Bb genotype, and there would be $2pq$, or $(2 \times 0.6 \times 0.4) \times 100$, or 48 heterozygous Bb individuals.

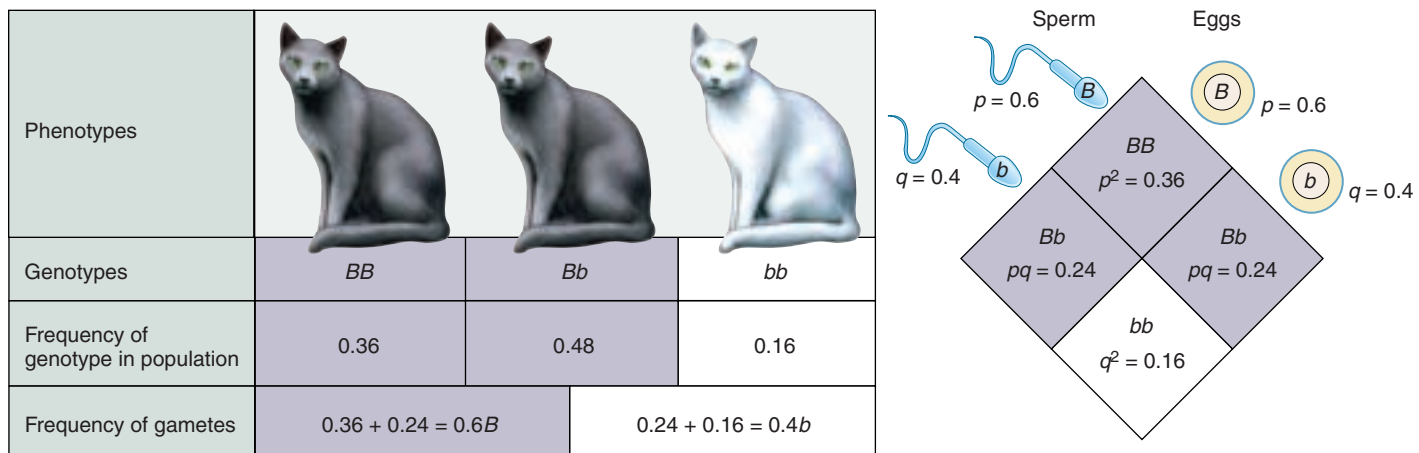


FIGURE 20.4

The Hardy–Weinberg equilibrium. In the absence of factors that alter them, the frequencies of gametes, genotypes, and phenotypes remain constant generation after generation.

Using the Hardy–Weinberg Equation

The Hardy–Weinberg equation is a simple extension of the Punnett square described in chapter 13, with two alleles assigned frequencies p and q . Figure 20.4 allows you to trace genetic reassortment during sexual reproduction and see how it affects the frequencies of the B and b alleles during the next generation. In constructing this diagram, we have assumed that the union of sperm and egg in these cats is random, so that all combinations of b and B alleles occur. For this reason, the alleles are mixed randomly and represented in the next generation in proportion to their original representation. Each individual egg or sperm in each generation has a 0.6 chance of receiving a B allele ($p = 0.6$) and a 0.4 chance of receiving a b allele ($q = 0.4$).

In the next generation, therefore, the chance of combining two B alleles is p^2 , or 0.36 (that is, 0.6×0.6), and approximately 36% of the individuals in the population will continue to have the BB genotype. The frequency of bb individuals is q^2 (0.4×0.4) and so will continue to be about 16%, and the frequency of Bb individuals will be $2pq$ ($2 \times 0.6 \times 0.4$), or approximately 48%. Phenotypically, if the population size remains at 100 cats, we will still see approximately 84 black individuals (with either BB or Bb genotypes) and 16 white individuals (with the bb genotype) in the population. Allele, genotype, and phenotype frequencies have remained unchanged from one generation to the next.

This simple relationship has proved extraordinarily useful in assessing actual situations. Consider the recessive allele responsible for the serious human disease cystic fibrosis. This allele is present in North Americans of Caucasian descent at a frequency q of about 22 per 1000 individuals, or 0.022. What proportion of North American Caucasians, therefore, is expected to express this trait? The frequency of double recessive individuals (q^2) is expected to be 0.022×0.022 , or 1 in every 2000 individuals. What proportion is expected to be heterozygous carriers? If the frequency of the recessive allele q is 0.022, then the frequency of the dominant allele p must be $1 - 0.022$, or 0.978. The frequency of heterozygous individuals ($2pq$) is thus expected to be $2 \times 0.978 \times 0.022$, or 43 in every 1000 individuals.

How valid are these calculated predictions? For many genes, they prove to be very accurate. As we will see, for some genes the calculated predictions do *not* match the actual values. The reasons they do not tell us a great deal about evolution.

Why Do Allele Frequencies Change?

According to the Hardy–Weinberg principle, both the allele and genotype frequencies in a large, random-mating population will remain constant from generation to generation if no mutation, no gene flow, and no selection occur. The stipulations tacked onto the end of the state-

Table 20.1 Agents of Evolutionary Change

Factor	Description
Mutation	The ultimate source of variation. Individual mutations occur so rarely that mutation alone does not change allele frequency much.
Gene flow	A very potent agent of change. Populations exchange members.
Nonrandom mating	Inbreeding is the most common form. It does not alter allele frequency but decreases the proportion of heterozygotes.
Genetic drift	Statistical accidents. Usually occurs only in very small populations.
Selection	The only form that produces <i>adaptive</i> evolutionary changes.

ment are important. In fact, they are the key to the importance of the Hardy–Weinberg principle, because individual allele frequencies often change in natural populations, with some alleles becoming more common and others decreasing in frequency. The Hardy–Weinberg principle establishes a convenient baseline against which to measure such changes. By looking at how various factors alter the proportions of homozygotes and heterozygotes, we can identify the forces affecting particular situations we observe.

Many factors can alter allele frequencies. Only five, however, alter the proportions of homozygotes and heterozygotes enough to produce significant deviations from the proportions predicted by the Hardy–Weinberg principle: mutation, gene flow (including both immigration into and emigration out of a given population), nonrandom mating, genetic drift (random change in allele frequencies, which is more likely in small populations), and selection (table 20.1). Of these, only selection produces adaptive evolutionary change because only in selection does the result depend on the nature of the environment. The other factors operate relatively independently of the environment, so the changes they produce are not shaped by environmental demands.

The Hardy–Weinberg principle states that in a large population mating at random and in the absence of other forces that would change the proportions of the different alleles at a given locus, the process of sexual reproduction (meiosis and fertilization) alone will not change these proportions.

Five Agents of Evolutionary Change

1. Mutation

Mutation from one allele to another can obviously change the proportions of particular alleles in a population. Mutation rates are generally so low that they have little effect on the Hardy–Weinberg proportions of common alleles. A single gene may mutate about 1 to 10 times per 100,000 cell divisions (although *some* genes mutate much more frequently than that). Because most environments are constantly changing, it is rare for a population to be stable enough to accumulate changes in allele frequency produced by a process this slow. Nonetheless, mutation is the ultimate source of genetic variation and thus makes evolution possible. It is important to remember, however, that the likelihood of a particular mutation occurring is not affected by natural selection; that is, mutations do not occur more frequently in situations in which they would be favored by natural selection.

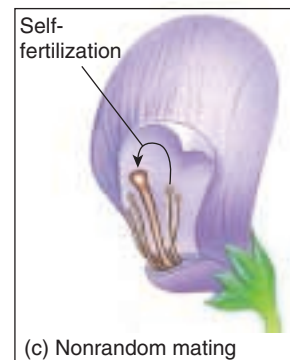
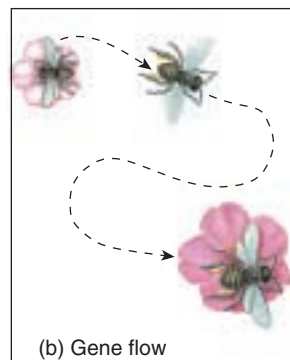
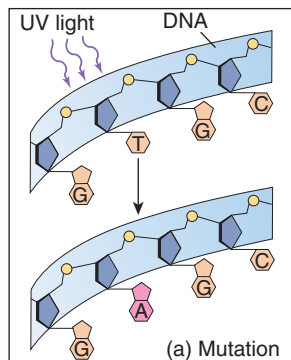


FIGURE 20.5
Five agents of evolutionary change. (a) Mutation, (b) gene flow, (c) nonrandom mating, (d) genetic drift, and (e) selection.

2. Gene Flow

Gene flow is the movement of alleles from one population to another. It can be a powerful agent of change because members of two different populations may exchange genetic material. Sometimes gene flow is obvious, as when an animal moves from one place to another. If the characteristics of the newly arrived animal differ from those of the animals already there, and if the newcomer is adapted well enough to the new area to survive and mate successfully, the genetic composition of the receiving population may be altered. Other important kinds of gene flow are not as obvious. These subtler movements include the drifting of gametes or immature stages of plants or marine animals from one place to another (figure 20.5). Male gametes of flowering plants are often carried great distances by insects and other animals that visit their flowers. Seeds may also blow in the wind or be carried by animals or other agents to new populations far from their place of origin. In addition, gene flow may also result from the mating of individuals belonging to adjacent populations.

However it occurs, gene flow can alter the genetic characteristics of populations and prevent them from maintaining Hardy–Weinberg equilibrium. In addition, even low levels of gene flow tend to homogenize allele frequencies

among populations and thus keep the populations from diverging genetically. In some situations, gene flow can counter the effect of natural selection by bringing an allele into a population at a rate greater than that at which the allele is removed by selection.

3. Nonrandom Mating

Individuals with certain genotypes sometimes mate with one another more commonly than would be expected on a random basis, a phenomenon known as nonrandom mating. **Inbreeding** (mating with relatives) is a type of nonrandom mating that causes the frequencies of particular genotypes to differ greatly from those predicted by the Hardy–Weinberg principle. Inbreeding does not change the frequency of the alleles, but rather increases the proportion of homozygous individuals because relatives are likely to be genetically similar and thus produce offspring with two copies of the same allele. This is why populations of self-fertilizing plants consist primarily of homozygous individuals, whereas **outcrossing** plants, which interbreed with individuals different from themselves, have a higher proportion of heterozygous individuals.

By increasing homozygosity in a population, inbreeding increases the expression of recessive alleles. It is for this reason that marriage between close relatives is discouraged and to some degree outlawed—it increases the possibility of producing children homozygous for an allele associated with one or more of the recessive genetic disorders discussed in chapter 13.

4. Genetic Drift

In small populations, frequencies of particular alleles may change drastically by chance alone. Such changes in allele frequencies occur randomly, as if the frequencies were drifting, and are thus known as **genetic drift**. For this reason, a population must be large to be in Hardy–Weinberg equilibrium. If the gametes of only a few individuals form the next generation, the alleles they carry may by chance not be representative of the parent population from which they were drawn, as illustrated in figure 20.6, where a small number of individuals are removed from a bottle containing many. By chance, most of the individuals removed are blue, so the new population has a much higher population of blue individuals than the parent one had.

A set of small populations that are isolated from one another may come to differ strongly as a result of genetic drift even if the forces of natural selection do not differ between the populations. Indeed, because of genetic drift, harmful alleles may increase in frequency in small populations, despite selective disadvantage, and favorable alleles may be lost even though selectively advantageous. It is interesting to realize that humans have lived in small groups for much of the course of their evolution; consequently, genetic drift may have been a particularly important factor in the evolution of our species.

Even large populations may feel the effect of genetic drift. Large populations may have been much smaller in the past, and genetic drift may have greatly altered allele frequencies at that time. Imagine a population containing only two alleles of a gene, B and b , in equal frequency (that is, $p = q = 0.5$). In a large Hardy–Weinberg population, the genotype frequencies are expected to be 0.25 BB , 0.50 Bb , and 0.25 bb . If only a small sample produces the next generation, large deviations in these genotype frequencies can occur by chance. Imagine, for example, that four individuals form the next generation, and that by chance they are two Bb heterozygotes and two BB homozygotes—the allele frequencies in the next generation are $p = 0.75$ and $q = 0.25$! If you were to replicate this experiment 1000 times, each time randomly drawing four individuals from the parental population, one of the two alleles would be missing entirely from about 8 of the 1000 populations. This leads to an important conclusion: genetic drift leads to the loss of alleles in isolated populations. Two related causes of decreases in a population's size are founder effects and bottlenecks.

Founder Effects. Sometimes one or a few individuals disperse and become the founders of a new, isolated population at some distance from their place of origin. These pioneers are not likely to have all the alleles present in the source population. Thus, some alleles may be lost from the new population and others may change drastically in frequency. In some cases, previously rare alleles in the source population may be a significant fraction of the new population's genetic endowment. This phenomenon is called the founder effect. Founder effects are not rare in nature.

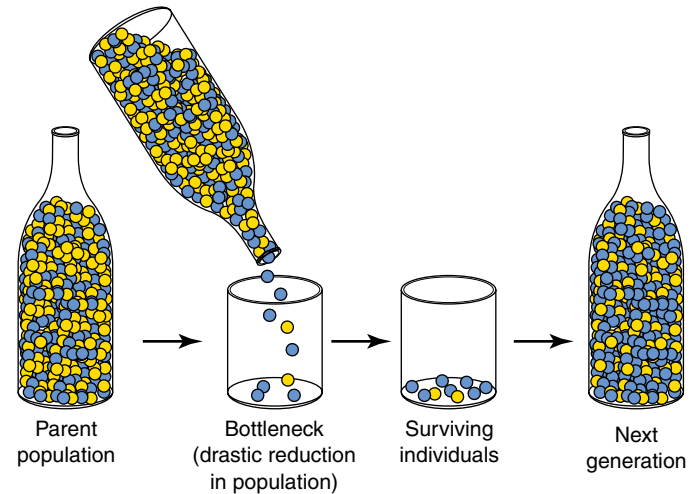


FIGURE 20.6

Genetic drift: The bottleneck effect. The parent population contains roughly equal numbers of blue and yellow individuals. By chance, the few remaining individuals that comprise the next generation are mostly blue. The bottleneck occurs because so few individuals form the next generation, as might happen after an epidemic or catastrophic storm.

Many self-pollinating plants start new populations from a single seed.

Founder effects have been particularly important in the evolution of organisms on distant oceanic islands, such as the Hawaiian Islands and the Galápagos Islands visited by Darwin. Most of the organisms in such areas probably derive from one or a few initial “founders.” In a similar way, isolated human populations are often dominated by genetic features characteristic of their particular founders.

The Bottleneck Effect. Even if organisms do not move from place to place, occasionally their populations may be drastically reduced in size. This may result from flooding, drought, epidemic disease, and other natural forces, or from progressive changes in the environment. The few surviving individuals may constitute a random genetic sample of the original population (unless some individuals survive specifically because of their genetic makeup). The resultant alterations and loss of genetic variability has been termed the **bottleneck effect**.

Some living species appear to be severely depleted genetically and have probably suffered from a bottleneck effect in the past. For example, the northern elephant seal, which breeds on the western coast of North America and nearby islands, was nearly hunted to extinction in the nineteenth century and was reduced to a single population containing perhaps no more than 20 individuals on the island of Guadalupe off the coast of Baja, California. As a result of this bottleneck, even though the seal populations have rebounded and now number in the tens of thousands, this species has lost almost all of its genetic variation.

5. Selection

As Darwin pointed out, some individuals leave behind more progeny than others, and the rate at which they do so is affected by phenotype and behavior. We describe the results of this process as **selection** and speak of both **artificial selection** and **natural selection**. In artificial selection, the breeder selects for the desired characteristics. In natural selection, environmental conditions determine which individuals in a population produce the most offspring. For natural selection to occur and result in evolutionary change, three conditions must be met:

- 1. Variation must exist among individuals in a population.** Natural selection works by favoring individuals with some traits over individuals with alternative traits. If no variation exists, natural selection cannot operate.
- 2. Variation among individuals results in differences in number of offspring surviving in the next generation.** This is the essence of natural selection. Because of their phenotype or behavior, some individuals are more successful than others in producing offspring and thus passing their genes on to the next generation.
- 3. Variation must be genetically inherited.** For natural selection to result in evolutionary change, the selected differences must have a genetic basis. However, not all variation has a genetic basis—even genetically identical individuals may be phenotypically quite distinctive if they grow up in different environments. Such environmental effects are common in nature. In many turtles, for example, individuals that hatch from eggs laid in moist soil are heavier, with longer and wider shells, than individuals from nests in drier areas. As a result of these environmental effects, variation within a population does not always indicate the existence of underlying genetic variation. When phenotypically different individuals do not differ genetically, then differences in the number of their offspring will not alter the genetic composition of the population in the next generation and, thus, no evolutionary change will have occurred.

It is important to remember that natural selection and evolution are not the same—the two concepts often are incorrectly equated. Natural selection is a process, whereas evolution is the historical record of change through time. Evolution is an outcome, not a process. Natural selection (the process) can lead to evolution (the outcome), but natural selection is only one of several processes that can produce evolutionary change. Moreover, natural selection can occur without producing evolutionary change; only if variation is genetically based will natural selection lead to evolution.

Selection to Avoid Predators. Many of the most dramatic documented instances of adaptation involve genetic changes which decrease the probability of capture by a predator. The caterpillar larvae of the common sulphur butterfly *Colias eurytheme* usually exhibit a dull Kelly green color, providing excellent camouflage on the alfalfa plants on which they feed. An alternative bright blue color morph is kept at very low frequency because this color renders the larvae highly visible on the food plant, making it easier for bird predators to see them. In a similar fashion, the way the shell markings in the land snail *Cepaea nemoralis* match its background habitat reflects the same pattern of avoiding predation by camouflage.

One of the most dramatic examples of background matching involves ancient lava flows in the middle of deserts in the American southwest. In these areas, the black rock formations produced when the lava cooled contrasts starkly to the surrounding bright glare of the desert sand. Populations of many species of animals—including lizards, rodents, and a variety of insects—occurring on these rocks are dark in color, whereas sand-dwelling populations in surrounding areas are much lighter (figure 20.7). Predation is the likely cause selecting for these differences in color. Laboratory studies have confirmed that predatory birds are adept at picking out individuals occurring on backgrounds to which they are not adapted.

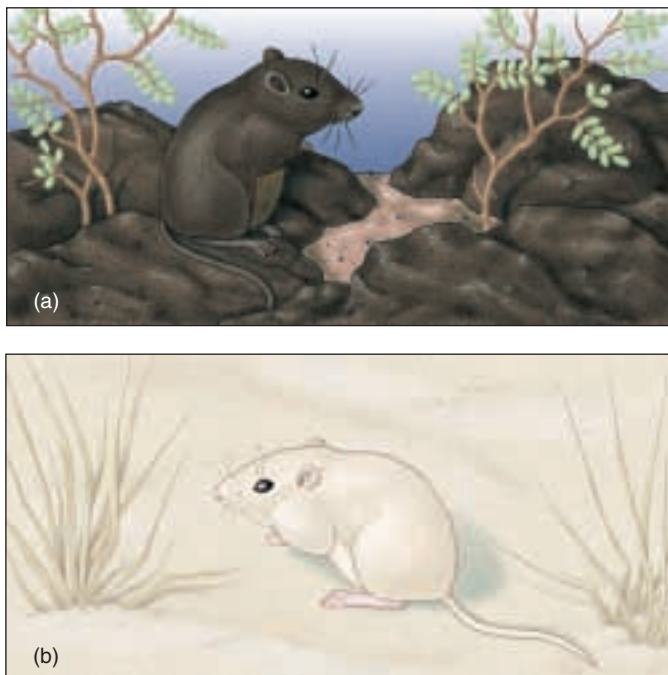


FIGURE 20.7
Pocket mice from the Tularosa Basin of New Mexico whose color matches their background. (a) The rock pocket mouse lives on lava, (b) while the Apache pocket mouse lives on white sand.

Selection to Match Climatic Conditions. Many studies of selection have focused on genes encoding enzymes because in such cases the investigator can directly assess the consequences to the organism of changes in the frequency of alternative enzyme alleles. Often investigators find that enzyme allele frequencies vary latitudinally, with one allele more common in northern populations but progressively less common at more southern locations. A superb example is seen in studies of a fish, the mummichog, *Fundulus heteroclitus*, which ranges along the eastern coast of North America. In this fish, allele frequencies of the gene that produces the enzyme lactase dehydrogenase, which catalyzes the conversion of pyruvate to lactate, vary geographically (figure 20.8). Biochemical studies show that the enzymes formed by these alleles function differently at different temperatures, thus explaining their geographic distributions. For example, the form of the enzyme that is more frequent in the north is a better catalyst at low temperatures than the enzyme from the south. Moreover, functional studies indicate that at low temperatures, individuals with the northern allele swim faster, and presumably survive better, than individuals with the alternative allele.

Selection for Pesticide Resistance. A particularly clear example of selection in action in natural populations is provided by studies of pesticide resistance in insects. The widespread use of insecticides has led to the rapid evolution of resistance in more than 400 pest species. For example, the resistance allele at the *pen* gene decreases the uptake of insecticide, whereas alleles at the *kdr* and *dld-r* genes decrease the number of target sites, thus decreasing the binding ability of the insecticide (figure 20.9). Other alleles enhance the ability of the insects' enzymes to identify and detoxify insecticide molecules.

Single genes are also responsible for resistance in other organisms. The pigweed, *Amaranthus hybridus*, is one of about 28 agricultural weeds that have evolved resistance to the herbicide Triazine. Triazine inhibits photosynthesis by binding to a protein in the chloroplast membrane. Single amino acid substitutions in the gene encoding the protein diminish the ability of Triazine to decrease the plant's photosynthetic capabilities. Similarly, Norway rats are normally susceptible to the pesticide Warfarin, which diminishes the clotting ability of the rat's blood and leads to fatal hemorrhaging. However, a resistance allele at a single gene alters a metabolic pathway and renders Warfarin ineffective.

Five factors can bring about a deviation from the proportions of homozygotes and heterozygotes predicted by the Hardy-Weinberg principle. Only selection regularly produces adaptive evolutionary change, but the genetic constitution of individual populations, and thus the course of evolution, can also be affected by mutation, gene flow, nonrandom mating, and genetic drift.

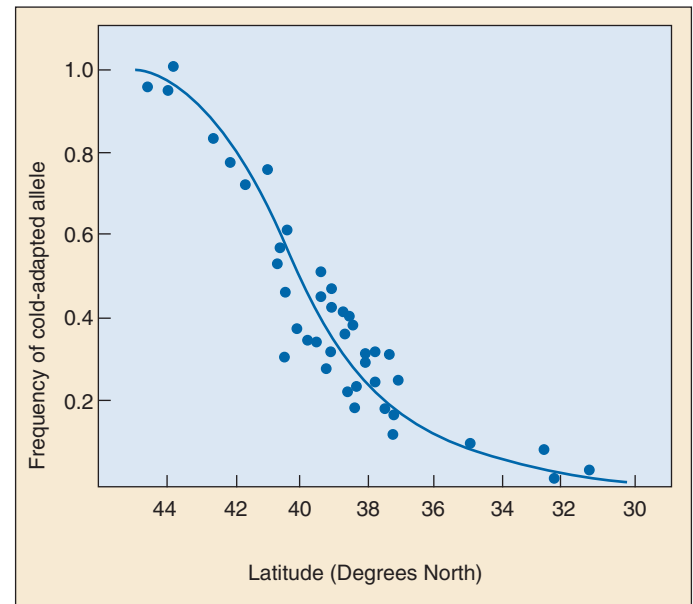


FIGURE 20.8
Selection to match climatic conditions. Frequency of the cold-adapted allele for lactase dehydrogenase in the mummichog (*Fundulus heteroclitus*) decreases at lower latitudes, which are warmer.

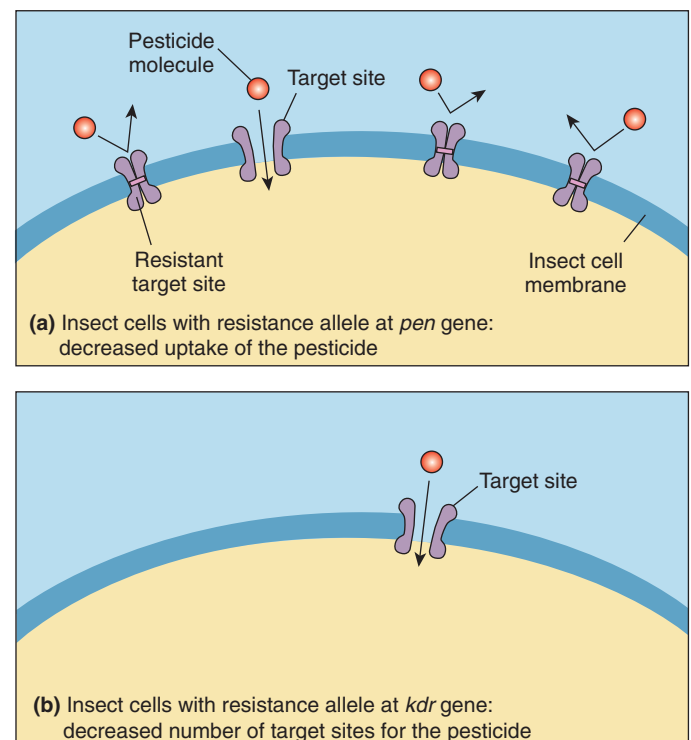


FIGURE 20.9
Selection for pesticide resistance. Resistance alleles at genes like *pen* and *kdr* allow insects to be more resistant to pesticides. Insects that possess these resistance alleles have become more common through selection.

Identifying the Evolutionary Forces Maintaining Polymorphism

The Adaptive Selection Theory

As evidence began to accumulate in the 1970s that natural populations exhibit a great deal of genetic polymorphism (that is, many alleles of a gene exist in the population), the question arose: What evolutionary force is maintaining the polymorphism? As we have seen, there are in principle five processes that act on allele frequencies: mutation, migration, nonrandom mating, genetic drift, and selection. Because migration and nonrandom mating are not major influences in most natural populations, attention focused on the other three forces.

The first suggestion, advanced by R. C. Lewontin (one of the discoverers of enzyme polymorphism) and many others, was that selection was the force acting to maintain the polymorphism. Natural environments are often quite heterogeneous, so selection might reasonably be expected to pull gene frequencies in different directions within different microhabitats, generating a condition in which many alleles persist. This proposal is called the **adaptive selection theory**.

The Neutral Theory

A second possibility, championed by the great Japanese geneticist Moto Kimura, was that a balance between mutation and genetic drift is responsible for maintaining polymorphism. Kimura used elegant mathematics to demonstrate that, even in the absence of selection, natural populations could be expected to contain considerable polymorphism if mutation rates (generating the variation) were high enough and population sizes (promoting genetic drift) were small enough. In this proposal, selection is not acting, differences between alleles being “neutral to selection.” The proposal is thus called the **neutral theory**.

Kimura’s theory, while complex, can be stated simply:

$$\bar{H} = 1/(4N_e\mu + 1)$$

$\bar{H}$, the mean heterozygosity, is the likelihood that a randomly selected member of the population will be heterozygous at a randomly selected locus. In a population without selection, this value is influenced by two variables, the effective population size (N_e) and the mutation rate (μ).

The peculiar difficulty of the neutral theory is that the level of polymorphism, as measured by $\bar{H}$, is determined by the product of a very large number, N_e , and a very small number, μ , both very difficult to measure with precision. As a result, the theory can account for almost any value of $\bar{H}$, making it very difficult to prove or disprove. As you might expect, a great deal of controversy has resulted.

Testing the Neutral Theory

Choosing between the adaptive selection theory and the neutral theory is not simple, for they both appear to account for much of the data on gene polymorphism in natural populations. A few well-characterized instances where selection acts on enzyme alleles do not settle the more general issue. An attempt to test the neutral theory by examining large-scale patterns of polymorphism sheds light on the difficulty of choosing between the two theories:

Population size: According to the neutral theory, polymorphism as measured by $\bar{H}$ should be proportional to the effective population size N_e , assuming the mutation rate among neutral alleles μ is constant. Thus, $\bar{H}$ should be much greater for insects than humans, as there are far more individuals in an insect population than in a human one. When DNA sequence variation is examined, the fruit fly *Drosophila melanogaster* indeed exhibits sixfold higher levels of variation, as the theory predicts; but when enzyme polymorphisms are examined, levels of variation in fruit flies and humans are similar. If the level of DNA variation correctly mirrors the predictions of the neutral theory, then something (selection?) is increasing variation at the enzyme level in humans. These sorts of patterns argue for rejection of the neutral theory.

The nearly neutral model: One way to rescue the neutral theory from these sorts of difficulties is to retreat from the assumption of strict neutrality, modifying the theory to assume that many of the variants are slightly deleterious rather than strictly neutral to selection. With this adjustment, it is possible to explain many of the population-size-dependent large-scale patterns. However, little evidence exists that the wealth of enzyme polymorphism in natural populations is in fact slightly deleterious.

As increasing amounts of DNA sequence data become available, a detailed picture of variation at the DNA level is emerging. It seems clear that most nucleotide substitutions that change amino acids are disadvantageous and are eliminated by selection. But what about the many protein alleles that are seen in natural populations? Are they nearly neutral or advantageous? No simple answer is yet available, although the question is being actively investigated. Levels of polymorphism at enzyme-encoding genes may depend on both the action of selection on the gene (the adaptive selection theory) and on the population dynamics of the species (the nearly neutral theory), with the relative contribution varying from one gene to the next.

Adaptive selection clearly maintains some enzyme polymorphisms in natural populations. Genetic drift seems to play a major role in producing the variation we see at the DNA level. For most enzyme-level polymorphism, investigators cannot yet choose between the selection theory and the nearly neutral theory.

Interactions among Evolutionary Forces

When alleles are not selectively neutral, levels of variation retained in a population may be determined by the relative strength of different evolutionary processes. In theory, for example, if allele *B* mutates to allele *b* at a high enough rate, allele *b* could be maintained in the population even if natural selection strongly favored allele *B*. In nature, however, mutation rates are rarely high enough to counter the effects of natural selection.

The effect of natural selection also may be countered by genetic drift. Both processes may act to remove variation from a population. However, whereas selection is a deterministic process that operates to increase the representation of alleles that enhance survival and reproductive success, drift is a random process. Thus, in some cases, drift may lead to a decrease in the frequency of an allele that is favored by selection. In some extreme cases, drift may even lead to the loss of a favored allele from a population. Remember, however, that the magnitude of drift is negatively related to population size; consequently, natural selection is expected to overwhelm drift except when populations are very small.

Gene Flow versus Natural Selection

Gene flow can be either a constructive or a constraining force. On one hand, gene flow can increase the adaptedness of a species by spreading a beneficial mutation that arises in one population to other populations within a species. On the other hand, gene flow can act to impede adaptation within a population by continually importing inferior alleles from other populations. Consider two populations of a species that live in different environments. In this situation, natural selection might favor different alleles—*B* and *b*—in the different populations. In the absence of gene flow and other evolutionary processes, the frequency of *B* would be expected to reach 100% in one population and 0% in the other. However, if gene flow were going on between the two populations, then the less favored allele would continually be reintroduced into each population. As a result, the frequency of the two alleles in each population would reflect a balance between the rate at which gene flow brings the inferior allele into a population versus the rate at which natural selection removes it.

A classic example of gene flow opposing natural selection occurs on abandoned mine sites in Great Britain. Although mining activities ceased hundreds of years ago, the concentration of metal ions in the soil is still much greater

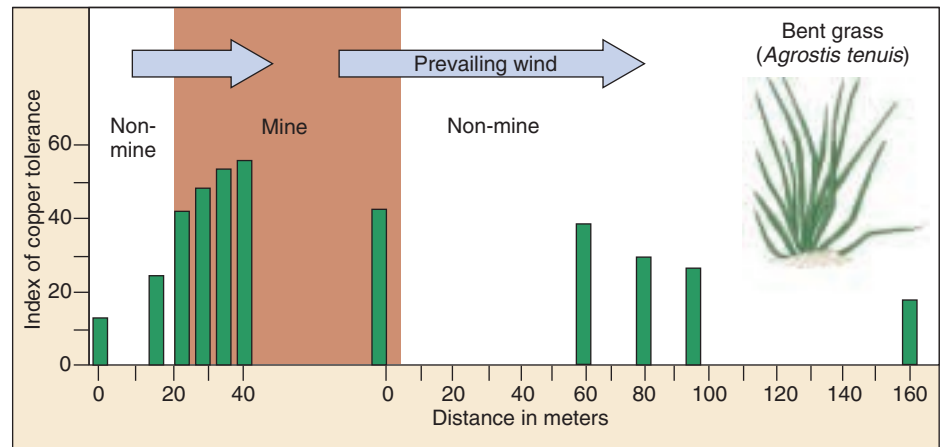


FIGURE 20.10

Degree of copper tolerance in grass plants on and near ancient mine sites. Prevailing winds blow pollen containing nontolerant alleles onto the mine site and tolerant alleles beyond the site's borders.

than in surrounding areas. Heavy metal concentrations are generally toxic to plants, but alleles at certain genes confer resistance. The ability to tolerate heavy metals comes at a price, however; individuals with the resistance allele exhibit lower growth rates on non-polluted soil. Consequently, we would expect the resistance allele to occur with a frequency of 100% on mine sites and 0% elsewhere. Heavy metal tolerance has been studied particularly intensively in the slender bent grass, *Agrostis tenuis*, in which researchers have found that the resistance allele occurs at intermediate levels in many areas (figure 20.10). The explanation relates to the reproductive system of this grass in which pollen, the male gamete (that is, the floral equivalent of sperm), is dispersed by the wind. As a result, pollen—and the alleles it carries—can be blown for great distances, leading to levels of gene flow between mine sites and unpolluted areas high enough to counteract the effects of natural selection.

In general, the extent to which gene flow can hinder the effects of natural selection should depend on the relative strengths of the two processes. In species in which gene flow is generally strong, such as birds and wind-pollinated plants, the frequency of the less favored allele may be relatively high, whereas in more sedentary species which exhibit low levels of gene flow, such as salamanders, the favored allele should occur at a frequency near 100%.

Evolutionary processes may act to either remove or maintain genetic variation within a population. Allele frequency sometimes may reflect a balance between opposed processes, such as gene flow and natural selection. In such cases, observed frequencies will depend on the relative strength of the processes.

Heterozygote Advantage

In the previous pages, natural selection has been discussed as a process that removes variation from a population by favoring one allele over others at a genetic locus. However, if heterozygotes are favored over homozygotes, then natural selection actually will tend to maintain variation in the population. The reason is simple. Instead of tending to remove less successful alleles from a population, such **heterozygote advantage** will favor individuals with copies of both alleles, and thus will work to maintain both alleles in the population. Some evolutionary biologists believe that heterozygote advantage is pervasive and can explain the high levels of polymorphism observed in natural populations. Others, however, believe that it is relatively rare.

Sickle Cell Anemia

The best documented example of heterozygote advantage is sickle cell anemia, a hereditary disease affecting hemoglobin in humans. Individuals with sickle cell anemia exhibit symptoms of severe anemia and contain abnormal red blood cells which are irregular in shape, with a great number of long and sickle-shaped cells. The disease is particularly common among African Americans. In chapter 13, we noted that this disorder, which affects roughly 3 African Americans out of every 1000, is associated with a particular recessive allele. Using the Hardy-Weinberg equation, you can calculate the frequency of the sickle cell allele in the African-American population; this frequency is the square root of 0.003, or approximately 0.054. In contrast, the frequency of the allele among white Americans is only about 0.001.

Sickle cell anemia is often fatal. Until therapies were developed to more effectively treat its symptoms, almost all affected individuals died as children. Even today, 31% of patients in the United States die by the age of 15. The disease occurs because of a single amino acid change, repeated in the two beta chains of the hemoglobin molecule. In this change, a valine replaces the usual glutamic acid at a location on the surface of the protein near the oxygen-binding site. Unlike glutamic acid, valine is nonpolar (hydrophobic). Its presence on the surface of the molecule creates a “sticky” patch that attempts to escape from the polar water environment by binding to another similar patch. As long as oxygen is bound to the hemoglobin molecule there is no problem, because the hemoglobin atoms shield the critical area of the surface. When oxygen levels fall, such as after exercise or when an individual is stressed, oxygen is not so readily bound to hemoglobin and the exposed sticky patch binds to similar patches on other hemoglobin molecules, eventually producing long, fibrous clumps (figure 20.11). The result is a deformed, “sickle-shaped” red blood cell.

Individuals who are heterozygous or homozygous for the valine-specifying allele (designated allele *S*) are said to possess the sickle cell trait. Heterozygotes produce some sickle-shaped red blood cells, but only 2% of the number seen in homozygous individuals. The reason is that in heterozygotes, one-half of the molecules do not contain valine at the critical location. Consequently, when a molecule produced by the non-sickle cell allele is added to the chain, there is no further “sticky” patch available to add additional molecules and chain elongation stops. Hence, most chains in heterozygotes are too short to produce sickling of the cell.

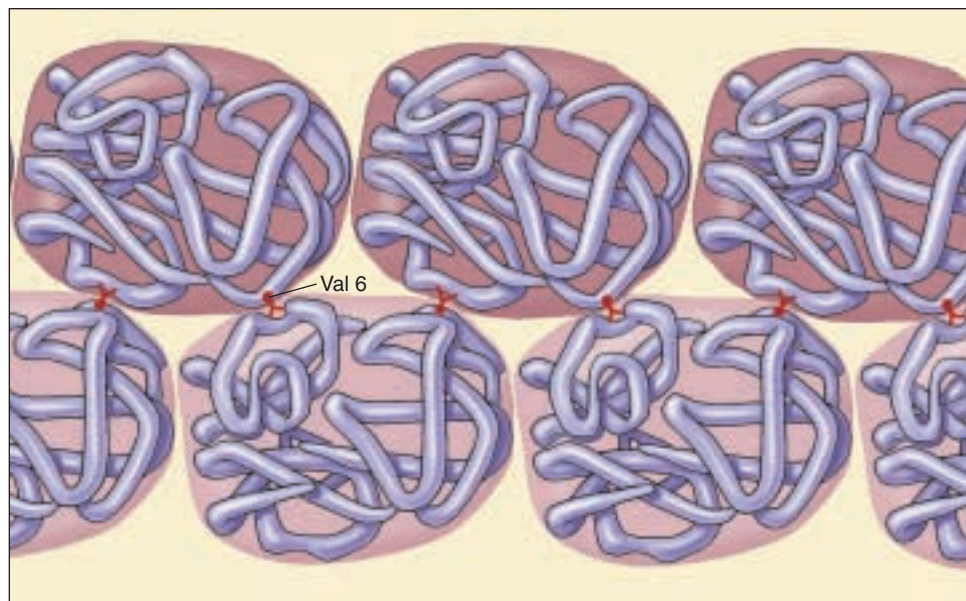


FIGURE 20.11

Why the sickle cell mutation causes hemoglobin to clump. The sickle cell mutation changes the sixth amino acid in the hemoglobin β chain (position B6) from glutamic acid (very polar) to valine (nonpolar). The unhappy result is that the nonpolar valine at position B6, protruding from a corner of the hemoglobin molecule, fits into a nonpolar pocket on the opposite side of another hemoglobin molecule, causing the two molecules to clump together. As each molecule has both a B6 valine and an opposite nonpolar pocket, long chains form. When polar glutamic acid (the normal allele) occurs at position B6, it is not attracted to the nonpolar pocket, and no clumping occurs.
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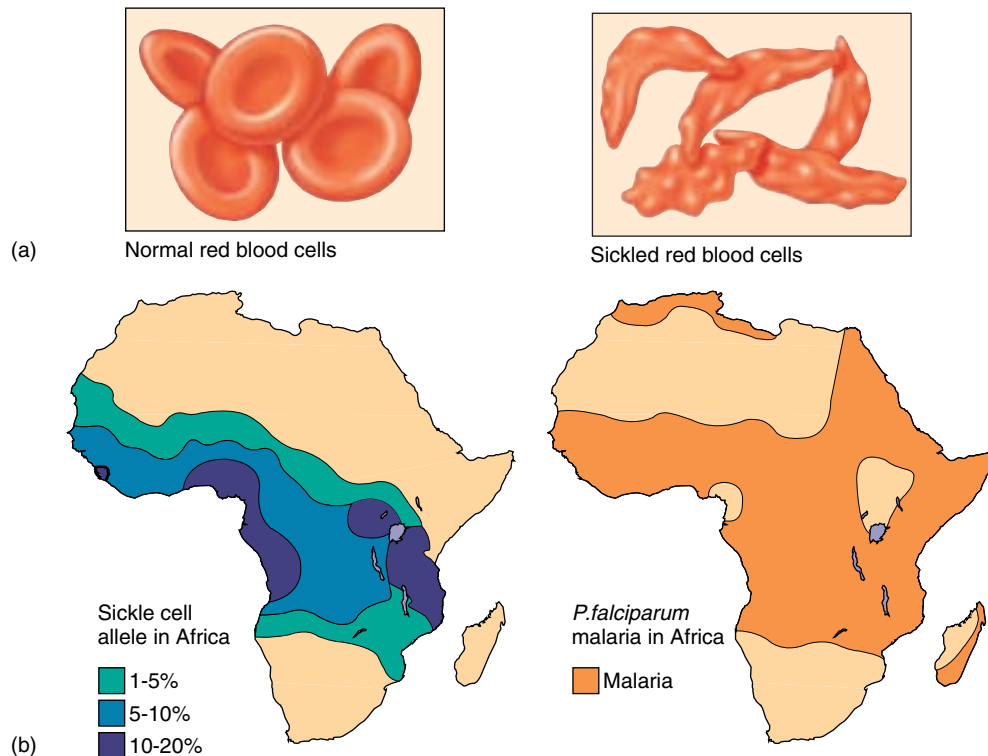


FIGURE 20.12

Frequency of sickle cell allele and distribution of *Plasmodium falciparum* malaria. (a) The red blood cells of people homozygous for the sickle cell allele collapse into sickled shapes when the oxygen level in the blood is low. (b) The distribution of the sickle cell allele in Africa coincides closely with that of *P. falciparum* malaria.

Malaria and Heterozygote Advantage

The average incidence of the *S* allele in the Central African population is about 0.12, far higher than that found among African Americans. From the Hardy–Weinberg principle, you can calculate that 1 in 5 Central African individuals are heterozygous at the *S* allele, and 1 in 100 develops the fatal form of the disorder. People who are homozygous for the sickle cell allele almost never reproduce because they usually die before they reach reproductive age. Why is the *S* allele not eliminated from the Central African population by selection, rather than being maintained at such high levels? People who are heterozygous for the sickle cell allele are much less susceptible to malaria—one of the leading causes of illness and death in Central Africa, especially among young children—in the areas where the allele is common. The reason is that when the parasite that causes malaria, *Plasmodium falciparum*, enters a red blood cell, it causes extremely low oxygen tension in the cell, which leads to cell sickling even in heterozygotes. Such cells are quickly filtered out of the bloodstream by the spleen, thus eliminating the parasite (the spleen’s filtering effect is what leads to anemia in homozygotes as large numbers of red blood cells are removed).

Consequently, even though most homozygous recessive individuals die before they have children, the sickle cell allele is maintained at high levels in these populations (it is selected for) because of its association with resistance to malaria in heterozygotes and also, for reasons not yet fully understood, with increased fertility in female heterozygotes.

For people living in areas where malaria is common, having the sickle cell allele in the heterozygous condition has adaptive value (figure 20.12). Among African Americans, however, many of whose ancestors have lived for some 15 generations in a country where malaria has been relatively rare and is now essentially absent, the environment does not place a premium on resistance to malaria. Consequently, no adaptive value counterbalances the ill effects of the disease; in this nonmalarial environment, selection is acting to eliminate the *S* allele. Only 1 in 375 African Americans develop sickle cell anemia, far less than in Central Africa.

The hemoglobin allele *S*, responsible for sickle cell anemia in homozygotes, is maintained by heterozygote advantage in Central Africa, where heterozygotes for the *S* allele are resistant to malaria.

20.3 Selection can act on traits affected by many genes.

Forms of Selection

In nature many traits, perhaps most, are affected by more than one gene. The interactions between genes are typically complex, as you saw in chapter 13. For example, alleles of many different genes play a role in determining human height (see figure 13.18). In such cases, selection operates on all the genes, influencing most strongly those that make the greatest contribution to the phenotype. How selection changes the population depends on which genotypes are favored.

Disruptive Selection

In some situations, selection acts to eliminate rather than to favor intermediate types. A clear example is the different

beak sizes of the African fire-bellied seedcracker finch *Pyronestes ostrinus*. Populations of these birds contain individuals with large and small beaks, but very few individuals with intermediate-sized beaks. As their name implies, these birds feed on seeds, and the available seeds fall into two size categories: large and small. Only large-beaked birds can open the tough shells of large seeds, whereas birds with the smallest beaks are most adept at handling small seeds. Birds with intermediate-sized beaks are at a disadvantage with both seed types: unable to open large seeds and too clumsy to efficiently process small seeds. Consequently, selection acts to eliminate the intermediate phenotypes, in effect partitioning the population into two phenotypically distinct groups. This form of selection is called **disruptive selection** (figure 20.13a).

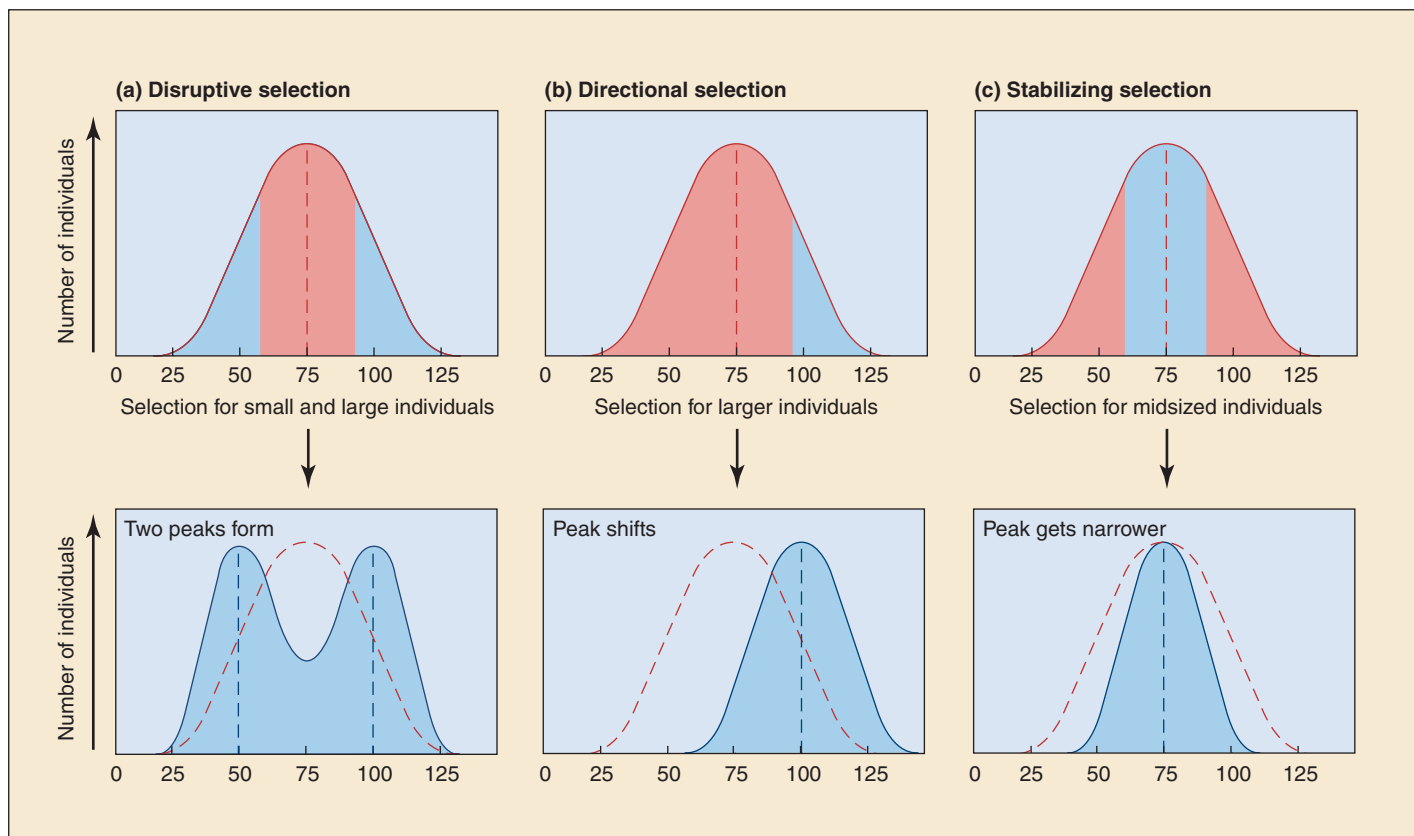


FIGURE 20.13

Three kinds of natural selection. The top panels show the populations before selection has occurred, with the forms that will be selected against shaded red and the forms that will be favored shaded blue. The bottom panels indicate what the populations will look like after selection has occurred. (a) In *disruptive selection*, individuals in the middle of the range of phenotypes of a certain trait are selected against (*red*), and the extreme forms of the trait are favored (*blue*). (b) In *directional selection*, individuals concentrated toward one extreme of the array of phenotypes are favored. (c) In *stabilizing selection*, individuals with midrange phenotypes are favored, with selection acting against both ends of the range of phenotypes.

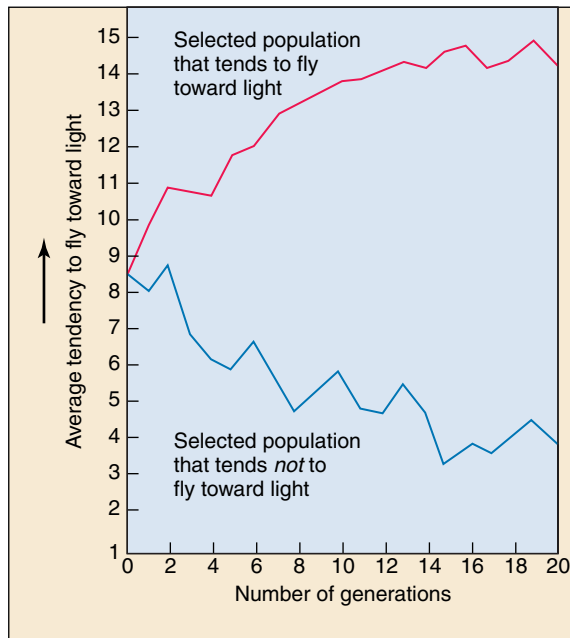


FIGURE 20.14

Directional selection for phototropism in *Drosophila*. In generation after generation, individuals of the fly *Drosophila* were selectively bred to obtain two populations. When flies with a strong tendency to fly toward light (positive phototropism) were used as parents for the next generation, their offspring had a greater tendency to fly toward light (top curve). When flies that tended *not* to fly toward light were used as parents for the next generation, their offspring had an even greater tendency *not* to fly toward light (bottom curve).

Directional Selection

When selection acts to eliminate one extreme from an array of phenotypes (figure 20.13*b*), the genes promoting this extreme become less frequent in the population. Thus, in the *Drosophila* population illustrated in figure 20.14, the elimination of flies that move toward light causes the population to contain fewer individuals with alleles promoting such behavior. If you were to pick an individual at random from the new fly population, there is a smaller chance it would spontaneously move toward light than if you had selected a fly from the old population. Selection has changed the population in the direction of lower light attraction. This form of selection is called **directional selection**.

Stabilizing Selection

When selection acts to eliminate *both* extremes from an array of phenotypes (figure 20.13*c*), the result is to increase the frequency of the already common intermediate type. In effect, selection is operating to prevent change away from this middle range of values. Selection does not change the most common phenotype of the population, but rather makes it even more common by eliminating extremes.

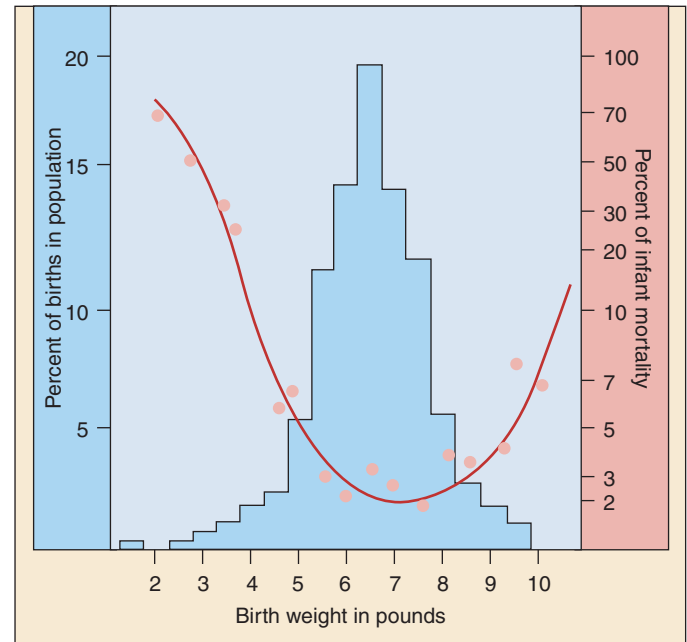


FIGURE 20.15

Stabilizing selection for birth weight in human beings. The death rate among babies (red curve; right y-axis) is lowest at an intermediate birth weight; both smaller and larger babies have a greater tendency to die than those around the optimum weight (blue area; left y-axis) of between 7 and 8 pounds.

Many examples are known. In humans, infants with intermediate weight at birth have the highest survival rate (figure 20.15). In ducks and chickens, eggs of intermediate weight have the highest hatching success. This form of selection is called **stabilizing selection**.

Components of Fitness

Natural selection occurs when individuals with one phenotype leave more surviving offspring in the next generation than individuals with an alternative phenotype. Evolutionary biologists quantify reproductive success as **fitness**, the number of surviving offspring left in the next generation. Although selection is often characterized as “survival of the fittest,” differences in survival are only one component of fitness. Even if no differences in survival occur, selection may operate if some individuals are more successful than others in attracting mates. In many territorial animal species, large males mate with many females and small males rarely get to mate. In addition, the number of offspring produced per mating is also important. Large female frogs and fish lay more eggs than smaller females and thus may leave more offspring in the next generation.

Selection on traits affected by many genes can favor both extremes of the trait, or intermediate values, or only one extreme.

Limits to What Selection Can Accomplish

Although selection is perhaps the most powerful of the five principal agents of genetic change, there are limits to what it can accomplish. These limits arise because alternative alleles may interact in different ways with other genes and because alleles often affect multiple aspects of the phenotype (the phenomena of epistasis and pleiotropy discussed in chapter 13). These interactions tend to set limits on how much a phenotype can be altered. For example, selecting for large clutch size in barnyard chickens eventually leads to eggs with thinner shells that break more easily. For this reason, we do not have gigantic cattle that yield twice as much meat as our leading strains, chickens that lay twice as many eggs as the best layers do now, or corn with an ear at the base of every leaf, instead of just at the base of a few leaves.

Evolution Requires Genetic Variation

Over 80% of the gene pool of the thoroughbred horses racing today goes back to 31 known ancestors from the late eighteenth century. Despite intense directional selection on thoroughbreds, their performance times have not improved for the last 50 years (figure 20.16). Years of intense selection presumably have removed variation from the population at a rate greater than it could be replenished by mutation such that now no genetic variation remains and evolutionary change is not possible.

In some cases, phenotypic variation for a trait may never have had a genetic basis. The compound eyes of insects are made up of hundreds of visual units, termed ommatidia. In some individuals, the left eye contains more ommatidia than the right eye. However, despite intense selection in the laboratory, scientists have never been able to produce a line of fruit flies that consistently have more ommatidia in the left eye than in the right. The reason is that separate genes do not exist for the left and right eyes. Rather, the same genes affect both eyes, and differences in the number of ommatidia result from differences that occur as the eyes are formed in the development process (figure 20.17). Thus, despite the existence of phenotypic variation, no genetic variation is available for selection to favor.

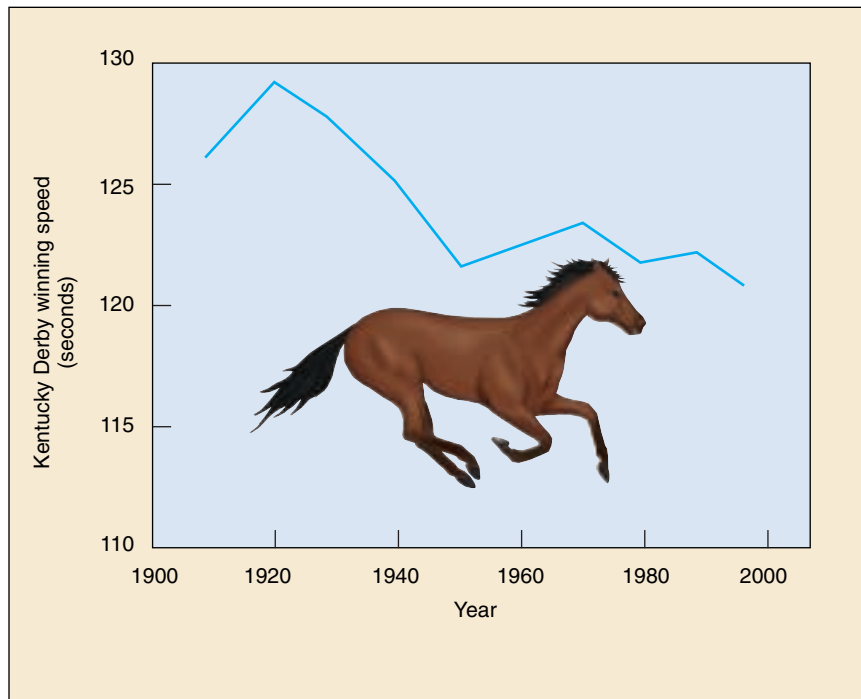


FIGURE 20.16

Selection for increased speed in racehorses is no longer effective. Kentucky Derby winning speeds have not improved significantly since 1950.

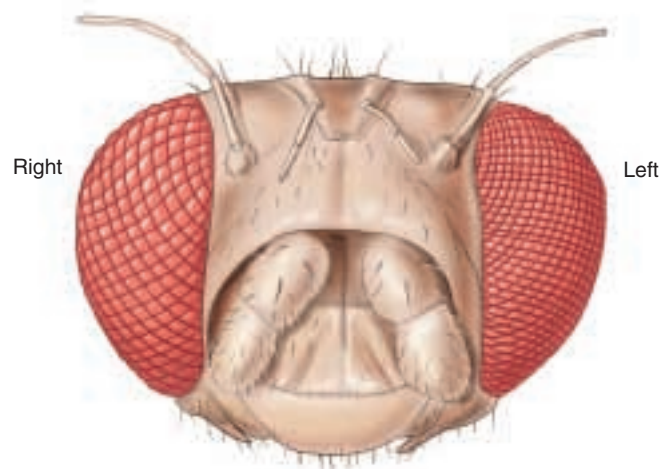


FIGURE 20.17

Phenotypic variation in insect ommatidia. In some individuals, the number of ommatidia in the left eye is greater than the number in the right eye. However, this difference is not genetically based; developmental processes cause the difference.

Selection against Rare Alleles

A second factor limits what selection can accomplish: selection acts only on phenotypes. For this reason, selection does not operate efficiently on rare recessive alleles, simply because there is no way to select against them unless they come together as homozygotes. For example, when a recessive allele a is present at a frequency q equal to 0.2, only four out of a hundred individuals (q^2) will be double recessive and display the phenotype associated with this allele (figure 20.18). For lower allele frequencies, the effect is even more dramatic: if the frequency in the population of the recessive allele $q = 0.01$, the frequency of recessive homozygotes in that population will be only 1 in 10,000.

The fact that selection acts on phenotypes rather than genotypes means that selection against undesirable genetic traits in humans or domesticated animals is difficult unless the heterozygotes can also be detected. For example, if a particular recessive allele r ($q = 0.01$) was considered undesirable, and none of the homozygotes for this allele were allowed to breed, it would take 1000 generations, or about 25,000 years in humans, to lower the allele frequency by half to 0.005. At this point, after 25,000 years of work, the frequency of homozygotes would still be 1 in 40,000, or 25% of what it was initially.

Selection in Laboratory Environments

One way to assess the action of selection is to carry out artificial selection in the laboratory. Strains that are genetically identical except for the gene subject to selection can be crossed so that the possibility of linkage disequilibrium does not confound the analysis. Populations of bacteria provide a particularly powerful tool for studying selection in the laboratory because bacteria have a short generation time (less than an hour) and can be grown in huge numbers in growth vats called chemostats. In pioneering studies, Dan Hartl and coworkers backcrossed bacteria with different alleles of the enzyme 6-PGD into a homogeneous genetic background, and then compared the growth of the different strains when they were fed only gluconate, the enzyme's substrate. Hartl found that all of the alleles grew at the same rate! The different alle-

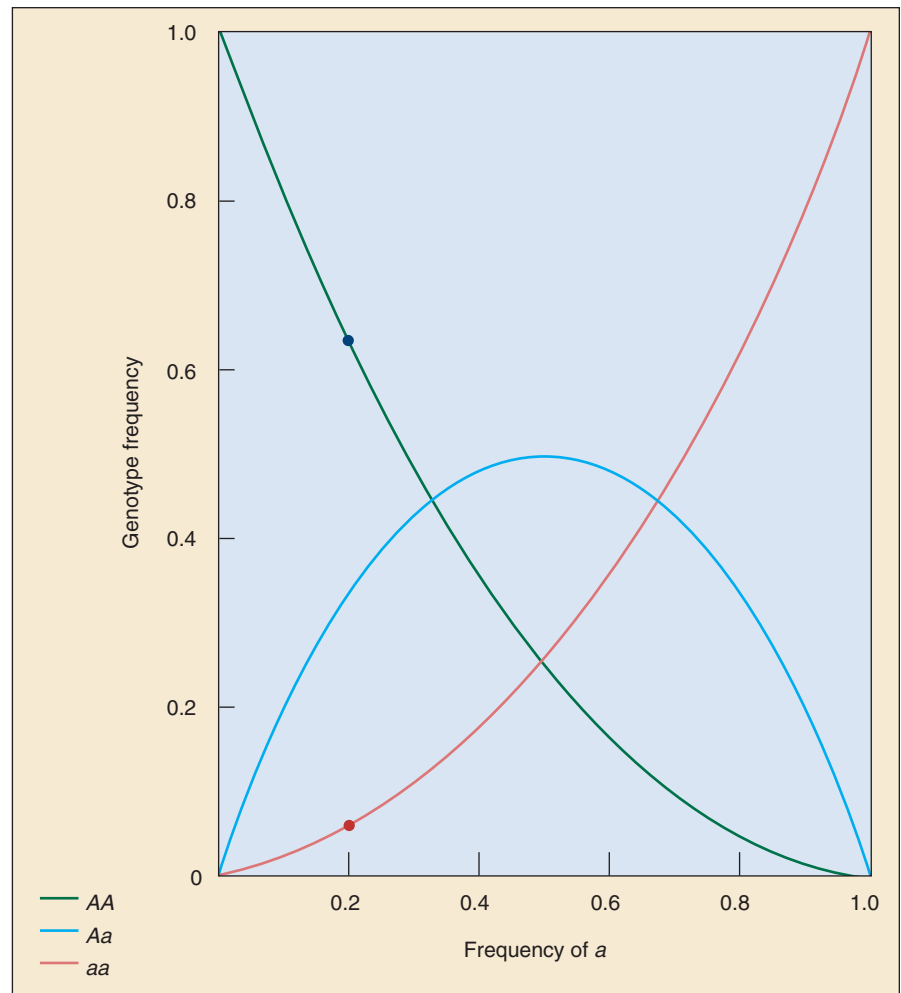


FIGURE 20.18

The relationship between allele frequency and genotype frequency. If allele a is present at a frequency of 0.2, the double recessive genotype aa is only present at a frequency of 0.04. In other words, only 4 in 100 individuals will have a homozygous recessive genotype, while 64 in 100 will have a homozygous dominant genotype.

les were thus selectively neutral in a normal genetic background. However, when Hartl disabled an alternative biochemical pathway for the metabolism of gluconate, so that only 6-PGD mediated the utilization of this sole source of carbon, he obtained very different results: several alleles were markedly superior to others. Selection was clearly able to operate on these alleles, but only under certain conditions.

The ability of selection to produce evolutionary change is hindered by a variety of factors, including multiple effects of single genes, gene interactions, and lack of genetic variation. Moreover, selection can only eliminate rare recessive alleles very slowly.



Summary

Questions

Media Resources

20.1 Genes vary in natural populations.

- Evolution is best defined as “descent with modification.”
- Darwin’s primary insight was to propose that evolutionary change resulted from the operation of natural selection.
- By the 1860s, natural selection was widely accepted as the correct explanation for the process of evolution. The field of evolution did not progress much further, however, until the 1920s because of the lack of a suitable explanation of how hereditary traits are transmitted.
- Invertebrates and outcrossing plants are often heterozygous at about 12 to 15% of their loci; the corresponding value for vertebrates is about 4 to 8%.

1. What is the difference between natural selection and evolution?
2. What is adaptation? How does it fit into Darwin’s concept of evolution?
3. What is genetic polymorphism? What has polymorphism to do with evolution?



- Scientists on Science: from Butterflies to Global Preservation
- Student Research: Cotton Boll Weevil

20.2 Why do allele frequencies change in populations?

- Studies of how allele frequencies shift within populations allow investigators to study evolution in action.
- Meiosis does not alter allele frequencies within populations. Unless selection or some other force acts on the genes, the frequencies of their alleles remain unchanged from one generation to the next.
- A variety of processes can lead to evolutionary change within a population, including genetic drift, inbreeding, gene flow, and natural selection.
- For evolution to occur by natural selection, three conditions must be met: 1. variation must exist in the population; 2. the variation must have a genetic basis; and 3. variation must be related to the number of offspring left in the next generation.
- Natural selection can usually overpower the effects of genetic drift, except in very small populations.
- Natural selection can overwhelm the effects of gene flow in some cases, but not in others.

4. Given that allele A is present in a large random-mating population at a frequency of 54 per 100 individuals, what is the proportion of individuals in that population expected to be heterozygous for the allele? homozygous dominant? homozygous recessive?

5. Why does the founder effect have such a profound influence on a population’s genetic makeup? How does the bottleneck effect differ from the founder effect?

6. What effect does inbreeding have on allele frequency? Why is marriage between close relatives discouraged?



- Hardy Weinberg Equilibrium



- Activity: Natural Selection
- Activity: Allele Frequencies
- Activity: Genetic Drift
- Types of Selection
- Evolutionary Variation
- Other Processes of Evolution
- Adaptation

20.3 Selection can act on traits affected by many genes.

- Directional selection acts to eliminate one extreme from an array of phenotypes; stabilizing selection acts to eliminate *both* extremes; and disruptive selection acts to eliminate rather than to favor the intermediate type.
- Natural selection is not all powerful; genetic variation is required for natural selection to produce evolutionary change.

7. Define *selection*. How does it alter allele frequencies? What are the three types of selection? Give an example of each.

8. Why are there limitations to the success of selection?



- Book Review: *The Evolution of Jane* by Schine

21

The Evidence for Evolution

Concept Outline

21.1 Fossil evidence indicates that evolution has occurred.

The Fossil Record. When fossils are arranged in the order of their age, a continual series of change is seen, new changes being added at each stage.

The Evolution of Horses. The record of horse evolution is particularly well-documented and instructive.

21.2 Natural selection can produce evolutionary change.

The Beaks of Darwin's Finches. Natural selection favors stouter bills in dry years, when large tough-to-crush seeds are the only food available to finches.

Peppered Moths and Industrial Melanism. Natural selection favors dark-colored moths in areas of heavy pollution, while light-colored moths survive better in unpolluted areas.

Artificial Selection. Artificial selection practiced in laboratory studies, agriculture, and domestication demonstrate that selection can produce substantial evolutionary change.

21.3 Evidence for evolution can be found in other fields of biology.

The Anatomical Record. When anatomical features of living animals are examined, evidence of shared ancestry is often apparent.

The Molecular Record. When gene or protein sequences from organisms are arranged, species thought to be closely related based on fossil evidence are seen to be more similar than species thought to be distantly related.

Convergent and Divergent Evolution. Evolution favors similar forms under similar circumstances.

21.4 The theory of evolution has proven controversial.

Darwin's Critics. Critics have raised seven objections to Darwin's theory of evolution by natural selection.



FIGURE 21.1

A window into the past. The fossil remains of the now-extinct reptile *Mesosaurus* found in Permian sediments in Africa and South America provided one of the earliest clues to a former connection between the two continents. *Mesosaurus* was a freshwater species and so clearly incapable of a transatlantic swim. Therefore, it must have lived in the lakes and rivers of a formerly contiguous landmass that later became divided as Africa and South America drifted apart in the Cretaceous.

Of all the major ideas of biology, the theory that today's organisms evolved from now-extinct ancestors (figure 21.1) is perhaps the best known to the general public. This is not because the average person truly understands the basic facts of evolution, but rather because many people mistakenly believe that it represents a challenge to their religious beliefs. Similar highly publicized criticisms of evolution have occurred ever since Darwin's time. For this reason, it is important that, during the course of your study of biology, you address the issue squarely: Just what is the evidence for evolution?

21.1 Fossil evidence indicates that evolution has occurred.

At its core, the case for evolution is built upon two pillars: first, evidence that natural selection can produce evolutionary change and, second, evidence from the fossil record that evolution has occurred. In addition, information from many different areas of biology—including fields as different as embryology, anatomy, molecular biology, and biogeography (the study of the geographic distribution of species)—can only be interpreted sensibly as the outcome of evolution.

The Fossil Record

The most direct evidence that evolution has occurred is found in the fossil record. Today we have a far more complete understanding of this record than was available in Darwin's time. Fossils are the preserved remains of once-living organisms. Fossils are created when three events occur. First, the organism must become buried in sediment; then, the calcium in bone or other hard tissue must mineralize; and, finally, the surrounding sediment must eventually harden to form rock. The process of fossilization probably occurs rarely. Usually, animal or plant remains will decay or be scavenged before the process can begin. In addition, many fossils occur in rocks that are inaccessible to scientists. When they do become available, they are often destroyed by erosion and other natural processes before they can be collected. As a result, only a fraction of the species that have ever existed (estimated by some to be as many as 500 million) are known from fossils. Nonetheless, the fossils that have been discovered are sufficient to provide detailed information on the course of evolution through time.

Dating Fossils

By dating the rocks in which fossils occur, we can get an accurate idea of how old the fossils are. In Darwin's day, rocks were dated by their position with respect to one another (*relative dating*); rocks in deeper strata are generally older. Knowing the relative positions of sedimentary rocks and the rates of erosion of different kinds of sedimentary rocks in different environments, geologists of the nineteenth century derived a fairly accurate idea of the relative ages of rocks.

Today, rocks are dated by measuring the degree of decay of certain radioisotopes contained in the rock (*absolute dating*); the older the rock, the more its isotopes have decayed. Because radioactive isotopes decay at a constant rate unaltered by temperature or pressure, the isotopes in a rock act as an internal clock, measuring the time since the rock was formed. This is a more accurate way of dating rocks and provides dates stated in millions of years, rather than relative dates.

A History of Evolutionary Change

When fossils are arrayed according to their age, from oldest to youngest, they often provide evidence of successive evolutionary change. At the largest scale, the fossil record documents the progression of life through time, from the origin of eukaryotic organisms, through the evolution of fishes, the rise of land-living organisms, the reign of the dinosaurs, and on to the origin of humans (figure 21.2).

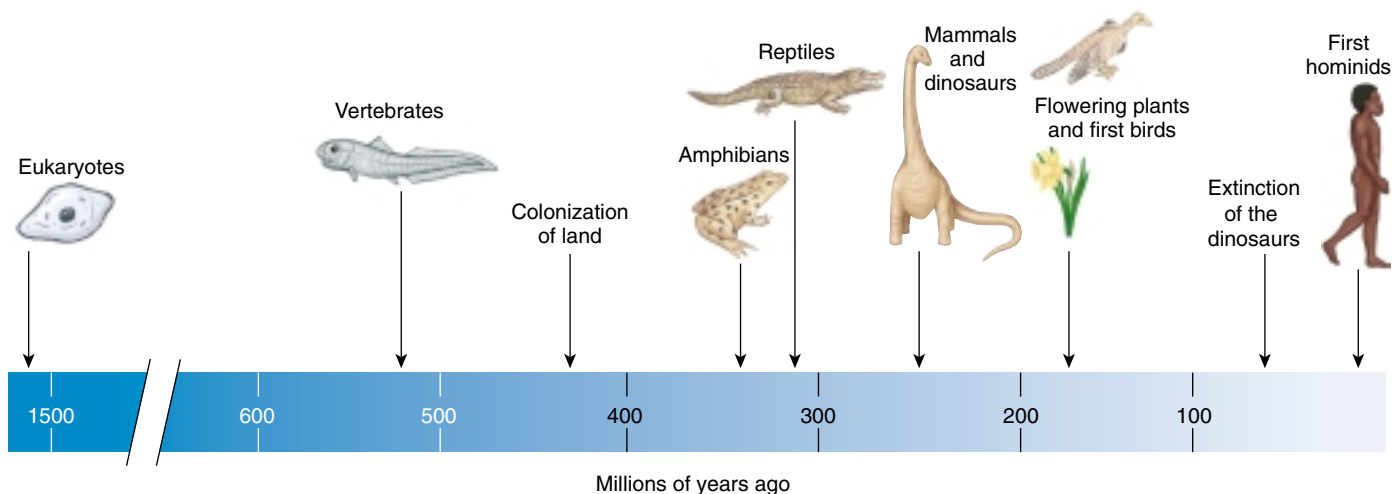


FIGURE 21.2
Timeline of the history of life as revealed by the fossil record.

Gaps in the Fossil Record

This is not to say that the fossil record is complete. Given the low likelihood of fossil preservation and recovery, it is not surprising that there are gaps in the fossil record. Nonetheless, paleontologists (the scientists who study fossils) continue to fill in the gaps in the fossil record. While many gaps interrupted the fossil record in Darwin's era, even then, scientists knew of the *Archaeopteryx* fossil transitional between dinosaurs and birds. Today, the fossil record is far more complete, particularly among the vertebrates; fossils have been found linking all the major groups. Recent years have seen spectacular discoveries closing some of the major remaining gaps in our understanding of vertebrate evolution. For example, recently a four-legged aquatic mammal was discovered that provides important insights concerning the evolution of whales and dolphins from land-living, hoofed ancestors (figure 21.3). Similarly, a fossil snake with legs has shed light on the evolution of snakes, which are descended from lizards that gradually became more and more elongated with simultaneous reduction and eventual disappearance of the limbs.

On a finer scale, evolutionary change within some types of animals is known in exceptional detail. For example, about 200 million years ago, oysters underwent a change from small curved shells to larger, flatter ones, with progressively flatter fossils being seen in the fossil record over a period of 12 million years (figure 21.4). A host of other examples all illustrate a record of successive change. The demonstration of this successive change is one of the strongest lines of evidence that evolution has occurred.

The fossil record provides a clear record of the major evolutionary transitions that have occurred through time.

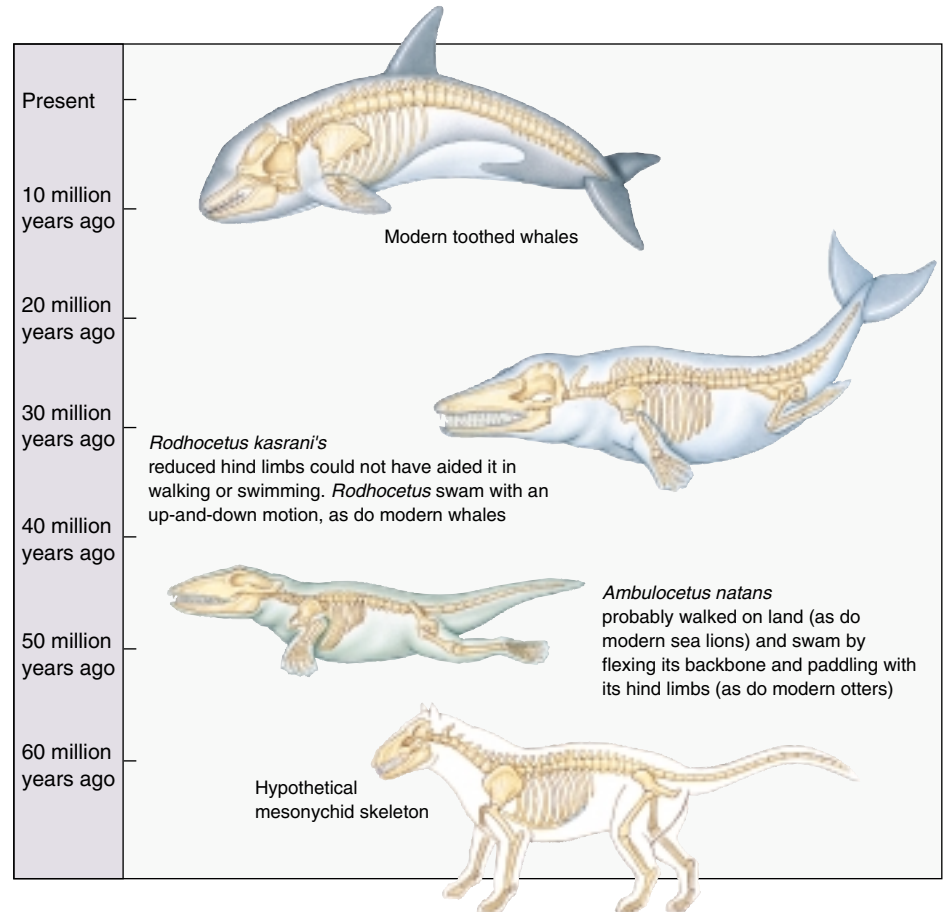


FIGURE 21.3

Whale “missing links.” The recent discoveries of *Ambulocetus* and *Rodhocetus* have filled in the gaps between the mesonychids, the hypothetical ancestral link between the whales and the hoofed mammals, and present-day whales.

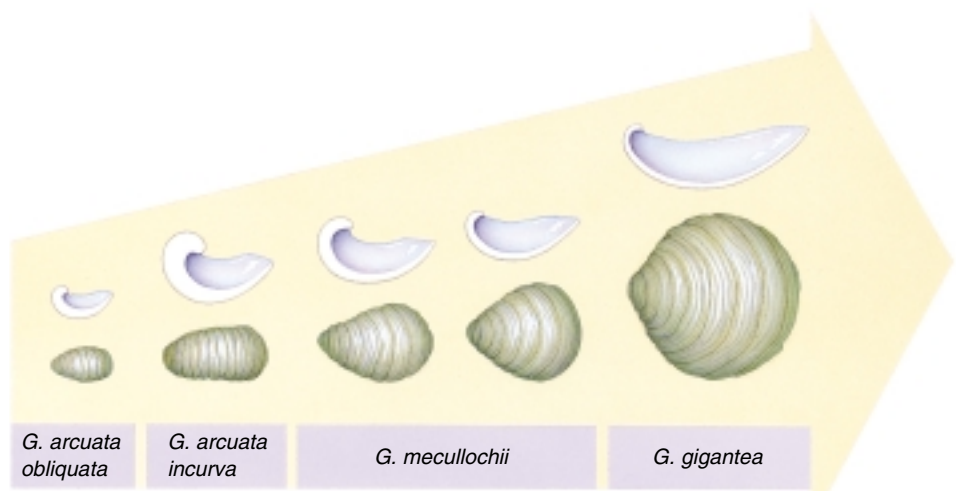


FIGURE 21.4

Evolution of shell shape in oysters. Over 12 million years of the Early Jurassic Period, the shells of this group of coiled oysters became larger, thinner, and flatter. These animals rested on the ocean floor in a special position called the “life position,” and it may be that the larger, flatter shells were more stable in disruptive water movements.

The Evolution of Horses

One of the best-studied cases in the fossil record concerns the evolution of horses. Modern-day members of the Equidae include horses, zebras, donkeys and asses, all of which are large, long-legged, fast-running animals adapted to living on open grasslands. These species, all classified in the genus *Equus*, are the last living descendants of a long lineage that has produced 34 genera since its origin in the Eocene Period, approximately 55 million years ago. Examination of these fossils has provided a particularly well-documented case of how evolution has proceeded by adaptation to changing environments.

The First Horse

The earliest known members of the horse family, species in the genus *Hyracotherium*, didn't look much like horses at all. Small, with short legs and broad feet (figure 21.5), these species occurred in wooded habitats, where they probably browsed on leaves and herbs and escaped predators by dodging through openings in the forest vegetation. The evolutionary path from these diminutive creatures to the workhorses of today has involved changes in a variety of traits, including:

Size. The first horses were no bigger than dogs, with some considerably smaller. By contrast, modern equids can weigh more than a half ton. Examination of the fossil record reveals that horses changed little in size for their first 30 million years, but since then, a number of different lineages exhibited rapid and substantial increases. However, trends toward decreased size were also exhibited among some branches of the equid evolutionary tree (figure 21.6).

Toe reduction. The feet of modern horses have a single toe, enclosed in a tough, bony hoof. By contrast, *Hyracotherium* had four toes on its front feet and three on its hindfeet. Rather than hooves, these toes were encased in fleshy pads. Examination of the fossils clearly shows the transition through time: increase in length of the central toe, development of the bony hoof, and reduction and loss of the other toes (figure 21.7). As with body size, these trends occurred concurrently on several different branches of the horse evolutionary tree. At the same time as these developments, horses were evolving changes in the length and skeletal structure of the limbs, leading to animals capable of running long distances at high speeds.

Tooth size and shape. The teeth of *Hyracotherium* were small and relatively simple in shape. Through time, horse teeth have increased greatly in length and have developed a complex pattern of ridges on their molars and premolars (figure 21.7). The effect of these changes is to produce teeth better capable of chewing tough and gritty vegetation, such as grass, which tends to wear

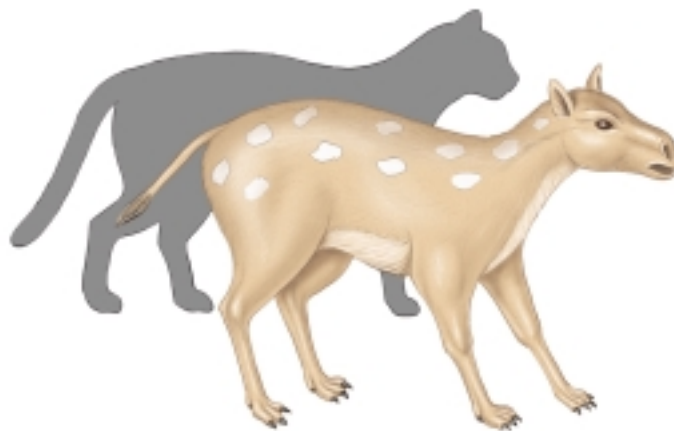


FIGURE 21.5
Hyracotherium sandrae, one of the earliest horses, was the size of a housecat.

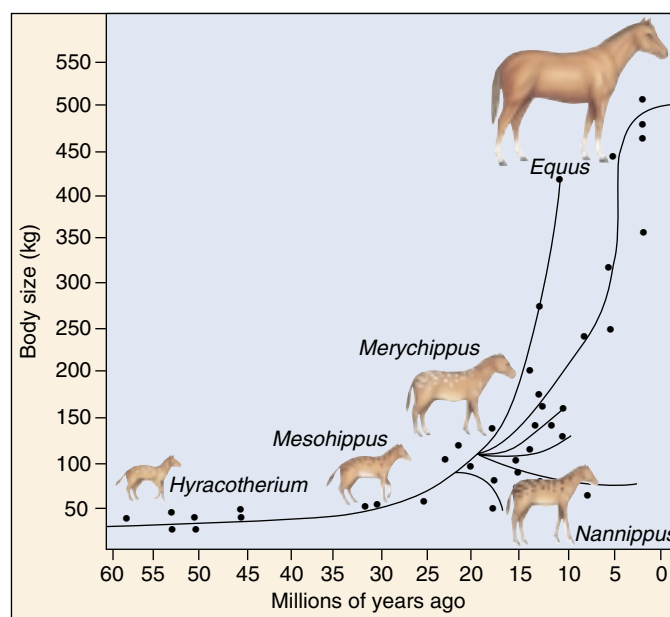


FIGURE 21.6
Evolutionary change in body size of horses. Lines show the broad outline of evolutionary relationships. Although most change involved increases in size, some decreases also occurred.

teeth down. Accompanying these changes have been alterations in the shape of the skull that strengthened the skull to withstand the stresses imposed by continual chewing. As with body size, evolutionary change has not been constant through time. Rather, much of the change in tooth shape has occurred within the past 20 million years.

All of these changes may be understood as adaptations to changing global climates. In particular, during the late

Miocene and early Oligocene (approximately 20 to 25 million years ago), grasslands became widespread in North America, where much of horse evolution occurred. As horses adapted to these habitats, long-distance and high-speed locomotion probably became more important to escape predators and travel great distances. By contrast, the greater flexibility provided by multiple toes and shorter limbs, which was advantageous for ducking through complex forest vegetation, was no longer beneficial. At the same time, horses were eating grasses and other vegetation that contained more grit and other hard substances, thus favoring teeth and skulls better suited for withstanding such materials.

Evolutionary Trends

For many years, horse evolution was held up as an example of constant evolutionary change through time. Some even saw in the record of horse evolution evidence for a progressive, guiding force, consistently pushing evolution to move in a single direction. We now know that such views are misguided; evolutionary change over millions of years is rarely so simple.

Rather, the fossils demonstrate that, although there have been overall trends evident in a variety of characteristics, evolutionary change has been far from constant and uniform through time. Instead, rates of evolution have varied widely, with long periods of little change and some periods of great change. Moreover, when changes happen, they often occur simultaneously in different lineages of the horse evolutionary tree. Finally, even when a trend exists, exceptions, such as the evolutionary decrease in body size exhibited by some lineages, are not uncommon. These patterns, evident in our knowledge of horse evolution, are usually discovered for any group of plants and animals for which we have an extensive fossil record, as we shall see when we discuss human evolution in chapter 23.

Horse Diversity

One reason that horse evolution was originally conceived of as linear through time may be that modern horse diversity is relatively limited. Thus, it is easy to mentally picture a straight line from *Hyracotherium* to modern-day *Equus*. However, today's limited horse diversity—only one surviving genus—is unusual. Indeed, at the peak of horse diversity in the Miocene, as many as 13 genera of horses could be found in North America alone. These species differed in body size and in a wide variety of other characteristics. Presumably, they lived in different habitats and exhibited different dietary preferences. Had this diversity existed to modern times, early workers presumably would have had a different outlook on horse evolution, a situation that is again paralleled by the evolution of humans.

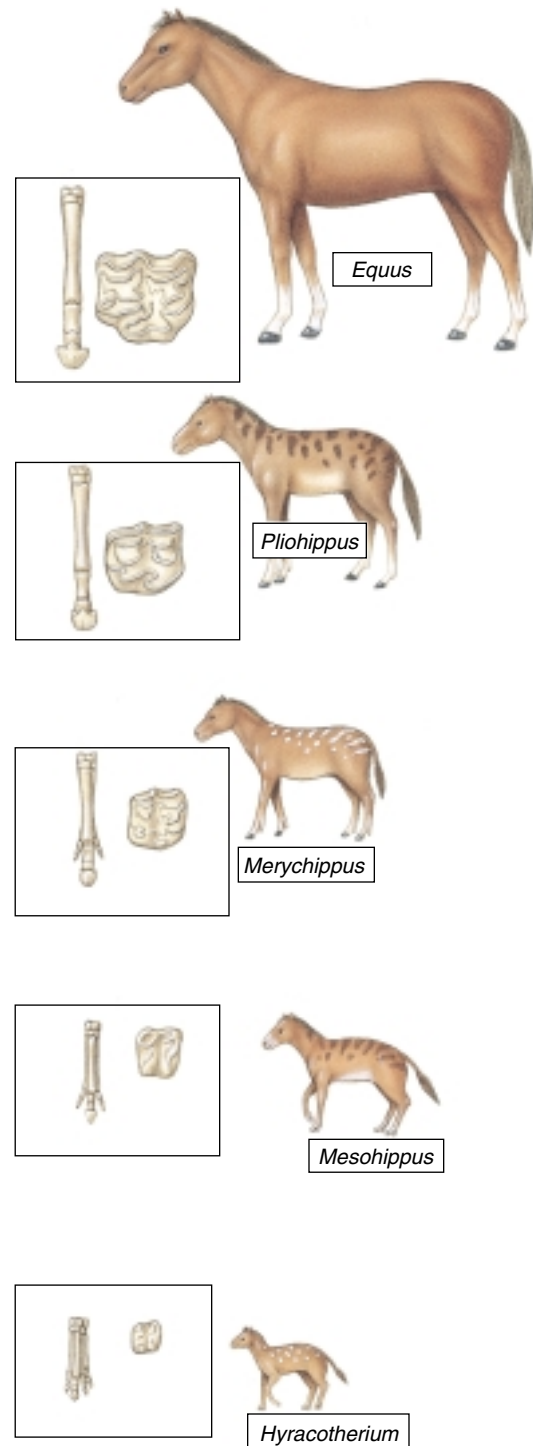


FIGURE 21.7
Evolutionary changes in horses through time.

The extensive fossil record for horses provides a detailed view of the evolutionary diversification of this group from small forest dwellers to the large and fast modern grassland species.

21.2 Natural selection can produce evolutionary change.

As we saw in chapter 20, a variety of different processes can result in evolutionary change. Nonetheless, in agreement with Darwin, most evolutionary biologists would agree that natural selection is the process responsible for most of the major evolutionary changes that have occurred through time. Although we cannot travel back through time, a variety of modern-day evidence confirms the power of natural selection as an agent of evolutionary change. These data come from both the field and the laboratory and from natural and human-altered situations.

The Beaks of Darwin's Finches

Darwin's finches are a classic example of evolution by natural selection. Darwin collected 31 specimens of finch from three islands when he visited the Galápagos Islands off the coast of Ecuador in 1835. Darwin, not an expert on birds, had trouble identifying the specimens, believing by examining their bills that his collection contained wrens, "gross-beaks," and blackbirds. You can see Darwin's sketches of four of these birds in figure 21.8.

The Importance of the Beak

Upon Darwin's return to England, ornithologist John Gould examined the finches. Gould recognized that Darwin's collection was in fact a closely related group of distinct species, all similar to one another except for their bills. In all, there were 13 species. The two ground finches

with the larger bills in figure 21.8 feed on seeds that they crush in their beaks, whereas the two with narrower bills eat insects. One species is a fruit eater, another a cactus eater, yet another a "vampire" that creeps up on seabirds and uses its sharp beak to drink their blood. Perhaps most remarkable are the tool users, woodpecker finches that pick up a twig, cactus thorn, or leaf stalk, trim it into shape with their bills, and then poke it into dead branches to pry out grubs.

The correspondence between the beaks of the 13 finch species and their food source immediately suggested to Darwin that evolution had shaped them:

"Seeing this gradation and diversity of structure in one small, intimately related group of birds, one might really fancy that from an original paucity of birds in this archipelago, one species has been taken and modified for different ends."

Was Darwin Wrong?

If Darwin's suggestion that the beak of an ancestral finch had been "modified for different ends" is correct, then it ought to be possible to see the different species of finches acting out their evolutionary roles, each using their bills to acquire their particular food specialty. The four species that crush seeds within their bills, for example, should feed on different seeds, those with stouter beaks specializing on harder-to-crush seeds.

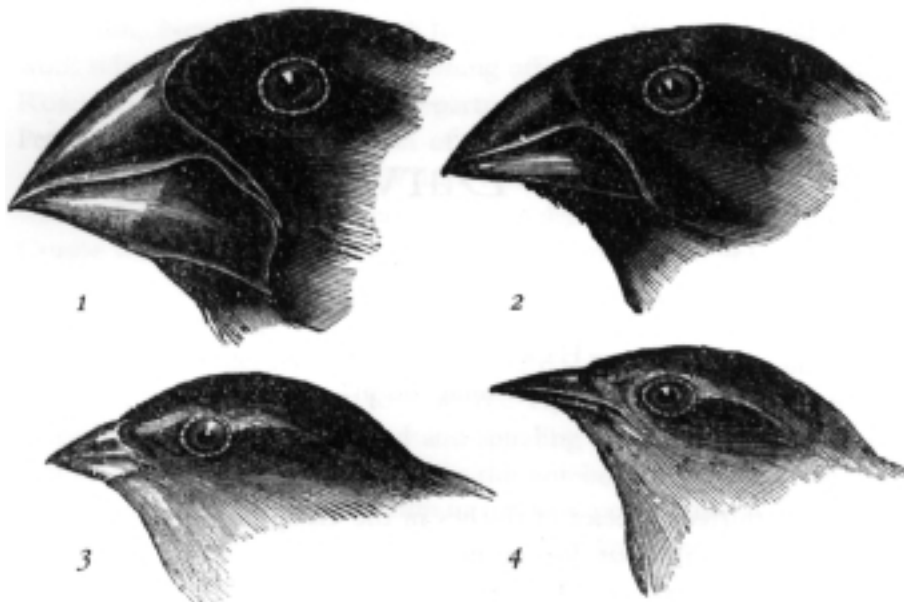


FIGURE 21.8
Darwin's own sketches of Galápagos finches. From Darwin's *Journal of Researches*: (1) large ground finch *Geospiza magnirostris*; (2) medium ground finch *Geospiza fortis*; (3) small tree finch *Camarhynchus parvulus*; (4) warbler finch *Certhidea olivacea*.

Many biologists visited the Galápagos after Darwin, but it was 100 years before any tried this key test of his hypothesis. When the great naturalist David Lack finally set out to do this in 1938, observing the birds closely for a full five months, his observations seemed to contradict Darwin's proposal! Lack often observed many different species of finch feeding together on the same seeds. His data indicated that the stout-beaked species and the slender-beaked species were feeding on the very same array of seeds.

We now know that it was Lack's misfortune to study the birds during a wet year, when food was plentiful. The finch's beak is of little importance in such flush times; small seeds are so abundant that birds of all species are able to get enough to eat. Later work has revealed a very different picture during leaner, dry years, when few seeds are available and the difference between survival and starvation depends on being able to efficiently gather enough to eat. In such times, having beaks designed to be maximally effective for a particular type of food becomes critical and the species diverge in their diet, each focusing on the type of food to which it is specialized.

A Closer Look

The key to successfully testing Darwin's proposal that the beaks of Galápagos finches are adaptations to different food sources proved to be patience. Starting in 1973, Peter and Rosemary Grant of Princeton University and generations of their students have studied the medium ground finch *Geospiza fortis* on a tiny island in the center of the Galápagos called Daphne Major. These finches feed preferentially on small tender seeds, produced in abundance by plants in wet years. The birds resort to larger, drier seeds, which are harder to crush, only when small seeds become depleted during long periods of dry weather, when plants produce few seeds.

The Grants quantified beak shape among the medium ground finches of Daphne Major by carefully measuring beak depth (width of beak, from top to bottom, at its base) on individual birds. Measuring many birds every year, they were able to assemble for the first time a detailed portrait of evolution in action. The Grants found that beak depth changed from one year to the next in a predictable fashion. During droughts, plants produced few seeds and all available small seeds quickly were eaten, leaving large seeds as the major remaining source of food. As a result, birds with large beaks survived better, because they were better able to break open these large seeds. Consequently, the average beak depth of birds in the population increased the next year, only to decrease again when wet seasons returned (figure 21.9).

Could these changes in beak dimension reflect the action of natural selection? An alternative possibility might be that the changes in beak depth do not reflect changes in

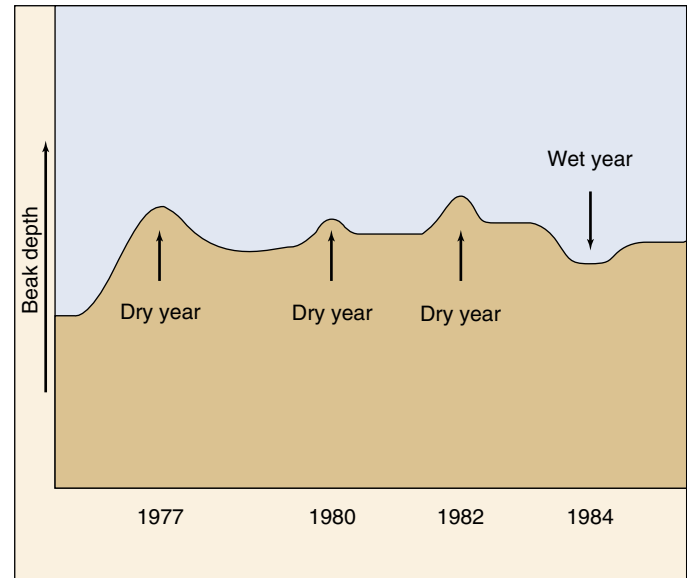


FIGURE 21.9

Evidence that natural selection alters beak size in *Geospiza fortis*. In dry years, when only large, tough seeds are available, the mean beak size increases. In wet years, when many small seeds are available, smaller beaks become more common.

gene frequencies, but rather are simply a response to diet—perhaps during lean times the birds become malnourished and then grow stouter beaks, for example. To rule out this possibility, the Grants measured the relation of parent bill size to offspring bill size, examining many broods over several years. The depth of the bill was passed down faithfully from one generation to the next, regardless of environmental conditions, suggesting that the differences in bill size indeed reflected genetic differences.

Darwin Was Right After All

If the year-to-year changes in beak depth indeed reflect genetic changes, as now seems likely, and these changes can be predicted by the pattern of dry years, then Darwin was right after all—natural selection does seem to be operating to adjust the beak to its food supply. Birds with stout beaks have an advantage during dry periods, for they can break the large, dry seeds that are the only food available. When small seeds become plentiful once again with the return of wet weather, a smaller beak proves a more efficient tool for harvesting the more abundant smaller seeds.

Among Darwin's finches, natural selection adjusts the shape of the beak in response to the nature of the available food supply, adjustments that can be seen to be occurring even today.

Peppered Moths and Industrial Melanism

When the environment changes, natural selection often may favor new traits in a species. The example of the Darwin's finches clearly indicates how natural variation can lead to evolutionary change. Humans are greatly altering the environment in many ways; we should not be surprised to see organisms attempting to adapt to these new conditions. One classic example concerns the peppered moth, *Biston betularia*. Until the mid-nineteenth century, almost every individual of this species captured in Great Britain had light-colored wings with black specklings (hence the name "peppered" moth). From that time on, individuals with dark-colored wings increased in frequency in the moth populations near industrialized centers until they made up almost 100% of these populations. Black individuals had a dominant allele that was present but very rare in populations before 1850. Biologists soon noticed that in industrialized regions where the dark moths were common, the tree trunks were darkened almost black by the soot of pollution. Dark moths were much less conspicuous resting on them than were light moths. In addition, the air pollution that was spreading in the industrialized regions had killed many of the light-colored lichens on tree trunks, making the trunks darker.

Selection for Melanism

Can Darwin's theory explain the increase in the frequency of the dark allele? Why did dark moths gain a survival advantage around 1850? An amateur moth collector named J. W. Tutt proposed what became the most commonly accepted hypothesis explaining the decline of the light-colored moths. He suggested that peppered forms were more visible to predators on sooty trees that have lost their lichens. Consequently, birds ate the peppered moths resting on the trunks of trees during the day. The black forms, in contrast, were at an advantage because they were camouflaged (figure 21.10). Although Tutt initially had no evidence, British ecologist Bernard Kettlewell tested the hypothesis in the 1950s by rearing populations of peppered moths with equal numbers of dark and light individuals. Kettlewell then released these populations into two sets of woods: one, near heavily polluted Birmingham, the other, in unpolluted Dorset. Kettlewell set up rings of traps around the woods to see how many of both kinds of moths survived. To evaluate his results, he had marked the released moths with a dot of paint on the underside of their wings, where birds could not see it.

In the polluted area near Birmingham, Kettlewell trapped 19% of the light moths, but 40% of the dark ones. This indicated that dark moths had a far better chance of surviving in these polluted woods, where the tree trunks were dark. In the relatively unpolluted Dorset woods, Kettlewell recovered 12.5% of the light moths but only 6% of



FIGURE 21.10
Tutt's hypothesis explaining industrial melanism. These photographs show color variants of the peppered moth, *Biston betularia*. Tutt proposed that the dark moth is more visible to predators on unpolluted trees (*top*), while the light moth is more visible to predators on bark blackened by industrial pollution (*bottom*).

the dark ones. This indicated that where the tree trunks were still light-colored, light moths had a much better chance of survival. Kettlewell later solidified his argument by placing hidden blinds in the woods and actually filming birds eating the moths. Sometimes the birds Kettlewell observed actually passed right over a moth that was the same color as its background.

Industrial Melanism

Industrial melanism is a term used to describe the evolutionary process in which darker individuals come to predominate over lighter individuals since the industrial revolution as a result of natural selection. The process is widely believed to have taken place because the dark organisms are better concealed from their predators in habitats that have been darkened by soot and other forms of industrial pollution, as suggested by Kettlewell's research.

Dozens of other species of moths have changed in the same way as the peppered moth in industrialized areas throughout Eurasia and North America, with dark forms becoming more common from the mid-nineteenth century onward as industrialization spread.

Selection against Melanism

In the second half of the twentieth century, with the widespread implementation of pollution controls, these trends are reversing, not only for the peppered moth in many areas in England, but also for many other species of moths throughout the northern continents. These examples provide some of the best documented instances of changes in allelic frequencies of natural populations as a result of natural selection due to specific factors in the environment.

In England, the pollution promoting industrial melanism began to reverse following enactment of Clean Air legislation in 1956. Beginning in 1959, the *Biston* population at Caldy Common outside Liverpool has been sampled each year. The frequency of the melanic (dark) form has dropped from a high of 94% in 1960 to its current (1994) low of 19% (figure 21.11). Similar reversals have been documented at numerous other locations throughout England. The drop correlates well with a drop in air pollution, particularly with tree-darkening sulfur dioxide and suspended particulates.

Interestingly, the same reversal of industrial melanism appears to have occurred in America during the same time that it was happening in England. Industrial melanism in the American subspecies of the peppered moth was not as widespread as in England, but it has been well-documented at a rural field station near Detroit. Of 576 peppered moths collected there from 1959 to 1961, 515 were melanic, a frequency of 89%. The American Clean Air Act, passed in 1963, led to significant reductions in air pollution. Resampled in 1994, the Detroit field station peppered moth population had only 15% melanic moths (see figure 21.11)! The moths in Liverpool and Detroit, both part of the same natural experiment, exhibit strong evidence of natural selection.

Reconsidering the Target of Natural Selection

Tutt's hypothesis, widely accepted in the light of Kettlewell's studies, is currently being reevaluated. The problem is that the recent selection against melanism does not

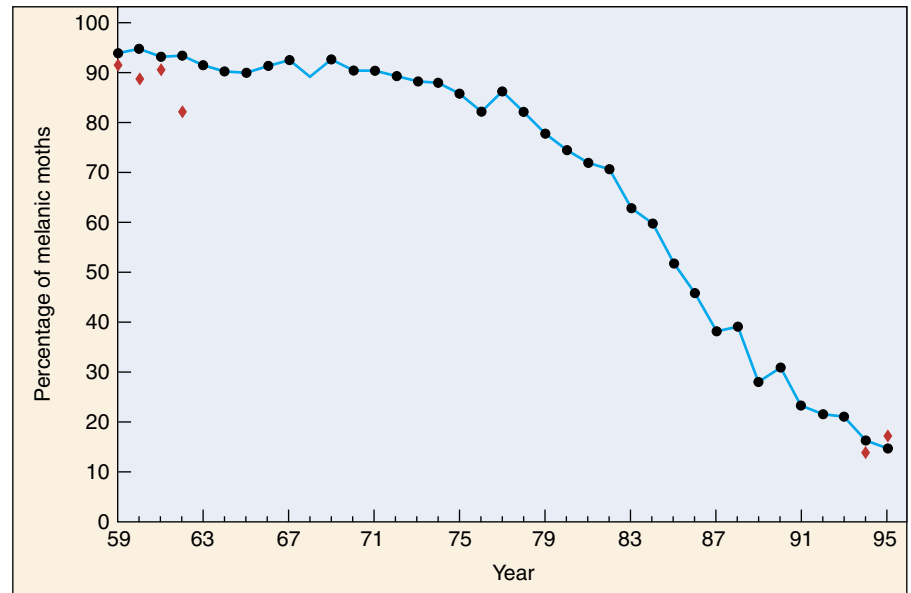


FIGURE 21.11

Selection against melanism. The circles indicate the frequency of melanic *Biston* moths at Caldy Common in England, sampled continuously from 1959 to 1995. Diamonds indicate frequencies in Michigan from 1959 to 1962 and from 1994 to 1995.

Source: Data from Grant, et al., "Parallel Rise and Fall of Melanic Peppered Moths" in *Journal of Heredity*, vol. 87, 1996, Oxford University Press.

appear to correlate with changes in tree lichens. At Caldy Common, the light form of the peppered moth began its increase in frequency long before lichens began to reappear on the trees. At the Detroit field station, the lichens never changed significantly as the dark moths first became dominant and then declined over the last 30 years. In fact, investigators have not been able to find peppered moths on Detroit trees at all, whether covered with lichens or not. Wherever the moths rest during the day, it does not appear to be on tree bark. Some evidence suggests they rest on leaves on the treetops, but no one is sure.

The action of selection may depend less on the presence of lichens and more on other differences in the environment resulting from industrial pollution. Pollution tends to cover all objects in the environment with a fine layer of particulate dust, which tends to decrease how much light surfaces reflect. In addition, pollution has a particularly severe effect on birch trees, which are light in color. Both effects would tend to make the environment darker and thus would favor darker color in moths.

Natural selection has favored the dark form of the peppered moth in areas subject to severe air pollution, perhaps because on darkened trees they are less easily seen by moth-eating birds. Selection has in turn favored the light form as pollution has abated.

Artificial Selection

Humans have imposed selection upon plants and animals since the dawn of civilization. Just as in natural selection, artificial selection operates by favoring individuals with certain phenotypic traits, allowing them to reproduce and pass their genes into the next generation. Assuming that phenotypic differences are genetically determined, such selection should lead to evolutionary change and, indeed, it has. Artificial selection, imposed in laboratory experiments, agriculture, and the domestication process, has produced substantial change in almost every case in which it has been applied. This success is strong proof that selection is an effective evolutionary process.

Laboratory Experiments

With the rise of genetics as a field of science in the 1920s and 1930s, researchers began conducting experiments to test the hypothesis that selection can produce evolutionary change. A favorite subject was the now-famous laboratory fruit fly, *Drosophila melanogaster*. Geneticists have imposed selection on just about every conceivable aspect of the fruit fly—including body size, eye color, growth rate, life span, and exploratory behavior—with a consistent result: selection for a trait leads to strong and predictable evolutionary response.

In one classic experiment, scientists selected for fruit flies with many bristles (stiff, hairlike structures) on their abdomen. At the start of the experiment, the average number of bristles was 9.5. Each generation, scientists picked out the 20% of the population with the greatest number of bristles and allowed them to reproduce, thus establishing the next generation. After 86 generations of such selection, the average number of bristles had quadrupled, to nearly 40. In a similar experiment, fruit flies were selected for either the most or the fewest numbers of bristles. Within 35 generations, the populations did not overlap at all in range of variation (figure 21.12).

Similar experiments have been conducted on a wide variety of other laboratory organisms. For example, by selecting for rats that were resistant to tooth decay, scientists were able to increase in less than 20 generations the average time for onset of decay from barely over 100 days to greater than 500 days.

Agriculture

Similar methods have been practiced in agriculture for many centuries. Familiar livestock, such as cattle and pigs, and crops, like corn and strawberries, are greatly different from their wild ancestors (figure 21.13). These differences have resulted from generations of selection for desirable traits like milk production and corn stalk size.

An experimental study with corn demonstrates the ability of artificial selection to rapidly produce major change in

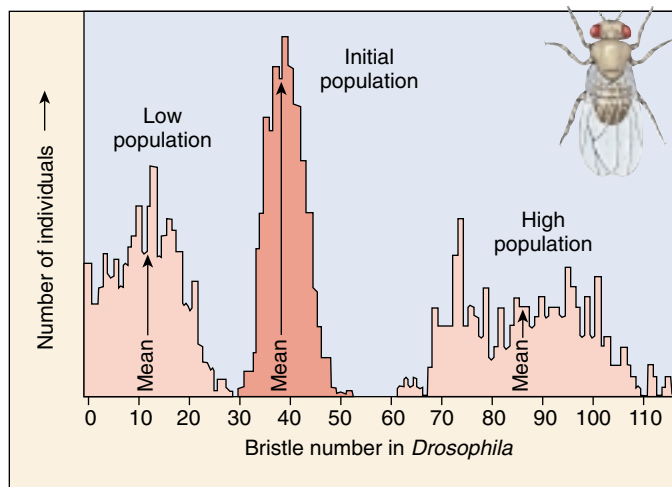


FIGURE 21.12

Artificial selection in the laboratory. In this experiment, one population of *Drosophila* was selected for low numbers of bristles and the other for high numbers. Note that not only did the means of the populations change greatly in 35 generations, but also that all individuals in both experimental populations lie outside the range of the initial population.

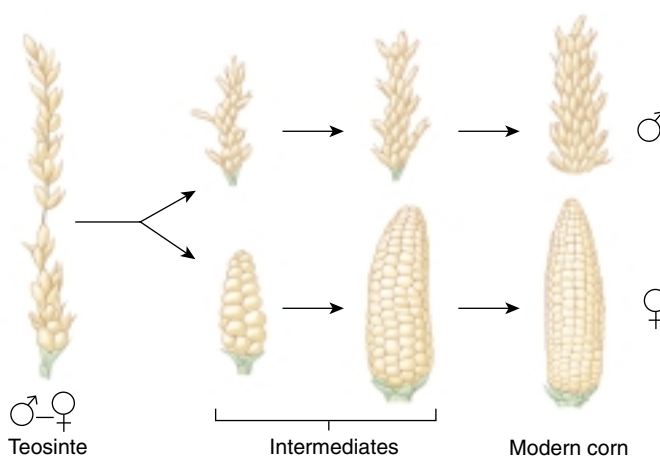


FIGURE 21.13

Corn looks very different from its ancestor. The tassels and seeds of a wild grass, such as teosinte, evolved into the male tassels and female ears of modern corn.

crop plants. In 1896, agricultural scientists began selecting on oil content of corn kernels, which initially was 4.5%. As in the fruit fly experiments, the top 20% of all individuals were allowed to reproduce. In addition, a parallel experiment selected for the individuals with the lowest oil content. By 1986, at which time 90 generations had passed, average oil content had increased approximately 450% in the high-content experiment; by contrast, oil content in the low experiment had decreased to about 0.5%, a level at which it is difficult to get accurate readings.

Domestication

Artificial selection has also been responsible for the great variety of breeds of cats, dogs (figure 21.14), pigeons, cattle and other domestic animals. In some cases, breeds have been developed for particular purposes. Greyhound dogs, for example, were bred by selecting for maximal running abilities, with the end result being an animal with long legs and tail (the latter used as a rudder), an arched back (to increase the length of its stride), and great muscle mass. By contrast, the odd proportions of the ungainly basset hound resulted from selection for dogs that could enter narrow holes in pursuit of rabbits and other small game. In other cases, breeds have been developed primarily for their appearance, such as the many colorful and ornamented varieties of pigeons or the breeds of cats.

Domestication also has led to unintentional selection for some traits. In recent years, as part of an attempt to domesticate the silver fox, Russian scientists each generation have chosen the most docile animals and allowed them to reproduce. Within 40 years, the vast majority of foxes born were exceptionally docile, not only allowing themselves to be petted, but also whimpering to get attention and sniffing and licking their caretakers. In many respects, they had become no different than domestic dogs! However, it was not only behavior that changed. These foxes also began to exhibit different color patterns, floppy ears, curled tails, and shorter legs and tails. Presumably, the genes responsible for docile behavior have other effects as well (the phenomenon of pleiotropy discussed in the last chapter); as selection has favored docile animals, it has also led to the evolution of these other traits.

Can Selection Produce Major Evolutionary Changes?

Given that we can observe the results of selection operating over relatively short periods of time, most scientists believe that natural selection is the process responsible for the evolutionary changes documented in the fossil record. Some critics of evolution accept that selection can lead to changes within a species, but contend that such changes are relatively minor in scope and not equivalent to the substantial changes documented in the fossil record. In other words, it is one thing to change the number of bristles on a fruit fly or the size of a corn stalk, and quite another to produce an entirely new species.

This argument does not fully appreciate the extent of change produced by artificial selection. Consider, for ex-

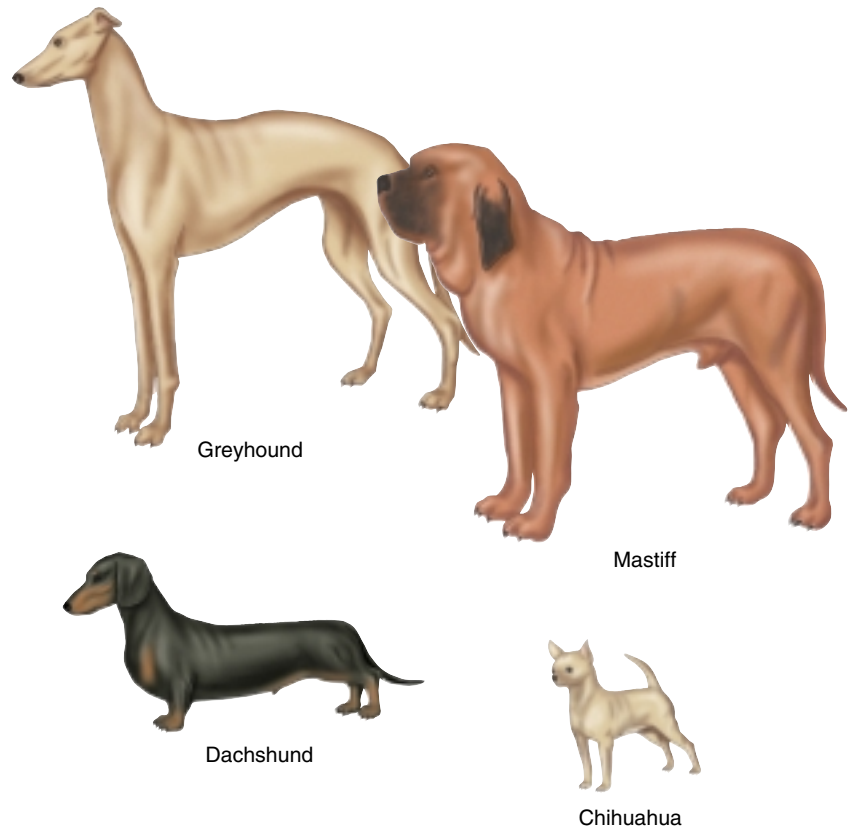


FIGURE 21.14

Breeds of dogs. The differences between these dogs are greater than the differences displayed between any wild species of canids.

ample, the breeds of dogs, all of which have been produced since wolves were first domesticated, perhaps 10,000 years ago. If the various dog breeds did not exist and a paleontologist found fossils of animals similar to dachshunds, greyhounds, mastiffs, Chihuahuas, and pomeranians, there is no question that they would be considered different species. Indeed, these breeds are so different that they would probably be classified in different genera. In fact, the diversity exhibited by dog breeds far outstrips the differences observed among wild members of the family Canidae—such as coyotes, jackals, foxes, and wolves. Consequently, the claim that artificial selection produces only minor changes is clearly incorrect. Indeed, if selection operating over a period of only 10,000 years can produce such substantial differences, then it would seem powerful enough, over the course of many millions of years, to produce the diversity of life we see around us today.

Artificial selection often leads to rapid and substantial results over short periods of time, thus demonstrating the power of selection to produce major evolutionary change.

21.3 Evidence for evolution can be found in other fields of biology.

The Anatomical Record

Much of the power of the theory of evolution is its ability to provide a sensible framework for understanding the diversity of life. Many observations from a wide variety of fields of biology simply cannot be understood in any meaningful way except as a result of evolution.

Homology

As vertebrates evolved, the same bones were sometimes put to different uses. Yet the bones are still seen, their presence betraying their evolutionary past. For example, the forelimbs of vertebrates are all **homologous structures**, that is, structures with different appearances and functions that all derived from the same body part in a common ancestor. You can see in figure 21.15 how the bones of the forelimb have been modified in different ways for different vertebrates. Why should these very different structures be composed of the same bones? If evolution had not occurred, this would indeed be a riddle. But when we consider that all of these animals are descended from a common ancestor, it is easy to understand that natural selection has modified the same initial starting blocks to serve very different purposes.

Development

Some of the strongest anatomical evidence supporting evolution comes from comparisons of how organisms develop. In many cases, the evolutionary history of an organism can be seen to unfold during its development, with the embryo exhibiting characteristics of the embryos of its ancestors (figure 21.16). For example, early in their development, human embryos possess gill slits, like a fish; at a later stage, every human embryo has a long bony tail, the vestige of which we carry to adulthood as the coccyx at the end of our spine. Human fetuses even

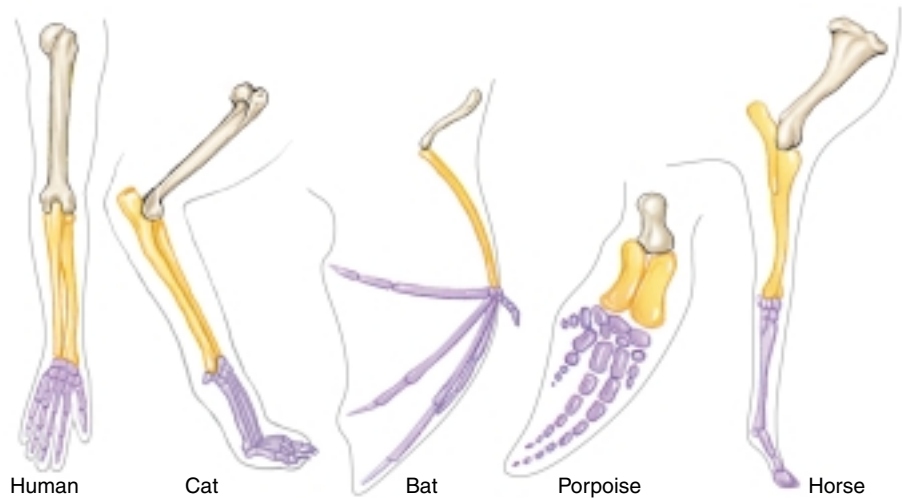


FIGURE 21.15

Homology among the bones of the forelimb. Although these structures show considerable differences in form and function, the same basic bones are present in the forelimbs of humans, cats, bats, porpoises, and horses.

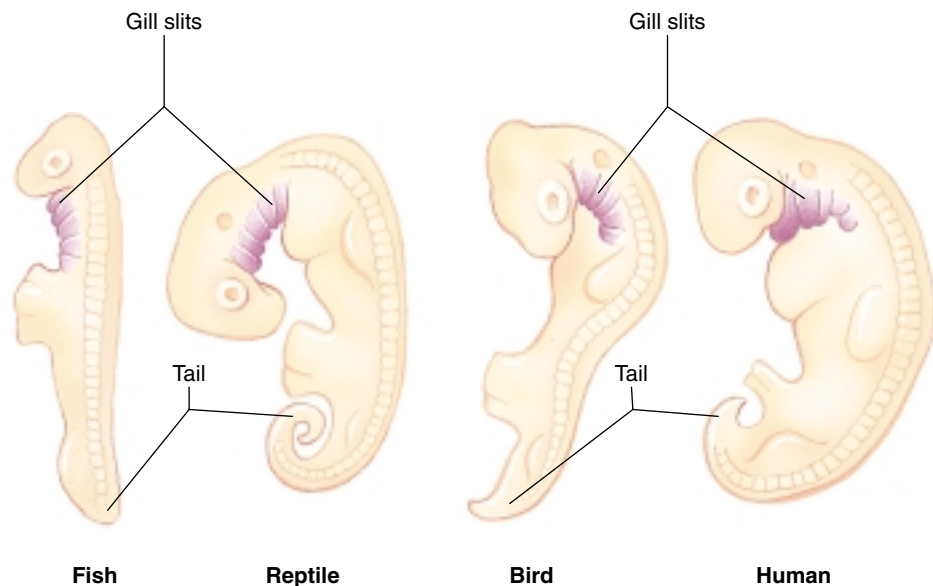


FIGURE 21.16

Our embryos show our evolutionary history. The embryos of various groups of vertebrate animals show the features they all share early in development, such as gill slits (*in purple*) and a tail.

possess a fine fur (called *lanugo*) during the fifth month of development. These relict developmental forms suggest strongly that our development has evolved, with new instructions layered on top of old ones.

The observation that seemingly different organisms may exhibit similar embryological forms provides indirect but convincing evidence of a past evolutionary relationship. Slugs and giant ocean squids, for example, do not bear much superficial resemblance to each other, but the similarity of their embryological forms provides convincing evidence that they are both mollusks.

Vestigial Structures

Many organisms possess **vestigial structures** that have no apparent function, but that resemble structures their presumed ancestors had. Humans, for example, possess a complete set of muscles for wiggling their ears, just as a coyote does (table 21.1). Boa constrictors have hip bones and rudimentary hind legs. Manatees (a type of aquatic mammal often referred to as “sea cows”) have fingernails on their fins (which evolved from legs). Figure 21.17 illustrates the skeleton of a baleen whale, which contains pelvic bones, as other mammal skeletons do, even though such bones serve no known function in the whale. The human vermiform appendix is apparently vestigial; it represents the degenerate terminal part of the cecum, the blind pouch or sac in which the large intestine begins. In other mammals such as mice, the cecum is the largest part of the large intestine and functions in storage—usually of bulk cellulose in herbivores. Although some suggestions have been made, it is difficult to assign any current function to the vermiform appendix. In many respects, it is a dangerous organ: quite often it becomes infected, leading to an inflammation called appendicitis; without surgical re-

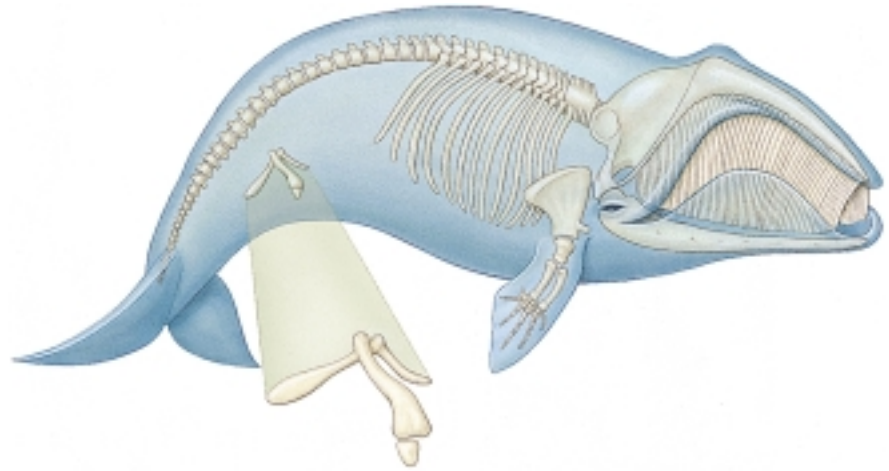


FIGURE 21.17

Vestigial features. The skeleton of a baleen whale, a representative of the group of mammals that contains the largest living species, contains pelvic bones. These bones resemble those of other mammals, but are only weakly developed in the whale and have no apparent function.

moval, the appendix may burst, allowing the contents of the gut to come in contact with the lining of the body cavity, a potentially fatal event. It is difficult to understand vestigial structures such as these as anything other than evolutionary relicts, holdovers from the evolutionary past. They argue strongly for the common ancestry of the members of the groups that share them, regardless of how different they have subsequently become.

Comparisons of the anatomy of different living animals often reveal evidence of shared ancestry. In some instances, the same organ has evolved to carry out different functions, in others, an organ loses its function altogether. Sometimes, different organs evolve in similar ways when exposed to the same selective pressures.

Table 21.1 Some Vestigial Traits in Humans

Trait	Description
Ear-wiggling muscles	Three small muscles around each ear that are large and important in some mammals, such as dogs, turning the ears toward a source of sound. Few people can wiggle their ears, and none can turn them toward sound.
Tail	Present in human and all vertebrate embryos. In humans, the tail is reduced; most adults only have three to five tiny tail bones and, occasionally, a trace of a tail-extending muscle.
Appendix	Structure which presumably had a digestive function in some of our ancestors, like the cecum of some herbivores. In humans, it varies in length from 5–15 cm, and some people are born without one.
Wisdom teeth	Molars that are often useless and sometimes even trapped in the jawbone. Some people never develop wisdom teeth.

Based on a suggestion by Dr. Leslie Dendy, Department of Science and Technology, University of New Mexico, Los Alamos.

The Molecular Record

Traces of our evolutionary past are also evident at the molecular level. If you think about it, the fact that organisms have evolved successively from relatively simple ancestors implies that a record of evolutionary change is present in the cells of each of us, in our DNA. When an ancestral species gives rise to two or more descendants, those descendants will initially exhibit fairly high overall similarity in their DNA. However, as the descendants evolve independently, they will accumulate more and more differences in their DNA. Consequently, organisms that are more distantly related would be expected to accumulate a greater number of evolutionary differences, whereas two species that are more closely related should share a greater portion of their DNA.

To examine this hypothesis, we need an estimate of evolutionary relationships that has been developed from data other than DNA (it would be a circular argument to use DNA to estimate relationships and then conclude that closely related species are more similar in their DNA than are distantly related species). Such an hypothesis of evolutionary relationships is provided by the fossil record, which indicates when particular types of organisms evolved. In addition, by examining the anatomical structures of fossils and of modern species, we can infer how closely species are related to each other.

When degree of genetic similarity is compared with our ideas of evolutionary relationships based on fossils, a close match is evident. For example, when the human hemoglobin polypeptide is compared to the corresponding molecule in other species, closely related species are found to be more similar. Chimpanzees, gorillas, orangutans, and macaques, vertebrates thought to be more closely related to humans, have fewer differences from humans in the 146-amino-acid hemoglobin β chain than do more distantly related mammals, like dogs. Nonmammalian vertebrates differ even more, and nonvertebrate hemoglobins are the most different of all (figure 21.18). Similar patterns are also evident when the DNA itself is compared. For example, chimps and humans, which are thought to have descended from a common ancestor that lived approximately 6 million years ago, exhibit few differences in their DNA.

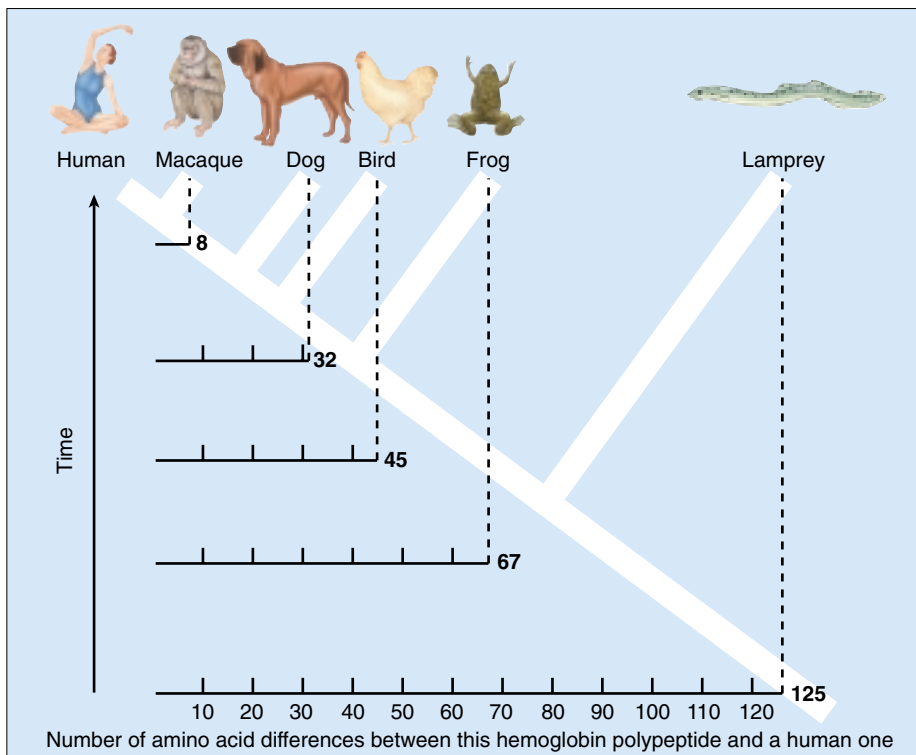


FIGURE 21.18

Molecules reflect evolutionary divergence. You can see that the greater the evolutionary distance from humans (white cladogram), the greater the number of amino acid differences in the vertebrate hemoglobin polypeptide.

Why should closely related species be similar in DNA? Because DNA is the genetic code that produces the structure of living organisms, one might expect species that are similar in overall appearance and structure, such as humans and chimpanzees, to be more similar in DNA than are more dissimilar species, such as humans and frogs. This expectation would hold true even if evolution had not occurred. However, there are some noncoding stretches of DNA (sometimes called “junk DNA”) that have no function and appear to serve no purpose. If evolution had not occurred, there would be no reason to expect similar-appearing species to be similar in their junk DNA. However, comparisons of such stretches of DNA provide the same results as for other parts of the genome: more closely related species are more similar, an observation that only makes sense if evolution has occurred.

Comparison of the DNA of different species provides strong evidence for evolution. Species deduced from the fossil record to be closely related are more similar in their DNA than are species thought to be more distantly related.

Convergent and Divergent Evolution

Different geographical areas sometimes exhibit groups of plants and animals of strikingly similar appearance, even though the organisms may be only distantly related. It is difficult to explain so many similarities as the result of coincidence. Instead, natural selection appears to have favored parallel evolutionary adaptations in similar environments. Because selection in these instances has tended to favor changes that made the two groups more alike, their phenotypes have converged. This form of evolutionary change is referred to as **convergent evolution**, or sometimes, **parallel evolution**.

The Marsupial-Placental Convergence

In the best-known case of convergent evolution, two major groups of mammals, marsupials and placentals, have evolved in a very similar way, even though the two lineages have been living independently on separate continents. Australia separated from the other continents more than 50 million years ago, after marsupials had evolved but before the appearance of placental mammals. As a result, the only mammals in Australia (other than bats and a few colonizing rodents) have been marsupials, members of a group in which the young are born in a very immature condition and held in a pouch until they are ready to emerge into the outside world. Thus, even though placental mammals are the dominant mammalian group throughout most of the world, marsupials retained supremacy in Australia.

What are the Australian marsupials like? To an astonishing degree, they resemble the placental mammals living today on the other continents (figure 21.19). The similarity between some individual members of these two sets of mammals argues strongly that they are the result of convergent evolution, similar forms having evolved in different, isolated areas because of similar selective pressures in similar environments.

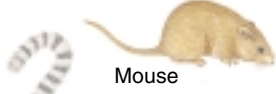
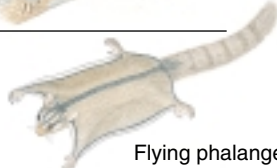
Niche	Placental Mammals	Australian Marsupials
Burrower	 Mole	 Marsupial mole
Anteater	 Lesser anteater	 Numbat (anteater)
Mouse	 Mouse	 Marsupial mouse
Climber	 Lemur	 Spotted cuscus
Glider	 Flying squirrel	 Flying phalanger
Cat	 Ocelot	 Tasmanian "tiger cat"
Wolf	 Wolf	 Tasmanian wolf

FIGURE 21.19

Convergent evolution. Marsupials in Australia resemble placental mammals in the rest of the world. They evolved in isolation after Australia separated from other continents.

Homology versus Analogy

How do we know when two similar characters are homologous and when they are analogous? As we have seen, adaptation favoring different functions can obscure homologies, while convergent evolution can create analogues that appear as similar as homologues. There is no hard-and-fast answer to this question; the determination of homologues is often a thorny issue in biological classification. As we have seen in comparing vertebrate embryos, and again in comparing slugs and squids, studies of embryonic development often reveal features not apparent when studying adult organisms. In general, the more complex two structures are, the less likely they evolved independently.



FIGURE 21.20

A Galápagos tortoise most closely resembles South American tortoises. Isolated on these remote islands, the Galápagos tortoise has evolved distinctive forms. This natural experiment is being terminated, however. Since Darwin's time, much of the natural habitat of the larger islands has been destroyed by human intrusion. Goats introduced by settlers, for example, have drastically altered the vegetation.

Darwin and Patterns of Recent Divergence

Darwin was the first to present evidence that animals and plants living on oceanic islands resemble most closely the forms on the nearest continent—a relationship that only makes sense as reflecting common ancestry. The Galápagos turtle in figure 21.20 is more similar to South American turtles than to those of any other continent. This kind of relationship strongly suggests that the island forms evolved from individuals that came from the adjacent mainland at some time in the past. Thus, the Galápagos finches of figure 21.8 have different beaks than their South American relatives. In the absence of evolution, there seems to be no logical explanation of why individual kinds of island plants and animals would be clearly related to others on the nearest mainland, but still have some divergent features. As Darwin pointed out, this relationship provides strong evidence that macroevolution has occurred.

A similar resemblance to mainland birds can be seen in an island finch Darwin never saw—a solitary finch species living on Cocos Island, a tiny, remote volcanic island located 630 kilometers to the northeast of the Galápagos. This finch does not resemble the finches of Europe, Australia, Africa, or North America. Instead, it resembles the finches of Costa Rica, 500 kilometers to the east.

Of course, because of adaptation to localized habitats, island forms are not identical to those on the nearby continents. The turtles have evolved different shell shapes, for example; those living in moist habitats have dome-shaped

shells while others living in dry places have low, saddle-backed shells with the front of the shell bent up to expose the head and neck. Similarly, the Galápagos finches have evolved from a single presumptive ancestor into 13 species, each specialized in a different way. These Galápagos turtles and finches have evolved in concert with the continental forms, from the same ancestors, but the two lineages have diverged rather than converged.

It is fair to ask how Darwin knew that the Galápagos tortoises and finches do not represent the convergence of unrelated island and continental forms (analogues) rather than the divergence of recently isolated groups (homologues). While either hypothesis would argue for natural selection, Darwin chose divergence of homologues as by far the simplest explanation, because the turtles and finches differ by only a few traits, and are similar in many.

In sum total, the evidence for macroevolution is overwhelming. In the next chapter, we will consider Darwin's proposal that microevolutionary changes have led directly to macroevolutionary changes, the key argument in his theory that evolution occurs by natural selection.

Evolution favors similar forms under similar circumstances. Convergence is the evolution of similar forms in different lineages when exposed to the same selective pressures. Divergence is the evolution of different forms in the same lineage when exposed to different selective pressures.

21.4 The theory of evolution has proven controversial.

Darwin's Critics

In the century since he proposed it, Darwin's theory of evolution by natural selection has become nearly universally accepted by biologists, but has proven controversial among the general public. Darwin's critics raise seven principal objections to teaching evolution:

1. **Evolution is not solidly demonstrated.** *"Evolution is just a theory,"* Darwin's critics point out, as if theory meant lack of knowledge, some kind of guess. Scientists, however, use the word theory in a very different sense than the general public does. Theories are the solid ground of science, that of which we are most certain. Few of us doubt the theory of gravity because it is "just a theory."
2. **There are no fossil intermediates.** *"No one ever saw a fin on the way to becoming a leg,"* critics claim, pointing to the many gaps in the fossil record in Darwin's day. Since then, however, most fossil intermediates in vertebrate evolution have indeed been found. A clear line of fossils now traces the transition between whales and hoofed mammals, between reptiles and mammals, between dinosaurs and birds, between apes and humans. The fossil evidence of evolution between major forms is compelling.
3. **The intelligent design argument.** *"The organs of living creatures are too complex for a random process to have produced—the existence of a clock is evidence of the existence of a clockmaker."* Biologists do not agree. The intermediates in the evolution of the mammalian ear can be seen in fossils, and many intermediate "eyes" are known in various invertebrates. These intermediate forms arose because they have value—being able to detect light a little is better than not being able to detect it at all. Complex structures like eyes evolved as a progression of slight improvements.
4. **Evolution violates the Second Law of Thermodynamics.** *"A jumble of soda cans doesn't by itself jump neatly into a stack—things become more disorganized due to random events, not more organized."* Biologists point out that this argument ignores what the second law really says: disorder increases in a closed system, which the earth most certainly is not. Energy continually enters the biosphere from the sun, fueling life and all the processes that organize it.
5. **Proteins are too improbable.** *"Hemoglobin has 141 amino acids. The probability that the first one would be leucine is 1/20, and that all 141 would be the ones they are by chance is (1/20)¹⁴¹, an impossibly rare event."* This is statistical foolishness—you cannot use probability to argue backwards. The probability that a student in a classroom has a particular birthday is 1/365; arguing

this way, the probability that everyone in a class of 50 would have the birthdays they do is $(1/365)^{50}$, and yet there the class sits.

6. **Natural selection does not imply evolution.** *"No scientist has come up with an experiment where fish evolve into frogs and leap away from predators."* Is microevolution (evolution within a species) the mechanism that has produced macroevolution (evolution among species)? Most biologists that have studied the problem think so. Some kinds of animals produced by artificial selection are remarkably distinctive, such as Chihuahuas, dachshunds, and greyhounds. While all dogs are in fact the same species and can interbreed, laboratory selection experiments easily create forms that cannot interbreed and thus would in nature be considered different species. Thus, production of radically different forms has indeed been observed, repeatedly. To object that evolution still does not explain really major differences, like between fish and amphibians, simply takes us back to point 2—these changes take millions of years, and are seen clearly in the fossil record.
7. **The irreducible complexity argument.** *The intricate molecular machinery of the cell cannot be explained by evolution from simpler stages. Because each part of a complex cellular process like blood clotting is essential to the overall process, how can natural selection fashion any one part?* What's wrong with this argument is that each part of a complex molecular machine evolves as part of the system. Natural selection can act on a complex system because at every stage of its evolution, the system functions. Parts that improve function are added, and, because of later changes, become essential. The mammalian blood clotting system, for example, has evolved from much simpler systems. The core clotting system evolved at the dawn of the vertebrates 600 million years ago, and is found today in lampreys, the most primitive fish. One hundred million years later, as vertebrates evolved, proteins were added to the clotting system making it sensitive to substances released from damaged tissues. Fifty million years later, a third component was added, triggering clotting by contact with the jagged surfaces produced by injury. At each stage as the clotting system evolved to become more complex, its overall performance came to depend on the added elements. Thus, blood clotting has become "irreducibly complex"—as the result of Darwinian evolution.

Darwin's theory of evolution has proven controversial among the general public, although the commonly raised objections are without scientific merit.



Summary

Questions

Media Resources

21.1 Fossil evidence indicates that evolution has occurred.

- Fossils of many extinct species have never been discovered. Nonetheless, the fossil record is complete enough to allow a detailed understanding of the evolution of life through time. The evolution of the major vertebrate groups is quite well known.
- Although evolution of groups like horses may appear to be a straight-line progression, in fact there have been many examples of parallel evolution, and even reversals from overall trends.

- Why do gaps exist in the fossil record? What lessons can be learned from the fossil record of horse evolution?
- How did scientists date fossils in Darwin's day? Why are scientists today able to date rocks more accurately?



- On Science Article: feathered Dinosaurs
- Book Reviews:
 - In Search of Deep Time* by Gee
 - Digging Dinosaurs* by Horner

21.2 Natural selection can produce evolutionary change.

- Natural populations provide clear evidence of evolutionary change.
- Darwin's finches have different-sized beaks, which are adaptations to eating different kinds of seeds. In particularly dry years, natural selection favors birds with stout beaks within one species, *Geospiza fortis*. As a result, the average bill size becomes larger in the next generation.
- The British populations of the peppered moth, *Biston betularia*, consisted mostly of light-colored individuals before the Industrial Revolution. Over the last two centuries, populations that occur in heavily polluted areas, where the tree trunks are darkened with soot, have come to consist mainly of dark-colored (melanic) individuals—a result of rapid natural selection.

- Why did the average beak size of the medium ground finch increase after a particularly dry year?
- Why did the frequency of light-colored moths decrease and that of dark-colored moths increase with the advent of industrialism? What is industrial melanism?
- What can artificial selection tell us about evolution? Is artificial selection a good analogy for the selection that occurs in nature?



- Activity: Evolution of Fish

21.3 Evidence for evolution can be found in other fields of biology.

- Several indirect lines of evidence argue that macroevolution has occurred, including successive changes in homologous structures, developmental patterns, vestigial structures, parallel patterns of evolution, and patterns of distribution.
- When differences in genes or proteins are examined, species that are thought to be closely related based on the fossil record may be more similar than species thought to be distantly related.

- What is homology? How does it support evolutionary theory?
- What is convergent evolution? Give examples.
- How did Darwin's studies of island populations provide evidence for evolution?



- Exploration: Evolution of the Heart



- Molecular Clock



- Activity: Divergence



- Student Research: Evolution of Insect Diets
- On Science Articles:
 - Darwinism at the Cellular Level
 - Was Darwin Wrong?

21.4 The theory of evolution has proven controversial.

- The objections raised by Darwin's critics are easily answered.

- Is "Darwinism" really science? Explain.



- On Science Article: Answering Evolution's Critics
- Bioethics Case Study: Creationism
- Book Reviews:
 - Mr. Darwin's Shooter* by McDonald

22

The Origin of Species

Concept Outline

22.1 Species are the basic units of evolution.

The Nature of Species. Species are groups of actually or potentially interbreeding natural populations which are reproductively isolated from other such groups and that maintain connectedness over geographic distances.

22.2 Species maintain their genetic distinctiveness through barriers to reproduction.

Prezygotic Isolating Mechanisms. Some breeding barriers prevent the formation of zygotes.

Postzygotic Isolating Mechanisms. Other breeding barriers prevent the proper development or reproduction of the zygote after it forms.

22.3 We have learned a great deal about how species form.

Reproductive Isolation May Evolve as a By-Product of Evolutionary Change. Speciation can occur in the absence of natural selection, but reproductive isolation generally occurs more quickly when populations are adapting to different environments.

The Geography of Speciation. Speciation occurs most readily when populations are geographically isolated. Sympatric speciation can occur by polyploidy and, perhaps, by other means.

22.4 Clusters of species reflect rapid evolution.

Darwin's Finches. Thirteen species of finches, all descendants of one ancestral finch, occupy diverse niches.

Hawaiian *Drosophila*. More than a quarter of the world's fruit fly species are found on the Hawaiian Islands.

Lake Victoria Cichlid Fishes. Isolation has led to extensive species formation among these small fishes.

New Zealand Alpine Buttercups. Repeated glaciations have fostered waves of species formation in alpine plants.

Diversity of Life through Time. The number of species has increased through time, despite a number of mass extinction events.

The Pace of Evolution. The idea that evolution occurs in spurts is controversial.

Problems with the Biological Species Concept. This concept is not as universal as previously thought.



FIGURE 22.1

A group of Galápagos iguanas bask in the sun on their isolated island. How does geographic isolation contribute to the formation of new species?

Although Darwin titled his book *On the Origin of Species*, he never actually discussed what he referred to as that “mystery of mysteries” of how one species gives rise to another. Rather, his argument concerned evolution by natural selection; that is, how one species evolves through time to adapt to its changing environment. Although of fundamental importance to evolutionary biology, the process of adaptation does not explain how one species becomes another (figure 22.1); much less can it explain how one species can give rise to many descendant species. As we shall see, adaptation may be involved in this process of **speciation**, but it need not be.

22.1 Species are the basic units of evolution.

The Nature of Species

Before we can discuss how one species gives rise to another, we need to understand exactly what a species is. Even though definition of what constitutes a species is of fundamental importance to evolutionary biology, this issue has still not been completely settled and is currently the subject of considerable research and debate. However, any concept of a species must account for two phenomena: the distinctiveness of species that occur together at a single locality, and the connection that exists among populations of the same species that are geographically separated.

The Distinctiveness of Sympatric Species

Put out a birdfeeder on your balcony or back porch and you will attract a wide variety of different types of birds (especially if you put out a variety of different kinds of foods). In the midwestern United States, for example, you might routinely see cardinals, blue jays, downy woodpeckers, house finches—even hummingbirds in the summer (figure 22.2). Although it might take a few days of careful observation, you would soon be able to readily distinguish the many different species. The reason is that species that occur together (termed **sympatric** from the Greek *sym* for “same” and *patria* for “species”) are distinctive entities that are phenotypically different, utilize different parts of the habitat, and behave separately. This observation is generally true not only for birds, but also for most other types of organisms in most places.

Occasionally, two species occur together that appear to be nearly identical, and are thus called **sibling species**. In most cases, however, our inability to distinguish the two reflects our own reliance on vision as our primary sense. When the mating calls or chemicals exuded by such species are examined, they usually reveal great differences. In other words, even though we have trouble separating them, the animals themselves have no such difficulties!

Geographic Variation within Species

Within the units classified as species, populations that occur in different areas may be more or less distinct from one another. Such groups of distinctive individuals may be classified taxonomically as **subspecies** or **varieties** (the vague term “race” has a similar connotation, but is no longer commonly used). In areas where these populations approach one another, individuals often exhibit combinations of features characteristic of both populations. In other words, even though geographically distant populations may appear distinct, they usually are connected by intervening populations that are intermediate in their characteristics (figure 22.3).

The Biological Species Concept

What can account both for the distinctiveness of sympatric species and the connectedness of geographic populations of the same species? One obvious possibility is that each species exchanges genetic material only with other members of its species. If sympatric species commonly ex-

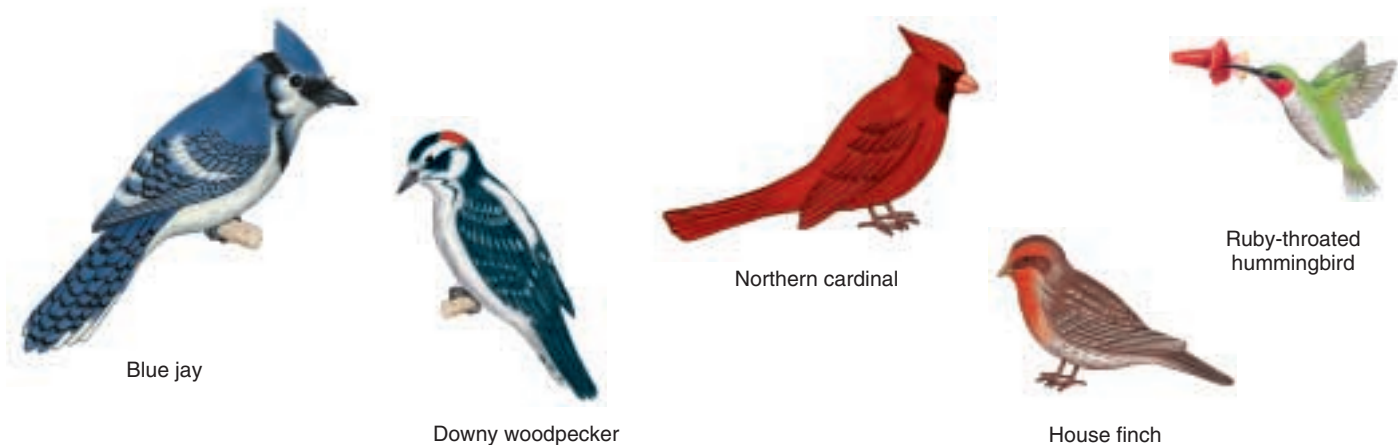


FIGURE 22.2

Common birds in the midwestern United States. No one would doubt that these birds are distinct species. Each can be distinguished from the others by many ecological, behavioral, and phenotypic traits.

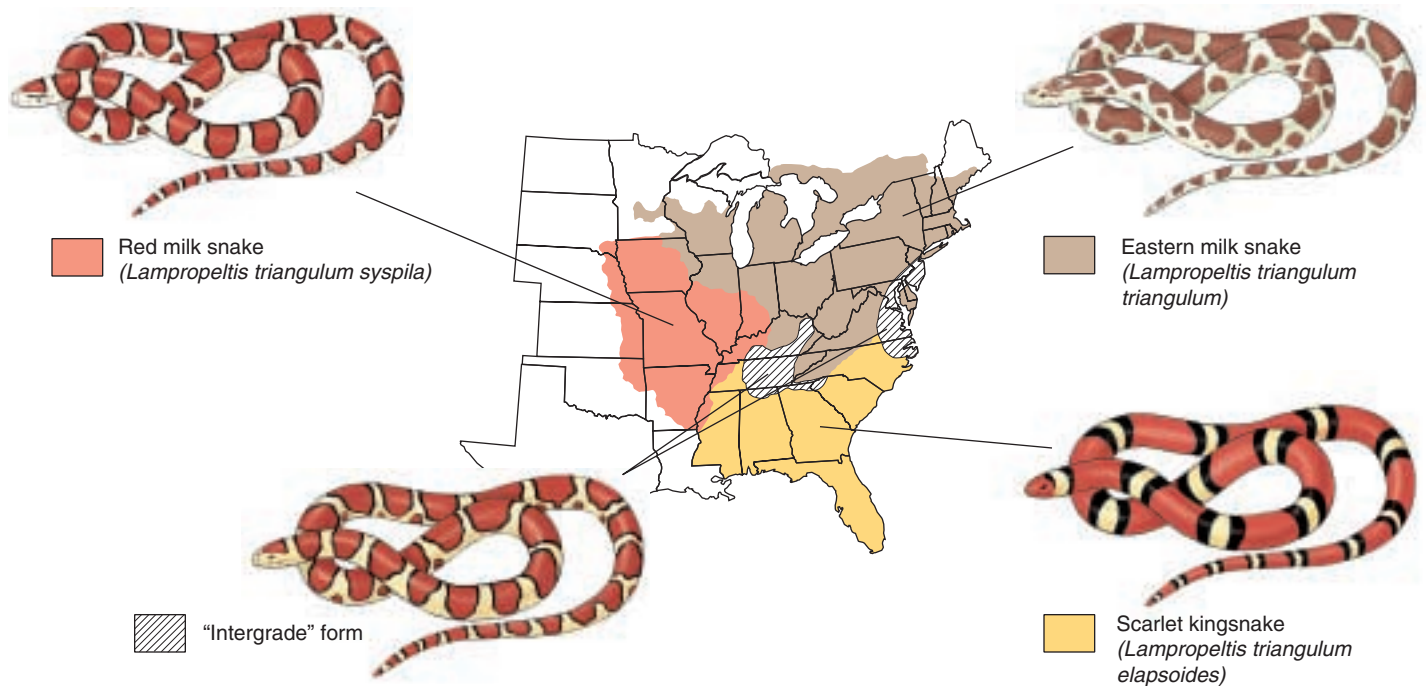


FIGURE 22.3

Geographic variation in the milk snake, *Lampropeltis triangulum*. Although each subspecies appears phenotypically quite distinctive from the others, they are connected by populations that are phenotypically intermediate.

changed genes, we might expect such species to rapidly lose their distinctions as the **gene pools** of the different species became homogenized. Conversely, the ability of geographically distant populations to share genes through the process of gene flow may keep these populations integrated as members of the same species. Based on these ideas, the evolutionary biologist Ernst Mayr coined the **biological species concept** which defines species as:

“... groups of actually or potentially interbreeding natural populations which are reproductively isolated from other such groups.”

In other words, the biological species concept says that a species is all individuals that are capable of interbreeding and producing fertile offspring. Conversely, individuals that cannot produce fertile offspring are said to be **reproductively isolated** and, thus, members of different species.

Occasionally, members of different species will interbreed, a process termed **hybridization**. If the species are reproductively isolated, either no offspring will result, or if offspring are produced, they will be either unhealthy or sterile. In this way, genes from one species generally will not be able to enter the gene pool of another species.

Problems with Applying the Biological Species Concept

The biological species concept has proven to be an effective way of understanding the existence of species in nature. Nonetheless, the concept has some practical difficulties.

For example, it can be difficult to apply the concept to populations that do not occur together in nature (and are thus said to be **allopatric**). Because individuals of these populations do not encounter each other, it is not possible to observe whether they would interbreed naturally. Although experiments can determine whether fertile hybrids can be produced, this information is not enough. The reason is that many species that will coexist without interbreeding in nature will readily hybridize in the artificial settings of the laboratory or zoo. Consequently, evaluating whether allopatric populations constitute different species is ultimately a judgment call.

In addition, the concept is more limited than its name would imply. Many organisms are asexual and reproduce without mating; reproductive isolation has no meaning for such organisms.

Moreover, despite its name, the concept is really a zoological species concept and applies less readily to plants. Even among animals, the biological species concept appears to apply more successfully to some groups than to others. As we will see in section 22.4, biologists are currently reevaluating this and other approaches to the study of species.

Species are groups of organisms that are distinct from other co-occurring species and that are interconnected geographically. The ability to exchange genes appears to be a hallmark of such species.

22.2 Species maintain their genetic distinctiveness through barriers to reproduction.

Prezygotic Isolating Mechanisms

How do species keep their separate identities? Reproductive isolating mechanisms fall into two categories: **prezygotic isolating mechanisms**, which prevent the formation of zygotes; and **postzygotic isolating mechanisms**, which prevent the proper functioning of zygotes after they form. In the following sections we will discuss various isolating mechanisms in these two categories and offer examples that illustrate how the isolating mechanisms operate to help species retain their identities.

Ecological Isolation

Even if two species occur in the same area, they may utilize different portions of the environment and thus not hybridize because they do not encounter each other. For example, in India, the ranges of lions and tigers overlapped until about 150 years ago. Even when they did, however, there were no records of natural hybrids. Lions stayed mainly in the open grassland and hunted in groups called prides; tigers tended to be solitary creatures of the forest (figure 22.4). Because of their ecological and behavioral differences, lions and tigers rarely came into direct contact with each other, even though their ranges overlapped over thousands of square kilometers.

In another example, the ranges of two toads, *Bufo woodhousei* and *B. americanus*, overlap in some areas. Although these two species can produce viable hybrids, they usually do not interbreed because they utilize different portions of the habitat for breeding. Whereas *B. woodhousei* prefers to breed in streams, *B. americanus* breeds in rainwater puddles. Similarly, the ranges of two species of dragonflies overlap in Florida. However, the dragonfly *Progomphus obscurus* lives near rivers and streams, and *P. alachue-nis* lives near lakes.

Similar situations occur among plants. Two species of oaks occur widely in California: the valley oak, *Quercus lobata*, and the scrub oak, *Q. dumosa*. The valley oak, a graceful deciduous tree that can be as tall as 35 meters, occurs in the fertile soils of open grassland on gentle slopes and valley floors. In contrast, the scrub oak is an evergreen shrub, usually only 1 to 3 meters tall, which often forms the kind of dense scrub known as chaparral. The scrub oak is found on steep slopes in less fertile soils. Hybrids between these different oaks do occur and are fully fertile, but they are rare. The sharply distinct habitats of their parents limit their occurrence together, and there is no intermediate habitat where the hybrids might flourish.



(a)



(b)



(c)

FIGURE 22.4

Lions and tigers are ecologically isolated. The ranges of lions and tigers used to overlap in India. However, lions and tigers do not hybridize in the wild because they utilize different portions of the habitat. (a) Lions live in open grassland. (b) Tigers are solitary animals that live in the forest. (c) Hybrids, such as this tiglon, have been successfully produced in captivity, but hybridization does not occur in the wild.

Behavioral Isolation

In chapter 27, we will consider the often elaborate courtship and mating rituals of some groups of animals. Related species of organisms such as birds often differ in their courtship rituals, which tends to keep these species distinct in nature even if they inhabit the same places (figure 22.5). For example, mallard and pintail ducks are perhaps the two most common freshwater ducks in North America. In captivity, they produce completely fertile offspring, but in nature they nest side-by-side and only rarely hybridize.

More than 500 species of flies of the genus *Drosophila* live in the Hawaiian Islands. This is one of the most remarkable concentrations of species in a single animal genus found anywhere. The genus occurs throughout the world, but nowhere are the flies more diverse in external appearance or behavior than in Hawaii. Many of these flies differ greatly from other species of *Drosophila*, exhibiting characteristics that can only be described as bizarre.

The Hawaiian species of *Drosophila* are long-lived and often very large compared with their relatives on the mainland. The females are more uniform than the males, which are often bizarrely distinctive. The males display complex territorial behavior and elaborate courtship rituals.

The mating behavior patterns among Hawaiian species of *Drosophila* are of great importance in maintaining the distinctiveness of the individual species. For example, despite the great differences between them, *D. heteroneura* and *D. silvestris* are very closely related. Hybrids between them are fully fertile. The two species occur together over a wide area on the island of Hawaii, yet hybridization has been observed at only one locality. The very different and complex behavioral characteristics of these flies obviously play a major role in maintaining their distinctiveness.

Other Prezygotic Isolating Mechanisms

Temporal Isolation. *Lactuca graminifolia* and *L. canadensis*, two species of wild lettuce, grow together along roadsides throughout the southeastern United States. Hybrids between these two species are easily made experimentally and are completely fertile. But such hybrids are rare in nature because *L. graminifolia* flowers in early spring and *L. canadensis* flowers in summer. When their blooming periods overlap, as they do occasionally, the two species do form hybrids, which may become locally abundant.

Many species of closely related amphibians have different breeding seasons that prevent hybridization between the species. For example, five species of frogs of the genus *Rana* occur together in most of the eastern United States, but hybrids are rare because the peak breeding time is different for each of them.



FIGURE 22.5

Differences in courtship rituals can isolate related bird species.

These Galápagos blue-footed boobies select their mates only after an elaborate courtship display. This male is lifting his feet in a ritualized high-step that shows off his bright blue feet. The display behavior of other species of boobies, some of which also occur in the Galápagos, is much different.

Mechanical Isolation. Structural differences prevent mating between some related species of animals. Aside from such obvious features as size, the structure of the male and female copulatory organs may be incompatible. In many insect and other arthropod groups, the sexual organs, particularly those of the male, are so diverse that they are used as a primary basis for classification.

Similarly, flowers of related species of plants often differ significantly in their proportions and structures. Some of these differences limit the transfer of pollen from one plant species to another. For example, bees may pick up the pollen of one species on a certain place on their bodies; if this area does not come into contact with the receptive structures of the flowers of another plant species, the pollen is not transferred.

Prevention of Gamete Fusion. In animals that shed their gametes directly into water, eggs and sperm derived from different species may not attract one another. Many land animals may not hybridize successfully because the sperm of one species may function so poorly within the reproductive tract of another that fertilization never takes place. In plants, the growth of pollen tubes may be impeded in hybrids between different species. In both plants and animals the operation of such isolating mechanisms prevents the union of gametes even following successful mating.

Prezygotic isolating mechanisms lead to reproductive isolation by preventing the formation of hybrid zygotes.

Postzygotic Isolating Mechanisms

All of the factors we have discussed up to this point tend to prevent hybridization. If hybrid matings do occur and zygotes are produced, many factors may still prevent those zygotes from developing into normally functioning, fertile individuals. Development in any species is a complex process. In hybrids, the genetic complements of two species may be so different that they cannot function together normally in embryonic development. For example, hybridization between sheep and goats usually produces embryos that die in the earliest developmental stages.

Leopard frogs (*Rana pipiens* complex) of the eastern United States are a group of similar species, assumed for a long time to constitute a single species (figure 22.6). However, careful examination revealed that although the frogs appear similar, successful mating between them is rare because of problems that occur as the fertilized eggs develop. Many of the hybrid combinations cannot be produced even in the laboratory.

Examples of this kind, in which similar species have been recognized only as a result of hybridization experi-

ments, are common in plants. Sometimes the hybrid embryos can be removed at an early stage and grown in an artificial medium. When these hybrids are supplied with extra nutrients or other supplements that compensate for their weakness or inviability, they may complete their development normally.

Even if hybrids survive the embryo stage, however, they may not develop normally. If the hybrids are weaker than their parents, they will almost certainly be eliminated in nature. Even if they are vigorous and strong, as in the case of the mule, a hybrid between a horse and a donkey, they may still be sterile and thus incapable of contributing to succeeding generations. Sterility may result in hybrids because the development of sex organs may be abnormal, the chromosomes derived from the respective parents may not pair properly, or from a variety of other causes.

Postzygotic isolating mechanisms are those in which hybrid zygotes fail to develop or develop abnormally, or in which hybrids cannot become established in nature.



FIGURE 22.6

Postzygotic isolation in leopard frogs. Numbers indicate the following species in the geographical ranges shown: (1) *Rana pipiens*; (2) *Rana blairi*; (3) *Rana utricularia*; (4) *Rana berlandieri*. These four species resemble one another closely in their external features. Their status as separate species was first suspected when hybrids between them produced defective embryos in some combinations. Subsequent research revealed that the mating calls of the four species differ substantially, indicating that the species have both pre- and postzygotic isolating mechanisms.

22.3 We have learned a great deal about how species form.

One of the oldest questions in the field of evolution is: how does one ancestral species become divided into two descendant species? If species are defined by the existence of reproductive isolation, then the process of speciation equates with the evolution of reproductive isolating mechanisms. How do reproductive isolating mechanisms evolve?

Reproductive Isolation May Evolve as a By-Product of Evolutionary Change

Most reproductive isolating mechanisms initially arise for some reason other than to provide reproductive isolation. For example, a population that colonizes a new habitat may evolve adaptations for living in that habitat. As a result, individuals from that population might never encounter individuals from the ancestral population. Even if they do meet, the population in the new habitat may have evolved new phenotypes or behavior so that members of the two populations no longer recognize each other as potential mates (figure 22.7). For this reason, some biologists believe that the term “isolating mechanisms” is misguided, because it implies that the traits evolved specifically for the purpose of genetically isolating a species, which in most cases is probably incorrect.

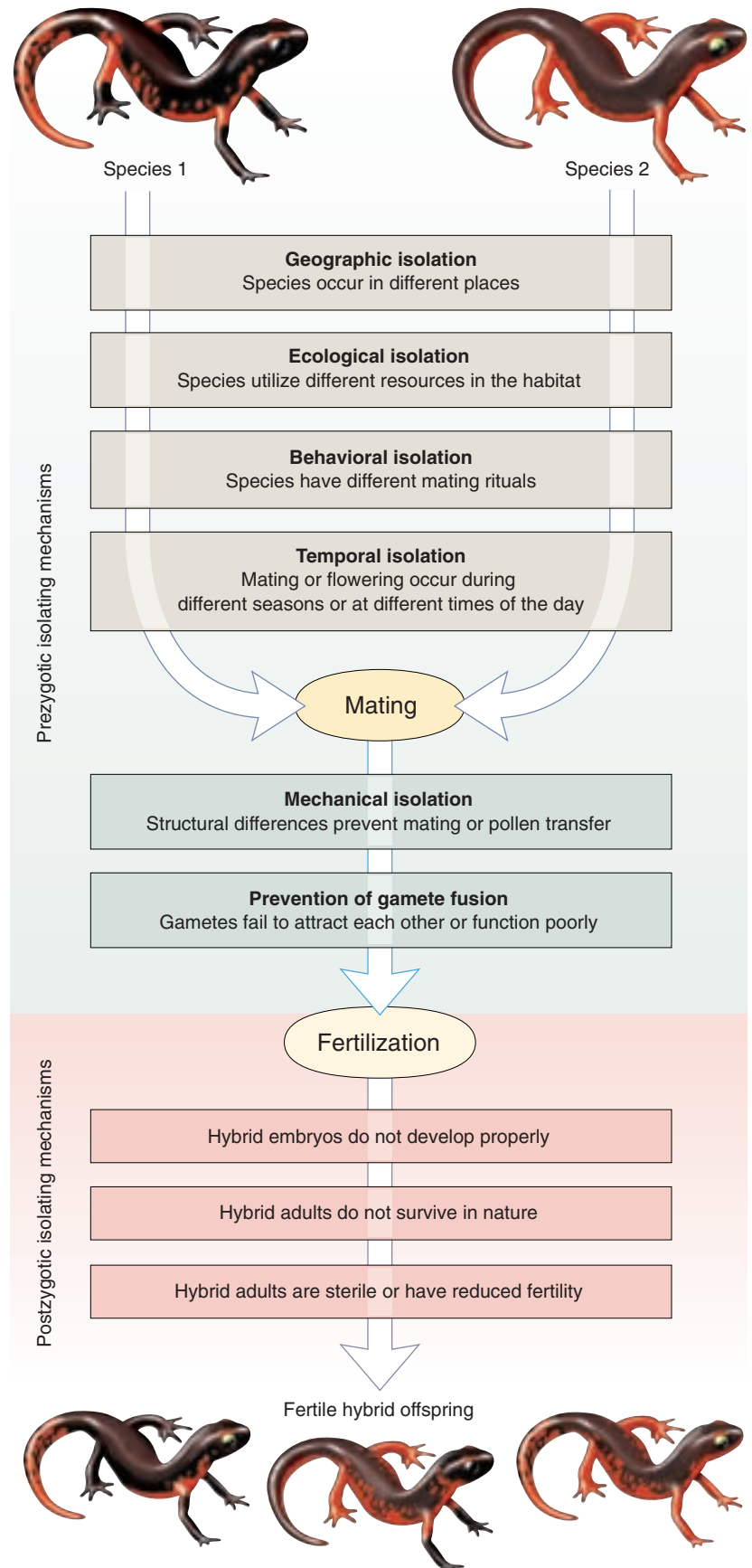


FIGURE 22.7
Reproductive isolating mechanisms. A variety of different mechanisms can prevent successful reproduction between individuals of different species.

Selection May Reinforce Isolating Mechanisms

The formation of species is a continuous process, one that we can understand because of the existence of intermediate stages at all levels of differentiation. If populations that are partly differentiated come into contact with one another, they may still be able to interbreed freely, and the differences between them may disappear over the course of time as genetic exchange homogenizes the populations. Conversely, if the populations are reproductively isolated, then no genetic exchange will occur and the two populations will be different species.

However, there is an intermediate situation in which reproductive isolation has partially evolved, but is not complete. As a result, hybridization will occur at least occasionally. If they are partly sterile, or not as well adapted to the existing habitats as their parents, these hybrids will be at a disadvantage. As a result, selection would favor any alleles in the parental populations that prevented hybridization because individuals that avoided hybridizing would be more successful in passing their genes on to the next generation. The result would be the continual improvement of prezygotic isolating mechanisms until the two populations were completely reproductively isolated. This process is termed **reinforcement** because initially incomplete isolating mechanisms are reinforced by natural selection until they are completely effective.

Reinforcement is by no means inevitable, however. When incompletely isolated populations come together, gene flow immediately begins to occur between the species. Although hybrids may be inferior, they are not, in this case, completely inviable or infertile (if they were, then the species would be reproductively isolated); hence, when these hybrids reproduce with members of either population, they will serve as a conduit of genetic exchange from one population to the other. As a result, the two populations will tend to lose their genetic distinctiveness. Thus, a race ensues: can reproductive isolation be perfected before gene flow destroys the differences between the populations? Experts disagree on the likely outcome, but many believe that reinforcement is the much less common outcome.

The Role of Natural Selection in Speciation

What role does natural selection play in the speciation process? Certainly, the process of reinforcement is driven by natural selection favoring the perfection of reproductive isolation. But, as we have seen, reinforcement may not be common. Is natural selection necessarily involved in the initial evolution of isolating mechanisms?

Random Changes May Cause Reproductive Isolation

As we discussed in chapter 20, populations may diverge for purely random reasons. Genetic drift in small populations, founder effects, and population bottlenecks all may lead to changes in traits that cause reproductive isolation. For example, in the Hawaiian Islands, closely related species of *Drosophila* often differ greatly in their courtship behavior. Colonization of new islands by these fruit flies probably involves a founder effect, in which one or a few fruit flies—perhaps only a single pregnant female—is blown by strong winds to a new island. Changes in courtship behavior between ancestor and descendant populations may be the result of such founder events. Given long enough periods of time, any two isolated populations will diverge due to genetic drift. In some cases, this random divergence may affect traits responsible for reproductive isolation, and speciation will have occurred.

Adaptation and Speciation

Nonetheless, adaptation and speciation are probably related in many cases. As species adapt to different circumstances, they will accumulate many differences that may lead to reproductive isolation. For example, if one population of flies adapts to wet conditions and another to dry conditions, then the populations will evolve a variety of differences in physiological and sensory traits; these differences may promote ecological and behavioral isolation and may cause any hybrids they produce to be poorly adapted to either habitat.

Selection might also act directly on mating behavior. Male *Anolis* lizards, for example, court females by extending a colorful flap of skin, called a “dewlap,” that is located under their throat (figure 22.8). The ability of one lizard to see the dewlap of another lizard depends not only on the color of the dewlap, but the environment in which they occur. As a result, a light-colored dewlap is most effective in reflecting light in a dim forest, whereas dark colors are more apparent in the bright glare of open habitats. As a result, when these lizards occupy new habitats, natural selection will favor evolutionary change in dewlap color because males whose dewlaps cannot be seen will attract few mates. However, the lizards also distinguish members of their own species from those of other species by the color of the dewlap. Hence, adaptive change in mating behavior could have the incidental consequence of causing speciation.

Laboratory scientists have conducted experiments on fruit flies and other organisms in which they isolate popula-



(a) *Anolis carolinensis*.



(b) *Anolis sagrei*.



(c) *Anolis grahami*.



(d) Species name to come.

FIGURE 22.8

Dewlaps of several different species of Caribbean *Anolis* lizards. Males use their dewlaps in both territorial and courtship displays. Coexisting species almost always differ in their dewlaps, which are used in species recognition. Some dewlaps are easier to see in open habitats, whereas others are more visible in shaded environments.

tions in different laboratory chambers and measure how much reproductive isolation evolves. These experiments indicate that genetic drift by itself can lead to some degree of reproductive isolation, but, in general, reproductive isolation evolves more rapidly when the populations are forced to adapt to different laboratory environments (such as temperature or food type).

Reproductive isolating mechanisms can evolve either through random changes or as an incidental by-product of adaptive evolution. Under some circumstances, however, natural selection can directly select for traits that increase the reproductive isolation of a species.

The Geography of Speciation

Speciation is a two-part process. First, initially identical populations must diverge and, second, reproductive isolation must evolve to maintain these differences. The difficulty with this process, as we have seen, is that the homogenizing effect of gene flow between populations will constantly be acting to erase any differences that may arise, either by genetic drift or natural selection. Of course, gene flow only occurs between populations that are in contact. Consequently, evolutionary biologists have long recognized that speciation is much more likely in geographically isolated populations.

Allopatric Divergence Is the Primary Means of Speciation

Ernst Mayr was the first biologist to strongly make the case for allopatric speciation. Marshalling data from a wide variety of organisms and localities, Mayr was clearly able to demonstrate that geographically separated populations appear much more likely to have evolved substantial differences leading to speciation. For example, the Papuan kingfisher, *Tanysiptera hydrocharis*, varies little throughout its wide range in New Guinea despite the

great variation in the island's topography and climate. By contrast, isolated populations on nearby islands are strikingly different from each other and from the mainland population (figure 22.9).

Many other examples indicate that speciation can occur in allopatry. Given that one would expect isolated populations to diverge over time by either drift or selection, this result is not surprising. Rather, the question becomes: Is geographic isolation required for speciation to occur?

Whether Speciation Can Occur in Sympatry Is Controversial

As we saw in chapter 20, disruptive selection can cause a population to contain individuals exhibiting two different phenotypes. One might think that if selection were strong enough, these two phenotypes would evolve into different species. However, before the two phenotypes could become different species, they would have to evolve reproductive isolating mechanisms. Because the two phenotypes would initially not be reproductively isolated at all, genetic exchange between individuals of the two phenotypes would tend to prevent genetic divergence in mating preferences or other isolating mechanisms. As a result, the two pheno-

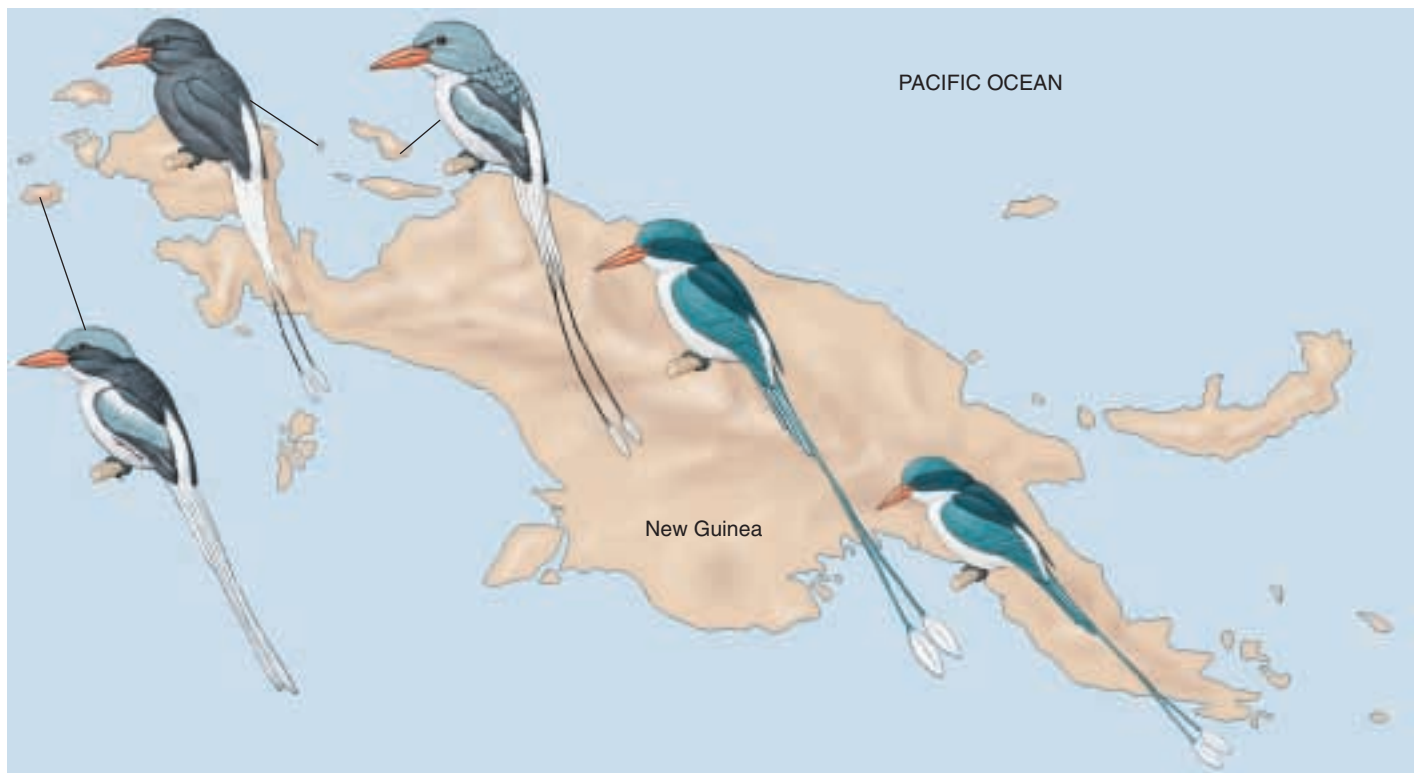


FIGURE 22.9

Phenotypic differentiation in the Papuan kingfisher in New Guinea. Isolated island populations (*above left*) are quite distinctive, showing variation in tail feather structure and length, plumage coloration, and bill size, whereas kingfishers on the mainland (*above right*) show little variation.

FIGURE 22.10

Differences between species can result from a few genes that have major effects. (a) *Mimulus lewisii* has pale pink flowers and concentrated nectar, which are optimal for attracting bumblebees to serve as pollinators. (b) By contrast, *M. cardinalis* has the red flowers and copious dilute nectar typical of hummingbird-pollinated plants. Differences in flower shape and color are the result of a few genes of large effect.



(a) *Mimulus lewisii*



(b) *Mimulus cardinalis*

types would be retained as polymorphisms within a single population. For this reason, most biologists consider sympatric speciation a rare event.

Nonetheless, in recent years, a number of cases have appeared that appear difficult to interpret in any way other than sympatric speciation. For example, the volcanic crater lake Barombi Mbo in Cameroon is extremely small and ecologically homogeneous, with no opportunity for within-lake isolation. Nonetheless, 11 species of closely related cichlid fish occur in the lake; all of the species are more closely related evolutionarily to each other than to any species outside of the crater. The most reasonable explanation is that an ancestral species colonized the crater and subsequently speciated in sympatry multiple times.

Genetic Changes Underlying Speciation

How much divergence does it take to create a new species? How many gene changes does it take? Since Darwin, the traditional view has been that new species arise by the accumulation of many small genetic differences. While there is little doubt that many species have formed in this gradual way, new techniques of molecular biology suggest that in at least some cases, the evolution of a new species may involve very few genes. Studying two species of monkeyflower found in the western United States, researchers found that only a few genes separate the two species, even though at first glance the two species appear to be very different (figure 22.10). Using gene technologies like those described in chapter 19, the researchers found that all of the major differences in the flowers, including not only flower shape and color, but also nectar production, were attributable to several genes, each of which had great phenotypic effects. Because individual genes have such powerful effects, species as different as these two can evolve in relatively few steps.

The Role of Polyploidy in Species Formation

Among plants, fertile individuals often arise from sterile ones through **polyploidy**, which doubles the chromosome number of the original sterile hybrid individual. A polyploid cell, tissue, or individual has more than two sets of chromosomes. Polyploid cells and tissues occur spontaneously and reasonably often in all organisms, although in many they are soon eliminated. A hybrid may be sterile simply because its sets of chromosomes, derived from male and female parents of different species, do not pair with one another. If the chromosome number of such a hybrid doubles, the hybrid, as a result of the doubling, will have a duplicate of each chromosome. In that case, the chromosomes will pair, and the fertility of the polyploid hybrid individual may be restored. It is estimated that about half of the approximately 260,000 species of plants have a polyploid episode in their history, including many of great commercial importance, such as bread wheat, cotton, tobacco, sugarcane, bananas, and potatoes. As you might imagine, the advantages a polyploid plant offers for natural selection can be substantial as a result of their great levels of genetic variation; hence, the significance of polyploidy in the evolution of plants.

Because polyploid plants cannot reproduce with their ancestors, reproductive isolation can evolve in one step. Consequently, speciation by polyploidy is one uncontroversial means of sympatric speciation. Although much rarer than in plants, speciation by polyploidy is also known from a variety of animals, including insects, fish, and salamanders.

Speciation occurs much more readily in the absence of gene flow among populations. However, speciation can occur in sympatry by means of polyploidy, and perhaps in other cases also.

22.4 Clusters of species reflect rapid evolution.

Darwin's Finches

One of the most visible manifestations of evolution is the existence of groups of closely related species that have recently evolved from a common ancestor by occupying different habitats. This type of **adaptive radiation** occurred among the 13 species of Darwin's finches on the Galápagos Islands. Presumably, the ancestor of Darwin's finches reached these islands before other land birds, and all of the types of habitats where birds occur on the mainland were unoccupied. As the new arrivals moved into these vacant ecological niches and adopted new lifestyles, they were subjected to diverse sets of selective pressures. Under these circumstances, and aided by the geographic isolation afforded by the many islands of the Galápagos archipelago, the ancestral finches rapidly split into a series of diverse populations, some of which evolved into separate species. These species now occupy many different kinds of habitats on the Galápagos Islands (figure 22.11), habitats comparable to those several distinct groups of birds occupy on the mainland. The 13 species comprise four groups:

1. **Ground finches.** There are six species of *Geospiza* ground finches. Most of the ground finches feed on seeds. The size of their bills is related to the size of the seeds they eat. Some of the ground finches feed primarily on cactus flowers and fruits and have a longer, larger, more pointed bill than the others.
2. **Tree finches.** There are five species of insect-eating tree finches. Four species have bills that are suitable for feeding on insects. The woodpecker finch has a chisel-like beak. This unusual bird carries around a twig or a cactus spine, which it uses to probe for insects in deep crevices.
3. **Warbler finch.** This unusual bird plays the same ecological role in the Galápagos woods that warblers play on the mainland, searching continually over the leaves and branches for insects. It has a slender, warbler-like beak.
4. **Vegetarian finch.** The very heavy bill of this bud-eating bird is used to wrench buds from branches.

Darwin's finches, all derived from one similar mainland species, have radiated widely on the Galápagos Islands in the absence of competition.

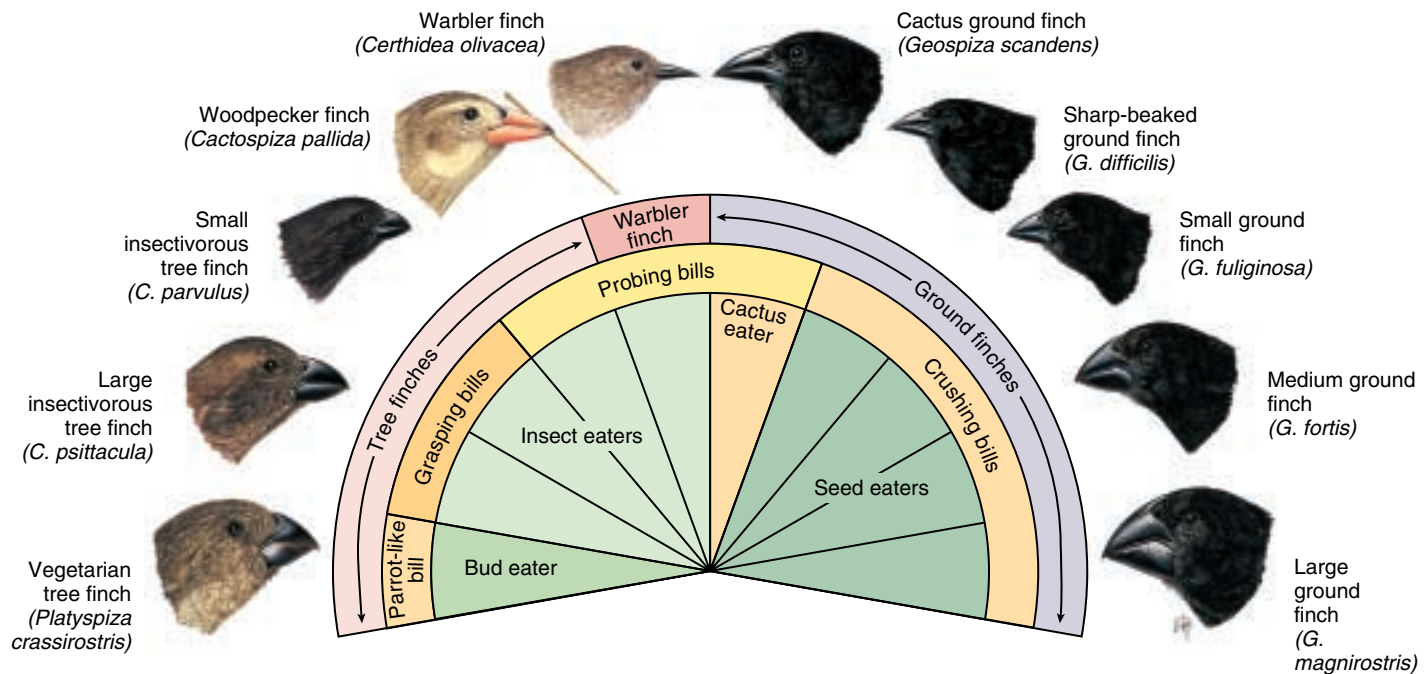


FIGURE 22.11

Darwin's finches. Ten of the 13 Galápagos species of Darwin's finches occur on Isla Santa Cruz, one of the Galápagos Islands. These species show differences in bills and feeding habits. The bills of several of these species resemble those of distinct families of birds on the mainland. This condition presumably arose when the finches evolved new species in habitats lacking small birds. The woodpecker finch uses cactus spines to probe in crevices of bark and rotten wood for food. Scientists believe all of these birds derived from a single common ancestor.

Hawaiian *Drosophila*

Our second example of a cluster of species is the fly genus *Drosophila* on the Hawaiian Islands, which we mentioned earlier as an example of behavioral isolation. There are at least 1250 species of this genus throughout the world, and more than a quarter are found only in the Hawaiian Islands (figure 22.12). New species of *Drosophila* are still being discovered in Hawaii, although the rapid destruction of the native vegetation is making the search more difficult. Aside from their sheer number, Hawaiian *Drosophila* species are unusual because of the morphological and behavioral traits discussed earlier. No comparable species of *Drosophila* are found anywhere else in the world.

A second, closely related genus of flies, *Scaptomyza*, also forms a species cluster in Hawaii, where it is represented by as many as 300 species. A few species of *Scaptomyza* are found outside of Hawaii, but the genus is better represented there than elsewhere. In addition, species intermediate between *Scaptomyza* and *Drosophila* exist in Hawaii, but nowhere else. The genera are so closely related that scientists have suggested that all of the estimated 800 species of these two genera that occur in Hawaii may have derived from a single common ancestor.

The native Hawaiian flies are closely associated with the remarkable native plants of the islands and are often abundant in the native vegetation. Evidently, when their ancestors first reached these islands, they encountered many “empty” habitats that other kinds of insects and other animals occupied elsewhere. The evolutionary opportunities the ancestral *Drosophila* flies found were similar to those the ancestors of Darwin’s finches in the Galápagos Islands encountered, and both groups evolved in a similar way. Many of the Hawaiian *Drosophila* species are highly selective in their choice of host plants for their larvae and in the part of the plant they use. The larvae of various species live in rotting stems, fruits, bark, leaves, or roots, or feed on sap.

New islands have continually arisen from the sea in the region of the Hawaiian Islands. As they have done so, they appear to have been invaded successively by the various *Drosophila* groups present on the older islands. New species have evolved as new islands have been colonized. The Hawaiian species of *Drosophila* have had even greater evolutionary opportunities than Darwin’s finches because of their restricted ecological niches and the variable ages of the islands. They clearly tell one of the most unusual evolutionary stories found anywhere in the world.



(a) *Drosophila mulli*



(b) *Drosophila digressa*

FIGURE 22.12

Hawaiian *Drosophila*. The hundreds of species that have evolved on the Hawaiian Islands are extremely variable in appearance, although genetically almost identical.

The adaptive radiation of about 800 species of the flies *Drosophila* and *Scaptomyza* on the Hawaiian Islands, probably from a single common ancestor, is one of the most remarkable examples of intensive species formation found anywhere on earth.

Lake Victoria Cichlid Fishes

Lake Victoria is an immense shallow freshwater sea about the size of Switzerland in the heart of equatorial East Africa, until recently home to an incredibly diverse collection of over 300 species of cichlid fishes.

Recent Radiation

This cluster of species appears to have evolved recently and quite rapidly. By sequencing the cytochrome *b* gene in many of the lake's fish, scientists have been able to estimate that the first cichlids entered Lake Victoria only 200,000 years ago, colonizing from the Nile. Dramatic changes in water level encouraged species formation. As the lake rose, it flooded new areas and opened up new habitat. Many of the species may have originated after the lake dried down 14,000 years ago, isolating local populations in small lakes until the water level rose again.

Cichlid Diversity

These small, perchlike fishes range from 2 to 10 inches in length, and the males come in endless varieties of colors. The most diverse assembly of vertebrates known to science, the Lake Victoria cichlids defy simple description. We can gain some sense of the vast range of types by looking at how different species eat. There are mud biters, algae scrapers, leaf chewers, snail crushers, snail shellers (who pounce on slow-crawling snails and spear

their soft parts with long curved teeth before the snail can retreat into its shell), zooplankton eaters, insect eaters, prawn eaters, and fish eaters. Scale-scraping cichlids rasp slices of scales off of other fish. There are even cichlid species that are “pedophages,” eating the young of other cichlids.

Cichlid fish have a remarkable trait that may have been instrumental in this evolutionary radiation: a second set of functioning jaws occurs in the throats of cichlid fish (figure 22.13)! The ability of these jaws to manipulate and process food has freed the oral jaws to evolve for other purposes, and the result has been the incredible diversity of ecological roles filled by these fish.

Abrupt Extinction

Much of this diversity is gone. In the 1950s, the Nile perch, a commercial fish with a voracious appetite, was introduced on the Ugandan shore of Lake Victoria. Since then it has spread through the lake, eating its way through the cichlids. By 1990 all the open-water cichlid species were extinct, as well as many living in rocky shallow regions. Over 70% of all the named Lake Victoria cichlid species had disappeared, as well as untold numbers of species that had yet to be described.

Very rapid speciation occurred among cichlid fishes isolated in Lake Victoria, but widespread extinction followed when the isolation ended.

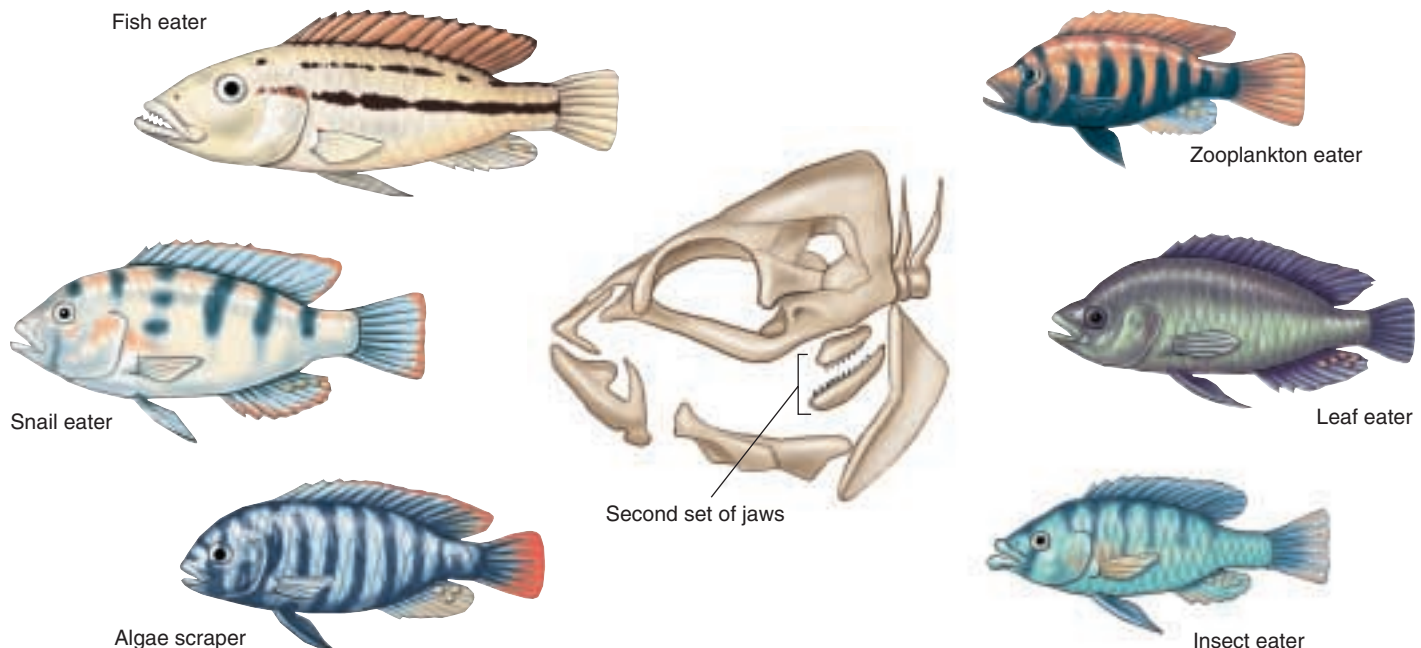


FIGURE 22.13

Cichlid fishes of Lake Victoria. These fishes have evolved adaptations to use a variety of different habitats. The second set of jaws located in the throat of these fish has provided evolutionary flexibility, allowing oral jaws to be modified in many ways.

New Zealand Alpine Buttercups

Adaptive radiations as we have described in Galápagos finches, Hawaiian *Drosophila*, and cichlid fishes seem to be favored by *periodic isolation*. Finches and *Drosophila* invade new islands, local species evolve, and they in turn reinvade the home island, in a cycle of expanding diversity. Similarly, cichlids become isolated by falling water levels, evolving separate species in isolated populations that later are merged when the lake's water level rises again.

A clear example of the role periodic isolation plays in species formation can be seen in the alpine buttercups (genus *Ranunculus*) which grow among the glaciers of New Zealand (figure 22.14). More species of alpine buttercup grow on the two islands of New Zealand than in all of North and South America combined. Detailed studies by the Canadian taxonomist Fulton Fisher revealed that the evolutionary mechanism responsible for inducing this diversity is recurrent isolation associated with the recession of glaciers. The 14 species of alpine *Ranunculus* occupy five distinctive habitats within glacial areas: *snowfields* (rocky crevices among outcrops in permanent snowfields at 7000 to 9000 feet elevation); *snowline fringe* (rocks at lower margin of snowfields between 4000 and 7000 ft); *stony debris* (scree slopes of exposed loose rocks at 2000 to 6000 ft); *sheltered situations* (shaded by rock or shrubs at 1000 to 6000 ft); and *boggy habitats* (sheltered slopes and hollows, poorly drained tussocks at elevations between 2500 and 5000 ft).

Ranunculus speciation and diversification has been promoted by repeated cycles of glacial advance and retreat. As the glaciers retreat, populations become isolated on mountain peaks, permitting speciation (figure 22.15). In the next advance, these new species can expand throughout the mountain range, coming into contact with their close relatives. In this way, one initial species could give rise to many



FIGURE 22.14

A New Zealand alpine buttercup. Fourteen species of alpine *Ranunculus* grow among the glaciers and mountains of New Zealand, including this *R. lyallii*, the giant buttercup.

descendants. Moreover, on isolated mountaintops during glacial retreats, species have convergently evolved to occupy similar habitats; these distantly related but ecologically similar species have then been brought back into contact in subsequent glacial advances.

Recurrent isolation promotes species formation.

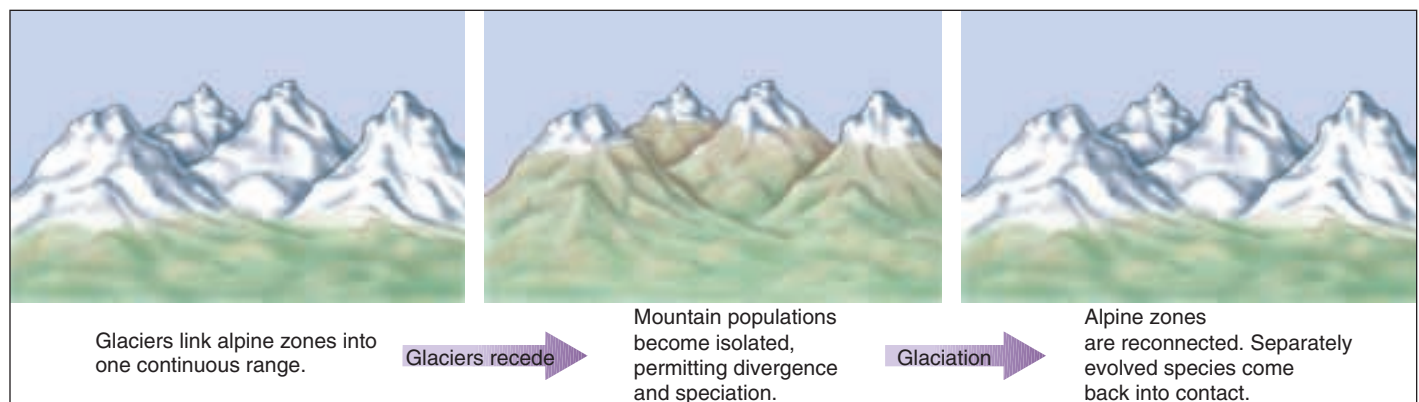


FIGURE 22.15

Periodic glaciation encouraged species formation among alpine buttercups in New Zealand. The formation of extensive glaciers during the Pleistocene linked the alpine zones of many mountains together. When the glaciers receded, these alpine zones were isolated from one another, only to become reconnected with the advent of the next glacial period. During periods of isolation, populations of alpine buttercups diverged in the isolated habitats.

Diversity of Life through Time

Although eukaryotes evolved nearly 3 billion years ago, the diversity of life didn't increase substantially until approximately 550 million years ago. Then, almost all of the extant types of animals evolved in a geologically short period termed the "Cambrian explosion." In addition to organisms whose descendants are recognizable today, a wide variety of

other types of organisms also evolved (figure 22.16). The biology of these creatures, which quickly disappeared without leaving any descendants, is poorly understood. The Cambrian explosion seems to have been a time of evolutionary experimentation and innovation, in which many types of organisms appeared, but most were quickly weeded out. What prompted this explosion of diversity is still a subject of considerable controversy.



FIGURE 22.16

Diversity of animals that evolved during the Cambrian explosion. In addition to the appearance of the ancestors of many present-day groups, such as insects and vertebrates, a variety of bizarre creatures evolved that left no descendants, such as *Wiwaxia*, *Marrella*, *Opabinia*, and the aptly named *Hallucigenia*. The natural history of these species is open to speculation. Key: (1) *Amiskwia*, (2) *Odontogriphus*, (3) *Eldonia*, (4) *Halichondrites*, (5) *Anomalocaris canadensis*, (6) *Pikaia*, (7) *Canadia*, (8) *Marrella splendens*, (9) *Opabinia*, (10) *Ottoia*, (11) *Wiwaxia*, (12) *Yoboia*, (13) *Xiangiangia*, (14) *Aysbeia*, (15) *Sidneyia*, (16) *Dinomischus*, (17) *Hallucigenia*.

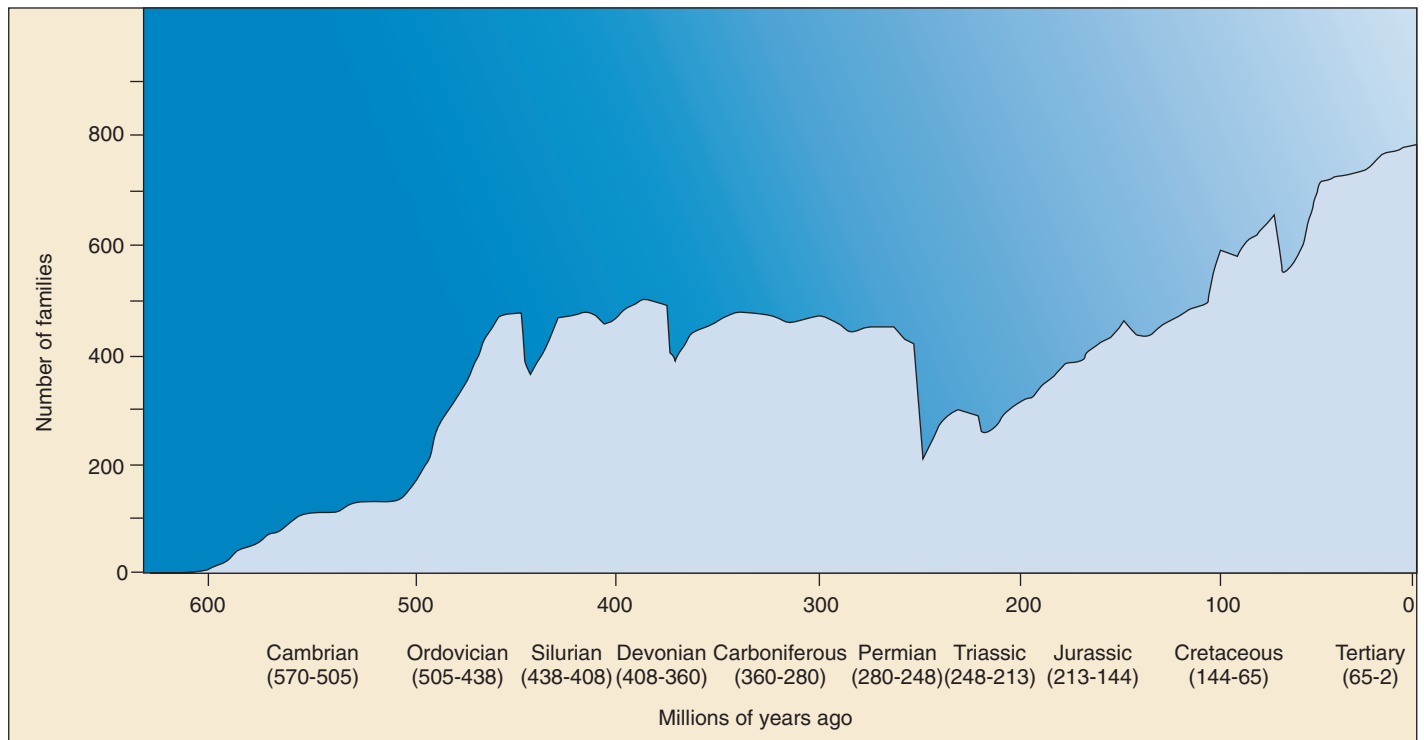


FIGURE 22.17

Diversity through time. Taxonomic diversity of families of marine animals since the Cambrian Period. The fossil record is most complete for marine organisms because they are more readily fossilized than terrestrial species. Families are shown, rather than species, because many species are known from only one specimen, thus introducing error into estimates of time of extinction.

Trends in Species Diversity

The number of species in the world has increased vastly since the Cambrian. However, the trend has been far from consistent (figure 22.17). After a rapid rise, the number of species reached a plateau for about 200 million years ago; since then, the number has risen steadily.

Interspersed in these patterns, however, have been a number of major setbacks, termed **mass extinctions**, in which the number of species has greatly decreased. Five major mass extinctions have been identified, the most severe of which occurred at the end of the Permian Period, approximately 225 million years ago, at which time more than half of all families and as many as 96% of all species may have perished.

The most famous and well-studied extinction, though not as drastic, occurred at the end of the Cretaceous Period (63 million years ago), at which time the dinosaurs and a variety of other organisms went extinct. Recent studies have provided support for the hypothesis that this extinction event was triggered by a large asteroid which slammed into the earth, perhaps causing global forest fires and obscuring the sun for months by throwing particles into the air. This mass extinction did have one positive effect, though: with the disappearance of dinosaurs, mammals,

which previously had been small and inconspicuous, quickly experienced a vast evolutionary radiation, which ultimately produced a wide variety of organisms, including elephants, tigers, whales, and humans. Indeed, a general observation is that biological diversity tends to rebound quickly after mass extinctions, reaching comparable levels of species richness, even if the organisms making up that diversity are not the same.

A Sixth Extinction

The number of species in the world in recent times is greater than it has ever been. Unfortunately, that number is decreasing at an alarming rate due to human activities. Some estimate that as many as one-fourth of all species will become extinct in the next 50 years, a rate of extinction not seen on earth since the Cretaceous mass extinction.

The number of species has increased through time, although not at constant rates. Several major extinction events have substantially, though briefly, reduced the number of species.

The Pace of Evolution

Different kinds of organisms evolve at different rates. Mammals, for example, evolve relatively slowly. On the basis of a relatively complete fossil record, it has been estimated that an average value for the duration of a “typical” mammal species, from formation of the species to its extinction, might be about 200,000 years. American paleontologist George Gaylord Simpson has pointed out that certain groups of animals, such as lungfishes, are apparently evolving even more slowly than mammals. In fact, Simpson estimated that there has been little evolutionary change among lungfishes over the past 150 million years, and even slower rates of evolution occur in other groups.

Evolution in spurts?

Not only does the rate of evolution differ greatly from group to group, but evolution within a group apparently proceeds rapidly during some periods and relatively slowly during others. The fossil record provides evidence for such variability in evolutionary rates, and evolutionists are very interested in understanding the factors that account for it. In 1972, paleontologists Niles Eldredge of the American Museum of Natural History in New York and Stephen Jay Gould of Harvard University proposed that evolution normally proceeds in spurts. They claimed that the evolutionary process is a series of **punctuated equilibria**. Evolutionary innovations would occur and give rise to new lines; then these lines might persist unchanged for a long time, in “equilibrium.” Eventually there would be a new spurt of evolution, creating a “punctuation” in the fossil record. Eldredge and Gould contrast their theory of punctuated equilibrium with that of **gradualism**, or gradual evolutionary change, which they claimed was what Darwin and most earlier students of evolution had considered normal (figure 22.18).

Eldredge and Gould proposed that **stasis**, or lack of evolutionary change, would be expected in large populations experiencing stabilizing selection over long periods of time. In contrast, rapid evolution of new species might occur if populations colonized new areas. Such populations would be small, isolated, and possibly already differing from their parental population as a result of the founder effect. This, combined with selective pressures

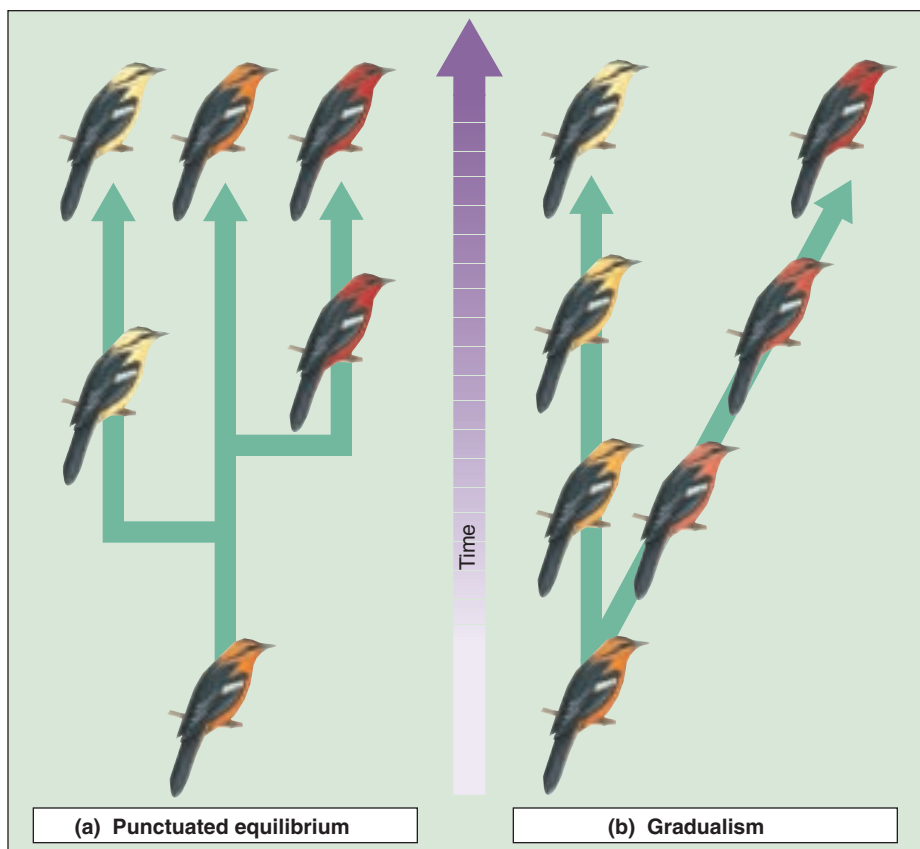


FIGURE 22.18

Two views of the pace of macroevolution. (a) Punctuated equilibrium surmises that species formation occurs in bursts, separated by long periods of quiet, while (b) gradualism surmises that species formation is constantly occurring.

from a new environment, could bring about rapid change.

Unfortunately, the distinctions are not as clear-cut as implied by this discussion. Some well-documented groups such as African mammals clearly have evolved gradually, and not in spurts. Other groups, like marine bryozoa, seem to show the irregular pattern of evolutionary change the punctuated equilibrium model predicts. It appears, in fact, that gradualism and punctuated equilibrium are two ends of a continuum. Although some groups appear to have evolved solely in a gradual manner and others only in a punctuated mode, many other groups appear to show evidence of both gradual and punctuated episodes at different times in their evolutionary history.

The punctuated equilibrium model assumes that evolution occurs in spurts, between which there are long periods in which there is little evolutionary change. The gradualism model assumes that evolution proceeds gradually, with successive change in a given evolutionary line.

Problems with the Biological Species Concept

Since the biological species concept was first proposed by Ernst Mayr in the 1940s, it has been the predominant idea of how to recognize and define species. However, in recent years, workers from a variety of fields have begun to question how universally applicable the concept really is.

The Extent of Hybridization

The crux of the matter concerns hybridization. Biological species are reproductively isolated, so that hybridization should be rare. If hybridization is common, one would expect one of two quick outcomes: either reinforcement would occur, leading to the perfection of isolating mechanisms and an end to hybridization, or the two populations would merge together into a single homogeneous gene pool.

However, in recent years biologists have detected much greater amounts of hybridization than previously realized between populations that seem to neither be experiencing reinforcement nor losing their specific identities. Botanists have always been aware that species can often experience substantial amounts of hybridization. One study found that more than 50% of the plant species surveyed in California were not well defined by genetic isolation. For example, the fossil record indicates that balsam poplars and cottonwoods have been phenotypically distinct for 12 million years, but throughout this time, they have routinely produced hybrids. Consequently, for many years, many botanists have felt that the biological species concept only applies to animals.

What is becoming increasingly evident, however, is that hybridization is not all that uncommon in animals, either. One recent survey indicated that almost 10% of the world's 9500 bird species are known to have hybridized in nature. Recent years have seen the documentation of more and more cases in which substantial hybridization occurs between animal species. Again, the Galápagos finches provide a particularly well-studied example. Three species on the island of Daphne Major—the medium ground finch, the cactus finch, and the small ground finch—are clearly distinct morphologically and occupy different ecological niches. Careful studies over the past 20 years by Peter and Rosemary Grant found that, on average, 2% of the medium ground finches and 1% of the cactus finches mated with other species every year. Furthermore, hybrid offspring appeared to be at no

disadvantage either in terms of survival or subsequent reproduction. This is not a trivial amount of genetic exchange, and one might expect to see the species coalescing into one variable population, but the species are nonetheless maintaining their distinctiveness.

Alternatives to the Biological Species Concept

This is not to say hybridization is rampant throughout the animal world. As the bird survey indicated, 90% of bird species are not known to hybridize, and even fewer probably experience significant amounts of hybridization. Still, it is a common enough occurrence to cast doubt about whether reproductive isolation is the only force maintaining the integrity of species.

An alternative hypothesis is that the distinctions among species are maintained by natural selection. The idea is that each species has adapted to its own specific part of the environment. Stabilizing selection then maintains the species' adaptations; hybridization has little effect because alleles introduced into the gene pool from other species quickly would be eliminated by natural selection.

We have already seen in chapter 20 that the interaction between gene flow and natural selection can have many outcomes. In some cases, strong selection can overwhelm any effects of gene flow, but in other situations, gene flow can prevent populations from eliminating less successful alleles from a population. As a general explanation, then, natural selection is not likely to have any fewer exceptions than the biological species concept, although it may prove more successful for certain types of organisms or habitats.

A variety of other ideas have been put forward to establish criteria for defining species. Many of these are specific to a particular type of organism and none has universal applicability. In truth, it may be that there is no single explanation for what maintains the identity of species. Given the incredible variation evident in plants, animals, and microorganisms in all aspects of their biology, it is perhaps not surprising that different processes are operating in different organisms. This is an area of active research that demonstrates the dynamic nature of the field of evolutionary biology.

Hybridization has always been recognized to be widespread among plants, but recent research reveals that it is surprisingly high in animals, too. Because of the diversity of living organisms, no single definition of what constitutes a species may be universally applicable.



Summary

Questions

Media Resources

22.1 Species are the basic units of evolution.

- Species are groups of organisms that differ from one another in one or more characteristics and do not hybridize freely when they come into contact in their natural environment. Many species cannot hybridize with one another at all.
- Species exhibit geographic variation, yet phenotypically distinctive populations are connected by intermediate forms.

- Define the term *sympatry*. Why is sympatric speciation thought by many to be unlikely?
- What is the biological species concept?

22.2 Species maintain their genetic distinctiveness through barriers to reproduction.

- Among the factors that separate populations and species are geographical, ecological, temporal, behavioral, and mechanical isolation, as well as factors that inhibit the fusion of gametes or the normal development of the hybrid organisms.
- Some isolating mechanisms (prezygotic) prevent hybrid formation; others (postzygotic) prevent hybrids from surviving and reproducing.

- What is the difference between prezygotic and postzygotic isolating mechanisms?
- What barriers exist to hybrid formation and success? Which are prezygotic and which are postzygotic isolating mechanisms? Why do some people think the term “isolating mechanism” is misleading?



- Activity: Allopatric Speciation
- Introduction to Speciation
- Sympatric Speciation
- Allopatric Speciation
- Constructing Phylogenies

22.3 We have learned a great deal about how species form.

- Reproductive isolation can arise as populations differentiate by adaptation to different environments, as well as by random genetic drift, founder effects, or population bottlenecks.
- Natural selection may favor changes in the mating system when a species occupies a new habitat, so that the species becomes reproductively isolated from other species.
- When two species are not completely reproductively isolated, natural selection may favor the evolution of more effective isolating mechanisms to prevent hybridization, a process termed “reinforcement.”

- How does selection relate to population divergence?
- How many genes are involved in the speciation process?
- When are hybrids at a disadvantage? What can be the result of this disadvantage?
- Define the term *polyploidy*.



- Student Research: Evolution in Ferns

22.4 Clusters of species reflect rapid evolution.

- Clusters of species arise when populations differentiate to fill several niches. On islands, differentiation is often rapid because of numerous open habitats.
- The pace of evolution is not constant among all organisms. Some scientists believe it occurs in spurts, others argue that it proceeds gradually.
- Hybridization occurs commonly among plants and even among animals. The biological species concept may not apply to all organisms.

- What is adaptive radiation? What types of habitats encourage it? Why?
- What is the difference between gradualism and punctuated equilibrium?
- Why is the biological species concept no longer considered to be universally applicable?



- Evolutionary Trends
- Book Reviews
 - Darwin's Dreampond* by Goldschmidt
 - The Beak of the Finch* by Weiner

23

How Humans Evolved

Concept Outline

23.1 The evolutionary path to humans starts with the advent of primates.

The Evolutionary Path to Apes. Primates first evolved 65 million years ago, giving rise first to prosimians and then to monkeys.

How the Apes Evolved. Apes, including our closest relatives, the chimpanzees, arose from an ancestor common to Old World monkeys.

23.2 The first hominids to evolve were australopithecines.

An Evolutionary Tree with Many Branches. The first hominids were australopithecines, of which there were several different kinds.

The Beginning of Hominid Evolution. The ability to walk upright on two legs marks the beginning of hominid evolution. One can draw the hominid family tree in two very different ways, either lumping variants together or splitting them into separate species.

23.3 The genus *Homo* evolved in Africa.

African Origin: Early *Homo*. There may have been several species of early *Homo*, with brains significantly larger than those of australopithecines.

Out of Africa: *Homo erectus*. The first hominid species to leave Africa was the relatively large-brained *H. erectus*, the longest lived species of *Homo*.

23.4 Modern humans evolved quite recently.

The Last Stage of Hominid Evolution. Modern humans evolved within the last 600,000 years, our own species within the last 200,000 years.

Our Own Species: *Homo sapiens*. Our species appears to have evolved in Africa, and then migrated to Europe and Asia.

Human Races. Our species is unique in evolving culturally. Differences in populations in skin color reflect adaptation to different environments, rather than genetic differentiation among populations.



FIGURE 23.1

The trail of our ancestors. These fossil footprints, made in Africa 3.7 million years ago, look as if they might have been left by a mother and child walking on the beach. But these tracks, preserved in volcanic ash, are not human. They record the passage of two individuals of the genus *Australopithecus*, the group from which our genus, *Homo*, evolved.

In 1871 Charles Darwin published another groundbreaking book, *The Descent of Man*. In this book, he suggested that humans evolved from the same African ape ancestors that gave rise to the gorilla and the chimpanzee. Although little fossil evidence existed at that time to support Darwin's case, numerous fossil discoveries made since then strongly support his hypothesis (figure 23.1). Human evolution is the part of the evolution story that often interests people most, and it is also the part about which we know the most. In this chapter we follow the evolutionary journey that has led to humans, telling the story chronologically. It is an exciting story, replete with controversy.

23.1 The evolutionary path to humans starts with the advent of primates.

The Evolutionary Path to Apes

The story of human evolution begins around 65 million years ago, with the explosive radiation of a group of small, arboreal mammals called the Archonta. These primarily insectivorous mammals had large eyes and were most likely nocturnal (active at night). Their radiation gave rise to different types of mammals, including bats, tree shrews, and primates, the order of mammals that contains humans.

The Earliest Primates

Primates are mammals with two distinct features that allowed them to succeed in the arboreal, insect-eating environment:

1. **Grasping fingers and toes.** Unlike the clawed feet of tree shrews and squirrels, primates have grasping hands and feet that let them grip limbs, hang from branches, seize food, and, in some primates, use tools. The first digit in many primates is opposable and at least some, if not all, of the digits have nails.
2. **Binocular vision.** Unlike the eyes of shrews and squirrels, which sit on each side of the head so that the two fields of vision do not overlap, the eyes of primates are shifted forward to the front of the face. This produces overlapping binocular vision that lets the brain judge distance precisely—important to an animal moving through the trees.

Other mammals have binocular vision, but only primates have both binocular vision and grasping hands, making them particularly well adapted to their environment. While early primates were mostly insectivorous, their dentition began to change from the shearing, triangular-shaped molars specialized for insect eating to the more flattened, square-shaped molars and rodentlike incisors specialized for plant eating. Primates that evolved later also show a continuous reduction in snout length and number of teeth.

The Evolution of Prosimians

About 40 million years ago, the earliest primates split into two groups: the prosimians and the anthropoids. The prosimians (“before monkeys”) looked something like a cross between a squirrel and a cat and were common in North America, Europe, Asia, and Africa. Only a few prosimians survive today, lemurs, lorises and tarsiers (figure 23.2). In addition to having grasping digits and binocular vision, prosimians have large eyes with increased visual acuity. Most prosimians are nocturnal, feeding on fruits, leaves, and flowers, and many lemurs have long tails for balancing.



FIGURE 23.2

A prosimian. This tarsier, a prosimian native to tropical Asia, shows the characteristic features of primates: grasping fingers and toes and binocular vision.

Origin of the Anthropoids

The anthropoids, or higher primates, include monkeys, apes, and humans (figure 23.3). Anthropoids are almost all diurnal—that is, active during the day—feeding mainly on fruits and leaves. Evolution favored many changes in eye design, including color vision, that were adaptations to daytime foraging. An expanded brain governs the improved senses, with the braincase forming a larger portion of the head. Anthropoids, like the relatively few diurnal prosimians, live in groups with complex social interactions. In addition, the anthropoids tend to care for their young for prolonged periods, allowing for a long childhood of learning and brain development.

The early anthropoids, now extinct, are thought to have evolved in Africa. Their direct descendants are a very successful group of primates, the monkeys.

New World Monkeys. About 30 million years ago, some anthropoids migrated to South America, where they evolved in isolation. Their descendants, known as the **New World monkeys**, are easy to identify: all are arboreal, they have flat spreading noses, and many of them grasp objects with long prehensile tails (figure 23.4a).

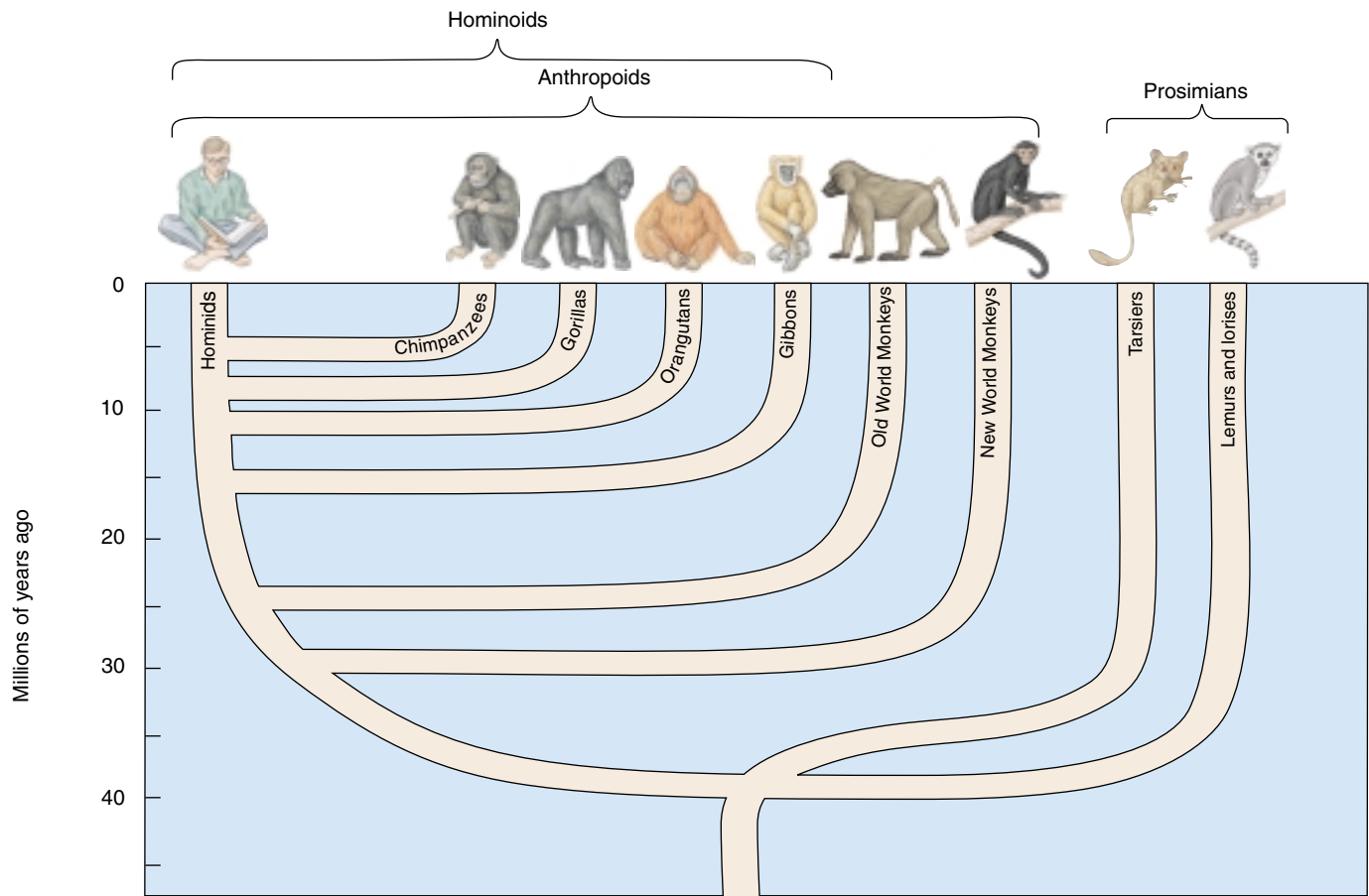


FIGURE 23.3

A primate evolutionary tree. The most ancient of the primates are the prosimians, while the hominids were the most recent to evolve.

Old World Monkeys. Around 25 million years ago, anthropoids that remained in Africa split into two lineages: one gave rise to the **Old World monkeys** and one gave rise to the hominoids (see page 480). Old World monkeys include ground-dwelling as well as arboreal species. None of the Old World monkeys have prehensile tails. Their nostrils are close together, their noses point downward, and some have toughened pads of skin for prolonged sitting (figure 23.4b).

The earliest primates arose from small, tree-dwelling, insect-eaters and gave rise to prosimians and then anthropoids. Early anthropoids gave rise to New World monkeys and Old World monkeys.



(a)



(b)

FIGURE 23.4

New and Old World monkeys. (a) New World monkeys, such as this golden lion tamarin, are arboreal, and many have prehensile tails. (b) Old World monkeys lack prehensile tails, and many are ground dwellers.

How the Apes Evolved

The other African anthropoid lineage is the **hominoids**, which includes the **apes** and the **hominids** (humans and their direct ancestors). The living apes consist of the gibbon (genus *Hylobates*), orangutan (*Pongo*), gorilla (*Gorilla*), and chimpanzee (*Pan*) (figure 23.5). Apes have larger brains than monkeys, and they lack tails. With the exception of the gibbon, which is small, all living apes are larger than any monkey. Apes exhibit the most adaptable behavior of any mammal except human beings. Once widespread in Africa and Asia, apes are rare today, living in relatively small areas. No apes ever occurred in North or South America.

The First Hominoid

Considerable controversy exists about the identity of the first hominoid. During the 1980s it was commonly believed that the common ancestor of apes and hominids was a late Miocene ape living 5 to 10 million years ago. In 1932, a candidate fossil, an 8-million-year-old jaw with teeth, was unearthed in India. It was called *Ramapithecus* (after the Hindi deity Rama). However, these fossils have never been found in Africa, and more complete fossils discovered in 1981 made it clear that *Ramapithecus* is in fact closely related to the orangutan. Attention has now shifted to an earlier Miocene ape, *Proconsul*, which has many of the characteristics of Old World monkeys but lacks a tail and has apelike hands, feet, and pelvis. However, because very few fossils have been recovered from the period 5 to 10 million years ago, it is not yet possible to identify with certainty the first hominoid ancestor.



(a)



(b)



(c)



(d)

FIGURE 23.5

The living apes. (a) Mueller gibbon, *Hylobates muelleri*. (b) Orangutan, *Pongo pygmaeus*. (c) Gorilla, *Gorilla gorilla*. (d) Chimpanzee, *Pan troglodytes*.

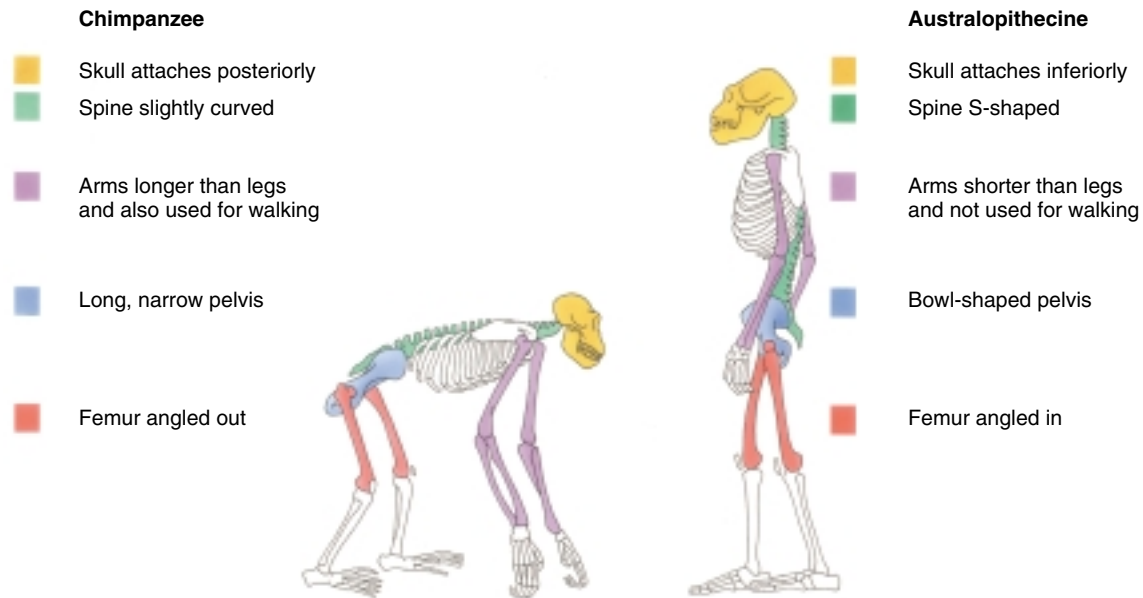


FIGURE 23.6

A comparison of ape and hominid skeletons. Early humans, such as australopithecines, were able to walk upright because their arms were shorter, their spinal cord exited from the bottom of the skull, their pelvis was bowl-shaped and centered the body weight over the legs, and their femurs angled inward, directly below the body, to carry its weight.

Which Ape Is Our Closest Relative?

Studies of ape DNA have explained a great deal about how the living apes evolved. The Asian apes evolved first. The line of apes leading to gibbons diverged from other apes about 15 million years ago, while orangutans split off about 10 million years ago (see figure 23.3). Neither are closely related to humans.

The African apes evolved more recently, between 6 and 10 million years ago. These apes are the closest living relatives to humans; some taxonomists have even advocated placing humans and the African apes in the same zoological family, the Hominidae. Fossils of the earliest hominids, described later in the chapter, suggest that the common ancestor of the hominids was more like a chimpanzee than a gorilla. Based on genetic differences, scientists estimate that gorillas diverged from the line leading to chimpanzees and humans some 8 million years ago.

Sometime after the gorilla lineage diverged, the common ancestor of all hominids split off from chimpanzee line to begin the evolutionary journey leading to humans. Because this split was so recent, the genes of humans and chimpanzees have not had time to evolve many genetic differences. For example, a human hemoglobin molecule differs from its chimpanzee counterpart in only a single amino acid. In general, humans and chimpanzees exhibit a level of genetic similarity normally found between closely related sibling species of the same genus!

Comparing Apes to Hominids

The common ancestor of apes and hominids is thought to have been an arboreal climber. Much of the subsequent evolution of the hominoids reflected different approaches to locomotion. Hominids became **bipedal**, walking upright, while the apes evolved knuckle-walking, supporting their weight on the back sides of their fingers (monkeys, by contrast, use the palms of their hands).

Humans depart from apes in several areas of anatomy related to bipedal locomotion (figure 23.6). Because humans walk on two legs, their vertebral column is more curved than an ape's, and the human spinal cord exits from the bottom rather than the back of the skull. The human pelvis has become broader and more bowl-shaped, with the bones curving forward to center the weight of the body over the legs. The hip, knee, and foot (in which the human big toe no longer splays sideways) have all changed proportions. Being bipedal, humans carry much of the body's weight on the lower limbs, which comprise 32 to 38% of the body's weight and are longer than the upper limbs; human upper limbs do not bear the body's weight and make up only 7 to 9% of human body weight. African apes walk on all fours, with the upper and lower limbs both bearing the body's weight; in gorillas, the longer upper limbs account for 14 to 16% of body weight, the somewhat shorter lower limbs for about 18%.

Hominoids, the apes and hominids, arose from Old World monkeys. Among living apes, chimpanzees seem the most closely related to humans.

23.2 The first hominids to evolve were australopithecines.

An Evolutionary Tree with Many Branches

Five to 10 million years ago, the world's climate began to get cooler, and the great forests of Africa were largely replaced with savannas and open woodland. In response to these changes, a new kind of hominoid was evolving, one that was bipedal. These new hominoids are classified as hominids—that is, of the human line.

There are two major groups of hominids: three to seven species of the genus *Homo* (depending how you count them) and seven species of the older, smaller-brained genus *Australopithecus*. In every case where the fossils allow a determination to be made, the hominids are bipedal, walking upright. Bipedal locomotion is the hallmark of hominid evolution. We will first discuss *Australopithecus*, and then *Homo*.

Discovery of *Australopithecus*

The first hominid was discovered in 1924 by Raymond Dart, an anatomy professor at Johannesburg in South Africa. One day, a mine worker brought him an unusual chunk of rock—actually, a rock-hard mixture of sand and soil. Picking away at it, Professor Dart uncovered a skull unlike that of any ape he had ever seen. Beautifully pre-

served, the skull was of a five-year-old individual, still with its milk teeth. While the skull had many apelike features such as a projecting face and a small brain, it had distinctly human features as well—for example, a rounded jaw unlike the pointed jaw of apes. The ventral position of the foramen magnum (the hole at the base of the skull from which the spinal cord emerges) suggested that the creature had walked upright. Dart concluded it was a human ancestor.

What riveted Dart's attention was that the rock in which the skull was embedded had been collected near other fossils that suggested that the rocks and their fossils were several million years old! At that time, the oldest reported fossils of hominids were less than 500,000 years old, so the antiquity of this skull was unexpected and exciting. Scientists now estimate Dart's skull to be 2.8 million years old. Dart called his find *Australopithecus africanus* (from the Latin *australo*, meaning “southern” and the Greek *pithecus*, meaning “ape”), the ape from the south of Africa.

Today, fossils are dated by the relatively new process of single-crystal laser-fusion dating. A laser beam melts a single potassium feldspar crystal, releasing argon gas, which is measured in a gas mass spectrometer. Because the argon in the crystal has accumulated at a known rate, the amount released reveals the age of the rock and thus of nearby fossils. The margin of error is less than 1%.



FIGURE 23.7

Nearly human. These four skulls, all photographed from the same angle, are among the best specimens available of the key *Australopithecus* species.

Other Kinds of *Australopithecus*

In 1938, a second, stockier kind of *Australopithecus* was unearthed in South Africa. Called *A. robustus*, it had massive teeth and jaws. In 1959, in East Africa, Mary Leakey discovered a third kind of *Australopithecus*—*A. boisei* (after Charles Boise, an American-born businessman who contributed to the Leakeys' projects)—who was even more stockily built. Like the other australopithecines, *A. boisei* was very old—almost 2 million years. Nicknamed “Nutcracker man,” *A. boisei* had a great bony ridge—a Mohawk haircut of bone—on the crest of the head to anchor its immense jaw muscles (figure 23.7).

In 1974, anthropologist Don Johanson went to the remote Afar Desert of Ethiopia in search of early human fossils and hit the jackpot. He found the most complete, best preserved australopithecine skeleton known. Nicknamed “Lucy,” the skeleton was 40% complete and over 3 million years old. The skeleton and other similar fossils have been assigned the scientific name *Australopithecus afarensis* (from the Afar Desert). The shape of the pelvis indicated that Lucy was a female, and her leg bones proved she walked upright. Her teeth were distinctly hominid, but her head was shaped like an ape's, and her brain was no larger than that of a chimpanzee, about 400 cubic centimeters, about the size of a large orange. More than 300 specimens of *A. afarensis* have since been discovered.

In the last 10 years, three additional kinds of australopithecines have been reported. These seven species provide ample evidence that australopithecines were a diverse group, and additional species will undoubtedly be described by future investigators. The evolution of hominids seems to have begun with an initial radiation of numerous species.

Early Australopithecines Were Bipedal

We now know australopithecines from hundreds of fossils. The structure of these fossils clearly indicates that australopithecines walked upright. These early hominids weighed about 18 kilograms and were about 1 meter tall. Their dentition was distinctly hominid, but their brains were not any larger than those of apes, generally 500 cc or less. *Homo* brains, by comparison, are usually larger than 600 cc; *modern H. sapiens* brains average 1350 cc. Australopithecine fossils have been found only in Africa. Although all the fossils to date come from sites in South and East Africa (except for one specimen from Chad), it is probable that they lived over a much broader area of Africa. Only in South and East Africa, however, are sediments of the proper age exposed to fossil hunters.

The australopithecines were hominids that walked upright and lived in Africa over 3 million years ago.



FIGURE 23.7 (continued).

The Beginning of Hominid Evolution

The Origins of Bipedalism

For much of this century, biologists have debated the sequence of events that led to the evolution of hominids. A key element may have been bipedalism. Bipedalism seems to have evolved as our ancestors left dense forests for grasslands and open woodland (figure 23.8). One school of thought proposes that hominid brains enlarged first, and then hominids became bipedal. Another school sees bipedalism as a precursor to bigger brains. Those who favor the brain-first hypothesis speculate that human intelligence was necessary to make the decision to walk upright and move out of the forests and onto the grassland. Those who favor the bipedalism-first hypothesis argue that bipedalism freed the forelimbs to manufacture and use tools, favoring the subsequent evolution of bigger brains.

A treasure trove of fossils unearthed in Africa has settled the debate. These fossils demonstrate that bipedalism extended back 4 million years ago; knee joints, pelvis, and leg bones all exhibit the hallmarks of an upright stance. Substantial brain expansion, on the other hand, did not appear until roughly 2 million years ago. In hominid evolution, upright walking clearly preceded large brains.

Remarkable evidence that early hominids were bipedal is a set of some 69 hominid footprints found at Laetoli, East Africa. Two individuals, one larger than the other, walked upright side-by-side for 27 meters, their footprints preserved in 3.7-million-year-old volcanic ash (see figure 23.1). Importantly, the big toe is not splayed out to the side as in a monkey or ape—the footprints were clearly made by a hominid.

The evolution of bipedalism marks the beginning of the hominids. The reason why bipedalism evolved in hominids remains a matter of controversy. No tools appeared until 2.5 million years ago, so toolmaking seems an unlikely cause. Alternative ideas suggest that walking upright is faster and uses less energy than walking on four legs; that an upright posture permits hominids to pick fruit from trees and see over tall grass; that being upright reduces the body surface exposed to the sun's rays; that an upright stance aided the wading of aquatic hominids, and that bipedalism frees the forelimbs of males to carry food back to females, encouraging pair-bonding. All of these suggestions have their proponents, and none are universally accepted. The origin of bipedalism, the key event in the evolution of hominids, remains a mystery.

The Root of the Hominid Tree

The Oldest Known Hominid. In 1994, a remarkable, nearly complete fossil skeleton was unearthed in Ethiopia. The skeleton is still being painstakingly assembled, but it seems almost certainly to have been bipedal; the foramen magnum, for example, is situated far forward, as in other bipedal hominids. Some 4.4 million years old, it is the most ancient hominid yet discovered. It is significantly more apelike than any australopithecine and so has been assigned to a new genus, *Ardipithecus* from the local Afar language *ardi* for “ground” and the Greek *pithecus* for “ape” (figure 23.9a).

The First Australopithecine. In 1995, hominid fossils of nearly the same age, 4.2 million years old, were found in the Rift Valley in Kenya. The fossils are fragmentary, but they include complete upper and lower jaws, a piece of the



FIGURE 23.8

A reconstruction of an early hominid walking upright. These articulated plaster skeletons, made by Owen Lovejoy and his students at Kent State University, depict an early hominid (*Australopithecus afarensis*) walking upright.

FIGURE 23.9

Hominid fossils. (a) Our earliest known ancestor. A tooth from *Ardipithecus ramidus*, discovered in 1994. The name *ramidus* is from the Latin word for “root,” as this is thought to be the root of the hominid family tree. The earliest known hominid, at 4.4 million years old, *A. ramidus* was about the size of a chimpanzee and apparently could walk upright. (b) The earliest australopithecine. This fossil jaw of *Australopithecus* is about 4.2 million years old, making *A. anamensis* the oldest known australopithecine.



(a)



(b)

skull, arm bones, and a partial leg bone. The fossils were assigned to the species *Australopithecus anamensis* (figure 23.9b); *anam* is the Turkana word for lake. They were categorized in the genus *Australopithecus* rather than *Ardipithecus* because the fossils have bipedal characteristics and are much less apelike than *A. ramidus*. Although clearly australopithecine, the fossils are intermediate in many ways between apes and *A. afarensis*. Numerous fragmentary specimens of *A. anamensis* have since been found. Most researchers agree that these slightly built *A. anamensis* individuals represent the true base of our family tree, the first members of the genus *Australopithecus*, and thus ancestor to *A. afarensis* and all other australopithecines.

Differing Views of the Hominid Family Tree

Investigators take two different philosophical approaches to characterizing the diverse group of African hominid fossils. One group focuses on common elements in different fossils, and tends to lump together fossils that share key characters. Differences between the fossils are attributed to diversity within the group. Other investigators focus more pointedly on the differences between hominid fossils. They are more inclined to assign fossils that exhibit differences to different species. The hominid phylogenetic tree in figure 23.10 presents such a view. Where the “lumpers” tree presents three species of *Homo*, for example, the “splitters” tree presents no fewer than seven! At this point, it is not possible to decide which view is correct; more fossils are needed to determine how much

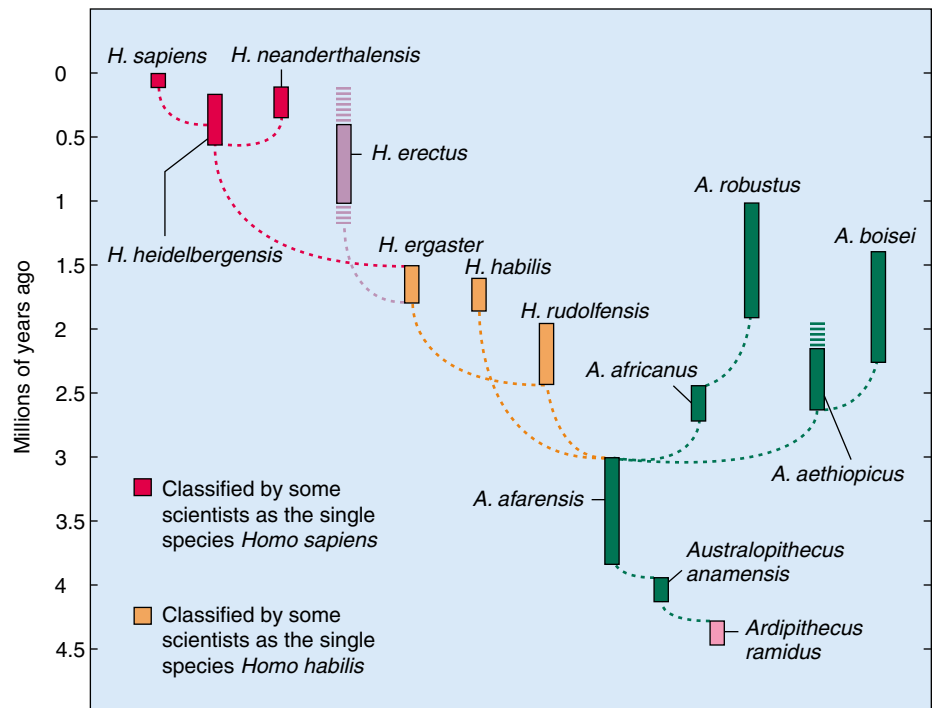


FIGURE 23.10

A hominid evolutionary tree. In this tree, the most widely accepted, the vertical bars show the known dates of first and last appearances of proposed species; bars are broken where dates are uncertain. Six species of *Australopithecus* and seven of *Homo* are included.

of the differences between fossils represents within-species variation and how much characterizes between-species differences.

The evolution of bipedalism—walking upright—marks the beginning of hominid evolution, although no one is quite sure why bipedalism evolved. The root of the hominid evolutionary tree is only imperfectly known. The earliest australopithecine yet described is *A. anamensis*, over 4 million years old.

23.3 The genus *Homo* evolved in Africa.

African Origin: Early *Homo*

The first humans evolved from australopithecine ancestors about 2 million years ago. The exact ancestor has not been clearly defined, but is commonly thought to be *A. afarensis*. Only within the last 30 years have a significant number of fossils of early *Homo* been uncovered. An explosion of interest has fueled intensive field exploration in the last few years, and new finds are announced regularly; every year, our picture of the base of the human evolutionary tree grows clearer. The account given here will undoubtedly be outdated by future discoveries, but it provides a good example of science at work.

Homo habilis

In the early 1960s, stone tools were found scattered among hominid bones close to the site where *A. boisei* had been unearthed. Although the fossils were badly crushed, painstaking reconstruction of the many pieces suggested a skull with a brain volume of about 680 cubic centimeters, larger than the australopithecine range of 400 to 550 cubic centimeters. Because of its association with tools, this early human was called *Homo habilis*, meaning “handy man.” Partial skeletons discovered in 1986 indicate that *H. habilis* was small in stature, with arms longer than legs and a skeleton much like *Australopithecus*. Because of its general similarity to australopithecines, many researchers at first questioned whether this fossil was human.

Homo rudolfensis

In 1972, Richard Leakey, working east of Lake Rudolf in northern Kenya, discovered a virtually complete skull about the same age as *H. habilis*. The skull, 1.9 million years old, had a brain volume of 750 cubic centimeters and many of the characteristics of human skulls—it was clearly human and not australopithecine. Some anthropologists assign this skull to *H. habilis*, arguing it is a large male. Other anthropologists assign it to a separate species, *H. rudolfensis*, because of its substantial brain expansion.

Homo ergaster

Some of the early *Homo* fossils being discovered do not easily fit into either of these species (figure 23.11). They tend to have even larger brains than *H. rudolfensis*, with skeletons less like an australopithecine and more like a modern human in both size and proportion. Interestingly, they also have small cheek teeth, as modern humans do. Some anthropologists have placed these specimens in a third species of early *Homo*, *H. ergaster* (*ergaster* is from the Greek for “workman”).



FIGURE 23.11

Early *Homo*. This skull of a boy, who apparently died in early adolescence, is 1.6 million years old and has been assigned to the species *Homo ergaster* (a form of *Homo habilis* recognized by some as a separate species). Much larger than earlier hominids, he was about 1.5 meters in height and weighed approximately 47 kilograms.

How Diverse Was Early *Homo*?

Because so few fossils have been found of early *Homo*, there is lively debate about whether they should all be lumped into *H. habilis* or split into the three species *H. rudolfensis*, *H. habilis*, and *H. ergaster*. If the three species designations are accepted, as increasing numbers of researchers are doing, then it would appear that *Homo* underwent an adaptive radiation (as described in chapter 22) with *H. rudolfensis* the most ancient species, followed by *H. habilis* and then *H. ergaster*. Because of its modern skeleton, *H. ergaster* is thought the most likely ancestor to later species of *Homo* (see figure 23.10).

Early species of *Homo*, the oldest members of our genus, had a distinctly larger brain than australopithecines and most likely used tools. There may have been several different species.

Out of Africa: *Homo erectus*

Our picture of what early *Homo* was like lacks detail, because it is based on only a few specimens. We have much more information about the species that replaced it, *Homo erectus*.

Java Man

After the publication of Darwin's book *On the Origin of Species* in 1859, there was much public discussion about "the missing link," the fossil ancestor common to both humans and apes. Puzzling over the question, a Dutch doctor and anatomist named Eugene Dubois decided to seek fossil evidence of the missing link in the home country of the orangutan, Java. Dubois set up practice in a river village in eastern Java. Digging into a hill that villagers claimed had "dragon bones," he unearthed a skull cap and a thighbone in 1891. He was very excited by his find, informally called Java man, for three reasons:

1. The structure of the thigh bone clearly indicated that the individual had long, straight legs and was an excellent walker.
2. The size of the skull cap suggested a very large brain, about 1000 cubic centimeters.
3. Most surprisingly, the bones seemed as much as 500,000 years old, judged by other fossils Dubois unearthed with them.

The fossil hominid that Dubois had found was far older than any discovered up to that time, and few scientists were willing to accept that it was an ancient species of human.

Peking Man

Another generation passed before scientists were forced to admit that Dubois had been right all along. In the 1920s a skull was discovered near Peking (now Beijing), China, that closely resembled Java man. Continued excavation at the site eventually revealed 14 skulls, many excellently preserved. Crude tools were also found, and most important of all, the ashes of campfires. Casts of these fossils were distributed for study to laboratories around the world. The originals were loaded onto a truck and evacuated from Peking at the beginning of World War II, only to disappear into the confusion of history. No one knows what happened to the truck or its priceless cargo. Fortunately, Chinese scientists have excavated numerous additional skulls of Peking man since 1949.

A Very Successful Species

Java man and Peking man are now recognized as belonging to the same species, *Homo erectus*. *Homo erectus* was a lot larger than *Homo habilis*—about 1.5 meters tall. It had a large brain, about 1000 cubic centimeters (figure 23.12),

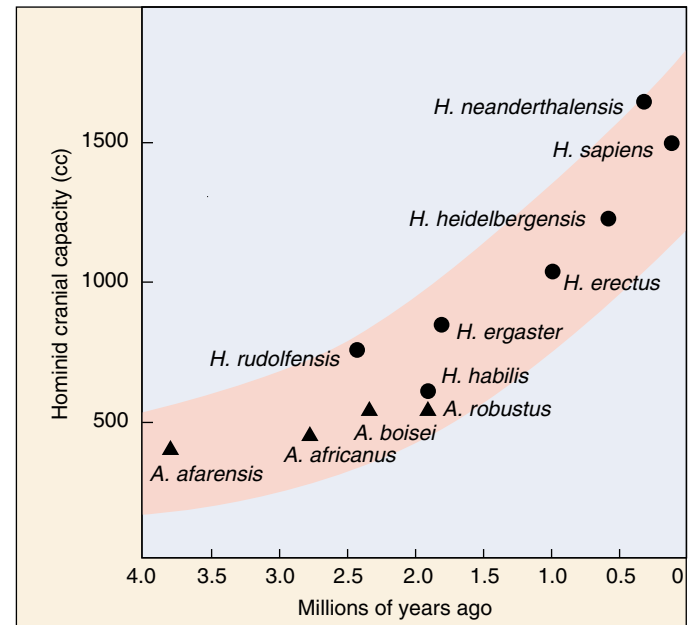


FIGURE 23.12

Brain size increased as hominids evolved. *Homo erectus* had a larger brain than early *Homo*, which in turn had larger brains than those of the australopithecines with which they shared East African grasslands. Maximum brain size (and apparently body size) was attained by *H. neanderthalensis*. Both brain and body size appear to have declined some 10% in recent millennia.

and walked erect. Its skull had prominent brow ridges and, like modern humans, a rounded jaw. Most interesting of all, the shape of the skull interior suggests that *H. erectus* was able to talk.

Where did *H. erectus* come from? It should come as no surprise to you that it came out of Africa. In 1976 a complete *H. erectus* skull was discovered in East Africa. It was 1.5 million years old, a million years older than the Java and Peking finds. Far more successful than *H. habilis*, *H. erectus* quickly became widespread and abundant in Africa, and within 1 million years had migrated into Asia and Europe. A social species, *H. erectus* lived in tribes of 20 to 50 people, often dwelling in caves. They successfully hunted large animals, butchered them using flint and bone tools, and cooked them over fires—the site in China contains the remains of horses, bears, elephants, deer, and rhinoceroses.

Homo erectus survived for over a million years, longer than any other species of human. These very adaptable humans only disappeared in Africa about 500,000 years ago, as modern humans were emerging. Interestingly, they survived even longer in Asia, until about 250,000 years ago.

Homo erectus evolved in Africa, and migrated from there to Europe and Asia.

23.4 Modern humans evolved quite recently.

The Last Stage of Hominid Evolution

The evolutionary journey to modern humans entered its final phase when modern humans first appeared in Africa about 600,000 years ago. Investigators who focus on human diversity consider there to have been three species of modern humans: *Homo heidelbergensis*, *H. neanderthalensis*, and *H. sapiens* (see figure 23.10). Other investigators lump the three species into one, *H. sapiens* (“wise man”). The oldest modern human, *Homo heidelbergensis*, is known from a 600,000-year-old fossil from Ethiopia. Although it coexisted with *H. erectus* in Africa, *H. heidelbergensis* has more advanced anatomical features, such as a bony keel running along the midline of the skull, a thick ridge over the eye sockets, and a large brain. Also, its forehead and nasal bones are very like those of *H. sapiens*.

As *H. erectus* was becoming rarer, about 130,000 years ago, a new species of human arrived in Europe from Africa. *Homo neanderthalensis* likely branched off from the ancestral line leading to modern humans as long as 500,000 years ago. Compared with modern humans, Neanderthals were short, stocky, and powerfully built. Their skulls were massive, with protruding faces, heavy, bony ridges over the brows (figure 23.13), and larger brain-cases.

Out of Africa—Again?

The oldest fossil known of *Homo sapiens*, our own species, is from Ethiopia and is about 130,000 years old. Other fossils from Israel appear to be between 100,000 and 120,000 years old. Outside of Africa and the Middle East, there are no clearly dated *H. sapiens* fossils older than roughly 40,000 years of age. The implication is that *H. sapiens* evolved in Africa, then migrated to Europe and Asia, the Out-of-Africa model. An opposing view, the Multiregional model, argues that the human races independently evolved from *H. erectus* in different parts of the world.

Recently, scientists studying human mitochondrial DNA have added fuel to the fire of this controversy. Because DNA accumulates mutations over time, the oldest populations should show the greatest genetic diversity. It turns out that the greatest number of different mitochondrial DNA sequences occur among modern Africans. This result is consistent with the hypothesis that humans have been living in Africa longer than on any other continent, and from there spread to all parts of the world, retracing the path taken by *H. erectus* half a million years before (figure 23.14).



FIGURE 23.13

Our own genus. These four skulls illustrate the changes that have occurred during the evolution of the genus *Homo*. The *Homo sapiens* is essentially the same as human skulls today. The skulls were photographed from the same angle.

A clearer analysis is possible using chromosomal DNA, segments of which are far more variable than mitochondrial DNA, providing more “markers” to compare. When a variable segment of DNA from human chromosome 12 was analyzed in 1996, a clear picture emerged. A total of 24 different versions of this segment were found. Fully 21 of them were present in human populations in Africa, while three were found in Europeans and only two in Asians and in Americans. This result argues strongly that chromosome 12 has existed in Africa far longer than among non-African humans, strongly supporting an African origin of *H. sapiens*. Recently discovered fossils of early *H. sapiens* from Africa also lend strong support to this hypothesis.

Homo sapiens, our species, seems to have evolved in Africa and then, like *H. erectus* before it, migrated to Europe and Asia.

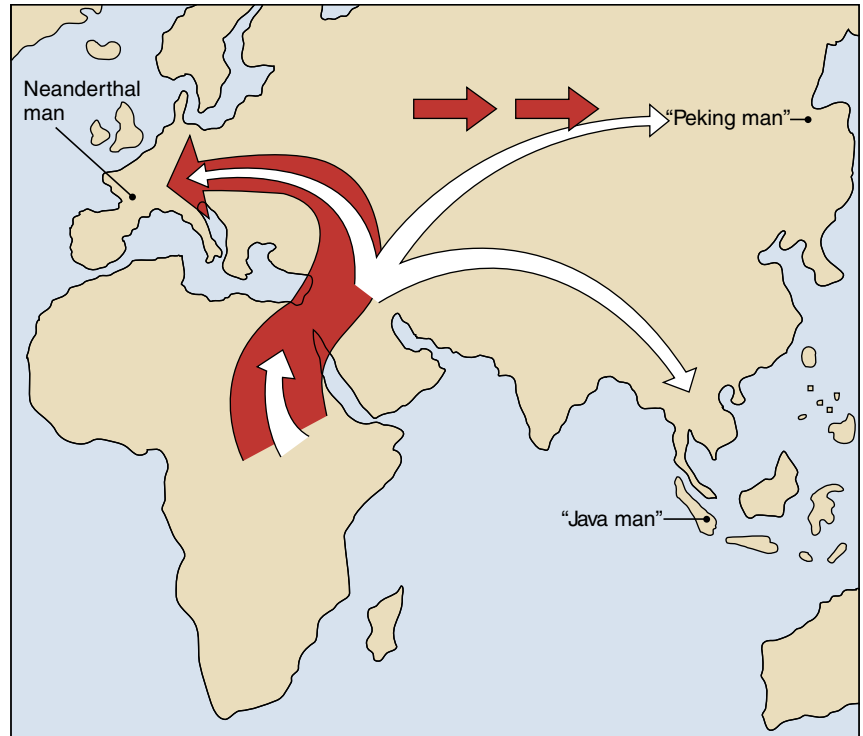


FIGURE 23.14

Out of Africa—many times. A still-controversial theory suggests that *Homo* spread from Africa to Europe and Asia repeatedly. First, *Homo erectus* (white arrow) spread as far as Java and China. Later, *H. erectus* was followed and replaced by *Homo neanderthalensis*, a pattern repeated again still later by *Homo sapiens* (red arrow).



FIGURE 23.13 (continued)

Our Own Species: *Homo sapiens*

H. sapiens is the only surviving species of the genus *Homo*, and indeed is the only surviving hominid. Some of the best fossils of *Homo sapiens* are 20 well-preserved skeletons with skulls found in a cave near Nazareth in Israel. Modern dating techniques date these humans to between 90,000 and 100,000 years old. The skulls are modern in appearance, with high, short braincases, vertical foreheads with only slight brow ridges, and a cranial capacity of roughly 1550 cc, well within the range of modern humans.

Cro-Magnons Replace the Neanderthals

The Neanderthals (classified by many paleontologists as a separate species *Homo neanderthalensis*) were named after the Neander Valley of Germany where their fossils were first discovered in 1856. Rare at first outside of Africa, they became progressively more abundant in Europe and Asia, and by 70,000 years ago had become common. The Neanderthals made diverse tools, including scrapers, spearheads, and handaxes. They lived in huts or caves. Neanderthals took care of their injured and sick and commonly buried their dead, often placing food, weapons, and even flowers with the bodies. Such attention to the dead strongly suggests that they believed in a life after death. This is the first evidence of the symbolic thinking characteristic of modern humans.

Fossils of *H. neanderthalensis* abruptly disappear from the fossil record about 34,000 years ago and are replaced by fossils of *H. sapiens* called the Cro-Magnons (named after the valley in France where their fossils were first discovered). We can only speculate why this sudden replacement occurred, but it was complete all over Europe in a short period. There is some evidence that the Cro-Magnons came from Africa—fossils of essentially modern aspect but as much as 100,000 years old have been found there. Cro-Magnons seem to have replaced the Neanderthals completely in the Middle East by 40,000 years ago, and then spread across Europe, coexisting and possibly even interbreeding with the Neanderthals for several thousand years. The Cro-Magnons had a complex social organization and are thought to have had full language capabilities. They lived by hunting. The world was cooler than it is now—the time of the last great ice age—and Europe was covered with grasslands inhabited by large herds of grazing animals. Pictures of them can be seen in elaborate and often beautiful cave paintings made by Cro-Magnons throughout Europe (figure 23.15).

Humans of modern appearance eventually spread across Siberia to North America, which they reached at least 13,000 years ago, after the ice had begun to retreat and a land bridge still connected Siberia and Alaska. By 10,000 years ago, about 5 million people inhabited the entire world (compared with more than 6 billion today).



FIGURE 23.15

Cro-Magnon art. Rhinoceroses are among the animals depicted in this remarkable cave painting found in 1995 near Vallon-Pont d'Arc, France.

Homo sapiens Are Unique

We humans are animals and the product of evolution. Our evolution has been marked by a progressive increase in brain size, distinguishing us from other animals in several ways. First, humans are able to make and use tools effectively—a capability that, more than any other factor, has been responsible for our dominant position in the animal kingdom. Second, although not the only animal capable of conceptual thought, we have refined and extended this ability until it has become the hallmark of our species. Lastly, we use symbolic language and can with words shape concepts out of experience. Our language capability has allowed the accumulation of experience, which can be transmitted from one generation to another. Thus, we have what no other animal has ever had: extensive cultural evolution. Through culture, we have found ways to change and mold our environment, rather than changing evolutionarily in response to the demands of the environment. We control our biological future in a way never before possible—an exciting potential and frightening responsibility.

Our species, *Homo sapiens*, is good at conceptual thought and tool use, and is the only animal that uses symbolic language.

Human Races

Human beings, like all other species, have differentiated in their characteristics as they have spread throughout the world. Local populations in one area often appear significantly different from those that live elsewhere. For example, northern Europeans often have blond hair, fair skin, and blue eyes, whereas Africans often have black hair, dark skin, and brown eyes. These traits may play a role in adapting the particular populations to their environments. Blood groups may be associated with immunity to diseases more common in certain geographical areas, and dark skin shields the body from the damaging effects of ultraviolet radiation, which is much stronger in the tropics than in temperate regions.

All human beings are capable of mating with one another and producing fertile offspring. The reasons that they do or do not choose to associate with one another are purely psychological and behavioral (cultural). The number of groups into which the human species might logically be divided has long been a point of contention. Some contemporary anthropologists divide people into as many as 30 “races,” others as few as three: Caucasoid, Negroid, and Oriental. American Indians, Bushmen, and Aborigines are examples of particularly distinctive subunits that are sometimes regarded as distinct groups.

The problem with classifying people or other organisms into races in this fashion is that the characteristics used to define the races are usually not well correlated with one another, and so the determination of race is always somewhat arbitrary. Humans are visually oriented; consequently, we have relied on visual cues—primarily skin color—to define races. However, when other types of characters, such as blood groups, are examined, patterns of variation correspond very poorly with visually determined racial classes. Indeed, if one were to break the human species into subunits based on overall genetic similarity, the groupings would be very different than those based on skin color or other visual features (figure 23.16).

In human beings, it is simply not possible to delimit clearly defined races that reflect biologically differentiated and well-defined groupings. The reason is simple: different groups of people have constantly intermingled and interbred with one another during the entire course of history. This constant gene flow has prevented the human species from fragmenting in highly differentiated subspecies. Those characteristics that are differentiated among populations, such as skin color, represent classic examples of the antagonism between gene flow and natural selection. As we saw in chapter 20, when selection is strong enough, as it is for dark coloration in tropical regions, populations can differentiate even in the presence of gene flow. However, even in cases such as this, gene flow will still ensure that populations are relatively homogeneous for genetic variation at other loci.

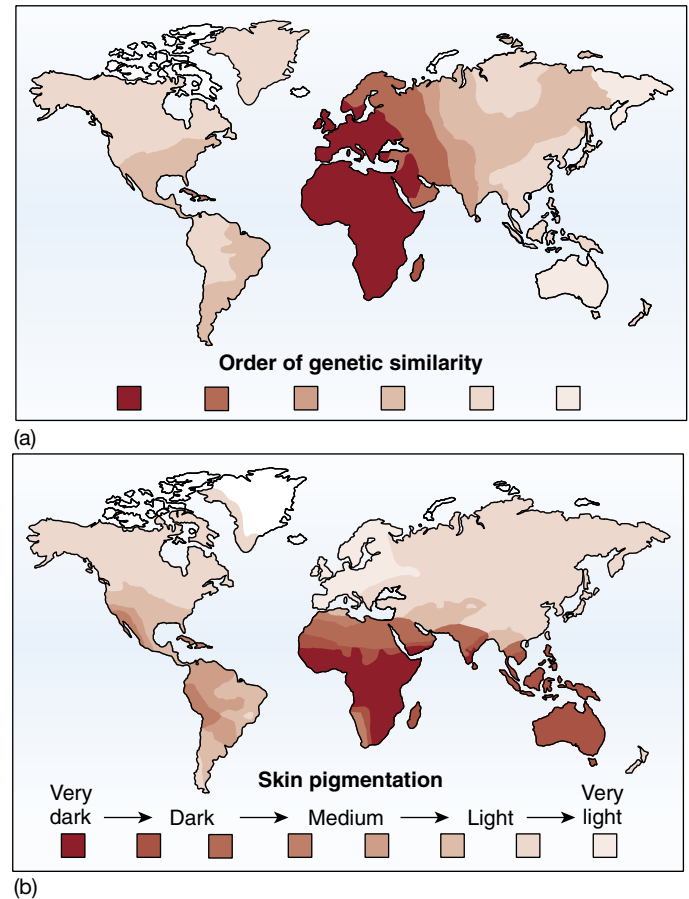


FIGURE 23.16
Patterns of genetic variation in human populations differ from patterns of skin color variation. (a) Genetic variation among *Homo sapiens*. Eight categories of humans were recognized based on overall similarity at many enzyme and blood group genetic loci. The code below the figure is arranged in order of similarity. (b) Similarity among *Homo sapiens* based on skin color. The categories are arranged by amount of pigmentation in the skin.

For this reason, relatively little of the variation in the human species represents differences between the described races. Indeed, one study calculated that only 8% of all genetic variation among humans could be accounted for as differences that exist among racial groups; in other words, the human racial categories do a very poor job in describing the vast majority of genetic variation that exists in humans. For this reason, most modern biologists reject human racial classifications as reflecting patterns of biological differentiation in the human species. This is a sound biological basis for dealing with each human being on his or her own merits and not as a member of a particular “race.”

Human races do not reflect significant patterns of underlying biological differentiation.



Summary

Questions

Media Resources

23.1 The evolutionary path to humans starts with the advent of primates.

- Prehensile (grasping) fingers and toes and binocular vision were distinct adaptations that allowed early primates to be successful in their particular environments.
- Mainly diurnal (day-active) anthropoids and mainly nocturnal (night-active) prosimians diverged about 40 million years ago. Anthropoids include monkeys, apes, and humans, and all exhibit complex social interactions and enlarged brains.
- The hominoids evolved from anthropoid ancestors about 25 million years ago. Hominoids consist of the apes (gibbons, orangutans, gorillas, and chimpanzees) and upright-walking hominids (human beings and their direct ancestors).

1. Which characteristics were selected for in the earliest primates to allow them to become successful in their environment?
2. How do monkeys differ from prosimians?
3. How are apes distinguished from monkeys?
4. What is the best explanation for why humans and chimpanzees are so similar genetically?



• Evolution of Primates

23.2 The first hominids to evolve were australopithecines.

- Early hominids belonging to the genus *Australopithecus* were ancestral to humans. They exhibited bipedalism (walking upright on two feet) and lived in Africa over 4 million years ago.

5. When did the first hominids appear? What were they called? What distinguished them from the apes?

23.3 The genus *Homo* evolved in Africa.

- Hominids with an enlarged brain and the ability to use tools belong to the genus *Homo*. Species of early *Homo* appeared in Africa about 2 million years ago and became extinct about 1.5 million years ago.
- *Homo erectus* appeared in Africa at least 1.5 million years ago and had a much larger brain than early species of *Homo*. *Homo erectus* also walked erect and presumably was able to talk. Within a million years, *Homo erectus* migrated from Africa to Europe and Asia.

6. Why is there some doubt in the scientific community that *Homo habilis* was a true human?

7. How did *Homo erectus* differ from *Homo habilis*?



• On Science Article: Human Evolution

23.4 Modern humans evolved quite recently.

- The modern species of *Homo* appeared about 600,000 years ago in Africa and about 350,000 years ago in Eurasia.
- The Neanderthals appeared in Europe about 130,000 years ago. They made diverse tools and showed evidence of symbolic thinking.
- Studies of mitochondrial DNA suggest (but do not yet prove) that all of today's human races originated from Africa.
- Categorization of humans into races does not adequately reflect patterns of genetic differentiation among people in different parts of the world.

8. The greatest number of different mitochondrial DNA sequences in humans occurs in Africa. What does this tell us about human evolution?

9. How did Cro-Magnons differ from Neanderthals? Is there any evidence that they coexisted with Neanderthals? If so, where and when?

10. Are the commonly recognized human races equivalent to subspecies of other plant and animal species?



• Hominid History

Ecology and Behavior

Why do tropical songbirds lay fewer eggs?

Sometimes odd generalizations in science lead to unexpected places. Take, for example, a long obscure monograph published in 1944 by British ornithologist (bird expert) Reginald Moreau in the journal *Ibis* on bird eggs. Moreau had worked in Africa for many years before moving to a professorship in England in the early 1940s. He was not in England long before noting that the British songbirds seemed to lay more eggs than he was accustomed to seeing in nests in Africa. He set out to gather information on songbird clutch size (that is, the number of eggs in a nest) all over the world.

Wading through a mountain of data (his *Ibis* paper is 51 pages long!), Moreau came to one of these odd generalizations: songbirds in the tropics lay fewer eggs than their counterparts at higher latitudes (see above *right*). Tropical songbirds typically lay a clutch of 2 or 3 eggs, on average, while songbirds in temperate and subarctic regions generally lay clutches of 4 to 6 eggs, and some species as many as 10. The trend is general, affecting all groups of songbirds in all regions of the world.

What is a biologist to make of such a generalization? At first glance, we would expect natural selection to maximize evolutionary fitness—that is, songbirds the world over should have evolved to produce as many eggs as possible. Clearly, the birds living in the tropics have not read Darwin, as they are producing only half as many eggs as they are capable of doing.

A way out of this quandary was proposed by ornithologist Alexander Skutch in 1949. He argued that birds produced just enough offspring to offset deaths in the population. Any extra offspring would be wasteful of individuals, and so minimized by natural selection. An interesting idea, but it didn't hold water. Bird populations are not smaller in the tropics, or related to the size of the populations there.

A second idea, put forward a few years earlier in 1947 by a colleague of Moreau's, David Lack, was more promising. Lack, one of the twentieth-century's great biologists, argued that few if any birds ever produce as many eggs as they might under ideal conditions, for the simple reason that conditions in nature are rarely ideal. Natural selection will indeed tend to maximize reproduc-



This Kentucky warbler is tending her nest of eggs. A similar species in the tropics would lay fewer eggs. Why?

tive rate (that is, the number of eggs laid in clutches) as Darwin predicted, but only to the greatest level possible within the limits of resources. There is nothing here that would have surprised Darwin. Birds lay fewer eggs in the tropics simply because parents can gather fewer resources to provide their young there—competition is just too fierce, resources too scanty.

Lack went on to construct a general theory of clutch size in birds. He started with the sensible assumption that in a resource-limited environment birds can supply only so much food to their young. Thus, the more offspring they have, the less they can feed each nestling. As a result, Lack proposed that natural selection will favor a compromise between offspring number and investment in each offspring, which maximizes the number of offspring which are fed enough to survive to maturity.

The driving force behind Lack's theory of optimal clutch size is his idea that broods with too many offspring would be undernourished, reducing the probability that the chicks would survive. In Lack's own words:

"The average clutch-size is ultimately determined by the average maximum number of young which the parents can successfully raise in the region and at the season in question, i.e. ... natural selection eliminates a disproportionately large number of young in those clutches which are higher than the average, through the inability of the parents to get enough food for their young, so that some or all of the brood die before or soon after fledging (leaving the nest), with the result that few or no descendants are left with their parent's propensity to lay a larger clutch."